

Bell
MODEL **412**



ROTORCRAFT FLIGHT MANUAL

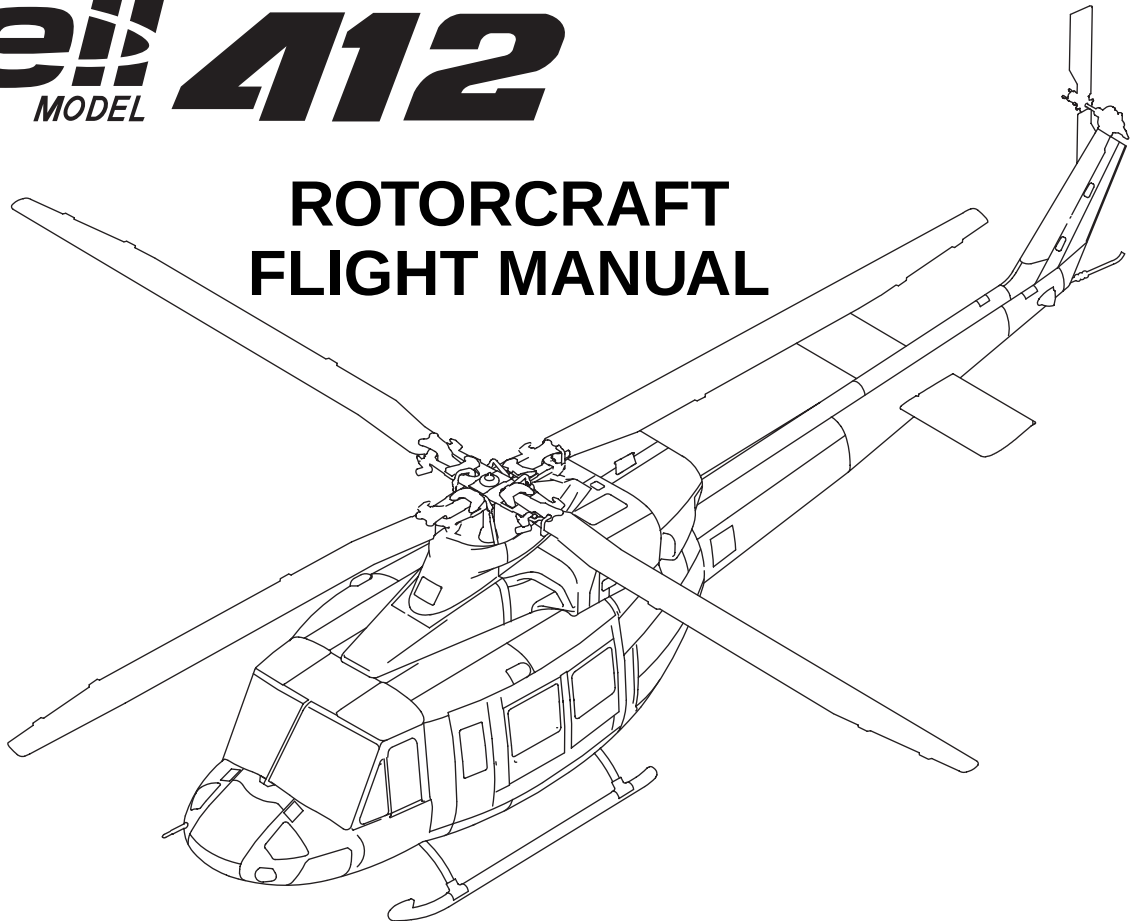
**BHT 33108 THROUGH 33213
AND
BHT36001 THROUGH 36019**

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ROTORCRAFT FLIGHT MANUAL



BHT 33108 — 33213 AND 36001 — 36019

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ROTORCRAFT CERTIFICATION OFFICE
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FT. WORTH, TEXAS 76193-0170

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ROTORCRAFT FLIGHT MANUAL

S/N 33108 — 33213

S/N 36001 — 36019

TEMPORARY REVISION FOR AIRSPEED RESTRICTION

This Flight Manual Temporary Revision mandates a reduction of airspeed until after compliance with ALERT SERVICE BULLETIN No. 412-96-89 Installation and/or Inspection of Tail Rotor Flapping Stop.

Insert these temporary revision pages opposite like numbered pages in Flight Manual.

DO NOT remove existing pages from Flight Manual.

Information contained herein supplements information of basic Flight Manual. For Limitations, Procedures, and Performance Data not contained in this supplement, consult basic Flight Manual.

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APRIL 29, 1992
TEMPORARY REVISION — 16 AUGUST 1996



ROTOR CRAFT FLIGHT MANUAL

S/N 33108 — 33213

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TEMPORARY REVISION FOR MAIN ROTOR DROOP RESTRAINT PREFLIGHT CHECK

This Temporary Revision addresses Main Rotor Droop Restraint Preflight Check Procedures Per ASB 412-97-91 Reissue A.

Insert these temporary revision pages opposite like numbered pages in Flight Manual.

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GENERAL INFORMATION

ORGANIZATION

The Rotorcraft Flight Manual is divided into six sections as follows:

- Section 1 — LIMITATIONS
- Section 2 — NORMAL PROCEDURES
- Section 3 — EMERGENCY AND
MALFUNCTION
PROCEDURES
- Section 4 — PERFORMANCE
- Section 5 — OPTIONAL EQUIPMENT
SUPPLEMENTS
- Section 6 — CATEGORY "A"
OPERATIONS

Sections 1 through 4 contain the FAA approved data necessary to operate the basic helicopter in a safe and efficient manner.

Section 5 contains the FAA approved supplements for optional equipment, which shall be used in conjunction with the basic Flight Manual when the respective optional equipment kits are installed.

Section 6 contains limitations, procedures and performance data for Category "A" Operations.

The Manufacturer's Data (BHT-412-MD-2) manual, contains information to be used in conjunction with the Flight Manual. The manual is divided into four sections.

- Section 1 — WEIGHT AND BALANCE
- Section 2 — SYSTEMS DESCRIPTION

Section 3 — OPERATIONAL
INFORMATION

Section 4 — HANDLING/SERVICING/
MAINTENANCE

TERMINOLOGY

WARNINGS, CAUTIONS, AND NOTES

Warnings, cautions, and notes are used throughout this manual to emphasize important and critical instructions as follows:

WARNING

AN OPERATING PROCEDURE, PRACTICE, ETC., WHICH, IF NOT CORRECTLY FOLLOWED, COULD RESULT IN PERSONAL INJURY OR LOSS OF LIFE.

CAUTION

AN OPERATING PROCEDURE, PRACTICE, ETC., WHICH, IF NOT STRICTLY OBSERVED, COULD RESULT IN DAMAGE TO OR DESTRUCTION OF EQUIPMENT.

NOTE

An operating procedure, condition, etc., which is essential to highlight.

USE OF PROCEDURAL WORDS

The concept of procedural word usage and intended meaning which has been adhered to in preparing this manual is as follows:

“Shall” has been used only when application of a procedure is mandatory.

“Should” has been used only when application of a procedure is recommended.

“May” and “need not” have been used only when application of a procedure is optional.

“Will” has been used only to indicate futurity, never to indicate a mandatory procedure.

ABBREVIATIONS AND ACRONYMS

Abbreviations and acronyms used throughout this manual are defined as follows:

AC	—	Alternating Current
ADI	—	Attitude Director Indicator
AFCS	—	Automatic Flight Control System
AGL	—	Above Ground Level
ALTN	—	Alternate
ANTI COLL	—	Anticollision
API	—	Actuator Position Indicator
ATC	—	Air Traffic Control
ATT	—	Attitude
AUTO	—	Automatic

AUX SYS	—	Auxiliary System
BAT	—	Battery
C	—	Celsius
C BOX	—	Combining Gearbox
CG	—	Center of Gravity
cm	—	Centimeter(s)
CYC CTR	—	Cyclic Center
DC	—	Direct Current
DECR	—	Decrease
DME	—	Distance Measuring Equipment
ELT	—	Emergency Locator Transmitter
EMERG	—	Emergency
ENG	—	Engine
ENG RPM	—	Engine Power Turbine RPM (N2)
F	—	Fahrenheit
FAR	—	Federal Aviation Regulation
FT	—	Force Trim or Foot/Feet
FUEL PRESS	—	Fuel Pressure
FUEL TRANS	—	Fuel Transfer
GAS PROD	—	Gas Producer (N1)
GEN	—	Generator
GOV	—	Governor
H _D	—	Density Altitude
H _P	—	Pressure Altitude

HP 1/HP 2 — Helipilot 1/Helipilot 2

H-V — Height-Velocity

HYDR SYS — Hydraulic System

IFR — Instrument Flight Rules

IGE — In Ground Effect

IGN — Ignition

IN — Inch(es)

INCR — Increase

INTCON — Interconnect

INV — Inverter

IMC — Instrument Meteorological
Conditions

ITT — Interturbine Temperature

IVSI — Instantaneous Vertical
Speed Indicator

KCAS — Knots Calibrated Airspeed

kg — Kilograms

KIAS — Knots Indicated Airspeed

KTAS — Knots True Airspeed

LB — Pound(s)

LRC — Long Range Cruise

LT — Light

MAG — Magnetic

MAX END — Maximum Endurance

MCP — Maximum Continuous
Power

mm — Millimeter(s)

NON-
ESNTL — Non Essential

NORM — Normal

OAT — Outside Air Temperature

OEI — One Engine Inoperative

OGE — Out of Ground Effect

OVRD — Override

PART SEP — Particle Separator

PNL — Panel

PRI — Primary

PSI — Pounds per Square Inch

RPM — Revolutions Per Minute

SAS — Stability Augmentation
System

SL — Sea Level

sq — Square

STBY — Standby

TEMP — Temperature

VFR — Visual Flight Rules

VG — Vertical Gyro

VHF — Very High Frequency

VMC — Visual Meteorological
Conditions V_{NE} — Never Exceed Speed V_{TOCS} — Takeoff Climbout Speed V_Y — Best Rate of Climb Speed

WSHLD — Windshield

XFEED — Crossfeed

XMSN — Transmission

Section 1

LIMITATIONS

Section 2

NORMAL PROCEDURES

Section 3

EMERGENCY AND MALFUNCTION PROCEDURES

Section 4

PERFORMANCE

Section 5

OPTIONAL EQUIPMENT SUPPLEMENTS

Section 6

CATEGORY A OPERATIONS

Section 1

LIMITATIONS

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Section 1

LIMITATIONS

GENERAL

Compliance with the limitations in this section is required by appropriate operating rules.

BASIS OF CERTIFICATION

This helicopter is certified under FAR Part 29, Category “A” and “B”.

TYPE OF OPERATION

The basic configured helicopter is approved as a fifteen-place helicopter and is certified for operation under day or night VFR nonicing conditions.

The IFR configured helicopter is certified for Category I IFR operation during day or night nonicing conditions.

Refer to Section 6 for additional limitation procedures and performance data for Category “A” operations.

REQUIRED EQUIPMENT

AFCS

AFCS shall be disengaged or operated in SAS mode during prolonged ground operation, except as required for AFCS check.

REQUIRED EQUIPMENT — IFR

In addition to the basic equipment required for certification, the 412-705-006 IFR Kit

shall be installed and the following equipment shall be operational for IFR flight:

Both helipilots HP 1 and HP 2 shall be engaged in ATT mode during IFR flight.

Heated pitot-static system

Pilot windshield wiper

3-inch standby attitude indicator

Two VHF communications radios

Two navigation receivers with auxiliary equipment appropriate to intended IFR route of flight

DME equipment

ATC transponder

Marker beacon receiver

Pilot IVSI

Force trim

Roof window blackout curtains

EMERGENCY COMM panel, if installed, (single pilot only)

OPTIONAL EQUIPMENT

Refer to appropriate Flight Manual Supplement(s) for additional limitations,

procedures, and performance data with optional equipment installed.

reduces V_{NE} . Refer to BHT-412-MD-2 and to Airspeed Limitations.

FLIGHT CREW

The minimum flight crew consists of one pilot who shall operate the helicopter from the right crew seat.

The left crew seat may be used for an additional pilot when the approved dual controls and copilot instrument kits are installed.

INTERNAL CARGO OPERATION

NOTE

Refer to applicable operating rules for internal cargo operations.

DOORS OPEN OR REMOVED

Helicopter may be flown with doors open or removed only with Standard Interior or Deluxe Interior installed. Flight operation is approved for the following alternative configurations during VFR conditions only:

Both crew doors removed.

Both sliding doors locked open or removed with both hinged panels installed or removed.

In all cases, door configuration shall be symmetrical for both sides of the fuselage.

NOTE

Opening or removing doors shifts helicopter center of gravity and

WEIGHT/CG

WEIGHT

Maximum gross weight for takeoff and landing is 11,900 pounds (5398 kilograms).

Refer to Weight-Altitude-Temperature Limitations chart (figure 1-1) for maximum allowable weight for takeoff, landing, and IGE hover operation.

Minimum gross weight for flight is 6400 pounds (2903 kilograms).

Minimum combined crew weight at fuselage station 47.0 is 170 pounds (77.1 kilograms).

CENTER OF GRAVITY — LONGITUDINAL

Longitudinal center of gravity limits vary from station 130 to 144, depending on gross weight. Refer to Gross Weight Center of Gravity Chart (figure 1-2).

CENTER OF GRAVITY — LATERAL

Lateral center of gravity limits are 4.5 inches (114.3 millimeters) left and right of fuselage centerline for all gross weights.

LOADING

NOTE

Refer to BHT-412-MD-2 for loading tables to be used in weight/CG computations.

NOTE: ALLOWABLE GROSS WEIGHTS OBTAINED FROM THIS CHART MAY EXCEED CONTINUOUS HOVER CAPABILITY UNDER CERTAIN AMBIENT CONDITIONS. REFER TO HOVER CEILING CHARTS IN SECTION 4.

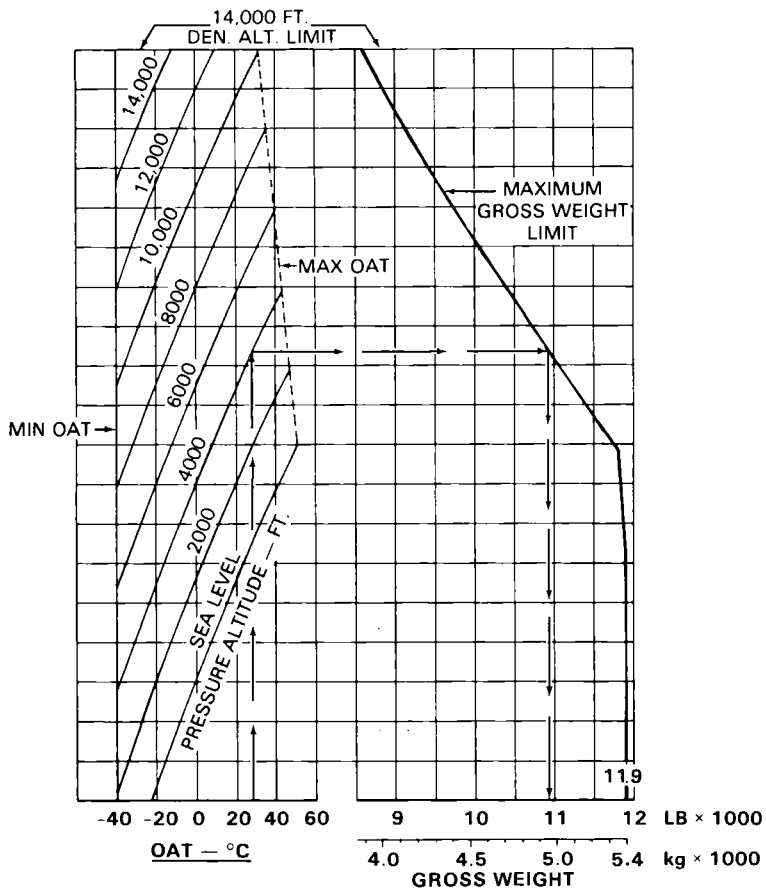


Figure 1-1. Weight-altitude-temperature limitations for takeoff, landing, and in-ground-effect maneuvers.

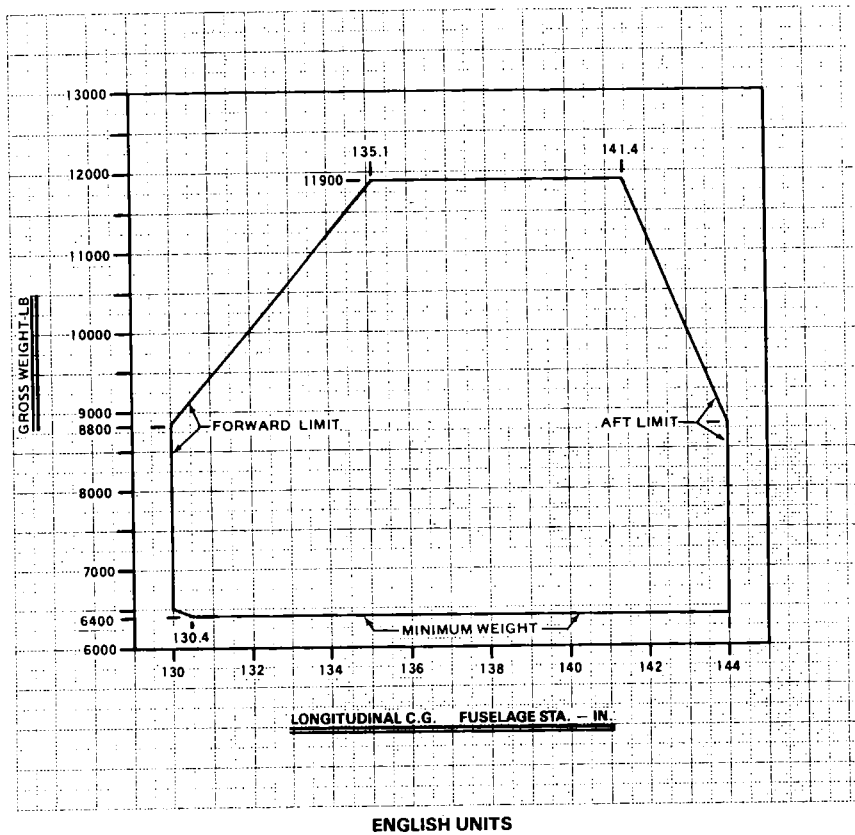


Figure 1-2. Gross weight center of gravity charts (Sheet 1 of 2).

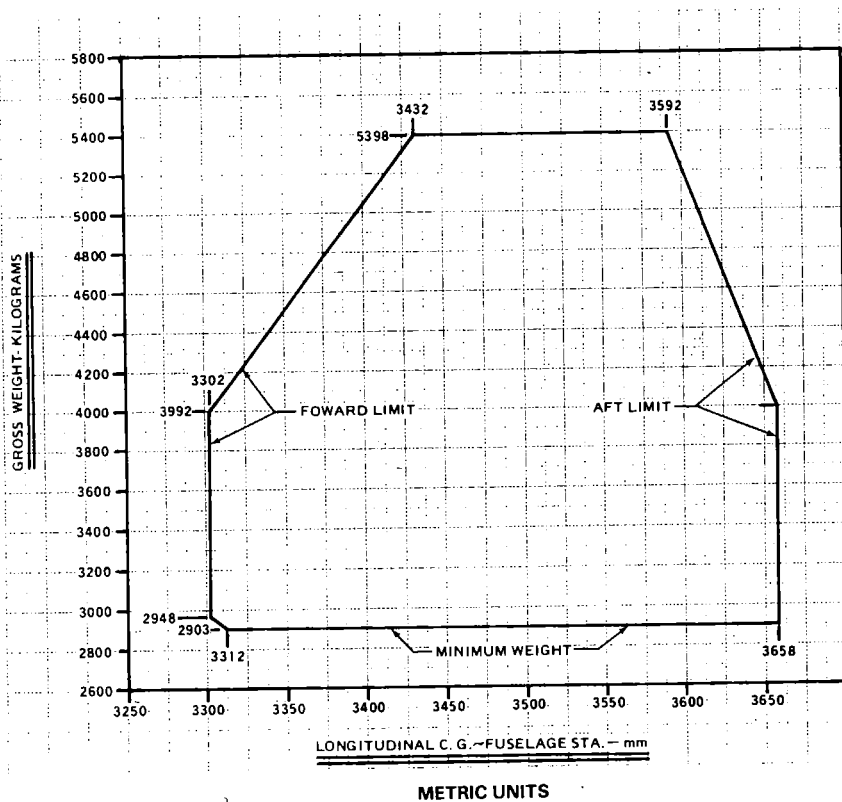


Figure 1-2. Gross weight center of gravity charts (Sheet 2 of 2).

AIRSPEED

NOTE

This limitation shall remain in effect until Alert Service Bulletin (ASB) 412-96-89 has been fully implemented.

V_{NE} is 120 KIAS. Reduce all placard V_{NE} airspeeds by 20 KIAS.



PASSENGER LOADING

The outboard facing seats should not be occupied unless at least four of the forward or aft facing passenger seats are occupied.

The above loading does not apply if cargo or a combination of cargo and passengers are being transported. It shall then be pilot responsibility to ensure helicopter is properly loaded so entire flight is conducted within the limits of gross weight center of gravity charts (figure 1-2).

INTERNAL CARGO LOADING

Maximum allowable deck loading for cargo is 100 pounds per square foot (4.9 kg/100 sq cm). Deck mounted cargo tiedown fittings are provided and have an airframe structural capacity of 1250 pounds (567.0 kilograms) vertical and 500 pounds (226.8 kilograms) horizontal per fitting. Provisions for installation of cargo tiedown fittings are incorporated in the aft cabin bulkhead and transmission support structure and have an airframe structural capacity of 1250 pounds (567.0 kilograms) at 90 degrees to the bulkhead and 500 pounds (226.8 kilograms) in any direction parallel to the bulkhead. Cargo shall be secured by an approved restraint method that will not impede access to the cargo in the event of an emergency.

Maximum allowable baggage compartment loading is 400 pounds (181 kilograms), not to exceed 100 pounds per square foot (4.9 kg/100 sq cm).

AIRSPEED

NOTE

All indicated airspeed values in this manual require instrument part number 412-075-009-105 be installed.

Minimum IFR airspeed is 60 KIAS.

Basic V_{NE} is 140 KIAS from sea level to 3000 feet density altitude at all gross weights. V_{NE} decreases for ambient conditions in accordance with airspeed limitations placard (figure 1-3).

Airspeed shall not exceed 105 KIAS (or placarded V_{NE} , if less) when operating above maximum continuous transmission torque (81%).

V_{NE} with only one helipilot engaged is 115 KIAS (or placarded V_{NE} , if less). If both helipilots are disengaged, basic V_{NE} applies.

V_{NE} for steady state autorotation is:

105 KIAS at or below 10,000 feet pressure altitude;

80 KIAS above 10,000 feet pressure altitude.

V_{NE} with doors open or removed is 60 KIAS with energy attenuating passenger seats (412-706-002) installed.

V_{NE} with doors open or removed is 100 KIAS with Blanket Interior (412-705-501 or 412-705-510), Deluxe Interior (412-705-500) or Utility Seats (412-706-018 or 205-706-043) installed.

Maximum allowable airspeed for sideward or rearward flight at or below 3000 feet H_D is 35 knots. Refer to figure 1-4 for additional limitations.

Maximum allowable tailwind or crosswind speeds for hover operations at or below 3000 feet H_D is 35 knots. Refer to figure 1-4 for additional limitations.

Refer to Critical Relative Wind Azimuths diagram in Section 4.

CLIMB/DESCENT

Maximum IFR rate of climb or descent is 1000 feet per minute.

Maximum IFR approach slope is 5 degrees.

OBSERVE TEMPORARY MAXIMUM NEVER EXCEED (V_{NE})
AIRSPEED REDLINE (MARKED AT 120 KIAS). V_{NE} IS 20 KIAS
LESS THAN THE VALUE PRESENTED ON THE AIRSPEED
LIMITATION PLACARD FOR EACH AMBIENT CONDITION.

(TYPICAL)

Figure 1-3. Placards and decals



TWIN & 30 MIN OEI 100.8%
2 1/2 MIN OEI 102.4%

TWIN & 30 MIN OEI 101.8%
2 1/2 MIN OEI 103.4%

FOR GAS PRODUCER
INSTRUMENT PART NUMBER
212-075-037-101

FOR GAS PRODUCER
INSTRUMENT PART NUMBER
212-075-037-113

OAT °C	PRESSURE ALTITUDE IN FT×1000										
	0	2	4	6	8	10	12	14	16	18	20
	INDICATED VNE KNOTS										
51.7	137	—	—	—	—	—	—	—	—	—	—
40	140	134	128	122	—	—	—	—	—	—	—
20	140	139	133	127	121	115	109	103	97	—	—
0	140	140	140	133	127	121	115	109	103	96	91
-10	140	140	140	140	131	124	118	112	106	100	94
-20	140	140	140	138	133	127	121	115	108	102	96
-30	140	139	134	129	124	120	115	110	106	101	97
-40	134	129	124	120	116	111	107	102	98	94	90
AUTOROTATION VNE 80 KIAS ABOVE 10,000 FT.											

DO NOT OPERATE
HEATER ABOVE 21
DEG C OUT AIR TEMP

DO NOT APPLY ROTOR BRAKE
ABOVE 40% RPM

BASIC FUEL CAP
2148 LBS
WITH AUX FUEL KIT
412-708-007
3212 LBS
412-708-009
2389 LBS

THIS HELICOPTER MUST BE OPERATED
IN COMPLIANCE WITH THE OPERATING
LIMITATIONS SPECIFIED IN THE FAA
APPROVED ROTORCRAFT FLIGHT MAN

IN ALTN POSITION MAINTAIN INSTRUMENT ACCURACY BY
CLOSING WINDOWS AIRVENTS AND TURNING HEATER OFF

(if installed)

412-F2-1-3

Figure 1-3. Placards and decals.

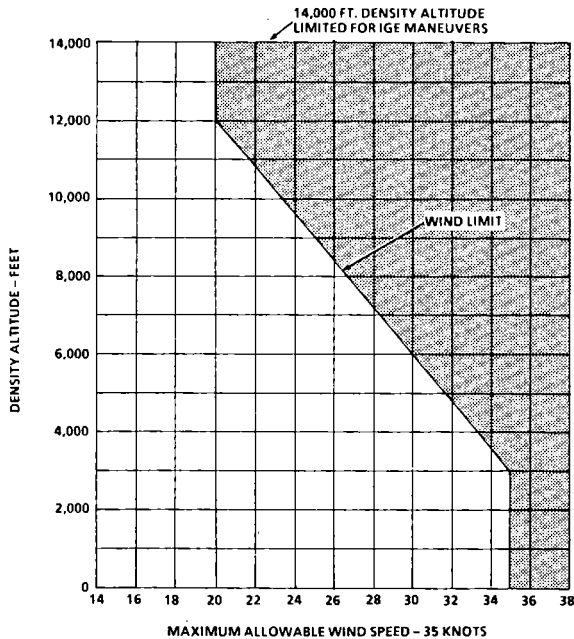
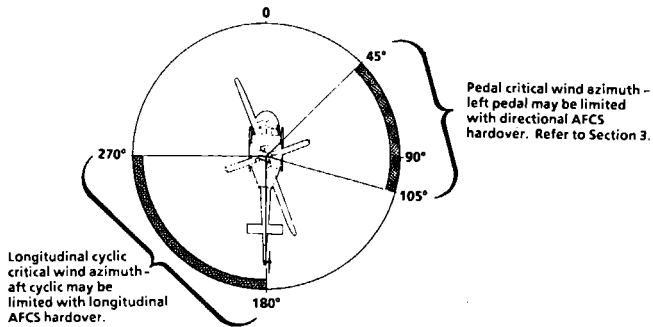


Figure 1-4. Maximum speed — sideward and rearward flight, crosswind and tailwind at a hover.

ALTITUDE

Maximum operating pressure altitude is 20,000 feet.

Maximum density altitude for takeoff, landing, and in-ground-effect maneuvers is 14,000 feet. Refer to Weight-Altitude-Temperature Limitations chart (figure 1-1).

NOTE

Refer to applicable operating rules for high altitude oxygen requirements.

AMBIENT AIR TEMPERATURE

The maximum sea level ambient air temperature for operation is +51.7°C (+125°F) and decreases with pressure altitude at the standard lapse rate of 2°C (3.6°F)/1000 feet to 20,000 feet.

The minimum ambient temperature for operation at all altitudes is -40°C (-40°F).

HEIGHT — VELOCITY

The height-velocity limitations are critical in the event of single engine failure during takeoff, landing, or other operation near the surface (figure 1-5). The AVOID area of the Height-Velocity diagram defines the combinations of airspeed and height above ground from which a safe single engine landing on a smooth, level, firm surface cannot be assured.

The H-V diagram is valid only when the Weight-Altitude-Temperature limitations are not exceeded (figure 1-1). The diagram does not define the conditions which assure continued flight following an engine failure nor the conditions from which a safe power off landing can be made.

MANEUVERING

Aerobatic maneuvers are prohibited.

SLOPE LANDINGS

Slope landings are limited to side slopes not to exceed 10 degrees.

ELECTRICAL

BATTERY

Maximum battery case temperature is 54.5°C (130°F), as indicated by illumination of BATTERY TEMP warning light.

WARNING

BATTERY SHALL NOT BE USED FOR ENGINE START AFTER ILLUMINATION OF BATTERY TEMP LIGHT. BATTERY SHALL BE REMOVED AND SERVICED IN ACCORDANCE WITH MANUFACTURER'S INSTRUCTIONS PRIOR TO RETURN TO SERVICE.

Minimum ambient temperature for battery start when battery and helicopter have been cold soaked is -25°C (-13°F).

GENERATOR

Continuous operation	0 to 75 amps
Caution	75 to 150 amps

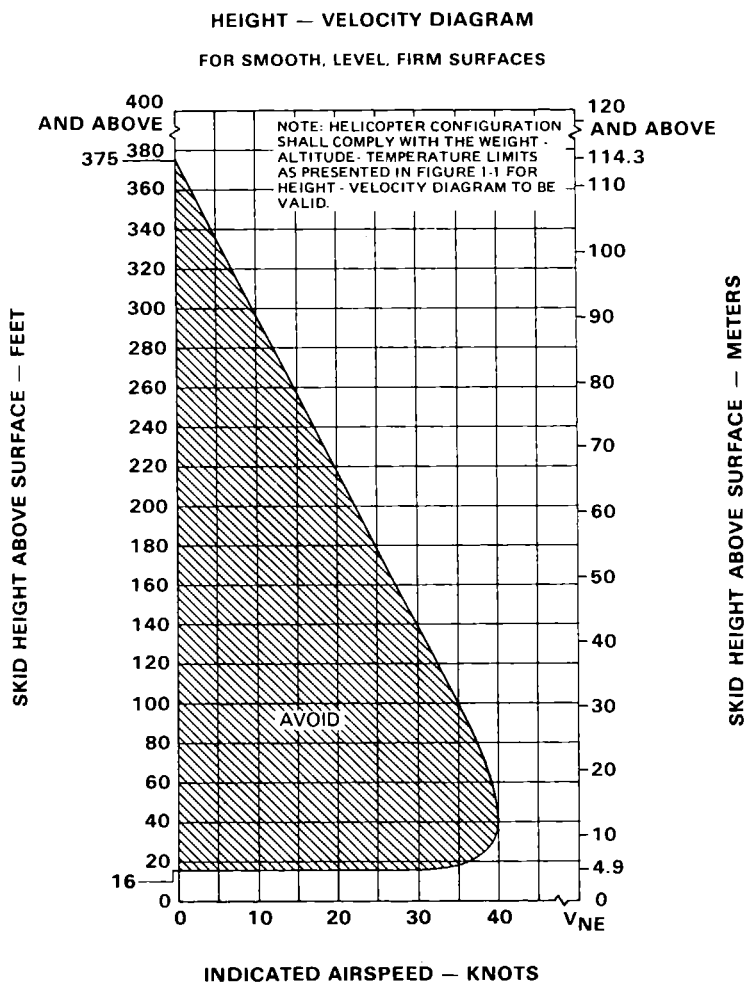


Figure 1-5. Height-velocity diagram (OEI).

NOTE

During OEI operation electrical loads may have to be reduced to remain below 150 amps.

Maximum continuous 150 amps (each)

NOTE

Ammeter needle may deflect full scale momentarily during generator assisted start of second engine.

ENGINE STARTER

Starter energizing times shall be limited as follows:

30 seconds ON

60 seconds OFF

30 seconds ON

5 minutes OFF

30 seconds ON

15 minutes OFF

GROUND POWER STARTS

28 vdc ground power units for starting shall be limited to 1000 amps maximum.

POWERPLANT

Pratt and Whitney Aircraft of Canada, Ltd. PT6T-3B.

NOTE

Operation in 2 1/2 minute or 30 minute OEI range is intended for emergency use only, when one engine becomes inoperative due to an actual malfunction.

Anytime an engine is operated in an OEI range, an entry shall be made in the helicopter logbook detailing the extent of operation in excess of twin engine takeoff power limits. This does not apply to approved ITT limits for starting.

GAS PRODUCER RPM

TWIN ENGINE OPERATION

(Instrument P/N 212-075-037-101)

Continuous operation	61 to 100.8%
Maximum continuous	100.8%
Maximum for takeoff	100.8%

(Instrument P/N 212-075-037-113)

Continuous operation	61 to 101.8%
Maximum continuous	101.8%
Maximum for takeoff	101.8%

ONE ENGINE INOPERATIVE (OEI)

(Instrument P/N 212-075-037-101)

30 minute OEI	100.8
2 1/2 minute OEI range	100.8 to 102.4%
Maximum OEI	102.4%

(Instrument P/N 212-075-037-113)

30 minute OEI	101.8
2 1/2 minute OEI range	101.8 to 103.4%
Maximum OEI	103.4%

POWER TURBINE RPM (ENG RPM)

Minimum	97%
Continuous operation	97 to 100%
Maximum continuous	100%
Operation with ENG TORQUE at or below 30%	100 to 104.5%
Maximum with ENG TORQUE at or below 30%	104.5%

INTERTURBINE TEMPERATURE (ITT)**TWIN ENGINE OPERATION**

Maximum continuous	765°C
Takeoff range (5 minutes maximum)	765 to 810°C
Maximum transient (5 seconds maximum)	850°C
Maximum for starting (2 seconds maximum above 960°C)	1090°C

CAUTION

INTENTIONAL USE OF *ITT* ABOVE 810°C IS PROHIBITED DURING NORMAL OPERATIONS EXCEPT DURING START.

ONE ENGINE INOPERATIVE (OEI)

Maximum continuous OEI	765°C
30 minute OEI range	765 to 822°C

2 1/2 minute OEI range

Maximum OEI	850°C
-------------	-------

FUEL PRESSURE

Minimum	4 psi
Continuous operation	4 to 35 psi
Maximum	35 psi

OIL PRESSURE**ENGINE**

Minimum for idle	40 psi
Operation below 79% GAS PROD RPM (N1)	40 to 80 psi
Continuous operation	80 to 115 psi
Maximum	115 psi

COMBINING GEARBOX

Minimum for idle	40 psi
Operation below 94% ENG RPM (N2)	40 to 60 psi
Continuous operation	60 to 80 psi
Maximum	80 psi

OIL TEMPERATURE**ENGINE**

Minimum	0°C
---------	-----

Continuous operation 0 to 115°C
Maximum 115°C

COMBINING GEARBOX

Minimum 0°C
Continuous operation 0 to 115°C
Maximum 115°C

ENGINE RESTART

Above 15,000 feet pressure altitude, restart shall be attempted in manual fuel control mode only.

Below 15,000 feet pressure altitude, restart may be attempted in either manual or automatic fuel control mode.

ENGINE TORQUE**TWIN ENGINE OPERATION**

Maximum allowable ENG TORQUE differential is 4% during normal operation. Refer to TRANSMISSION TORQUE LIMITS.

ONE ENGINE INOPERATIVE (OEI)

Maximum continuous OEI 58.9%
30 minute OEI range 58.9 to 73.2%
Maximum OEI 73.2%

TRANSMISSION**TRANSMISSION TORQUE****TWIN ENGINE OPERATION**

Maximum continuous 81%
Takeoff range (5 minutes maximum) 81 to 100%

WARNING

TAKEOFF POWER SHALL NOT BE USED ABOVE 105 KIAS.

Maximum 100%

TRANSMISSION OIL PRESSURE

Minimum for idle 30 psi
Idle range 30 to 40 psi
Continuous operation 40 to 70 psi
Maximum 70 psi

TRANSMISSION OIL TEMPERATURE

Continuous operation 15 to 110°C
Maximum 110°C

ROTOR

ROTOR RPM — POWER ON

Minimum	97%
Continuous operation	97 to 100%
Maximum continuous	100%
Operation with ENG TORQUE at or below 30%	100 to 104.5%
Maximum with ENG TORQUE at or below 30%	104.5%

ROTOR RPM — POWER OFF

Minimum for autorotation with gross weight below 8000 pounds (3629 kg)	80%
Power off operation with gross weight below 8000 pounds (3629 kg)	80 to 104.5%
Minimum for autorotation with gross weight at or above 8000 pounds (3629 kg)	91%
Maximum	104.5%

ROTOR RPM — GROUND OPERATION

Minimum	77%
---------	-----

Minimum with stick centering indicator system inoperative 97%

Transient (avoid steady state operations) 26 to 77%

ROTOR BRAKE

Engine starts with rotor brake engaged are prohibited. Rotor brake application is limited to ground operation and shall not be applied until both engines are shut down and ROTOR has decreased to 40% RPM or below.

FUEL AND OIL

NOTE

Refer to BHT-412-MD-2 for fuel capacity and lists of approved fuels, oils, and vendors.

FUEL

Fuel conforming to ASTM D-1655 Type B, NATO F-40, or MIL-T-5624 Grade JP-4 may be used at all ambient temperatures.

Fuel conforming to ASTM D-1655 Type A or A-1, NATO F-44, MIL-T-5624 Grade JP-5, NATO F-34, or MIL-T-83133 Grade JP-8, limited to ambient temperatures above -30°C (-22°F).

ENGINE AND COMBINING GEARBOX OIL

Oil conforming to PWA Specification No. 521 Type I and MIL-L-7808 (NATO 0-148) may be used at all ambient temperatures.

Oil conforming to PWA Specification No. 521 Type II and MIL-L-23699 (NATO 0-156), or DOD-L-85734AS limited to ambient temperatures above -40°C (-40°F).

TRANSMISSION, INTERMEDIATE AND TAIL ROTOR GEARBOX OIL

Oil conforming to DOD-L-85734AS (Turbine Oil 555), MIL-L-23699 (NATO 0-156), or MIL-L-7808 (NATO 0-148) may be used at all approved ambient temperatures.

NOTE

DOD-L-85734AS or MIL-L-23699 is recommended.

HYDRAULIC

NOTE

Refer to BHT-412-MD-2 for approved fluids and vendors.

Hydraulic fluid type MIL-H-5606 (NATO H-515) shall be used at all ambient temperatures.

WARNING

THE HELICOPTER IS NOT CONTROLLABLE WITH BOTH HYDRAULIC BOOST SYSTEMS INOPERATIVE.

Both hydraulic systems shall be operative prior to takeoff.

HYDRAULIC PRESSURE

Minimum 600 psi

Caution	600 to 900 psi
Continuous operation	900 to 1100 psi
Maximum	1100 psi

HYDRAULIC TEMPERATURE

Maximum 88°C

HEATER OPERATION

Heater shall not be operated when OAT is above 21°C (69.8°F).

HOIST PENALTY REGION

Pilot shall know C.G. at time of hoist operation to determine if C.G. is within penalty region of figure 1-7, Hoist C.G. envelope.

Each hoist operation performed is defined as an extension and retraction of hoist cable while hovering with any weight attached.

Refer to BHT-412-FMS-7 for BHT-412-FMS-26 for Bell Helicopter approved Hoists.

WARNING

THIS PENALTY REGION IS VALID FOR ALL HOIST INSTALLATIONS.

OPERATION IN PENALTY REGION AFFECTS AIRWORTHINESS LIMITATIONS OF ROTOR COMPONENTS (REFER TO BHT-412-MM).



AIRSPEED

	0 to 30 knots	Indicator unreliable
	30 to 140 knots	Continuous operation
	105 knots	Maximum for autorotation at or below 10,000 ft. Hp
	140 knots	VNE

DUAL TORQUE INDICATOR

TRANSMISSION (TWIN ENGINE OPERATION)

	10 to 81%	Continuous operation
	81 to 100%	5 minute takeoff range
	100%	Maximum

ENG (ONE ENGINE INOPERATIVE)

	5 to 58.9%	Continuous OEI operation
	58.9 to 73.2%	30 minute OEI range
	73.2%	Maximum OEI

TRIPLE TACHOMETER

ROTOR RPM

	26 to 77%	Transient ground operation
	80%	Minimum for autorotation below 8000 Lb (3629 kg) gross weight
	80 to 91%	Power off operation below 8000 Lb (3629 kg) gross weight
	91 to 104.5%	Continuous operation (91% minimum power off)
	104.5%	Maximum

ENG RPM (N2)

	97%	Minimum
	97 to 100%	Continuous operation
	100 to 104.5%	Operation at or below 30% ENG TORQUE
	104.5%	Maximum at or below 30% ENG TORQUE

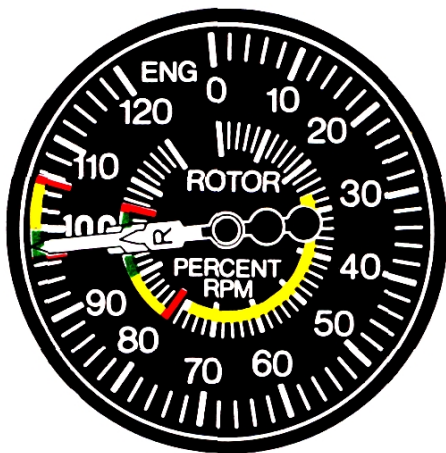
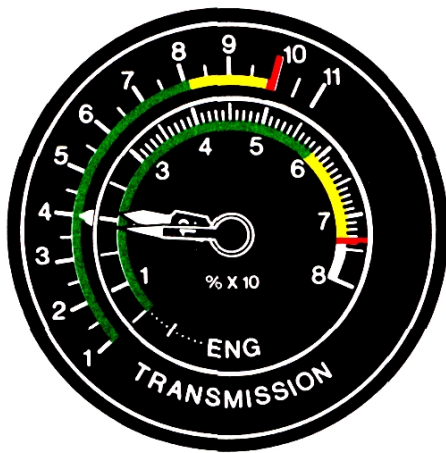


Figure 1-6. Instrument markings (Sheet 1 of 4).

INSTRUMENT PART NUMBER

212-075-037-101



INSTRUMENT PART NUMBER

212-075-037-113



GAS PRODUCER RPM (N1)

EITHER GAGE MAY BE INSTALLED IN PAIRS



12%

Minimum for opening
throttle during start

61%

Idle RPM



61 to 100.8%

Continuous operation



100.8 %

Maximum for takeoff/
30 minutes OEI

100.8 to 102.4%

2 1/2 minute OEI range



102.4%

Maximum OEI



12%

Minimum for opening
throttle during start

61%

Idle RPM



61 to 101.8%

Continuous operation



101.8%

Maximum for takeoff/
30 minute OEI

101.8 to 103.4%

2 1/2 minute OEI range



103.4%

Maximum OEI

TRANSMISSION OIL TEMPERATURE



15 to 110 °C

Continuous operation



110 °C

Maximum

TRANSMISSION OIL PRESSURE



30 PSI

Minimum for idle



30 to 40 PSI

Idle range



40 to 70 PSI

Continuous operation



70 PSI

Maximum

FUEL PRESSURE



4 PSI

Minimum



4 to 35 PSI

Continuous operation



35 PSI

Maximum

412-F2-1-6-2

Figure 1-6. Instrument markings (Sheet 2 of 4).



ENGINE OIL TEMPERATURE

	0°C	Minimum
	0 to 115°C	Continuous operation
	115°C	Maximum

ENGINE OIL PRESSURE

	40 PSI	Minimum for idle
	40 to 80 PSI	Operation below 79% GAS PROD RPM (N1)
	80 to 115 PSI	Continuous operation
	115 PSI	Maximum

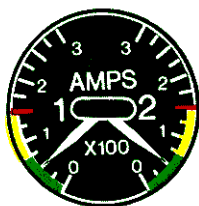
COMBINING GEARBOX
OIL TEMPERATURE

	0°C	Minimum
	0 to 115°C	Continuous operation
	115°C	Maximum

COMBINING GEARBOX
OIL PRESSURE

	40 PSI	Minimum for idle
	40 to 60 PSI	Operation below 94% ENG RPM (N2)
	60 to 80 PSI	Continuous operation
	80 PSI	Maximum

Figure 1-6. Instrument markings (Sheet 3 of 4).

**AMMETER**

	0 to 75 AMPS	Continuous operation
	75 to 150 AMPS	Caution
	150 AMPS	Maximum continuous

**INTERTURBINE TEMPERATURE (ITT)**

	300 to 765°C	Continuous operation
	765 to 810°C	5 minute takeoff range
	810°C	Maximum for takeoff
	822°C	Maximum 30 minute OEI
	850°C	Maximum 2½ minute OEI
	1090°C	Maximum for starting (2 seconds maximum above 960°C)

**HYDRAULIC OIL TEMPERATURE**

	88°C	Maximum
--	------	---------

HYDRAULIC OIL PRESSURE

	600 PSI	Minimum
	600 to 900 PSI	Caution
	900 to 1100 PSI	Continuous operation
	1100 PSI	Maximum

Figure 1-6. Instrument markings (Sheet 4 of 4).

Longitudinal/Lateral C.G. Envelope for Hoist Operations

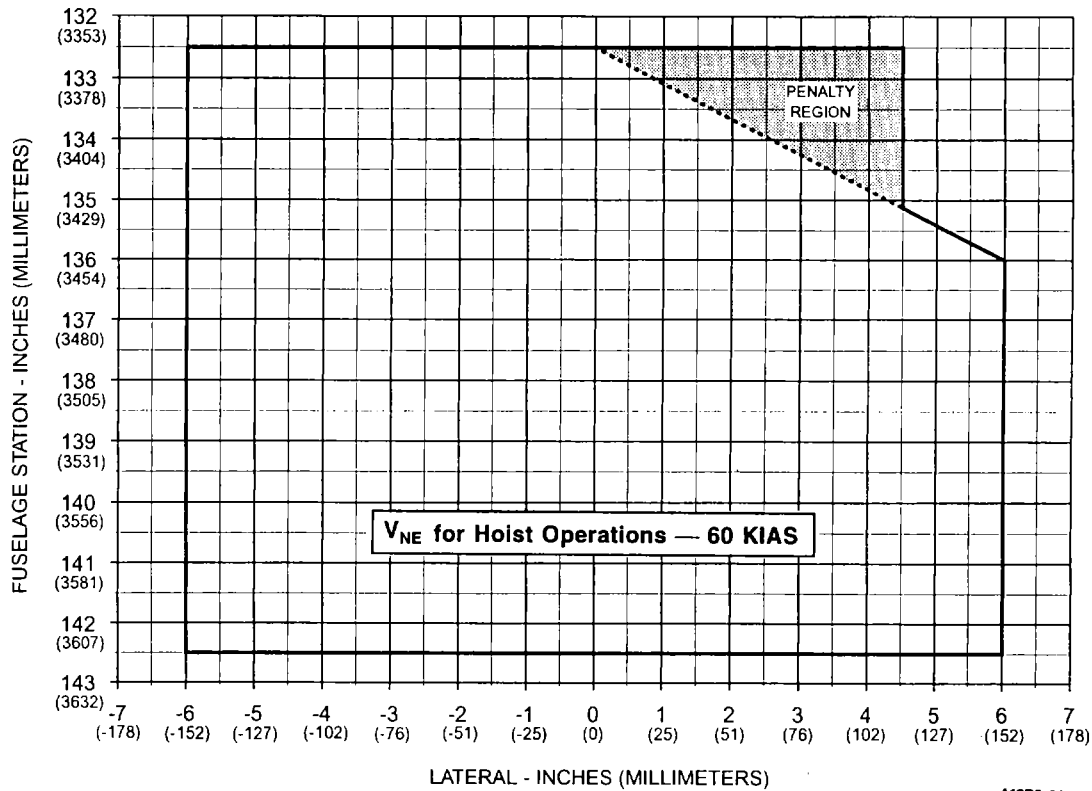


Figure 1-7. Hoist C.G. envelope

Section 2

NORMAL PROCEDURES

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Section 2

NORMAL PROCEDURES

INTRODUCTION

This section contains instructions and procedures for operating the helicopter from the planning stage, through actual flight conditions, to securing the helicopter after landing.

Normal and standard conditions are assumed in these procedures. Pertinent data in other sections is referenced when applicable.

The instructions and procedures contained herein are written for the purpose of standardization and are not applicable to all situations.

OPERATING LIMITATIONS

The minimum, normal, maximum, and cautionary operation ranges for helicopter systems and subsystems are indicated by instrument markings and placards.

Anytime an operating limitation is exceeded, an appropriate entry shall be made in the helicopter logbook. The entry shall state which limit was exceeded, the duration of time, the extreme value attained, and any additional information essential in determining the maintenance action required.

The limits depicted on instrument markings and placards represent careful aerodynamic calculations that are substantiated by flight test data.

Refer to Section 1, LIMITATIONS, for subsystems restrictions.

FLIGHT PLANNING

Each flight should be planned adequately to ensure safe operations and to provide the pilot with the data to be used during flight. Essential weight, balance, and performance information should be compiled as follows:

Check type of flight to be performed and destination.

Select appropriate performance charts to be used.

TAKEOFF AND LANDING DATA

Refer to the LIMITATIONS section for takeoff and landing weight limits and to the Performance section for takeoff and landing distance information.

WEIGHT AND BALANCE

Determine proper weight and balance of the helicopter as follows:

Consult BHT-412-MD-2 for instructions.

Compute takeoff and anticipated landing gross weight, check helicopter center of gravity (CG) locations, and determine weight of fuel, oil, payload, etc.

Ensure loading limitations listed in Section 1 are not exceeded.

PREFLIGHT CHECK

The pilot is responsible for determining whether the helicopter is in condition for safe flight. Refer to figure 2-1 for preflight check sequence.

NOTE

The pilot walk-around and interior checks are outlined in the following procedures. The preflight check is not intended to be a detailed mechanical check, but simply a guide to help the pilot check the condition of the helicopter. It may be made as comprehensive as conditions warrant, at the discretion of the pilot.

All areas checked shall include a visual check for evidence of corrosion, particularly when helicopter is flown near or over salt water, or in areas of high industrial emissions.

BEFORE EXTERIOR CHECK

Flight planning — Completed.

Gross weight and CG — Compute (refer to BHT-412-MD-2).

Publications — Checked.

Portable fire extinguishers — Condition and security.

Aft fuel sumps — Drain samples as follows:

FUEL TRANS switches — OFF.

BOOST PUMP switches — OFF.

ENGINE 1 and ENGINE 2 FUEL switches — OFF.

BAT BUS 1 switch — ON.

Aft fuel sump drain buttons (left and right) — Press.

NOTE

If aft sumps fail to drain, the sump valves may be operated manually.

Forward and middle fuel sumps — Drain samples as follows:

Press-to-drain valves — Press.

Fuel filters — Drain before first flight of day as follows:

BOOST PUMP switches — ON.

ENGINE 1 and ENGINE 2 FUEL switches — ON.

Fuel filter (left and right) — Drain samples.

ENGINE 1 and ENGINE 2 FUEL switches — OFF.

BOOST PUMP switches — OFF.

BAT BUS 1 switch — OFF.

Rotor tie downs — Removed and secured.

EXTERIOR CHECK

Refer to figure 2-1 for areas.



IF HELICOPTER HAS BEEN EXPOSED TO SNOW OR ICING CONDITIONS, SNOW AND ICE SHALL BE REMOVED PRIOR TO FLIGHT.

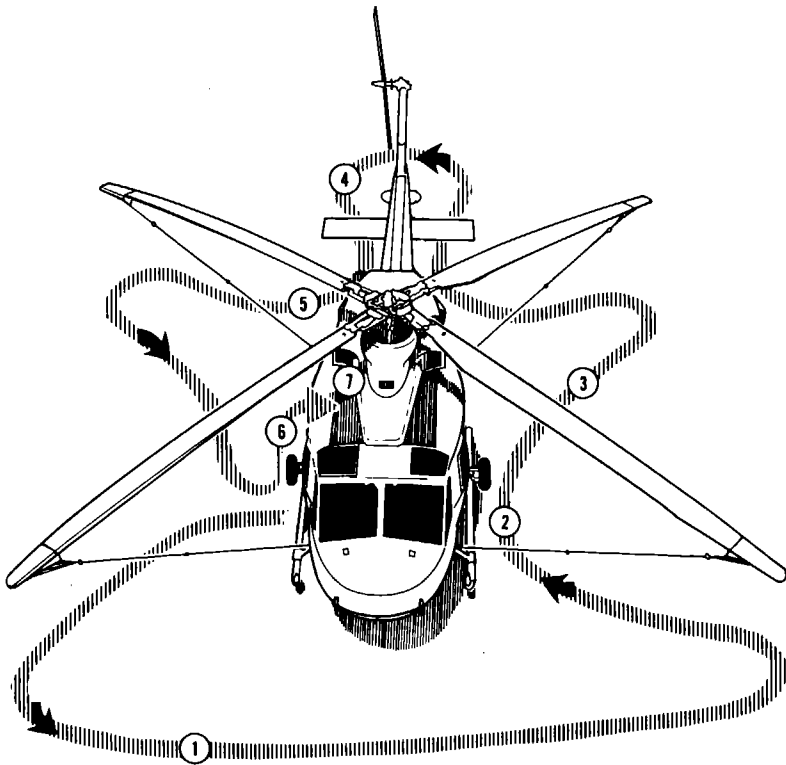


Figure 2-1. Preflight check sequence

1. FUSELAGE — FRONT

Cabin nose — Condition; all glass clean; wipers stowed.

Remote hydraulic filter bypass indicator — Verify green.

Pitot tube(s) — Cover(s) removed; unobstructed.

Static ports (left and right) — Unobstructed.

Rotor blade (forward) — Remove tiedown. Visually check condition and cleanliness.

Cabin nose ventilators — Unobstructed.

Nose compartment door — Secure.

Battery vent and drain tubes — Unobstructed.

Searchlight and landing light — Stowed.

Antennas — Condition and security.

2. FUSELAGE — CABIN LEFT SIDE

Copilot door — Condition and operation; glass clean. Check security of emergency release handles.

Position lights — Condition.

Passenger door — Condition and operation; glass clean. Condition of pop-out windows.

Landing gear — Condition; handling wheels removed.

Passenger step (if installed) — Condition and security.

3. FUSELAGE — AFT LEFT SIDE

Rotor blade (aft) — Remove tiedown. Visually check condition and cleanliness.

No. 1 engine compartment — Check.

No. 1 engine oil level — Verify actual presence of oil in sight gage. Visually check oil level. Filler cap secured.

Governor spring — Check condition.

Engine fire extinguisher — Check bottle pressure gage and temperature range.

Combining gearbox filter — Check bypass indicator retracted.

Oil cooler blower — Unobstructed.

Access doors and engine cowling — Secured.

Drain lines — Unobstructed.

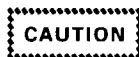
Engine exhaust ejectors — Covers removed; unobstructed.

Oil coolers — Unobstructed.

4. TAILBOOM

Tailboom — Condition; access covers secured.

Tail rotor driveshaft covers — Secured.



**DO NOT BEND ELEVATOR
TRAILING EDGE TAB.**

Elevator — Condition and security. Check for spring condition by moving elevator toward the leading edge down position.

Tail rotor (90°) gearbox — Verify actual presence of oil in sight gage. Visually

7. CABIN TOP

Droop restrainers — Security and condition. Verify droop stop clevis is in lower position of cam plate.

FAA APPROVED



BHT-412-FM-2

check oil level, check filler cap and chip detector plug for security.

Tail rotor blade — Remove tiedown. Visually check condition and cleanliness.

Tail rotor — Condition and free movement on flapping axis.

Tail rotor yoke — Evidence of static stop contact damage (deformed static stop yield indicator).

Tail skid — Condition and security.

Intermediate (42°) gearbox — Verify actual presence of oil in sight gauge. Visually check oil level. Check filler cap and chip detector plug for security.

Elevator — Condition and security.

Tailboom — Condition.

Baggage compartment — Cargo secured, smoke detector condition, door secured.

5. FUSELAGE — AFT RIGHT SIDE

Aft compartment — Check unobstructed.

Tail rotor actuator — Check.

AFCS computers — Secured, compartment door secured.

Engine fire extinguisher — Check bottle pressure gauge and temperature range.

Combining gearbox oil level — Verify actual presence of oil in sight gauge. Visually check oil level.

Oil cooler blower — Unobstructed.

No. 2 engine compartment — Check.

No. 2 engine oil level — Verify actual presence of oil in sight gauge. Visually check oil level. Filler cap secured.

Access doors and engine cowling — Secured.

Fuel filler — Visually check quantity, secure cap.

6. FUSELAGE — CABIN RIGHT SIDE

Passenger door — Condition and operation, glass clean, condition of pop-out windows.

Transmission oil — Verify actual presence of oil in sight gauge. Visually check oil level.

Position lights — Condition.

Landing gear — Condition, handling wheels removed.

Passenger step (if installed) — Condition and security.

Pilot door — Condition and operation, glass clean. Check security of emergency release handles.

7. CABIN TOP

Hub and sleeve assembly — Check condition.

Swashplate, support assembly, and collective lever — Check condition.

Main rotor pitch links — Security and condition.

Main rotor hub — Check general condition:

Mast retaining nut — Secured.

Yoke assembly — Condition.

Pitch horns — Security and condition.

Elastomeric bearings, lead-lag dampers — Check general condition.

Blade retention bolts — Security and proper latching.

Droop restrainers — Security and condition.

Simple pendulum absorbers (if installed) — Security and condition.

Rotor blades — Visually check condition and cleanliness.

Main driveshaft and couplings — Condition and security, and grease leakage. Check Temp-Plates (four places each coupling) for evidence of elevated temperature indicated by dot changing color to black.

CAUTION

IF ANY TEMP-PLATE IS MISSING OR HAS BLACK DOTS, MAINTENANCE PERSONNEL SHALL ASSIST IN DETERMINING AIRWORTHINESS PER ALERT SERVICE BULLETIN 412-93-79.

Transmission oil filler cap — Secured.

No. 1 and No. 2 hydraulic reservoirs — Visually check fluid levels; caps secured.

Antenna(s) — Condition and security.

Combining gearbox oil filler cap — Secured.

Anticollision light — Condition and security.

No. 1 and No. 2 engine air intakes — Covers removed, unobstructed; check particle separator doors closed.

Engine and transmission cowling — Secured.

Fresh air inlet screen — Unobstructed.

Rotor brake reservoir cap — Security.

INTERIOR CHECK

Cabin interior — Cleanliness and security of equipment.

Cargo and baggage (if applicable) — Check security.

Protective breathing equipment (if installed) — Condition and properly serviced.

NOTE

Opening or removing doors shifts helicopter center of gravity and reduces V_{NE} . Refer to BHT-412-MD-2 and to Section 1.

Passenger doors — Secured.

PRESTART CHECK

DELETED

Seat and pedals — Adjust.

Seat belt and shoulder harness — Fasten and adjust.

Shoulder harness inertia reel and lock — Check.

Directional control pedals — Check freedom of movement; position for engine start.

Flight controls — Position for start; friction as desired.

Transmission chip detector indicators — Check; reset if required.

Collective switches — OFF.

Lower pedestal circuit breakers — IN.

Radio equipment — OFF.

COMPASS CONTROL switch(es) — MAG (slave position).

FUEL INTCON switch — NORM.

FUEL TRANS switches — OFF.

BOOST PUMP switches — OFF.

FUEL XFEED switch — NORM.

ENGINE 1 and ENGINE 2 FUEL switches — OFF.

PART SEP switches — NORM.

ENGINE 1 and ENGINE 2 GOV switches — AUTO.

HYDR SYS NO.1 and NO.2 switches — ON.

STEP switch (If installed) — As desired.

FORCE TRIM switch — ON, cover down.

Instruments — Static check.

STATIC SOURCE switch (if installed) — PRI.

APPROACH PLATE AND MAP LIGHT knob(s) — OFF.

AUX SYS PITOT and STATIC switches (if installed) — NORM.

Altimeter(s) — Set.

Clock — Set and running.

FIRE EXT switch — OFF.

FIRE PULL handles — In (forward).

AFT DOME LIGHT rheostat and switch — OFF.

PITOT STATIC HEATERS switch — OFF.

WIPERS switches — OFF.

CARGO RELEASE switch (if installed) — OFF.

HEATER switch — OFF.

AFT OUTLET switch — OFF.

VENT BLOWER switch — OFF.

EMERG LT switch (if installed) — DISARM.

STBY ATT switch (if installed) — TEST; check standby attitude instrument light illuminates and OFF flag retracts momentarily, then switch OFF.

WSHLD HEAT switches (if installed) — OFF.

Overhead circuit breakers — In.

All LT rheostats — OFF.

UTILITY LIGHT switch — OFF.

POSITION light — OFF.

ANTI COLL light — ON.

EMERG LOAD switch — NORMAL.

NON-ESNTL BUS switch — Spring loaded to NORMAL.

INV 1 and 2 switches — OFF.

GEN 1 and 2 switches — OFF.

IF EXTERNAL POWER IS USED —
CONNECT (1000 AMPS MAXIMUM).
CHECK 27 ± 1 VOLTS DC; ADJUST
POWER SOURCE IF REQUIRED.

BATTERY BUS 1 and BUS 2 switches — ON; check BATTERY caution light illuminates.

NOTE

Test all lights when night flights are planned or anticipated. Accomplish light tests with external power connected or during engine runup.

ROTOR BRAKE lights — Test. Pull brake ON and check that both caution lights illuminate; return to OFF and check lights extinguish.

NOTE

Rotor brake shall be off at all times when the engines are running.

FIRE 1 and 2 warning lights test button — Press to test.

BAGGAGE FIRE warning light test button — Press to test (verify light flashes).

CYC CTR caution lights — Press to test.

Caution panel TEST switch — PNL (All segments extinguish except CAUTION PANEL).

Caution panel TEST switch — LT (All segments illuminate).

Caution Panel RESET button — Press (MASTER CAUTION light extinguishes).

FUEL SYS test switch — FWD TANK, then MID TANK; note digital and needle indications.

FUEL SYS DIGITS TEST button — Press (Digital display reads 888).

INV 1 and 2 switches — ON.

ENGINE STARTING

NOTE

If the helicopter has been cold soaked in ambient temperatures of -18°C (0°F) or less, both throttles will be difficult to move and follow through coupling may be increased.

Throttles — Rotate engine 1 throttle full open, then back against idle stop. Actuate ENG 1 IDLE STOP release, roll engine 1 throttle to full closed, then apply friction as desired. Repeat procedure using engine 2 throttle and ENG 2 IDLE STOP release.

NOTE

When either IDLE STOP release is activated, the appropriate idle stop plunger will not release if pressure is applied toward the closed position of the throttle.

Moderate frictions should be applied to overcome follow-through coupling between throttles.

RPM INCR/DECR switch — DECR for 8 seconds.

NOTE

Either engine may be started first; however, the following procedure is provided for starting engine 1 first.

ENGINE 1 START

ENGINE 1 FUEL TRANS switch — ON; check NO. 1 FUEL TRANS caution light extinguished.

ENGINE 1 BOOST PUMP switch — ON; check NO. 1 FUEL BOOST light extinguished.

ENGINE 1 FUEL switch — ON. (FUEL VALVE caution light will illuminate momentarily.)

Engine 1 FUEL PRESS — Check.

Rotor — Clear.

CAUTION

PROLONGED EXPOSURE TO AMBIENT TEMPERATURES OF 0°C (32°F) OR LESS MAY FREEZE MOISTURE IN THE ENGINE FUEL CONTROL SYSTEM. MONITOR ENG RPM (N2) DURING COLD WEATHER STARTING FOR OVERSPEED. IF AN OVERSPEED APPEARS IMMINENT, ABORT START AND CLOSE THROTTLE TO OFF POSITION.

START switch — ENG 1 position. Observe starter limitations.

Engine 1 ENGINE OIL pressure — Indicating.

Engine 1 throttle — Open to idle at 12% GAS PROD RPM (N1) minimum.

Engine 1 ITT — Monitor to avoid a hot start. Maximum ITT during start is 1090°C, not to exceed two seconds above 960°C. If

ITT continues to rise, abort start by activating idle stop release and rolling throttle fully closed. Starter should remain engaged until ITT decreases. Do not attempt restart until corrective maintenance has been accomplished.

NOTE

If engine fails to start, refer to FALSE START procedures this section.

Collective pitch — Ensure in full down position.

CAUTION

IF STICK CENTERING INDICATOR SYSTEM IS INOPERATIVE, GROUND OPERATION SHALL BE CONDUCTED AT 97% ROTOR RPM OR ABOVE.

NOTE

On side slopes greater than five degrees, disregard CYC CTR caution lights and position cyclic as required.

Cyclic — Position as necessary to extinguish CYC CTR caution lights.

NOTE

CYC CTR caution lights are inhibited between 95 and 105% ROTOR RPM.

START switch — Off at 55% GAS PROD RPM (N1).

GAS PROD — Check $61 \pm 1\%$ RPM (N1) when throttle is on idle stop.

NOTE

During extremely cold ambient temperatures, idle rpm will be high and the ENGINE, XMSN, and GEAR BOX OIL pressures may exceed maximum limits for up to

two minutes after starting. Warm up shall be conducted at 77 - 85% ROTOR RPM at flat pitch.

Do not increase ROTOR above 85% RPM until XMSN OIL temperature is above 15°C.

ENGINE, XMSN, and GEAR BOX OIL pressures — Check.

Engine 1 PART SEP OFF caution light — Check extinguished.

CAUTION

DURING RPM INCREASE, ANY ABNORMAL INCREASE IN ONE-PER-REV VIBRATION MAY INDICATE ONE OR MORE MAIN ROTOR DROOP RESTRAINERS FAILED TO DISENGAGE FROM STATIC POSITION. VERIFY PROPER OPERATION PRIOR TO FLIGHT.

Engine 1 throttle — Increase to 77 - 85% ENG RPM (N2). Friction as desired.

NOTE

For ground operation, maintain ROTOR RPM within allowable range. Higher minimum ROTOR RPM reduces blade flapping.

ROTOR RPM — Maintain 77 - 85%, as desired.

IF EXTERNAL POWER IS USED, PROCEED TO ENGINE 2 START. IF BATTERY WAS USED, PROCEED AS FOLLOWS:

GEN 1 switch — ON.

AMPS 1 — Check at or below 150 amps.

ENGINE 2 START

ENGINE 2 FUEL TRANS switch — ON. Check NO. 2 FUEL TRANS caution light extinguished.

ENGINE 2 BOOST PUMP switch — ON. Check NO. 2 FUEL BOOST light out (FUEL XFEED caution light will illuminate momentarily).

ENGINE 2 FUEL switch — ON (FUEL VALVE caution light will illuminate momentarily).

Engine 2 FUEL PRESS — Check.

START switch — ENG 2 position. Observe starter limitations.

Engine 2 ENGINE OIL pressure — Indicating.

Engine 2 throttle — Open to idle at 12% GAS PROD RPM (N1) minimum.

Engine 2 ITT — Monitor. Observe ITT limitations.

START switch — Off at 55% GAS PROD RPM (N1).

GAS PROD — Check $61 \pm 1\%$ RPM (N1) when engine 2 throttle is on idle stop.

CAUTION

ENSURE SECOND ENGINE ENGAGES AS THROTTLE IS INCREASED. A NONENGAGED ENGINE INDICATES 10 TO 15% ENG RPM (N2) HIGHER THAN THE ENGAGED ENGINE AND NEAR ZERO TORQUE. IF A NONENGAGEMENT OCCURS, CLOSE THROTTLE OF THE NONENGAGED ENGINE. WHEN THE NONENGAGED ENGINE HAS STOPPED, SHUT DOWN THE ENGAGED ENGINE.

IF A SUDDEN (HARD) ENGAGEMENT OCCURS, SHUT DOWN BOTH ENGINES. MAINTENANCE ACTION IS REQUIRED.

Engine 2 throttle — Increase slowly to match Engine 1 N₂ RPM. Monitor tachometer and torque meter to verify engagement of second engine.

Engine 2 ENGINE OIL pressure — Check.

ENG 2 PART SEP OFF caution light — Check extinguished.

IF EXTERNAL POWER WAS USED — DISCONNECT. GEN 1 SWITCH — ON.

GEN 2 switch — ON. (BATTERY BUS 1 will switch OFF automatically.)

NOTE

Only one BATTERY BUS switch (1 or 2) should remain on with both generators operating.

Caution lights — Check all extinguished (except AFCS).

ENGINE, XMSN, and GEAR BOX OIL temperatures and pressures — Within limits.

AMPS 1 and 2 — Within limits.

NOTE

AMPS 2 will indicate a higher load than AMPS 1 until battery is fully charged.

Radios — ON as required.

ELT (if installed) — Check for inadvertent transmission.

FALSE START

ATTEMPTED ENGINE START WITH NO LIGHTOFF

When the engine fails to light off within 15 seconds after the throttle has been opened

to idle, the following action is recommended:

IDLE STOP release — Actuate.

Throttle — Fully closed.

Starter — Disengage.

FUEL switch — OFF.

BOOST PUMP switch — OFF.

After GAS PROD RPM (N1) has decreased to zero, allow 30 seconds for fuel to drain from engine. Conduct a DRY MOTORING RUN before attempting another start.

DRY MOTORING RUN

The following procedure is used to clear an engine whenever it is deemed necessary to remove internally trapped fuel and vapor.

Throttle — Fully closed.

BOOST PUMP switch — ON.

FUEL switch — ON.

IGN circuit breaker — Pull out.

Starter — Engage for 15 seconds, then disengage.

FUEL switch — OFF.

BOOST PUMP switch — OFF.

IGN circuit breaker — Push in.

Allow the required cooling period for the starter before proceeding. Follow normal start sequence as described on preceding pages.

SYSTEMS CHECKS

STICK CENTERING INDICATOR CHECK



DURING EXTREME COLD AMBIENT TEMPERATURES LIMIT CYCLIC MOVEMENTS UNTIL XMSN OIL TEMPERATURE REACHES 15°C.



DO NOT DISPLACE CYCLIC MORE THAN 1.5 INCHES FROM CENTER TO CHECK THE SYSTEM. IF *CYC CTR* CAUTION LIGHTS DO NOT ILLUMINATE WITHIN THE 1.5 INCH DISPLACEMENT, THE SYSTEM IS INOPERATIVE.

DO NOT DISPLACE CYCLIC BEYOND POINT AT WHICH *CYC CTR* CAUTION LIGHT ILLUMINATES.

NOTE

CYC CTR caution lights are inhibited between 95 and 105% ROTOR RPM.

Cyclic — Displace approximately 1.25 inch (31.7 mm) forward, aft, left and right. Check *CYC CTR* caution light illuminates each time when displaced and extinguishes when centered.

FORCE TRIM CHECK

Flight controls — Friction off; collective lock removed.

Cyclic and pedals — Move slightly each direction to check force gradients.

Cyclic **FORCE TRIM** release button — Press; check trim releases with button pressed, reengages with button released.

FORCE TRIM switch — OFF; check trim disengages and **FT OFF** caution light illuminates.

FORCE TRIM switch — ON, cover down.

PRELIMINARY HYDRAULIC CHECK

Throttles — Set to idle.

NOTE

Uncommanded control movement or motoring with either hydraulic system off may indicate hydraulic system malfunction.

HYDR SYS NO. 1 switch — OFF, then ON.

HYDR SYS NO. 2 switch — OFF, then ON.

ENGINE FUEL CONTROL CHECK

Throttles (both) — Idle.



DO NOT ALLOW *GAS PROD* TO DECREASE BELOW 50% RPM (N1).

NOTE

In the vicinity of 8000 feet pressure altitude, **GAS PROD RPM (N1)** may not change significantly when manual fuel control is selected.

GOV switch (**ENGINE 1** or **2**) — **MANUAL**; observe a change in **GAS PROD RPM (N1)** and **GOV MANUAL** caution light illuminates. Open respective throttle carefully to ensure **GAS PROD RPM (N1)** responds upward, then return to idle

position. Return **GOV** switch to **AUTO**. Check for a return to original **GAS PROD RPM (N1)** and **GOV MANUAL** caution light extinguishes. Check second governor in like manner.

Throttles (both) — Increase slowly to above 85% **ROTOR RPM**.

FUEL CROSSFEED AND INTERCONNECT VALVE CHECK

FUEL XFEED/INTCON test switch — **TEST BUS 1** and hold.

NOTE

After turning either boost pump off, **FUEL BOOST** caution light should illuminate on failed side only.

ENGINE 1 BOOST PUMP switch — OFF. Check engine 1 **FUEL PRESS** decreases, then returns to normal. (This indicates crossfeed valve has been opened by bus no. 1 power and check valve is functioning properly). Return switch to ON.

FUEL INTCON switch — OPEN. Check **FUEL INTCON** caution light illuminates then extinguishes. (This indicates interconnect valve has been opened by bus no. 1 power and valve is functioning properly).

FUEL INTCON switch — **OVRD CLOSE**. Check **FUEL INTCON** caution light illuminates then extinguishes.

FUEL XFEED/INTCON test switch — **TEST BUS 2** and hold.

ENGINE 2 BOOST PUMP switch — OFF. Check engine 2 **FUEL PRESS** decreases, then returns to normal. Return switch to ON.

FUEL INTCON switch — OPEN. Check **FUEL INTCON** caution light illuminates then extinguishes. (This indicates that the interconnect valve has been opened by

bus no. 2 power and that the valve is functioning properly).

FUEL INTCN switch — NORM. Check FUEL INTCN caution light illuminates then extinguishes.

FUEL XFEED/INTCN test switch — NORM.

FUEL XFEED switch — OVRD CLOSE.

ENGINE 1 BOOST PUMP switch — OFF. Check FUEL PRESS drops to zero on affected system. Return switch to ON. Repeat procedure for ENGINE 2 BOOST PUMP switch.

FUEL XFEED switch — NORM.

ELECTRICAL SYSTEMS CHECK

DC VOLTS — Check 27 ± 1 volts.

AC VOLTS — Check 104 to 122 volts.

AMPS 1 and 2 — Check within limits.

■ GEN 1 and 2 switches — OFF

EMERG LOAD switch — EMERG LOAD. Check that the following items remain operational:

One helicopter

One NAV-COM

Panel lights

ICS lights

Essential engine instruments

Essential navigation instruments

EMERG LOAD switch — NORMAL

■ GEN 1 and 2 switches — ON

INV 1 switch — OFF; check INVERTER 1 caution light illuminates. Check no. 1 and no. 2 AC VOLTS for indication that inverter 2 has assumed all ac loads. Return INV 1 switch to ON.

INV 2 switch — OFF; check INVERTER 2 caution light illuminates. Check no. 1 and no. 2 AC VOLTS for indication that inverter 1 has assumed all ac loads. Return INV 2 switch to ON.

EMERG LT switch (if installed) — TEST; check all emergency lights illuminate. Switch to ARM; check lights dim to faint glow.

STBY ATT switch (if installed) — ON.

AFCS CHECK

NOTE

Verification of AFCS actuator centering is necessary. Failure of the actuators to center could result in reduced control margins and abnormal control positions.

NOTE

If fast slaving is desired, center ADI roll trim knob, then push and hold VG FAST ERECT button until attitude indicator displays zero degrees bank angle. Use of VG FAST ERECT button will disengage the respective helicopter.

Pilot and copilot attitude indicators — Erect and set as necessary.

WARNING

IF AFCS IS LEFT ENGAGED IN ATT MODE DURING GROUND OPERATION, IT CAN DRIVE THE CYCLIC STICK TO A CONTROL STOP.

HP 1 and HP 2 buttons — ON. Observe ATT light illuminates, APIs center, and AFCS caution light extinguishes.

NOTE

CYC CTR caution lights may illuminate momentarily during cyclic control checks.

Move cyclic forward, aft, right, left.
Observe APIs do not move.

SYS 2 button — Press and hold.

Move cyclic forward, aft, right, left.
Observe APIs do not move.

SYS 2 button — Release.

Cyclic ATTD TRIM switch — Right for 2 seconds, then aft for 2 seconds. Observe APIs move right, up.

SYS 2 button — Press and hold. Observe SYS 2 actuators agree.

Cyclic FORCE TRIM release button — Press. Observe APIs move to center.

SYS 2 button — Release. Observe SYS 1 actuators centered.

SAS/ATT button — Press. Observe SAS light illuminates.

Move cyclic right, left, forward, and aft.
Observe APIs move in corresponding direction.

Displace right pedal, then left. Observe yaw API moves right, left.

SYS 2 button — Press and hold.

Move cyclic right, left, forward, and aft.
Observe APIs move in corresponding direction.

SYS 2 button — Release.

ENGINE RUNUP



IF HELICOPTER IS SITTING ON
ICE OR OTHER SLIPPERY OR

LOOSE SURFACE, ADVANCE
THROTTLES SLOWLY TO
PREVENT ROTATION OF
HELICOPTER.

Engine 1 throttle — Fully open.

ENG — Stabilized at $95 \pm 1\%$ RPM (N2).

Engine 2 throttle — Fully open. Check no. 1 engine increases 2% ENG RPM (N2) and both engines stabilize at $97 \pm 1\%$ ENG RPM (N2).

RPM INCR/DECR switch — Full INCR. Check ENG does not exceed 101.5% RPM (N2). Set at 100% ENG RPM (N2).

CABIN HEATER CHECK

GAS PROD — Check 75% RPM (N1) minimum (both engines).

Thermostat knob — Fully COLD.



HEATER SWITCH SHALL BE
TURNED OFF WHEN HEATED
AIRFLOW DOES NOT SHUT OFF
AFTER THERMOSTAT IS TURNED
TO FULLY COLD. HEATER AIR
LINE LIGHT ILLUMINATES, OR
CABIN HTR CIRCUIT BREAKER
TRIPS.



DO NOT OPERATE HEATER
ABOVE 21°C OAT.

HEATER switch — ON.

VENT BLOWER switch — ON.

Thermostat setting — Increase and observe heated airflow.

DEFOG lever — ON; check airflow is diverted from pedestal outlets to windshield nozzles. Return lever to OFF.

AFT OUTLET switch — ON; check airflow distributed equally between pedestal outlets and aft outlets. Return switch to OFF.

NOTE

Heater operation affects performance. Refer to Hover Ceiling and Rate of Climb charts for HEATER ON in Section 4.

HEATER switch — As desired.

VENT BLOWER switch — As desired.

HYDRAULIC SYSTEMS CHECK

NOTE

The HYDRAULIC SYSTEMS CHECK is to determine proper operation of the hydraulic actuators for each flight control system. If abnormal forces, unequal forces, control binding or motoring are encountered, it may be an indication of a malfunctioning flight control actuator.

FORCE TRIM switch — OFF.

Collective — Fully down, friction removed.

ROTOR — Set to 100% RPM.

Cyclic — Centered, friction removed.

HYDR SYS NO. 1 switch — OFF. Check NO. 1 HYDRAULIC caution light and MASTER

CAUTION light illuminate and system 1 pressure drops to zero.

Cyclic — Check normal operation by moving cyclic forward, aft, left, and right approximately one inch. Center cyclic.

Collective — Check for normal operation by increasing collective control 1 to 2 inches. Repeat 2 to 3 times as required. Return to fully down position.

Pedals — Displace slightly left and right. Note an increase in force required to move pedal in each direction.

HYDR SYS NO. 2 switch — OFF. Check hydraulic system 2 remains operational, and system 1 remains off.

HYDR SYS NO. 1 switch — ON. Check NO. 1 HYDRAULIC caution light extinguishes, and system 1 regains normal pressure. Check NO. 2 HYDRAULIC caution light illuminates and system 2 pressure drops to zero.

Cyclic — Check normal operation by moving cyclic forward, aft, left, and right approximately one inch. Center cyclic.

Collective — Check for normal operations by increasing collective control 1 to 2 inches. Repeat 2 to 3 times as required. Return to fully down position.

Pedals — Displace slightly left and right. Note the pedals are now hydraulically boosted.

HYDR SYS NO. 2 switch — ON. Check NO. 2 HYDRAULIC caution light extinguishes, system 2 pressure returns to normal, and hydraulic system 1 remains operational.

Cyclic and collective friction — Set as desired.

FORCE TRIM switch — ON.

WARNING

BOTH HYDRAULIC SYSTEMS SHALL BE OPERATIONAL PRIOR TO TAKEOFF.

NOTE

System 1 will normally operate 10 to 20°C cooler than system 2.

BEFORE TAKEOFF

Engine, gearbox, transmission, hydraulic, and electrical instruments — Within operating ranges.

Caution and warning lights — Extinguished.

WARNING

MODERATE FRICTION SHALL BE APPLIED TO OVERCOME FOLLOW-THROUGH COUPLING BETWEEN THROTTLES.

Throttles — Fully open. Adjust frictions.

ENG — 100% RPM (N2) for both engines.

Flight instruments — Check operation and set.

POSITION lights — As required.

ANTI COLL light — Check ON.

PITOT STATIC HEATERS switch — ON. Check ammeter for load indication. Leave ON in visible moisture when temperature is below 4.4°C (40°F), OFF if not required.

Radio(s) — Check functioning.

Cyclic control — Centered or slightly into the wind.

EMERGENCY COMM panel — (If installed) Check for single pilot operations.

AFCS — Select ATT or SAS mode as desired (ATT mode shall be used during

IFR flight. SAS mode recommended for ground operation, hover, and takeoff).

FORCE TRIM switch — ON in ATT mode, as desired in SAS mode.

STEP switch (if installed) — As desired.

Passenger seat belts — Fastened.

All doors — Secured.

POWER ASSURANCE CHECK

Power assurance check should be performed daily. (Refer to Section 4.)

P R O L O N G E D G R O U N D OPERATION**NOTE**

For prolonged ground operation, AFCS shall not be operated in ATT mode.

CAUTION

MINIMUM ROTOR — 97% RPM FOR GROUND OPERATION WITH STICK CENTERING INDICATOR SYSTEM INOPERATIVE.

NOTE

Minimize blade flapping by maintaining highest rotor RPM (N_R) within allowable range.

ROTOR RPM — 77 - 85% or above, as desired.

Cyclic — Position as necessary to extinguish CYC CTR caution lights.

NOTE

On side slopes greater than five degrees, maintain 100% ROTOR RPM. CYC CTR caution lights are inhibited.

TAKEOFF

CAUTION

DURING LIFTOFF TO HOVER, ANY ABNORMAL INCREASE IN ONE PER REV VIBRATION MAY INDICATE ONE OR MORE MAIN ROTOR DROOP RESTRAINERS FAILED TO DISENGAGE FROM STATIC POSITION. VERIFY PROPER OPERATION PRIOR TO FLIGHT.

NOTE

When AFCS is in ATT mode, the FORCE TRIM release button should be pressed before liftoff (to trim actuators to center positions) and should be held until desired climbout attitude is attained.

ENG — 100% RPM (N2).

Area — Clear.

Hover power — Check torque required to hover at four feet skid height.

NOTE

Downwind takeoffs are not recommended since the published takeoff distance performance will not be achieved.

During takeoff, pitch attitude must be adjusted commensurate with power application to prevent entering the AVOID area of the Height-Velocity diagram. Torque shall not exceed 15% above IGE hover power while accelerating to Takeoff Climbout Safety Speed. (Refer to Section 4.)

Cyclic control — Apply forward cyclic to accelerate smoothly.

Collective — Adjust as desired after reaching V_{TOCS} (45 KIAS).

AIRSPPEED — Within limits (60 KIAS minimum for IFR).

IN-FLIGHT OPERATION

NOTE

With the simple pendulum absorber kit, vibration isolation is most effective in cruise flight at 97% ENG RPM (N2).

ENG — 97 to 100% RPM (N2).

AIRSPPEED — Within limits.

Engine, gearbox, and transmission instruments — Within limits.

NOTE

Maximum pitch attitude capability of standby attitude indicator is ± 60 degrees.

Refer to applicable operating rules for high altitude oxygen requirements.

MANEUVERING WITH AFCS IN SAS MODE

Use normal pilot control techniques.

MANEUVERING WITH AFCS IN ATT MODE

Press cyclic FORCE TRIM release button and maneuver as desired. Release button when desired attitude is reached. Helipilot will hold attitude until retrimmed to new attitude. Attitude may also be adjusted with cyclic ATTD TRIM switch.

For momentary attitude changes, manual cyclic movement may be used; however, AFCS actuators may be saturated to limit authority when cyclic is moved manually.

NOTE

Inflight use of VG FAST ERECT button will disengage the respective helipilot and decouple the automatic flight control modes.

BEFORE LANDING

Flight controls — Adjust friction as desired.

AFCS — Engage ATT or SAS mode as desired.

FORCE TRIM switch — ON in ATT mode, as desired in SAS mode.

Throttles — Fully open.

ENG — 100% RPM (N2).

Flight path — Stay clear of AVOID area of Height-Velocity diagram (Refer to Section 1). For landing distance information in the event of engine failure during approach, refer to Section 4.

STEP switch (if installed) — As desired.

CAUTION

RUN ON LANDINGS MAY RESULT IN ROLL OSCILLATIONS WHILE ON THE GROUND. IF THIS OCCURS, LOWERING COLLECTIVE FULLY DOWN OR DISENGAGING HP 1 AND HP 2 WILL STOP THE OSCILLATIONS.

AFTER LANDING

Collective — Fully down.

Pedals — Centered.

FORCE TRIM switch — ON.

AFCS — SAS mode.

CAUTION

MINIMUM ROTOR — 97% RPM FOR GROUND OPERATION WITH

STICK CENTERING INDICATOR SYSTEM INOPERATIVE.

Stick centering check — Complete. Center cyclic and friction as necessary to extinguish CYC CTR caution lights.

NOTE

On side slopes greater than five degrees, disregard CYC CTR caution lights and position cyclic as required.

ENGINE SHUTDOWN

HP 1 and HP 2 — Disengage. Check helipilot lights extinguish, AFCS and MASTER CAUTION lights illuminate.

Cyclic — Frictioned as desired. Maintain cyclic stick as near center as possible at all rotor speeds.

NOTE

For ground operation, maintain ROTOR RPM within allowable range. Higher minimum ROTOR RPM reduces blade flapping.

Throttles — Reduce to 77 - 85% ROTOR RPM, as desired.

ITT — Stabilize for one minute.

ELT (if installed) — Check for inadvertent transmission.

STBY ATTD switch (if installed) — OFF.

EMERG LT switch (if installed) — DISARM.

Engine instruments — Within limits.

IDLE STOP release switch — ENG 1 position.

Engine 1 throttle — Fully closed. Check ITT and GAS PROD RPM (N1) decreasing.

BATTERY BUS 1 switch — ON.

IDLE STOP release switch — ENG 2 position.

Engine 2 throttle — Fully closed. Check ITT and GAS PROD RPM (N1) decreasing.

GEN 1 and 2 switches — OFF.

INV 1 and 2 switches — OFF.

ENGINE 1 and 2 FUEL switches — OFF.

ENGINE 1 and 2 BOOST PUMP switches — OFF.

ENGINE 1 and 2 FUEL TRANS switches — OFF.

Radios — OFF.

WARNING

DO NOT USE COLLECTIVE TO SLOW ROTOR RPM. USE OF COLLECTIVE TO SLOW ROTOR CAN CAUSE EXCESSIVE FLAPPING AND / OR CONING.

CAUTION

AVOID RAPID ENGAGEMENT OF ROTOR BRAKE IF HELICOPTER IS ON ICE OR OTHER SLIPPERY OR LOOSE SURFACE TO PREVENT ROTATION OF HELICOPTER.

Rotor brake — As desired. Apply at or below 40% ROTOR RPM. Return to stowed position after main rotor stops.

Pilot — Remain at flight controls until rotor has come to a complete stop.

Lighting and miscellaneous switches — OFF.

BATTERY BUS 1 and BUS 2 switches — OFF.

Collective down lock — Secured as desired.

AFTER EXITING HELICOPTER

If conditions require, perform the following:

NOTE

Refer to BHT-412-MD-2 for additional information.

Check general condition of droop restraint system and verify droop restraint arms are engaged in the lower detent of cam window.

Install main rotor blade tiedown socks on blades and secure to mooring points.

Install tail rotor tiedown strap and secure to vertical fin.

Install exhaust covers, engine inlet protective plugs, and pitot tube covers.

Section 3

EMERGENCY AND MALFUNCTION PROCEDURES

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Section 3

EMERGENCY AND MALFUNCTION PROCEDURES

INTRODUCTION

The following procedures contain the indications of equipment or system failure or malfunction, the use of emergency features of primary and backup systems, and appropriate warnings, cautions, and explanatory notes. Table 3-1 lists fault conditions and corrective actions required for illumination of red warning lights. Table 3-2 addresses malfunction procedures associated with yellow caution lights.

All corrective action procedures listed herein assume the pilot gives first priority to aircraft control and a safe flight path.

The helicopter should not be operated following any emergency landing or shutdown until the cause of the malfunction has been determined and corrective maintenance action taken.

LAND AS SOON AS PRACTICAL

The duration of the flight and landing site are at the discretion of the pilot. Extended flight beyond the nearest approved landing area is not recommended.

The following terms are used to describe the operating condition of a system, subsystem, assembly, or component.

Affected

Fails to operate in the intended or usual manner.

Normal

Operates in the intended or usual manner.

DEFINITIONS

The following terms indicate the degree of urgency in landing the helicopter.

LAND AS SOON AS POSSIBLE

Land without delay at the nearest suitable area (i.e., open field) at which a safe approach and landing is reasonably assured.

Table 3-1. Warning lights

PANEL WORDING	FAULT CONDITION	CORRECTIVE ACTION
FIRE PULL (1 or 2)	Fire indication in No. 1 or No. 2 engine compartment.	Pull illuminated FIRE PULL handle. Select MAIN fire extinguisher. Close throttle of affected engine. Select RESERVE fire extinguisher if necessary. Land as soon as possible.
BAGGAGE FIRE	Smoke in baggage compartment.	Reduce power to minimum required. Land as soon as possible. Inspect tailboom area for damage.
ENG OUT (1 or 2)	GAS PROD abnormally low, below $53 \pm 2\%$ RPM (N1), on No. 1 or No. 2 engine.	Check ENG TORQUE, GAS PROD RPM (N1), ENG RPM (N2), and ITT. Adjust power and AIRSPEED (65 KIAS). Reset remaining ENG RPM (N2) to normal range. Close throttle of affected engine. Refer to ENGINE FAILURES and RESTART IN FLIGHT procedures. Land as soon as practical.
X M S N PRESSURE	O I L Transmission oil pressure below limit.	Reduce power. Land as soon as possible.
XMSN OIL TEMP	Transmission oil temperature above limit.	Reduce power. Check XMSN OIL temperature. If not within limits, land as soon as possible.
C B O X PRESSURE	O I L Combining gearbox oil pressure below normal.	Reduce power. Land as soon as possible.
C BOX TEMP	Combining gearbox oil temperature above limit.	Reduce power. Check GEAR BOX OIL temperature. If not within limits, land as soon as possible.

Table 3-1. Warning lights (Cont)

PANEL WORDING	FAULT CONDITION	CORRECTIVE ACTION
BATTERY TEMP	Battery case temperature above limit.	BATTERY BUS 1 and BUS 2 switch — OFF. Land as soon as practical.
WARNING BATTERY SHALL NOT BE USED FOR ENGINE START AFTER ILLUMINATION OF <i>BATTERY TEMP</i> LIGHT. BATTERY SHALL BE REMOVED AND SERVICED IN ACCORDANCE WITH MANUFACTURER'S INSTRUCTIONS PRIOR TO RETURN TO SERVICE.		
ROTOR BRAKE	Rotor brake linings not retracted.	Check rotor brake handle fully up in detent. If light remains on, land as soon as possible.

Table 3-2. Caution lights

PANEL WORDING	FAULT CONDITION	CORRECTIVE ACTION
OIL PRESSURE (ENG 1 or 2)	Engine oil pressure below limit.	Shut down affected engine. FUEL INTCON switch — OPEN. Land as soon as practical.
DC GENERATOR (ENG 1 or 2)	Failure of dc generator.	GEN FIELD and GEN RESET circuit breakers — Check in. GEN switch (affected generator) — RESET, then ON. If light remains on, turn GEN switch OFF. If No. 2 generator failed: BATTERY BUS 2 switch — OFF. BATTERY BUS 1 switch — ON.

Table 3-2. Caution lights (Cont)

PANEL WORDING	FAULT CONDITION	CORRECTIVE ACTION
		If nonessential bus power is required: NON-ESNTL BUS switch — MANUAL. DC AMPS — Monitor. If both generators fail:
		CAUTION DO NOT SELECT <i>EMERG LOAD</i> AT PRESSURE ALTITUDES ABOVE 5000 FEET. BOTH FUEL BOOST PUMPS WILL BECOME INOPERATIVE, RESULTING IN POSSIBLE FUEL STARVATION. <i>EMERG LOAD</i> switch — As required. Land as soon as practical.
PART SEP OFF (ENG 1 or 2)	Particle separator bypass door closed or circuit breaker out. Ice and dust protection system inoperative.	Check ENG 1 (or 2) RPM and PART SEP circuit breakers in. Move PART SEP switch to OVRD ON.
NO. 1 FUEL BOOST/ NO. 2 FUEL BOOST	Fuel boost pump failure has occurred	If practical, descend below 5000 feet H_p to prevent possible fuel starvation in the event remaining boost pump fails.

Table 3-2. Caution lights (Cont)

PANEL WORDING	FAULT CONDITION	CORRECTIVE ACTION
	NOTE	FUEL INTCON switch — OPEN.
	If either fuel boost pump fails, and the FUEL XFEED switch is in NORM position, the crossfeed valve is opened automatically by a pressure switch, allowing either boost pump to furnish fuel to both engines.	CAUTION IF EITHER BOOST PUMP FAILS, USABLE FUEL WILL BE APPROXIMATELY 60 POUNDS LESS THAN INDICATED. Land as soon as practical.
NO. 1 FUEL FILTER/ NO. 2 FUEL FILTER	Fuel filter is partially blocked.	Land as soon as practical.
FUEL LOW (Less than 100 lbs. difference between No. 1 and No. 2 fuel quantities)	Fuel level in left or right cells at or below 190 pounds.	Plan landing.
	NOTE	NOTE
	The FUEL LOW light will not illuminate for the affected side when fuel quantity indication malfunction occurs. Refer to FUEL QUANTITY INDICATION MALFUNCTION.	Interconnect valve will open automatically when fuel level in opposite side decreases to 190 pounds (as indicated by illumination of FUEL INTCON caution light). This will allow the fuel quantity in the lower aft cells to equalize. If either boost pump fails, usable fuel will be approximately 60 pounds less than indicated. This fuel will be available to both engines through either boost pump. FUEL INTCON caution light can be extinguished by placing FUEL INTCON switch to OPEN position.
FUEL LOW (100 lbs. or more difference between No. 1 and No. 2 fuel quantities)	Possible fuel leak in cells with lower quantity.	FUEL INTCON switch — OVRD CLOSE. Land as soon as possible.

Table 3-2. Caution lights (Cont)

PANEL WORDING	FAULT CONDITION	CORRECTIVE ACTION
GOV MANUAL (ENG 1 or 2)	Engine governor in manual mode.	TORQUE, ITT, and RPM must be controlled with throttle.
ENGINE CHIP (ENG 1 or 2)	Metal particles in engine oil.	Reduce power and shut down engine as soon as practical to minimize engine damage. Land as soon as practical.
FUEL VALVE (ENG 1 or 2)	Fuel valve not properly seated or circuit breaker out.	Check FUEL VALVE circuit breakers in. Land as soon as practical. If on ground, cycle FUEL switch.
NO. 1 GEN OVHT/ NO. 2 GEN OVHT	Generator overheating.	GEN switch — OFF.
CAUTION PANEL	Caution panel inoperative.	Check MASTER CAUTION circuit breaker in. Monitor aircraft instruments. Land as soon as practical.
NO. 1 INVERTER/ NO. 2 INVERTER	Failure of ac power inverter;	Check both ac voltmeters to determine that remaining inverter automatically assumed load for failed inverter. Check INV PWR circuit breakers in. Reengage HP 1 or HP 2. During IFR flight, if both inverters fail, land as soon as practical; or continue flight under VFR, if desired.
or		
NO. 2 INVERTER	EMERG LOAD switch in EMERG LOAD position.	Place EMERG LOAD switch in NORMAL position, if electrical load shedding is not required.
EXTERNAL POWER	External power receptacle door open.	Check external power door closed.
DOOR LOCK	Passenger door(s) or baggage compartment door not secured.	Check doors secured.
BATTERY	Both BATTERY switches/ relays in the same position.	Turn one BATTERY switch ON, other OFF. If light remains on, reverse BATTERY switch positions.
C BOX CHIP	Metal particle in combining gearbox oil.	Reduce power. Land as soon as practical.

Table 3-2. Caution lights (Cont)

PANEL WORDING	FAULT CONDITION	CORRECTIVE ACTION
XMSN CHIP	Metal particles in transmission oil (one or more remote XMSN CHIP indicators tripped).	Reduce Power. Land as soon as practical.
42/90 BOX CHIP	Metal particles in 42° or 90° gearbox oil.	Land as soon as practical.
NO. 1 HYDRAULIC/ NO. 2 HYDRAULIC	Hydraulic pressure below limit or temperature above limit.	Verify fault and affected system from gage readings. Turn off affected system. Land as soon as possible.
NO. 1 FUEL TRANS/ NO. 2 FUEL TRANS	Fuel transfer pump or ejector pump malfunction (no fuel transfer from lower forward and middle cells to lower aft cell); or Check valve malfunction allowing fuel to leak from aft to mid cell after normal transfer is complete (total fuel 800 pounds or less).	Check FUEL TRANS circuit breaker — in. Check FUEL TRANS switch — ON. CAUTION IF EITHER TRANSFER PUMP FAILS, USABLE FUEL WILL BE 25 POUNDS LESS THAN INDICATED.
NOTE FUEL TRANS light will remain illuminated after fuel transfer with fuel quantity indication malfunction. Refer to FUEL QUANTITY INDICATION MALFUNCTION.		If light remains illuminated: FUEL TRANS switch — OFF. CAUTION FUEL TRAPPED IN MID CELL IS UNUSABLE AND MUST BE SUBTRACTED FROM TOTAL FUEL QTY INDICATION. Monitor MID TANK quantity periodically. Plan landing.

Table 3-2. Caution lights (Cont)

PANEL WORDING	FAULT CONDITION	CORRECTIVE ACTION
FUEL INTCON (Switch in NORM position).	Fuel interconnect valve not fully closed. (Automatic valve opening is normal if FUEL LOW light is also illuminated.)	Check FUEL INTCON circuit breakers (both) in. FUEL INTCON switch — OPEN, then NORM.
FUEL INTCON (Switch in OPEN position).	FUEL interconnect valve not fully open or FUEL INTCON circuit breakers out.	Check FUEL INTCON circuit breakers in. FUEL INTCON switch — OVRD CLOSE, then OPEN.
FUEL XFEED	Fuel crossfeed valve not fully open or closed, or FUEL XFEED circuit breakers out.	Check FUEL XFEED circuit breakers (both) in. Cycle FUEL XFEED switch.
HEATER AIR LINE	Heater mixing valve malfunction.	Turn HEATER switch OFF immediately.
AFCS	Automatic flight control system hardover;	Reduce AIRSPEED to 115 KIAS or below. Check AFCS control panel. If either helipilot is off, attempt to switch ON. (Refer to AFCS malfunction procedures.)
	or	
	Loss of ac power to HP 1 or HP 2;	
	or	
	Loss of attitude gyro input to HP 1 or HP 2 (possible disengagement of either or both helipilots.)	During IFR flight, if both HP 1 and HP 2 are failed and will not reset, land as soon as practical; or continue flight under VFR, if desired.
	or	
	Auto trim malfunction. Displacement between HP 1 and HP 2 actuators at least 50 percent travel.	Reduce AIRSPEED to 115 KIAS or below. Check actuator position panel. If APIs are centered, depress SYS 2 button to check HP 2 actuator displacement. Turn off affected system.

Table 3-2. Caution lights (Cont)

PANEL WORDING	FAULT CONDITION	CORRECTIVE ACTION
FT OFF	Force trim inoperative.	Check FORCE TRIM switch ON and FORCE TRIM circuit breaker in. During IFR flight, if system remains inoperative, land as soon as practical; or continue flight under VFR, if desired. Pilot may increase cyclic friction to provide additional cyclic stabilization.
CYC CTR	Cyclic not centered.	Center cyclic.
RPM (with audio) or RPM (without audio)	ROTOR RPM at or below 95%. or ROTOR RPM at or above 105%.	Adjust collective pitch and/or RPM INCR/DECR switch as required. Refer to ENGINE FUEL CONTROL MALFUNCTION procedures.

FIRE

If FIRE warning light remains on more than 10 seconds: FIRE EXT switch — RESERVE.

ENGINE FIRES

Complete engine shutdown.

INDICATIONS:

Exit helicopter.

FIRE 1 PULL or FIRE 2 PULL handle illuminated.

ENGINE FIRE DURING TAKEOFF OR LANDING**ENGINE FIRE DURING START****PROCEDURE:****PROCEDURE:**

Abort start of affected engine as follows:

Throttle — Closed.

FUEL XFEED switch — OVRD CLOSE.

BOOST PUMP switch — OFF.

FUEL switch — OFF.

Appropriate FIRE PULL handle — Pull.

FIRE EXT switch — MAIN.

The primary concern for the pilot is safety of the passengers and crew. The decision whether to begin an approach, or continue the takeoff is based on landing site availability. Proceed as follows:

AIRSPEED — 45 KIAS minimum.

Collective — Reduce (altitude permitting).

Appropriate FIRE PULL handle — Pull.

FIRE EXT switch — MAIN.

If FIRE warning light remains on more than 10 seconds: FIRE EXT switch — RESERVE.

ENG — Set at 100% RPM (N2) if possible.

Land as soon as possible.

Complete engine shutdown.

Exit helicopter.

ENGINE FIRE IN FLIGHT

PROCEDURE:

Initiate emergency descent immediately if possible.

Shut down affected engine as follows:

Collective — Reduce (altitude permitting).

Appropriate FIRE PULL handle — Pull.

Throttle — Closed.

FIRE EXT switch — MAIN.

FUEL XFEED switch — OVRD CLOSE.

BOOST PUMP switch — OFF.

FUEL switch — OFF.

FUEL INTCON switch — OPEN.

If FIRE warning light remains on more than 10 seconds: FIRE EXT switch — RESERVE.

ENG (unaffected engine)— Set at 100% RPM (N2) if possible.

Land as soon as possible.

If a landing site is not readily available, proceed as follows:

FIRE PULL handle — In (to provide fire protection for unaffected engine).

GEN switch (affected engine) — OFF.

NON-ESNTL BUS switch — As desired.

If No. 2 engine was shut down:

BATTERY BUS 2 switch — OFF;

BATTERY BUS 1 switch — ON.

After landing, proceed as follows:

Complete engine shutdown.

Exit helicopter.

SMOKE OR FUMES IN CABIN

INDICATIONS:

Smoke, toxic fumes, etc., in cabin.

PROCEDURE:

VENT BLOWER switch — ON.

Vents and windows — Open.

If additional ventilation is required:

AIRSPEED — Reduce to 60 KIAS or less.

Passenger doors — Open.

If time and altitude permit and the source is suspected to be electrical, attempt to identify and isolate the affected system.

Land as soon as possible.

ENGINE FAILURES

SINGLE ENGINE FAILURE

ENG RPM (N2) of the normally operating engine is allowed to droop to 97% during transition from twin engine operation to single engine operation. When the best rate of climb airspeed (70 KIAS) is obtained, ENG RPM (N2) should be increased to 100% if possible.

Flight can be continued on remaining engine until a desirable landing site is available. There are certain combinations of gross weight, altitude, and cold ambient temperatures at which a single engine approach will result in the OEI torque limit being exceeded. A run-on landing at 20 to 30 KIAS is recommended.

CAUTION

RUN ON LANDINGS MAY RESULT IN ROLL OSCILLATIONS WHILE ON THE GROUND. IF THIS OCCURS, LOWERING COLLECTIVE FULLY DOWN OR DISENGAGING HP 1 AND HP 2 WILL STOP THE OSCILLATIONS.

Loss of an engine while hovering at high gross weight and extremely cold conditions will most likely result in exceeding the OEI torque limit. If an overtorque is observed or suspected, an appropriate log book entry shall be made. Refer to Performance charts in Section 4.

NOTE

If an engine restart is to be attempted, refer to ENGINE RESTART in MALFUNCTION PROCEDURES.

INDICATIONS:

ENG 1 OUT or ENG 2 OUT warning light illuminated.

GAS PROD below 53% RPM (N1) and decreasing.

ENG below 85% RPM (N2) and decreasing.

ITT below 400°C and decreasing.

ENG 1 or ENG 2 OIL PRESSURE, DC GENERATOR, and PART SEP OFF caution lights illuminated.

PROCEDURE:

WARNING

IF CORRECTIVE ACTION IS NOT INITIATED IMMEDIATELY, ROTOR RPM CAN DECAY EXCESSIVELY.

CAUTION

DURING COLD WEATHER OPERATIONS, CAREFULLY MONITOR *TORQUE* OF THE NORMAL ENGINE WHEN ONE ENGINE FAILS OR IS SHUT DOWN IN FLIGHT.

Collective — Reduce as required to maintain ROTOR RPM and power within OEI limits.

AIRSPEED — 70 KIAS.

RPM switch — INCR; set ENG RPM (N2) at 100% if possible.

Throttle (affected engine) — Closed.

BOOST PUMP switch (affected engine) — OFF.

FUEL switch (affected engine) — OFF.

FUEL XFEED switch — OVRD CLOSE.

FUEL INTCON switch — OPEN.

GEN switch (affected engine) — OFF.

NON-ESNTL BUS switch — As desired.

If No. 2 engine failed:

BATTERY BUS 2 switch — OFF;

BATTERY BUS 1 switch — ON.

MASTER CAUTION light — Reset.

Altitude — Descend below 5000 ft H_p (if possible).

Land as soon as practical.

DUAL ENGINE FAILURE

INDICATIONS:

ENG 1 OUT and ENG 2 OUT warning lights illuminated.

RPM caution light illuminated.

Rotor rpm audio on.

GAS PROD below 53% RPM (N1) and decreasing (both engines).

ENG below 85% RPM (N2) and decreasing (both engines).

ITT below 400°C and decreasing (both engines).

ENG 1 and ENG 2 OIL PRESSURE, DC GENERATOR, and PART SEP OFF caution lights illuminated.

PROCEDURE:

WARNING

IF CORRECTIVE ACTION IS NOT INITIATED IMMEDIATELY, ROTOR RPM CAN DECAY EXCESSIVELY.

Collective pitch — Reduce. Establish autorotative glide at 70 to 90 KIAS.

NOTE

AIRSPEED for best angle of glide in autorotation is 90 KIAS, and AIRSPEED for minimum rate of descent is 70 KIAS. Autorotational rate of descent is a function of AIRSPEED and ROTOR RPM and is virtually unaffected by gross weight and density altitude.

Accomplish autorotative landing.

If time permits before landing and a restart will not be attempted, proceed as follows:

Throttles (both) — Closed.

FUEL switches (both) — OFF.

BOOST PUMP switches (both) — OFF.

FUEL TRANS switches (both) — OFF.

After landing, complete shutdown.

TAIL ROTOR FAILURES

The key to successful handling of a tail rotor emergency lies in the pilot's ability to quickly recognize the type of malfunction and to select the proper emergency procedure. Following is a discussion of some types of tail rotor malfunctions and their probable effects.

COMPLETE LOSS OF TAIL ROTOR THRUST

INDICATIONS:

This is a situation involving a break in the drive system, such as a severed driveshaft, wherein the tail rotor stops turning and delivers no thrust. A failure of this type in powered flight will result in the nose of the helicopter swinging to the right (left side slip) and usually a roll of the fuselage. Nose down attitude may also be present. The severity of the initial reaction will be affected by airspeed, density altitude, gross weight center of gravity, and power being used.

LOSS OF T/R THRUST AT HOVER

PROCEDURE:

Close throttles immediately and make a hovering autorotation landing. Yawing can be expected on touchdown.

LOSS OF T/R THRUST IN CLIMB

The degree of right yaw upon failure will be greater than that experienced in level flight due to the higher power and anti-torque settings.

PROCEDURES:

Close throttles and lower collective pitch immediately. Establish a glide speed slightly above normal autorotation approach speed.

If a turn is required to reach a more desirable place to land or to align into the wind, make it to the right if possible. A turn to the right can be more nearly streamlined by the use of a little power.

Once aligned for landing, yaw can be controlled in the following manner:

Right Yaw

If the nose yaws right with power off, a pulse of up-collective will produce more friction in the mast thrust bearings, creating a left moment. The greater the input of the pulse, the more the response will be.

WARNING

DO NOT ALLOW ROTOR RPM TO DECAY BELOW MINIMUM LIMITS.

Moving the collective upward abruptly increases rotor loading. Do not hold collective up, as rotor rpm will decrease lower than desirable. It is essential that the collective be returned to the down position for autorotation. This cycle is one pulse. The pulse should be rapid (up and down) but should not be used at low altitudes.

Left Yaw

If the nose yaws left with power off, a slight addition of power should arrest it. Further increase in power results in more right yaw response.

Landing

CAUTION

RUN ON LANDINGS MAY RESULT IN ROLL OSCILLATIONS WHILE ON THE GROUND. IF THIS OCCURS, LOWERING COLLECTIVE FULLY DOWN OR DISENGAGING HP 1 AND HP 2 WILL STOP THE OSCILLATIONS.

During the final stages of the approach, a mild flare should be executed and all power to the rotor should be off. Maintain helicopter in a slight flare and use collective smoothly to execute a soft, slightly nose-high landing. Landing on the

aft portion of the skids will tend to correct side drift. If helicopter starts to turn, move cyclic as necessary to follow the turn until helicopter comes to a complete stop. This technique will, in most cases, result in a run-on type landing.

CAUTION

FOR ZERO GROUND SPEED LANDING, THE FLARE AND THE ABRUPT USE OF COLLECTIVE MAY CAUSE THE NOSE TO YAW LEFT. DO NOT CORRECT WITH THROTTLE. ALTHOUGH APPLICATION OF THROTTLE WILL RESULT IN YAWING TO THE RIGHT, ADDITION OF POWER IS A VERY STRONG RESPONSE MEASURE AND IS TOO SENSITIVE FOR THE PILOT TO MANAGE PROPERLY. DO NOT ADD POWER AT THIS TIME. SLIGHT YAWING UPON TOUCHDOWN AT ZERO GROUND SPEED MAY BE EXPECTED.

LOSS OF T/R THRUST IN LEVEL FLIGHT OR DESCENT

PROCEDURE:

Close throttles and reduce collective pitch immediately. Attain an airspeed slightly above the normal autorotative glide speed.

If altitude permits with AIRSPEED above 60 KIAS, throttle and collective may be gently applied to determine if some degree of powered flight can be resumed. If unacceptable yawing is experienced, re-enter autorotation and continue descent to a landing.

The landing technique is the same as prescribed for the climb condition above.

LOSS OF TAIL ROTOR COMPONENTS

The loss of any tail rotor components will result in a forward center of gravity shift. Other than additional nose down pitching, this situation would be quite similar to complete loss of tail rotor thrust, as discussed above.

TAIL ROTOR FIXED PITCH FAILURES

INDICATIONS:

Tail rotor pitch change control failures are characterized either by a lack of directional response when a pedal is pushed or by locked pedals. If pedals cannot be moved with a moderate amount of force, do not attempt to apply a maximum effort, since a more serious malfunction could result.

FIXED PITCH FAILURE AT HOVER

PROCEDURE:

Do not close throttles unless a severe right yaw occurs. If pedals lock in any position at a hover, landing from a hover can be accomplished with greater safety under power controlled flight rather than by closing throttles and entering autorotation.

FIXED PITCH FAILURE IN FLIGHT

If tail rotor fixed pitch failure occurs during climb (left pedal applied), cruise (approximately neutral pedals), and descent (right pedal applied), a descent and landing can be effected safely by use of power and throttle changes.

PROCEDURES:

If the helicopter is in a trimmed condition when the malfunction is discovered,

engine power and airspeed should be noted and aircraft flown to a suitable landing area.

Combinations of ENG TORQUE, ROTOR RPM, and AIRSPEED will correct or aggravate yaw attitude and these should be adjusted as required to control yaw during landing.

Right Pedal Locked Forward of Neutral

Power should be reduced and ENG RPM (N2) maintained within the green arc. This will help streamline the helicopter in flight. Right turns are easier than left turns. AIRSPEED should be maintained at or above 60 KIAS.

Execute a normal to steep approach adjusting the power as necessary to minimize or prevent right yaw. Maintain ENG RPM (N2) and an AIRSPEED of 60 KIAS during the initial part of the approach.

At 60 to 75 feet AGL and when the landing area can be made, start a slow deceleration to arrive at the intended landing point with AIRSPEED at about 25 KIAS.

At 2 to 5 feet AGL, slowly reduce throttle to overcome yaw effect and allow helicopter to settle. When aligned with the landing area, allow helicopter to touch down.

CAUTION

RUN ON LANDINGS MAY RESULT IN ROLL OSCILLATIONS WHILE ON THE GROUND. IF THIS OCCURS, LOWERING COLLECTIVE FULLY DOWN OR DISENGAGING HP 1 AND HP 2 WILL STOP THE OSCILLATIONS.

After ground contact, use collective and throttle as necessary to maintain alignment with landing strip, and to

minimize forward speed. If helicopter starts to turn, move cyclic as necessary to follow the turn until helicopter comes to a complete stop.

Left Pedal Locked Forward of Neutral

Reduce power and maintain ENG RPM (N2) within the green arc. Normal turns can be safely made under these conditions, although the nose may be displaced to the left.

On final approach, begin a slow deceleration so as to arrive at a point about four to five feet above the intended touchdown area as effective translational lift is lost.

Apply collective pitch to stop the rate of descent and forward speed, and to align the helicopter with the intended landing path. Allow helicopter to touch down at near-zero ground speed, maintaining alignment with throttle.

Pedals Locked In Neutral

Reduce power and maintain ENG RPM (N2) within the green arc. Normal turns can be safely made under these conditions.

Execute a normal to shallow approach, holding AIRSPEED at 60 KIAS during the initial part of the approach. Adjust power as necessary to minimize or prevent right yaw.

At 50 to 75 feet AGL and when the landing area can be made, start a deceleration to arrive at the intended landing point with AIRSPEED at 25 KIAS.

At 2 to 5 feet AGL, use throttle slowly as necessary to maintain alignment with the landing area and to control yaw; do not allow helicopter to settle until alignment is assured, then touch down.

CAUTION

RUN ON LANDINGS MAY RESULT IN ROLL OSCILLATIONS WHILE ON THE GROUND. IF THIS OCCURS, LOWERING COLLECTIVE FULLY DOWN OR DISENGAGING *HP 1* AND *HP 2* WILL STOP THE OSCILLATIONS.

After ground contact, use collective and throttle as necessary to minimize forward speed and to maintain alignment. Move cyclic as necessary to follow the turn until helicopter has come to a complete stop.

LOSS OF PITCH CHANGE CONTROL LINKAGE**INDICATIONS:**

In this type of failure, the pitch-change mechanism is broken at some point and the tail rotor will assume a blade angle determined by the aerodynamic and counterbalance forces.

PROCEDURES:

The corrective action procedures are described in **FIXED PITCH FAILURES** above. The specific procedure to be used depends on the yaw change experienced.

MAIN DRIVESHAFT FAILURE**WARNING**

FAILURE OF THE MAIN DRIVESHAFT TO THE TRANSMISSION WILL RESULT IN COMPLETE LOSS OF POWER TO THE MAIN ROTOR. ALTHOUGH THE COCKPIT INDICATIONS FOR A DRIVESHAFT FAILURE ARE SOMEWHAT COMPARABLE TO A DUAL ENGINE FAILURE, IT IS

IMPERATIVE THAT AUTOROTATIVE FLIGHT PROCEDURES BE ESTABLISHED IMMEDIATELY. FAILURE TO REACT IMMEDIATELY TO THE LOW ROTOR RPM AUDIO SIGNAL, CAUTION LIGHT AND TACHOMETER INDICATION WILL RESULT IN LOSS OF CONTROL.

INDICATIONS:

Left yaw.

Rapid decrease in ROTOR RPM.

Rapid Increase in ENG RPM (N2).

Illumination of rotor RPM caution light with audio.

Possible increase in noise due to:

Overspeeding engine turbines.

Overspeeding combining gearbox.

Driveshaft breakage.

PROCEDURE:

Collective — As required to establish autorotative descent.

Airspeed — Establish airspeed for minimum rate of descent (70 KIAS) or maximum glide (90 KIAS).

Throttles — Close, if time permits.

Controls — As required for autorotative landing.

ENGINE HOT START**INDICATIONS:**

A hot start is caused by a combination of excessive fuel in the combustion chamber and delayed fuel ignition. A hot start may be evidenced by flames emitting from the

exhaust and/or excessive ITT indication. Internal and external damage can result.

PROCEDURE:

Abort start of affected engine as follows:

Throttle — Closed; keep starter engaged.

FUEL switch — OFF.

BOOST PUMP switch — OFF.

Starter — Continue to energize until ITT decreases.

Complete shutdown.

Exit helicopter and check for damage.

If ITT limits for starting were exceeded, refer to Engine Maintenance Manual for inspection requirements.

ENGINE RESTART IN FLIGHT

The conditions which would warrant an attempt to restart the engine would probably be a flameout, caused by a malfunction of the automatic mode of the fuel control unit. The decision to attempt an engine restart during flight is the pilots responsibility.

CAUTION

IF THE CAUSE OF ENGINE FAILURE IS OBVIOUSLY MECHANICAL, AS EVIDENCED BY ABNORMAL SOUNDS, DO NOT ATTEMPT A RESTART.

PROCEDURE:

Position controls of affected engine to attempt restart as follows:

Throttle — Closed.

BOOST PUMP switch — ON.

FUEL XFEED switch — NORM.

FUEL switch — ON.

GOV switch — MANUAL.

GEN switch — OFF.

CAUTION

OEI PERFORMANCE CAN BE AFFECTED DURING GENERATOR ASSISTED START (WITH BOTH BATTERY SWITCHES ON).

For nonassisted battery start (If No. 1 engine failed):

BATTERY BUS 2 switch — OFF.

BATTERY BUS 1 switch — ON.

START switch — ENG 1; observe starter limitations.

ENGINE OIL pressure — Indicating a rise.

CAUTION

WHEN RESTARTING ENGINE IN MANUAL FUEL CONTROL MODE, CAREFULLY MONITOR ITT.

Throttle — Open slowly at 12% GAS PROD RPM (N1) until ITT begins to rise. Do not open throttle further until ITT and GAS PROD RPM (N1) stabilize.

START switch — Centered at 55% GAS PROD RPM (N1).

CAUTION

WHEN OPERATING IN *MANUAL FUEL CONTROL MODE*, MAKE SLOW, SMOOTH, COORDINATED THROTTLE AND COLLECTIVE MOVEMENTS TO AVOID COMPRESSOR STALL, OVERTEMP, UNDERSPEED/OVERSPEED, AND POSSIBLE DRIVETRAIN DAMAGE.

Throttle — Increase slowly; adjust as required to control TORQUE, ITT, and GAS PROD RPM (N1).

NOTE

If TORQUE of affected engine is controlled slightly (approximately 4%) below TORQUE of normal engine, ROTOR RPM will be governed within limits automatically by normal engine.

GEN switches (both) — ON.

BATTERY BUS 2 switch — ON.

FUEL TRANS switch (affected engine) — ON.

FUEL INTCON switch — NORM.

Land as soon as practical.

If restart was unsuccessful, secure affected engine as prescribed in SINGLE ENGINE FAILURE procedure.

ENGINE FUEL CONTROL MALFUNCTIONS

Components of each engine fuel control system subject to malfunction are the manual fuel control unit, the automatic fuel control unit (containing the gas producer turbine governor), the power turbine governor, and the torque control unit. In-flight determination of which component

has malfunctioned is virtually impossible and is irrelevant to the required corrective action. The pilot, therefore, is tasked with interpreting the abnormal indications only so far as to determine which engine has been affected, and which way, in order to perform the proper corrective action.

The primary indications of a fuel control failure usually will be a TORQUE split and an accompanying increase or decrease in ENG RPM (N2) and ROTOR RPM. Normal deviations of ROTOR RPM from the governed setting may occur when large collective changes are made but should not be confused with fuel control failure, unless a large steady-state TORQUE split occurs. The indications of TORQUE, GAS PROD RPM (N1), and ITT alone will not distinguish a high side failure from a low side failure. The triple tachometer must be checked for high or low ENG/ROTOR RPM indications.

The indications of a high side or a low side fuel control failure will vary in accordance with the specific cause of failure and the total power demand at the time of failure.

HIGH SIDE FUEL CONTROL FAILURE

If there is a low power demand (less than single engine power available) at the time of high side failure, ROTOR RPM and ENG RPM (N2) of the affected engine will increase considerably above the governed value. TORQUE, ITT, and GAS PROD RPM (N1) of the affected engine will also increase. As ENG RPM (N2) and ROTOR RPM increase above the governed value, the normal engine will reduce power to keep itself from overspeeding, and will indicate significantly lower TORQUE, ITT, and GAS PROD RPM (N1) than the affected engine.

If there is a high power demand (greater than single engine power available) at the time of high side failure, ROTOR RPM and ENG RPM (N2) of the affected engine will surge initially, along with TORQUE, ITT, and GAS PROD RPM (N1). As ENG RPM

(N2) and ROTOR RPM increase, the normal engine will reduce power to keep itself from overspeeding. The affected engine then tries to assume all of the load, which is beyond its capability due to the high power demand. ENG RPM (N2) of the affected engine (and ROTOR RPM) will then decrease and rejoin the ENG RPM (N2) of the normal engine, stabilizing at or slightly above the governed value as the normal engine adjusts power output to share the load.

INDICATIONS:

High ENG RPM (N2) and ROTOR RPM, possibly with RPM caution light.

Definite TORQUE split (proportional to power demand).

High GAS PROD RPM (N1), ITT, and TORQUE on affected engine.

Return of ENG RPM (N2) and ROTOR RPM to governed value (if power demand is very high).

PROCEDURE:

CAUTION

IF CORRECTIVE ACTION IS NOT INITIATED IMMEDIATELY, *ROTOR* RPM CAN OVERSPEED EXCESSIVELY.

Collective — Adjust as necessary to maintain ROTOR RPM.

Affected engine — Identify.

Throttle (affected engine) — Reduce to maintain TORQUE at or slightly below TORQUE of normal engine.

Throttle frictions — Tighten on normal engine; reduce on affected engine.

Throttle (affected engine) — Reduce to idle.

GOV switch (affected engine) — MANUAL.

CAUTION

WHEN OPERATING IN *MANUAL* FUEL CONTROL MODE, MAKE SLOW, SMOOTH, COORDINATED THROTTLE AND COLLECTIVE MOVEMENTS TO AVOID COMPRESSOR STALL, OVERTEMP, UNDERSPEED/ OVERSPEED, AND POSSIBLE DRIVETRAIN DAMAGE.

Throttle (affected engine) — Increase slowly. Adjust throttle and collective as required to maintain TORQUE of affected engine slightly below TORQUE of normal engine.

MASTER CAUTION light — Reset.

Land as soon as practical.

LOW SIDE FUEL CONTROL FAILURE

If there is a low power demand (less than single engine power available) at the time of low side failure, ROTOR RPM and ENG RPM (N2) of the affected engine will decrease and stabilize at or slightly below the governed value. TORQUE, ITT, and GAS PROD RPM (N1) of the affected engine will also decrease. As ROTOR RPM decreases, the normal engine will increase TORQUE output to assume the load. If power demand is near zero, there may not be a significant TORQUE split.

If there is a high power demand (greater than single engine power available) at the time of low side failure, ROTOR RPM will decrease along with ENG RPM (N2), TORQUE, ITT, and GAS PROD RPM (N1) of the affected engine. As ROTOR RPM

decreases, the normal engine will increase to maximum power to assume the load, causing significant increases in TORQUE, ITT, and GAS PROD RPM (N1), while ENG RPM (N2) will remain below the governed value.

INDICATIONS:

Low ENG RPM (N2) and ROTOR RPM (possibly with RPM caution light and audio if power demand is in excess of single engine power available).

TORQUE split (proportional to power demand).

Low GAS PROD RPM (N1), ITT, and TORQUE on affected engine.

PROCEDURE:**WARNING**

IF CORRECTIVE ACTION IS NOT INITIATED IMMEDIATELY, *ROTOR* RPM CAN DECAY EXCESSIVELY.

Collective — Adjust as necessary to maintain ROTOR RPM.

AIRSPEED — 65 KIAS.

Affected engine — Identify.

Throttle frictions — Tighten on normal engine; reduce on affected engine.

Throttle (affected engine) — Reduce to idle.

GOV switch (affected engine) — MANUAL.

CAUTION

WHEN OPERATING IN *MANUAL* FUEL CONTROL MODE, MAKE

SLOW, SMOOTH, COORDINATED THROTTLE AND COLLECTIVE MOVEMENTS TO AVOID COMPRESSOR STALL, OVERTEMP, UNDERSPEED/ OVERSPEED, AND POSSIBLE DRIVETRAIN DAMAGE.

Throttle (affected engine) — Increase slowly. Adjust throttle and collective as required to maintain torque of affected engine slightly below torque of normal engine.

MASTER CAUTION light — Reset.

Land as soon as practical.

GOVERNOR ACTUATOR FAILURE (FULL INCREASE)**INDICATIONS:**

ENG RPM (N2) and ROTOR RPM increase to approximately 101%.

RPM INCR/DECR switch inoperative.

PROCEDURE:

If this failure occurs during takeoff or landing, no immediate corrective action is necessary to complete either maneuver.

As soon as practical, roll back both throttles to maintain 97 to 100% ENG RPM (N2). Further adjustment of collective and throttles simultaneously will allow full power at pilot's discretion.

Land as soon as practical.

ELECTRICAL POWER FAILURES

DC POWER FAILURE

INDICATIONS:

DC GENERATOR caution light illuminates.

All lighting and avionics on nonessential buses inoperative.

PROCEDURE:

GEN FIELD and GEN RESET circuit breakers — Check in.

GEN switch (affected generator) — RESET, then ON.

If generator remains inoperative, proceed as follows:

GEN switch (affected generator) — OFF.

MASTER CAUTION light — Reset.

If No. 2 generator failed:

BATTERY BUS 2 switch — OFF;

BATTERY BUS 1 switch — ON.

NON-ESNTL BUS switch — MANUAL.

DC AMPS — Monitor; if load exceeds limit:

NON-ESNTL BUS switch — As desired. Switch off unnecessary equipment as required.

If both generators fail and neither will reset, proceed as follows:



DO NOT SELECT *EMERG LOAD* AT PRESSURE ALTITUDES ABOVE 5000 FEET. BOTH FUEL BOOST PUMPS WILL BECOME INOPERATIVE, RESULTING IN POSSIBLE FUEL STARVATION.

EMERG LOAD switch — As desired.

NOTE

A fully charged battery provides electrical power for approximately 30 minutes under normal conditions. With EMERG LOAD switch in EMERG LOAD position, the battery provides approximately 90 minutes of electrical power.

Land as soon as practical.

AC POWER FAILURE

INDICATIONS:

NO. 1 or NO. 2 INVERTER caution light illuminates.

Possible loss of power to certain ac instruments (with no INVERTER caution light).

PROCEDURES:

If either INVERTER caution light illuminates, proceed as follows:

AC VOLTS — Check to determine that remaining inverter has assumed all ac loads.

INV PWR circuit breakers — Check in.

HP 1 or HP 2 button (affected system) — Press to reengage helipilot.

If power is lost only to certain ac instruments, but INVERTER caution lights remain out, proceed as follows:

AC FEEDERS circuit breakers (8 each) — Check in.

During IFR flight, if both inverters fail, land as soon as practical; or continue flight under VFR, if desired.

HYDRAULIC SYSTEM FAILURE

This helicopter has two independent hydraulic boost systems, both of which supply power to the flight control system for the main rotor. The tail rotor control system is powered by system 1 only.

If a hydraulic system failure occurs shortly after the helicopter has been cold soaked at or below -25°C (-13°F), some resistance may occur when the cyclic is near control position extremes. This resistance can be overcome by increased pilot effort.

INDICATIONS:

NO. 1 or NO. 2 HYDRAULIC caution light illuminates.

Abnormal (low, high, or fluctuating) hydraulic pressure in affected system.

Possible high temperature in affected system.

Increased pedal forces (If system 1 failed).

Increased cyclic forces near control extremes (cold weather only).

PROCEDURE:

If either hydraulic system fails, or if system temperature or pressure exceeds limits, proceed as follows:

WARNING

DO NOT EXTEND FLIGHT WITH FAILED HYDRAULIC SYSTEM. THE HELICOPTER IS NOT CONTROLLABLE WITH BOTH HYDRAULIC SYSTEMS INOPERATIVE.

DURING COLD WEATHER OPERATION AVOID HIGH RATES OF CLIMB. MAKE APPROACHES AND LANDINGS INTO THE WIND. AVOID EXTENDED HOVERING AND DO NOT HOVER WITH THE WIND COMING FROM THE AFT LEFT QUADRANT.

Affected system — Identify positively.

HYDR SYS switch (affected system) — OFF.

MASTER CAUTION light — Reset.

Land as soon as possible.

AFCS MALFUNCTIONS

The automatic flight control system can be affected by malfunctions of pilot or copilot attitude gyro, either inverter, or by other electrical malfunctions. Failure of No. 1 hydraulic system will render yaw SAS inoperative but will not affect pitch or roll SAS or ATT mode functions. Failure of No. 2 hydraulic system will not affect AFCS.

If both helipilots are disengaged, the following procedures do not apply.

AFCS FAILS TO ENGAGE OR DISENGAGES

INDICATIONS:

AFCS caution light illuminated.

HP 1 or HP 2 off (button not illuminated).

Possible erratic API indications on HP 1 or HP 2.

Possible ATT flag displayed on pilot or copilot attitude indicator.

Possible Illumination of NO. 1 or NO. 2 INVERTER caution light.

NOTE

If inverter 1 or 2 fails, HP 1 or HP 2 will disengage, but can be reengaged by pressing respective button on AFCS control panel.

PROCEDURE:

AIRSPPEED — Reduce to 115 KIAS or less.

INV 1 and 2 switches — ON; check NO. 1 and NO. 2 INVERTER caution lights extinguished.

Pilot and copilot ADIs — Check ATT flags retracted, Indicators functioning properly.

Check the following circuit breakers in:

CAUTION

DO NOT ATTEMPT TO RESET ANY CIRCUIT BREAKER MORE THAN ONCE.

INV 1 PWR and INV 2 PWR

AC FEEDERS (8)

NO. 1 and NO. 2 ESNTL BUS FEEDERS (ON MAIN DC)

AFCS (No. 1 and No. 2)

AFCS 26V (No. 1 and No. 2)

AFCS 115V (No. 1 and No. 2)

PILOT ATT SYS and CPLT ATT SYS

HP 1 or HP 2 button (affected system) — Press to reengage.

If either helipilot will not reengage, or if abnormal control disturbance occurs, proceed as follows:

Affected helipilot — Disengage.

If IFR, land as soon as practical; or continue flight under VFR, if desired.

If both helipilots fail to reengage, proceed as follows:

AIRSPPEED — As desired.

If IFR, land as soon as practical; or continue flight under VFR, if desired.

AFCS FAILS TO HOLD ATTITUDE

PROCEDURE:

FORCE TRIM switch — Check ON.

SAS/ATT button — Check ATT light illuminated.

If malfunction persists, follow procedure for AFCS FAILS TO ENGAGE OR DISENGAGES.

AFCS HARDOVER OR ABNORMAL CONTROL DISTURBANCE

PROCEDURE:

WARNING

IF HP 1 OR HP 2 FAILS OR IS DISENGAGED, REDUCE AIRSPEED TO 115 KIAS OR LESS.

1. **Cyclic FORCE TRIM release button** — Press; correct helicopter attitude with cyclic and pedals, then release button.
2. **AIRSPEED** — Reduce to 115 KIAS or less.
3. **Actuator position indicators** — Check both systems. If any API shows maximum displacement or erratic operation of any actuator, switch affected helicopter OFF.
4. If IFR, land as soon as practical, or continue flight under VFR, if desired.

AUTOTRIM RUNAWAY

An autotrim runaway can occur only when both HP 1 and HP 2 are ON in ATT mode.

INDICATIONS:

An autotrim runaway in flight will be evidenced by the cyclic stick being driven in a direction opposite to the actuator position indications (HP 1 or HP 2). This condition occurs because the series actuators will be driven to limit authority to compensate for the autotrim runaway. When the actuators are saturated (on stops), the helicopter will respond to the runaway trim command; however, with both HP 1 and HP 2 operative, the autotrim will be cut off automatically two seconds after actuator saturation.

PROCEDURE:

1. **Cyclic FORCE TRIM release button** — Depress to center actuators and retrim to desired attitude.
2. **AIRSPEED** — Reduce to 115 KIAS or less.

NOTE

It is preferable to turn HP 2 off to retain yaw stabilization.

3. HP 2 or HP 1 — OFF.
4. **APIs** — Monitor for proper operation.
5. If IFR, land as soon as practical; or continue flight under VFR, if desired.

STICK CENTERING INDICATOR FAILURE

INDICATIONS:

CYC CTR caution lights fail to illuminate when cyclic is displaced 1.5 inches or more from center position while RPM caution light is illuminated.

PROCEDURE:

Maintain ROTOR between 97 and 100% RPM for ground operation before beginning ENGINE SHUTDOWN procedures.

COMMUNICATION SYSTEM

INTERCOM FAILURE

INDICATION:

Weak reception in headset.

No reception in headset.

PROCEDURE:

Check headset connection.

Verify volume and ICS controls set properly.

Cycle ICS circuit breaker out and in.

For single pilot operations only with Emergency Communications panel installed:

Plug headset into EMERGENCY COMM jack (above and behind pilot position).

Select desired radio on copilot ICS panel.

Key selected radio with EMERGENCY COMM switch (on center pedestal).

COMMUNICATION RADIO FAILURE

INDICATION:

Weak reception in headset.

No reception in headset.

PROCEDURE:

Verify proper radio selected.

Verify volume properly adjusted.

Verify frequency properly set.

Cycle appropriate circuit breaker out and in.

power failure to the fuel signal conditioner.)

NOTE

A power failure to the signal conditioner will disable the FUEL LOW caution light and alter the FUEL TRANS caution indication for affected fuel system. Refer to table 3-2.

PROCEDURE:

FUEL QTY circuit breaker — Recycle. (Affected side.)

FUEL INTCON switch — OPEN.

CABIN HEATER MALFUNCTION

A malfunction in the bleed air heater controls may or may not cause heater to become inoperative.

INDICATIONS:

1. HEATER AIR LINE caution light illuminates.
2. Heated airflow does not shut off when thermostat knob is turned to fully COLD position.

PROCEDURE:

1. HEATER switch — OFF immediately.
2. CABIN HTR circuit breaker — Check; if out, do not reset.

FUEL QUANTITY INDICATIONS MALFUNCTION

INDICATION:

FUEL QTY indication goes to zero from a previously normal condition. (Possible

NOTE

Allow sufficient time for fuel levels to equalize. Approximate fuel loads may be obtained by doubling remaining fuel quantity indicated.

STATIC PORT OBSTRUCTION

INDICATION:

Erratic readings from the AIRSPEED indicator, VERTICAL SPEED indicator, and altimeter when operating helicopter in rain with STATIC SOURCE switch in PRI position.

PROCEDURE

1. Windows and vents — Close.
2. HEATER switch — OFF.
3. STATIC SOURCE switch — ALTN.

NOTE

This procedure selects an alternate static source (cabin air) for pilot side instruments only.

Section 4

PERFORMANCE

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Section 4

PERFORMANCE

INTRODUCTION

The performance data presented herein are derived from the engine manufacturer's specification power for the engine less installation losses. These data are applicable to the basic helicopter without any optional equipment which would appreciably affect lift, drag, or power available.

POWER ASSURANCE CHECKS

Power Assurance Check charts (figure 4-1) are provided to determine if the engines can produce installed specification power.

A power assurance check should be performed daily. Additional checks should be made if unusual operating conditions or

indications arise. The hover check is performed prior to takeoff, and the in-flight check is provided for periodic in-flight monitoring of engine performance. Either power assurance check method may be selected at the discretion of the pilot. It is the pilots responsibility to accomplish the procedure safely, considering passenger load, terrain being overflown, and the qualifications of persons on board to assist in watching for other air traffic and to record power check data.

If either engine does not meet the requirements of the hover or the in-flight power assurance check, published performance may not be achievable. The cause of engine power loss, or excessive ITT or GAS PROD RPM (N1) should be determined as soon as practical. Refer to Engine Maintenance Manual.

DENSITY ALTITUDE

A Density Altitude Chart (figure 4-2) is provided to aid in calculation of performance and limitations. Density altitude (H_D) is an expression of the density of the air in terms of height above sea level; hence, the less dense the air, the higher the density altitude. For standard conditions of temperature and pressure, density altitude is the same as pressure altitude (H_P). As temperature increases above standard for any altitude, the density altitude will also increase to values higher than pressure altitude. The chart expresses density altitude as a function of pressure altitude and temperature.

The chart also includes the inverse of the square root of the density ratio ($1/\sigma$), which is used to calculate KTAS by the relation:

$$KTAS = KCAS \times 1/\sigma$$

EXAMPLE:

If the ambient temperature is -15°C and the pressure altitude is 6000 feet, find the density altitude, $1/\sigma$, and true airspeed for 100 KCAS.

Solution:

- Enter the bottom of the chart at -15°C .
- Move vertically upward to the 6000 foot pressure altitude line.
- From this point, move horizontally to the left and read a density altitude of 4000 feet and move horizontally to the right and read $1/\sigma$ equals 1.06.
- True airspeed = $KCAS \times 1/\sigma = 100 \times 1.06 = 106 \text{ KTAS}$.

HOVER CEILING IGE

Adequate cyclic and directional control are available at the gross weights allowed by the Hover Ceiling IGE charts in relative winds up to 35 knots from any direction at or below 3000 feet H_D . Improved control margins will be achieved by avoiding winds in the critical relative wind azimuth areas (figure 4-3).

The Hover Ceiling In Ground Effect charts (figure 4-4) provide the maximum allowable gross weights for hovering IGE at all pressure altitude and outside air temperature conditions with heater on or off. Conversely, the hover ceiling altitude can be determined for any given gross weight.

HOVER CEILING OGE

The Hover Ceiling Out of Ground Effect charts (figure 4-5) provide maximum weights for hovering OGE at all pressure altitude and outside air temperature conditions with heater on or off.

CAUTION

**OGE HOVER OPERATION MAY
RESULT IN VIOLATION OF
HEIGHT-VELOCITY LIMITATIONS.**

Some of the OGE hover ceiling charts are divided into two areas as follows:

AREA A (unshaded area) as shown on the hover ceiling charts presents hover performance for which satisfactory cyclic and directional control have been demonstrated in relative winds of 35 knots from any direction at or below 3000 feet H_D . Improved control margins will be achieved by avoiding winds in the critical relative wind azimuth areas (figure 4-3).

AREA B (shaded area) as shown on hover ceiling charts presents additional hover performance which can be achieved in

**MODEL 412
POWER ASSURANCE CHECK (HOVER)
PT6T-3B ENGINE (WITH GAS PRODUCER GAGE P/N 212-075-037-101)**

HEATER/ECU — OFF.

THROTTLES:

TEST ENGINE — FULL OPEN, FRICTIONED.
OTHER ENGINE — IDLE.

ENG — 97% RPM (N2).

COLLECTIVE PITCH — INCREASE UNTIL LIGHT ON
SKIDS OR HOVERING. DO NOT EXCEED 810° ITT OR
100.8% GAS PROD RPM (N1).

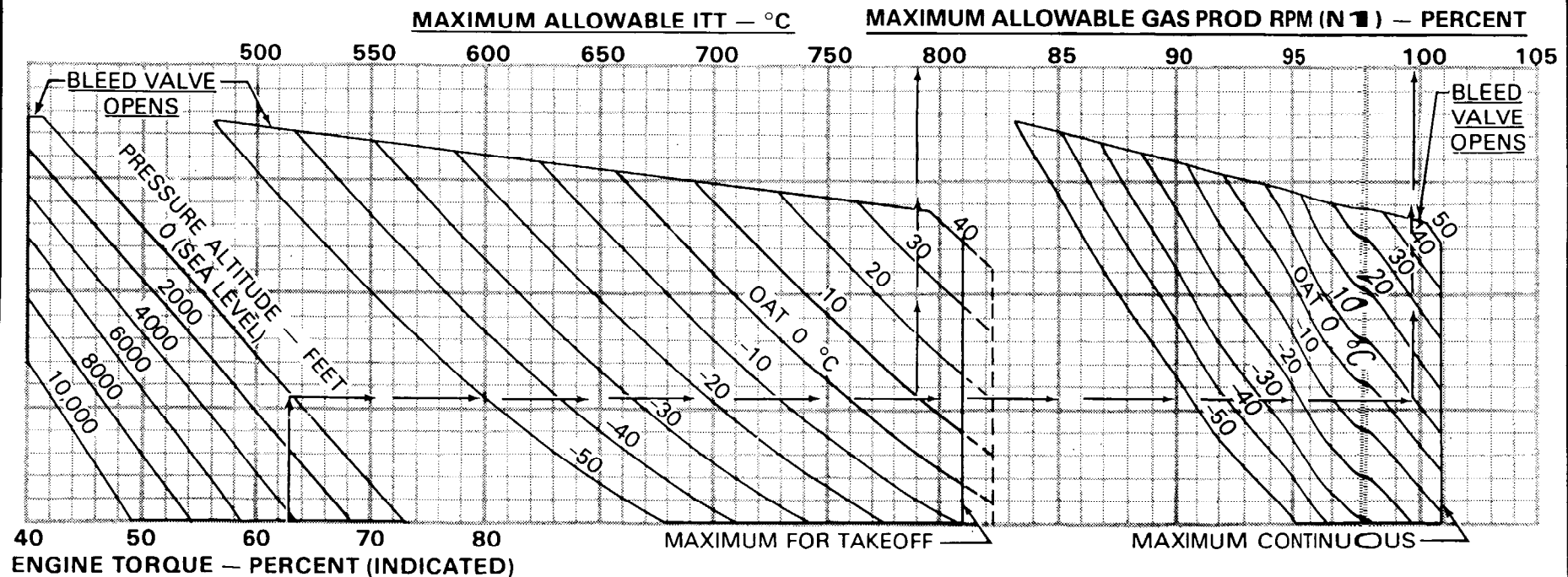
STABILIZE POWER ONE MINUTE, THEN RECORD
PRESSURE ALTITUDE, OAT, ENGINE TORQUE, ITT,
AND GAS PROD RPM (N1).

ENTER CHART AT INDICATED ENGINE TORQUE,
MOVE UP TO INTERSECT PRESSURE ALTITUDE,
PROCEED TO THE RIGHT TO INTERSECT OUTSIDE
AIR TEMPERATURE, THEN MOVE UP TO READ
VALUES FOR MAXIMUM ALLOWABLE ITT AND GAS
PROD RPM (N1).

IF INDICATED ITT OR GAS PROD RPM (N1) EXCEEDS
MAX ALLOWABLE, REPEAT CHECK, STABILIZING
POWER FOUR MINUTES.

REPEAT CHECK USING OTHER ENGINE.

IF EITHER ENGINE EXCEEDS ALLOWABLE ITT OR GAS
PROD RPM (N1) AFTER STABILIZING FOUR MINUTES,
PUBLISHED PERFORMANCE MAY NOT BE
ACHIEVABLE. CAUSE SHOULD BE DETERMINED AS
SOON AS PRACTICAL.



412-F2-4-1-1

Figure 4-1. Power assurance check (Sheet 1 of 4)

**MODEL 412
POWER ASSURANCE CHECK (IN-FLIGHT)
PT6T-3B ENGINE (WITH GAS PRODUCER GAGE P/N 212-075-037-101)**

ESTABLISH LEVEL FLIGHT ABOVE 1000 FEET AGL.

AIRSPEED — 100 KIAS (OR VNE, IF LESS).

HEATER/ECU — OFF.

THROTTLES:

TEST ENGINE — FULL OPEN, FRICTIONED

OTHER ENGINE — DECREASE SLOWLY UNTIL TEST ENGINE TORQUE IS WITHIN TEST RANGE. DO NOT EXCEED 810°C ITT OR 100.8% GAS PROD RPM (N1).

ENG — 97% RPM (N2).

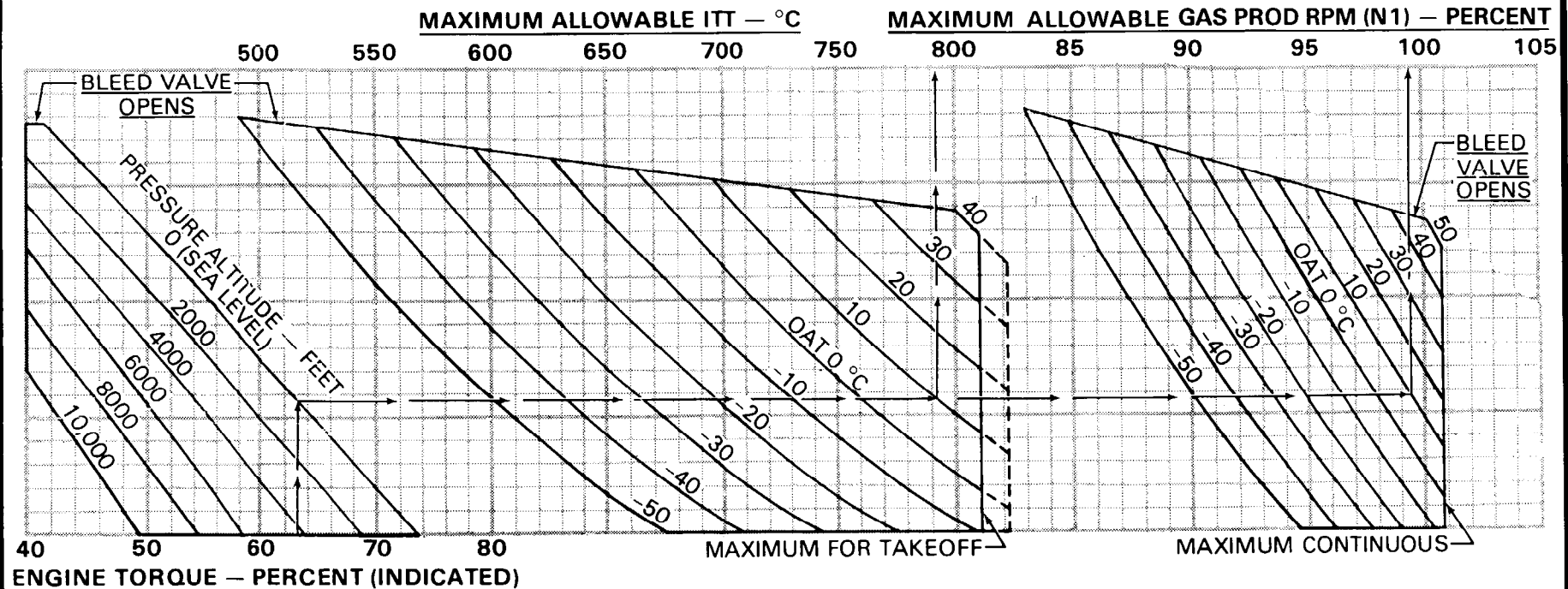
STABILIZE POWER ONE MINUTE IN LEVEL FLIGHT, THEN RECORD PRESSURE ALTITUDE, OAT, ENGINE TORQUE, ITT, AND GAS PROD RPM (N1).

ENTER CHART AT INDICATED ENGINE TORQUE, MOVE UP TO INTERSECT PRESSURE ALTITUDE, PROCEED TO THE RIGHT TO INTERSECT OUTSIDE AIR TEMPERATURE, THEN MOVE UP TO READ VALUES FOR MAXIMUM ALLOWABLE ITT AND GAS PROD RPM (N1).

IF INDICATED ITT OR GAS PROD RPM EXCEEDS MAX ALLOWABLE, REPEAT CHECK, STABILIZING POWER FOUR MINUTES.

REPEAT CHECK USING OTHER ENGINE.

IF EITHER ENGINE EXCEEDS ALLOWABLE ITT OR GAS PROD RPM (N1) AFTER STABILIZING FOUR MINUTES, PUBLISHED PERFORMANCE MAY NOT BE ACHIEVABLE. CAUSE SHOULD BE DETERMINED AS SOON AS PRACTICAL.



412-F2-4-1-2

Figure 4-1. Power assurance check (Sheet 2 of 4)

**MODEL 412
POWER ASSURANCE CHECK (HOVER)
PT6T-3B ENGINE (WITH GAS PRODUCER GAGE P/N 212-075-037-113)**

HEATER/ECU — OFF.

THROTTLES:

TEST ENGINE — FULL OPEN, FRICTIONED.
OTHER ENGINE — IDLE.

ENG — 97% RPM (N2).

COLLECTIVE PITCH — INCREASE UNTIL LIGHT ON
SKIDS OR HOVERING. DO NOT EXCEED 810° ITT OR
101.8% GAS PROD RPM (N1).

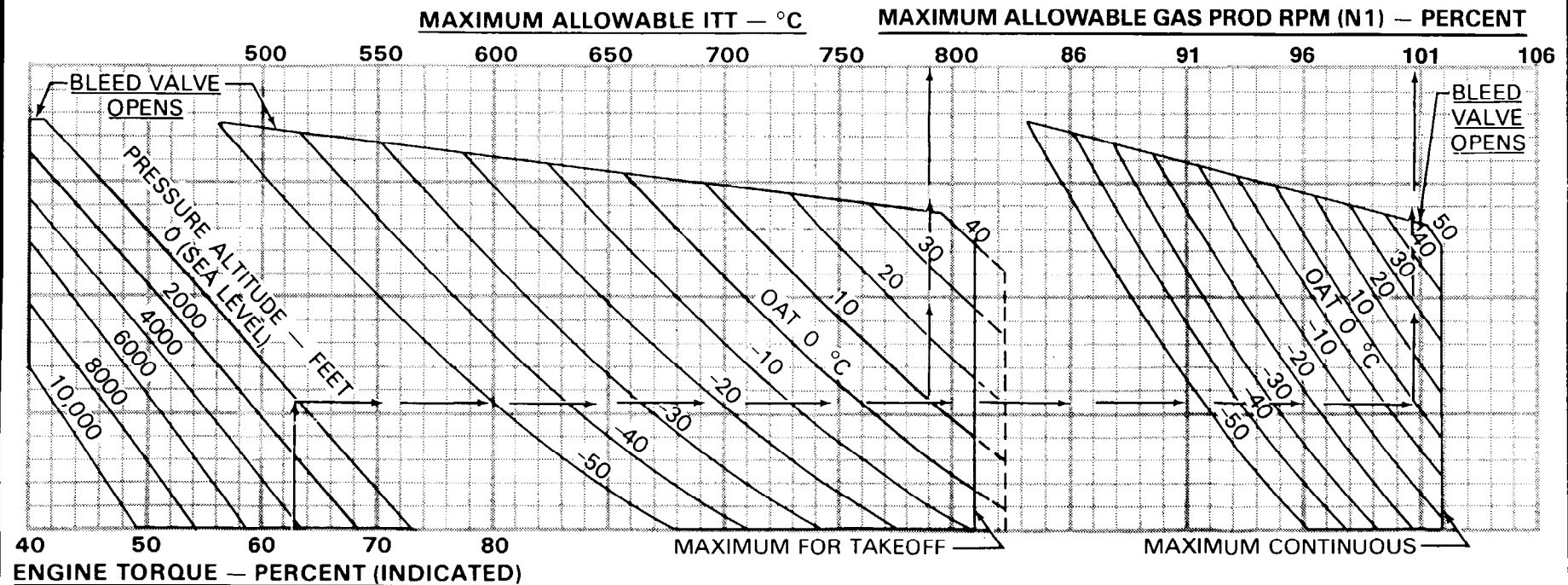
STABILIZE POWER ONE MINUTE, THEN RECORD
PRESSURE ALTITUDE, OAT, ENGINE TORQUE, ITT,
AND GAS PROD RPM (N1).

ENTER CHART AT INDICATED ENGINE TORQUE,
MOVE UP TO INTERSECT PRESSURE ALTITUDE,
PROCEED TO THE RIGHT TO INTERSECT OUTSIDE
AIR TEMPERATURE, THEN MOVE UP TO READ
VALUES FOR MAXIMUM ALLOWABLE ITT AND GAS
PROD RPM (N1).

IF INDICATED ITT OR GAS PROD RPM (N1) EXCEEDS
MAX ALLOWABLE, REPEAT CHECK, STABILIZING
POWER FOUR MINUTES.

REPEAT CHECK USING OTHER ENGINE.

IF EITHER ENGINE EXCEEDS ALLOWABLE ITT OR GAS
PROD RPM (N1) AFTER STABILIZING FOUR MINUTES,
PUBLISHED PERFORMANCE MAY NOT BE
ACHIEVABLE. CAUSE SHOULD BE DETERMINED AS
SOON AS PRACTICAL.



412-F2-4-1-3

Figure 4-1. Power assurance check (Sheet 3 of 4)

**MODEL 412
POWER ASSURANCE CHECK (IN-FLIGHT)
PT6T-3B ENGINE (WITH GAS PRODUCER GAGE P/N 212-075-037-113)**

ESTABLISH LEVEL FLIGHT ABOVE 1000 FEET AGL.

AIRSPED — 100 KIAS (OR VNE, IF LESS).

HEATER/ECU — OFF.

THROTTLES:

TEST ENGINE — FULL OPEN, FRICTIONED.

OTHER ENGINE — DECREASE SLOWLY UNTIL TEST ENGINE TORQUE IS WITHIN TEST RANGE. DO NOT EXCEED 810°C ITT OR 101.8% GAS PROD RPM (N1).

ENG — 97% RPM (N2).

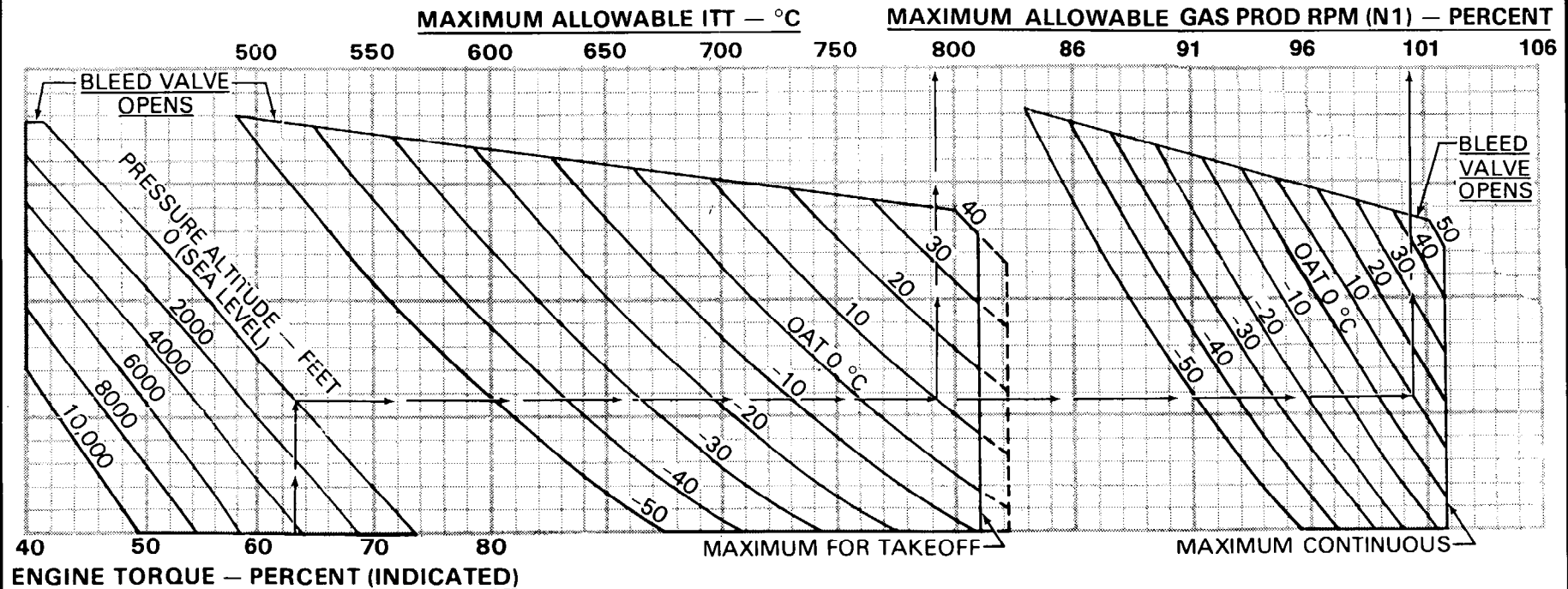
STABILIZE POWER ONE MINUTE IN LEVEL FLIGHT, THEN RECORD PRESSURE ALTITUDE, OAT, ENGINE TORQUE, ITT, AND GAS PROD RPM (N1).

ENTER CHART AT INDICATED ENGINE TORQUE, MOVE UP TO INTERSECT PRESSURE ALTITUDE, PROCEED TO THE RIGHT TO INTERSECT OUTSIDE AIR TEMPERATURE, THEN MOVE UP TO READ VALUES FOR MAXIMUM ALLOWABLE ITT AND GAS PROD RPM (N1).

IF INDICATED ITT OR GAS PROD RPM EXCEEDS MAX ALLOWABLE, REPEAT CHECK, STABILIZING POWER FOUR MINUTES.

REPEAT CHECK USING OTHER ENGINE.

IF EITHER ENGINE EXCEEDS ALLOWABLE ITT OR GAS PROD RPM (N1) AFTER STABILIZING FOUR MINUTES, PUBLISHED PERFORMANCE MAY NOT BE ACHIEVABLE. CAUSE SHOULD BE DETERMINED AS SOON AS PRACTICAL.



412-F2-4-1-4

Figure 4-1. Power assurance check (Sheet 4 of 4)

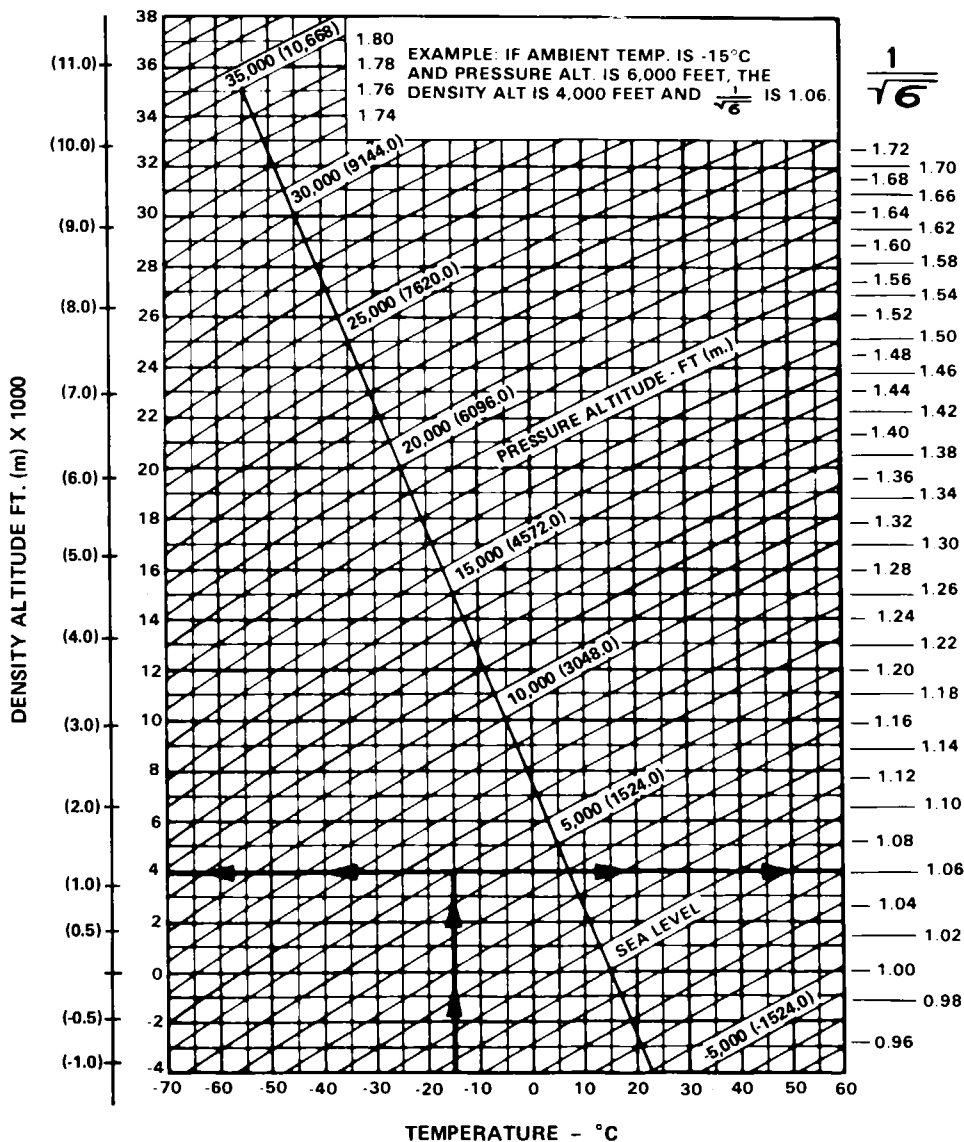


Figure 4-2. Density altitude

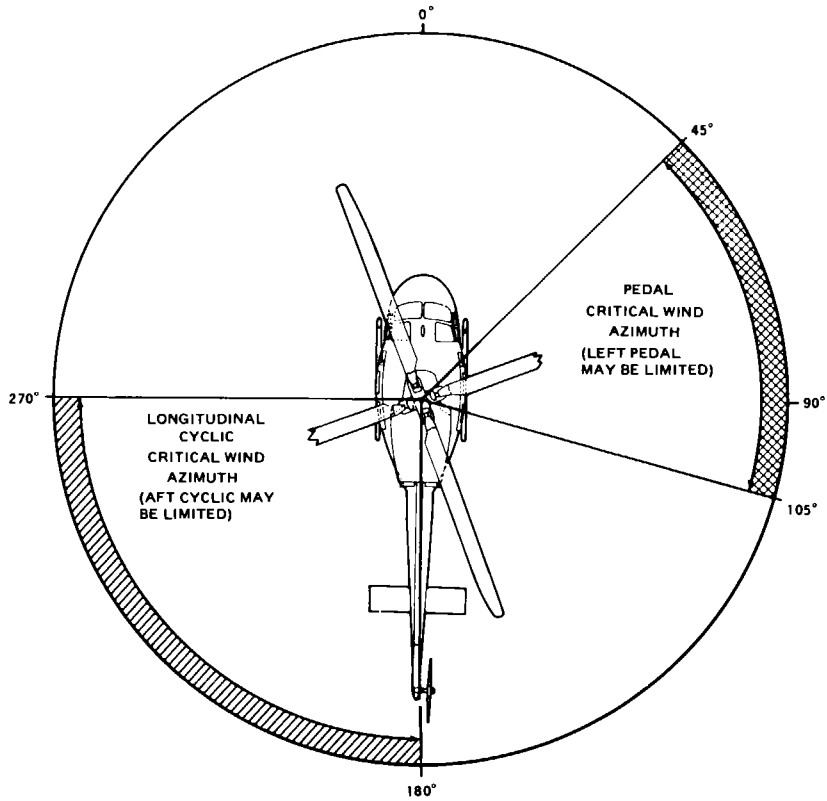


Figure 4-3. Critical relative wind azimuths

HOVER CEILING IN GROUND EFFECT

POWER: SEE NOTE BELOW
ENG — 100% RPM (N2)
GENERATOR 150 AMPS (EA.)

SKID HEIGHT 4 FEET
HEATER ON OR OFF
-40° TO 52°C

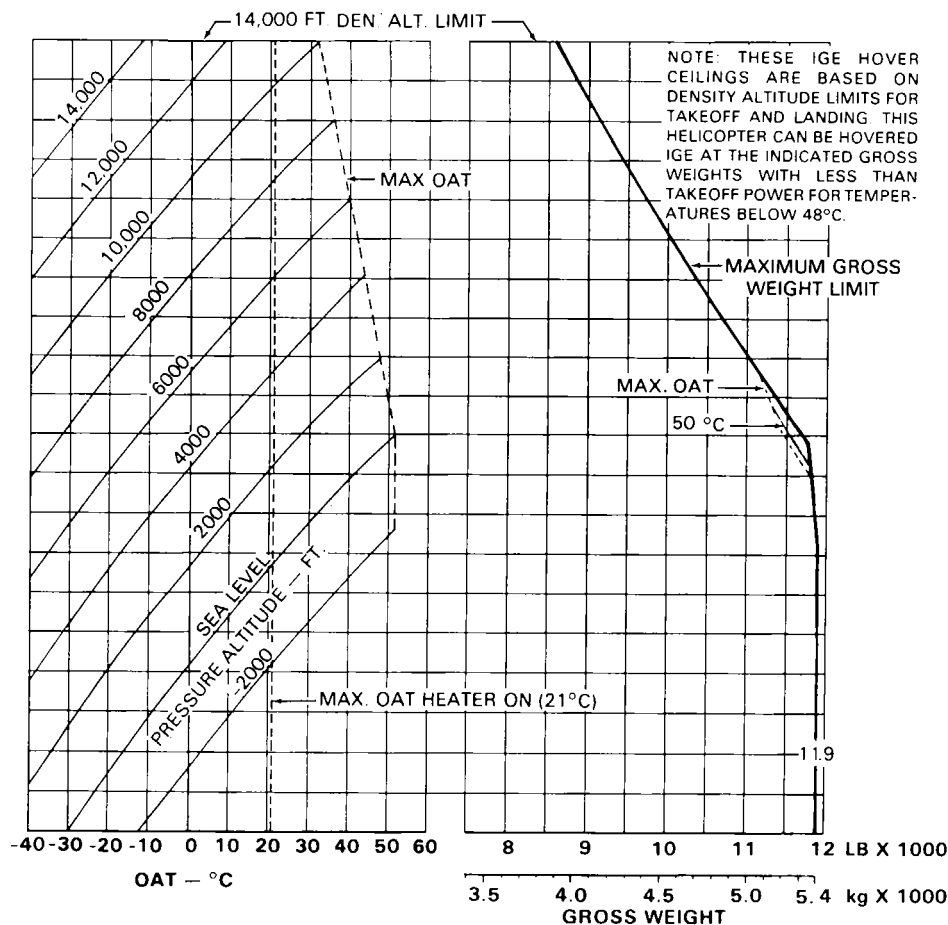


Figure 4-4. Hover ceiling in ground effect (Sheet 1 of 2)

HOVER CEILING IN GROUND EFFECT

MAXIMUM CONTINUOUS POWER
 ENG - 100% RPM (N2)
 GENERATOR 150 AMPS (EA.)

SKID HEIGHT 4 FEET
 HEATER ON OR OFF
 -40° TO 52°C

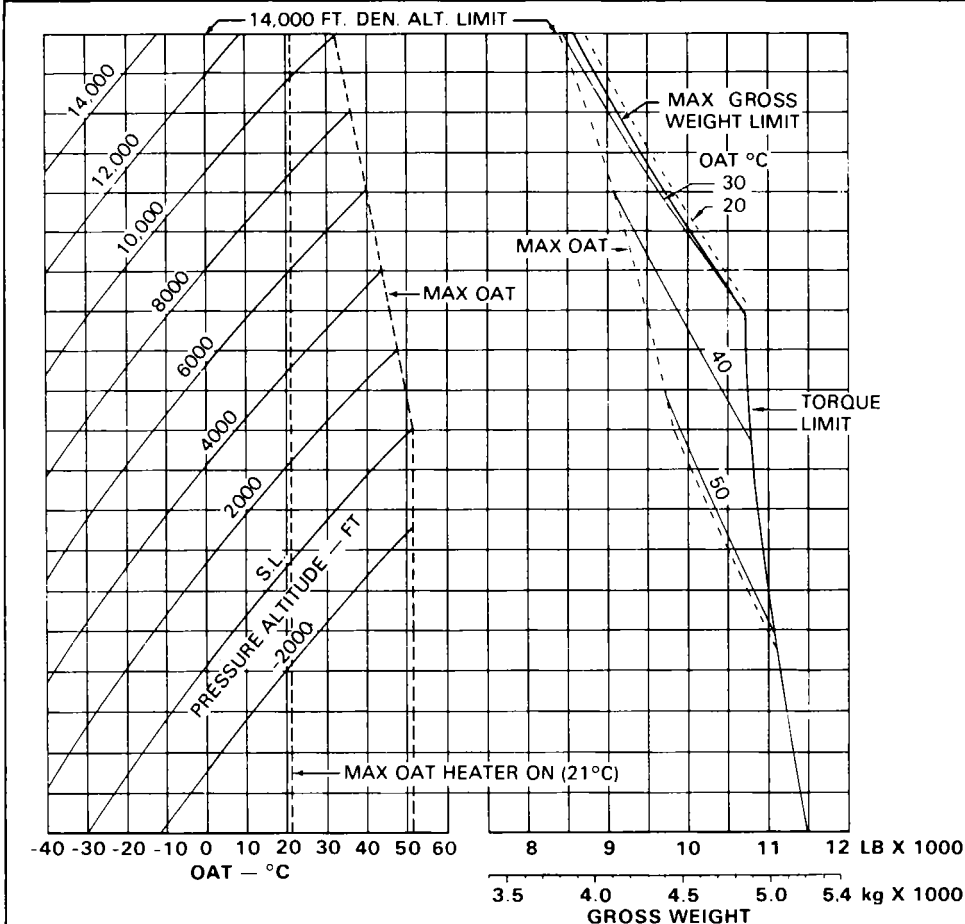


Figure 4-4. Hover ceiling in ground effect (Sheet 2 of 2)

HOVER CEILING OUT OF GROUND EFFECT

TAKEOFF POWER
ENG - 100% RPM (N2)
GENERATOR 150 AMPS (EA.)

SKID HEIGHT 60 FT.
HEATER OFF
0 TO 52°C

CAUTION: OGE HOVER OPERATION MAY RESULT IN VIOLATION OF H-V LIMITATIONS.

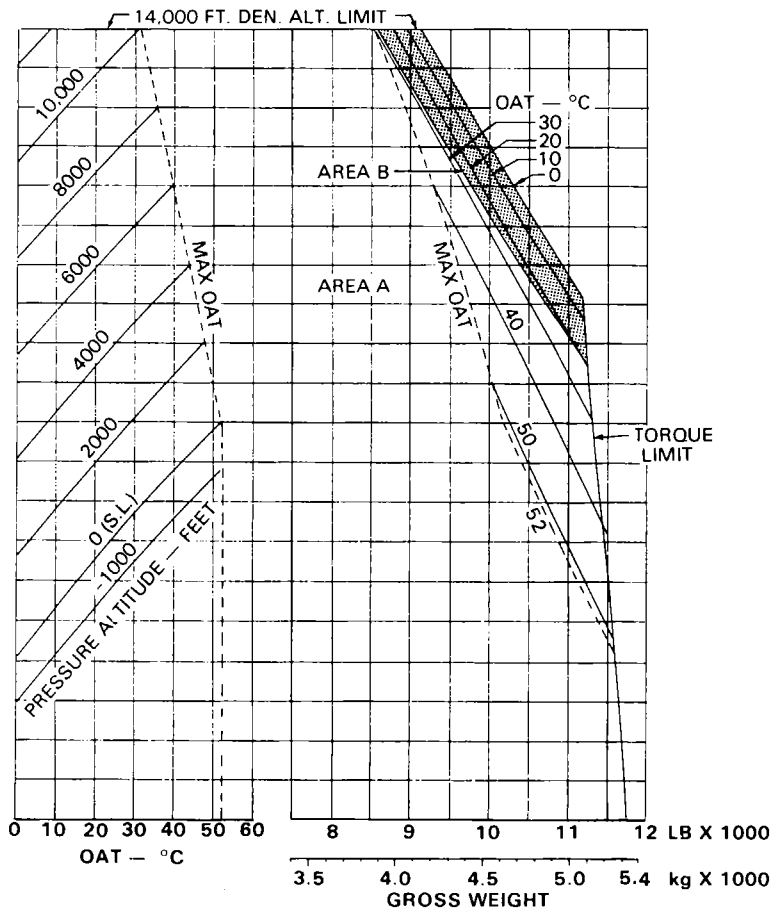


Figure 4-5. Hover ceiling out of ground effect (Sheet 1 of 8)

HOVER CEILING OUT OF GROUND EFFECT

TAKEOFF POWER
ENG — 100% RPM (N2)
GENERATOR 150 AMPS (EA.)

SKID HEIGHT 60 FT.
HEATER OFF
-40 TO 0°C

CAUTION: OGE HOVER OPERATION MAY RESULT IN VIOLATION OF H-V LIMITATIONS.

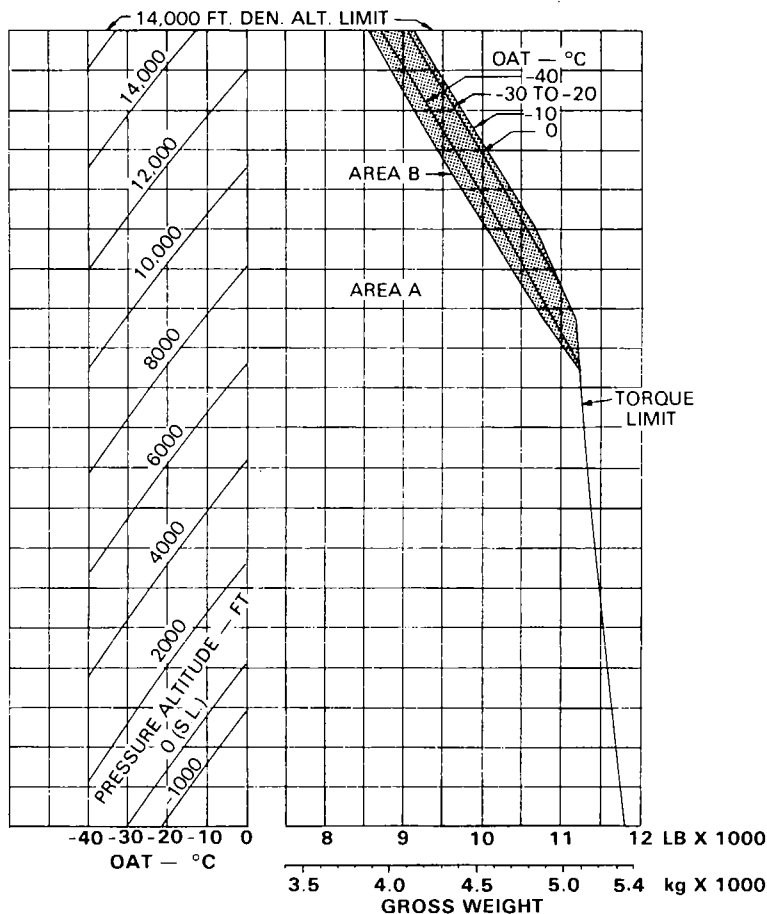


Figure 4-5. Hover ceiling out of ground effect (Sheet 2 of 8)

HOVER CEILING OUT OF GROUND EFFECT

TAKEOFF POWER
ENG — 100% RPM (N2)
GENERATOR 150 AMPS (EA.)

SKID HEIGHT 60 FT.
HEATER ON
0 TO 20°C

CAUTION: OGE HOVER OPERATION MAY RESULT IN VIOLATION OF H-V LIMITATIONS.

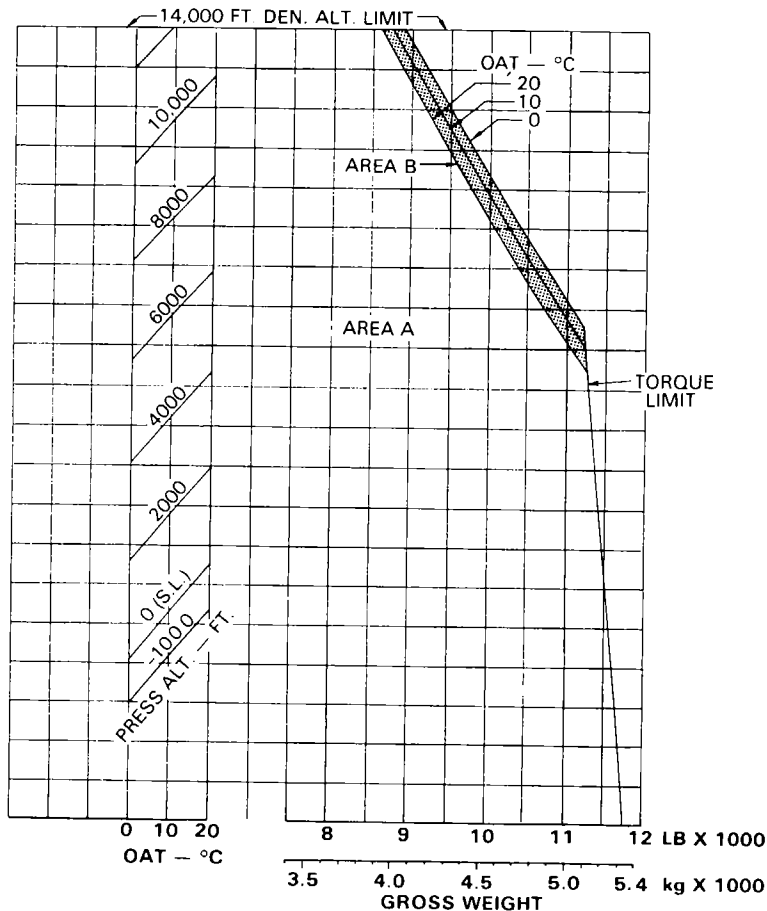


Figure 4-5. Hover ceiling out of ground effect (Sheet 3 of 8)

HOVER CEILING OUT OF GROUND EFFECT

TAKEOFF POWER
ENG - 100% RPM (N2)
GENERATOR 150 AMPS (EA.)

SKID HEIGHT 60 FT.
HEATER ON
-40 TO 0°C

CAUTION: OGE HOVER OPERATION MAY RESULT IN VIOLATION OF H-V LIMITATIONS.

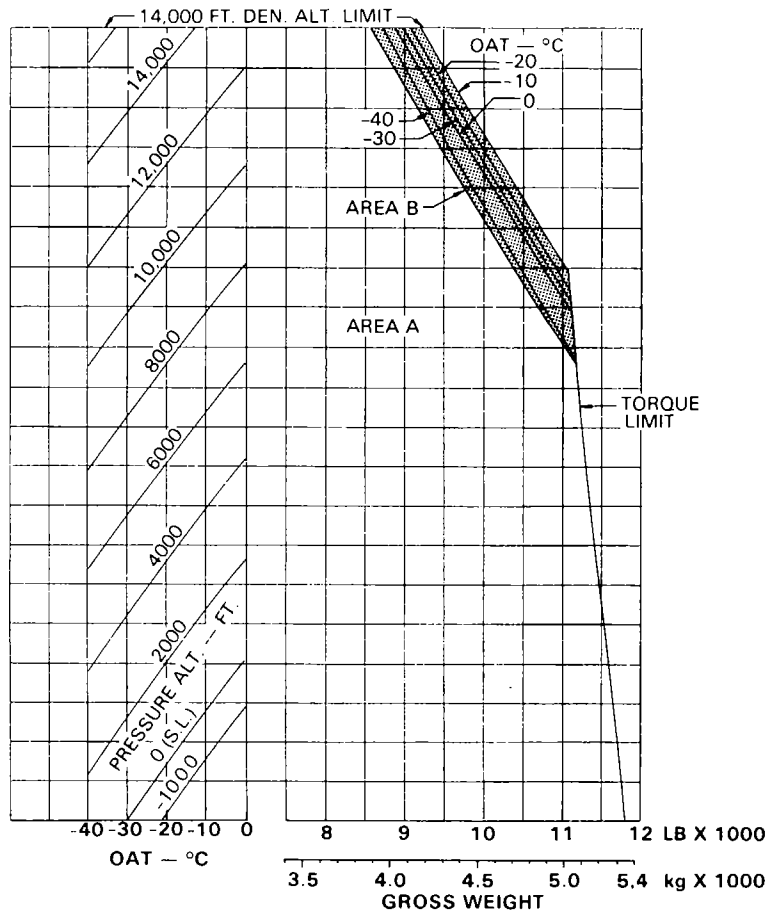


Figure 4-5. Hover ceiling out of ground effect (Sheet 4 of 8)

HOVER CEILING OUT OF GROUND EFFECT

MAXIMUM CONTINUOUS POWER
ENG - 100% RPM (N2)
GENERATOR 150 AMPS (EA.)

SKID HEIGHT 60 FT.
HEATER OFF
0 TO 52°C

CAUTION: OGE HOVER OPERATION MAY RESULT IN VIOLATION OF H-V LIMITATIONS.

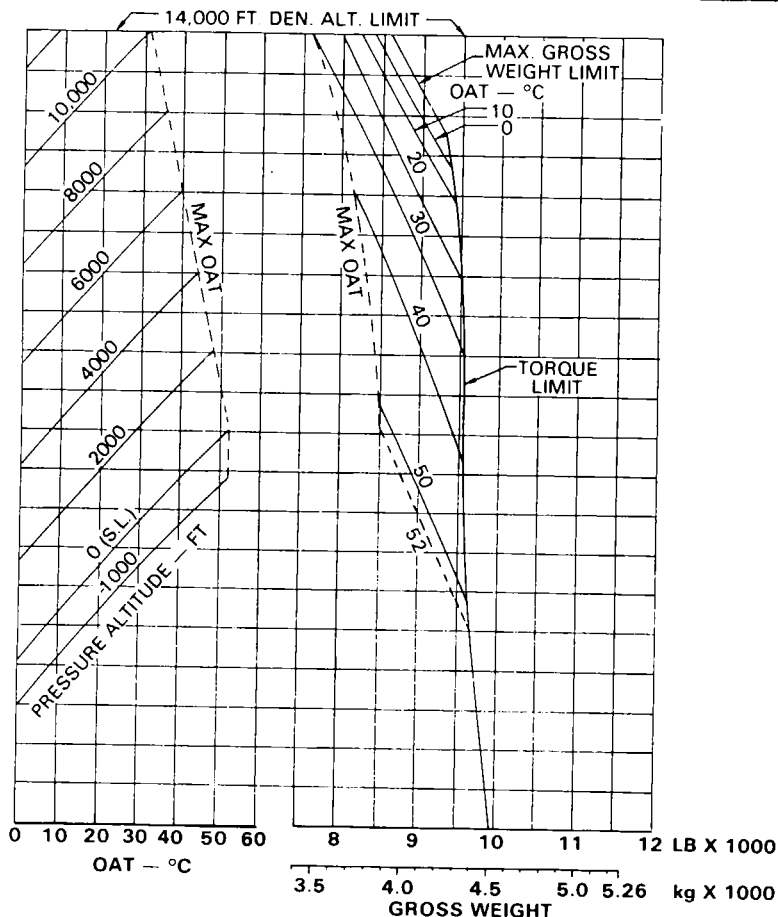


Figure 4-5. Hover ceiling out of ground effect (Sheet 5 of 8)

HOVER CEILING OUT OF GROUND EFFECT

MAXIMUM CONTINUOUS POWER
ENG — 100% RPM (N2)
GENERATOR 150 AMPS (EA.)

SKID HEIGHT 60 FT.
HEATER OFF
-40 TO 0°C

CAUTION: OGE HOVER OPERATION MAY RESULT IN VIOLATION OF H-V LIMITATIONS.

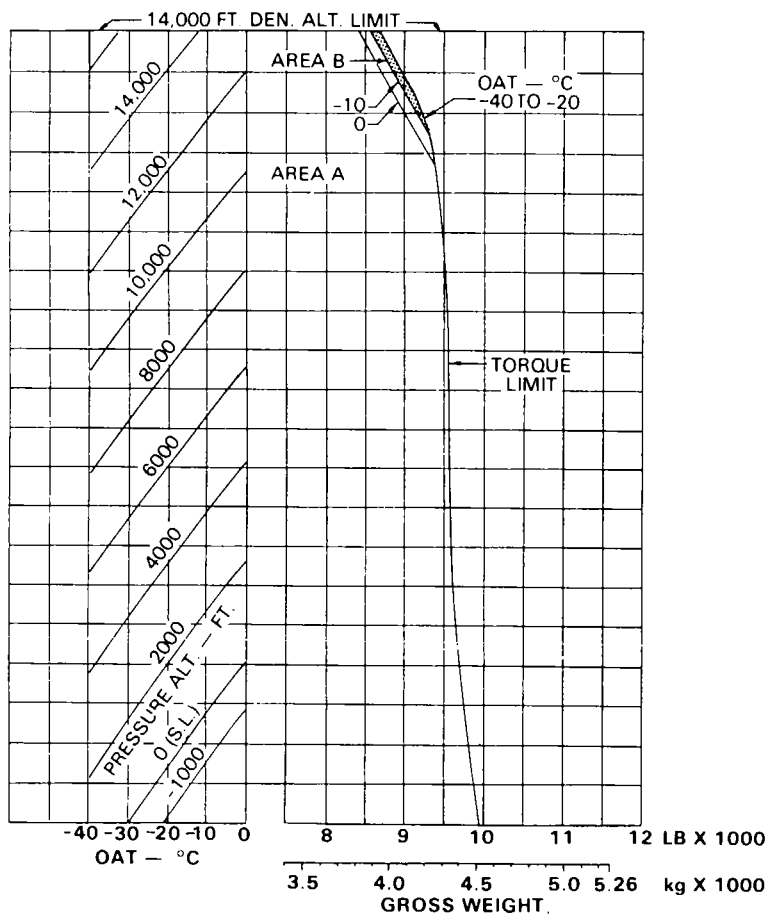


Figure 4-5. Hover ceiling out of ground effect (Sheet 6 of 8)

HOVER CEILING OUT OF GROUND EFFECT

MAXIMUM CONTINUOUS POWER
 ENG — 100% RPM (N2)
 GENERATOR 150 AMPS (EA.)

SKID HEIGHT 60 FT.
 HEATER ON
 0 TO 20°C

CAUTION: OGE HOVER OPERATION MAY RESULT IN VIOLATION OF H-V LIMITATIONS.

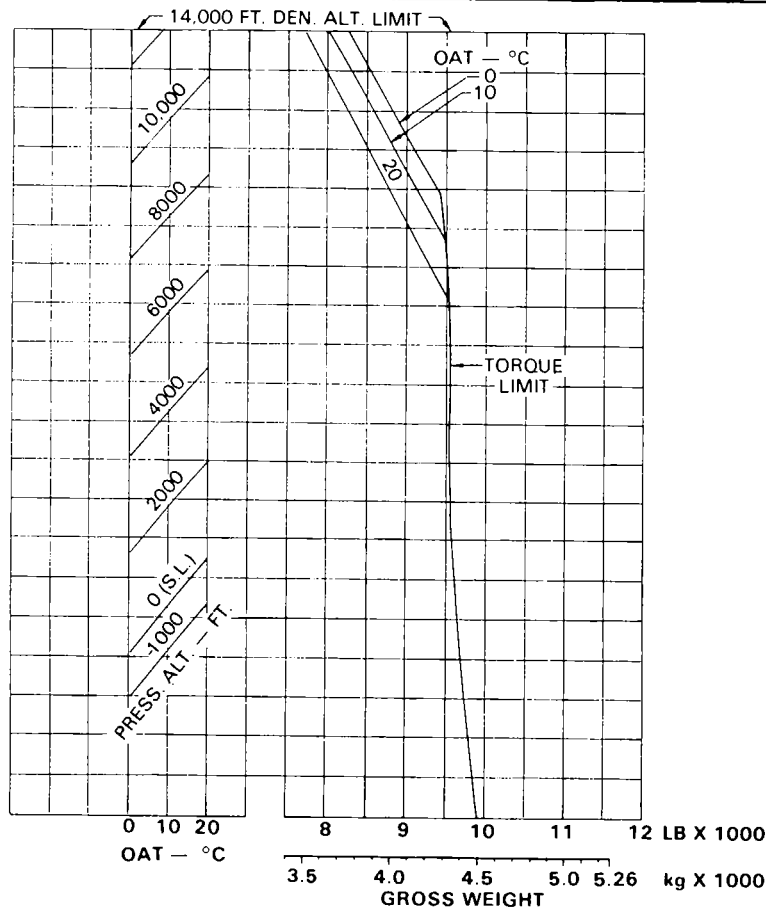


Figure 4-5. Hover ceiling out of ground effect (Sheet 7 of 8)

HOVER CEILING OUT OF GROUND EFFECT

MAXIMUM CONTINUOUS POWER
ENG — 100% RPM (N2)
GENERATOR 150 AMPS (EA.)

SKID HEIGHT 60 FT.
HEATER ON
-40 TO 0°C

CAUTION: OGE HOVER OPERATION MAY RESULT IN VIOLATION OF H-V LIMITATIONS.

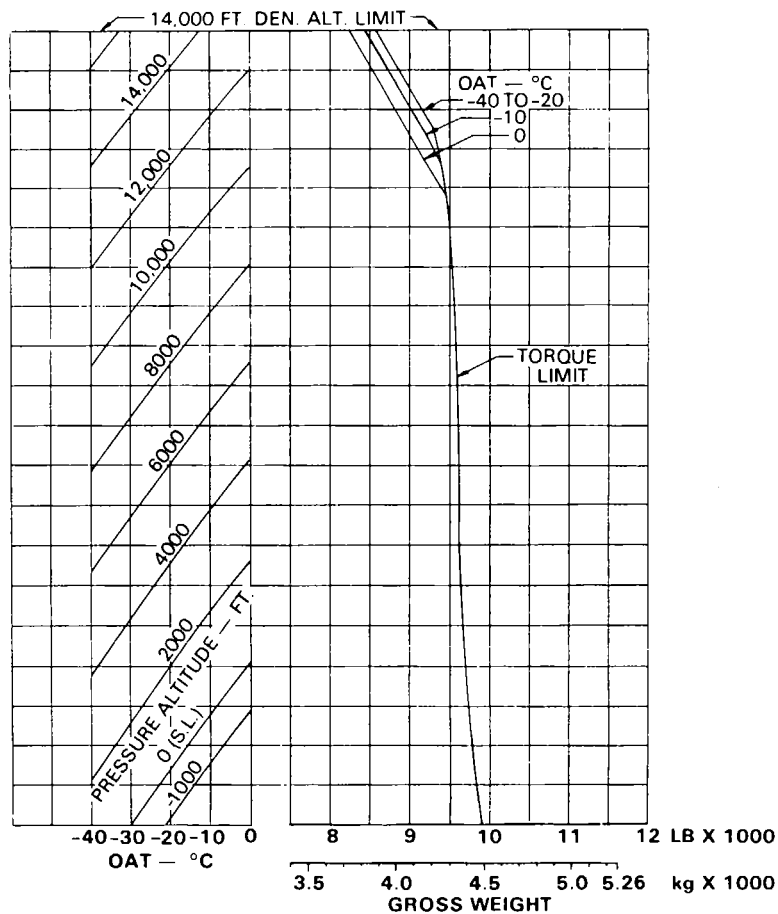


Figure 4-5. Hover ceiling out of ground effect (Sheet 8 of 8)

calm winds or winds outside the critical relative wind azimuth areas.

NOTE

Tail rotor or cyclic control margin may preclude operation in AREA B of the hover ceiling charts when the relative wind is in the respective critical wind azimuth area.

TAKEOFF DISTANCE

The Takeoff Distance charts (figure 4-6) provide takeoff distances required to clear a 50 foot or 15 meter obstacle in a zero wind condition, using a takeoff flight path which will avoid the critical areas of the Height-Velocity diagram (Section 1). Takeoff is initiated from a hover at 4 feet (1.2 meters) skid height with climbout speed of 45 knots.

NOTE

Downwind takeoffs are not recommended because the published takeoff distance performance cannot be achieved.

LANDING DISTANCE

The Single Engine Landing Distance chart (figure 4-7) provides the landing distances required to clear a 50 foot (15 meter) obstacle for all outside air temperatures, pressure altitudes, and gross weights. Landing distances are based on an approach condition of 45 KIAS and 500 feet per minute rate of descent and zero wind.

TWIN ENGINE RATE OF CLIMB

The Twin Engine Rate of Climb charts (figure 4-8) provide the rates of climb that

can be obtained at all outside air temperatures/pressure altitudes/gross weight combinations with heater on or off at maximum continuous power and takeoff power.

NOTE

All rate of climb data are based on changes in true altitude (pressure altitude corrected for nonstandard temperature).

SINGLE ENGINE RATE OF CLIMB

The Single Engine Rate of Climb charts (figure 4-9) provide the rates of climb that can be obtained at all outside air temperatures/pressure altitudes/gross weight combinations with heater off at maximum continuous power and 30 minute OEI power.

NOTE

Published single engine performance is intended for emergency use only when one engine becomes inoperative due to an actual malfunction. Routine operation in 2 1/2 minute or 30 minute OEI range can affect engine service life.

AIRSPEED CALIBRATION

The Airspeed Calibration chart (figure 4-10) provides calibrated airspeeds for all indicated airspeeds during level flight, climb and autorotation.

TAKEOFF DISTANCE OVER 50 FOOT OBSTACLE

HOVER POWER + 15% TORQUE
ENG - 100% RPM (N2)
GENERATOR 150 AMPS (EA.)

INITIATED FROM 4 FT. SKID HEIGHT
 $V_{LOCS} = 45$ KIAS
HEATER ON OR OFF

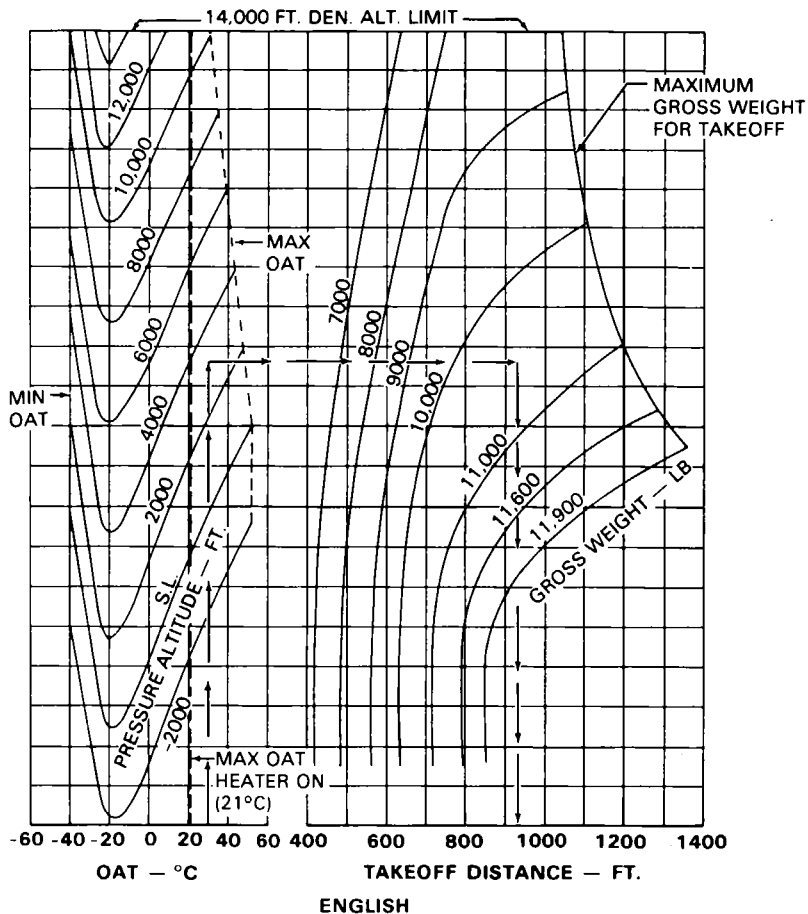


Figure 4-6. Takeoff distance (Sheet 1 of 2)

TAKEOFF DISTANCE OVER 15 METER OBSTACLE

HOVER POWER + 15% TORQUE
ENG - 100% RPM (N2)
GENERATOR 150 AMPS (EA.)

INITIATED FROM 1.2 meter SKID HEIGHT
 $V_{TOCS} = 45$ KIAS
HEATER ON OR OFF

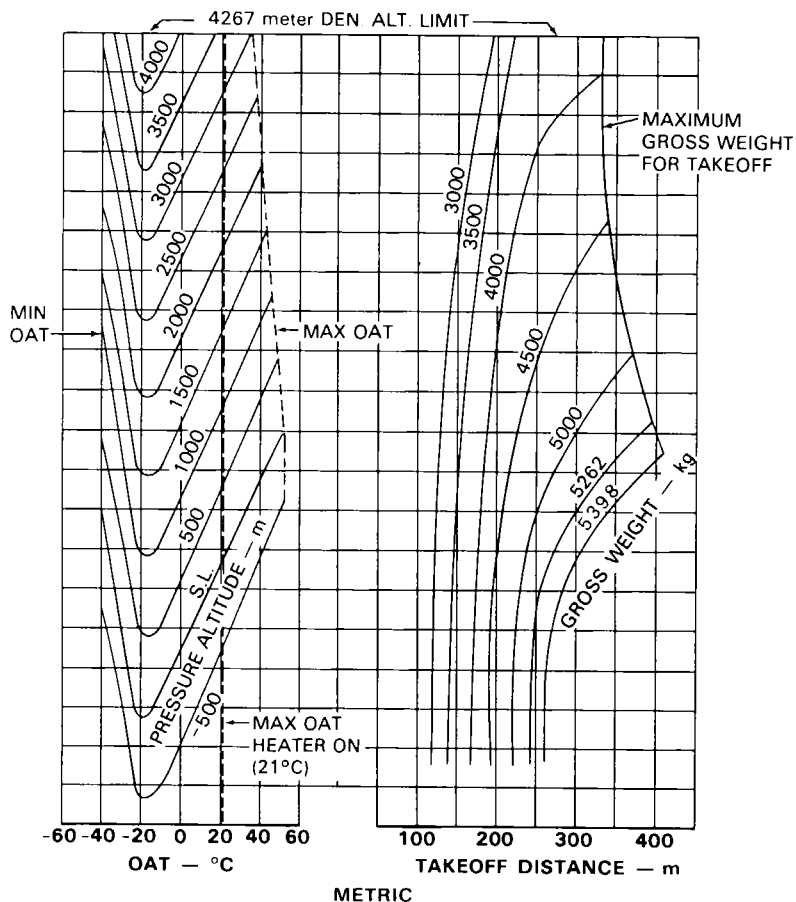


Figure 4-6. Takeoff distance (Sheet 2 of 2)

SINGLE ENGINE LANDING DISTANCE OVER 50 FT. (15 M) OBSTACLE

2 1/2 MINUTE OEI
POWER AS REQUIRED
ENG - 97% RPM (N2)
GENERATOR 150 AMPS

45 KIAS
RATE OF DESCENT 500 FT/MIN
HARD SURFACED RUNWAY
INOPERATIVE ENGINE SECURED

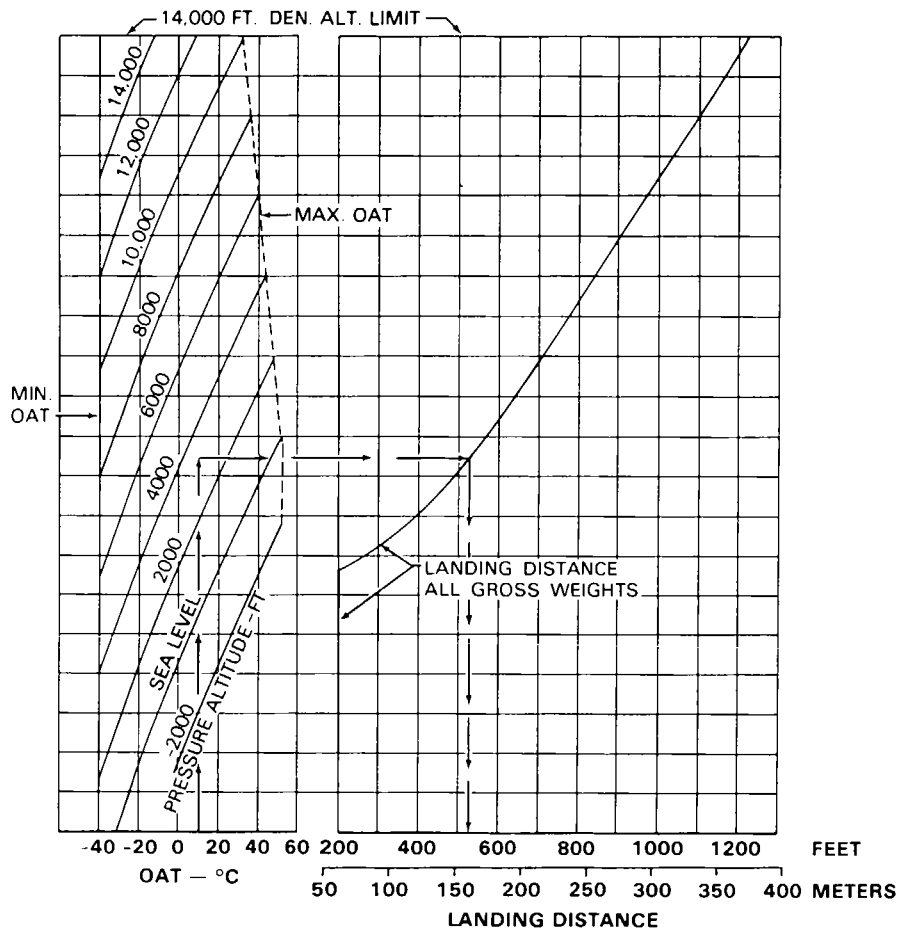


Figure 4-7. Single engine landing distance

TWIN ENGINE RATE OF CLIMB

TAKEOFF POWER
ENGINE RPM 100%
GENERATOR 150 AMPS (EA.)

70 KIAS
HEATER OFF

WITH ALL DOORS OPEN OR REMOVED:

1. CLIMB SPEED IS 60 KIAS
2. RATE OF CLIMB WILL DECREASE 275 FT/MIN

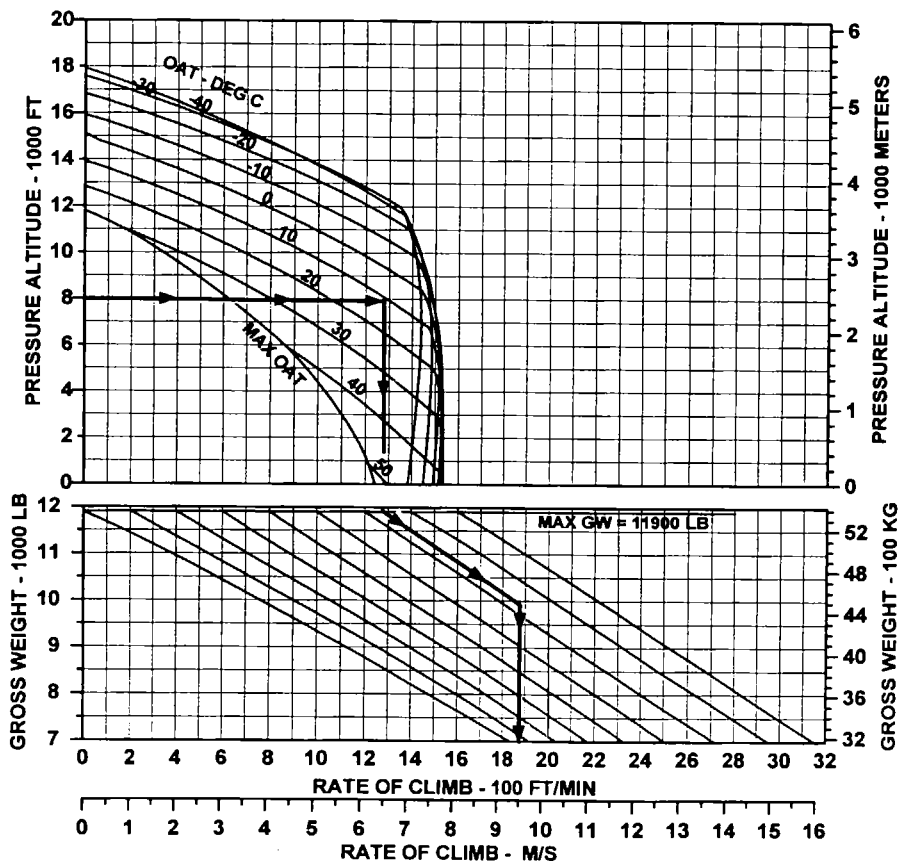


Figure 4-8. Twin engine rate of climb (Sheet 1 of 4)

TWIN ENGINE RATE OF CLIMB

TAKEOFF POWER
ENGINE RPM 100%
GENERATOR 150 AMPS (EA.)

70 KIAS
HEATER ON

WITH ALL DOORS OPEN OR REMOVED:

1. CLIMB SPEED IS 60 KIAS
2. RATE OF CLIMB WILL DECREASE 275 FT/MIN

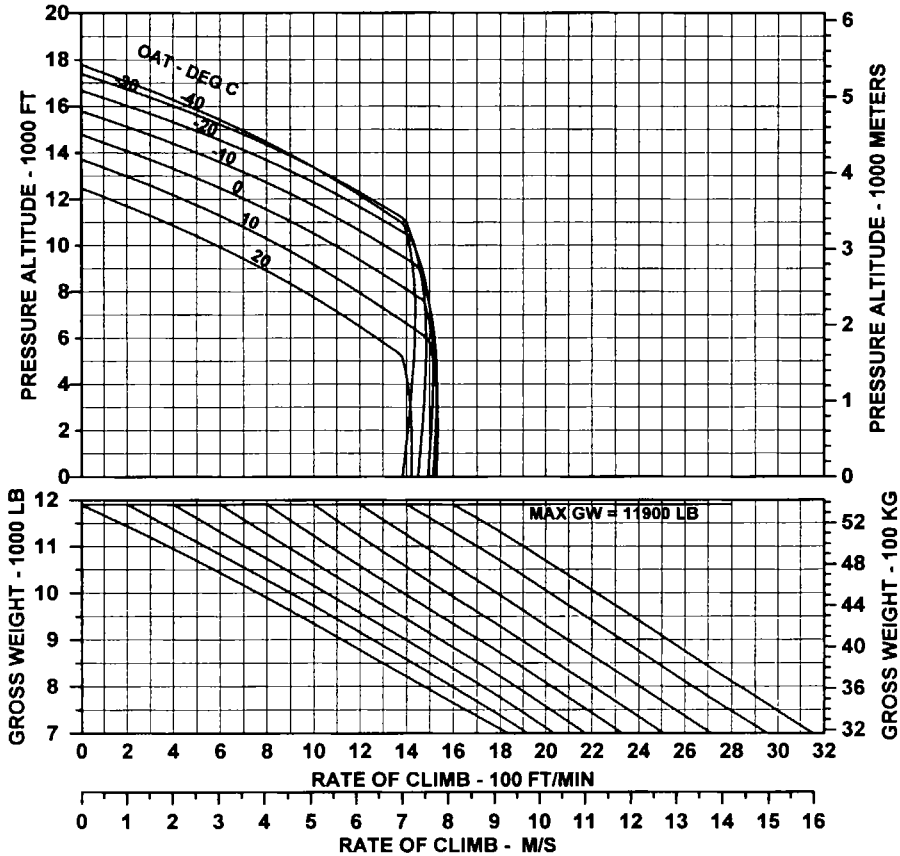


Figure 4-8. Twin engine rate of climb (Sheet 2 of 4)

TWIN ENGINE RATE OF CLIMB

**MAXIMUM CONTINUOUS POWER
ENGINE RPM 100%
GENERATOR 150 AMPS (EA.)**

**70 KIAS
HEATER OFF**

WITH ALL DOORS OPEN OR REMOVED:

1. CLIMB SPEED IS 60 KIAS
2. RATE OF CLIMB WILL DECREASE 275 FT/MIN

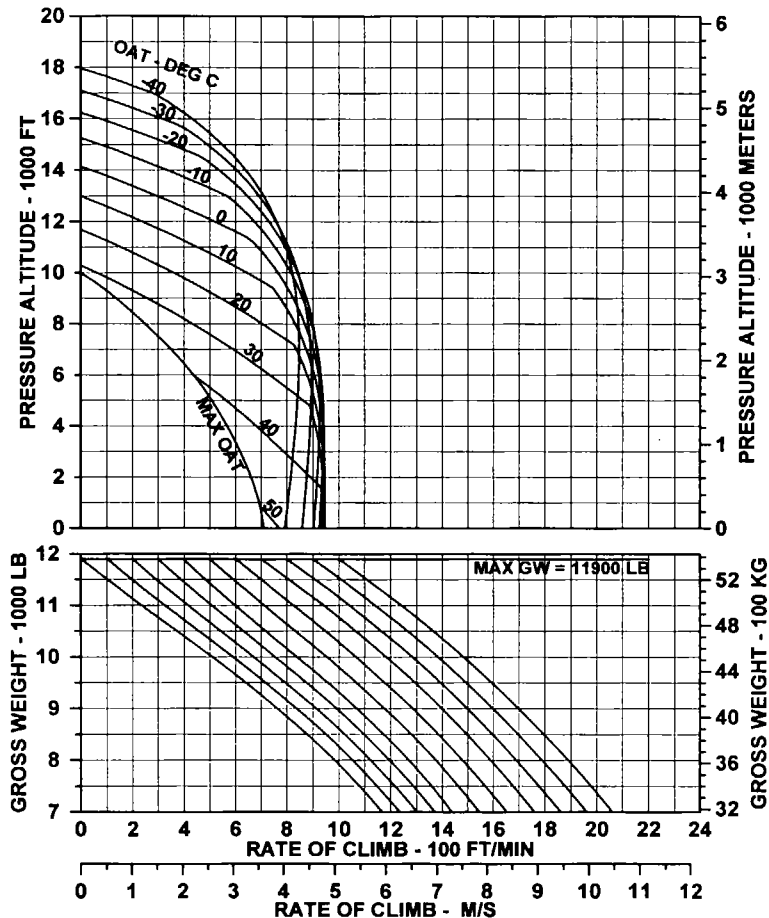


Figure 4-8. Twin engine rate of climb (Sheet 3 of 4)

TWIN ENGINE RATE OF CLIMB

**MAXIMUM CONTINUOUS POWER
ENGINE RPM 100%
GENERATOR 150 AMPS (EA.)**

**70 KIAS
HEATER ON**

WITH ALL DOORS OPEN OR REMOVED:

1. CLIMB SPEED IS 60 KIAS
2. RATE OF CLIMB WILL DECREASE 275 FT/MIN

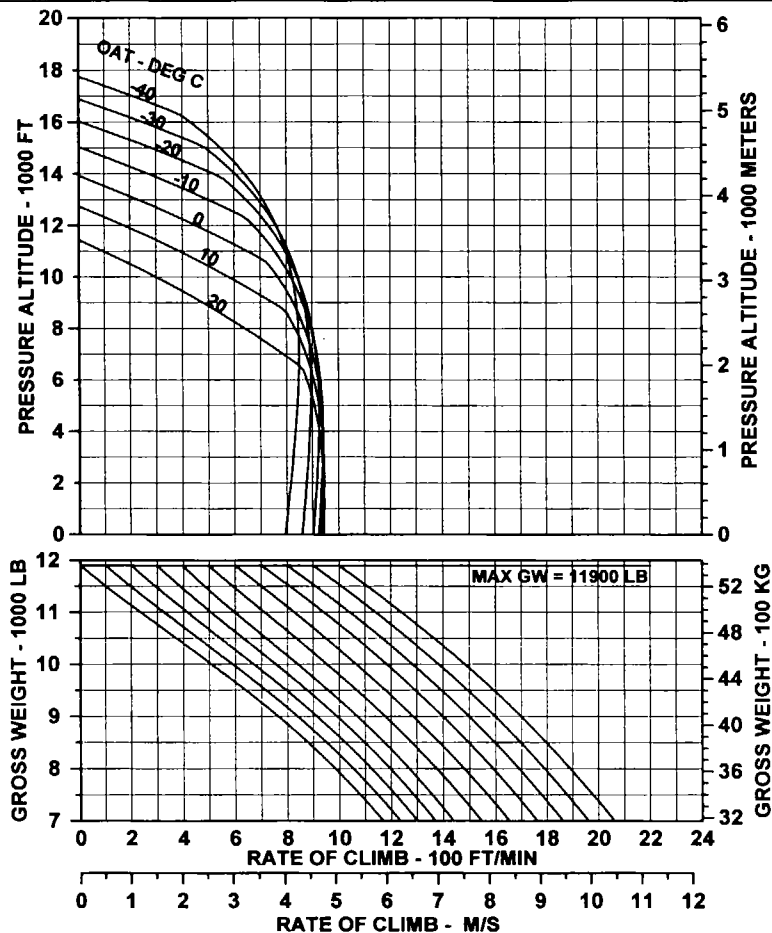


Figure 4-8. Twin engine rate of climb (Sheet 4 of 4)

SINGLE ENGINE RATE OF CLIMB

30 MINUTE POWER
ENGINE RPM 97%
GENERATOR 150 AMPS

70 KIAS
HEATER OFF

WITH ALL DOORS OPEN OR REMOVED:

1. CLIMB SPEED IS 60 KIAS
2. RATE OF CLIMB WILL DECREASE 275 FT/MIN

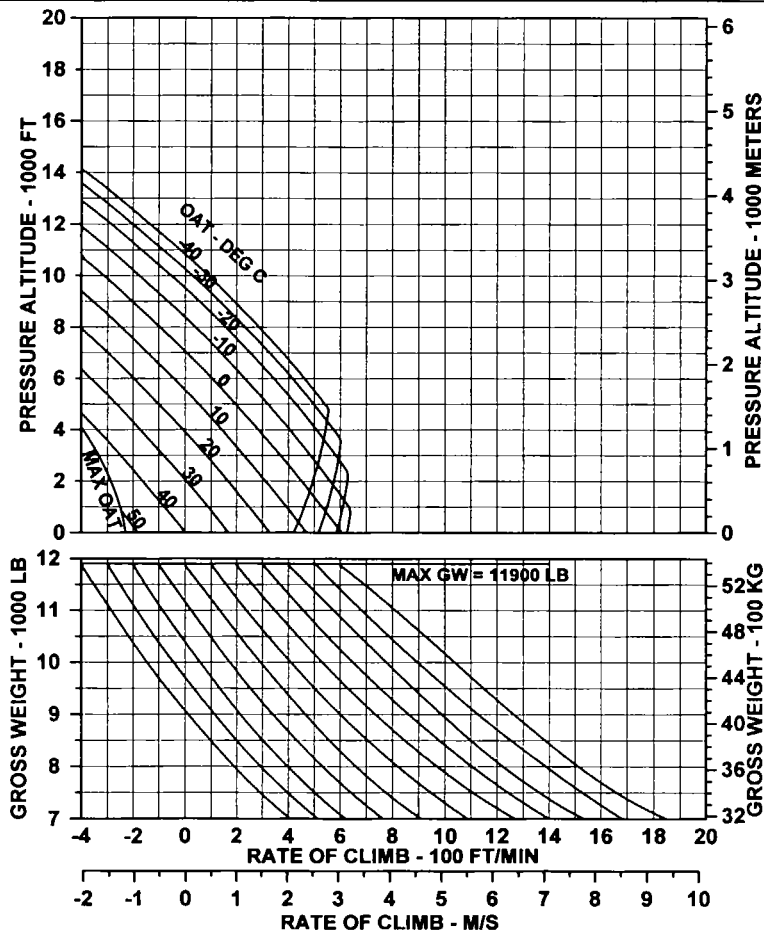


Figure 4-9. Single engine rate of climb (Sheet 1 of 2)

SINGLE ENGINE RATE OF CLIMB

CONTINUOUS POWER
ENGINE RPM 97%
GENERATOR 150 AMPS

70 KIAS
HEATER OFF

WITH ALL DOORS OPEN OR REMOVED:

1. CLIMB SPEED IS 60 KIAS
2. RATE OF CLIMB WILL DECREASE 275 FT/MIN

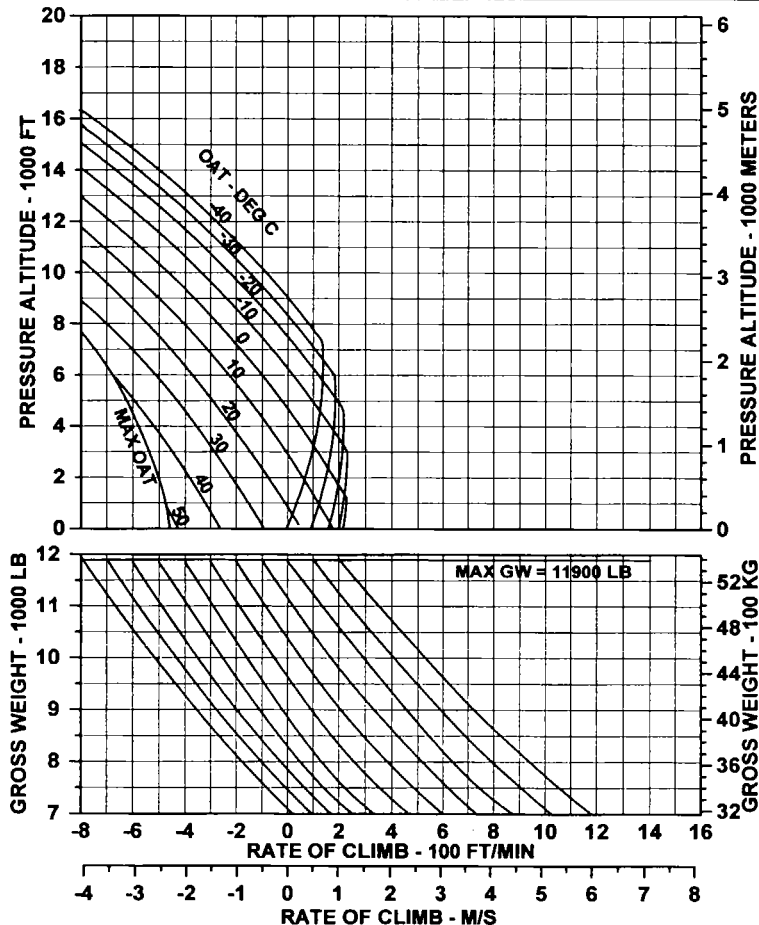


Figure 4-9. Single engine rate of climb (Sheet 2 of 2)

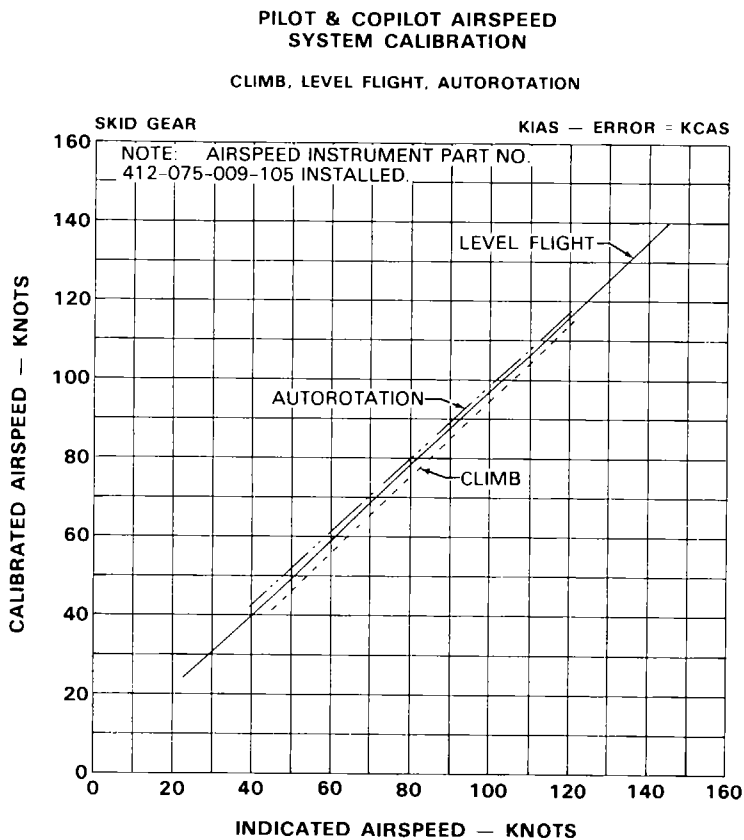


Figure 4-10. Airspeed calibration

4-11. NOISE LEVELS

4-11-A. CERTIFICATED FAR PART 36 STAGE 2 NOISE LEVELS

This aircraft is certified as a Stage 2 helicopter as prescribed in FAR Part 36, Subpart H, for gross weights up to and including the certificated maximum takeoff and landing weight of 11,900 lbs. There are no operating limitations in meeting the takeoff, flyover, or approach noise requirements.

The following noise levels comply with FAR Part 36, Appendix H, Stage 2 noise level requirements. They were obtained by analysis of approved data from noise tests conducted under the provisions of FAR Part 36, Amendment 36-14.

The certified noise levels are:

<u>Flight Condition</u>	<u>EPNL (EPNdB)</u>
Takeoff	93.2
Flyover	93.4
Approach	95.6

NOTE

No determination has been made by the Federal Aviation

Administration that the noise levels of this aircraft are or should be acceptable or unacceptable for operation at, into, or out of, any airport.

V_H is defined as the airspeed in level flight obtained using the minimum specification engine torque corresponding to maximum continuous power available for sea level, 25°C ambient conditions at the relevant maximum certificated weight. The value of V_H thus defined for this aircraft is 122 KTAS.

4-11-B. SUPPLEMENTAL ICAO ANNEX 16, CHAPTER 8 NOISE LEVEL INFORMATION

The test and analysis procedures used to obtain these noise levels are essentially equivalent to those required by the International Civil Aviation Organization (ICAO) in Annex 16, Volume 1, Chapter 8. ICAO Annex 16, Volume 1, Chapter 8 approval is applicable only after endorsement by the Civil Aviation Authority of the country of aircraft registration.

Section 5

OPTIONAL EQUIPMENT SUPPLEMENTS

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Section 5

OPTIONAL EQUIPMENT SUPPLEMENTS

OPTIONAL EQUIPMENT

Bell Helicopter Textron's policy is one of continuous product improvement and Bell reserves the right to incorporate design changes, make additions to and improve its products without imposing any obligation upon the company to furnish for or install such changes, additions, improvements, etc., on its products previously manufactured.

The following items may be installed on the basic helicopter by authorized personnel. Only the optional equipment listed in this section require a Flight Manual supplement.

NOTE

Flight Manual Supplement numbers ending with .1 or -1 are applicable to Model 412 S/N 33001 — 33107 only. Supplement numbers ending with .2 or -2 are applicable to Model 412 S/N 33108 — 33213 and 36001 — 36019. Supplement numbers ending with a .3 or -3 are applicable to Model 412 S/N 33214 — 33999 and 36020 and SUB. Flight manual supplements not ending with a decimal or dash number are applicable to all 412 helicopters.

Table 5-1. Flight Manual Supplements for Optional Equipment

NAME OF EQUIPMENT	KIT NUMBER	DATE CERTIFIED	CURRENT REVISION
BHT-412-FMS-1.2 Winterization Heater Operations	212-706-008	20 January 1981	Rev. 8 14 MAY 93
BHT-412-FMS-2 Emergency Floats	412-706-004	20 January 1981	Rev. 1 15 APR 96
BHT-412-FMS-3 Heated Windshield	412-706-010	20 January 1981	Reissued 18 OCT 94
BHT-412-FMS-4.1	Effectivity S/N 33001-33107		
BHT-412-FMS-5.1	Effectivity S/N 33001-33107		
BHT-412-FMS-6 Flight Director	412-706-111	13 February 1981	Reissued 8 MAY 89
BHT-412-FMS-7 Internal Hoist	214-706-003	2 October 1981	Reissued 11 MAY 95

Table 5-1. Flight Manual Supplements for Optional Equipment (Cont)

NAME OF EQUIPMENT	KIT NUMBER	DATE CERTIFIED	CURRENT REVISION
BHT-412-FMS-8 Litter Kit Operations	412-706-006	29 September 1981	Reissued 5 OCT 94
BHT-412-FMS-9.2 External Cargo Operation	212-706-103	14 May 1981	Reissued 15 SEP 95
BHT-412-FMS-10 Category A Operations		Data incorporated into Section 6 of basic Flight Manual	
BHT-412-FMS-11.1	Effectivity S/N 33001-33107		
BHT-412-FMS-12 Nightsun Searchlight	212-899-333	4 December 1981	Reissued 8 MAY 89
BHT-412-FMS-13 Cold Weather Operations	412-703-004	Data incorporated into basic Flight Manual	
BHT-412-FMS-14 Thailand Special Avionics	412-899-003	11 February 1982	Not Printed
BHT-412-FMS-15 Fixed Step	212-706-057	6 February 1982	Reissued 23 JUN 94
BHT-412-FMS-16 Droop Restraint, Main Rotor and Stick Centering Indicator	412-704-114/412-704- 115	Data incorporated into basic Flight Manual	
BHT-412-FMS-17.2 Auxiliary Fuel Operations	412-706-007	5 January 1984	Reissued 23 JUN 94
BHT-412-FMS-18.2 Loudhailer Operations	412-899-143	17 November 1983	Reissued 8 OCT 91
BHT-412-FMS-19.2 Soft Interior	412-705-510	28 March 1985	Rev. 3 XX SEP 98
BHT-412-FMS-20 Weather Radar Kit	412-899-107	16 June 1986	Reissued 5 OCT 94
BHT-412-FMS-21 Global Nav. System GNS- 500A/S3 with NAV switching	412-899-141	16 June 1986	Reissued 8 MAY 89
BHT-412-FMS-22.2 Category A Operations		6 June 1986	Rev. 1 11 NOV 89
BHT-412-FMS-23 (Reserved)			Original

Table 5-1. Flight Manual Supplements for Optional Equipment (Cont)

NAME OF EQUIPMENT	KIT NUMBER	DATE CERTIFIED	CURRENT REVISION
BHT-412-FMS-24 Seat Cushion Kit	412-706-019	24 July 1987	Reissued 8 DEC 95
BHT-412-FMS-25.2 Auxiliary Fuel Operations	412-706-009	10 March 1988	Reissued 23 JUN 94
BHT-412-FM-CTA-2 Brazilian Registered Helicopters		19 February 1988	Original
BHT-412-RNoAF-FMS Royal Norwegian Air Force Configuration	412-899-022	1 July 1987	Rev. 2
BHT-412-FMS-26 Two-Speed Internal Hoist	412-899-223/214-706-003	19 September 1988	Reissued 11 MAY 95
BHT-412-FMS-27 Litter Kit Operation	205-706-047	14 October 1988	Reissued 23 JUN 94
BHT-412-FMS-28.2 Dual Battery Installation	412-899-225	5 April 19 89	Reissued 8 OCT 91
BHT-412-FMS-29.2 Removal of Upper Aft Center Fuel Cell	412-899-227	23 May 1989	Reissued 8 OCT 91
BHT-412-FMS-CAN-30 Canadian Addendum to the Supplements for Internal Hoist and External Cargo Operation		9 November 1989	Reissued 23 JUN 94
BHT-412-FMS-31 Category B Operations with Approved Configuration of Nine or Less Passenger Seats STC No. SH7727SW		8 February 1990	Original
BHT-412-FMS-32.2 Improved Transmission	412-570-001-101	29 June 1990	Not Printed
BHT-412-FMS-33 Loran C Navigation System (King KLN-88)	412-899-231	22 June 1990	Original
BHT-412-FMS-34.2 Improved Hover Performance with PT6T3BE Engines and 5- Minute Takeoff Power Rating	412-570-001-103	12 October 1990	Not Printed

Table 5-1. Flight Manual Supplements for Optional Equipment (Cont)

NAME OF EQUIPMENT	KIT NUMBER	DATE CERTIFIED	CURRENT REVISION
BHT-412-FMS-35.2 Category B Operations when Configured with Nine or Less Passenger Seats		10 April 1991	Rev. 1 23 APR 98
BHT-412-FMS-36.3 Dual Digital Automatic Flight Control System, Search and Rescue (SAR)	Effectivity S/N 33214- 33999 and 36020 and SUB		Not Printed
BHT-412-FMS-37.4	Effectivity S/N 36087 and SUB		
BHT-412-FMS-38.4	Effectivity S/N 36087 and SUB		
BHT-412-FMS-39.3 and 39.4	Effectivity S/N 36024-36086 S/N 36087 and SUB		
BHT-412-FMS-40 Increases Generator Capacity	412-706-026	29 October 1992	Reissued 5 OCT 94
BHT-412-FMS-41.3	Effectivity S/N 36020-36086		
BHT-412-FMS-43.3 and 43.4	Effectivity S/N 36020-36086		
BHT-412-FMS-44.3 and 44.4	Effectivity S/N 36020-36086		
BHT-412-FMS-45.3 and 45.4	Effectivity S/N 36020-36086		
BHT-412-FMS-46.3	Effectivity S/N 36020-36086		
BHT-412-FMS-47 Folding Step	412-899-287	25 October 1993	Original
BHT-412-FMS-48.2 Engine No. 2 Gov Trim Switch	TB 412-93-118	28 July 1994	Original
BHT-412-FMS-49.4	Effectivity S/N 36087 and SUB		

Table 5-1. Flight Manual Supplements for Optional Equipment (Cont)

NAME OF EQUIPMENT	KIT NUMBER	DATE CERTIFIED	CURRENT REVISION
BHT-412-FMS-53.4	Effectivity S/N 36119, 36122,36123,36126,36127, and 36133 ONLY		
BHT-412-FMS-54	TBD		
BHT-412-FMS-55.4	Effectivity S/N 36122, 36123, 36125, and SUB		
BHT-412-FMS-56.3 and 56.4	Effectivity S/N 36020 - 36086 S/N 36087 and SUB		
BHT-412-FMS-CAA-57.3 and 57.4	Effectivity S/N 36087 and SUB		
BHT-412-FMS-58.4	Effectivity S/N 36087 and SUB		
BHT-412-FMS-59.4	Effectivity S/N 36087 and SUB		
BHT-412-FMS-60.4	Effectivity S/N 36087 and SUB		
BHT-412-FMS-61.3 and 61.4	Effectivity S/N 36020 - 36086 S/ N 36087 and SUB		
BHT-412-FMS-63.2, 63.3, and 63.4 Self Sealing Fuel Cells	Effectivity S/N 33108 - 33213 S/N 36001 - 36019 S/N 36020 - 36086 S/N 36087 - SUB	19 September 1997	Rev. 1 22 OCT 97
BHT-412-FMS-65.2, 65.3, and 65.4 Ten Cell — Self Sealing Fuel	Effectivity S/N 33108 - 33213 S/N 36001 - 36019 S/N 36020 - 36086 S/N 36087 - SUB	22 June 1998	Rev. 1 2 JUL 98
BHT-412-FMS-66.4	Effectivity S/N 36087 and SUB		

Bell
MODEL **412**

ROTORCRAFT FLIGHT MANUAL

**33008 — 33213
36001 — 36019**

SUPPLEMENT FOR WINTERIZATION HEATER OPERATIONS (212-706-008)

**CERTIFIED
JANUARY 20, 1981**

This supplement shall be attached to the Model 412 Flight Manual (BHT-412-FM-2) when the 212-706-008 Winterization Heater has been installed.

The information contained herein supplements the information of the basic Flight Manual. For Limitations, Procedures, and Performance Data not contained in this supplement, consult the basic Flight Manual.

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ROTORCRAFT CERTIFICATION OFFICE
FEDERAL AVIATION ADMINISTRATION
FT. WORTH, TX 76193-0170

DEC 18 1998

NOTE

Revised text is indicated by a black vertical line. Insert latest revision pages; dispose of superseded pages.

INTRODUCTION

The Winterization Heater is installed to provide increased cabin heating capacity. The heater basically consists of a larger mixing valve, a larger noise suppressor, extra outlets, ducts, and hardware necessary for installation. Operation of the Winterization Heater is identical to the basic heater.

Section 1

LIMITATIONS

WEIGHT C/G LIMITATIONS

Actual weight change shall be determined after the heater is installed and ballast readjusted if necessary to return empty weight CG within allowable limits.

HEATER OPERATION

Heater shall not be operated when OAT is above 21°C (69.8°F).

Section 2

NORMAL PROCEDURES

PRESTART CHECK

Battery switch — ON.

Heater circuit breaker switch — In.

Heater switch — ON.

Check "Heater Air Line" light illuminates.

Heater switch — OFF.

Heater switch — ON.

Increase thermostat setting and observe heated air-flow.

Return thermostat to full cold and observe heater airflow shutoff. If heater airflow shutoff is observed, reset thermostat to desired temperature if heater operation is desired.

CAUTION

HEATER OPERATION CHECK

Operation check may be accomplished at this time or at any time heater operation is desired.

100% (N₂) rpm and at least 75% N₁ on both engines.

Thermostat — Full cold.

WARNING

**DO NOT OPERATE HEATER
ABOVE 21°C (69.8°F) OAT.**

TURN HEATER OFF WHEN:

**THE HEATER AIRFLOW DOES
NOT SHUT OFF WHEN
THERMOSTAT IS TURNED TO
FULL COLD.**

**THE "HEATER AIR LINE" LIGHT
ILLUMINATES.**

**THE HEATER CIRCUIT BREAKER
TRIPS.**

NOTE

If heater is on for takeoff, refer to appropriate performance chart in Section 4.

Section 3

EMERGENCY AND MALFUNCTION PROCEDURES

No change from basic Flight Manual.

Section 4

PERFORMANCE

Performance with Winterization Heater switched OFF is the same as that shown in basic Flight Manual for heater OFF.

Performance with Winterization Heater switched ON is presented as follows:

Refer to figure 4-1 for out-of-ground-effect hover performance.

In-ground-effect hover performance is the same as that shown in basic Flight Manual for heater ON.

Refer to figure 4-2 for takeoff performance.

Refer to figure 4-3 for climb performance.

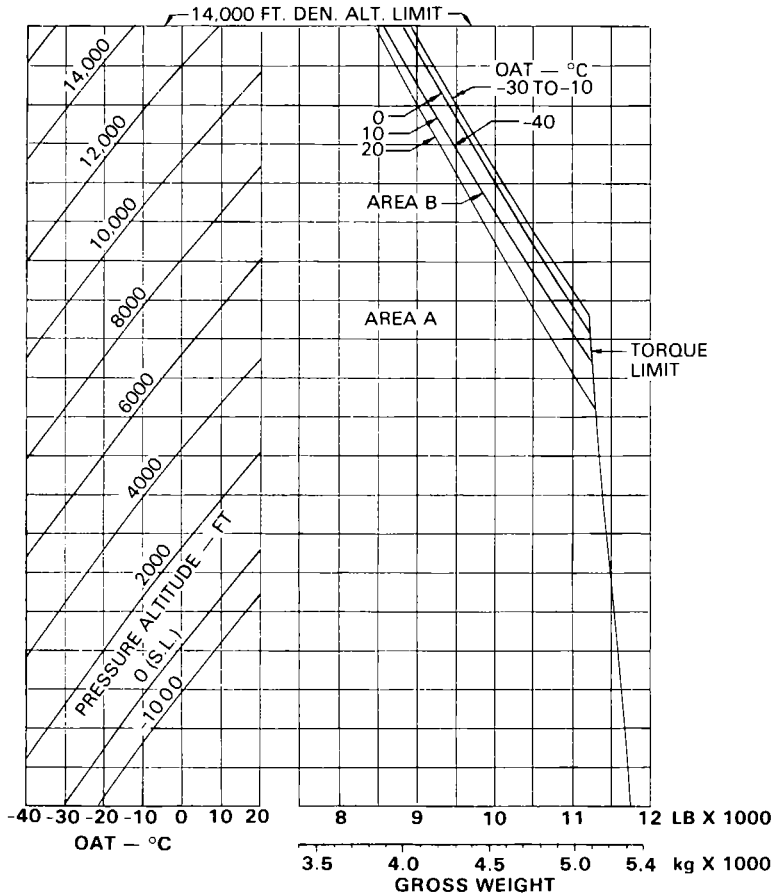
BHT-412-FMS-1.2

HOVER CEILING OUT OF GROUND EFFECT

TAKEOFF POWER
ENGINE RPM 100%
GENERATOR 150 AMPS (EA.)

SKID HEIGHT 60 FT
WINTERIZATION HEATER ON
-40° TO 20°C

CAUTION: OGE HOVER OPERATION MAY RESULT IN VIOLATION OF H-V LIMITATIONS.



412FS1.2-4-1-1

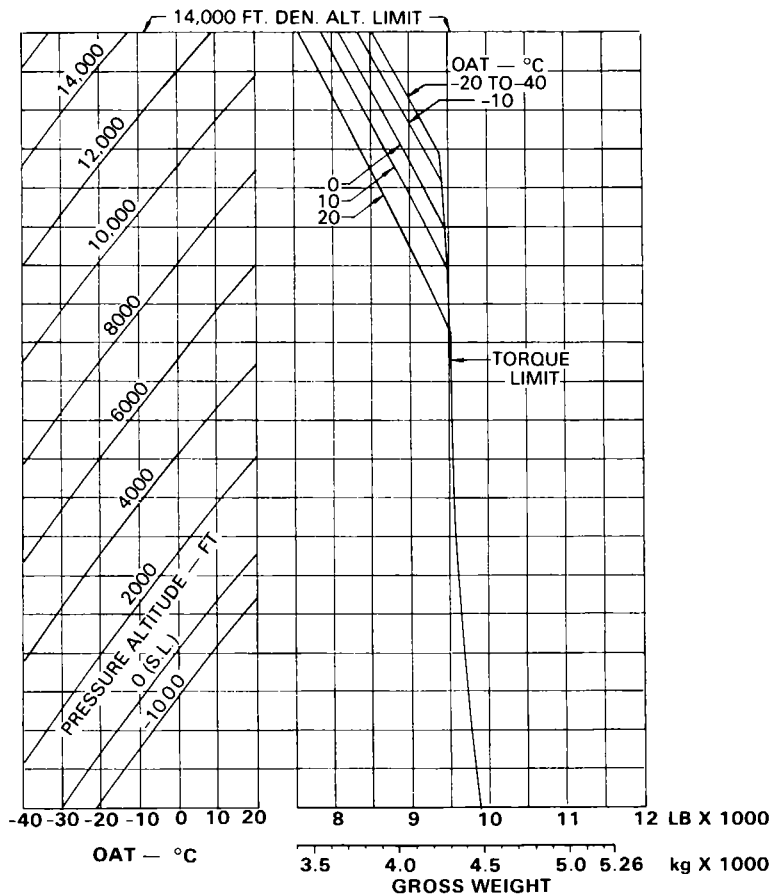
Figure 4-1. (Sheet 1 of 2)

**HOVER CEILING
OUT OF GROUND EFFECT**

MAXIMUM CONTINUOUS POWER
ENGINE RPM 100%
GENERATOR 150 AMPS (EA.)

SKID HEIGHT 60 FT
WINTERIZATION HEATER ON
-40° TO 20°C

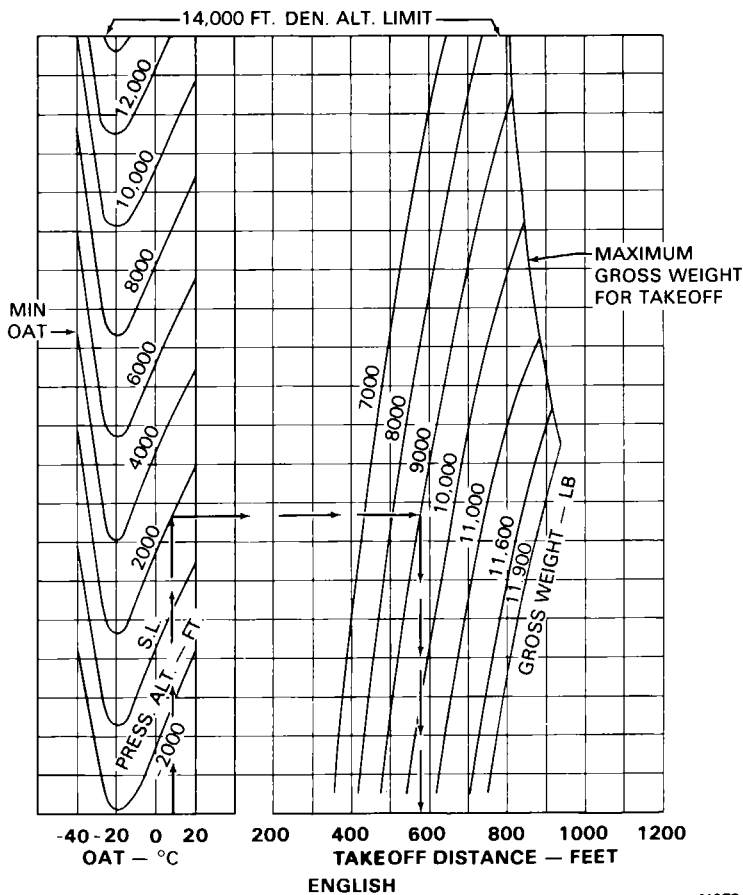
CAUTION: OGE HOVER OPERATION MAY RESULT IN VIOLATION OF H-V LIMITATIONS.



412FS1.2-4-1-2

Figure 4-1. (Sheet 2 of 2)

BHT-412FMS-1.2

TAKEOFF DISTANCE
OVER 50 FT. OBSTACLEHOVER POWER + 15% TORQUE
ENGINE RPM 100%
GENERATOR 150 AMPS (EA.)INITIATED FROM 4 FT SKID HEIGHT
 $V_{LOCS} = 45$ KIAS
WINTERIZATION HEATER ON

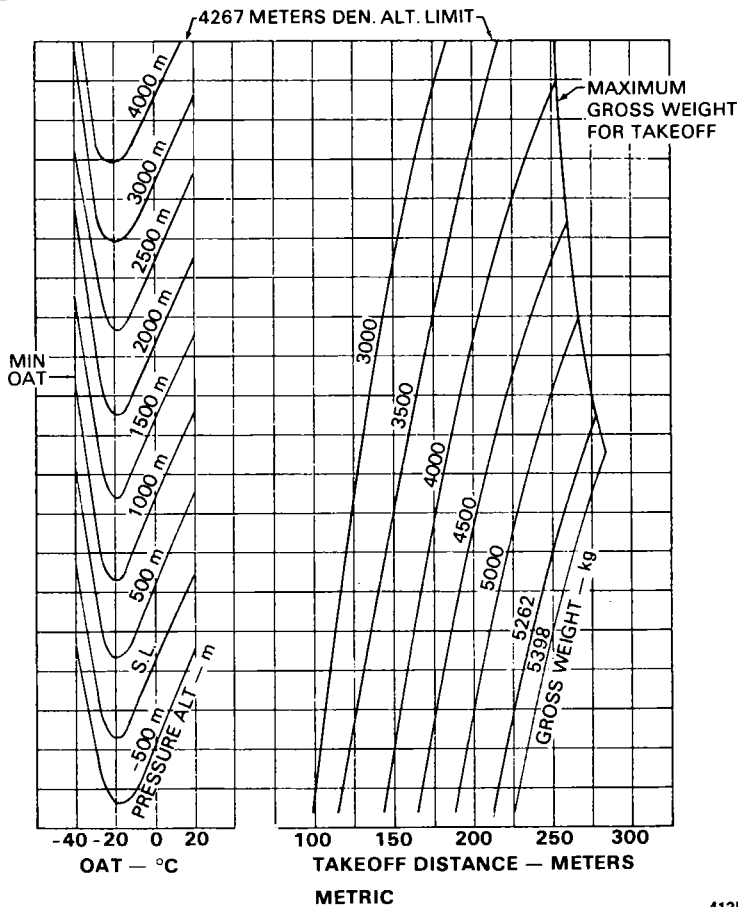
412FS1.2-4-2-1

Figure 4-2. (Sheet 1 of 2)

TAKEOFF DISTANCE
OVER 15 METER OBSTACLE

HOVER POWER + 15% TORQUE
ENGINE RPM 100%
GENERATOR 150 AMPS (EA.)

INITIATED FROM 1.2m SKID HEIGHT
 $V_{TOCS} = 45$ KIAS
WINTERIZATION HEATER ON



412FS1.2-4-2-2

Figure 4-2. (Sheet 2 of 2)

BHT-412FMS-1.2

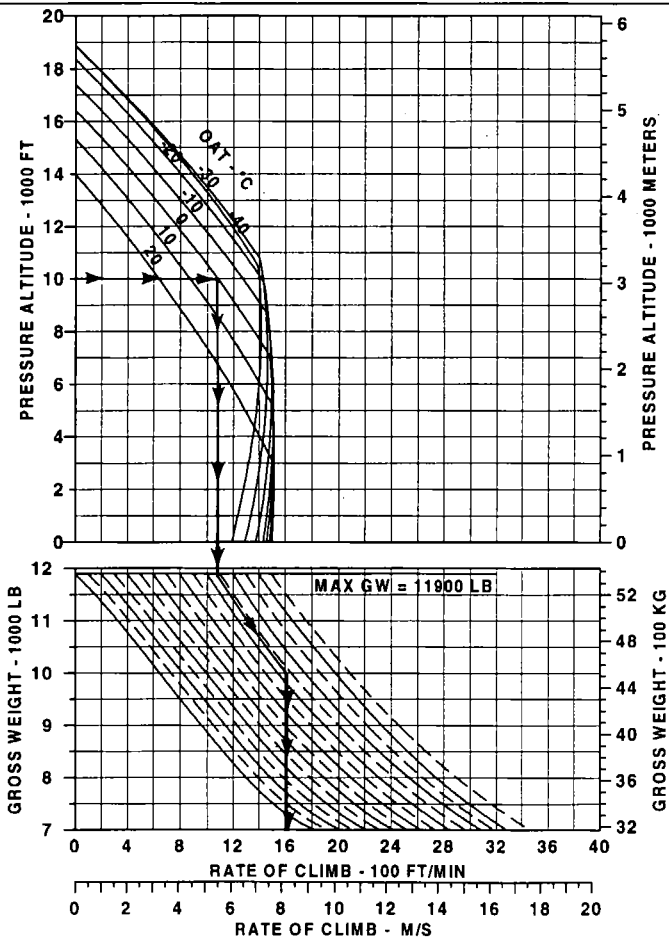
TWIN ENGINE RATE OF CLIMB

TAKEOFF POWER
ENGINE RPM 100%
GENERATOR 150 AMPS (EA.)

70 KIAS
WINTERIZATION HEATER ON

WITH ALL DOORS OPEN OR REMOVED:

1. CLIMB SPEED IS 60 KIAS
2. RATE OF CLIMB WILL DECREASE 275 FT/MIN



412FS1.2-4-3-1

Figure 4-3. (Sheet 1 of 2)

BHT-412FMS-1.2

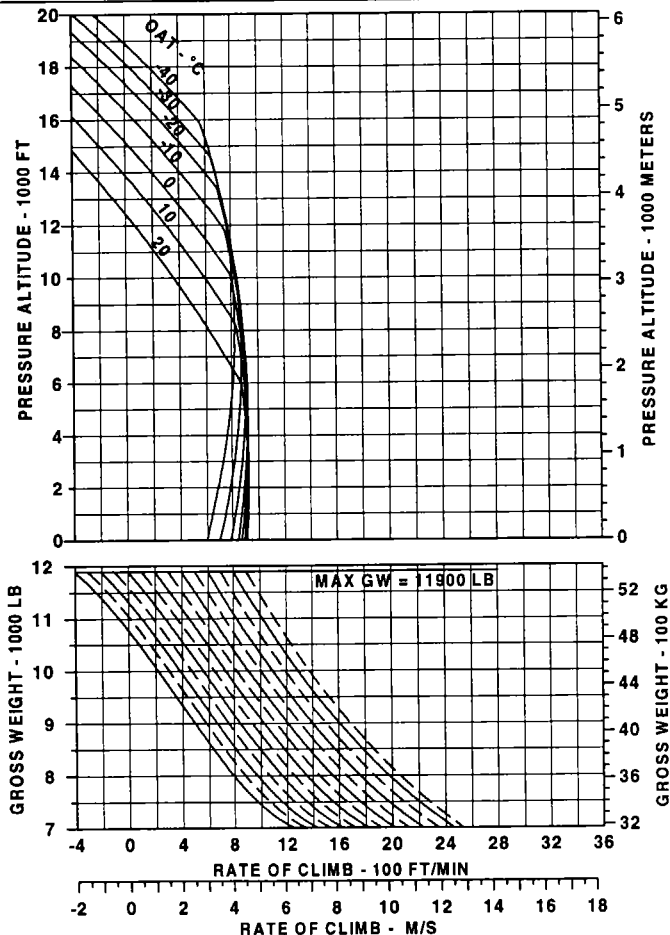
TWIN ENGINE RATE OF CLIMB

MAXIMUM CONTINUOUS POWER
ENGINE RPM 100%
GENERATOR 150 AMPS (EA.)

70 KIAS
WINTERIZATION HEATER ON

WITH ALL DOORS OPEN OR REMOVED:

1. CLIMB SPEED IS 60 KIAS
2. RATE OF CLIMB WILL DECREASE 275 FT/MIN



412FS1.2-4-3-2

Figure 4-3. (Sheet 2 of 2)

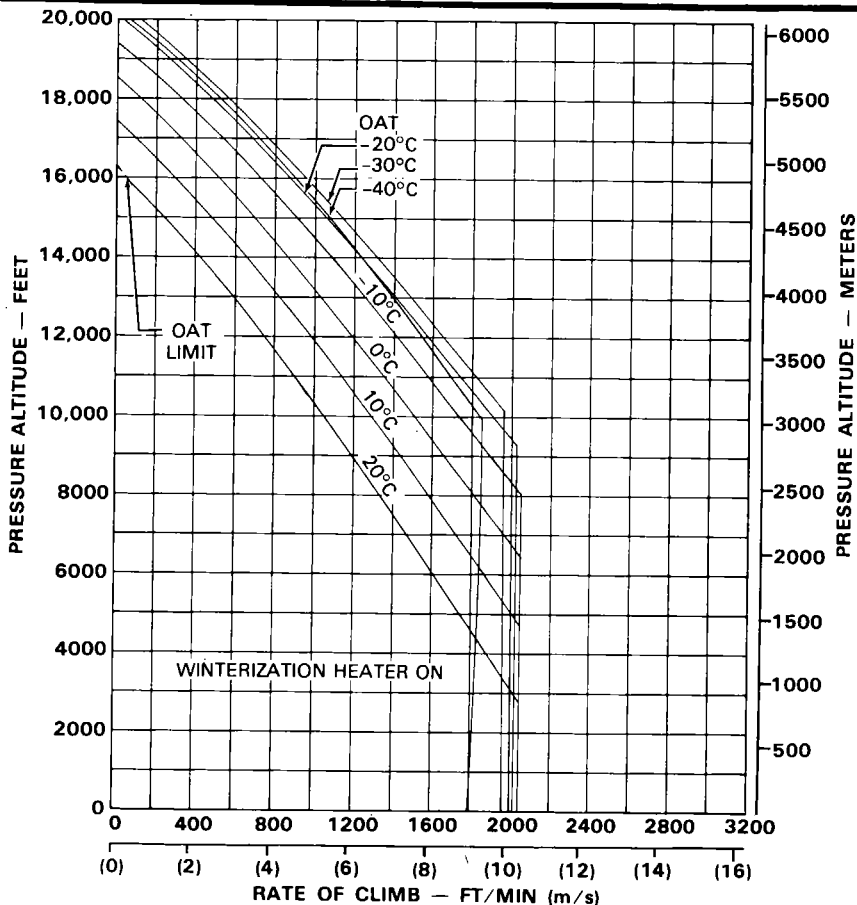
BHT-412-FMS-1.2

TWIN ENGINE RATE OF CLIMB
GROSS WEIGHT 10,000 LB. (4536 kg)

TAKEOFF POWER
ENGINE RPM 100%
GENERATOR 150 AMPS (EA.)

70 KIAS
WINTERIZATION HEATER ON

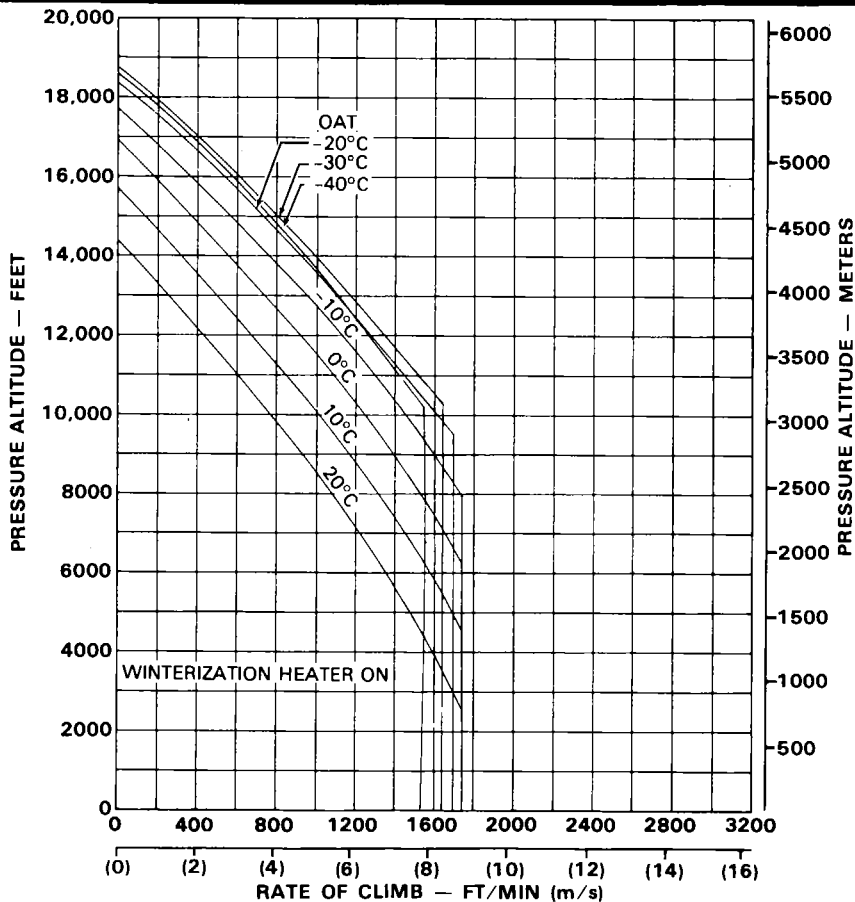
WITH ALL DOORS OPEN OR REMOVED: 1 CLIMB SPEED IS 60 KIAS
2 RATE OF CLIMB WILL DECREASE
275 FT./MIN



412900-36-11F

Figure 4-3. (Sheet 4 of 12)

BHT-412-FMS-1.2

TWIN ENGINE RATE OF CLIMB
GROSS WEIGHT 11,000 LB (4990 kg)TAKEOFF POWER
ENGINE RPM 100%
GENERATOR 150 AMPS (EA.)70 KIAS
WINTERIZATION HEATER ONWITH ALL DOORS OPEN OR REMOVED: 1. CLIMB SPEED IS 60 KIAS
2. RATE OF CLIMB WILL DECREASE
275 FT / MIN.

412900-36-12F

Figure 4-3. (Sheet 5 of 12)

BHT-412-FMS-1.2

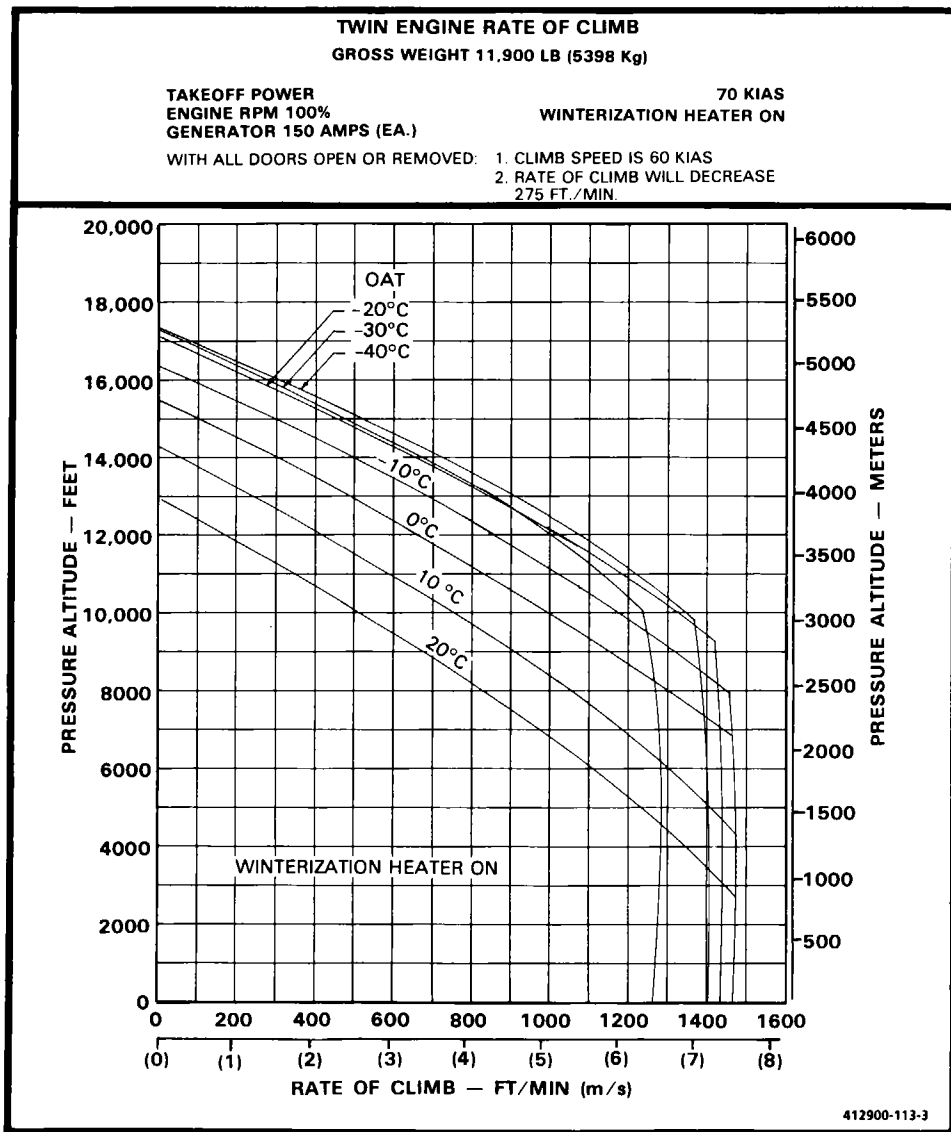


Figure 4-3. (Sheet 6 of 12)

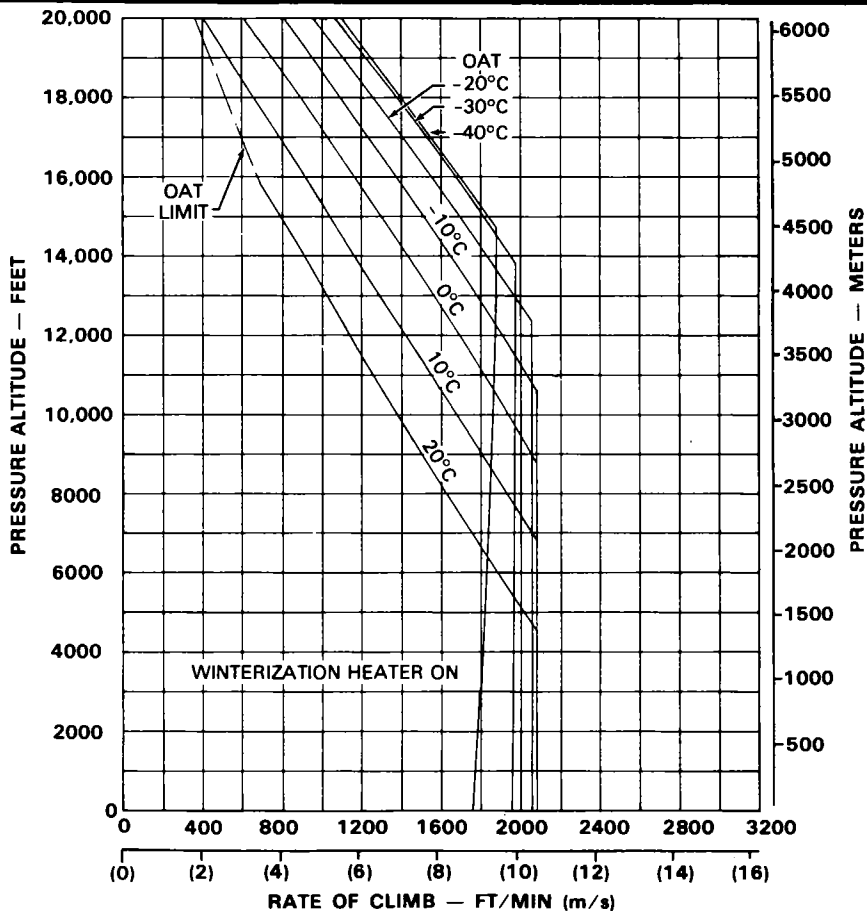
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BHT-412-FMS-1.2

TWIN ENGINE RATE OF CLIMB
GROSS WEIGHT 7000 LB. (3175 kg)MAXIMUM CONTINUOUS POWER
ENGINE RPM 100%
GENERATOR 150 AMPS (EA.)

70 KIAS

WINTERIZATION HEATER ON

WITH ALL DOORS OPEN OR REMOVED: 1. CLIMB SPEED IS 60 KIAS
2. RATE OF CLIMB WILL DECREASE
275 FT./MIN

412900-36-2E

Figure 4-3. (Sheet 7 of 12)

BHT-412-FMS-1.2

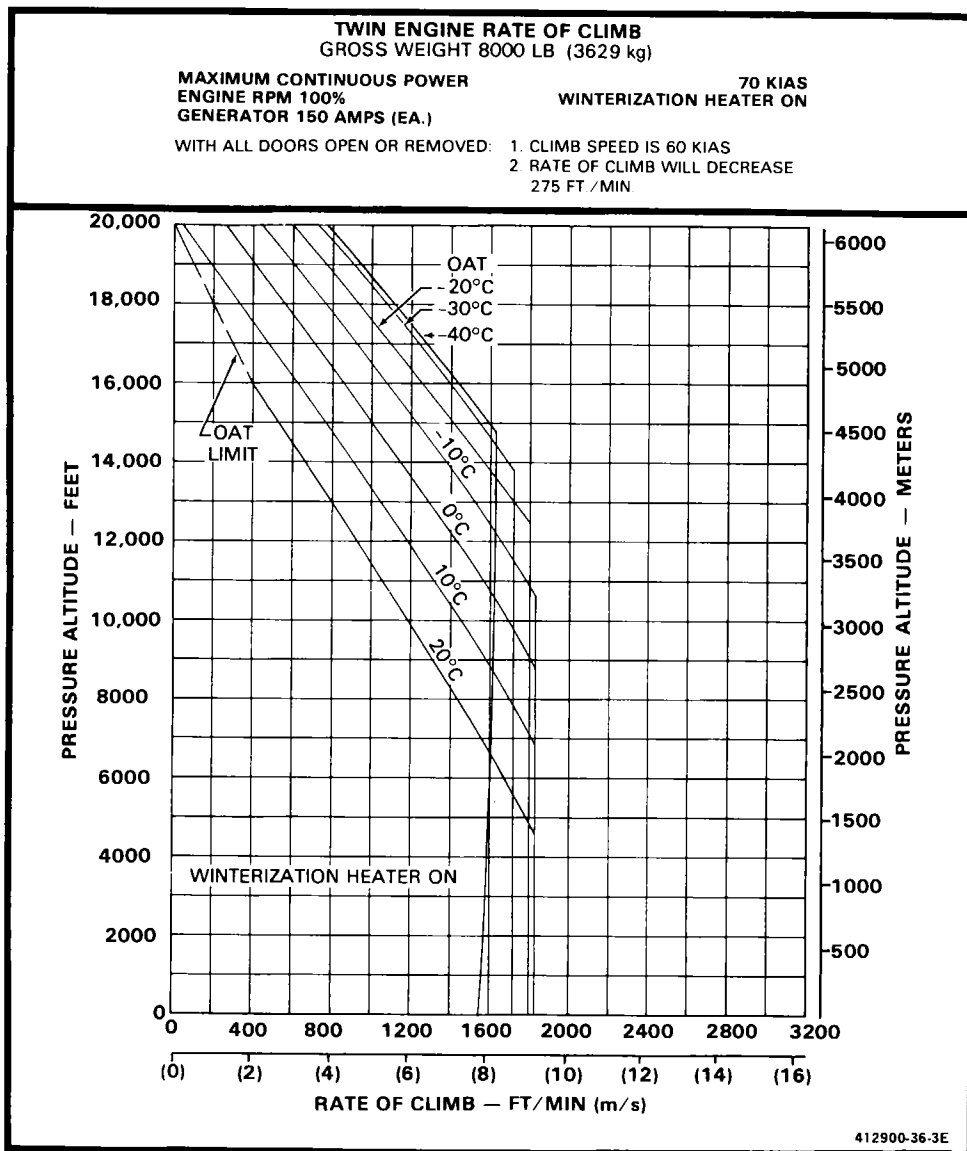


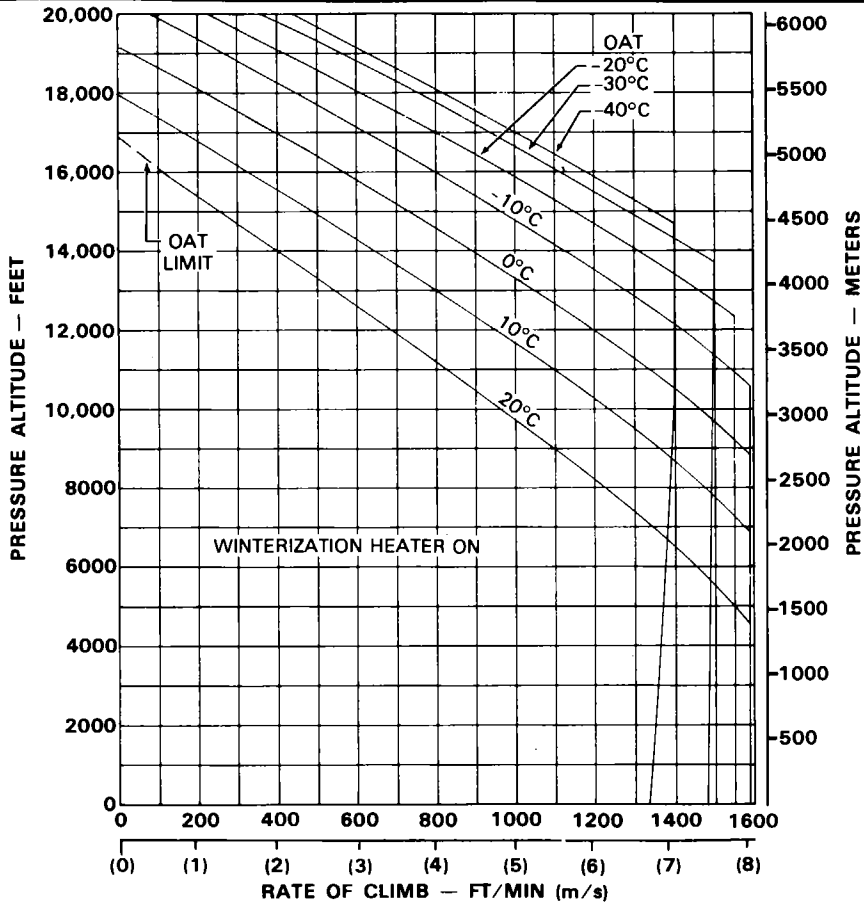
Figure 4-3. (Sheet 8 of 12)

BHT-412-FMS-1.2

TWIN ENGINE RATE OF CLIMB
GROSS WEIGHT 9000 LB (4082 kg)MAXIMUM CONTINUOUS POWER
ENGINE RPM 100%
GENERATOR 150 AMPS (EA.)

70 KIAS

WINTERIZATION HEATER ON

WITH ALL DOORS OPEN OR REMOVED: 1. CLIMB SPEED IS 60 KIAS
2. RATE OF CLIMB WILL DECREASE
275 FT./MIN.

412900-36-4E

Figure 4-3. (Sheet 9 of 12)

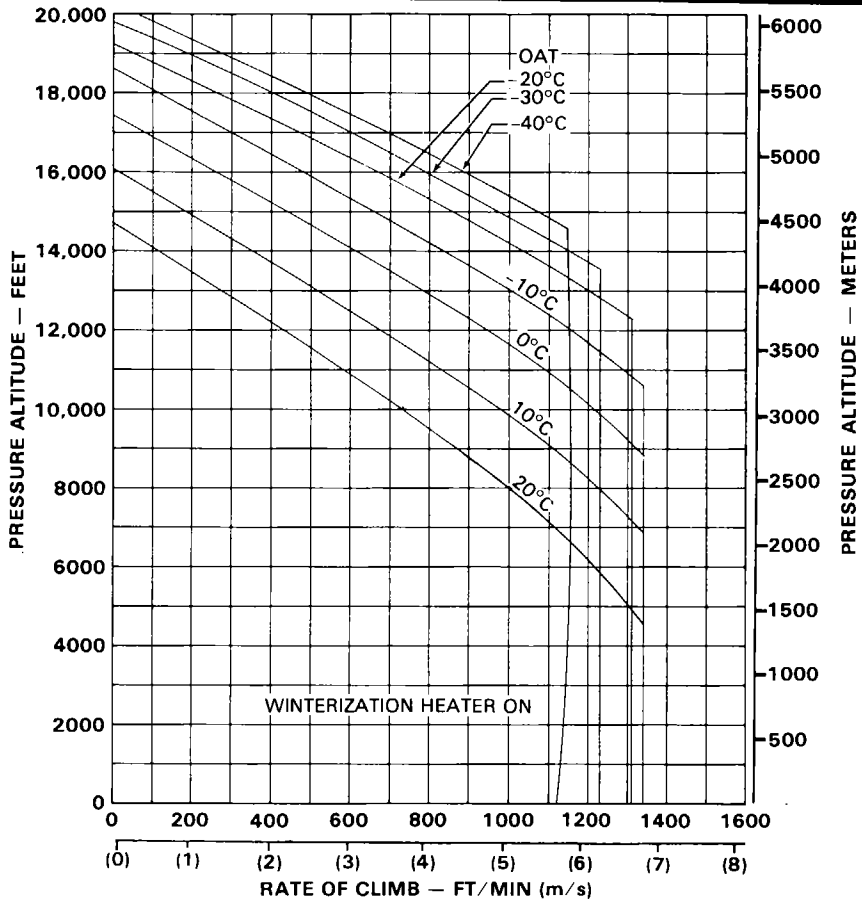
BHT-412-FMS-1.2

TWIN ENGINE RATE OF CLIMB
GROSS WEIGHT 10,000 LB (4536 kg)

MAXIMUM CONTINUOUS POWER
ENGINE RPM 100%
GENERATOR 150 AMPS (EA.)

70 KIAS
WINTERIZATION HEATER ON

- WITH ALL DOORS OPEN OR REMOVED:
1. CLIMB SPEED IS 60 KIAS
 2. RATE OF CLIMB WILL DECREASE 275 FT./MIN.



412900-36-5E

Figure 4-3. (Sheet 10 of 12)

BHT-412-FMS-1.2

TWIN ENGINE RATE OF CLIMB
GROSS WEIGHT 11,000 LB (4990 kg)

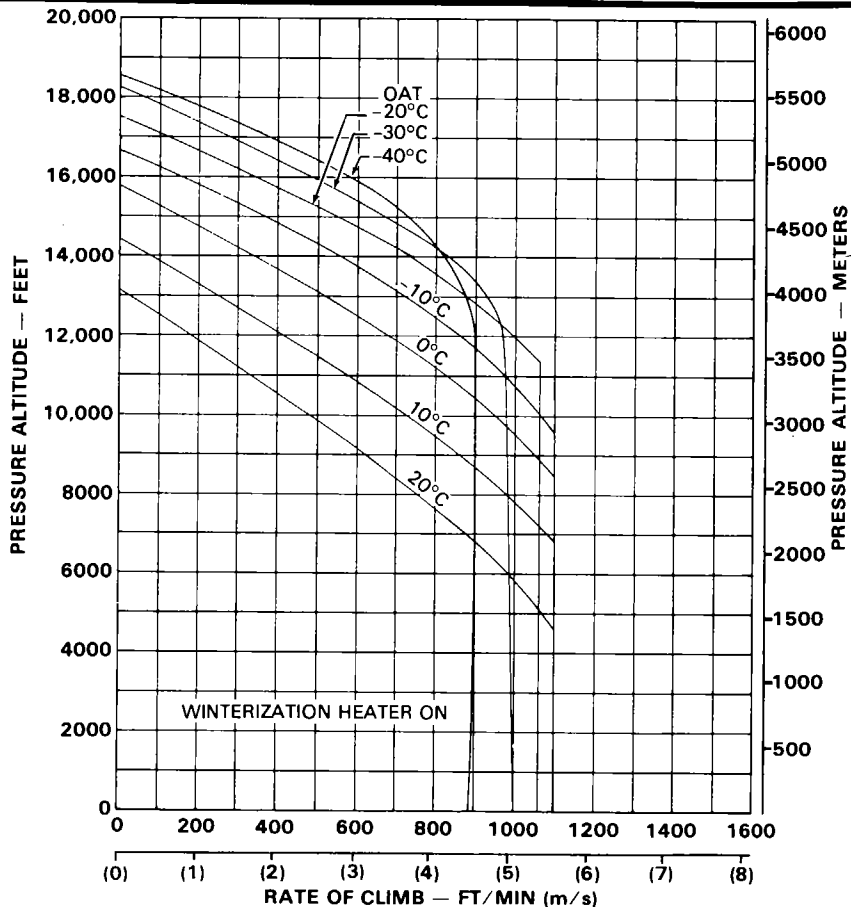
70 KIAS

MAXIMUM CONTINUOUS POWER
ENGINE RPM 100%
GENERATOR 150 AMPS (EA.)

WINTERIZATION HEATER ON

WITH ALL DOORS OPEN OR REMOVED

1. CLIMB SPEED IS 60 KIAS
2. RATE OF CLIMB WILL DECREASE
275 FT./MIN.



412900-36-6E

Figure 4-3. (Sheet 11 of 12)

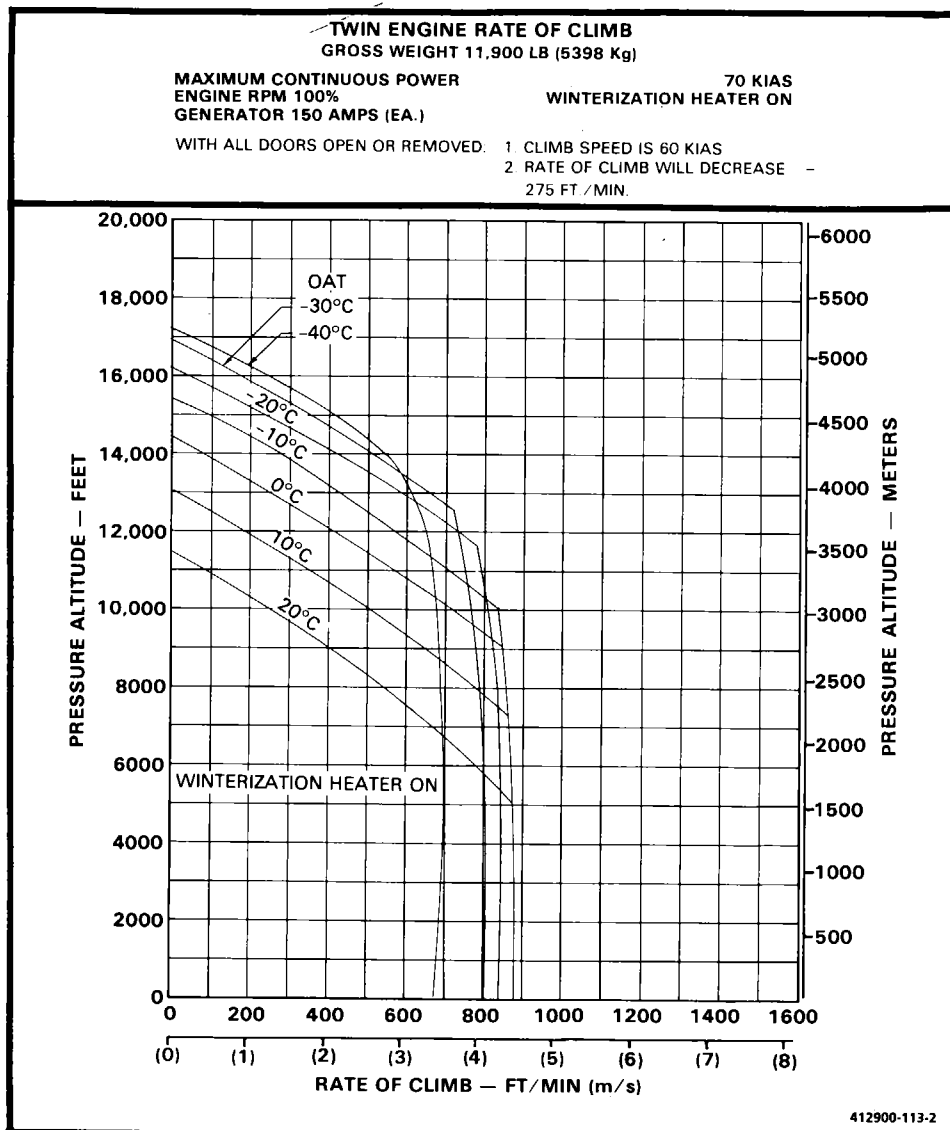


Figure 4-3. (Sheet 12 of 12)

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Bell **MODELS 412/412EP**

ROTORCRAFT FLIGHT MANUAL

SUPPLEMENT EMERGENCY FLOATS

(412-706-004)

CERTIFIED
20 JANUARY 1981

This supplement shall be attached to the Models 412 and 412 EP Flight Manual when the 412-706-004 Emergency Floats have been installed.

Refer to BHT-412-FMS-55.4 for S/N 36122, 36123, 36125 and subsequent.

Information contained herein supplements information of basic Flight Manual. For Limitations, Procedures, and Performance Data not contained in this supplement, consult basic Flight Manual.

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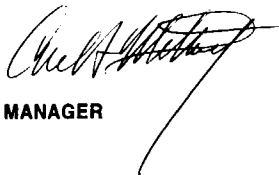
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APPROVED:



MANAGER

ROTORCRAFT CERTIFICATION OFFICE
FEDERAL AVIATION ADMINISTRATION
FT. WORTH, TX 76193-0170

GENERAL INFORMATION

The emergency floats will allow the helicopter to land in water during an emergency situation. The kit consists of six skid mounted float bags, four water activated float switches that are mounted on the underside of the fuselage, a pneumatic bottle located in the nose compartment, a collective mounted EMERG FLTS switch, a pedestal mounted EMER INFLATION handle, and the necessary hardware to complete the installation. Also, the kit includes provisions for installing life rafts and vests.

The floats automatically inflate when the EMERG FLTS switch is in the ARMED position and the water activated float switches are submerged. If the system or one of the float bags rupture after inflation, check valves in each float bag will maintain float integrity. An instrument panel mounted EMER FLOATS caution light illuminates whenever the EMERG FLTS switch is in the ARMED position. System security is protected by an EMERG FLOATS circuit breaker located on the overhead console.

The EMER INFLATION handle provides for manual activation of the emergency flotation system in the event the floats fail to inflate automatically.

Section 1

LIMITATIONS

TYPE OF OPERATION

The emergency floats are installed for assistance during emergency ditching and use is approved for VFR and IFR operation.

FLOATS switch shall be OFF and FLOATS caution light extinguished when operating over land.

Emergency float safety pin shall be removed prior to flight over water.

FLIGHT WITH OPTIONAL EQUIPMENT INSTALLED

Fixed passenger step shall not be installed concurrently with emergency floats.

Retractable passenger step shall be stowed prior to flight over water.

When internal hoist is installed, hoist cable anti-chafing guard shall be installed on same side of helicopter as hoist.

AIRSPEED LIMITATIONS

Inflation of emergency floats during forward flight is prohibited. Flight after landing with floats inflated is prohibited.

WARNING

**SEVERE NOSE UP PITCHING
WILL OCCUR IF EMERGENCY
FLOATS ARE INFLATED IN**

**FORWARD FLIGHT OR DESCENT.
REFER TO INADVERTENT FLOAT
INFLATION IN FLIGHT (SECTION
3).**

Maximum autorotation airspeed (floats stowed):

105 knots below 10,000 feet pressure altitude

80 knots above 10,000 feet pressure altitude

Maximum forward speed for ditching:

33 knots in calm water

15 knots in rough water

Rate of descent should be reduced as low as possible upon water contact.

ALTITUDE LIMITATIONS

Maximum pressure altitude for inflation of emergency floats is 10,000 feet. Helicopter operation above 10,000 feet is permitted provided the FLOATS switch is in the OFF position and FLOATS caution light is extinguished.

WEIGHT/CG LIMITATIONS

Actual weight change shall be determined after floats are installed and ballast readjusted, if necessary, to return empty weight cg within allowable limits.

The emergency float kit does not change the CG limits of the helicopter.

FLOAT ARMING Floats shall not be armed:

1. Over land.
2. Above 60 KIAS.
3. Above 600 feet AGL.

PLACARDS

FLOAT INFLATION OR OPERATION IN
FORWARD FLIGHT IS PROHIBITED

(Located above pilot airspeed indicator.)

Section 2

NORMAL PROCEDURES

EXTERIOR CHECK

NOTE

Ensure that emergency floats have had periodic inflation and inspection.

Emergency floats — Stowed.

Emergency float covers and supports — Clean and secured.

INTERIOR CHECK

EMER INFLATION handle — Down and safetied.

Nitrogen bottle — Secured and pressure within allowable limits for pressure altitude and ambient temperature as shown on chart decal, located on nitrogen bottle.

Emergency floats safety pin — Remove for over-water flight.

PRESTART CHECK

FLOATS switch — OFF.

EMER FLOATS circuit breaker — IN.

BEFORE TAKEOFF OVER WATER

STEP switch — STOW.

FLOATS switch — ARMED; check FLOATS caution light illuminated.

IN-FLIGHT OPERATION

FLOATS switch — OFF upon reaching 600 feet above water and airspeed of 60 KIAS or above, or when over land; check FLOATS caution light extinguished.

BEFORE LANDING OVER WATER

FLOATS switch — ARMED when below 600 feet above water and airspeed of 60 KIAS or below; check FLOATS caution light illuminated.

AFTER LANDING

FLOATS switch — OFF; check FLOATS caution light extinguished.

BEFORE LEAVING HELICOPTER

Emergency floats safety pin — Installed.

Section 3

EMERGENCY AND MALFUNCTION PROCEDURES

INADVERTENT FLOAT INFLATION IN FLIGHT

In event of unintended inflation, reduce airspeed, adjust rate of descent to 200 fpm or less, and land as soon as possible. Avoid landing on terrain which could damage float bags.

ELECTRICAL MALFUNCTIONS

FLOATS CAUTION LIGHT ON WITH FLOATS SWITCH OFF

If FLOATS caution light illuminates with FLOATS switch OFF, pull EMER FLOATS circuit breaker.

NOTE

If EMER FLOATS circuit breaker is pulled floats will not deploy automatically.

If FLOATS caution light remains illuminated after pulling circuit breaker, land as soon as practical.

ELECTRICAL SYSTEM FAILURE

Pull EMER INFLATION handle upon water contact to inflate floats.

EMERGENCY DITCHING

WARNING

CREW AND PASSENGER DOORS SHALL REMAIN CLOSED DURING DITCHING.

FLOATS switch — ARMED; check FLOATS caution light illuminated. Establish an autorotative glide speed of 65 to 70 knots for all gross weight.

WARNING

SEVERE NOSE UP PITCHING WILL OCCUR IF EMERGENCY FLOATS ARE INFLATED IN FORWARD FLIGHT OR DESCENT.

At 100 feet altitude execute a moderate cyclic flare to reduce airspeed. Adjust collective and cyclic pitch sufficiently to touchdown in a nose-up attitude with forward speed and rate of descent as low as possible.

If emergency floats do not inflate immediately upon water contact, pull EMER INFLATION handle.

After water touchdown, complete shutdown, check for damage, and determine if helicopter should be abandoned.

NOTE

Takeoff after emergency landing
with floats inflated is prohibited.

Crew egress should be accomplished by
jettisoning crew doors.

Passenger egress should be accomplished
by pushing out passenger door windows
at corners marked **EMERGENCY EXIT —
PUSH HERE.**

EGRESS AFTER DITCHING

Ensure rotor is stopped prior to egress.

Section 4

PERFORMANCE

No change from basic manual.

Bell **412/412EP**
MODELS

ROTORCRAFT FLIGHT MANUAL

SUPPLEMENT FOR HEATED WINDSHIELD

(412-706-010)

**CERTIFIED
20 JANUARY 1981**

This supplement shall be attached to the Models 412 and 412 EP Flight Manual when the 412-706-010 Heated Windshield has been installed.

The information contained herein supplements the information of the basic Flight Manual. For Limitations, Procedures, and Performance Data not contained in this supplement, consult the basic Flight Manual.

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FEDERAL AVIATION ADMINISTRATION
FT. WORTH, TX 76193-0170

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INTRODUCTION

The Heated Windshield Kit allows the crew to electrically defrost/defog the windshield. The kit consists of two heated windshield panels, switches, circuit breakers, caution/advisory lights, and the necessary hardware and wiring to complete the installation.

The ON/OFF switches are located on the overhead console and identified as WSHLD HEAT LH and RH. Placing either switch in the ON position activates the corresponding heated windshield panel and illuminates the respective ON advisory light. The caution/advisory lights are located on the instrument panel. The caution portion of the light illuminates HOT whenever the respective heated windshield panel overheats. Electrical power is provided by the 28 Vdc NON ESNTL bus. CONT and PWR circuit breakers, located on the overhead console and identified as WINDSHIELD HEAT LH and RH, provide circuit protection.

Section 1

LIMITATIONS

PLACARDS AND DECALS

CAUTION

STAND-BY COMPASS UNRELIABLE
WITH WINDSHIELD HEAT ON

WSHLD HEAT

LH

RH

INDICATOR

ON

ON

HOT

HOT

GREEN

YELLOW

Section 2

NORMAL PROCEDURES

PRESTART CHECK

WSHLD HEAT LH switch — OFF.

WSHLD HEAT RH switch — OFF.

WINDSHIELD HEAT circuit breakers — In.

BATTERY BUS 1 switch — ON.

WINDSHIELD HEAT caution/advisory light segments — PRESS TO TEST; check ON lights illuminate green, HOT lights illuminate amber.

NOTE

The intensity of the heated windshield lights is controlled by the caution panel BRIGHT/DIM switch when PILOT INSTR LT rheostat is on.

BEFORE TAKEOFF

WSHLD HEAT LH switch — ON (if desired);
check ON light illuminates.

WSHLD HEAT RH switch — ON (if desired);
check ON light illuminates.

NOTE

The windshield heat green advisory lights will illuminate and extinguish as the windshield heat cycles on and off during normal operation.

Section 3

EMERGENCY AND MALFUNCTION PROCEDURES

DC GENERATOR FAILURE

NON-ESNTL BUS switch — MANUAL.

NOTE

DC power for the heated windshield is supplied by the nonessential bus.

WINDSHIELD HEAT CAUTION LIGHT

If either WSHLD HEAT caution light illuminates (HOT), turn respective WSHLD HEAT switch OFF.

WINDSHIELD HEAT CIRCUIT BREAKER

If LH or RH WINDSHIELD HEAT CONT or PWR circuit breaker pops out, turn respective WSHLD HEAT switch OFF.

Section 4

PERFORMANCE DATA

No change from basic Flight Manual.

Bell *Model 412*

ROTORCRAFT FLIGHT MANUAL

SUPPLEMENT FOR
FLIGHT DIRECTOR
(412-706-111)

CERTIFIED
FEBRUARY 13, 1981

This supplement shall be attached to the Model 412 Flight Manual when the 412-706-111 Flight Director has been installed.

The information contained herein supplements the information of the basic Flight Manual. For Limitations, Procedures, and Performance Data not contained in this supplement, consult the basic Flight Manual.

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APPROVED:

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for

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DEPARTMENT OF TRANSPORTATION
SOUTHWEST REGION, FORT WORTH, TEXAS

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INTRODUCTION

The Flight Director assists the flight crew in control and navigation of the helicopter. Nine Flight Director modes, in addition to standby (SBY) mode, are available for selection by the flight crew: Altitude (ALT), Indicated Airspeed (IAS), Vertical Speed (VS), Heading (HDG), Navigation (NAV), Localizer and Glideslope (ILS), Back Course (BC), VOR Approach (VOR APR), and Go-Around (GA). Flight director operation is controlled by the mode selector on the instrument panel.

The Flight Director can be coupled to the Automatic Flight Control System (AFCS) for fully automatic (hands off) flight path control.

Section 1

LIMITATIONS

TYPE OF OPERATION

The Flight Director may be used during VFR or IFR nonicing conditions.

NOTE

Flight Director bars are repeated on copilot ADI.

FLIGHT DIRECTOR LIMITATIONS

During VOR approaches, except for VORs collocated at the airport, the Flight Director shall not be coupled in VOR APR mode prior to VOR station passage inbound.

Maximum approach gradient is 5 degrees.

Flight Director modes may not be commanded from the copilot position.

WEIGHT/CG LIMITATIONS

Actual weight change shall be determined after the Flight Director is installed and ballast readjusted, if necessary, to return empty weight CG within allowable limits.

AIRSPEED LIMITATIONS

Minimum airspeed for coupled operation of Flight Director is 60 KIAS.

Section 2

NORMAL PROCEDURES

PRESTART CHECK

PILOT INSTR LT knob — OFF (day operation).

AUX SYS PITOT and STATIC switches — NORMAL.

SYSTEMS CHECKS

FLIGHT DIRECTOR CHECK

FORCE TRIM switch — ON.

HP1 and HP2 buttons — ON.

Flight director SBY button — Depress and hold; check DCPL light and all mode selector and helipilot controller lights illuminate and FD fail flag on ADI appears; then release button.

HSI heading marker — Set to aircraft heading.

Flight director HDG button — Depress; check HDG and CPL lights illuminate, SBY light extinguishes.

ADI roll command bar — Check centered.

HSI heading marker — Move right; check ADI roll command bar moves right and roll actuator position indicator (API) moves right; check cyclic stick moves right in approximately 2 seconds.

HSI heading marker — Reset to aircraft heading.

Flight director VS button — Depress; check VS light illuminates.

ADI pitch command bar — Check centered.

Cyclic ATTD TRIM switch — Move aft; check ADI pitch command bar moves up and pitch API moves up; check cyclic stick moves aft in approximately 2 seconds.

Cyclic NAV STBY button — Depress momentarily and release. Check HDG, VS, and CPL lights extinguish, SBY light illuminates; check all APIs center, pitch and roll command bars retract from view.

HSI heading marker — Set to takeoff heading.

NOTE

For prolonged ground operation, AFCS shall be operated in SAS mode only.

TAKEOFF

SAS/ATT button — ATT or SAS as desired during hover and takeoff; ATT prior to entering Instrument Meteorological Conditions (IMC).

NOTE

It is recommended that the cyclic FORCE TRIM release button be depressed before liftoff to trim actuators to center positions.

Cyclic FORCE TRIM release button — Depress and hold until desired climbout attitude is attained, then release.

Flight director — Select modes as desired after reaching 60 KIAS.

Collective pitch — Adjust to desired climb power setting.

Airspeed — Adjust to desired climb speed.

Flight director — Select modes as desired.

IN-FLIGHT OPERATION

Flight director — Select modes as desired.

DESCENT AND APPROACH

Flight director — Select mode for type of approach to be flown.

Collective pitch — Adjust to maintain desired approach speed.

GO-AROUND

Collective GO-AROUND button — Depress at Missed Approach Point.

LANDING

Cyclic NAV STBY button — Depress at or above 60 KIAS to decouple flight director.

AFCS — SAS or ATT mode as desired.

NOTE

It is recommended that the cyclic FORCE TRIM release button be depressed before touchdown to trim actuators to center positions.

For prolonged ground operation, AFCS shall be operated in SAS mode only.

Section 3

EMERGENCY AND MALFUNCTION PROCEDURES

Table 3-1. Caution lights

CAUTION LIGHT WORDING	FAULT CONDITION	CORRECTIVE ACTION
DCPL	Flight director not coupled.	Ensure that HP1 and HP2 are engaged in ATT mode. Engage flight director modes as desired. Depress CPL button if CPL light not illuminated.

BHT-412-FMS-6

Table 3-2. Warning flags

FLAG WORDING	FLAG LOCATION	FAULT CONDITION	CORRECTIVE ACTION
ATT	ADI	Vertical gyro not turning at full speed/attitude information unreliable. (Flight director may or may not decouple.)	If only pilot ATT flag is displayed, pilot shall monitor standby attitude indicator and flight director. If only the copilot ATT flag is displayed, the copilot shall monitor the standby attitude indicator. Flight director is functional and flight director indications repeated on copilots ADI are valid. Check PILOT and CPLT ATTD SYS circuit breakers — in. If flag does not retract, continue flight in ATT or SAS mode.
FD	ADI	Flight director not coupled due to flight director failure. (Pitch and roll command bars may or may not retract from view.)	Check FLT DIR (AC and DC) circuit breakers in. Check desired mode(s) engaged. If flag does not retract from view, continue flight in ATT or SAS mode. (ATT mode for IMC conditions.)
GS (No legend)	HSI	Glideslope signal unreliable. (ADI pitch command bar retracts from view. AFCS holds pitch attitude present at time of signal failure.)	Continue flight in any mode, using cyclic ATTD TRIM switch and collective to maintain airspeed and glidepath. (Monitor raw glideslope data.)
NAV (No legend)	HSI	VOR or localizer signal unreliable. (ADI roll command bar retracts from view. HSI course deviation bar and No. 1 bearing pointer unreliable. AFCS holds roll level attitude.)	Check NAV 1 (DC) circuit breaker in. Check VHF NAV 1 tuned properly. signal identified. Continue flight in HDG or ATT mode.
OFF	HSI	Directional gyro failure or HSI failure. Heading information unreliable. (AFCS holds roll level attitude.)	Check PILOT and CPLT HSI and GYRO CMPS (AC) circuit breakers in. Continue flight in any mode except HDG.

FLIGHT DIRECTOR FAILS TO COUPLE

HP1 and HP2 buttons — ON.

SAS/ATT button — ATT.

CPL button — Depress if CPL light not illuminated.

NOTE

The Flight Director will not couple if HP1 or HP2 is inoperative.

PITOT-STATIC SYSTEM MALFUNCTION

In the event of an apparent malfunction of the copilot altimeter, vertical speed indicator, and/or airspeed indicator, proceed as follows:

AUX SYS PITOT or STATIC switch (as applicable) — OFF.

Flight director — Disengage vertical modes.

Section 4

PERFORMANCE DATA

No change from basic Flight Manual.

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Section 1

MANUFACTURER'S DATA

WEIGHT AND BALANCE

No change from basic Flight Manual

Section 2

MANUFACTURER'S DATA

SYSTEMS DESCRIPTION

FLIGHT DIRECTOR (412-706-111)

The flight director is designed for use as a workload reliever to assist the pilot in control and navigation of the helicopter. The flight director has nine modes of operation, any of which may be coupled to the helipilot system for fully automatic hands off flight path control. When decoupled from the helipilots, the flight director provides automatic flight path computation and visual pitch and roll command indications to direct the pilot in maneuvering the helicopter to maintain the selected flight path. When the flight director is coupled in the appropriate modes, the automatic flight control system will maneuver the helicopter to perform the following functions:

Maintain a constant pressure altitude (ALT).

Maintain a constant indicated airspeed (IAS).

Maintain a constant vertical speed climb or descent (VS).

Turn to and maintain a selected magnetic heading (HDG).

Capture and track a selected VOR radial (NAV or VOR APR).

Capture and track an ILS localizer and glideslope (ILS).

Capture and track a localizer back course (BC).

Initiate a missed approach (go-around) climbout (GA).

The flight director computer analyzes vertical and lateral flight and navigational data to generate pitch and roll steering commands which are displayed visually on the attitude director indicator (ADI). The vertical channel combines pitch attitude, airspeed, altitude, vertical speed, and glideslope deviation information to produce computed pitch command signals. The lateral channel combines roll attitude, heading, and course deviation information to produce computed roll command signals. Automatic flight path control is achieved when the pitch and roll commands from the flight director computer are coupled to the helipilot computers.

Should a flight or navigation data signal become invalid, the affected pitch or roll channel will revert to attitude hold mode and the respective command bar on the ADI will retract from view. If either helipilot fails or is disengaged, the flight director will decouple automatically.

Use or disclosure of data contained on this page is subject to the restriction on the title page of this document.

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FLIGHT DIRECTOR CONTROLS**MODE SELECTOR**

The mode selector (figure 2-1) enables the pilot to select the desired flight director mode by depressing the appropriate push-on/push-off button. Selected modes are annunciated by illumination of the respective mode select buttons. The NAV, BC, and VOR APR buttons have two lights each (ARM and CAP) to advise the pilot of the status of the flight director computer. The ILS button legends (ARM and GS) advise when the computer has armed or captured the glideslope. Depressing the SBY button disengages all modes and tests the illumination of all mode selector lights. Turning the PILOT INSTR LT knob clockwise dims all mode selector lights for night operation.

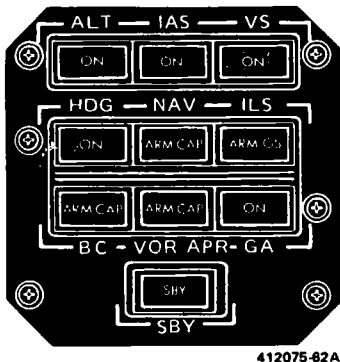


Figure 2-1. Flight director mode selector

COUPLE BUTTON

The CPL button is a push-on/push-off button located on the helipilot control panel on the pedestal. When both helipilots (HP1 and HP2) are engaged in the attitude retention mode (ATT), selecting any valid flight director mode will couple the flight director to the helipilots automatically, as indicated by illumination of the ON legend of the CPL button. The pilot may decouple the flight director by depressing the

CPL button. When decoupled, the flight director will continue functioning in the selected mode, providing visual pitch and roll commands to the pilot on the attitude director indicator (ADI). Once depressed, the CPL button must be depressed again to recouple any flight director mode.

FORCE TRIM SWITCHES

The FORCE TRIM switch on the pedestal must be ON anytime the flight director is coupled for automatic flight control.

The cyclic mounted FORCE TRIM release button can be depressed to allow the pilot to reposition the cyclic control and pedals manually for large scale pitch, roll, and yaw corrections. Upon depressing the button, the rotary trim actuators are de-energized, the flight director modes are decoupled momentarily, the helipilot pitch, roll, and yaw linear actuators return to center positions, and the helipilot computers are placed in a fast follow-up mode to track flight control positions. Upon releasing the button, the helipilots and flight director will resume functioning in the preselected modes. If previously decoupled in the ATT mode, the helipilots will maintain the attitude existing at the time the button is released.

ATTITUDE TRIM SWITCH

The cyclic mounted ATTD TRIM switch can be moved fore and aft to adjust pitch attitude and laterally to adjust roll attitude during decoupled operation in attitude retention mode. The switch is also used to make small pitch attitude changes when coupled in any mode except ALT and ILS after glideslope capture. Roll attitude can also be adjusted by the ATTD TRIM switch, except when a lateral mode is engaged. (Large attitude changes should be made by depressing the cyclic FORCE TRIM release button.)

FLIGHT DIRECTOR STANDBY BUTTON

The cyclic mounted NAV STBY button is a remote switch having the same function as the SBY button on the flight director mode selector. Depressing the button disengages all flight

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director modes, tests the illumination of all mode selector lights, retracts the pitch and roll command bars on the ADI, and places the flight director in a standby status. The pilot must then reselect the modes if continued flight director operation is desired.

GO-AROUND BUTTON

The GO-AROUND button, located on the collective control head, is a remote switch having the same function as the GA button on the flight director mode selector. Depressing the button places the flight director in go-around mode and disengages all other modes. In GA mode the flight director commands a roll level attitude and a pitch attitude which will provide a 750 feet-per-minute rate of climb. (The pilot must adjust collective pitch to maintain desired climb airspeed.)

AUXILIARY PITOT-STATIC SWITCHES

The AUX SYS PITOT and STATIC switches provide a means for isolating the flight director airspeed and altitude sensors from the copilot pitot-static system in the event of leakage or other system malfunction. When the PITOT switch is OFF the flight director airspeed sensor is disconnected, rendering the IAS mode inoperative. Placing the STATIC switch in the OFF position disconnects both the airspeed and altitude sensors of the flight director and thereby disables the ALT, IAS, and VS modes. The pilot should disengage the affected vertical modes to prevent undesirable flight control inputs when either switch is in the OFF position.

Bell **412/412EP**
MODELS

ROTORCRAFT FLIGHT MANUAL

SUPPLEMENT SINGLE-SPEED INTERNAL HOIST

214-706-003

CERTIFIED
OCTOBER 2, 1981

This supplement shall be attached to the Model 412 or 412EP Flight Manual when the 214-706-003 Single Speed Internal Hoist has been installed.

Information contained herein supplements information of basic Flight Manual. For Limitations, Procedures, and Performance Data not contained in this supplement, consult basic Flight Manual.

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PAGE	REVISION NO.	PAGE	REVISION NO.
FLIGHT MANUAL		35/36	0
Title — NP	0	MANUFACTURER'S DATA	
A — B	0	37 — 38	0
I/II	0		
1 — 34	0		

NOTE

Revised text is indicated by a black vertical line. Insert latest revision pages; dispose of superseded pages.

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fa MANAGER

ROTORCRAFT CERTIFICATION OFFICE
FEDERAL AVIATION ADMINISTRATION
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GENERAL INFORMATION

The Single Speed Internal Hoist enables cargo and emergency rescue operations in areas where landings cannot be accomplished. The hoist can raise or lower loads up to 600 pounds (272 kilograms). The hoist cable is 250 usable feet (76.2 meters) in length. Each of the four cabin mounting locations allows the hoist to be extended 90 degrees outboard. Caution

lights on each side of the hoist illuminate when the hook reaches 20 feet (6 meters) below the skids during retraction. An electrically actuated cable cutting device allows pilot or hoist operator to sever the cable in an emergency. A manually operated cable cutter, accessible to the hoist operator, may be used to sever the cable if the electrical cable cutter fails.

Section 1

LIMITATIONS

1-3. TYPES OF OPERATION

Hoist operations shall be conducted under appropriate operating rules for external loads.

Passenger operations with hoist installed are approved if hoist is stowed and electrical system is deactivated.

Hoist operations are prohibited during instrument meteorological conditions.

1-4. FLIGHT CREW

A crewmember wearing an approved safety harness in passenger compartment is required during all phases of hoist operations. Crewmember shall wear protective gloves for guiding cable during operation. The hoist operator shall be familiar with hoist operating procedures and limitations.

1-5. CONFIGURATION

1-5-A. REQUIRED EQUIPMENT

Hoist cable antichafing guard shall be installed on standard or high skid landing gear (with or without floats) on same side of helicopter as hoist.

1-5-B. OPTIONAL EQUIPMENT

Fixed passenger step shall not be installed concurrently with internal hoist.

Retractable passenger steps shall be stowed during hoist operations.

Hoist operation with flight director in coupled mode is prohibited.

Hoisting or lowering an empty litter in open position (except Stokes litter) is prohibited.

Refer to appropriate Flight Manual Supplement(s) for additional limitations, procedures, and performance data.

1-6. WEIGHT AND CENTER OF GRAVITY

Actual weight change shall be determined after hoist is installed and ballast readjusted, if necessary.

For maximum gross weight, including hoist load, refer to applicable Flight Manual or BHT-412-FMS-19.1 when Increased Gross Weight and Takeoff Horsepower kit is installed.

Maximum hoist load is 600 pounds (272 kilograms). This is a structural limitation only and does not ensure that longitudinal or lateral CG will remain within approved limits. Maximum allowable hoist load varies with gross weight, center of gravity, and hoist location. Refer to appropriate Hoist Loading Schedule.

NOTE

The center of gravity of hoist load in forward position is F.S. 82 (2083 mm) and B.L. 60 (1524 mm). The center of gravity of hoist load in aft position is F.S. 131 (3327 mm) and B.L. 64.4 (1636 mm).

For Longitudinal vs. Lateral CG limits with internal hoist refer to Internal hoist CG envelope figure 1-1.

1-7. AIRPEED

VNE with asymmetrical door configuration is 20 KIAS.

VNE with hinged panels locked open and cargo doors open is 20 KIAS.

VNE with hinged panels removed and cargo doors removed or secured open is 60 KIAS.

1-24. HOIST DUTY CYCLE LIMITATIONS

The hoist is approved for continuous operation with loads not to exceed 600 pounds (272 kilograms).

1-25. ALLOWABLE HOIST LOAD

Select hoist loading schedules (figures 1-2 through 1-5) appropriate for position in which hoist is installed.

NOTE

Hoist loading schedules are based on most adverse loading combinations of pilot, copilot, and hoist operator, each weighing 170 or 200 pounds (77.1 or 90.7 kilograms), and on a weight empty CG of 0.3 inches (7.3 mm) to right of centerline prior to adding hoist. If lateral CG is appreciably different or crewmember weights are out of this range, allowable hoist load shall be computed. For computation, assume hoist operator in forward position to be located at F.S. 87 (2210 mm) and B.L. 40 (1016mm), and in aft position F.S. 125 (3175mm) and B.L. 40 (1016mm).

1-25-A. LEFT HOIST INSTALLATIONS

Enter appropriate schedule, figures 1-2 through 1-5 at gross weight of helicopter prior to hoisting. Proceed vertically to intersect with diagonal line representing number of crewmembers on board, top of schedule, or right cutoff line. Proceed horizontally to left to read maximum allowable hoist load. Intersecting with right cutoff line gives maximum load which does not cause helicopter to exceed gross weight limitations.

Using Weight empty chart, Section 5 and left hoist loading schedules ensures that both longitudinal and lateral limits are not exceeded during first hoist operation. However, for subsequent hoisting, additional precautions must be taken to avoid exceeding forward longitudinal limits.

1-25-A-1. LEFT FORWARD HOIST LOCATION

To continue using maximum allowable hoist capability: (Refer to figure 1-2 through 1-5)

- a. put hoisted load (people or cargo) along side of island, or
- b. when hoisted load is put immediately forward of island, reduce maximum hoist load to 300 pounds.

WARNING

DO NOT PUT HOISTED LOAD IN FORWARD AREA OF PASSENGER COMPARTMENT UNLESS MAXIMUM HOIST LOADS ARE COMPUTED FOR THAT CONFIGURATION.

1-25-A-2. LEFT AFT HOIST LOCATION

To continue using maximum allowable hoist capability: (Refer to figure 1-2 through 1-5)

- a. put hoisted load along island or immediately forward of island, or
- b. ensure empty weight CG is within Area A. Refer to Weight empty chart, Section 5.

1-25-B. RIGHT HOIST INSTALLATIONS

Right lateral limit for hoist operations varies with longitudinal center of gravity of the helicopter. The loading schedules have been modified to account for this variation.

- a. Starting with appropriate schedule for number of crewmembers on board, enter at gross weight of helicopter prior to hoisting.
- b. Proceed vertically to intersect with diagonal line representing helicopter center of gravity prior to hoisting, top of schedule, or right cutoff line.
- c. Proceed horizontally to left to read maximum allowable hoist load.

When helicopter center of gravity is between STA. lines, interpolate to determine CG.

Intersecting right cutoff line gives maximum load which does not cause helicopter to exceed gross weight limitations or forward longitudinal limits.

For multiple hoists during a single flight, after each hoist operation enter appropriate schedule at revised gross weight and proceed to new center of gravity to determine maximum allowable hoist load.

EXAMPLE 1: NORMAL

Determine Hoist Load when hoist is in R/H FWD POSITION and crew consist of Pilot, Copilot and Hoist Operator.

GIVEN:

Gross Weight — 9,500 lbs.

CG — STA. 135.5 before hoisting

From appropriate 11,600 lb. GW schedule obtain hoist load as follows:

Enter gross weight at 9,500 lbs.

Proceed up GW line to interpolated STA. 135.5

Proceed left to read hoist load of 210 lbs. Point (A).

EXAMPLE 2: NORMAL

Determine Hoist Load when hoist is in R/H FWD POSITION and crew consist of Pilot, Copilot and Hoist Operator.

GIVEN:

Gross Weight — 9,500 lbs.

CG — STA. 138.5 before hoisting

From appropriate 11,600 lb. GW schedule obtain hoist load as follows:

Enter gross weight at 9,500 lbs.

Proceed up GW line to STA. 138.5

Proceed left to read hoist load of 550 lbs. Point (B).

1-25-C. RIGHT HOIST INSTALLATION - PENALTY REGION OPERATION

The dashed line on schedules represents longitudinal center of gravity prior to hoisting which will result in a gross weight center of gravity at Sta. 135.2 and B.L. 4.5 during hoist operations with maximum hoist loads derived using this line. This center of gravity is the corner of but not in Penalty Region shown in Limitations.

Hoist loads derived for Normal Operations may be increased when GW/CG combinations are forward of those

represented by dashed line. Loads may be increased up to but not greater than those defined by dashed line. However, this procedure will result in operations within Penalty Region. Refer to Section 1, Internal Hoist CG Envelope, for Penalty Region.

EXAMPLE 3: PENALTY REGION

Determine Hoist Load when hoist is in R/H FWD POSITION and crew consist of Pilot, Copilot and Hoist Operator.

GIVEN:

Gross Weight — 9,500 lbs.

CG — STA. 135.5 before hoisting

From appropriate 11,600 lb. GW schedule obtain hoist load as previously determined in Example 1 the maximum hoist load for normal operations is 210 lbs. Point (A).

To increase hoist load to maximum for condition without exceeding GW/CG limits, proceed up to dashed line and read left to find 435 lbs. Point (C).

The Penalty Region is any load greater than Point (A) up to maximum load at Point (C).

For GW vs. CG combinations aft of the CG represented by the dashed line (see Example 2), there is no Penalty Region.

1-26. WEIGHT EMPTY CHART

The Weight empty chart for internal hoisting operations is shown in Section 5. Refer to the maintenance manual for additional information.

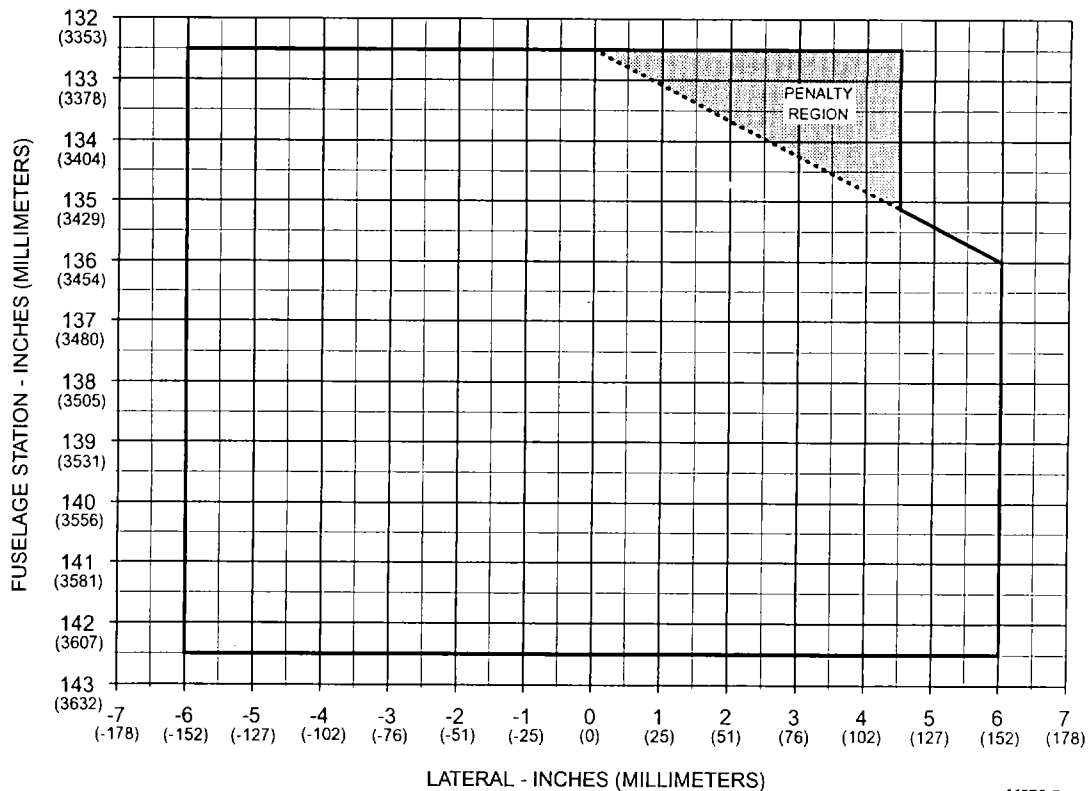
NOTE

Allowable hoist load must be computed when weight empty is not within specified guidelines, shown in Section 5.

NOTE

Allowable hoist loads must be computed when AUX Fuel kits are installed.

Longitudinal/Lateral C.G. Envelope for Internal Hoist Operations

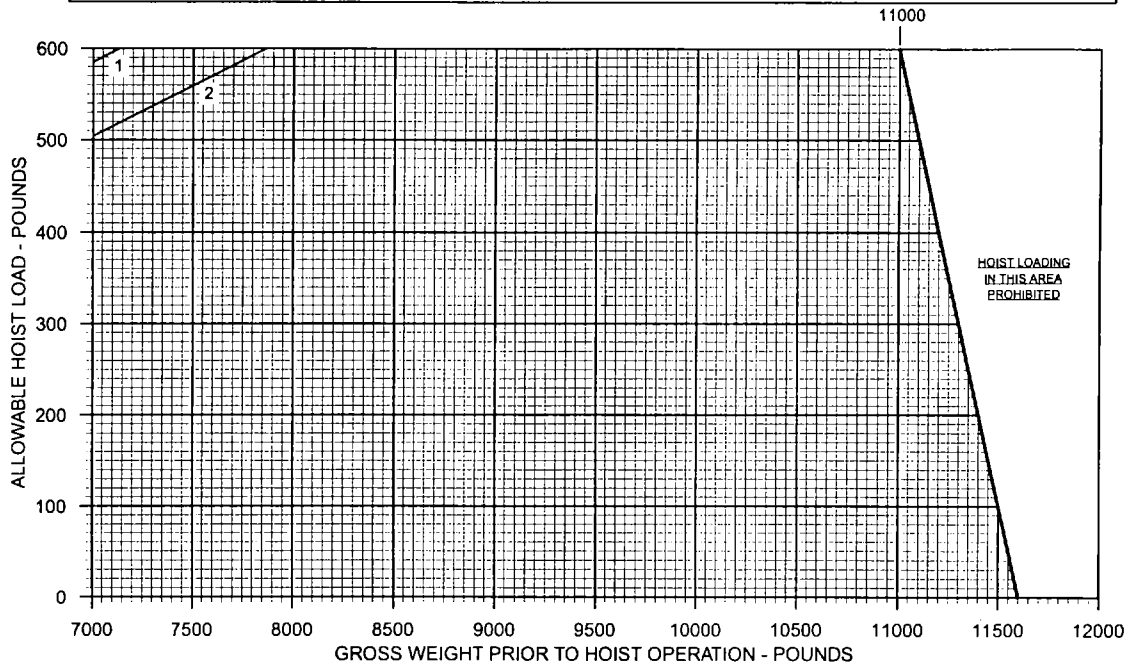


412FS-7-1-1

Figure 1-1. Internal hoist CG envelope

L/H FWD POSITION

- 1 PILOT AND HOIST OPERATOR - 600 LB HOIST LOAD IS ALLOWED AT GROSS WEIGHTS FROM 7133 TO 11000 LBS.
 2 PILOT, COPILOT, AND HOIST OPERATOR - 600 LB HOIST LOAD IS ALLOWED AT GROSS WEIGHTS FROM 7866 TO 11000 LBS.

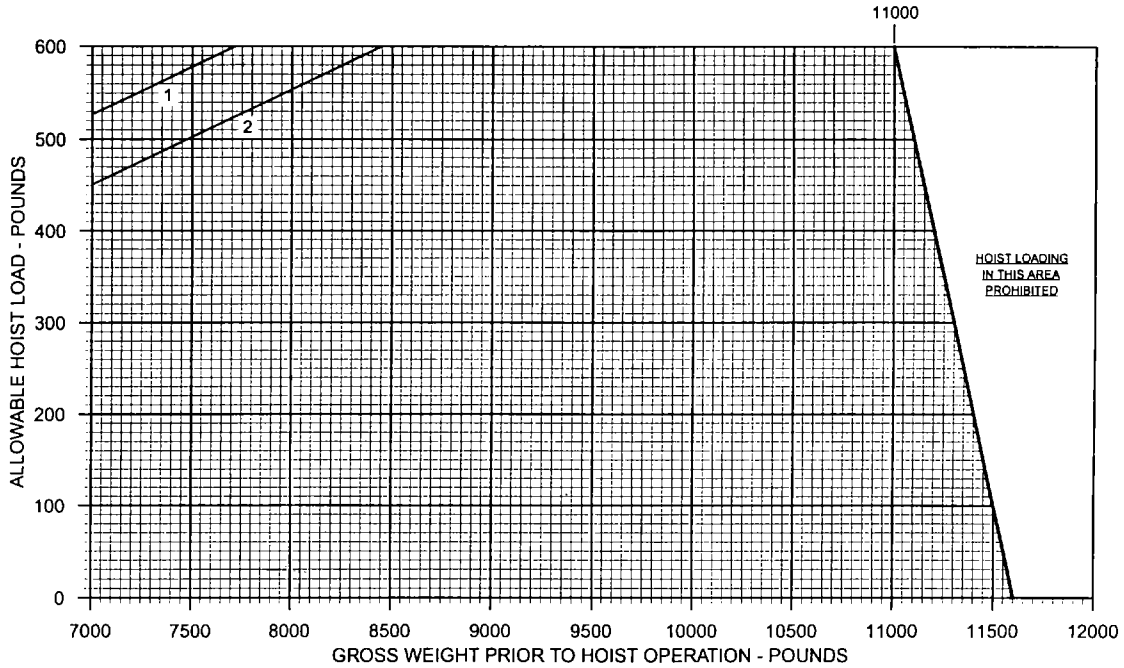


412FS-7-1-2-1

Figure 1-2. Hoist loading schedules 11,600 lb. GW (English) (Sheet 1 of 6)

L/H AFT POSITION

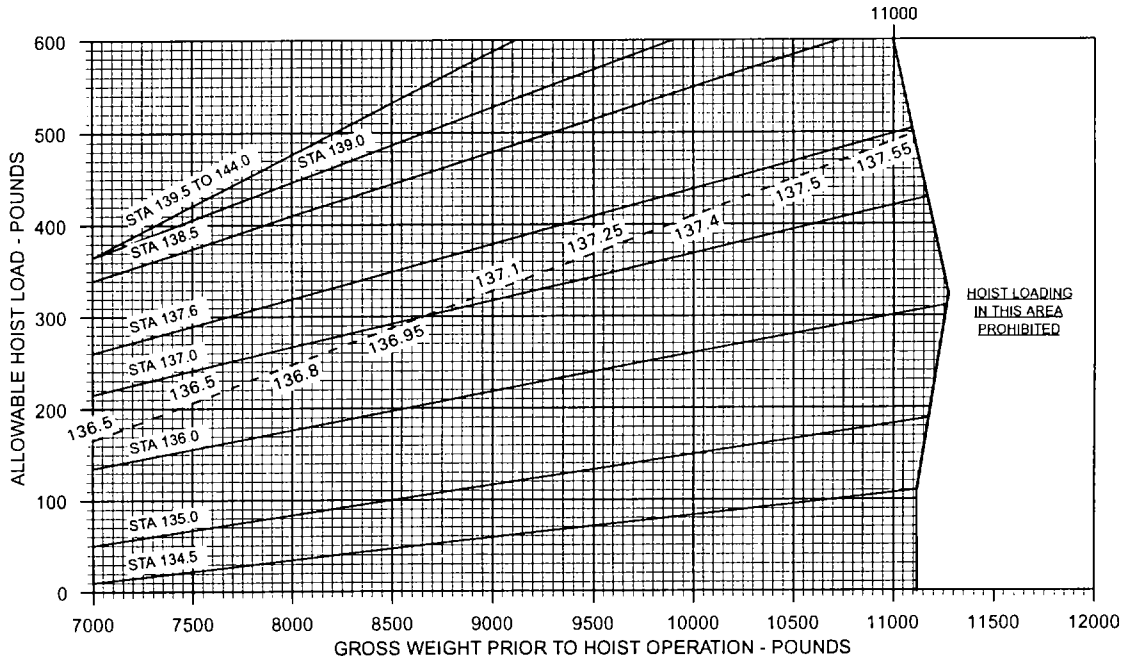
- 1 PILOT AND HOIST OPERATOR - 600 LB HOIST LOAD IS ALLOWED AT GROSS WEIGHTS FROM 7133 TO 11000 LBS.
- 2 PILOT, COPILOT, AND HOIST OPERATOR - 600 LB HOIST LOAD IS ALLOWED AT GROSS WEIGHTS FROM 7866 TO 11000 LBS.



412FS-7-1-2-2

Figure 1-2. Hoist loading schedules 11,600 lb. GW (English) (Sheet 2 of 6)

R/H FWD POSITION - PILOT AND HOIST OPERATOR

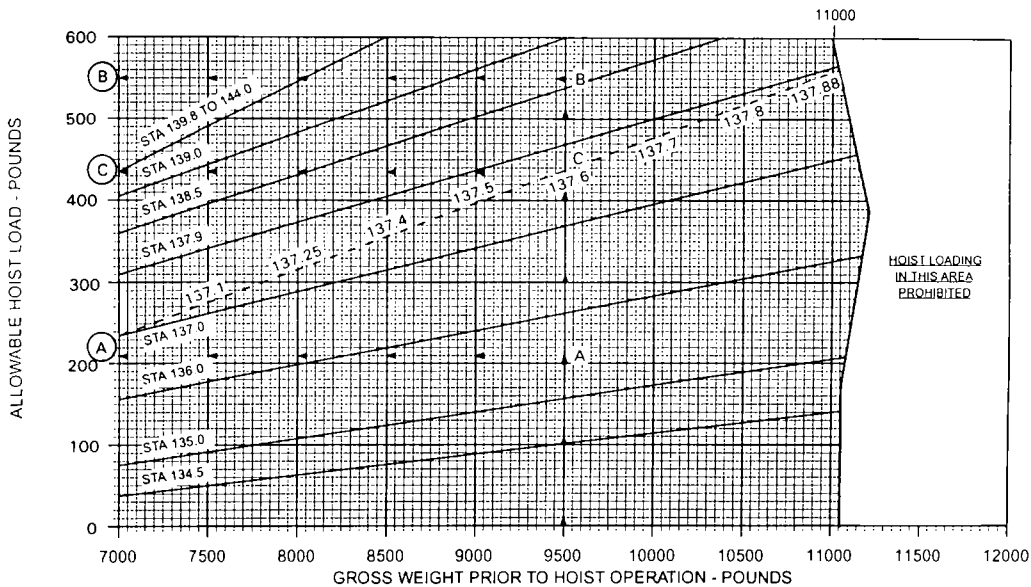


NOTE: THE DASHED LINE INDICATES A DIFFERENT LONGITUDINAL C.G. FOR EACH GROSS WEIGHT.

412FS-7-1-2-3

Figure 1-2. Hoist loading schedules 11,600 lb. GW (English) (Sheet 3 of 6)

R/H FWD POSITION - PILOT, COPILOT AND HOIST OPERATOR

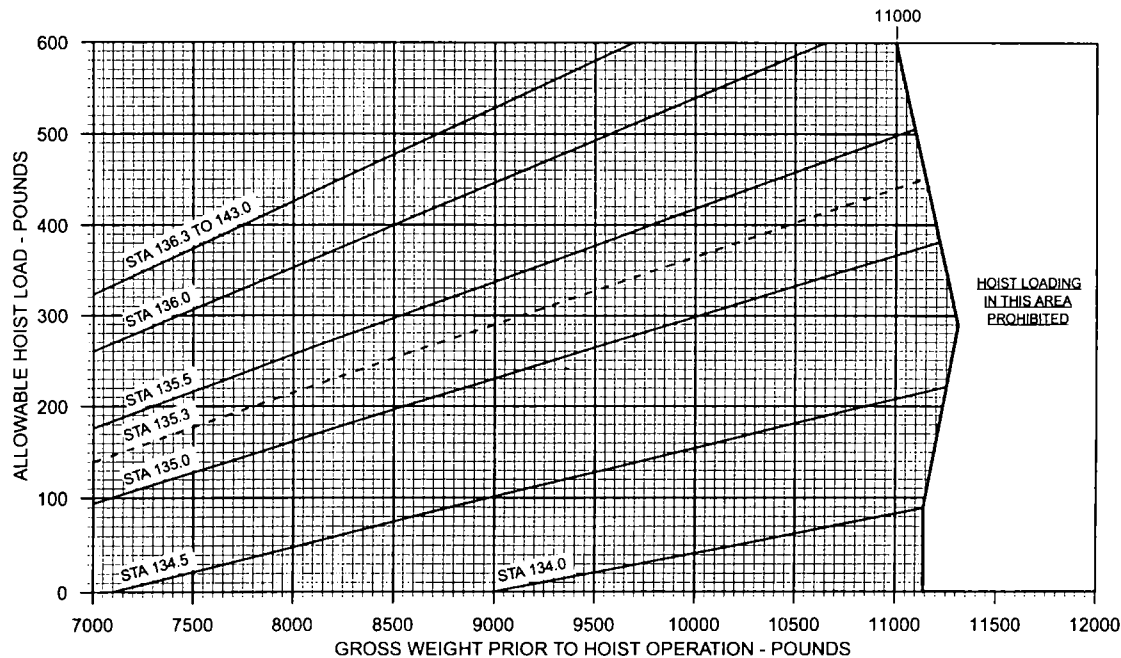


NOTE: THE DASHED LINE INDICATES A DIFFERENT LONGITUDINAL C.G. FOR EACH GROSS WEIGHT.

412FS-7-1-2-4

Figure 1-2. Hoist loading schedules 11,600 lb. GW (English) (Sheet 4 of 6)

R/H AFT POSITION - PILOT AND HOIST OPERATOR

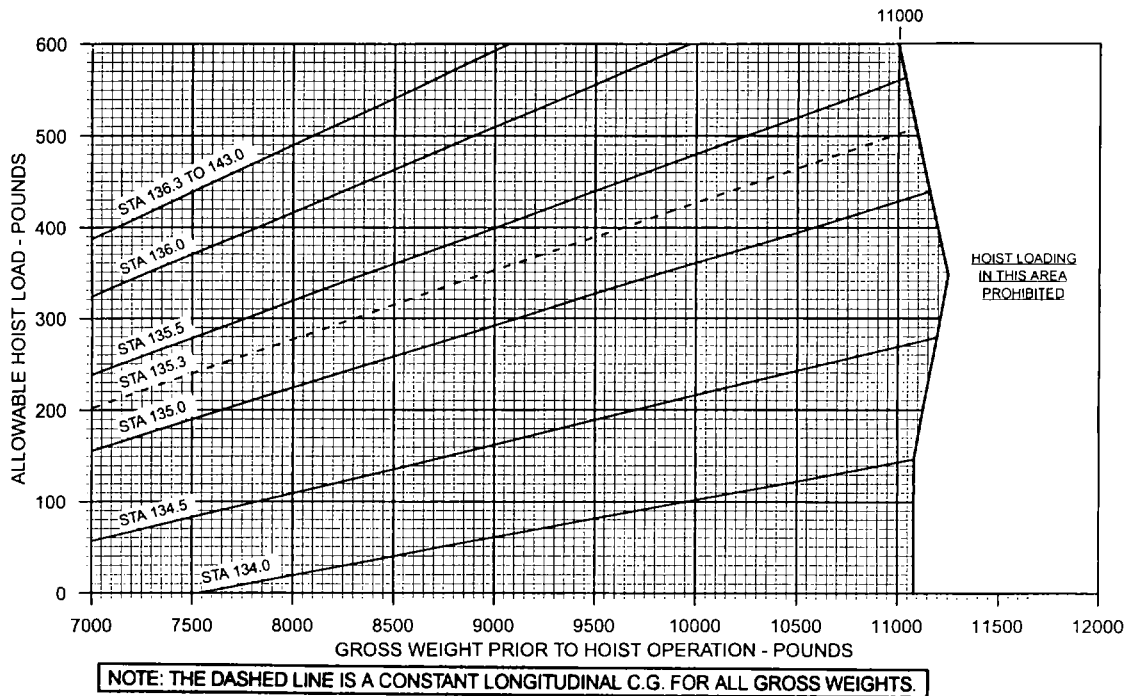


412FS-7-1-2-5

Figure 1-2. Hoist loading schedules 11,600 lb. GW (English) (Sheet 5 of 6)

R/H AFT POSITION - PILOT, COPILOT AND HOIST OPERATOR

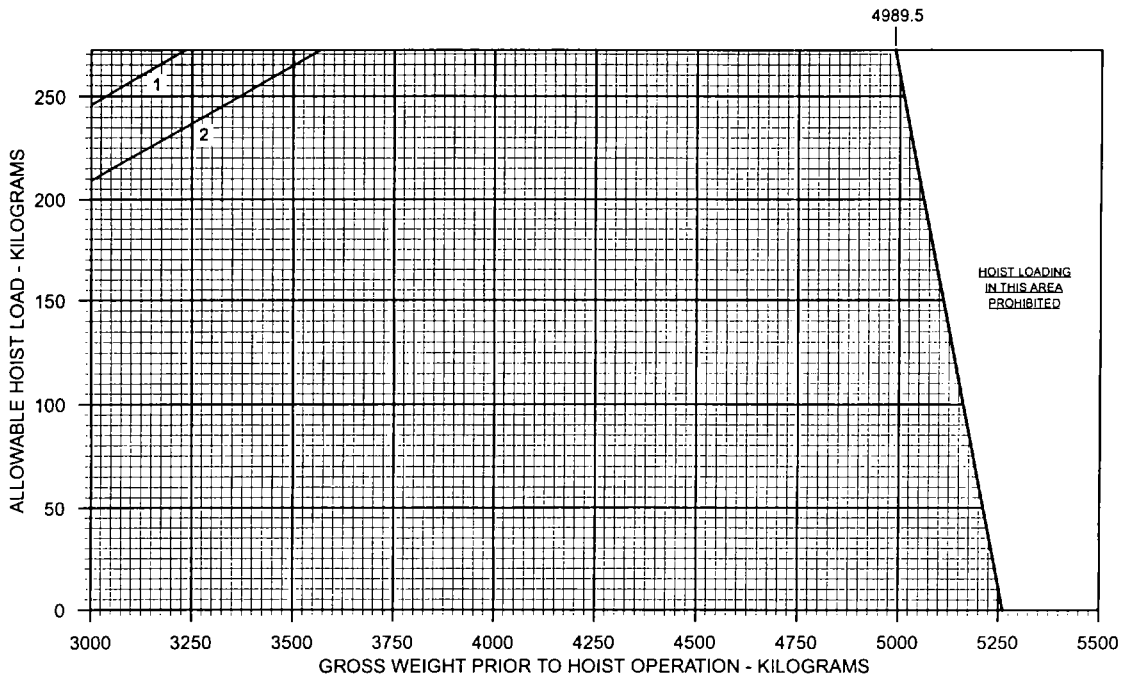
Figure 1-2. Hoist loading schedules 11,600 lb. GW (English) (Sheet 6 of 6)



412FS-7-1-2-6

L/H FWD POSITION

- 1 PILOT AND HOIST OPERATOR - 600 LB HOIST LOAD IS ALLOWED AT GROSS WEIGHTS FROM 3233.8 TO 4989.5 KGS.
 2 PILOT, COPILOT, AND HOIST OPERATOR - 600 LB HOIST LOAD IS ALLOWED AT GROSS WEIGHTS FROM 3566.5 TO 4989.5 KGS.

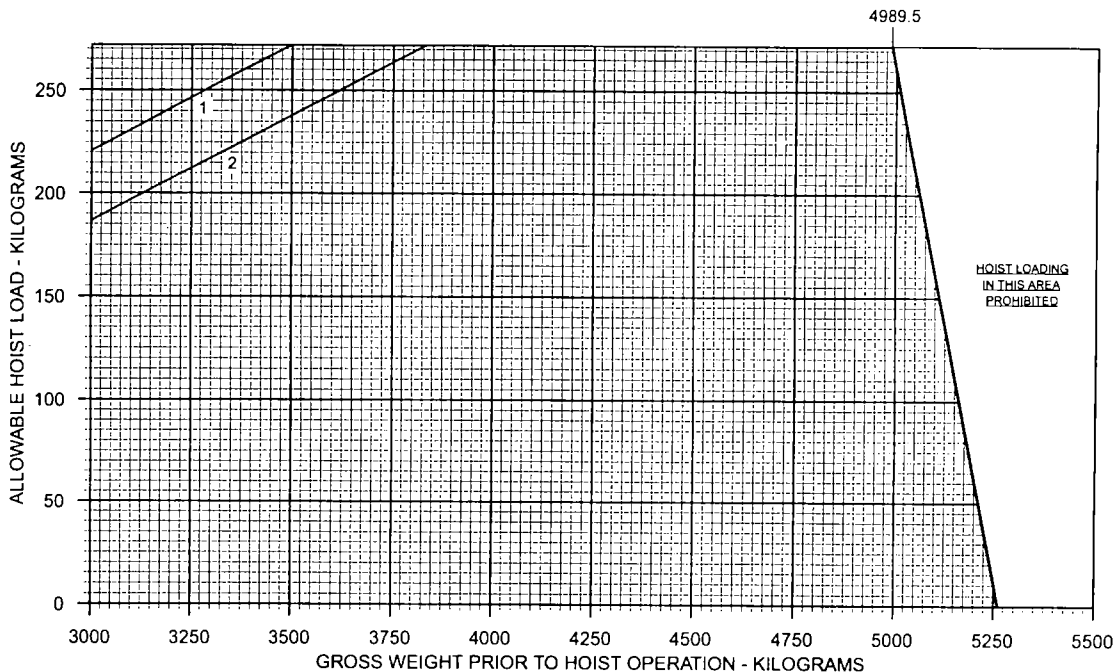


412FS-7-1-3-1

Figure 1-3. Hoist loading schedules 5261 kg. GW (Metric) (Sheet 1 of 6)

L/H AFT POSITION

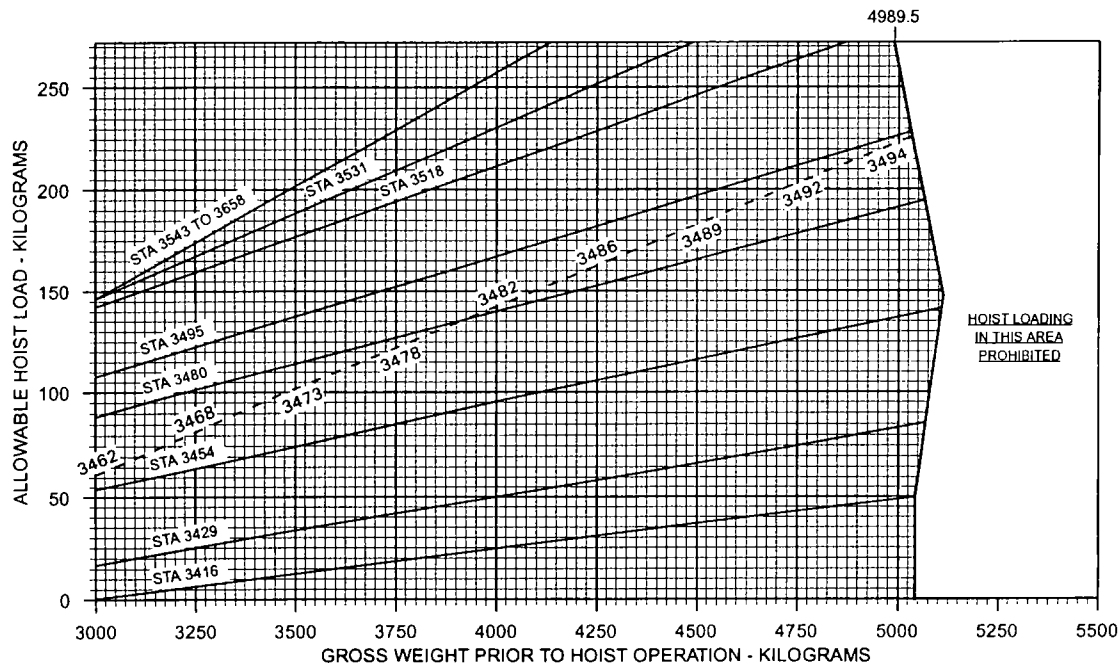
- 1 PILOT AND HOIST OPERATOR - 600 LB HOIST LOAD IS ALLOWED AT GROSS WEIGHTS FROM 3498.2 TO 4989.5 KGS.
- 2 PILOT, COPILOT, AND HOIST OPERATOR - 600 LB HOIST LOAD IS ALLOWED AT GROSS WEIGHTS FROM 3830.8 TO 4989.5 KGS.



412FS-7-1-3-2

Figure 1-3. Hoist loading schedules 5261 kg. GW (Metric) (Sheet 2 of 6)

R/H FWD POSITION - PILOT AND HOIST OPERATOR



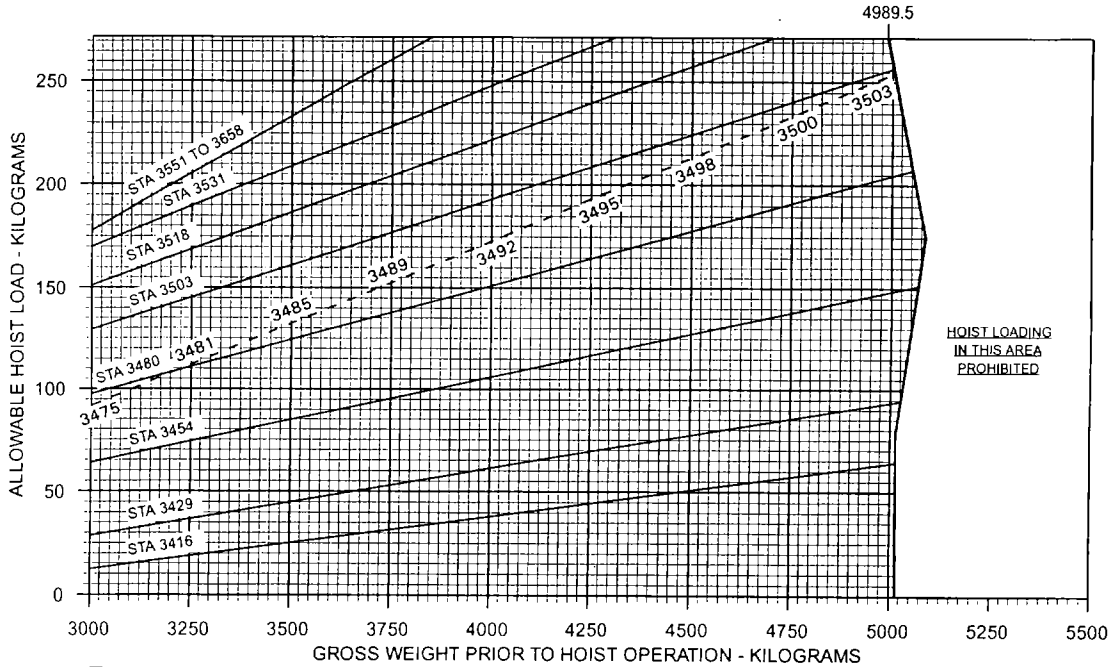
NOTE: THE DASHED LINE INDICATES A DIFFERENT LONGITUDINAL C.G. FOR EACH GROSS WEIGHT.

412FS-7-1-3-3

Figure 1-3. Hoist loading schedules 5261 kg. GW (Metric) (Sheet 3 of 6)

R/H FWD POSITION - PILOT, COPILOT AND HOIST OPERATOR

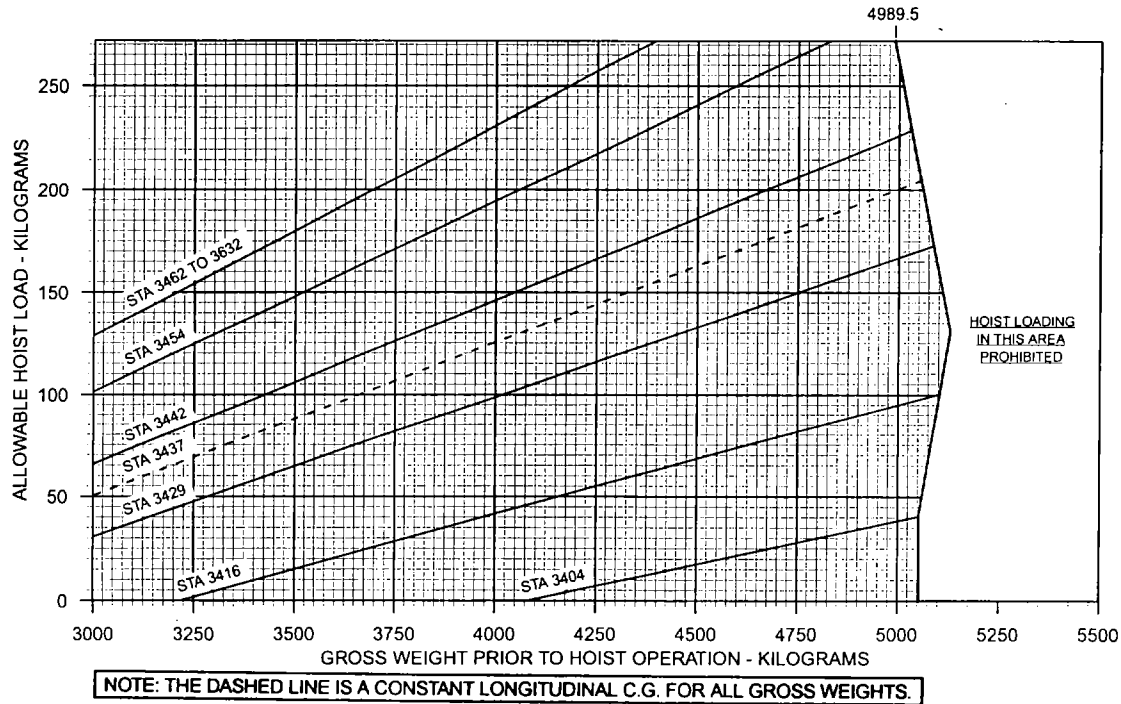
Figure 1-3. Hoist loading schedules 5261 kg. GW (Metric) (Sheet 4 of 6)



NOTE: THE DASHED LINE INDICATES A DIFFERENT LONGITUDINAL C.G. FOR EACH GROSS WEIGHT.

412FS-7-1-3-4

R/H AFT POSITION - PILOT AND HOIST OPERATOR

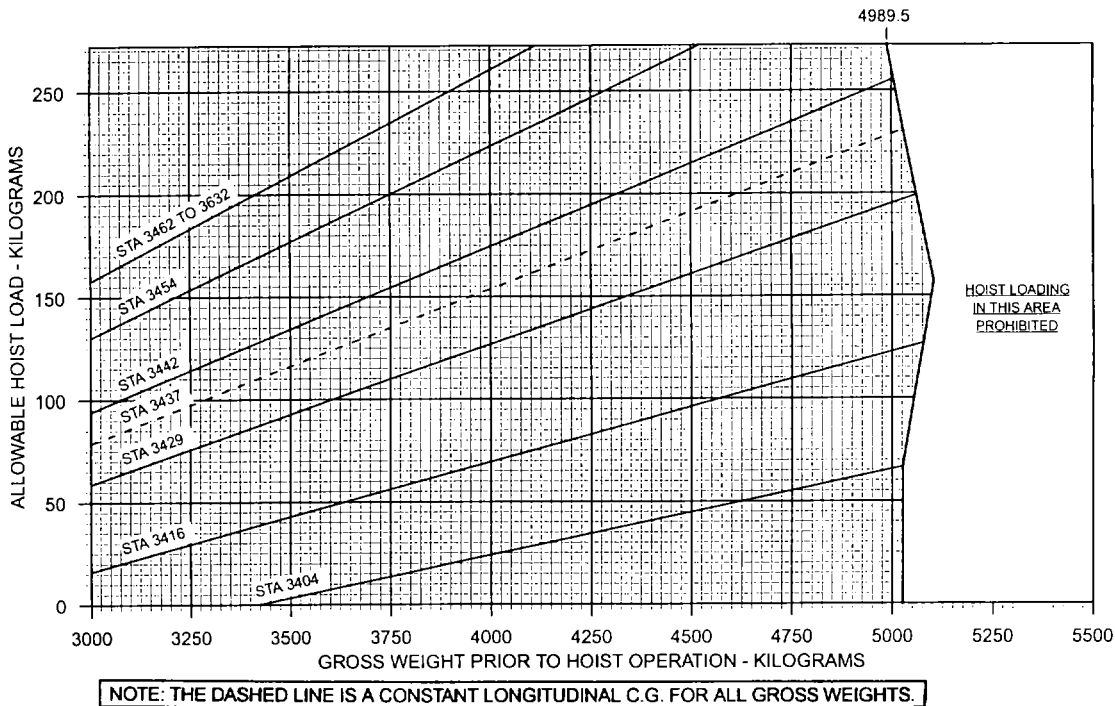


412FS-7-1-3-6



R/H AFT POSITION - PILOT, COPILOT AND HOIST OPERATOR

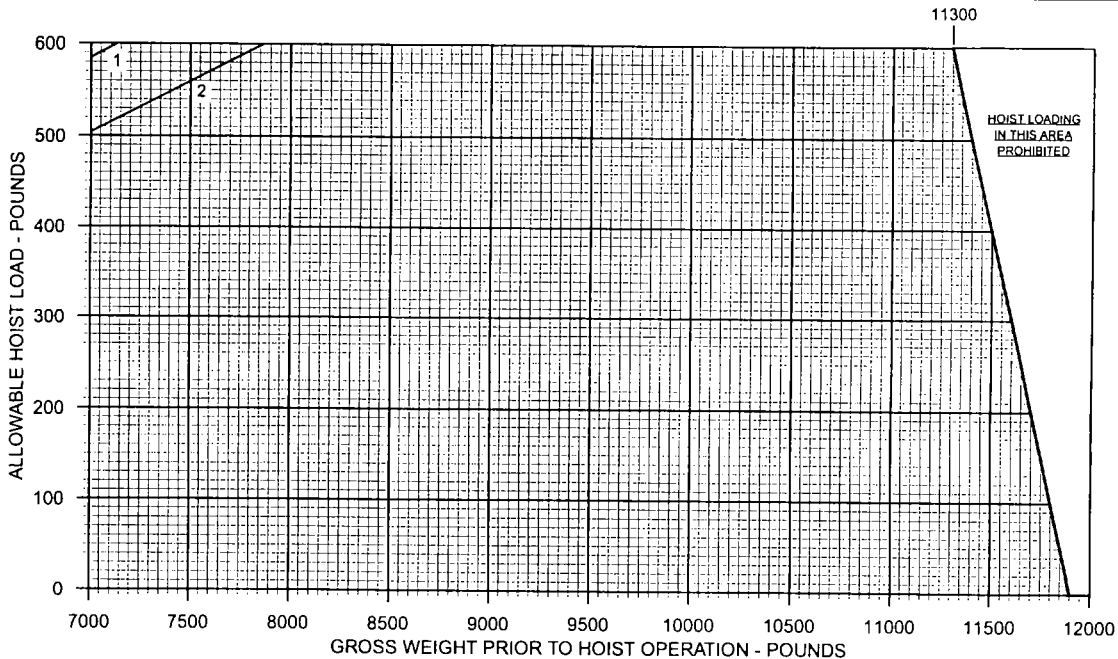
Figure 1-3. Hoist loading schedules 5261 kg. GW (Metric) (Sheet 6 of 6)



412FS-7-1-3-8

L/H FWD POSITION

- 1 PILOT AND HOIST OPERATOR - 600 LB HOIST LOAD IS ALLOWED AT GROSS WEIGHTS FROM 7133 TO 11300 LBS.
 2 PILOT, COPILOT, AND HOIST OPERATOR - 600 LB HOIST LOAD IS ALLOWED AT GROSS WEIGHTS FROM 7866 TO 11300 LBS.

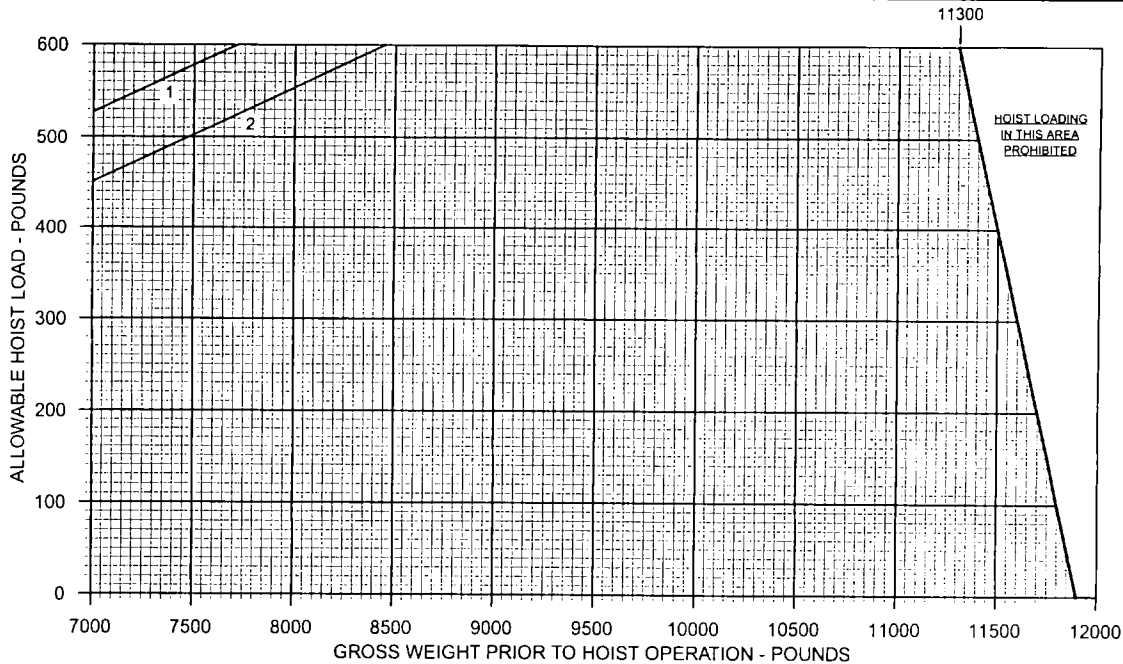


412FS-7-1-4-1

Figure 1-4. Hoist loading schedules 11,900 lb. GW (English) (Sheet 1 of 6)

L/H AFT POSITION

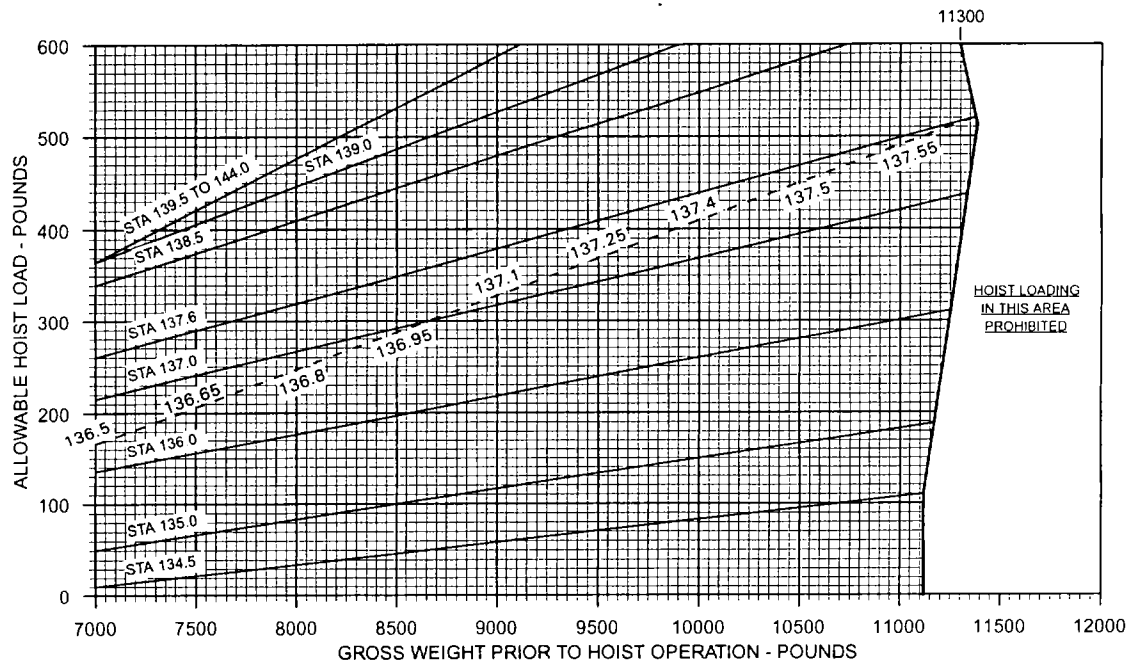
- 1 PILOT AND HOIST OPERATOR - 600 LB HOIST LOAD IS ALLOWED AT GROSS WEIGHTS FROM 7715 TO 11300 LBS.
- 2 PILOT, COPILOT, AND HOIST OPERATOR - 600 LB HOIST LOAD IS ALLOWED AT GROSS WEIGHTS FROM 8449 TO 11300 LBS.



412FS-7-1-4-2

Figure 1-4. Hoist loading schedules 11,900 lb. GW (English) (Sheet 2 of 6)

R/H FWD POSITION - PILOT AND HOIST OPERATOR

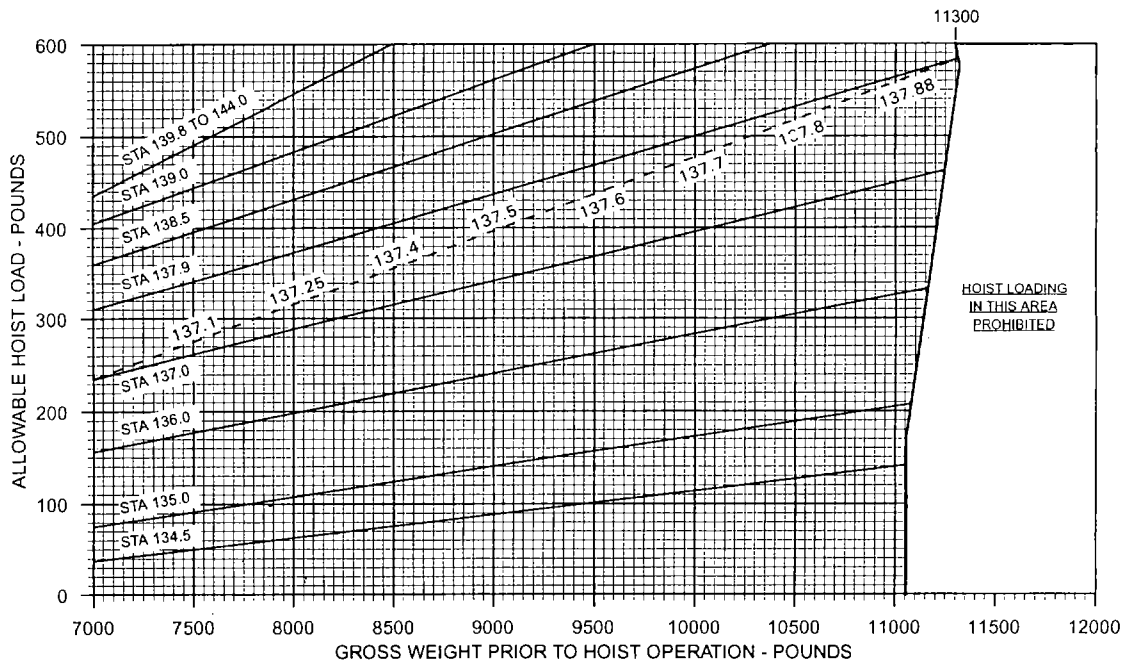


NOTE: THE DASHED LINE INDICATES A DIFFERENT LONGITUDINAL C.G. FOR EACH GROSS WEIGHT.

412FS-7-14-3

Figure 1-4. Hoist loading schedules 11,900 lb. GW (English) (Sheet 3 of 6)

R/H FWD POSITION - PILOT, COPILOT AND HOIST OPERATOR

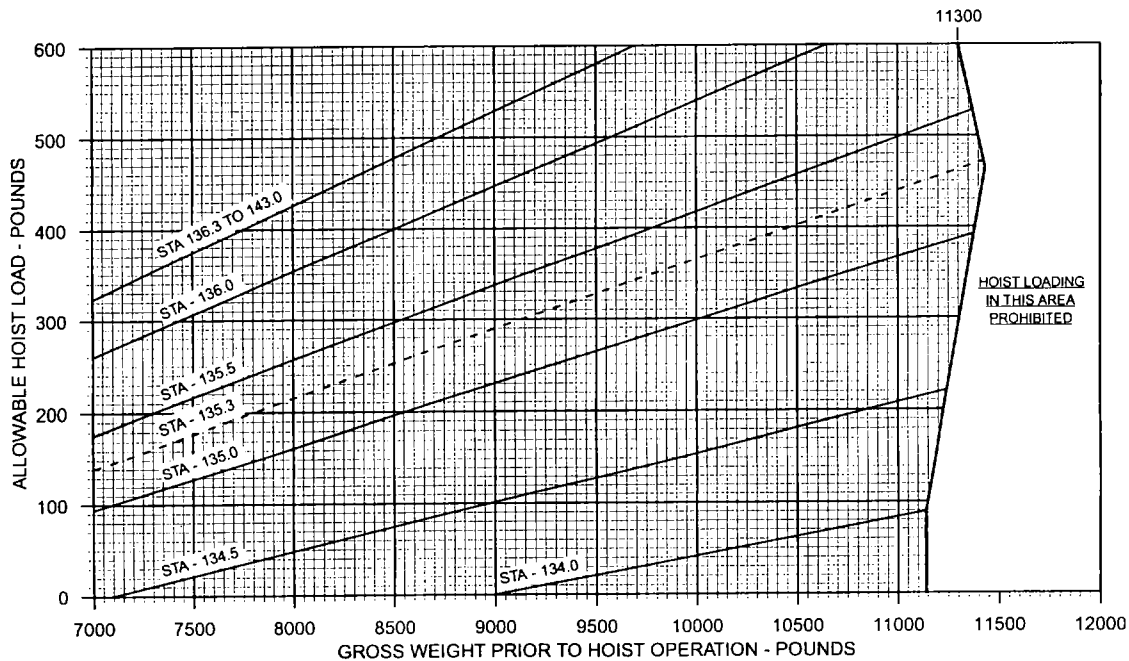


NOTE: THE DASHED LINE INDICATES A DIFFERENT LONGITUDINAL C.G. FOR EACH GROSS WEIGHT.

412FS-7-1-4-4

Figure 1-4. Hoist loading schedules 11,900 lb. GW (English) (Sheet 4 of 6)

R/H AFT POSITION - PILOT AND HOIST OPERATOR



NOTE: THE DASHED LINE IS A CONSTANT LONGITUDINAL C.G. FOR ALL GROSS WEIGHTS.

412FS-7-1-4-6

Figure 1-4. Hoist loading schedules 11,900 lb. GW (English) (Sheet 5 of 6)

R/H AFT POSITION - PILOT, COPILOT AND HOIST OPERATOR

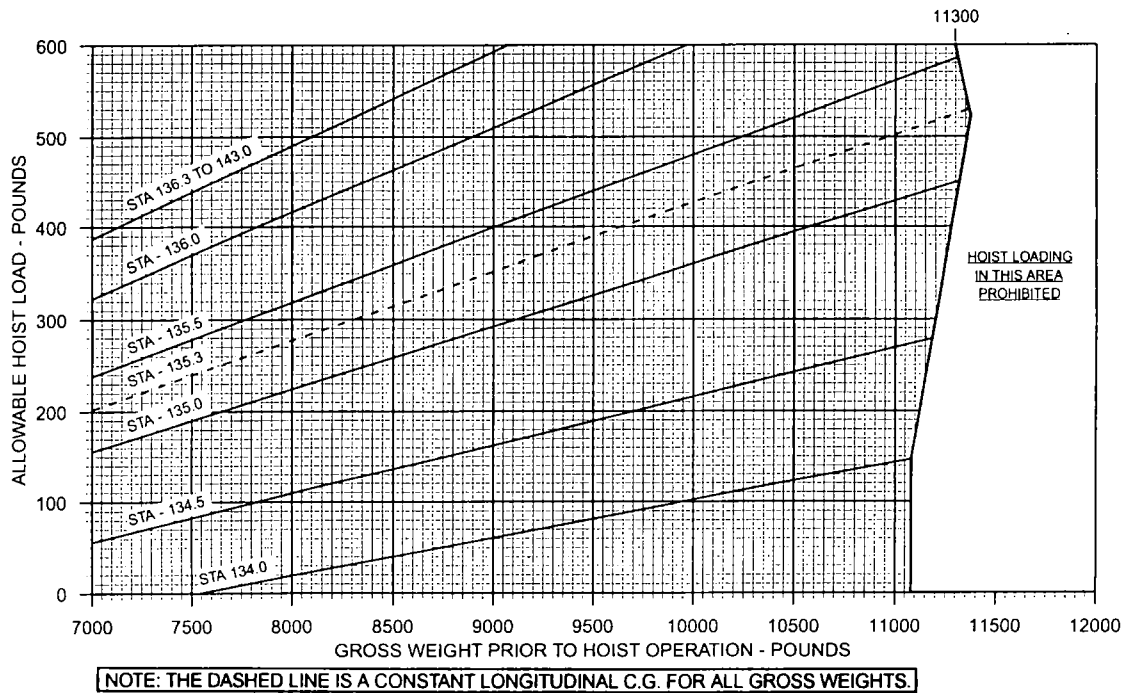
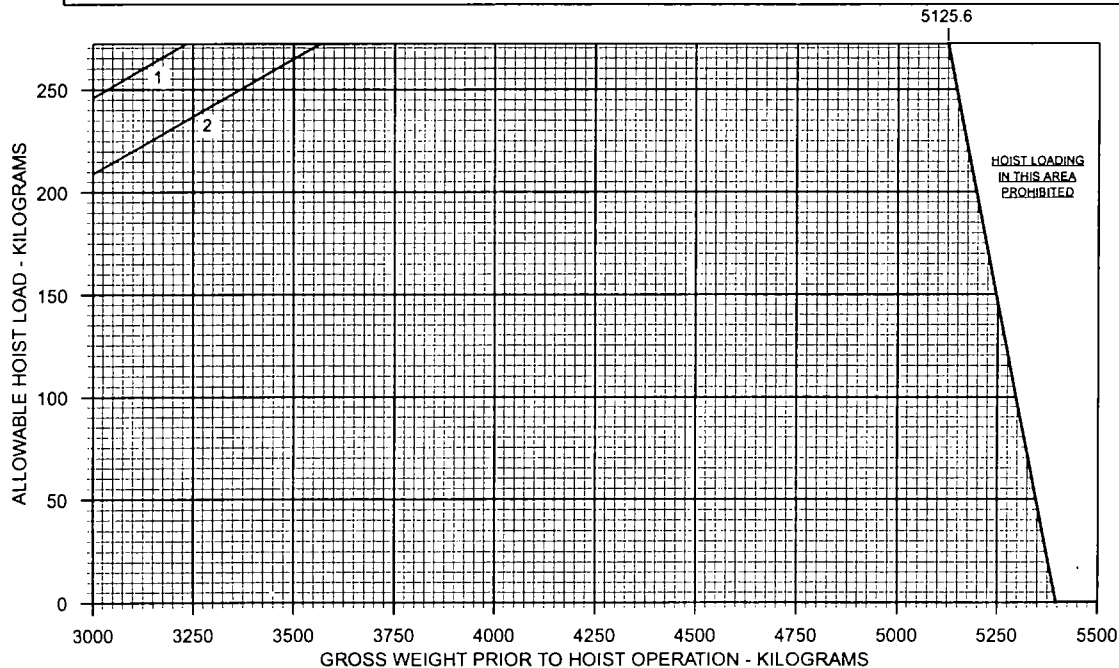


Figure 1-4. Hoist loading schedules 11,900 lb. GW (English) (Sheet 6 of 6)

L/H FWD POSITION

- 1 PILOT AND HOIST OPERATOR - 272 KG HOIST LOAD IS ALLOWED AT GROSS WEIGHTS FROM 3233.8 TO 5125.6 KGS.
 2 PILOT, COPILOT, AND HOIST OPERATOR - 272 KG HOIST LOAD IS ALLOWED AT GROSS WEIGHTS FROM 3566.5 TO 5125.6 KGS.

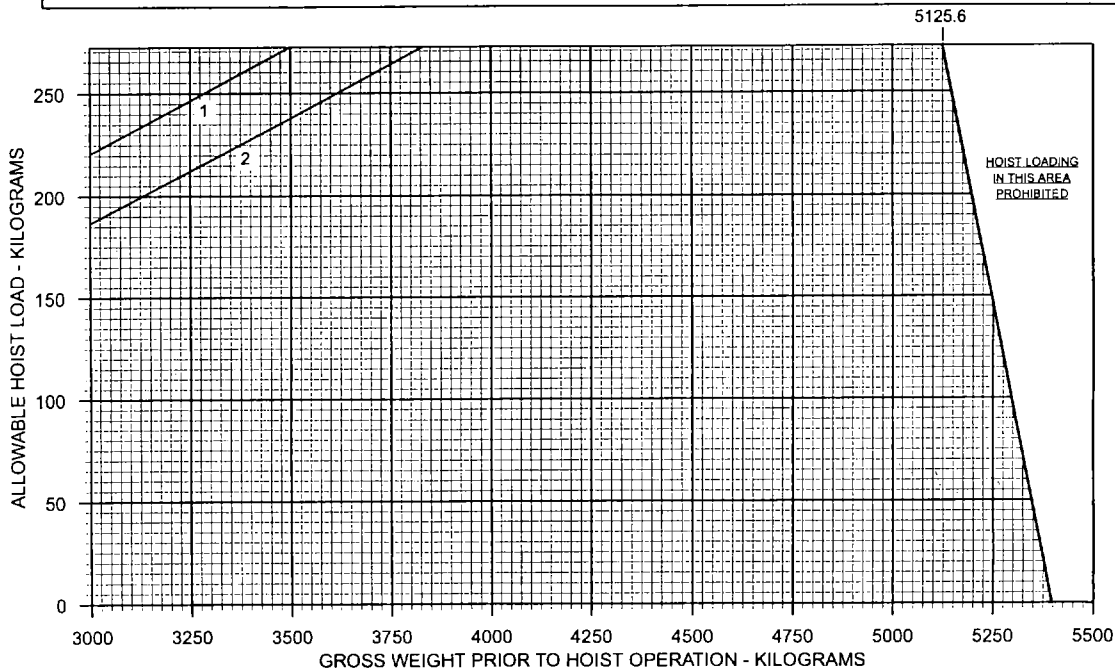


412FS-7-1-6-1

Figure 1-5. Hoist loading schedules 5397 kg. GW (Metric) (Sheet 1 of 6)

L/H AFT POSITION

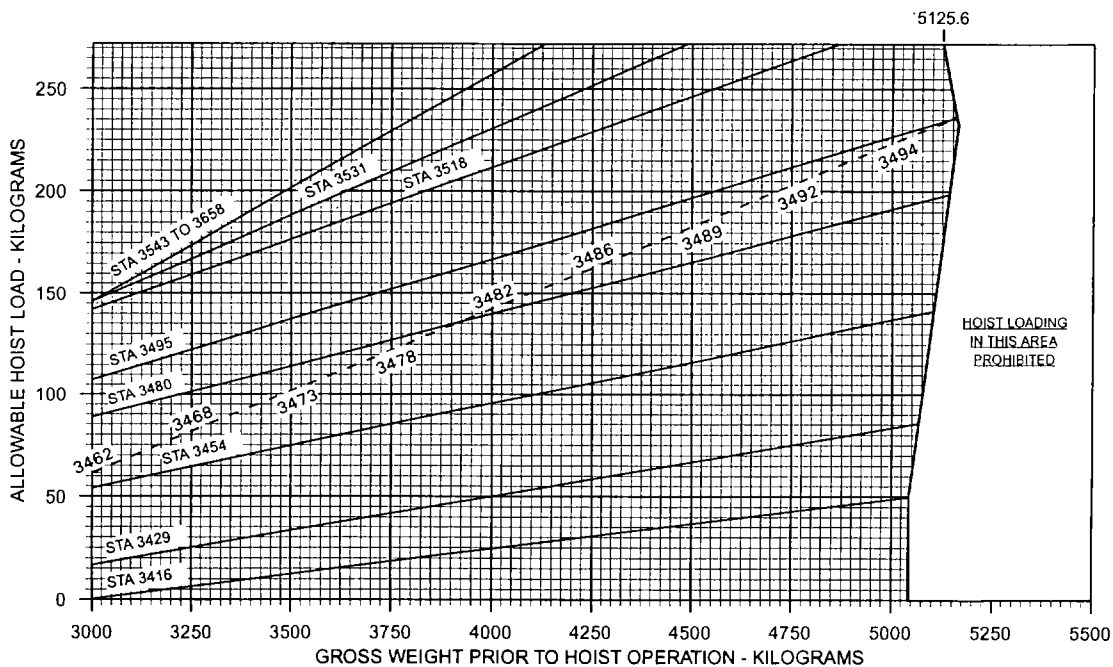
- 1 PILOT AND HOIST OPERATOR - 272 KG HOIST LOAD IS ALLOWED AT GROSS WEIGHTS FROM 3498.2 TO 5125.6 KGS.
 2 PILOT, COPILOT, AND HOIST OPERATOR - 272 KG HOIST LOAD IS ALLOWED AT GROSS WEIGHTS FROM 3830.8 TO 5125.6 KGS.



412FS-7-1-6-2

Figure 1-5. Hoist loading schedules 5397 kg. GW (Metric) (Sheet 2 of 6)

R/H FWD POSITION - PILOT AND HOIST OPERATOR



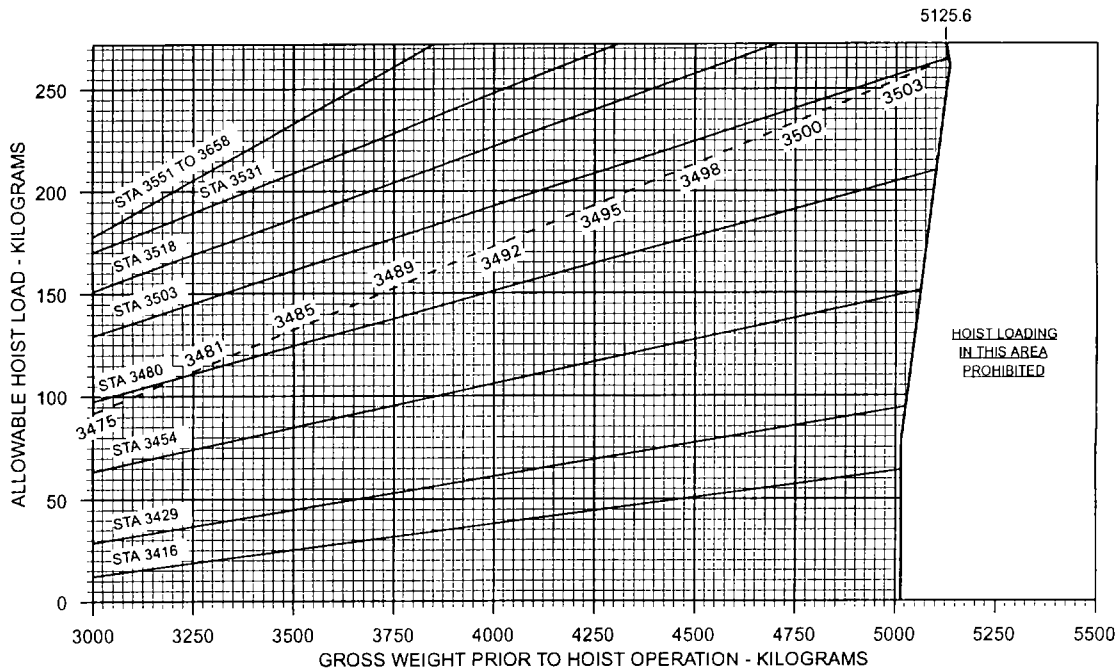
NOTE: THE DASHED LINE INDICATES A DIFFERENT LONGITUDINAL C.G. FOR EACH GROSS WEIGHT.

412FS-7-1-6-3

Figure 1-5. Hoist loading schedules 5397 kg. GW (Metric) (Sheet 3 of 6)

R/H FWD POSITION - PILOT, COPILOT AND HOIST OPERATOR

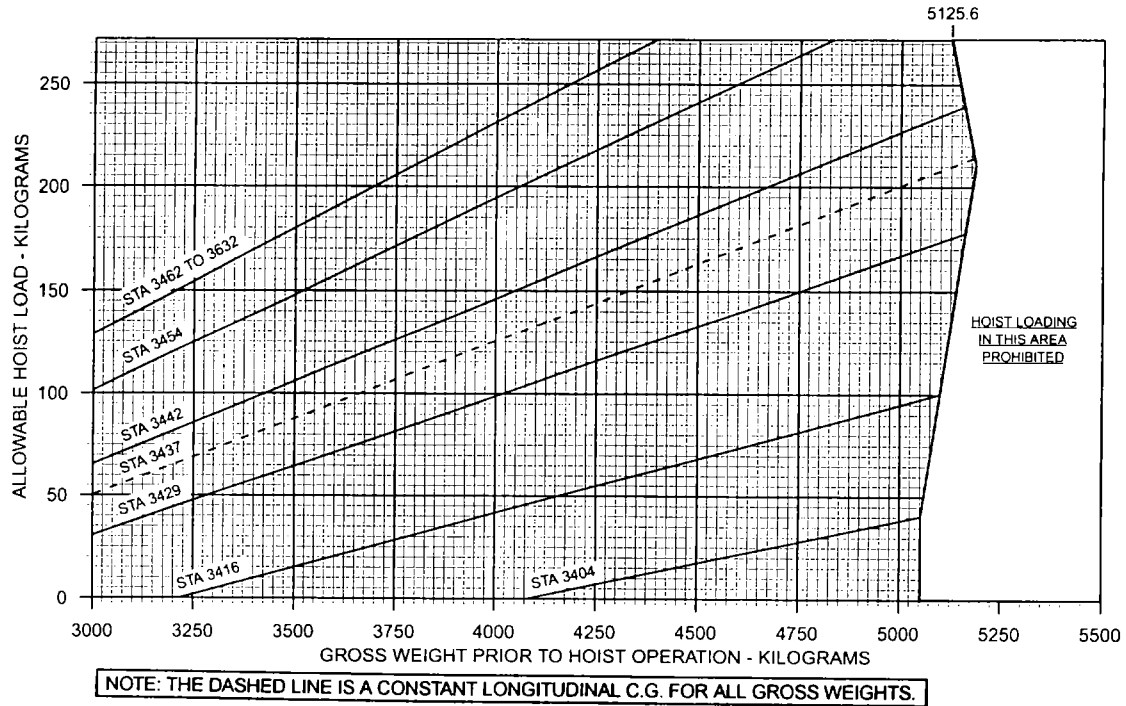
Figure 1-5. Hoist loading schedules 5397 kg. GW (Metric) (Sheet 4 of 6)



NOTE: THE DASHED LINE INDICATES A DIFFERENT LONGITUDINAL C.G. FOR EACH GROSS WEIGHT.

412FS-7-1-5-4

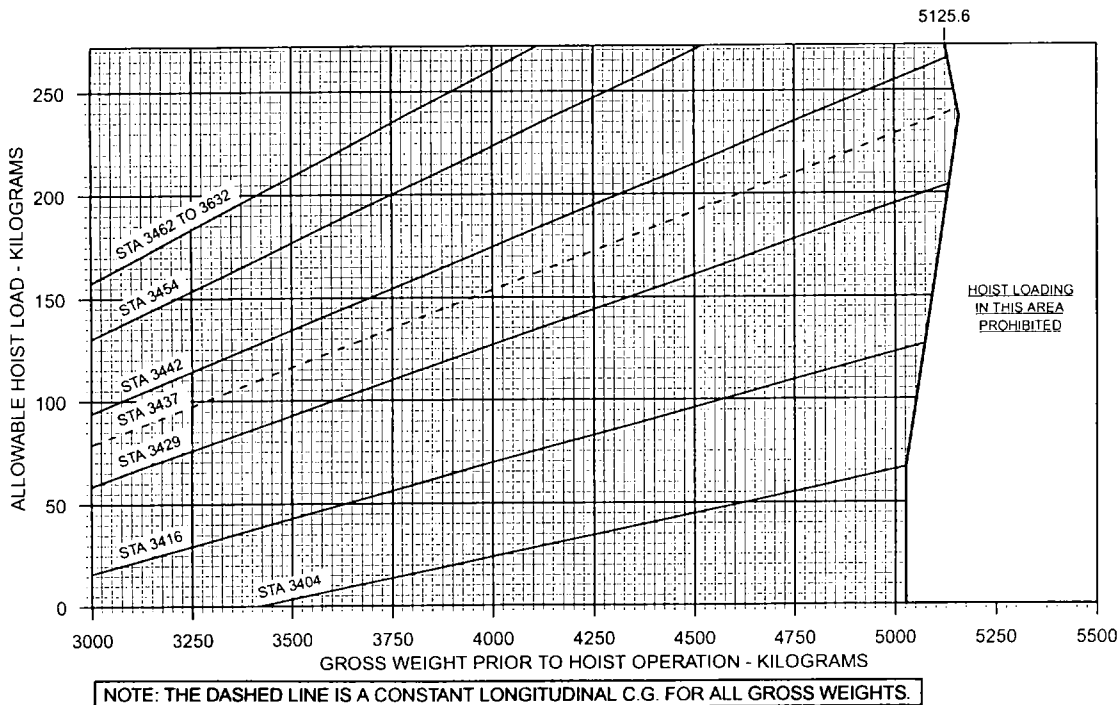
R/H AFT POSITION - PILOT AND HOIST OPERATOR



412FS-7-1-5-6

Figure 1-5. Hoist loading schedules 5397 kg. GW (Metric) (Sheet 5 of 6)

R/H AFT POSITION - PILOT, COPILOT AND HOIST OPERATOR



412FS-7-1-6-6

Figure 1-5. Hoist loading schedules 5397 kg. GW (Metric) (Sheet 6 of 6)

Section 2

NORMAL PROCEDURES

2-2. FLIGHT PLANNING

WARNING

HOIST LOAD CAN CAUSE LONGITUDINAL OR LATERAL CG LIMITS TO BE EXCEEDED. GROSS WEIGHT AND CENTER OF GRAVITY SHALL BE COMPUTED TO ASSURE LOADING WITHIN APPROVED LIMITS.

CAUTION

IF ADDITIONAL LOADS ARE CARRIED DURING HOISTING OPERATIONS, LOADS SHOULD BE PLACED ON SIDE OF HELICOPTER OPPOSITE HOIST POSITION.

Gross weight and CG — Compute with and without hoist load.

2-4. INTERIOR AND PRESTART CHECK

2-4-A. HOIST INSTALLATION CHECK

NOTE

If pilot plans to operate hoist with other crewmember in passenger compartment, hoist shall be installed in forward right position.

Hoist — Installed in desired position; check roof and floor stud adapters and locking collars properly secured.

Boom actuator — Installed in proper position; all fittings secured.

AIRCRAFT POSITION switch (on hoist control box, figure 2-1) — Set in proper position.

Hook — Rotates freely on cable.

Cable — Check proper routing through guide rollers, pulleys, and drums.

Gearbox oil levels — Check sight glasses.

Hoist operators pendant — Installed; connectors secured.

Electrical power cables — Condition; connectors secured.

WARNING

ACTUATION OF CABLE CUT SWITCH ON PEDESTAL CAN CUT CABLE REGARDLESS OF HOIST PWR SWITCH POSITION. ACTUATION OF CABLE CUT SWITCH ON HOIST CONTROL BOX CAN CUT CABLE, EVEN IF CABLE CUT CIRCUIT BREAKER IS OUT.

CABLE CUT switches (pedestal and hoist) — Off; covers safetied.

Safety vests, tether straps, hoisting slings, and litters — Condition; secured or stored.

2-4-B. HOIST OPERATION CHECK

Cargo doors and hinged panels — Secured open or removed.

HOIST PWR. CONT and CABLE CUT circuit breakers — In.

BATTERY switches — ON (or connect external power).

NON ESNTL BUS switch — MANUAL.

ICS — Check intercom between pilot and hoist operator using hoist pendant ICS trigger and HOT MIC switch (right ICS box only).

HOIST PWR switch — ON; check that green (power on) light and amber 20 FOOT CAUTION lights on hoist control box illuminate.

Hoist OVERTEMP warning lights — Press to test.



MAINTAIN TENSION ON HOIST CABLE WHILE REELING IN AND OUT TO PREVENT SLACK.

HOIST and BOOM switches (pilot and operator) — Actuate to check all hoist functions for proper operation. Check that pilot HOIST switch overrides operator pendant HOIST switch.

Hoist cable — Check for corrosion, kinks, flat spots, fraying, or broken strands.

Up limit switch actuator - Raise while hoist is reeling in and check hoist motor stops; then release and check hoist resumes operation.

Reduce hoist speed as cable approaches up limit. Check that hoist stops when hook reaches up limit without excess tension on cable.

Hoist - Stowed for flight; hook restraint secured.

HOIST PWR switch - OFF.

NON ESNTL BUS switch - NORMAL.

BATTERY switches - OFF.

NOTE

Ground crewmember should be instructed to discharge helicopter static electricity before attaching load to hoist when possible.

2-6. SYSTEMS CHECK

Cargo doors and hinged panels - Secured open or removed.

CABLE CUT switches (pedestal and hoist) - Off; covers safetied.

HOIST PWR, CONT, and CABLE CUT circuit breakers - In.

2-6-A. BEFORE TAKEOFF

Safety vests and straps - On and secured to helicopter.

Gloves - On.

STEP switch (if installed) - STOW.

2-9. IN-FLIGHT OPERATIONS

Maximum hoist load shall be determined prior to each hoist operation.

NOTE

The High-Velocity Diagram is not a limitation for internal hoist operations under an appropriate operating certificate.

HOIST PWR switch - ON.

WARNING

HOIST OPERATOR SHALL BE SECURED TO HELICOPTER WITH AN APPROVED SAFETY HARNESS DURING HOIST OPERATIONS.

Establish hover over hoist operation area.

Hoist hook restraint - Removed.

BOOM switch (or pilots HOIST switch) - OUT.

NOTE

Each hoist operation performed is defined as reeling hoist cable out and then in while hovering with any weight on hoist, regardless of whether the hoist was used for training or an actual rescue.

The pilot must record each operation in the penalty CG region. For each hoist operation performed within penalty CG region, four (4) additional hours of usage must be logged against the main rotor yoke, mast and lower cone seat.

HOIST switch - DOWN.

Discharge static electricity when possible, and connect hook to load, observing allowable hoist load.

NOTE

As hook nears the up or down limits, hoist speed automatically slows.

HOIST switch - UP.

CAUTION

USE CARE TO PREVENT CABLE, HOOK, AND LOAD FROM

FOULING ON FUSELAGE OR LANDING GEAR.

Maintain zero ground speed until load is clear of obstructions.

BOOM switch - IN to swing hoist boom and load into cabin, if possible.

Takeoff into wind, if possible, allowing adequate hoist load clearance over obstacles if load is not internal.

CAUTION

AIRSPPEED WITH EXTERNAL LOAD IS LIMITED BY CONTROLLABILITY. CAUTION SHOULD BE EXERCISED WHEN CARRYING AN EXTERNAL LOAD. HANDLING CHARACTERISTICS MAY BE AFFECTED BY THE SIZE, WEIGHT, AND SHAPE OF LOAD.

Airspeed - As required for adequate controllability, not to exceed limits for hoist operations (20 or 60 KIAS, as applicable).

2-13. LITTER HOISTING

When emergency transportation of a patient by litter is essential, every effort should be made to land the helicopter for litter loading. Litter hoisting can be hazardous and should be accomplished only when a landing is not feasible and the condition of the patient precludes the use of the personnel hoisting sling.

In addition to all other procedures contained herein, the following shall apply to litter hoisting operations.

2-13-A. EMPTY LITTER**WARNING**

HOISTING OR LOWERING AN EMPTY LITTER IN OPEN POSITION IS PROHIBITED. AN EMPTY LITTER CAN OSCILLATE UNCONTROLLABLY IN ROTOR WASH AND FLY UPWARD, STRIKING FUSELAGE OR TAIL ROTOR.

Prior to hoisting or lowering an empty litter, litter shall be closed and secured with straps. Litter should be suspended in a near-vertical position and sling straps should be drawn tight.

2-13-B. LOADED LITTER**WARNING**

LITTER PATIENT SHALL BE SECURED TO LITTER WITH

SAFETY STRAPS.

HOIST HOOK CATCH SHALL BE SECURED WITH SAFETY PIN PRIOR TO HOISTING LITTER PATIENT.

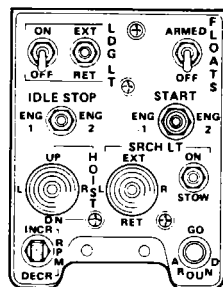
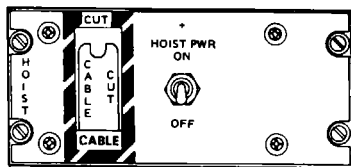
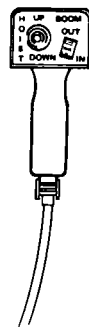
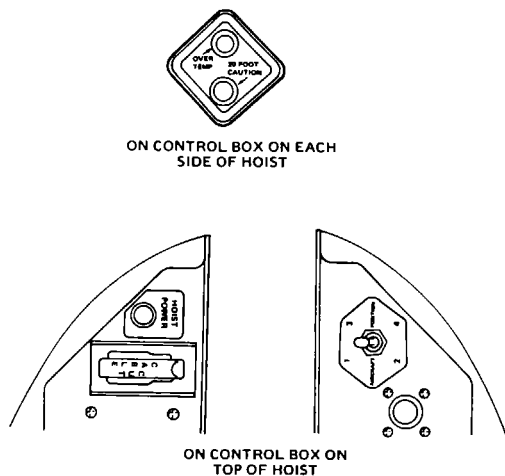
Litter sling straps should be adjusted so that litter is 24 to 28 inches (61 to 71 centimeters) below hoist hook.

NOTE

If litter is suspended too far below hook, litter cannot be loaded in helicopter with hoist hook at up limit.

CAUTION

A LOADED LITTER CAN ROTATE ABOUT CABLE DURING HOISTING. HOIST OPERATOR MAY HAVE TO GRASP LITTER SLING STRAPS TO CONTROL ROTATION AS LITTER APPROACHES LANDING GEAR.



412FS-7-2-1

Figure 2-1. Internal hoist controls

Section 3

EMERGENCY/MALFUNCTION PROCEDURES

3-15. HOIST LOAD JETTISON

To jettison hoist load in an emergency, actuate CABLE CUT switch (located on pedestal or hoist).

In the event of failure of CABLE CUT switch, sever cable with manual cable cutter (stowed in pouch on hoist).

3-15-A. HOIST OVERTEMP WARNING LIGHT

In the event that the OVERTEMP warning light (located on hoist control box) illuminates, continue present operation until hoist is reeled in. Leave HOIST PWR switch ON (for cooling fan operation) and allow hoist to cool. When OVERTEMP light extinguishes, hoisting may be resumed as desired.

Section 4

PERFORMANCE

No change from basic manual.

Section 5

WEIGHT AND BALANCE

5-11. WEIGHT EMPTY CHART

The Weight empty chart for internal hoisting operations is shown in figure 5-1. Refer to the maintenance manual for additional information.

NOTE

Allowable hoist load must be computed when weight empty is not within specified guidelines.

NOTE

Allowable hoist loads must be computed when AUX Fuel kits are installed.

412 Weight Empty Chart for Internal Hoist Operations

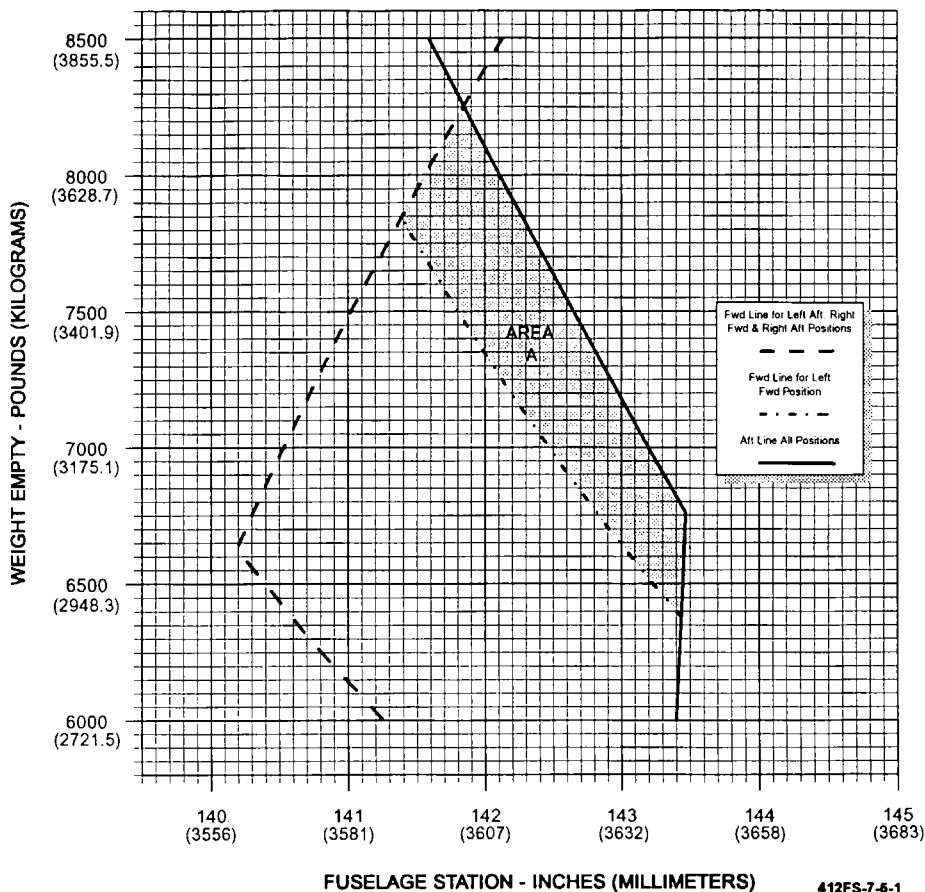


Figure 5-1. Weight empty chart

Bell **MODELS 412/412EP**

ROTORCRAFT FLIGHT MANUAL

SUPPLEMENT FOR LITTER KIT OPERATIONS

(412-706-006)

CERTIFIED
SEPTEMBER 29, 1981

This supplement shall be attached to the Models 412 or 412EP Flight Manual when the 412-706-006 Litter Kit has been installed.

The information contained herein supplements the information of the basic Flight Manual. For Limitations, Procedures, and Performance Data not contained in this supplement, consult the basic Flight Manual.

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REISSUE — 5 OCTOBER 1994

BHT-412-FMS-8

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FLIGHT MANUAL

MANUFACTURER'S DATA

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APPROVED:



MANAGER

ROTORCRAFT CERTIFICATION OFFICE
FEDERAL AVIATION ADMINISTRATION
FT. WORTH, TEXAS 76193-0170

NOTE: Revised text is indicated by a black vertical line.
Insert latest revision pages; dispose of superseded pages.

INTRODUCTION

The Litter Kit provides three litters and the provisions for installing up to three litters in the helicopter. A cabin attendant seat is also included in the kit.

Section 1

LIMITATIONS

WEIGHT/CG LIMITATIONS

Actual weight change shall be determined after the litter(s) are installed and ballast readjusted if necessary, to return empty weight CG within allowable limits.

MINIMUM FLIGHT CREW

The minimum flight crew for litter operations shall consist of a pilot and a second crewmember or cabin attendant, both of whom shall be trained in and capable of assisting in litter patient emergency evacuation procedures.

Section 2

NORMAL PROCEDURES

LITTER LOADING

Secure patients to litters, then load litters aboard the helicopter in sequence from top to bottom. When only two patients are carried, they should occupy the top and center litter positions. When only one patient is carried, the center litter should be used.

opened. Refer to Section 3, Emergency Procedures, for unloading procedures when cabin door cannot be opened.

Open cabin door and unload litters and patients from the helicopter in sequence from bottom to top.

LITTER UNLOADING

NOTE

Normal unloading procedures apply when either passenger door can be

Litters to be handled by one person inside cabin and one person outside cabin.

BHT-412-FMS-8

Section 3

EMERGENCY AND MALFUNCTION PROCEDURES

UNLOADING THROUGH EMERGENCY EXITS

NOTE

In the event that cabin doors cannot be opened, litter patients shall be unloaded through emergency pop-out windows. After all litter patients have been removed, ambulatory patients may then exit.

Remove emergency pop-out window by pushing at corners as marked.

Unstrap patient on center litter and remove patient through window opening.

Disconnect top litter at end near open window and lower end to rest on center litter. Remove patient retention straps and slide patient down litter and out through window opening.

Raise top and center litter ends near open window and engage center litter in brackets for top litter. Disconnect bottom litter. Raise bottom litter at end near open window and rest handles on the lower surface of the window opening. Unstrap patient and slide patient up litter and through window opening.

Section 4

PERFORMANCE

No change from basic Flight Manual.

MANUFACTURER'S DATA

BHT-412-FMS-8

Section 1

MANUFACTURER'S DATA

WEIGHT AND BALANCE

TABLE OF MOMENTS (IN-LB)		TABLE OF MOMENTS <u>(kg • mm)</u> 100	
Weight (Pounds)	LITTER PATIENT	Weight (K.G.)	LITTER PATIENT
	Loaded Laterally F.S. 117		Loaded Laterally 2972 mm
100	11700	50	1486.0
110	12870	55	1634.6
120	14040	60	1783.2
130	15210	65	1931.8
140	16380	70	2080.4
150	17550	75	2229.0
160	18720	77.1	2291.4
170	19890	80	2377.6
180	21060	85	2526.2
190	22230	90	2674.8
200	23400	95	2823.4
210	24570	100	2972.0
220	25740	105	3120.6
		110	3269.2

Bell **412**
MODEL

ROTORCRAFT FLIGHT MANUAL

33108 — 33213
36001 — 36019

SUPPLEMENT FOR EXTERNAL CARGO OPERATION (212-706-103)

CERTIFIED
MAY 14, 1981

This supplement shall be attached to the Model 412 Flight Manual (BHT-412-FM-2) when the 212-706-103 External Cargo Suspension Hook has been installed.

The information contained herein supplements the information of the basic Flight Manual. For Limitations, Procedures, and Performance Data not contained in this supplement, consult the basic Flight Manual.

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FLIGHT MANUAL

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NOTE

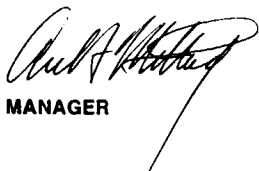
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Reissue 0 ... September 15, 1995

APPROVED:



MANAGER

ROTORCRAFT CERTIFICATION OFFICE
FEDERAL AVIATION ADMINISTRATION
FT. WORTH, TX 76193-0170

INTRODUCTION

The External Cargo Suspension Hook, when installed, will permit the operator to utilize the helicopter for transportation of external cargo, when operated by a qualified pilot.

Section 1

LIMITATIONS

TYPE OF OPERATION

Operation of the helicopter with no load on the external cargo suspension hook is authorized under the standard airworthiness certificate under VFR or IFR conditions without removing the unit from the helicopter.

The installation and use of the rear view mirror contained in the kit is left to the operators discretion.

The rear view mirror shall be covered or removed for night flight.

VFR OPERATION

With a load attached to the suspension assembly, operation shall be conducted in accordance with appropriate operating rules for external loads under VFR conditions.

IFR OPERATION

External load operations are permitted provided the operator substantiates to the Administrator that the rotorcraft — load combination meets IFR handling requirements and insures that the Rotorcraft External Load Operator Certificate reflects same with appropriate restrictions.

WEIGHT — CG LIMITATIONS

Actual weight change shall be determined after cargo hook is installed and ballast readjusted, if necessary, to retain empty weight CG within allowable limits.

Maximum gross weight including external cargo load is 11,900 pounds (5398 kilograms). Maximum external cargo load is 4500 pounds (2041 kilograms).

AIRSPEED LIMITATIONS

VNE is 80 KIAS at or below 10,000 feet density altitude for all gross weights with external cargo on suspension unit. Above 10,000 feet decrease VNE 2.5 knots per 1000 feet.

CAUTION

THE AIRSPEED WITH EXTERNAL CARGO IS LIMITED BY CONTROLLABILITY. CAUTION SHOULD BE EXERCISED WHEN CARRYING EXTERNAL CARGO, AS THE HANDLING CHARACTERISTICS MAY BE AFFECTED BY THE SIZE, WEIGHT, AND SHAPE OF THE CARGO LOAD.

Light weight, high drag loads require a swivel connector between the cargo hook and the sling to prevent unstable oscillations in flight above 20 KIAS.

PLACARDS AND MARKINGS

OCCUPANCY LIMITED
TO CREW WITH
EXTERNAL LOAD
CLASS B LOADING
APPROVED

(Located on forward right side of overhead console)

DO NOT OPERATE
HEATER ABOVE 21
DEG C OUT AIR TEMP

FLOATS CARGO RELEASE

ARMED ARMED

OPERATIONAL LIMITS

(Located on upper center part of instrument panel)

EXTERNAL LOAD LIMIT
4500 LBS
SEE FLIGHT MANUAL FOR
LOADING INSTRUCTIONS
AND FLIGHT LIMITATIONS

(Located on under side of helicopter near suspension assembly)

412-FS9.2-1-1

Figure 1-1. Placards and markings

Section 2

NORMAL PROCEDURES

GROUND CREW INSTRUCTIONS

Instruct ground crewmember to discharge helicopter static electricity before attaching cargo by touching the airframe with a ground wire, or if a metal sling is used, the hookup ring can be struck against the cargo hook. If contact has been lost after initial grounding, the helicopter should be electrically regrounded and, if possible, contact maintained until hookup is completed.

Instruct ground personnel to check primary and secondary load rings for condition and proper size

(Table 2-1). Check for proper rigging and configuration (Figure 2-1).

WARNING

USE OF INAPPROPRIATELY SIZED LOAD RINGS MAY RESULT IN LOAD HANG-UP WHEN LOAD RING IS TOO SMALL OR INADVERTENT LOAD RELEASE IF LOAD RING IS TOO LARGE.

Check that only one primary ring is captured in load beam and only one secondary ring with correct cross-section dimension is captured in primary ring.

Table 2-1. Ring Size — Cargo Hook P/N SP1709-62

PRIMARY RING INSIDE DIAMETER	PRIMARY RING CROSS-SECTION	MAXIMUM CROSS-SECTION OF SECONDARY MEMBER
3.0 to 3.1 in. (76.2 to 78.74 mm)	1.0 in. (25.4 mm)	0.625 in. (15.9 mm)
3.1 to 4.0 in. (78.74 to 101.6 mm)	1.0 in. (25.4 mm)	0.750 in. (19.0 mm)

EXTERIOR CHECK

Cargo suspension assembly — Condition and security.

Rear view mirror (if installed) — Secure and clean.

INTERIOR CHECK

CARGO HOOK REL circuit breaker — In.

Battery BUS 1 switch — ON.

CARGO REL switch (overhead) — ARM; check CARGO RELEASE ARMED caution light illuminates.

Cyclic CARGO RELEASE button — Depress and hold; pull down on cargo hook; hook should open. Release button and cargo hook; hook should close and lock.

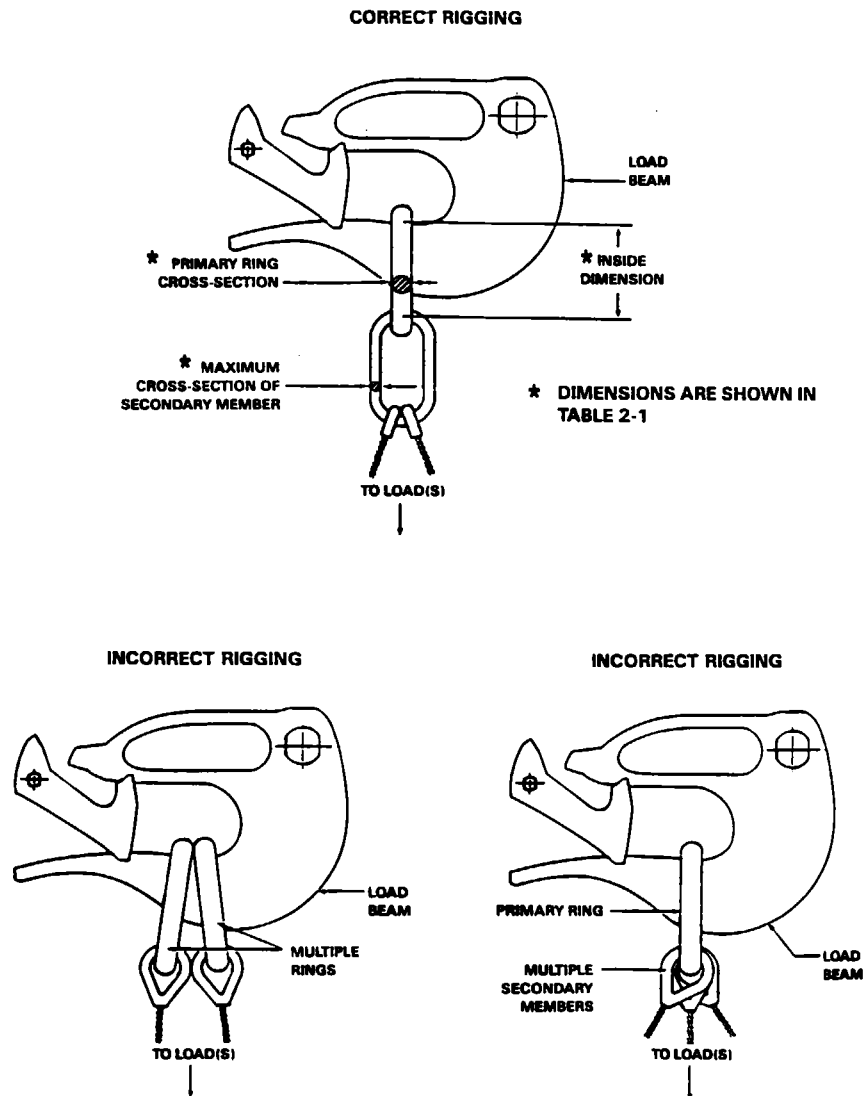
Cargo release pedal (between directional control pedal(s)) — PUSH and hold; pull down on cargo hook; hook should open. Release pedal and cargo hook; hook should close and lock.

NOTE

The pedal release will function regardless of CARGO REL switch position.

CARGO REL switch — OFF.

Battery BUS 1 switch — OFF.



412-FS9.2-2-1

Figure 2-1. Effective loading practices

BEFORE TAKEOFF

CARGO REL switch — ARM; check CARGO RELEASE ARMED caution light illuminates.

TAKEOFF**NOTE**

Better directional control may be realized by avoiding relative winds from the right front quadrant while performing external cargo operations.

Hover helicopter at sufficient height to allow crewmember to discharge static electricity and to attach cargo sling to cargo hook.

NOTE

Attachment of cargo sling to the hook can be observed by means of the rear view mirror.

Ascend vertically directly over cargo, then slowly lift cargo from surface.

Pedals — Check for adequate directional control.

Hover power — Check torque required to hover with external load.

NOTE

The Height-Velocity Diagram is not a limitation for external cargo operations under an appropriate operating certificate.

Take off into the wind if possible, allowing adequate sling load clearance over obstacles.

IN-FLIGHT OPERATION**NOTE**

Control movements should be made smoothly and kept to a minimum to prevent oscillation of sling load.

CARGO REL switch (overhead) — As desired.

NOTE

The pedal release will function regardless of CARGO REL switch position.

Airspeed — Within limits for adequate controllability of rotorcraft — load combination.

Flight path — As required to avoid flight with external load over any person, vehicle or structure.

DESCENT AND LANDING

CARGO REL switch (overhead) — ARM prior to final approach.

Flight path and approach angle — As required for wind direction and obstacle clearance.

Execute approach to a hover with cargo clear of the surface. When stabilized at a hover, descend slowly until cargo contacts surface. Maintain tension on sling.

Cyclic CARGO RELEASE button — Depress to release sling from hook.

NOTE

Release of sling load from the hook can be confirmed visually through rear view mirror.

Section 3

EMERGENCY AND MALFUNCTION PROCEDURES

CARGO FAILS TO RELEASE ELECTRICALLY

In the event that cargo hook will not release the sling when the CARGO RELEASE button is depressed, proceed as follows:

Maintain tension on sling.

Cargo release pedal (between directional control pedals) — PUSH.

Section 4

PERFORMANCE

INTRODUCTION

No change from basic Flight Manual performance with no load attached to cargo hook.

NOTE

Performance may be affected by the size and shape of the external load.

HOVER CEILING IGE

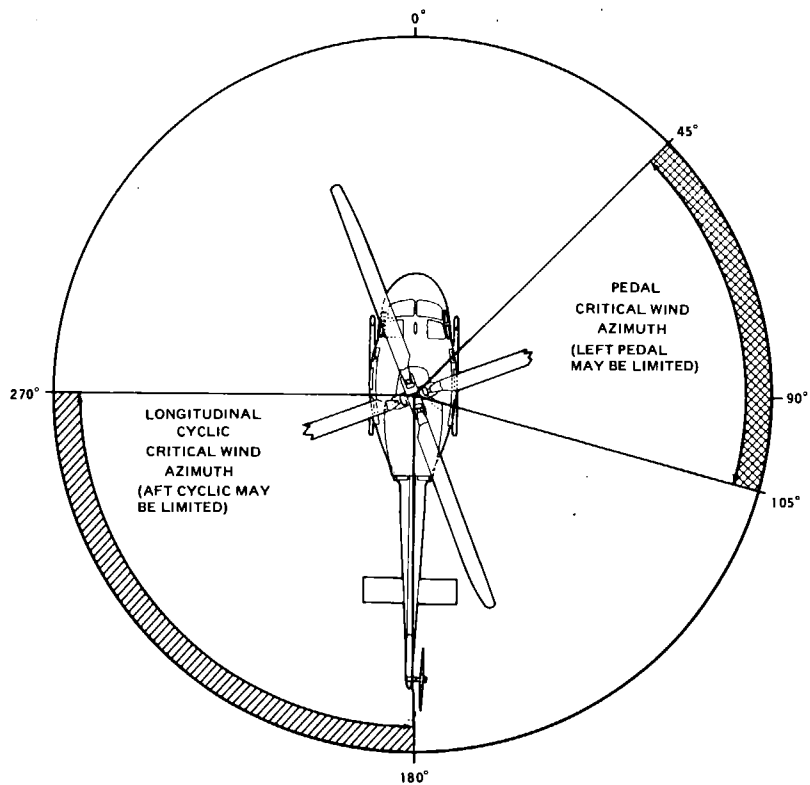
NOTE

When using any hover ceiling chart for external cargo operations, refer to figure 4-1, critical relative wind azimuths.

In ground effect hover ceiling charts for external cargo operations are presented in figure 4-2.

HOVER CEILING OGE

Refer to basic Flight Manual for out of ground effect hover ceiling charts during external cargo operations.



412-FS9.2-4-1

Figure 4-1. Critical relative wind azimuths

HOVER CEILING IN GROUND EFFECT

TAKEOFF POWER
ENGINE RPM 100%
GENERATOR 150 AMPS (EA.)

SKID HEIGHT 4 FEET
HEATER ON OR OFF
-40° TO 52°C

NOTE: THIS DATA VALID FOR ZERO WIND OR WIND OUT SIDE OF THE CRITICAL WIND AZIMUTH AREAS

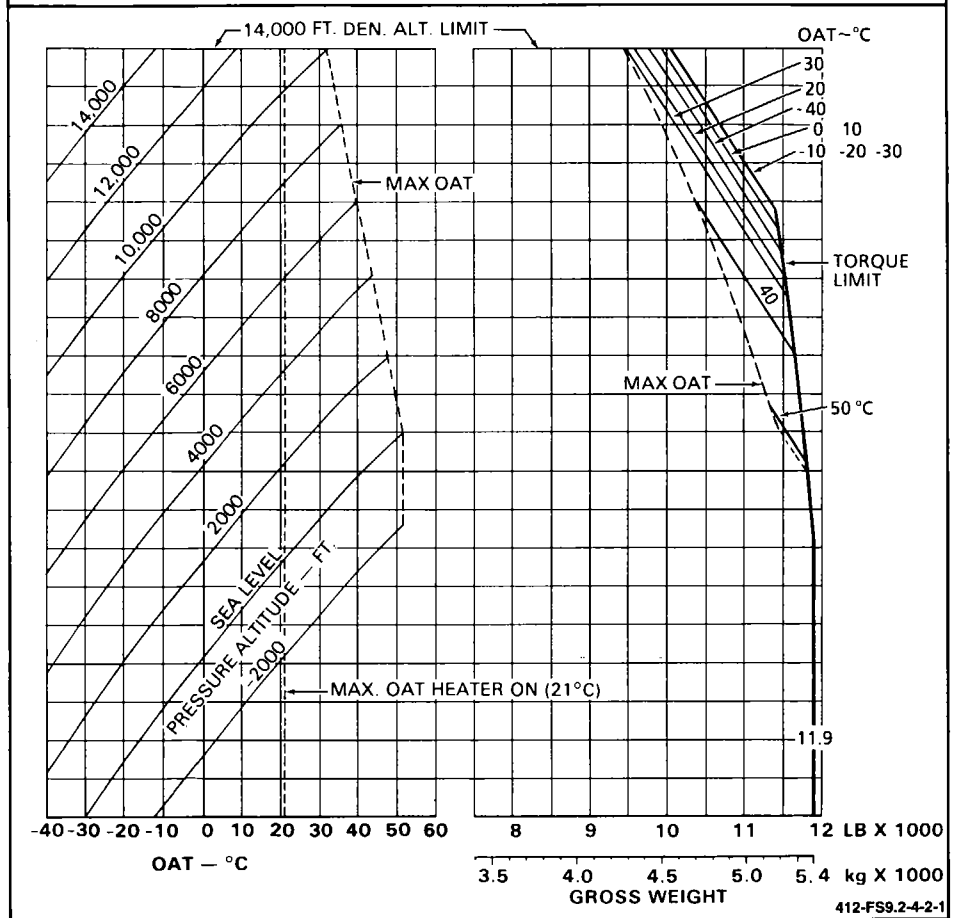


Figure 4-2. Hover ceiling in ground effect (Sheet 1 of 2)

HOVER CEILING IN GROUND EFFECT

MAXIMUM CONTINUOUS POWER
ENGINE RPM 100%
GENERATOR 150 AMPS (EA.)

SKID HEIGHT 4 FEET
HEATER ON OR OFF
-40° TO 52°C

NOTE: THIS DATA VALID FOR ZERO WIND OR WIND OUT SIDE OF THE CRITICAL WIND AZIMUTH AREAS

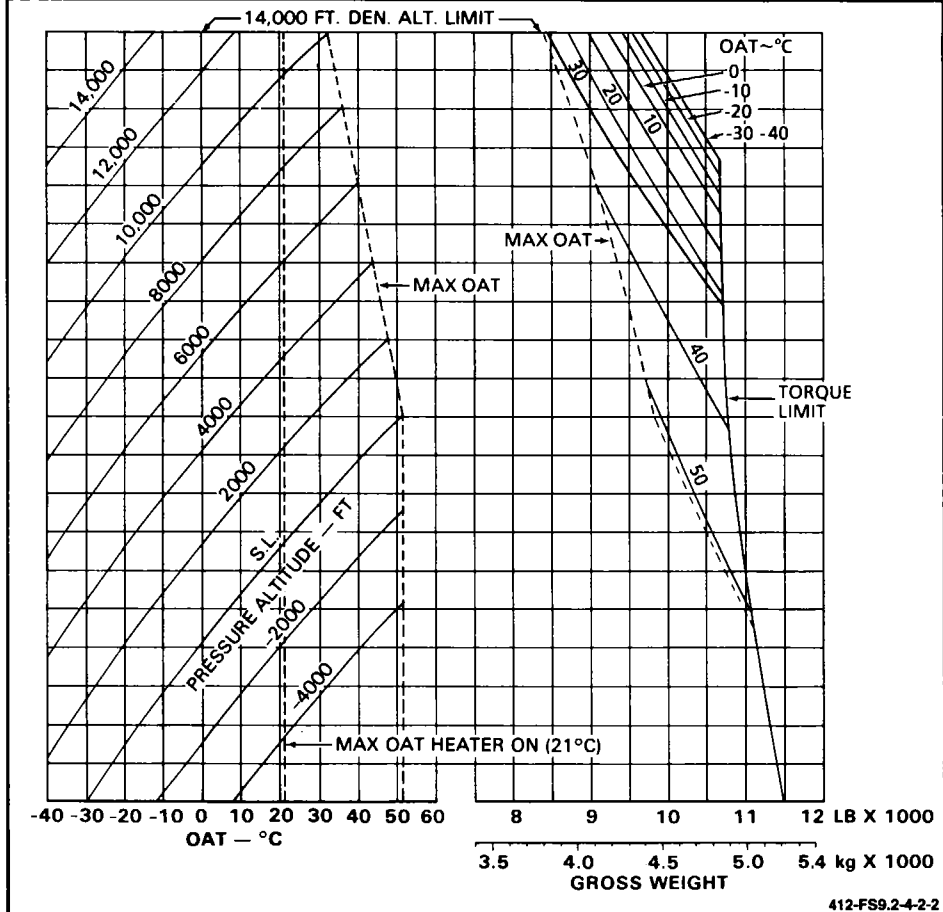


Figure 4-2. Hover ceiling in ground effect (Sheet 2 of 2)

Section 1

MANUFACTURER'S DATA

WEIGHT AND BALANCE

EXTERNAL CARGO LOADING

The External Cargo Loading Tables (tables 1-1 and 1-1M) present moments for external loads suspended from the cargo hook at fuselage station 138.0 (3505 mm).

Table 1-1. External cargo loading table (English)

EXTERNAL CARGO LOADING TABLE — ENGLISH					
Cargo Weight (Lb)	Moment F.S. 138.0	Cargo Weight (Lb)	Moment F.S. 138.0	Cargo Weight (Lb)	Moment F.S. 138.0
50	6900	1800	248400	3550	489900
100	13800	1850	255300	3600	496800
150	20700	1900	262200	3650	503700
200	27600	1950	269100	3700	510600
250	34500	2000	276000	3750	517500
300	41400	2050	282900	3800	524400
350	48300	2100	289800	3850	531300
400	55200	2150	296700	3900	538200
450	62100	2200	303600	3950	545100
500	69000	2250	310500	4000	552000
550	75900	2300	317400	4050	558900
600	82800	2350	324300	4100	565800
650	89700	2400	331200	4150	572700
700	96600	2450	338100	4200	579600
750	103500	2500	345000	4250	586500
800	110400	2550	351900	4300	593400
850	117300	2600	358800	4350	600300
900	124200	2650	365700	4400	607200
950	131100	2700	372600	4450	614100
1000	138000	2750	379500	4500	621000
1050	144900	2800	386400	4550	627900
1100	151800	2850	393300	4600	634800
1150	158700	2900	400200	4650	641700
1200	165600	2950	407100	4700	648600
1250	172500	3000	414000	4750	655500
1300	179400	3050	420900	4800	662400
1350	186300	3100	427800	4850	669300
1400	193200	3150	434700	4900	676200
1450	200100	3200	441600	4950	683100
1500	207000	3250	448500	5000	690000
1550	213900	3300	455400		
1600	220800	3350	462300		
1650	227700	3400	469200		
1700	234600	3450	476100		
1750	241500	3500	483000		

412900-43

Table 1-1M. External cargo loading table (Metric)

EXTERNAL CARGO LOADING TABLE — METRIC			
Cargo Weight (kg)	Moment 3505 mm (kg — mm) 100	Cargo Weight (kg)	Moment 3505 mm (kg — mm) 100
40	1402	1240	43462
80	2804	1280	44864
120	4206	1320	46266
160	5608	1360	47668
200	7010	1400	49070
240	8412	1440	50472
280	9814	1480	51874
320	11216	1520	53276
360	12618	1560	54678
400	14020	1600	56080
440	15422	1640	57482
480	16824	1680	58884
520	18226	1720	60286
560	19628	1760	61688
600	21030	1800	63090
640	22432	1840	64492
680	23834	1880	65894
720	25236	1920	67296
760	26638	1960	68698
800	28040	2000	70100
840	29442	2040	71502
880	30844	2080	72904
920	32246	2120	74306
960	33648	2160	75708
1000	35050	2200	77110
1040	36452	2240	78512
1080	37854	2268	79493
1120	39256		
1160	40658		
1200	42060		

412900-44

Bell *Model 412*

ROTORCRAFT FLIGHT MANUAL

SUPPLEMENT FOR
NIGHTSUN SEARCHLIGHT
(212-899-333)

CERTIFIED
DECEMBER 4, 1981

This supplement shall be attached to the Model 412 Flight Manual when the 212-899-333 Nightsun Searchlight has been installed.

The information contained herein supplements the information of the basic Flight Manual. For Limitations, Procedures, and Performance Data not contained in this supplement, consult the basic Flight Manual.

Bell Helicopter **TEXTRON**

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4 DECEMBER 1981

REISSUED 8 MAY 1989

BHT-412-FMS-12

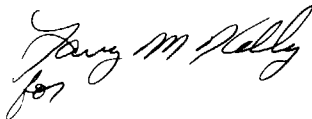
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MANAGER

ROTORCRAFT CERTIFICATION DIRECTORATE
AIRCRAFT CERTIFICATION SERVICE
DEPARTMENT OF TRANSPORTATION
SOUTHWEST REGION, FORT WORTH, TEXAS

NOTE: Revised text is indicated by a black vertical line.
Insert latest revision pages; dispose of superseded pages.

INTRODUCTION

The Nightsun Searchlight is a high intensity light which mounts on the lower nose of the helicopter. The xenon arc light may be started, aimed, and focused from the operator's panel inside the helicopter. The kit consists of the Nightsun Searchlight, mounts, hardware, cable, and operator's panel.

Section 1

LIMITATIONS

OPERATING LIMITATIONS

IFR operation is prohibited with Nightsun Searchlight installed.

FLIGHT CREW LIMITATIONS

Operation of the Nightsun Searchlight is restricted to the copilot or operator position.

WEIGHT/CG LIMITATIONS

Actual weight changes shall be determined after searchlight is installed and ballast readjusted, if

necessary, to return empty weight CG within allowable limits.

PLACARDS AND DECALS

CAUTION

**DO NOT USE NIGHTSUN SEARCHLIGHT
BELOW 50 FT AGL OR IN FOG CONDITIONS.
MONITOR LOADMETER WHEN USING
NIGHTSUN SEARCHLIGHT.**

412099-5

Section 2

NORMAL PROCEDURES

EXTERIOR CHECK

Nightsun Searchlight — Security and wiring.
Lens for cleanliness.

Aim and focus — As desired.

CAUTION

PRESTART CHECK

SCHLT PWR and SCHLT CONT circuit
breakers — IN.

HOLDING SWITCH IN START
POSITION AFTER IGNITION MAY
DAMAGE EQUIPMENT.

CAUTION

INFLIGHT OPERATION

NIGHTSUN SEARCHLIGHT MASTER switch
— ON.

DO NOT AIM THE BEAM TOWARD
OTHER AIRCRAFT OR VEHICLES
BECAUSE OF TEMPORARY BLIND-
ING EFFECT.

NIGHTSUN SEARCHLIGHT START switch —
START, hold in start position approximately 5
seconds, or until ignition has occurred.

BEFORE LANDING

NIGHTSUN SEARCHLIGHT MASTER
switch — OFF.

Section 3

EMERGENCY AND MALFUNCTION PROCEDURES

No change from basic Flight Manual.

Section 4

PERFORMANCE DATA

No change from the basic Flight Manual.

Bell
MODELS **412/412EP**

ROTORCRAFT FLIGHT MANUAL

SUPPLEMENT FOR FIXED STEP (212-706-057)

CERTIFIED
6 FEBRUARY 1982

This supplement shall be attached to the Models 412 and 412 EP Flight Manual when the 412-706-057 fixed step has been installed.

The information contained herein supplements the information of the basic Flight Manual. For Limitations, Procedures, and Performance Data not contained in this supplement, consult the basic Flight Manual.

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REISSUE — 23 JUNE 1994

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BHT-412-FMS-15

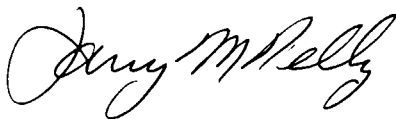
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APPROVED:



MANAGER

ROTORCRAFT CERTIFICATION OFFICE
FEDERAL AVIATION ADMINISTRATION
FT. WORTH, TX 76193-0170

NOTE: Revised text is indicated by a black vertical line.
Insert latest revision pages; dispose of superseded pages.

INTRODUCTION

The fixed steps mount to the sides of the fuselage to facilitate passenger entry and exit.

Section 1

LIMITATIONS

OPERATING LIMITATIONS

The contents of this supplement shall be used in conjunction with the basic Flight Manual for helicopters equipped with the fixed step.

The 412-706-004 Emergency Float Kit shall not be installed in conjunction with the fixed step.

The 212-706-105 Passenger Step shall not be installed in conjunction with the fixed step.

The 212-706-057 Fixed Step shall be removed when the 214-706-003 Internal Hoist is installed.

WEIGHT — CG LIMITATIONS

Actual weight change shall be determined after kit is installed and ballast readjusted, if necessary, to return empty weight CG within allowable limits.

BHT-412-FMS-15

Section 2

NORMAL PROCEDURES

No change from basic Flight Manual.

Section 3

EMERGENCY AND MALFUNCTION PROCEDURES

No change from basic Flight Manual.

Section 4

PERFORMANCE

No change from basic Flight Manual.



ROTORCRAFT FLIGHT MANUAL

SUPPLEMENT AUXILIARY FUEL SUPPLEMENT 412-706-007

33108 — 33213
36001 — 36019
AND
36020 — 36086
AND
36087 AND SUB

CERTIFIED
5 JANUARY 1984

This supplement shall be attached to Bell Helicopter Model 412 and 412EP Flight Manuals when 412-706-007 Auxiliary Fuel Kit has been installed.

Information contained herein supplements information of basic Flight Manual. For Limitations, Procedures, and Performance Data not contained in this supplement, consult basic Flight Manual.

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MANAGER



ROTORCRAFT CERTIFICATION OFFICE
FEDERAL AVIATION ADMINISTRATION
FT. WORTH, TX 76193-0170

BHT-412-FMS-17.2, 17.3 AND 17.4

INTRODUCTION

The Auxiliary Fuel Kit provides additional fuel capacity to extend the range of the helicopter. The kit consists of a left and right auxiliary fuel tank and the hardware and wiring necessary to complete the installation. The left or right auxiliary fuel tank may be removed as operational requirements dictate.

One fuel tank provides an additional 81.7 U.S. gallons (309.2 liters) of fuel. Both fuel tanks combined, provide an additional 163.4 U.S. gallons (618.5 liters) of fuel.

Section 1

LIMITATIONS

WEIGHT/CG LIMITATIONS

Actual weight changes shall be determined after installation of auxiliary fuel tanks(s), and ballast shall be readjusted, if necessary, to return empty weight CG within allowable limits.

The gross weight center of gravity limits as presented in the basic Flight Manual do not change when either or both auxiliary fuel tanks are installed.

Refer to Manufacturer's Data, Section 1, for weight and balance data and loading example.

WARNING

INDISCRIMINATE LOADING OF THE HELICOPTER MAY RESULT IN VIOLATION OF THE PERMISSIBLE CENTER OF GRAVITY LIMITATIONS WHEN THE HELICOPTER IS EQUIPPED WITH THE 412-706-007 AUXILIARY FUEL KIT.

Section 2

NORMAL PROCEDURES

IN-FLIGHT OPERATION

CAUTION

WHEN ONLY ONE CABIN MOUNTED AUXILIARY FUEL TANK IS USED, THE FUEL INTCON SWITCH MUST BE REPLACED TO OPEN WHEN A FUEL QUANTITY INDICATION OF 500 LBS. IS ON EITHER SIDE. FAILURE TO MANUALLY OPEN THE FUEL INTCON SWITCH WILL RESULT IN FUEL EXHAUSTION TO THE ENGINE OPPOSITE THE SIDE WHICH HAS THE AUXILIARY TANK. THE FEATURE WHICH AUTOMATICALLY OPENS THE INTERCONNECT VALVE WILL NOT HAVE A CHANCE TO FUNCTION WITH ONLY ONE AUXILIARY TANK INSTALLED. WITH TWO EQUALLY LOADED AUXILIARY TANKS INSTALLED, THE AUTOMATIC FEATURE WILL FUNCTION.

Section 3

EMERGENCY AND MALFUNCTION PROCEDURES

No change from basic Flight Manual.

Section 4

PERFORMANCE

No change from basic Flight Manual.

BHT-412-FMS-17.2, 17.3 AND 17.4

Section 1

MANUFACTURER'S DATA

WEIGHT AND BALANCE

AUXILIARY FUEL SYSTEM

AUXILIARY FUEL SYSTEM SERVICING

The auxiliary fuel tanks are interconnected with the basic fuel system to allow gravity flow of auxiliary fuel into main fuel cells as fuel is consumed. The auxiliary fuel system is serviced simultaneously with the basic fuel system through the single filler port located on the aft right side of the fuselage.

FUEL SYSTEM CAPACITIES

BASIC SYSTEM WITH LEFT OR RIGHT AUXILIARY TANK

Total capacity:
419.1 U.S. gallons (1586.5 liters)

Usable fuel:
412.1 U.S. gallons (1560.0 liters)

BASIC SYSTEM WITH BOTH AUXILIARY TANKS

Total capacity:
500.8 U.S. gallons (1895.7 liters)

Usable fuel:
493.8 U.S. gallons (1869.2 liters)

AUXILIARY FUEL LOADING TABLES

Fuel loading tables are presented for weight and balance computations in both English and Metric units. These tables shall be used in lieu of the tables for the basic fuel system when either or both auxiliary fuel tanks are installed. Weights and moments listed herein represent total fuel on board to include that contained in basic fuel cells. Refer to table 1-1 and 1-2 for English or 1-1M and 1-2M for Metric when both left and right auxiliary tanks are installed. Tables 1-3 and 1-4 for English or 1-3M and 1-4M for Metric apply to single auxiliary tank installed on left side, and tables 1-5 and 1-6 for English or 1-5M and 1-6M Metric, apply to single auxiliary tank installed on right side.

412 ROTORCRAFT MANUFACTURER'S DATA

BHT-412-FMS-17.2, 17.3 AND 17.4

Table 1-1. Fuel Loading With Left and Right Auxiliary Tanks — Longitudinal (English)

Longitudinal							
Jet B or JP-4 (6.5 Lb/U.S. Gallon)				Jet A, A1 or JP5 (6.8 Lb/U.S. Gallon)			
Quantity (U.S. Gal)	Weight (Pounds)	CG (Inches)	Moment (In-Lb)	Quantity (U.S. Gal)	Weight (Pounds)	CG (Inches)	Moment (In-Lb)
10	65	139.4	9061	10	68	139.4	9479
20	130	139.6	18148	20	136	139.6	18986
30	195	139.8	27261	30	204	139.8	28519
40	260	139.9	36374	40	272	139.9	38053
50	325	139.9	45468	50	340	139.9	47566
58.3	379	139.9	53022	58.3	396	139.9	55400
60	390	141.1	55029	60	408	141.1	57569
70	455	145.5	66203	70	476	145.5	69258
80	520	148.2	77064	80	544	148.2	80621
90	585	149.8	87633	90	612	149.8	91678
100	650	151.0	98150	100	680	151.0	102680
110	715	152.1	108752	110	748	152.1	113771
120	780	152.9	119262	120	816	152.9	124766
130	845	153.7	129877	130	884	153.7	135871
140	910	154.3	140413	140	952	154.3	146894
150	975	154.8	150930	150	1020	154.8	157896
160	1040	155.3	161512	160	1088	155.3	168966
170	1105	155.8	172159	170	1156	155.8	180105
180	1170	156.1	182637	180	1224	156.1	191066
190	1235	156.4	193154	190	1292	156.4	202069
200	1300	156.8	203840	200	1360	156.8	213248
207.9	1351	157.0	212107	207.9	1414	157.0	221998
210	1365	156.5	213623	210	1428	156.5	223482
220	1430	153.6	219648	220	1496	153.6	229786
230	1495	151.0	225745	230	1564	151.0	236164
240	1560	148.6	231816	240	1632	148.6	242515
241.0	1567	148.3	232386	241.0	1639	148.3	243064
250	1625	148.8	241800	250	1700	148.8	252960
260	1690	149.3	252317	260	1768	149.3	263962
270	1755	149.8	262899	270	1836	149.8	275033
280	1820	150.2	273364	280	1904	150.2	285981
290	1885	150.7	284070	290	1972	150.7	297180
300	1950	151.1	294645	300	2040	151.1	308244
310	2015	151.5	305273	310	2108	151.5	319362
320	2080	151.9	315952	320	2176	151.9	330534
330	2145	152.2	326469	330	2244	152.2	341537
340	2210	152.5	337025	340	2312	152.5	352580
350	2275	152.8	347620	350	2380	152.8	363664
360	2340	153.1	358254	360	2448	153.1	374789
370	2405	153.3	368867	370	2516	153.3	385703
374.5	2434	153.4	373376	374.5	2547	153.4	390710
380	2470	152.9	377663	380	2584	152.9	395904
390	2535	151.9	385067	390	2652	151.9	402839
400	2600	151.0	392600	400	2720	151.0	410720
410	2665	150.1	400017	410	2788	150.1	418479
420	2730	149.2	407316	420	2856	149.2	426115
426.4	2772	148.7	412196	426.4	2900	148.7	431230
430	2795	148.8	415896	430	2924	148.8	435091
440	2860	149.1	426426	440	2992	149.1	446107
450	2925	149.4	436995	450	3060	149.4	457164
460	2990	149.7	447603	460	3128	149.7	468262
470	3055	150.0	458250	470	3196	150.0	479400
480	3120	150.2	468624	480	3264	150.2	490253
490	3185	150.5	479343	490	3332	150.5	501466
493.8	3210	150.6	483426	493.8	3358	150.6	505715

*Critical fuel amount for most forward cg condition.

NOTE: All data above represents total fuel on board (basic and auxiliary), based on nominal density at 15°C.

412 ROTORCRAFT MANUFACTURER'S DATA

Section 1

BHT-412-FMS-17.2, 17.3 AND 17.4

Table 1-1M. Fuel Loading With Left and Right Auxiliary Tanks — Longitudinal (Metric)

Longitudinal							
Jet B or JP-4 (0.779 kg/liter)				Jet A, A1 or JP-5 (0.815 kg/liter)			
Quantity (liters)	Weight (kg)	CG (mm)	Moment (kg-mm)	Quantity (liters)	Weight (kg)	CG (mm)	Moment (kg-mm)
40	31.2	3542	110510	40	32.6	3541	115437
80	62.3	3547	220978	80	65.2	3547	231264
120	93.5	3551	332019	120	97.8	3551	347288
160	124.6	3552	442579	160	130.4	3552	463181
200	155.8	3552	553402	200	163.0	3552	578976
*220.7	171.9	3553	610761	*220.7	179.9	3553	639185
240	186.9	3635	679382	240	195.6	3635	711006
280	218.1	3727	812859	280	228.2	3727	850501
320	249.3	3783	943102	320	260.8	3783	986606
360	280.4	3825	1072530	360	293.3	3825	1121873
400	311.6	3852	1200283	400	325.9	3852	1255367
440	342.7	3877	1328648	440	358.5	3877	1389905
480	373.9	3898	1457462	480	391.1	3898	1524508
520	405.1	3915	1585967	520	423.7	3915	1658786
560	436.2	3930	1714266	560	456.3	3930	1793259
600	467.4	3944	1843426	600	488.9	3944	1928222
640	498.5	3955	1971568	640	521.5	3955	2062533
680	529.6	3965	2099864	680	554.1	3965	2197007
720	560.8	3974	2228619	720	586.7	3974	2331546
760	592.0	3982	2357344	760	619.3	3982	2466053
787.0	613.0	3988	2444644	787.0	641.3	3988	2557504
800	623.1	3965	2470592	800	651.9	3965	2584784
840	654.2	3887	2542875	840	684.4	3887	2660263
880	685.4	3817	2616172	880	717.0	3817	2736789
912.3	710.8	3766	2676873	912.3	743.4	3766	2799644
920	716.7	3770	2701959	920	749.8	3770	2826746
960	747.7	3785	2830045	960	782.2	3785	2960627
1000	778.9	3797	2957483	1000	814.8	3797	3093796
1040	810.0	3810	3086100	1040	847.4	3810	3228594
1080	841.2	3822	3215066	1080	880.0	3822	3363360
1120	872.3	3833	3343526	1120	912.9	3833	3499146
1160	903.5	3844	3473054	1160	945.2	3844	3633349
1200	934.7	3854	3602334	1200	977.8	3854	3768441
1240	965.8	3864	3731851	1240	1010.4	3864	3904186
1280	997.0	3872	3860384	1280	1043.0	3872	4038496
1320	1028.1	3880	3989028	1320	1075.6	3880	4173328
1360	1059.3	3887	4117499	1360	1108.1	3887	4307185
1400	1090.4	3895	4247108	1400	1140.7	3895	4443027
1417.6	1104.0	3897	4302288	1417.6	1155.1	3897	4501427
1440	1121.6	3882	4354051	1440	1173.4	3882	4555139
1480	1152.7	3856	4444811	1480	1205.9	3856	4649950
1520	1183.9	3832	4536705	1520	1238.5	3832	4745932
1560	1215.0	3808	4626720	1560	1271.1	3808	4840349
1600	1246.0	3784	4714864	1600	1303.7	3784	4933201
1614.1	1257.3	3776	4747565	1614.1	1315.2	3776	4966195
1640	1277.3	3781	4829471	1640	1336.3	3781	5052550
1680	1308.5	3789	4957907	1680	1368.9	3789	5186762
1720	1339.7	3797	5086841	1720	1401.5	3797	5321496
1760	1370.8	3805	5215894	1760	1434.1	3805	5456751
1800	1402.0	3813	5345826	1800	1466.7	3813	5592527
1840	1433.1	3820	5474442	1840	1499.3	3820	5727326
1869.2	1455.9	3825	5568818	1869.2	1523.3	3825	5826623

*Critical fuel amount for most forward cg condition.

NOTE: All data above represents total fuel on board (basic and auxiliary), based on nominal density at 15°C.

412 ROTORCRAFT MANUFACTURER'S DATA

BHT-412-FMS-17.2, 17.3 AND 17.4

Table 1-2. Fuel Loading With Left and Right Auxiliary Tanks — Lateral (English)

Lateral							
Jet B or JP-4 (6.5 Lb/U.S. Gallon)				Jet A, A1 or JP-5 (6.8 Lb/U.S. Gallon)			
Quantity (U.S. Gal.)	Weight (Pounds)	CG (Inches)	Moment (In-Lb)	Quantity (U.S. Gal.)	Weight (Pounds)	CG (Inches)	Moment (In-Lb)
10	65	0	0	10	68	0	0
20	130	0	0	20	136	0	0
30	195	0	0	30	204	0	0
40	260	0	0	40	272	0	0
50	325	0	0	50	340	0	0
58.3	379	0	0	58.3	397	0	0
60	390	-0.04	-16	60	408	-0.04	-16
70	455	-0.06	-27	70	476	-0.06	-29
80	520	-0.05	-26	80	544	-0.05	-27
90	585	-0.04	-23	90	612	-0.04	-25
100	650	-0.04	-26	100	680	-0.04	-27
110	715	-0.03	-21	110	748	-0.03	-22
120	780	-0.03	-23	120	816	-0.03	-24
130	845	-0.03	-25	130	884	-0.03	-27
140	910	-0.03	-27	140	952	-0.03	-29
150	975	-0.02	-20	150	1020	-0.02	-20
160	1040	-0.02	-21	160	1088	-0.02	-22
170	1105	-0.02	-22	170	1156	-0.02	-23
180	1170	-0.02	-23	180	1224	-0.02	-24
190	1235	-0.02	-25	190	1292	-0.02	-26
200	1300	-0.02	-26	200	1360	-0.02	-27
207.9	1351	-0.02	-27	207.9	1414	-0.02	-28
210	1365	-0.20	-27	210	1428	-0.20	-286
220	1430	-0.33	-472	220	1496	-0.33	-494
230	1495	-0.42	-628	230	1564	-0.42	-657
240	1560	-0.50	-780	240	1632	-0.50	-816
250	1625	-0.50	-813	250	1700	-0.50	-850
260	1690	-0.48	-811	260	1768	-0.48	-849
270	1755	-0.46	-807	270	1836	-0.46	-845
280	1820	-0.45	-819	280	1904	-0.45	-857
290	1885	-0.44	-829	290	1972	-0.44	-868
300	1950	-0.43	-839	300	2040	-0.43	-877
310	2015	-0.41	-826	310	2108	-0.41	-864
320	2080	-0.41	-853	320	2176	-0.41	-292
330	2145	-0.40	-858	330	2244	-0.40	-898
340	2210	-0.40	-884	340	2312	-0.40	-925
350	2275	-0.40	-910	350	2380	-0.40	-952
360	2340	-0.39	-913	360	2448	-0.39	-955
370	2405	-0.39	-938	370	2516	-0.39	-981
380	2470	-0.36	-889	380	2584	-0.36	930
390	2535	-0.35	-887	390	2652	-0.35	-928
400	2600	-0.34	-884	400	2720	-0.34	-925
410	2665	-0.33	-879	410	2788	-0.33	-920
420	2730	-0.31	-846	420	2856	-0.31	-885
430	2795	-0.30	-839	430	2924	-0.30	-877
440	2860	-0.30	-858	440	2992	-0.30	-898
450	2925	-0.29	-848	450	3060	-0.29	-887
460	2990	-0.29	-867	460	3128	-0.29	-907
470	3055	-0.28	-855	470	3196	-0.28	-895
480	3120	-0.27	-842	480	3264	-0.27	-881
490	3185	-0.25	-796	490	3332	-0.25	-833
493.8	3210	-0.25	-803	493.8	3358	-0.25	-840

*Critical fuel amount for most forward cg condition.

NOTE: All data above represents total fuel on board (basic and auxiliary), based on nominal density at 15°C.

412 ROTORCRAFT MANUFACTURER'S DATA

Section 1

BHT-412-FMS-17.2, 17.3 AND 17.4

Table 1-2M. Fuel Loading With Left and Right Auxiliary Tanks — Lateral (Metric)

Lateral							
Jet B or JP-4 (0.779 kg/liter)				Jet A, A1 or JP-5 (0.815 kg/liter)			
Quantity (liters)	Weight (kg)	CG (mm)	Moment (kg-mm)	Quantity (liters)	Weight (kg)	CG (mm)	Moment (kg-mm)
40	31.2	0	0	40	32.6	0	0
80	62.3	0	0	80	65.2	0	0
120	93.5	0	0	120	97.8	0	0
160	124.6	0	0	160	130.4	0	0
200	155.8	0	0	200	163.0	0	0
220.7	171.9	0	0	220.7	179.9	0	0
240	186.9	-2	-374	240	195.6	-2	-391
280	218.1	-2	-436	280	228.2	-2	-456
320	249.3	-1	-249	320	260.8	-1	-261
360	280.4	-1	-280	360	293.3	-1	-293
400	311.6	-1	-312	400	325.9	-1	-326
440	342.7	-1	-343	440	358.5	-1	-359
480	373.9	-1	-374	480	391.1	-1	-391
520	405.1	-1	-405	520	423.7	-1	-424
560	436.2	-1	-436	560	456.3	-1	-456
600	467.3	-1	-467	600	488.9	-1	-489
640	498.5	-1	-499	640	521.5	-1	-522
680	529.6	-1	-530	680	554.1	-1	-554
720	560.8	-1	-561	720	586.7	-1	-587
760	592.0	-1	-592	760	619.3	-1	-619
787.0	613.0	-1	-613	787.0	641.3	-1	-641
800	623.1	-5	-3116	800	651.9	-5	-3260
840	654.2	-9	-5888	840	684.4	-9	-6160
880	685.4	-11	-7539	880	717.0	-11	-7887
920	716.7	-13	-9317	920	749.8	-13	-9747
960	747.7	-13	-9720	960	782.2	-13	-10169
1000	778.9	-12	-9347	1000	814.8	-12	-9778
1040	810.0	-12	-9720	1040	847.4	-12	-10169
1080	841.2	-12	-10094	1080	880.0	-12	-10560
1120	872.3	-11	-9595	1120	912.9	-11	-10042
1160	903.5	-10	-9035	1160	945.2	-10	-9452
1200	934.7	-10	-9347	1200	977.8	-10	-9778
1240	965.8	-10	-9658	1240	1010.4	-10	-10104
1280	997.0	-10	-9970	1280	1043.0	-10	-10430
1320	1028.1	-10	-10281	1320	1075.6	-10	-10756
1360	1059.3	-10	-10593	1360	1108.1	-10	-11081
1400	1090.4	-10	-10904	1400	1140.7	-10	-11407
1440	1121.6	-9	-10094	1440	1173.4	-9	-10561
1480	1152.7	-9	-10374	1480	1205.9	-9	-10853
1520	1183.9	-8	-9471	1520	1238.5	-8	-9908
1560	1215.0	-8	-9720	1560	1271.1	-8	-10169
1600	1246.0	-8	-9968	1600	1303.7	-8	-10430
1640	1277.3	-8	-10218	1640	1336.3	-8	-10690
1680	1308.5	-8	-10468	1680	1368.9	-8	-10951
1720	1339.7	-7	-9378	1720	1401.5	-7	-9811
1760	1370.8	-7	-9596	1760	1434.1	-7	-10039
1800	1402.0	-7	-9814	1800	1466.7	-7	-10267
1840	1433.1	-6	-8599	1840	1499.3	-6	-8996
1869.2	1455.9	-6	-8735	1869.2	1523.2	-6	-9139

*Critical fuel amount for most forward cg condition.

NOTE: All data above represents total fuel on board (basic and auxiliary), based on nominal density at 15°C.

412 ROTORCRAFT MANUFACTURER'S DATA

BHT-412-FMS-17.2, 17.3 AND 17.4

Table 1-3. Fuel Loading With Left Auxiliary Tank — Longitudinal (English)

Longitudinal							
Jet B or JP-4 (6.5 Lb/U.S. Gallon)				Jet A, A1 or JP-5 (6.8 Lb/U.S. Gallon)			
Quantity (U.S. Gal)	Weight (Pounds)	CG (Inches)	Moment (In-Lb)	Quantity (U.S. Gal)	Weight (Pounds)	CG (Inches)	Moment (In-Lb)
10	65	139.4	9061	10	68	139.4	9479
20	130	139.6	18148	20	136	139.6	18986
30	195	139.8	27261	30	204	139.8	28519
40	260	139.9	36374	40	272	139.9	38053
50	325	139.9	45468	50	340	139.9	47566
*58.3	379	139.9	53022	*58.3	397	139.9	55540
60	390	141.1	55029	60	408	141.1	57569
70	455	145.8	66339	70	476	145.8	69401
80	520	148.7	77324	80	544	148.7	80893
90	585	150.7	88160	90	612	150.7	92228
100	650	152.3	98995	100	680	152.3	103564
110	715	153.6	109824	110	748	153.6	114893
120	780	154.7	120666	120	816	154.7	126235
130	845	155.7	131567	130	884	155.7	137639
140	910	156.5	142415	140	952	156.5	148988
150	975	157.2	153270	150	1020	157.2	160344
160	1040	157.8	164112	160	1088	157.8	171686
170	1105	158.4	175032	170	1156	158.4	183110
173.9	1130	158.6	179218	173.9	1183	158.6	187624
180	1170	156.2	182754	180	1224	156.2	191189
190	1235	153.1	189079	190	1292	153.1	197805
200	1300	150.1	195130	200	1360	150.1	204136
207.1	1346	148.2	199477	207.1	1408	148.2	208666
210	1365	148.4	202566	210	1428	148.4	211915
220	1430	149.2	213356	220	1496	149.2	223203
230	1495	150.0	224250	230	1564	150.0	234600
240	1560	150.7	235092	240	1632	150.7	245942
250	1625	151.4	246025	250	1700	151.4	257380
260	1690	152.0	256880	260	1768	152.0	268736
270	1755	152.6	267813	270	1836	152.6	280174
280	1820	153.1	278642	280	1904	153.1	291502
290	1885	153.6	289536	290	1972	153.6	302899
300	1950	154.0	300300	300	2040	154.0	314160
308.8	2007	154.4	309881	308.8	2100	154.4	324240
310	2015	154.2	310713	310	2108	154.2	325054
320	2080	153.0	318240	320	2176	153.0	332928
330	2145	151.8	325611	330	2244	151.8	340639
340	2210	150.7	333047	340	2312	150.7	348418
350	2275	149.7	340568	350	2380	149.7	356286
360	2340	148.6	347724	360	2448	148.6	363773
360.7	2345	148.6	348467	360.7	2453	148.6	364516
370	2405	149.0	358345	370	2516	149.0	374884
380	2470	149.6	369512	380	2584	149.6	386566
390	2535	150.1	380504	390	2652	150.1	398065
400	2600	150.5	391300	400	2720	150.5	409360
410	2665	150.8	401882	410	2788	150.8	420430
412.1	2679	150.9	404261	412.1	2802	150.9	422822

*Critical fuel amount for most forward cg condition.

NOTE: All data above represents total fuel on board (basic and auxiliary), based on nominal density at 15°C.

412 ROTORCRAFT MANUFACTURER'S DATA

Section 1

BHT-412-FMS-17.2, 17.3 AND 17.4

Table 1-3M. Fuel Loading With Left Auxiliary Tank — Longitudinal (Metric)

Longitudinal							
Jet B or JP-4 (0.779 kg/liter)				Jet A, A1 or JP-5 (0.815 kg/liter)			
Quantity (liters)	Weight (kg)	CG (mm)	Moment (kg-mm)	Quantity (liters)	Weight (kg)	CG (mm)	Moment (kg-mm)
40	31.2	3542	110510	40	32.6	3541	115437
80	62.3	3547	220978	80	65.2	3547	231264
120	93.5	3551	332019	120	97.8	3551	347288
160	124.6	3552	442579	160	130.4	3552	463181
200	155.8	3552	553402	200	163.0	3552	578976
*220.7	171.9	3553	610761	*220.7	179.9	3553	639185
240	186.9	3635	679382	240	195.6	3635	711006
280	218.1	3739	815476	280	228.2	3739	853240
320	249.3	3800	947340	320	260.8	3800	991040
360	280.4	3848	1078979	360	293.3	3848	1128618
400	311.6	3889	1211812	400	325.9	3889	1267425
440	342.7	3923	1344412	440	358.5	3923	1406396
480	373.9	3948	1476157	480	391.1	3948	1544063
520	405.0	3971	1608255	520	423.7	3971	1682513
560	436.2	4001	1745236	560	456.3	4001	1825656
600	467.3	4006	1872004	600	488.9	4006	1958533
640	498.5	4021	2004469	640	521.5	4021	2096952
658.3	512.5	4027	2063838	658.3	536.5	4027	2160486
680	529.6	3973	2104101	680	554.1	3973	2201439
720	560.8	3886	2179269	720	586.7	3886	2279916
760	592.0	3809	2254928	760	619.3	3809	2358914
784.0	610.4	3763	2296935	784.0	638.6	3763	2403052
800	623.1	3772	2350333	800	651.9	3772	2458967
840	654.2	3793	2481381	840	684.4	3793	2595929
880	685.4	3815	2614801	880	717.0	3815	2735355
920	716.7	3834	2747828	920	749.8	3834	2874733
960	747.7	3852	2880140	960	782.2	3852	3013034
1000	778.9	3868	3012785	1000	814.8	3868	3151646
1040	810.0	3882	3144420	1040	847.4	3882	3289607
1080	841.2	3896	3277315	1080	880.0	3896	3428480
1120	872.3	3909	3409821	1120	912.9	3909	3568526
1160	903.5	3919	3540817	1160	945.2	3919	3704239
1168.9	910.2	3922	3569804	1168.9	952.5	3922	3735705
1200	934.7	3895	3640657	1200	977.8	3895	3808531
1240	965.8	3863	3730885	1240	1010.4	3863	3903175
1280	997.0	3833	3821501	1280	1043.0	3833	3997819
1320	1028.1	3805	3911921	1320	1075.6	3805	4092658
1360	1059.3	3778	4002035	1360	1108.1	3778	4186402
1365.4	1063.5	3775	4014713	1365.4	1112.6	3775	4200065
1400	1090.4	3787	4129345	1400	1140.7	3787	4319831
1440	1121.6	3800	4262080	1440	1173.4	3800	4458920
1480	1152.7	3812	4394092	1480	1205.9	3812	4596891
1520	1183.9	3824	4527234	1520	1238.5	3824	4736024
1560.0	1215.0	3834	4658310	1560.0	1271.0	3834	4873014

*Critical fuel amount for most forward cg condition.

NOTE: All data above represents total fuel on board (basic and auxiliary), based on nominal density at 15°C.

412 ROTORCRAFT MANUFACTURER'S DATA

BHT-412-FMS-17.2, 17.3 AND 17.4

Table 1-4. Fuel Loading with Left Auxiliary Tank — Lateral (English)

Lateral							
Jet B or JP-4 (6.5 Lb/U.S. Gallon)				Jet A, A1 or JP-5 (6.8 Lb/U.S. Gallon)			
Quantity (U.S. Gal)	Weight (Pounds)	CG (Inches)	Moment (In-Lb)	Quantity (U.S. Gal)	Weight (Pounds)	CG (Inches)	Moment (In-Lb)
10	65	0	0	10	68	0	0
20	130	0	0	20	136	0	0
30	195	0	0	30	204	0	0
40	260	0	0	40	272	0	0
50	325	0	0	50	340	0	0
58.3	379	0	0	58.3	397	0	0
60	390	-0.03	-12	60	408	-0.03	-12
70	455	-0.45	-205	70	476	-0.45	-214
80	520	-1.45	-754	80	544	-1.45	-789
90	585	-2.25	-1316	90	612	-2.25	-1377
100	650	-2.97	-1931	100	680	-2.97	-2020
110	715	-3.55	-2538	110	748	-3.55	-2655
120	780	-4.04	-3151	120	816	-4.04	-3297
130	845	-4.45	-3760	130	884	-4.45	-3934
140	910	-4.80	-4368	140	952	-4.80	-4570
150	975	-5.10	-4973	150	1020	-5.10	-5202
160	1040	-5.34	-5554	160	1088	-5.34	-5810
170	1105	-5.55	-6133	170	1156	-5.55	-6416
175.4	1140	-5.79	-6601	175.4	1193	-5.79	-6907
180	1170	-5.70	-6669	180	1224	-5.70	-6977
190	1235	-5.58	-6891	190	1292	-5.58	-7209
200	1300	-5.40	-7020	200	1360	-5.40	-7344
207.1	1346	-5.32	-7161	207.1	1408	-5.32	-7491
210	1365	-5.38	-7344	210	1428	-5.38	-7683
220	1430	-5.55	-7937	220	1496	-5.55	-8303
230	1495	-5.70	-8522	230	1564	-5.70	-8915
240	1560	-5.85	-9126	240	1632	-5.85	-9547
250	1625	-5.97	-9701	250	1700	-5.97	-10149
260	1690	-6.10	-10309	260	1768	-6.10	-10785
270	1755	-6.20	-10881	270	1836	-6.20	-11383
280	1820	-6.30	-11466	280	1904	-6.30	-11995
290	1885	-6.40	-12064	290	1972	-6.40	-12621
300	1950	-6.47	-12617	300	2040	-6.47	-13199
308.8	2007	-6.54	-13126	308.8	2100	-6.54	-13734
310	2015	-6.50	-13098	310	2108	-6.50	-13702
320	2080	-6.30	-13104	320	2176	-6.30	-13709
330	2145	-6.10	-13085	330	2244	-6.10	-13688
340	2210	-5.95	-13150	340	2312	-5.95	-13756
350	2275	-5.80	-13195	350	2380	-5.80	-13804
360	2340	-5.60	-13104	360	2448	-5.60	-13709
360.7	2345	-5.60	-13132	360.7	2453	-5.60	-13737
370	2405	-5.68	-13660	370	2516	-5.68	-14291
380	2470	-5.75	-14203	380	2584	-5.75	-14858
390	2535	-5.85	-14830	390	2652	-5.85	-15514
400	2600	-5.94	-15444	400	2720	-5.94	-16157
410	2665	-6.03	-16070	410	2788	-6.03	-16812
412.1	2679	-6.03	-16154	412.1	2802	-6.03	-16896

* Critical fuel amount for most forward cg condition.

NOTE: All data above represents total fuel on board (basic and auxiliary), based on nominal density at 15°C.

412 ROTORCRAFT MANUFACTURER'S DATA

Section 1

BHT-412-FMS-17.2, 17.3 AND 17.4

Table 1-4M. Fuel Loading with Left Auxiliary Tank — Lateral (Metric)

Lateral							
Jet B or JP-4 (0.779 kg/liter)				Jet A, A1 or JP-5 (0.815 kg/liter)			
Quantity (liters)	Weight (kg)	CG (mm)	Moment (kg-mm)	Quantity (liters)	Weight (kg)	CG (mm)	Moment (kg-mm)
40	31.2	0	0	40	32.6	0	0
80	62.3	0	0	80	65.2	0	0
120	93.5	0	0	120	97.8	0	0
160	124.6	0	0	160	130.4	0	0
200	155.8	0	0	200	163.0	0	0
220.7	171.9	0	0	220.7	179.9	0	0
240	186.9	-1	-187	240	195.6	-1	-196
280	218.1	-21	-4580	280	228.2	-21	-4792
320	249.3	-35	-8726	320	260.8	-35	-9128
360	280.4	-67	-18787	360	293.3	-67	-19651
400	311.6	-84	-26174	400	325.9	-84	-27376
440	342.7	-99	-33927	440	358.5	-99	-35492
480	373.9	-110	-41129	480	391.1	-110	-43021
520	405.0	-119	-48195	520	423.7	-119	-50420
560	436.2	-128	-55834	560	456.3	-128	-58406
600	467.3	-135	-63086	600	488.9	-135	-66002
640	498.5	-140	-69790	640	521.5	-140	-73010
664.0	517.0	-147	-75999	664.0	541.0	-147	-79527
680	529.6	-145	-76792	680	554.1	-145	-80345
720	560.8	-142	-79634	720	586.7	-142	-83311
760	592.0	-138	-81696	760	619.3	-138	-85463
784.0	610.4	-135	-82404	784.0	638.6	-135	-86211
800	623.1	-137	-85365	800	651.9	-137	-89310
840	654.2	-141	-92242	840	684.4	-141	-96500
880	685.4	-145	-99383	880	717.0	-145	-103965
920	716.7	-150	-107505	920	749.8	-150	-112470
960	747.7	-152	-113650	960	782.2	-152	-118894
1000	778.9	-155	-120730	1000	814.8	-155	-126294
1040	810.0	-158	-127980	1040	847.4	-158	-133889
1080	841.2	-161	-135433	1080	880.0	-161	-141680
1120	872.3	-163	-142185	1120	912.9	-163	-148803
1160	903.5	-166	-149981	1160	945.2	-166	-156903
1168.9	910.2	-166	-151093	1168.9	952.5	-166	-158115
1200	934.7	-162	-151421	1200	977.8	-162	-158404
1240	965.8	-157	-151631	1240	1010.4	-157	-158633
1280	997.0	-152	-151544	1280	1043.0	-152	-158536
1320	1028.1	-147	-151131	1320	1075.6	-147	-158113
1360	1059.3	-142	-150421	1360	1108.1	-142	-157350
1365.4	1063.5	-142	-151017	1365.4	1112.6	-142	-157989
1400	1090.4	-144	-157018	1400	1140.7	-144	-164261
1440	1121.6	-146	-163754	1440	1173.4	-146	-171316
1480	1152.7	-149	-171752	1480	1205.9	-149	-179679
1520	1183.9	-151	-178769	1520	1238.5	-151	-187014
1560	1215.0	-153	-185895	1560	1271.0	-153	-194463

* Critical fuel amount for most forward cg condition.

NOTE: All data above represents total fuel on board (basic and auxiliary), based on nominal density at 15°C.

412 ROTORCRAFT MANUFACTURER'S DATA

BHT-412-FMS-17.2, 17.3 AND 17.4

Table 1-5. Fuel Loading with Right Auxiliary Tank — Longitudinal (English)

Longitudinal							
Jet B or JP-4 (6.5 Lb/U.S. Gallon)				Jet A, A1 or JP-5 (6.8 Lb/U.S. Gallon)			
Quantity (U.S. Gal)	Weight (Pounds)	CG (Inches)	Moment (In-Lb)	Quantity (U.S. Gal)	Weight (Pounds)	CG (Inches)	Moment (In-Lb)
10	65	139.4	9061	10	68	139.4	9479
20	130	139.6	18148	20	136	139.6	18986
30	195	139.8	27261	30	204	139.8	28519
40	260	139.9	36374	40	272	139.9	38053
50	325	139.9	45468	50	340	139.9	47566
*58.3	379	139.9	53022	*58.3	397	139.9	55540
60	390	141.1	55029	60	408	141.1	57569
70	455	145.8	66339	70	476	145.8	69401
80	520	148.7	77324	80	544	148.7	80893
90	585	150.7	88160	90	612	150.7	92228
100	650	152.3	98995	100	680	152.3	103564
110	715	153.6	109824	110	748	153.6	114893
120	780	154.7	120666	120	816	154.7	126235
130	845	155.7	131567	130	884	155.7	137639
140	910	156.5	142415	140	952	156.5	148988
150	975	157.2	153270	150	1020	157.2	160344
160	1040	157.8	164112	160	1088	157.8	171686
170	1105	158.4	175032	170	1156	158.4	183110
173.9	1130	158.6	179218	173.9	1183	158.6	187624
180	1170	156.2	182754	180	1224	156.2	191189
190	1235	153.1	189079	190	1292	153.1	197805
200	1300	150.1	195130	200	1360	150.1	204136
207.1	1346	148.2	199477	207.1	1408	148.2	208666
210	1365	148.4	202566	210	1428	148.4	211915
220	1430	149.2	213356	220	1496	149.2	223203
230	1495	150.0	224250	230	1564	150.0	234600
240	1560	150.7	235092	240	1632	150.7	245942
250	1625	151.4	246025	250	1700	151.4	257380
260	1690	152.0	256880	260	1768	152.0	268736
270	1755	152.6	267813	270	1836	152.6	280174
280	1820	153.1	278642	280	1904	153.1	291502
290	1885	153.6	289536	290	1972	153.6	302899
300	1950	154.0	300300	300	2040	154.0	314160
308.8	2007	154.4	309881	308.8	2100	154.4	324240
310	2015	154.2	310713	310	2108	154.2	325054
320	2080	153.0	318240	320	2176	153.0	332928
330	2145	151.8	325611	330	2244	151.8	340639
340	2210	150.7	333047	340	2312	150.7	348418
350	2275	149.7	340568	350	2380	149.7	356286
360	2340	148.6	347724	360	2448	148.6	363773
360.7	2345	148.6	348467	360.7	2453	148.6	364516
370	2405	149.0	358345	370	2516	149.0	374884
380	2470	149.6	369512	380	2584	149.6	386566
390	2535	150.1	380504	390	2652	150.1	398065
400	2600	150.5	391300	400	2720	150.5	409360
410	2665	150.8	401882	410	2788	150.8	420430
412.1	2679	150.9	404261	412.1	2802	150.9	422822

* Critical fuel amount for most forward cg condition.

NOTE: All data above represents total fuel on board (basic and auxiliary), based on nominal density at 15°C.

**412 ROTORCRAFT
MANUFACTURER'S DATA**

Section 1

BHT-412-FMS-17.2, 17.3 AND 17.4

Table 1-5M. Fuel Loading with Right Auxiliary Tank — Longitudinal (Metric)

Longitudinal							
Jet B or JP-4 (0.779 kg/liter)				Jet A, A1 or JP-5 (0.815 kg/liter)			
Quantity (liters)	Weight (kg)	CG (mm)	Moment (kg-mm)	Quantity (liters)	Weight (kg)	CG (mm)	Moment (kg-mm)
40	31.2	3542	110510	40	32.6	3541	115437
80	62.3	3547	220978	80	65.2	3547	231264
120	93.5	3551	332019	120	97.8	3551	347288
160	124.6	3552	442579	160	130.4	3552	463181
200	155.8	3552	553402	200	163.0	3552	578976
*220.7	171.9	3553	610761	*220.7	179.9	3553	639185
240	186.9	3635	679382	240	195.6	3635	711006
280	218.1	3739	815476	280	228.2	3739	853240
320	249.3	3800	947340	320	260.8	3800	991040
360	280.4	3848	1078979	360	293.3	3848	1128618
400	311.6	3889	1211812	400	325.9	3889	1267425
440	342.7	3923	1344412	440	358.5	3923	1406396
480	373.9	3948	1476157	480	391.1	3948	1544063
520	405.0	3971	1608255	520	437.7	3971	1682513
560	436.2	4001	1745236	560	456.3	4001	1825656
600	467.3	4006	1872004	600	488.9	4006	1958533
640	498.5	4021	2004469	640	521.5	4021	2096952
658.3	512.5	4027	2063838	658.3	536.5	4027	2160486
680	529.6	3973	2104101	680	554.1	3973	2201439
720	560.8	3886	2179269	720	586.7	3886	2279916
760	592.0	3809	2254928	760	619.3	3809	2358914
784.0	610.4	3763	2296935	784.0	638.6	3763	2403052
800	623.1	3772	2350333	800	651.9	3772	2458967
840	654.2	3793	2481381	840	684.4	3793	2595929
880	685.4	3815	2614801	880	717.0	3815	2735355
920	716.7	3834	2747828	920	749.8	3834	2874733
960	747.7	3852	2880140	960	782.2	3852	3013034
1000	778.9	3868	3012785	1000	814.8	3868	3151646
1040	810.0	3882	3144420	1040	847.4	3882	3289607
1080	841.2	3896	3277315	1080	880.0	3896	3428480
1120	872.3	3909	3409821	1120	912.9	3909	3568526
1160	903.5	3919	3540817	1160	945.2	3919	3704239
1168.9	910.2	3922	3569804	1168.9	952.5	3922	3735705
1200	934.7	3895	3640657	1200	977.8	3895	3808531
1240	965.8	3863	3730885	1240	1010.4	3863	3903175
1280	997.0	3833	3821501	1280	1043.0	3833	3997819
1320	1028.1	3805	3911921	1320	1075.6	3805	4092658
1360	1059.3	3778	4002035	1360	1108.1	3778	4186402
1365.4	1063.5	3775	4014713	1365.4	1112.6	3775	4200065
1400	1090.4	3787	4129345	1400	1140.7	3787	4319831
1440	1121.6	3800	4262080	1440	1173.4	3800	4458920
1480	1152.7	3812	4394092	1480	1205.9	3812	4596891
1520	1183.9	3824	4527234	1520	1238.5	3824	4736024
1560.0	1215.0	3834	4658310	1560.0	1271.0	3834	4873014

* Critical fuel amount for most forward cg condition.

NOTE: All data above represents total fuel on board (basic and auxiliary), based on nominal density at 15°C.

412 ROTORCRAFT MANUFACTURER'S DATA

BHT-412-FMS-17.2, 17.3 AND 17.4

Table 1-6. Fuel Loading with Right Auxiliary Tank — Lateral (English)

Lateral							
Jet B or JP-4 (6.5 Lb/U.S. Gallon)				Jet A, A1 or JP-5 (6.8 Lb/U.S. Gallon)			
Quantity (U.S. Gal)	Weight (Pounds)	CG (Inches)	Moment (In-Lb)	Quantity (U.S. Gal)	Weight (Pounds)	CG (Inches)	Moment (In-Lb)
10	65	0	0	10	68	0	0
20	130	0	0	20	136	0	0
30	195	0	0	30	204	0	0
40	260	0	0	40	272	0	0
50	325	0	0	50	340	0	0
58.3	379	0	0	58.3	397	0	0
60	390	-0.03	-12	60	408	-0.03	-12
70	455	0.34	155	70	476	0.34	162
80	520	1.30	676	80	544	1.30	707
90	585	2.22	1299	90	612	2.22	1359
100	650	2.90	1885	100	680	2.90	1972
110	715	3.48	2488	110	748	3.48	2603
120	780	3.95	3081	120	816	3.95	3223
130	845	4.43	3743	130	884	4.43	3916
140	910	4.71	4286	140	952	4.71	4484
150	975	5.00	4875	150	1020	5.00	5100
160	1040	5.28	5491	160	1088	5.28	5745
170	1105	5.50	6078	170	1156	5.50	6358
173.9	1130	5.60	6328	173.9	1183	5.60	6625
180	1170	5.15	6026	180	1224	5.15	6304
190	1235	4.73	5842	190	1292	4.73	6111
200	1300	4.36	5668	200	1360	4.36	5930
207.1	1346	4.12	5546	207.1	1408	4.12	5801
210	1365	4.20	5733	210	1428	4.20	5998
220	1430	4.45	6364	220	1496	4.45	6657
230	1495	4.63	6922	230	1564	4.63	7241
240	1560	4.81	7504	240	1632	4.81	7850
250	1625	5.00	8125	250	1700	5.00	8500
260	1690	5.13	8670	260	1768	5.13	9070
270	1755	5.28	9266	270	1836	5.28	9694
280	1820	5.40	9828	280	1904	5.40	10282
290	1885	5.60	10556	290	1972	5.60	11043
300	1950	5.65	11018	300	2040	5.65	11526
308.8	2007	5.74	11520	308.8	2100	5.74	12054
310	2015	5.72	11526	310	2108	5.72	12058
320	2080	5.53	11502	320	2176	5.53	12033
330	2145	5.36	11497	330	2244	5.36	12028
340	2210	5.20	11492	340	2312	5.20	12022
350	2275	5.05	11489	350	2380	5.05	12019
360	2340	4.95	11583	360	2448	4.95	12118
360.7	2345	4.92	11537	360.7	2453	4.92	12069
370	2405	5.01	12049	370	2516	5.01	12605
380	2470	5.10	12597	380	2584	5.10	13178
390	2535	5.20	13182	390	2652	5.20	13790
400	2600	5.30	13780	400	2720	5.30	14416
410	2665	5.41	14418	410	2788	5.41	15083
412.1	2679	5.43	14547	412.1	2802	5.43	15215

* Critical fuel amount for most forward cg condition.

NOTE: All data above represents total fuel on board (basic and auxiliary), based on nominal density at 15°C.

**412 ROTORCRAFT
MANUFACTURER'S DATA**

Section 1

BHT-412-FMS-17.2, 17.3 AND 17.4

Table 1-6M. Fuel Loading with Right Auxiliary Tank — Lateral (Metric)

Lateral							
Jet B or JP-4 (0.779 kg/liter)				Jet A, A1 or JP-5 (0.815 kg/liter)			
Quantity (liters)	Weight (kg)	CG (mm)	Moment (kg-mm)	Quantity (liters)	Weight (kg)	CG (mm)	Moment (kg-mm)
40	31.2	0	0	40	32.6	0	0
80	62.3	0	0	80	65.2	0	0
120	93.5	0	0	120	97.8	0	0
160	124.6	0	0	160	130.4	0	0
200	155.8	0	0	200	163.0	0	0
220.7	171.9	0	0	220.7	179.9	0	0
240	186.9	-1	-187	240	195.6	-1	-196
280	218.1	18	3926	280	228.2	18	4108
320	249.3	44	10969	320	260.8	44	11475
360	280.4	66	18506	360	293.3	66	19358
400	311.6	82	25551	400	325.9	82	26724
440	342.7	97	33242	440	358.5	97	34775
480	373.9	108	40381	480	391.1	108	42239
520	405.0	118	47790	520	423.7	118	49997
560	436.2	126	54961	560	456.3	126	57494
600	467.3	133	62151	600	488.9	133	65024
640	498.5	139	69292	640	521.5	139	72489
658.3	512.5	142	72775	658.3	536.5	142	76183
680	529.6	131	69378	680	554.1	131	72587
720	560.8	120	67296	720	586.7	120	70404
760	592.0	110	65120	760	619.3	110	68123
784.0	610.4	105	64092	784.0	638.6	105	67053
800	623.1	107	66672	800	651.9	107	69753
840	654.2	114	74579	840	684.4	114	78022
880	685.4	119	81563	880	717.0	119	85323
920	716.7	124	88871	920	749.8	124	92975
960	747.7	128	95706	960	782.2	128	100122
1000	778.9	132	102815	1000	814.8	132	107554
1040	810.0	136	110160	1040	847.4	136	115246
1080	841.2	140	117768	1080	880.0	140	123200
1120	872.3	142	123867	1120	912.9	142	129632
1160	903.5	145	131008	1160	945.2	145	137054
1168.9	910.2	146	132889	1168.9	952.5	146	139065
1200	934.7	142	132727	1200	977.8	142	138848
1240	965.8	137	132315	1240	1010.4	137	138425
1280	997.0	133	132601	1280	1043.0	133	138719
1320	1028.1	129	132625	1320	1075.6	129	138752
1360	1059.3	126	133472	1360	1108.1	126	139621
1365.4	1063.5	125	132938	1365.4	1112.6	125	139075
1400	1090.4	127	138481	1400	1140.7	127	144869
1440	1121.6	130	145808	1440	1173.4	130	152542
1480	1152.7	132	152156	1480	1205.9	132	159179
1520	1183.9	135	159827	1520	1238.5	135	167198
1560	1215.0	138	167670	1560	1271.0	138	175398

* Critical fuel amount for most forward cg condition.

NOTE: All data above represents total fuel on board (basic and auxiliary), based on nominal density at 15°C.

BHT-412-FMS-17.2, 17.3 AND 17.4

LOADING EXAMPLES

The CG examples shown below apply only to helicopters with standard seating which are equipped with both tanks of the 412-706-007 Auxiliary Fuel Kit. The loadings are based on standard 170 pound people for both crew and passengers.

At empty weights below 7400 pounds there are no restrictions on passenger or fuel loadings except as imposed by the gross weight CG limits of the helicopter.

With helicopter weights at or above 7400 pounds the following procedure should be applied:

Ballast the helicopter to the most aft weight empty line.

When flying with 2 crew members and 9 passengers add weight in the baggage compartment as required to maintain C.G. within the forward limits.

Examples:

With a ballasted weight empty of 7500 pounds, 20 pounds is required in the baggage compartment.

With a ballasted weight empty of 7700 pounds, 40 pounds is required in the baggage compartment.

WARNING

**BAGGAGE COMPARTMENT
WEIGHT MUST BE REMOVED
FOR SINGLE PILOT OPERATION.**

BHT-412-FMS-18.2 AND 18.3

Bell
MODEL **412**

**ROTORCRAFT
FLIGHT MANUAL**

33108 — 33213

36001 — 36019

AND

33214 — 33999

36020 AND SUB

**SUPPLEMENT FOR
LOUDHAILER OPERATIONS**

412-899-143

**CERTIFIED
NOVEMBER 17, 1983**

This supplement shall be attached to the Model 412 Flight Manual (BHT-412-FM-2 or -3) when the 412-899-143 Loudhailer has been installed.

The information contained herein supplements the information of the basic Flight Manual. For Limitations, Procedures, and Performance Data not contained in this supplement, consult the basic Flight Manual.

Bell Helicopter **TEXTRON**

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17 NOVEMBER 1983

REISSUED — 8 OCTOBER 1991

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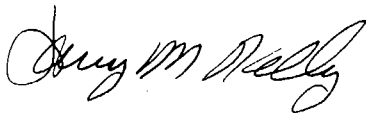
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1 — 3/4	0		

APPROVED:



MANAGER

ROTORCRAFT CERTIFICATION OFFICE
FEDERAL AVIATION ADMINISTRATION
FT. WORTH, TEXAS 76193-0170

NOTE

Revised text is indicated by a black vertical line. Insert latest revision pages; dispose of superseded pages.

INTRODUCTION

The Loudhailer, when installed, will permit the helicopter crew to direct ground personnel while remaining airborne. The kit contains a speaker assembly, amplifier, switches, and the necessary hardware to complete the installation. Use of Loudhailer is controlled through pilot or copilot control panel. Optional configurations allow use of a remote (hand held) microphone and/or a tape recorder.

Section 1

LIMITATIONS

WEIGHT/CG LIMITATIONS

Actual weight change shall be determined after Loudhailer is installed and ballast readjusted if necessary to return empty weight CG within allowable limits.

OPERATING LIMITATIONS

IFR operation is prohibited with Loudhailer installed.

Section 2

NORMAL PROCEDURES

BEFORE EXTERIOR CHECK

PA SYSTEM PWR switch — OFF.

EXTERIOR CHECK

FUSELAGE

Fuselage underside — Check security and wiring connections of Loudhailer.

LOUDHAILER OPERATION

WARNING

USE EXTREME CARE DURING GROUND OPERATION OF LOUDHAILER TO PREVENT INJURY TO PERSONNEL. GROUND SUPPORT PERSONNEL IN VICINITY OF HELICOPTER SHOULD WEAR PROTECTIVE HEARING DEVICES.

PA circuit breakers — Check in.

SIREN/MOM switch — OFF.

TRILL/MOM switch — OFF.

PA SYSTEM GAIN control switch — OFF.

NOTE

OFF position is the minimum gain preset at the remote amplifier located in the baggage compartment.

PA SYSTEM PWR switch — PWR.

Rotate control switch on communications control panel to HAIL/AUX.

PA mode select — As desired.

PA SYSTEM PWR switch — OFF, when Loudhailer operation is completed.

Section 3

EMERGENCY AND MALFUNCTION PROCEDURES

No change from basic manual.

Section 4

PERFORMANCE

No change from basic manual.

BHT-412-FMS-19.2, 19.3 AND 19.4

Bell **412/412EP**
MODELS

ROTORCRAFT FLIGHT MANUAL

33108 — 33213

36001 — 36019

AND

36020 — 36086

AND

36087 AND SUB

SUPPLEMENT FOR SOFT INTERIOR

412-705-510

CERTIFIED

28 MARCH 1985

This supplement shall be attached to the Model 412 Flight Manual (BHT-412-FM-2, or -3) or Model 412EP Flight Manual (BHT-412-FM-4) when the 412-705-510 Soft interior has been installed.

The information contained herein supplements the information of the basic Flight Manual. For Limitations, Procedures, and Performance Data not contained in this supplement, consult the basic Flight Manual.

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MANAGER

ROTORCRAFT CERTIFICATION OFFICE
FEDERAL AVIATION ADMINISTRATION
FT. WORTH, TEXAS 76193-0170

NOTE

Revised text is indicated by a black vertical line. Insert latest revision pages; dispose of superseded pages.

INTRODUCTION

The soft interior, when installed, will permit the helicopter to be flown with doors off/open with an airspeed limitation of 100 KIAS.

Section 1

LIMITATIONS

WEIGHT/CG LIMITATIONS

Actual weight change shall be determined after Soft Interior is installed and ballast readjusted if necessary to return empty weight CG within allowable limits.

DOORS OPEN OR REMOVED

Remove both seat backs from outboard facing seats on each side of transmission pylon.

Flight operation is approved for the following alternative configurations during VFR conditions only:

Both crew doors removed.

Both sliding doors locked open or removed with both hinged panels installed or removed.

In all cases, door configuration shall be symmetrical for both sides of the fuselage.

NOTE

Opening or removing doors shifts helicopter center of gravity and reduces V_{NE}. Refer to Weight and Balance section in Manufacturer's Data and to Airspeed Limitations.

AIRSPPEED LIMITATIONS

V_{NE} with doors open or removed is 100 KIAS.



412/412EP

ROTORCRAFT FLIGHT MANUAL

SUPPLEMENT FOR WEATHER RADAR KIT

412-899-107

**CERTIFIED
16 JUNE 1986**

This supplement shall be attached to the Model 412 Flight Manual (BHT-412-FM-1, -2 or -3) or Model 412EP Flight Manual (BHT-412-FM-4) when the 412-899-107 Weather Radar kit has been installed.

The information contained herein supplements the information of the basic Flight Manual. For Limitations, Procedures, and Performance Data not contained in this supplement consult the basic Flight Manual.

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MANAGERROTORCRAFT CERTIFICATION OFFICE
FEDERAL AVIATION ADMINISTRATION
FT. WORTH, TX 76193-0170

19 MAR 2003

C/D

INTRODUCTION

The primary purpose of the system is to detect storms along the flight path and give a visual indication in colors, of their intensity so a determination to avoid the storm can be made. The secondary purpose of the system is to interrogate and locate the surface-based transponder beacons. The system can be operated in one of three modes: radar, beacon, or both. In BOTH mode, the system performs both radar (weather or terrain) detection and beacon location simultaneously.

Section 1

LIMITATIONS

Contents of this supplement shall be used in conjunction with basic Flight Manual and Sperry-RCA Primus 500 Color Radar Pilot Handbook, for helicopters equipped with Weather Radar kit.

OPERATING LIMITATIONS

The minimum slant and horizontal range versus altitude at which ground targets can be mapped is shown in Figure 1.

Targets more than nineteen degrees (maximum depression) below helicopter centerline cannot be illuminated because of antenna tilt limitations (Figure 1).

Objects closer than 3/10 mile from radar antenna will not be detected because of system limitations.

The radar beam emitted is approximately seven and one-half degrees wide. The antenna may be raised or depressed fifteen degrees from helicopter centerline.

GROUND OPERATION

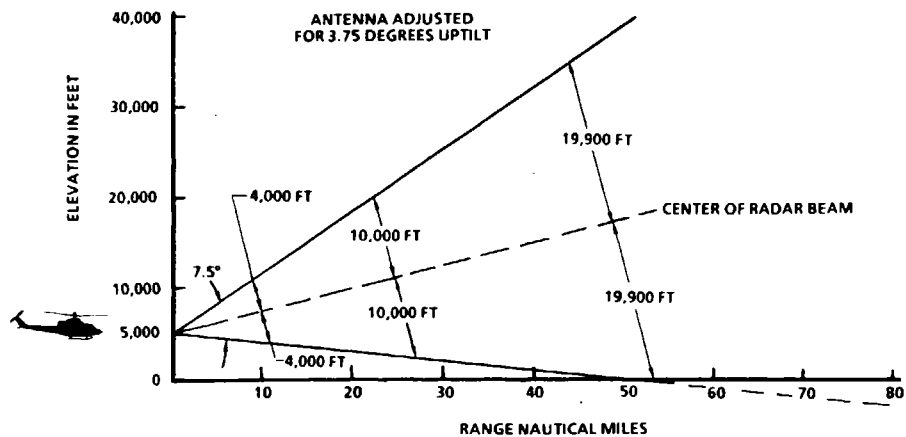
Radar system shall not be operated on ground when personnel are in the DANGER AREA (Figure 2).

Radar system shall not be operated within 100 feet of any fueling operation.

Radar system shall not be operated on ground anytime a large metallic object is forward of helicopter nose, within 60 degrees of centerline, and at a distance of less than 100 feet.

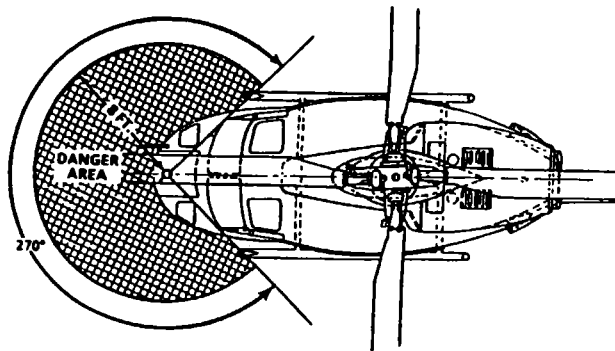
WEIGHT/CG LIMITATIONS

Actual weight change shall be determined after kit is installed and ballast added if necessary, to return empty weight CG within allowable limits.



412_FMS_20_0001
412-470-001

Figure 1. Radar Beam Illumination



412_FMS_20_0002
412-470-001

Figure 2. Personnel Danger Area

Section 2

NORMAL PROCEDURES

PREFLIGHT CHECK

1. NOSE AREA

Radome — Condition and cleanliness.

Antenna — Freedom of movement and security.

DO NOT OPERATE THE RADAR DURING REFUELING OPERATIONS OR WITHIN 100 FEET OF AIRCRAFT, VEHICLES, OR CONTAINERS CONTAINING FLAMMABLES OR EXPLOSIVES (WX, CYC OR MAP PUSHBUTTONS DEPRESSED).

CAUTION

INTERIOR CHECK

INSTRUMENT PANEL

Radar OFF pushbutton — Depress.

RAD GAIN control — PRESET.

BCN GAIN control — PRESET.

Mode control selector switch — RAD.

RANGE control — TEST.

TILT control — + 15 degrees.

INT control — Midpoint.

DO NOT OPERATE THE RADAR IN THE DIRECTION OF LARGE METALLIC OBJECTS THAT ARE WITHIN 100 FEET OF THE HELICOPTER (WX, CYC OR MAP PUSHBUTTONS DEPRESSED).

NOTE

120 degrees scan is automatically selected when system is activated.

Radar pushbutton — Depress WX.

NOTE

Radar requires approximately 60 seconds to warm-up.

BEFORE TAKEOFF CHECK

WARNING

DO NOT ALLOW PERSONNEL WITHIN 8 FEET AND 135 DEGREES EITHER SIDE OF THE HELICOPTER CENTERLINE DURING RADAR OPERATION (WX, CYC OR MAP PUSHBUTTONS DEPRESSED).

SEC SCAN pushbutton — Press, check for 60 degrees antenna scan. Press SEC SCAN again, antenna should return to 120 degrees antenna scan.

Radar STBY pushbutton — Depress.

INFLIGHT OPERATION

WARNING

THE SYSTEM PERFORMS ONLY THE FUNCTIONS OF WEATHER DETECTION, GROUND MAPPING, OR BEACON LOCATION. IT SHOULD NOT BE USED OR RELIED UPON FOR PROXIMITY OR ANTI-COLLISION WARNING.

Radar pushbutton — Depress WX, CYC or MAP. (Verify correct test pattern.)

INT control — As desired.

RANGE control — As desired.

TILT control — As desired.

SEC SCAN — As desired.

BEFORE LANDING

Radar RANGE control — TEST. (Verify correct test pattern.)

TILT control — + 15 degrees.

OFF pushbutton — Depress.

Section 3

EMERGENCY/MAINFUNCTION PROCEDURES

MODE FAILURE

Indication

1. Test display does not match test pattern.

Procedure:

1. OFF pushbutton — Depress.

POWER FAILURE

Indication:

1. No display on indicator.

Procedure:

1. WEATHER RDR AC/DC circuit breakers — Check in.

2. INT control — Rotate clockwise.

3. WX, CYC or MAP pushbutton — Depress.

DISPLAY DOES NOT STABILIZE

Indications:

1. Display follows changes in helicopter attitude.
2. STAB OFF light illuminated.

Procedures:

1. CPLT ATT circuit breaker — Check in.
2. STAB pushbutton — Depress.

Section 4

PERFORMANCE DATA

No change from basic manual.

***Bell* MODEL 412**

**ROTORCRAFT
FLIGHT MANUAL**

**SUPPLEMENT FOR
GLOBAL NAVIGATION SYSTEM
GNS-500A/S3, WITH NAV SWITCHING
(412-899-141)**

**CERTIFIED
JUNE 16, 1986**

This supplement shall be attached to the Model 412 Flight Manual when the 412-899-141 Global Navigation System GNS-500A/S3 has been installed.

The information contained herein supplements the information of the basic Flight Manual. For Limitations, Procedures, and Performance Data not contained in this supplement consult the basic Flight Manual.

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16 JUNE 1986

REISSUED 8 MAY 1989

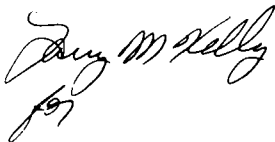
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MANAGER

ROTORCRAFT CERTIFICATION DIRECTORATE
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DEPARTMENT OF TRANSPORTATION
SOUTHWEST REGION, FORT WORTH, TEXAS

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INTRODUCTION

The GNS-500A/S3 is a very low frequency VLF/OMEGA radio navigation system that provides great circle point-to-point navigation on a worldwide basis. It also displays to the flight crew an assessment of its ability to navigate and will automatically revert to dead reckoning during periods of inadequate signal reception.

The GNS-500A/S3 consists of a control display unit (CDU) mounted on the pedestal, a computer receiver unit (CRU) and an optional equipment unit (OEU) located in the right avionics compartment.

A NAV-1/VLF switch is located on the pilot instrument panel next to the HSI. It is a combination 2 position pushbutton and annunciator. Pressing the pushbutton will couple the HSI to the NAV-1 receiver or GNS-500A/S3. The NAV-1 or VLF annunciator light will illuminate to identify the coupled mode. BRG PTR select switches are located on both the pilot and copilot instrument panels and allows selection of VLF or ADF bearing to be displayed on the pilot or copilot HSI.

Section 1

LIMITATIONS

OPERATIONAL LIMITATIONS

The following GNS Operators Manual must be immediately available to the flight crew whenever navigation is predicated on the use of the GNS-500A/S3:

Operator's manual Report 1080 dated 9-1-80 for the -3A Program and CRT/CDU.

Operator's Manual Report 1080 dated 1-19-81 for the -3B Program and CRT/CDU.

Provided the GNS-500A VLF/Omega navigation system is receiving usable signals from at least two Omega navigation stations, it is approved for:

VLF/IFR RNAV operation within the common boundaries of the United States and Alaska in accordance with the EN ROUTE criteria of AC 90-45A or the criteria of AC 20-101B.

Operation as a means to update self-contained navigation systems, such as INS or Doppler, in accordance with AC 120-31A in the areas between Latitudes 85°N to 55°S, with the exception of the area above 45°N Latitude bounded by Longitudes 30°E and 120°N extending across the Asian Continent.

Operation as sole means of long range navigation in accordance with AC 120-37 in the areas between Latitudes 85°N to 55°S, with the exception of the area above 45°N Latitude bounded by Longitudes 30°E and 120°E extending across the Asian Continent.

During RNAV operation of the GNS-500A/S3, additional navigation equipment required for the specific type of operation must be installed and operable.

The GNS-500A/S3 position information must be checked for accuracy (reasonableness) prior to use as a means of navigation and under the following conditions.

Prior to each compulsory reporting point during IFR operation when not under radar surveillance or control.

At or prior to arrival at each enroute waypoint during IFR operation along approved RNAV routes.

Prior to requesting off-airway routing, and at hourly intervals thereafter during RNAV operation off of approved RNAV routes.

During period of Dead Reckoning, navigation shall not be predicated on the use of the GNS-500A/S3 for RNAV operation.

Following a period of Dead Reckoning, the helicopter position should be verified by visually sighting ground reference points and/or by using other navigation equipment such as VOR, DME, Tacan, NDB, or radar fix.

The GNS-500A/S3 may not be used for navigation in terminal areas or during departures from, or approaches, to, airports.

OPERATIONAL LIMITATIONS (Cont)

Enroute navigation shall not be predicated on the GNS-500A/S3 during the period that DR is illuminated.

Section 2

NORMAL PROCEDURES

PREFLIGHT CHECK

NOTE

EXTERIOR CHECK

When the HSI is coupled to the VLF, the course set knob is disabled.

ENGINE AREA, AFT COMPARTMENTS AND TAILBOOM, RIGHT SIDE

BRG PTR 1 switches — As desired.

Avionic compartment — Check:

NOTE

CRU and OEU - Condition and security.

Selecting NAV 1, or NAV 2 will connect bearing pointer 1 to either the NAV 1 or NAV 2 receiver respectively.

NOTE

Refer to GNS Operators Manual for systems checks and operation.

BRG PTR 2 switches — As desired.

NOTE

BEFORE TAKEOFF

Selecting VLF or ADF will connect bearing pointer 2 to either the VLF or ADF receiver respectively.

NAV-1/VLF switch — As desired.

Section 3

EMERGENCY AND MALFUNCTION PROCEDURES

GNS-500A/S3 MALFUNCTIONS

NOTE

Refer to GNS Operators Manual for system malfunction indications and procedures.

Warning flags

FAIL FLAG	FLAG LOCATION	FAULT CONDITION	CORRECTIVE ACTION
Navigation	HSI	Navigation information unreliable.	NAV-1/VLF switch — Press to change navigation modes.

Section 4

PERFORMANCE DATA

No Change

Use or disclosure of data contained on this page is subject to the restriction on the title page of this document.

Bell
MODEL **412**

ROTORCRAFT FLIGHT MANUAL

BHT 33108 — 33213
AND
36001 — 36019

SUPPLEMENT FOR CATEGORY A OPERATIONS FOR HELICOPTERS EQUIPPED WITH PT6T-3B OR PT6T-3BF ENGINES

VERTICAL TAKEOFF

CERTIFIED
6 JUNE 1986

This supplement shall be attached to Model 412 Flight Manual (BHT-412-FM-2) when operating in Category A conditions and PT6T-3B or PT6T-3BF engines are installed.

The information contained herein supplements information of the basic Flight Manual. For limitations, Procedures, and Performance Data not contained in this supplement, consult the basic Flight Manual or other applicable supplements.

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Marty Saunders
for MANAGER

ROTORCRAFT CERTIFICATION OFFICE
FEDERAL AVIATION ADMINISTRATION
FT. WORTH, TX 76193-0170

INTRODUCTION

A "Category A" takeoff is defined as follows: operation of the helicopter in such a manner that if one engine fails at any time after the start of a takeoff, the helicopter can:

1. Return to, and safely stop on, the takeoff area; or
2. Continue the takeoff, climb out, and attain single engine forward flight.

A "Category A" landing is defined as follows: Operation of the helicopter in such a manner that if one engine fails at any point in the approach, the helicopter can —

1. Land, and stop safely on the intended landing area; or
2. Climb out from the point of failure and attain single engine forward flight.

This supplement is divided into limitations, procedures, and performance for a given set of conditions.

Section 1

OPERATING LIMITATIONS

ATTENTION

Mandatory compliance with the operating limitations in Section I of this Manual is required by law.

WEIGHT LIMITATIONS

Maximum Gross Weight 10,500 Pounds (4762.7 Kilograms).

TAKEOFF AND LANDING WEIGHT vs ALTITUDE LIMITATIONS

Maximum Takeoff and Landing Weight - Varies with temperature and altitude - See Gross Weight Limits for Takeoff and Landing Chart.

NOTE:

The minimum heliport width and length are 72 feet (22 meters) and 150 feet (46 meters) respectively for Category "A" vertical operations from ground level or elevated heliports.

ALTITUDE LIMIT FOR TAKE-OFF AND LANDING

The altitude limit for takeoff and landing is 2500 feet pressure altitude.

ALTIMETER (VERTICAL T. O. ALTIMETER)

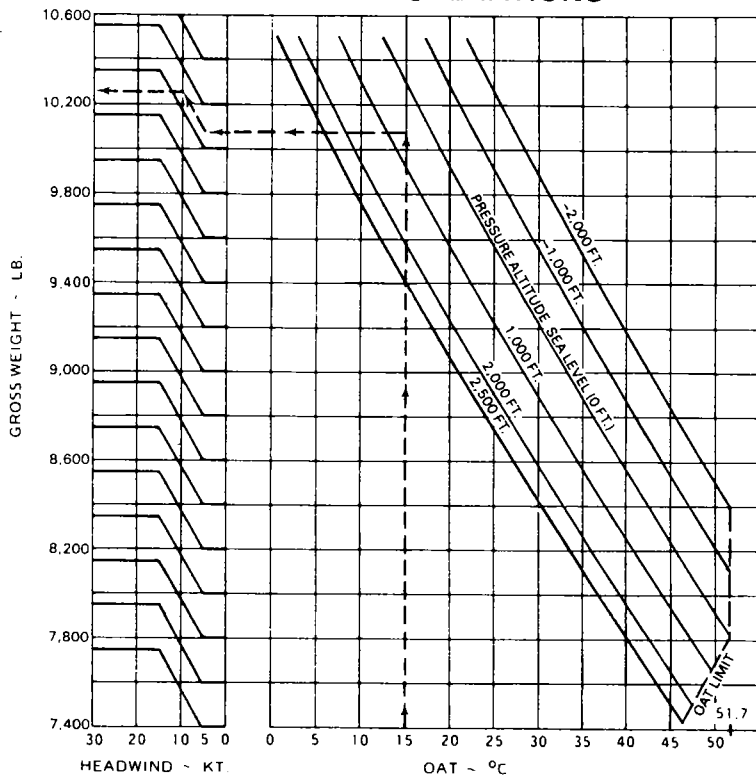
For vertical takeoff procedures the Vertical takeoff altimeter must be set, 100% N2, RPM-flat pitch, doors and windows closed, heater and vent off.

NOTE

Doors and windows remain closed, heater off, vent off until CDP is reached. This is required to prevent possible errors in the Vertical takeoff altimeter.

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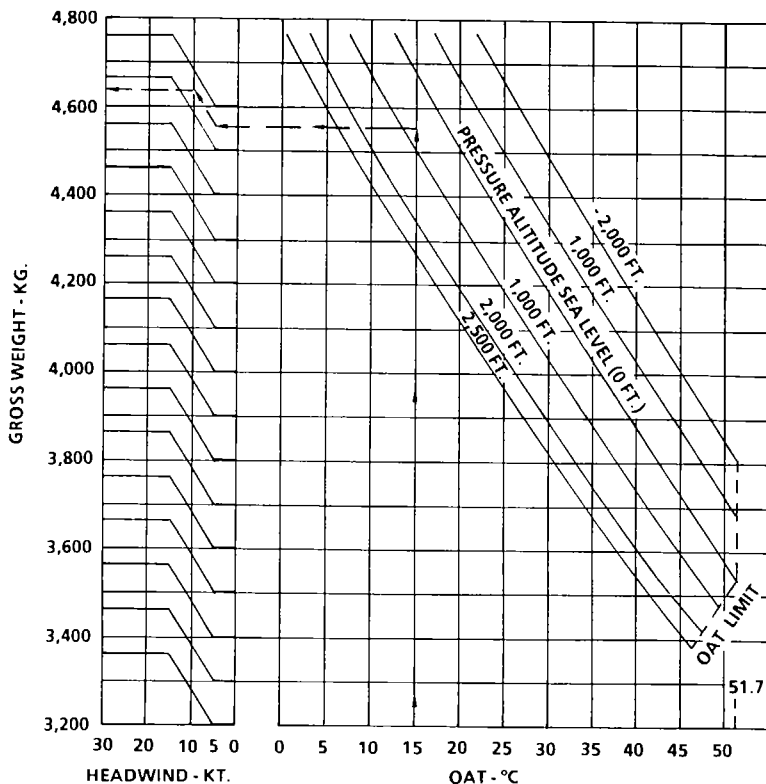
VERTICAL OPERATIONS



NOTE: SEE
HEADWIND
COMPONENT
CHART, SECTION 4.

GROSS WEIGHT LIMITS FOR TAKEOFF AND LANDING (ENGLISH)

VERTICAL OPERATIONS



NOTE: SEE
HEADWIND
COMPONENT
CHART, SECTION 4.

GROSS WEIGHT LIMITS FOR TAKEOFF AND LANDING
(METRIC)

AIRSPPEED LIMITATIONS

No change to basic envelope. For takeoff and landing airspeed refer to Performance Section.

AMBIENT AIR TEMPERATURE — OPERATING LIMITATIONS

See Performance Section in this Supplement.

CROSSWIND LIMITATIONS

The cross wind limit is that combination of wind velocity and direction where the cross wind component exceeds 15 knots. Refer to "Headwind Component Chart" in Section 4.

TYPE OF OPERATION

Category A operation is approved for day/night VMC, non-icing conditions. Night takeoff and landing may be accomplished with adequate lighting.

FLIGHT CREW

The minimum crew for vertical type takeoff and vertical type landing operations consists of two pilots.

CONFIGURATION

Skid landing gear only. All doors on.

Pratt and Whitney PT6T-3B or PT6T-3BF engines shall be installed.

CENTER OF GRAVITY LIMITS — FOR VERTICAL OPERATIONS ONLY

Center of gravity limits are from Station 130.0 (3302.0) to Station 142.0 (3606.8). The center of gravity operational range is variable, depending upon gross weight and shall be computed from the weight and balance data.

NOTE

Station 0 (datum) is located 20 inches (508.0 millimeters) aft of the most forward point of the cabin nose

Maximum asymmetric center of gravity limits are 3.5 inches (88.9 millimeters) to the left and right from the fuselage center line.

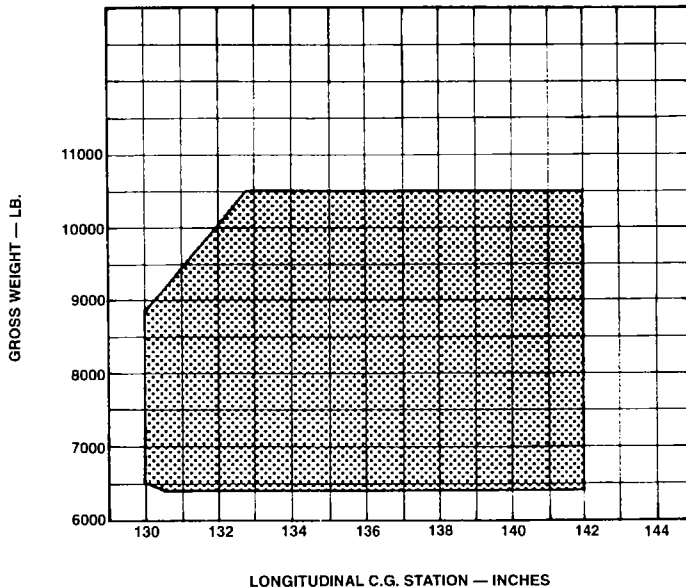
GENERATOR AMMETER

Maximum = 150 Amps per Ammeter

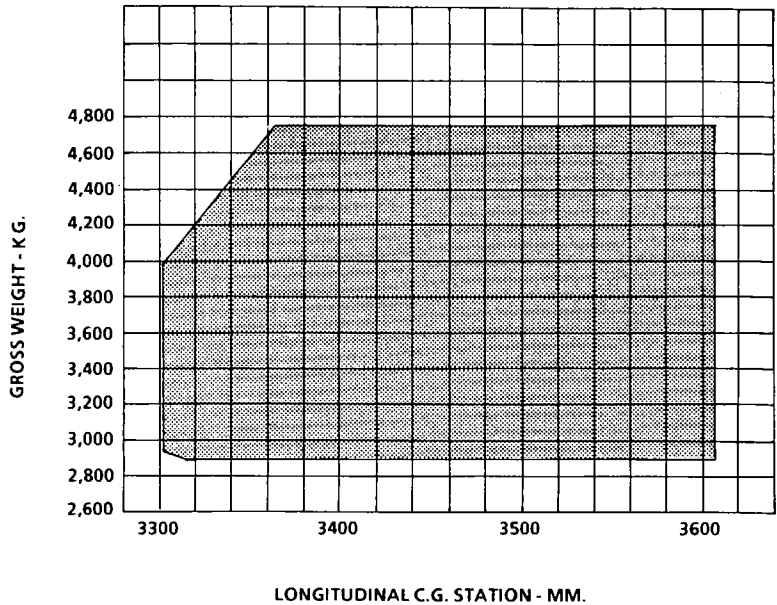
BHT-412-FMS-22.2

REQUIRED EQUIPMENT

This supplement requires the installation of an approved copilots instrument kit, an approved dual control kit, the 212-706-029 altimeter, and one operative SCAS.



CENTER OF GRAVITY LIMITS (ENGLISH)



CENTER OF GRAVITY LIMITS
(METRIC)

412099-14

CENTER OF GRAVITY LIMITS
(METRIC)

Section 2

NORMAL PROCEDURES

Power Assurance Check - Refer to Section 4 of this Supplement.

VERTICAL TYPE TAKEOFF

NOTE

See Vertical Takeoff Profile, Vertical Takeoff Figure 1, Vertical Takeoff Figure 2 and Vertical Takeoff Figure 3.

NOTE

Takeoff will be initiated with the helicopter positioned such that the takeoff index marks are directly opposite the crew doors and the helicopter centered on the heliport. This will assure that the tail rotor is within the confines of the heliport.

Triple Tachometer - 100%

Collective - Flat Pitch (full down).

Doors and Windows - Closed

Heater and Vent Blower - Off During Takeoff

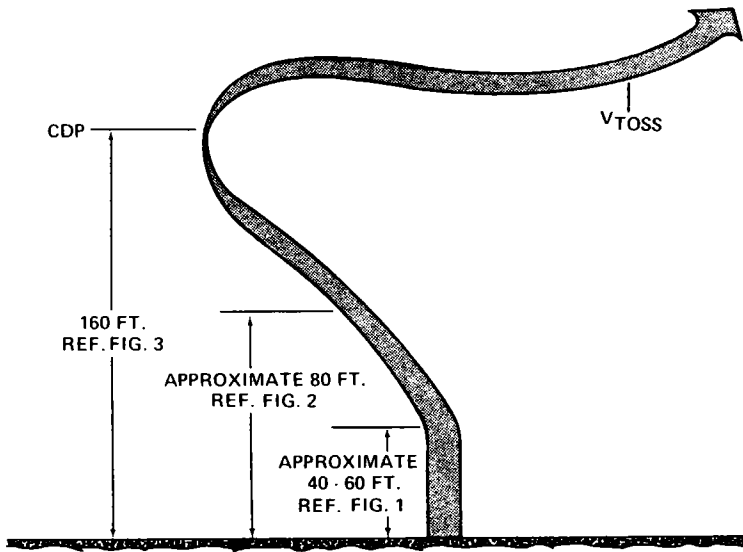
Vertical Takeoff Altimeter - Set to zero.

Flight Altimeter - Set to correct station pressure or elevation.

In Hover at Two to Four Feet - Note Transmission Torque.

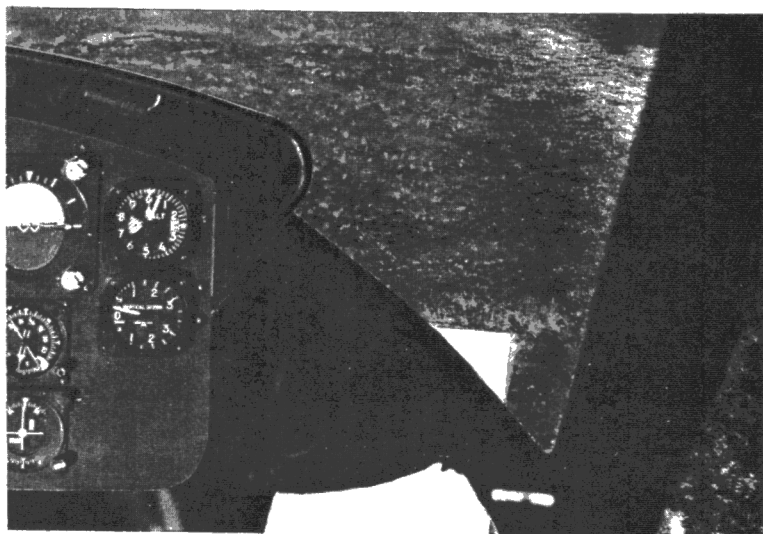
Collective - Apply smoothly to obtain a steady rate of climb along the takeoff flight path using a transmission torque not to exceed an additional 15% torque in excess of the value noted in hover.

BHT-412-FMS-22.2



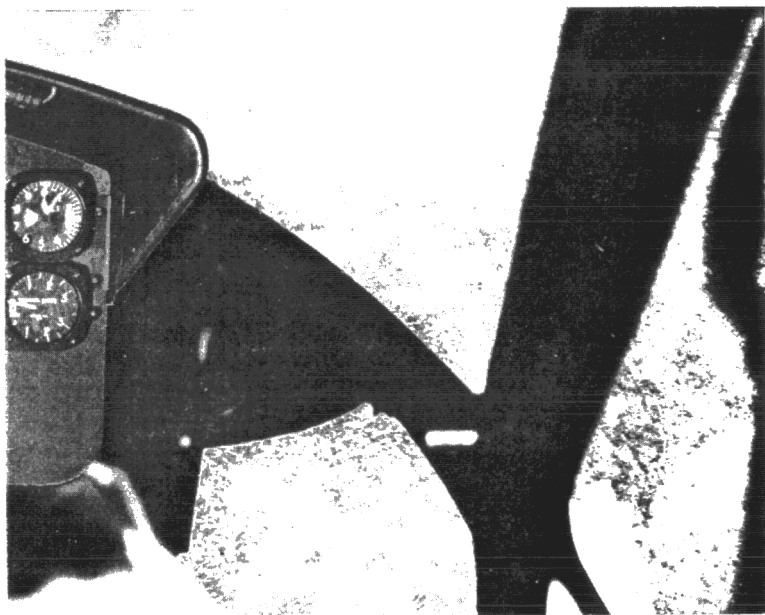
VERTICAL TAKEOFF PROFILE

BHT-412-FMS-22.2



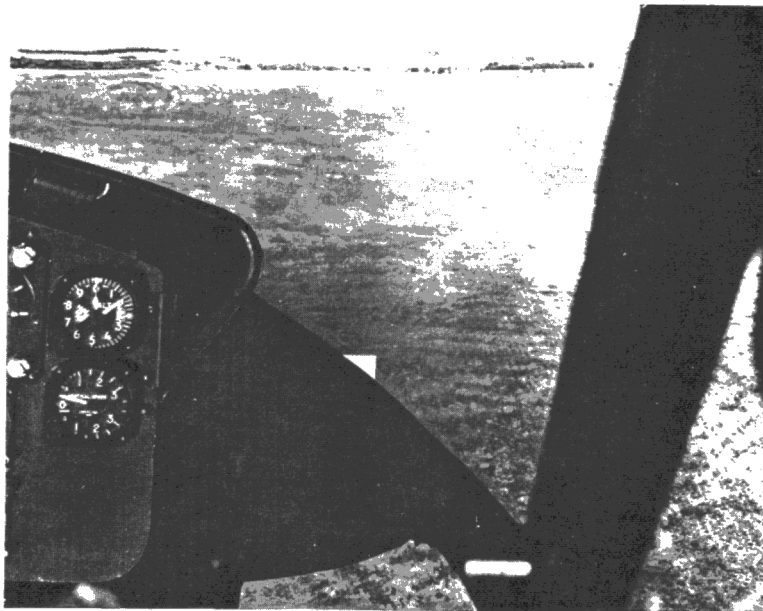
VERTICAL TAKEOFF
FIGURE 1

BHT-412-FMS-22.2



VERTICAL TAKEOFF
FIGURE 2

BHT-412-FMS-22.2



VERTICAL TAKEOFF
FIGURE 3

BHT-412-FMS-22.2

The pilot will control the flight path by visual reference to the far right corner of the heliport and the takeoff index marks. As altitude above the heliport is increased to approximately 40 to 60 feet, it will be necessary to transition into rearward flight at a very slow speed in order to maintain visual reference with the far right corner of the heliport (see Vertical Takeoff Figure 1). The takeoff will be continued by visual reference to the far right corner of the heliport (see Vertical Takeoff Figure 2) until the critical decision point (CDP) is reached (see Vertical Takeoff Figure 3).

NOTE

Visual reference with the heliport is defined as that position where the far right corner of the heliport is aligned approximately halfway between the edge of the instrument panel and the lower corner of the windshield. The amount of heliport area visible to the pilot will vary with height above the ground. At all points in the takeoff maintain visual contact with forward right hand corner of the heliport boundary.

Copilot will call out the following altitudes from the vertical takeoff altimeter during takeoff:

- a. At 60 feet indicated: "60 feet"
- b. At 80 feet indicated: "80 feet"
- c. At 100 feet indicated: "100 feet"
- d. At 120 feet indicated: "120 feet"
- e. At 140 feet indicated: "140 feet"
- f. At 160 feet indicated: "CDP, ROTATE"

At the indicated Critical Decision Point (CDP), translate into forward flight to obtain takeoff safety speed (VTOSS) of 30 knots plus wind (65 knots maximum). Apply power of not less than 73% torque and climb to 200 feet above takeoff point at VTOSS. Accelerate to best rate of climb airspeed (65 knots) and climb en route. Copilot should monitor power and systems parameters.

BHT-412-FMS-22.2

VERTICAL TYPE LANDING

NOTE

See Vertical Landing Profile, Vertical Landing Figure 1 and Vertical Landing Figure 2.

A vertical type landing is a landing that is initiated from the Landing Decision Point (LDP) which is 30 knots indicated airspeed plus reported wind velocity and at an altitude of 200 feet above the level of the heliport surface.

Flight Controls - Adjust frictions to desired levels.

Governor Switches - Automatic

Twist Grip N1 Controls - Full Open - Throttle Frictions adjusted to desired level.

Engine RPM - 100% N2.

Force Trim - As desired.

Flight Altimeter - Set to nearest reporting station

From an airspeed of 30 knots indicated plus wind speed and a height of 200 feet, the pilot will initiate the approach when the LDP is reached. The LDP is reached when the pilot obtains the correct sight picture of the heliport (see Vertical Landing Figure 1). Pilot calls "LDP" and indicates the approach.

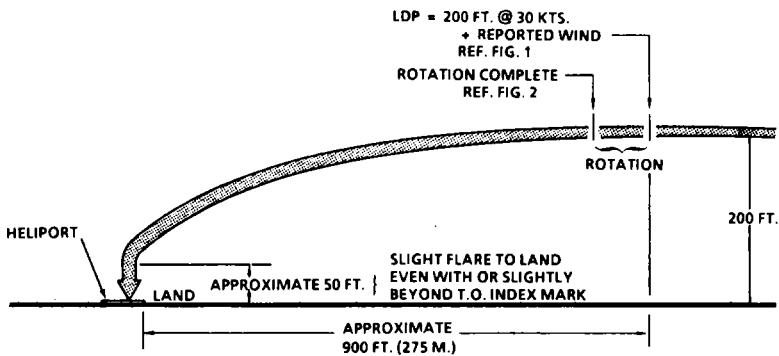
The approach is initiated by raising the nose of the aircraft to obtain the correct approach sight picture (see Vertical Landing Figure 2) and simultaneously lowering the collective to establish the approach angle.

During the descent, visual contact is maintained with the forward right hand corner of the heliport (see Vertical Landing Figure 1 and Vertical Landing Figure 2).

NOTE

The approach angle is such that the tail rotor will clear a 25-foot (7.6-meter) obstacle on the approach end of the heliport.

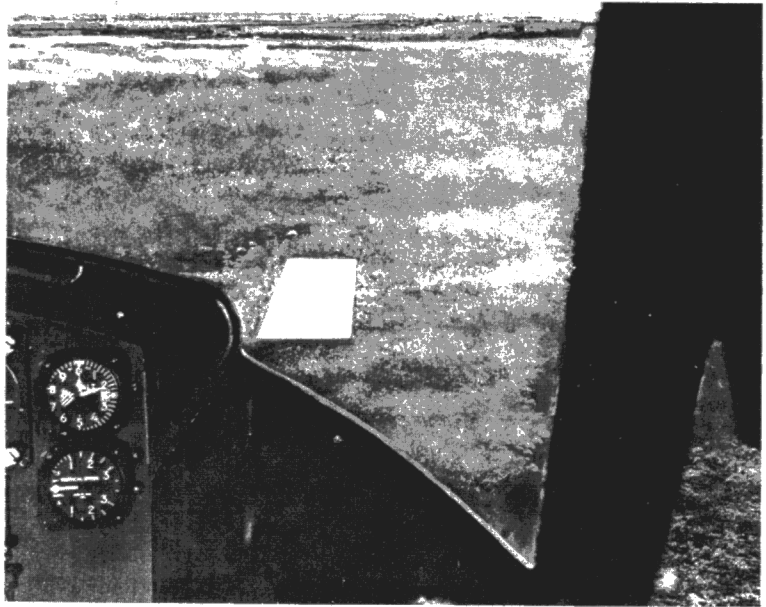
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VERTICAL LANDING PROFILE

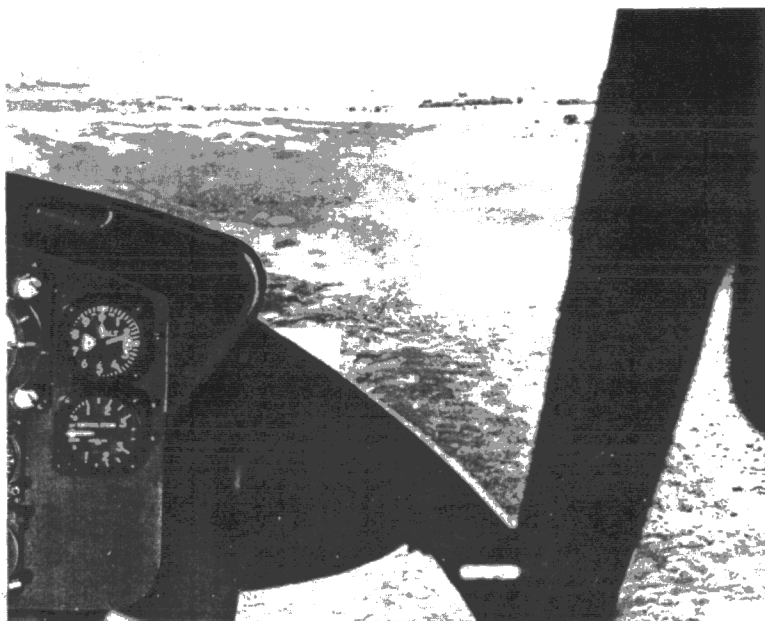
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VERTICAL LANDING
FIGURE 1

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VERTICAL LANDING
FIGURE 2

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As the helicopter crosses the approach end of the heliport with the required tail rotor obstacle clearance, a slight flare is initiated so that the helicopter is brought to a landing with the pilot's door even with or slightly forward of the takeoff mark on the heliport.

NOTE

The copilot will call out airspeed and altitude information prior to LDP (i.e., "Airspeed High" or "Airspeed Low," and "Altitude High" or "Altitude Low") and rotor rpm, torque, and Inter Turbine Temperature (ITT) during the approach from the LDP. In the event of an engine failure after LDP, the pilot will adjust the power as soon as possible to obtain the maximum single engine power available during the descent.

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Section 3

EMERGENCY AND MALFUNCTION PROCEDURES

PRIOR TO CDP

<u>SEGMENT WORDING</u>	<u>FAULT CONDITION</u>	<u>CORRECTIVE ACTION</u>
C BOX OIL PRESS	Combining gearbox oil pressure below normal.	Land
C BOX OIL TEMP	Combining gearbox oil temperature above limit.	Land
XMSN OIL TEMP	Transmission oil temperature is above limit.	Land
XMSN OIL PRESS	Transmission oil pressure is low.	Land
BAGGAGE FIRE	Smoke in baggage compartment.	Land
FIRE 1 PULL	Fire indication in No. 1 Engine com- partment.	Land, then pull No. 1 han- dle, select MAIN bottle, then RESERVE if neces- sary, close No. 1 twist grip.
FIRE 2 PULL	Fire indication in No. 2 engine com- partment.	Land, then pull No. 2 han- dle, select MAIN bottle, then RESERVE if neces- sary, close No. 2 twist grip.

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PRIOR TO CDP(Cont)

<u>SEGMENT WORDING</u>	<u>FAULT CONDITION</u>	<u>CORRECTIVE ACTION</u>
ENG 1 OUT	No. 1 Engine N1 RPM abnormally low.	See Engine Out Procedure.
ENG 2 OUT	No. 2 Engine N1 RPM abnormally low.	See Engine Out Procedure.

AFTER CDP

<u>SEGMENT WORDING</u>	<u>FAULT CONDITION</u>	<u>CORRECTIVE ACTION</u>
C BOX OIL PRESS	Combining gearbox oil pressure below normal.	Accelerate to VTOSS and land as soon as practicable.
C BOX OIL TEMP	Combining gearbox oil temperature above limit.	Accelerate to VTOSS and land as soon as practicable.
XMSN OIL TEMP	Transmission oil Temperature is above limit.	Accelerate to VTOSS and land as soon as practicable.
XMSN OIL PRESS	Transmission oil pressure is low.	Accelerate to VTOSS and land as soon as practicable.
BAGGAGE FIRE	Smoke in baggage compartment.	Accelerate to VTOSS and land as soon as practicable.
FIRE 1 PULL	Fire indication in No. 1 Engine compartment.	Accelerate to VTOSS and then pull No. 1 handle, select MAIN bottle, then RESERVE if necessary, close No. 1 twist grip.

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AFTER CDP(Cont)

<u>SEGMENT WORDING</u>	<u>FAULT CONDITION</u>	<u>CORRECTIVE ACTION</u>
FIRE 2 PULL	Fire indication in No. 2 Engine compartment.	Accelerate to VTOSS and then pull No. 2 handle, select MAIN bottle, then RESERVE if necessary, close No. 2 twist grip.
ENG 1 OUT	No. 1 Engine N1 RPM abnormally low.	See Engine Out Procedure.
ENG 2 OUT	No. 2 Engine N1 RPM abnormally low.	See Engine Out Procedure.

PRIOR TO LDP

<u>SEGMENT WORDING</u>	<u>FAULT CONDITION</u>	<u>CORRECTIVE ACTION</u>
C BOX OIL PRESS	Combining gearbox oil pressure below normal.	Reduce power. Land as soon as practicable.
C BOX OIL TEMP	Combining gearbox oil temperature above limit.	Reduce power. Observe temperature within limits, if not land as soon as possible.
XMSN OIL TEMP	Transmission oil temperature is above limit.	Reduce power. Observe temperature within limits, if not land as soon as possible.
XMSN OIL PRESS	Transmission oil pressure is low.	Reduce power. Land as soon as possible.
BAGGAGE FIRE	Smoke in baggage compartment.	Reduce power to minimum required. Land as soon as possible, and inspect tail boom area for damage.

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PRIOR TO LDP(Cont)

SEGMENT WORDING	FAULT CONDITION	CORRECTIVE ACTION
FIRE 1 PULL	Fire indication in No. 1 Engine com- partment	Pull No. 1 handle, select MAIN bottle, then RESERVE if necessary, close No. 1 twist grip.
FIRE 2 PULL	Fire indication in No. 2 Engine com- partment	Pull No. 2 handle, select MAIN bottle, then RESERVE if necessary, close No. 2 twist grip.
ENG 1 OUT	No. 1 Engine N1 RPM abnormally low.	See Engine Out Procedure.
ENG 2 OUT	No. 2 Engine N1 RPM abnormally low.	See Engine Out Procedure.

AFTER LDP

SEGMENT WORDING	FAULT CONDITION	CORRECTIVE ACTION
C BOX OIL PRESS	Combining gearbox oil pressure below normal.	Land.
C BOX OIL TEMP	Combining gearbox oil temperature above limit.	Land.
XMSN OIL TEMP	Transmission oil temperature is above limit.	Land
XMSN OIL PRESS	Transmission oil pressure is low.	Land.
BAGGAGE FIRE	Smoke in baggage compartment.	Land.

AFTER LDP (Cont)

<u>SEGMENT WORDING</u>	<u>FAULT CONDITION</u>	<u>CORRECTIVE ACTION</u>
FIRE 1 PULL	Fire indication in No. 1 Engine compartment.	Land, then pull No. 1 handle, select MAIN bottle, then RESERVE if necessary, close No. 1 twist grip.
FIRE 2 PULL	Fire indication in No. 2 Engine compartment.	Land, then pull No. 2 handle, select MAIN bottle, then RESERVE if necessary, close No. 2 twist grip.
ENG 1 OUT	No. 1 Engine N1 RPM abnormally low.	See Engine Out Procedure.
ENG 2 OUT	No. 2 Engine N1 RPM abnormally low.	See Engine Out Procedure.

ENGINE OUT PROCEDURE**NOTE**

After an engine failure, the power on the remaining engine shall be increased to the maximum permissible power limits (2.5 minute power rating for helicopters equipped with PT6T-3B engines or 30 minute power rating for helicopters equipped with PT6T-3BF engines). The rotor speed shall be maintained within limits.

During Takeoff Prior to Critical Decision Point (CDP)

An engine failure prior to reaching CDP (160 feet above the heliport) will necessitate a landing back to the heliport. The landing is accomplished by descending back toward the takeoff surface, while maintaining 97% rotor speed. While maintaining a level attitude, increase collective pitch as necessary to cushion the landing. Perform normal shutdown procedure.

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During Takeoff After Critical Decision Point (CDP)

If an engine fails during or following CDP, the helicopter shall be accelerated to takeoff safety speed (VTOSS) of 30 knots plus wind. When accelerating to VTOSS, rotor speed shall be maintained within limits. Accomplish climbout at an airspeed of 30 knots plus headwind. Use the 2.5 minute power rating (for helicopters equipped with PT6T-3B engines) or the 30 minute power rating (for helicopters equipped with PT6T-3BF engines) at 200 feet above the heliport. Then accelerate to the best rate of climb speed (65 knots). The copilot should monitor power. Shut down affected engine.

Twist Grip - Rotate to full closed.

Fuel - OFF

Boost Pump - Off

Crossfeed - Override Close

Interconnect Valve - Open

During Landing Prior to Landing Decision Point (LDP)

An emergency condition during landing prior to the LDP (200 feet above the heliport), the helicopter should be accelerated to best rate-of-climb speed for climbout, depending on the terrain and obstacles. Shut down affected engine.

Twist Grip - Rotate to full closed

Fuel - Off

Boost Pump - Off

Crossfeed - Override Close

Interconnect Valve - Open

OR, proceed to the LDP and use the procedure below.

During Landing at the Landing Decision Point (LDP)

Maintain rotor speed within limits and accomplish a climbout or accelerate to the best rate-of-climb speed of 65 knots, depending on the terrain and obstacles.

OR, proceed to a landing while using the procedure below.

During Landing After the Landing Decision Point (LDP)

If an emergency occurs after LDP, the helicopter is committed to land. The landing is accomplished using the 2.5 minute power rating (for helicopters equipped with PT6T-3B engines) or the 30 minute power rating (for helicopters equipped with PT6T-3BF engines). The rotor speed shall be maintained within limits. Visual contact with the sight picture of the two sides of the heliport shall be maintained. After landing, perform normal shutdown procedures.

MALFUNCTION PROCEDURES

Section 3 No change.

Section 4

PERFORMANCE DATA

The performance data presented in this section are based on the engine manufacturer's minimum specification power for the PT6T-3B or PT6T-3BF engines with installation losses.

An engine power assurance check chart is presented which is also based on a minimum specification engine with installation losses and proper rigging of engine controls.

If engine performance does not meet that shown in the "ENGINE POWER ASSURANCE CHART" steps should be taken to ascertain the causes of engine power loss.

The minimum heliport size, using the vertical takeoff and landing procedure, is 72 feet x 150 feet (22 meters x 46 meters). A chart is presented showing the MINIMUM HELIPORT SIZE AND AN APPROVED HELIPORT MARKING. The heliport marking shown was used during the type tests.

The gross weight limits for takeoff and landing varies with the component of head wind directly opposed to the flight path. The Headwind Component Chart presents a method of obtaining the headwind component for use on the Gross Weight Limits for Takeoff and Landing Chart, Section 1. The maximum cross wind demonstrated was 15 Knots and is shown as a limit on the Headwind Component Chart.

Interpolation of all data is allowable but extrapolation is not permitted.

HOVER PERFORMANCE

The Hover Performance chart is presented to show the percent torque required to hover in-ground-effect (IGE) at a four-foot (1.2 meters) skid height, and the percent torque available as shown in the following example:

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Torque Required to Hover

1. Determine Ambient Air Temperature (OAT) 29°C
2. Determine Pressure Altitude (H_p) 1000 ft.
3. Determine Gross Weight (G.W.) 8500 lb. (3855.6 kg.)
4. Enter chart at OAT, proceed vertically upward to H_p , proceed horizontally to the right to G.W., then proceed vertically downward to the TORQUEMETER scale and read percent torque required to hover (Transmission Torque). 59.5%

Torque Available (Twin Engine)

1. Enter chart at OAT, proceed vertically upward to H_p , proceed horizontally to the right to the TAKEOFF POWER AVAILABLE curves, then proceed vertically downward to the TORQUEMETER scale and read percent torque available. 100%

TAKEOFF PERFORMANCE

Takeoff Safety Speed (V_{toss})

The takeoff safety speed varies with wind ($V_{toss} = 30 \text{ kt. IAS} + \text{wind}$).

Altimeter Calibration

Takeoff performance is based on the altimeter calibration shown in the Takeoff Altimeter Calibration Chart.

Takeoff Distance

1. Vertical Takeoff Distance. 150 ft. (46 m.)

Using the vertical takeoff procedure, the takeoff distance measured from the aft end of the heliport when in takeoff position, is the maximum

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distance needed to land after a rejected takeoff (one engine failure prior to or at critical decision point for the vertical takeoff procedure). This takeoff distance applies to all conditions of GROSS WEIGHT, PRESSURE ALTITUDE, AND AMBIENT AIR TEMPERATURE When operating within allowable limits.

Takeoff Flight Path, Obstacle Clearance

The takeoff flight path begins at the end of the takeoff distance, at 35 feet above the takeoff surface and V_{toss} . Two charts are involved in the determination of the takeoff flight path and are titled as below:

Takeoff Flight Path, Climb Index

Takeoff Flight Path, Obstacle Clearance.

These charts are used as in the following example.

	Example
1. Determine Ambient Air Temperature	15°C
2. Determine Pressure Altitude	Sea Level (0 ft)
3. Determine VTOSS (30 kt. IAS + Headwind Component of 10 Knots)	40 kts.
4. Determine Actual Gross Weight	8300 lb. (3765 kg.)
5. Determine height above and distance from the Takeoff Surface of known obstacles along the Flight Path.	

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6. Check Maximum Allowable Gross Weight 10,500 lb.
(4762.7 kg.)
If actual gross weight is less than allowable,
proceed.
7. Enter Takeoff Flight Path, Climb Index Chart at
Ambient Air Temperature.
8. Move Vertically up to Pressure Altitude.
9. Move Horizontally Right to Actual Gross Weight
for Takeoff.
10. Move Vertically Down to VTOSS Correction
Curves.
11. Move Diagonally, Parallel to VTOSS Correction
Curves to VTOSS.
12. Move Vertically Down to Climb Index Scale and
Read Climb Index. 46.0
13. On the TAKEOFF FLIGHT PATH OBSTACLE
CLEARANCE CHART locate CLIMB INDEX
point, which has just been determined, at a
height above takeoff surface of 200 feet. A line
from this point through 35 foot height at 0
Horizontal Distance represents the minimum
height flight path from the end of the takeoff
distance and should be compared with the height
of known obstacles along the flight path for
obstacle clearance.

CLIMB PERFORMANCE

Single Engine at Minimum Vtoss (30 kts IAS)

Single engine rate of climb at minimum VTOSS (30 IAS) is shown for 7000, 8000, 9000, 10,000, and 10,500 pounds gross weight. In addition, chart is shown for 3200, 3600, 4000, 4400, and 4762.7 kilograms gross weight. These curves are for general information only, since takeoff flight path chart is presented for flight path determination.

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Single Engine at Best Rate of Climb Speed (65 kts Vcal)

Single engine rate of climb at best rate of climb speed (65 kts Vcal) is unchanged from the basic Rotorcraft Flight Manual Data (Cat. B).

LANDING PERFORMANCE

Landing Distance, Vertical Procedure:

Actual Landing Distance	110 ft. (33.5 m.)
Scheduled Landing Distance	138 ft. (42.0 m.)

Using the vertical procedure, the landing distance is the actual distance needed for the tail rotor to clear a 25 foot (8 meter) height and come to a stop on the landing surface with only one engine operating.

The actual landing distance is 110 feet (33.5 meters). For scheduled landings the landing distance is 138 feet (42.0 meters). These landing distances apply to all conditions of GROSS WEIGHT, PRESSURE ALTITUDE, AND AMBIENT TEMPERATURE when operating within allowable limits.

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**MODEL 412
POWER ASSURANCE CHECK (HOVER)
PT6T-3B ENGINE**

HEATER/ECU — OFF.

THROTTLES:

TEST ENGINE — FULL OPEN, FRICTIONED.
OTHER ENGINE — FLIGHT IDLE.

NII RPM — 97%.

COLLECTIVE PITCH — INCREASE UNTIL LIGHT ON
SKIDS OR HOVERING. DO NOT EXCEED 810° ITT OR
100.8% NI RPM.

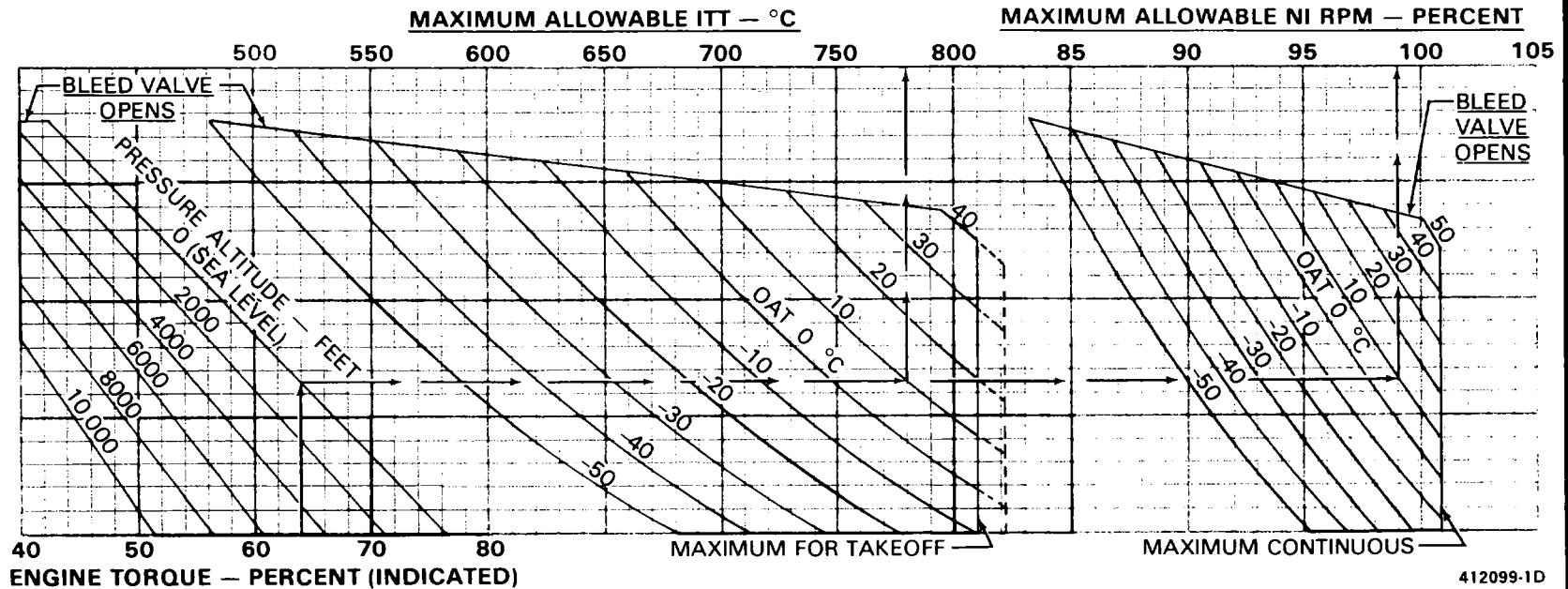
STABILIZE POWER ONE MINUTE, THEN RECORD
PRESSURE ALTITUDE, OAT, ENGINE TORQUE, ITT,
AND NI RPM.

ENTER CHART AT INDICATED ENGINE TORQUE, MOVE
UP TO INTERSECT PRESSURE ALTITUDE, PROCEED
TO THE RIGHT TO INTERSECT OUTSIDE AIR
TEMPERATURE, THEN MOVE UP TO READ VALUES
FOR MAXIMUM ALLOWABLE ITT AND NI RPM.

IF INDICATED ITT OR NI RPM EXCEEDS MAX
ALLOWABLE, REPEAT CHECK, STABILIZING POWER
FOUR MINUTES.

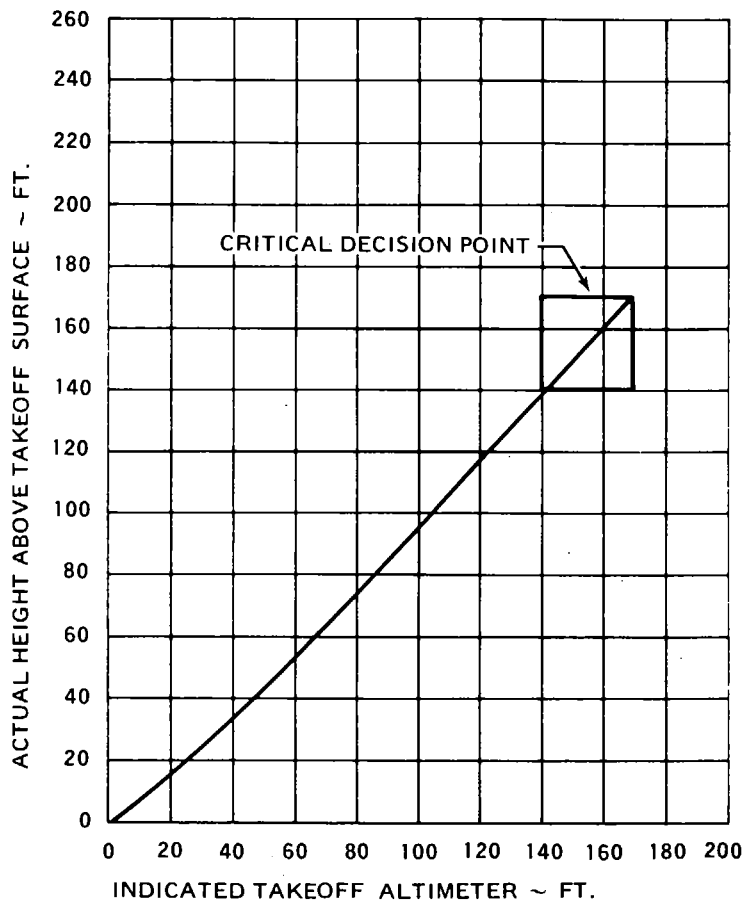
REPEAT CHECK USING OTHER ENGINE.

IF EITHER ENGINE EXCEEDS ALLOWABLE ITT OR NI
RPM AFTER STABILIZING FOUR MINUTES, PUBLISHED
PERFORMANCE MAY NOT BE ACHIEVABLE. CAUSE
SHOULD BE DETERMINED AS SOON AS PRACTICAL.



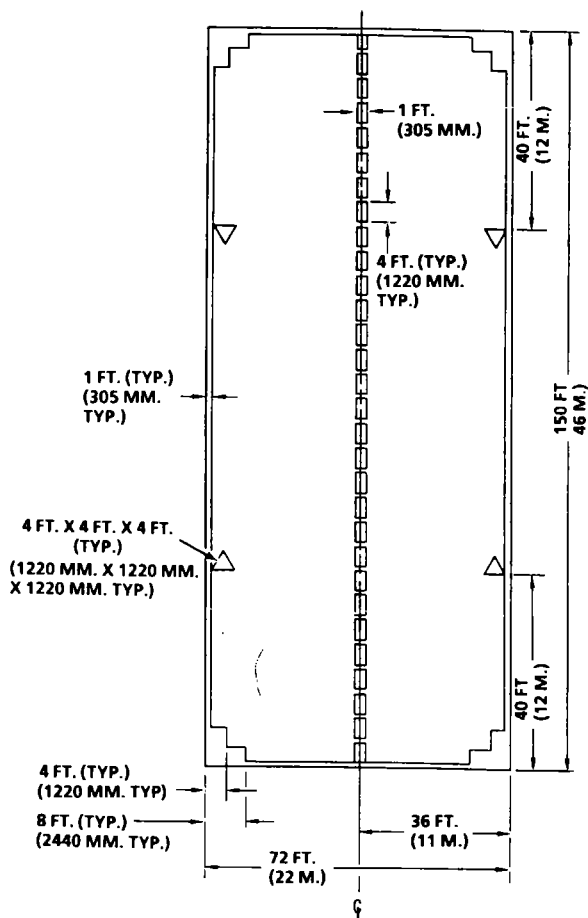
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TAKEOFF ALTIMETER CALIBRATION

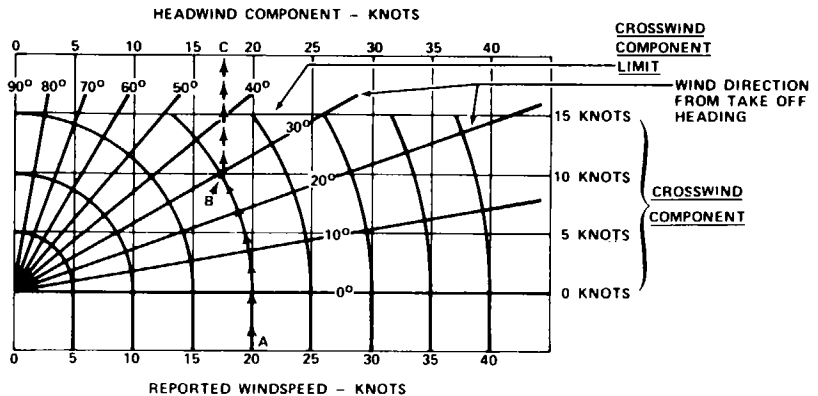
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MINIMUM HELIPORT SIZE AND
APPROVED HELIPORT MARKING

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EXAMPLE:

1. TAKE OFF HEADING.....170°
2. REPORTED WIND DIRECTION.....200°
3. WIND DIRECTION, DEGREES FROM TAKE OFF HEADING.....30°
4. REPORTED WIND SPEED.....20 KNOTS
5. ENTER CHART AT REPORTED WIND SPEED, POINT A
6. PROCEED UPWARD, FOLLOWING THE SHAPE OF THE CURVED LINES, TO WIND DIRECTION, DEGREES FROM TAKE OFF HEADING, POINT B.
7. PROCEED VERTICALLY UPWARD TO THE HEADWIND COMPONENT SCALE AND READ HEADWIND COMPONENT.....17.5 KNOTS

HEADWIND COMPONENT CHART

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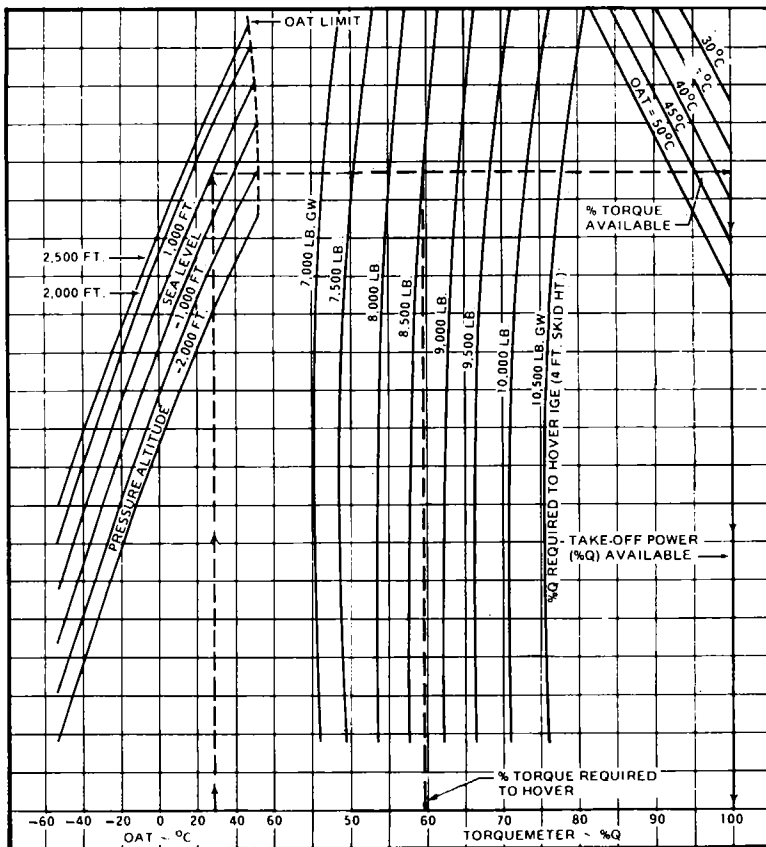
HOVER PERFORMANCE

(ENGLISH)

**IGE (4 FT. SKID HT.) POWER REQUIRED TO HOVER AND
TAKE-OFF POWER AVAILABLE**

ENGINE RPM 100%
GENERATOR 150 AMPS

**HEATER OFF
NO WIND**

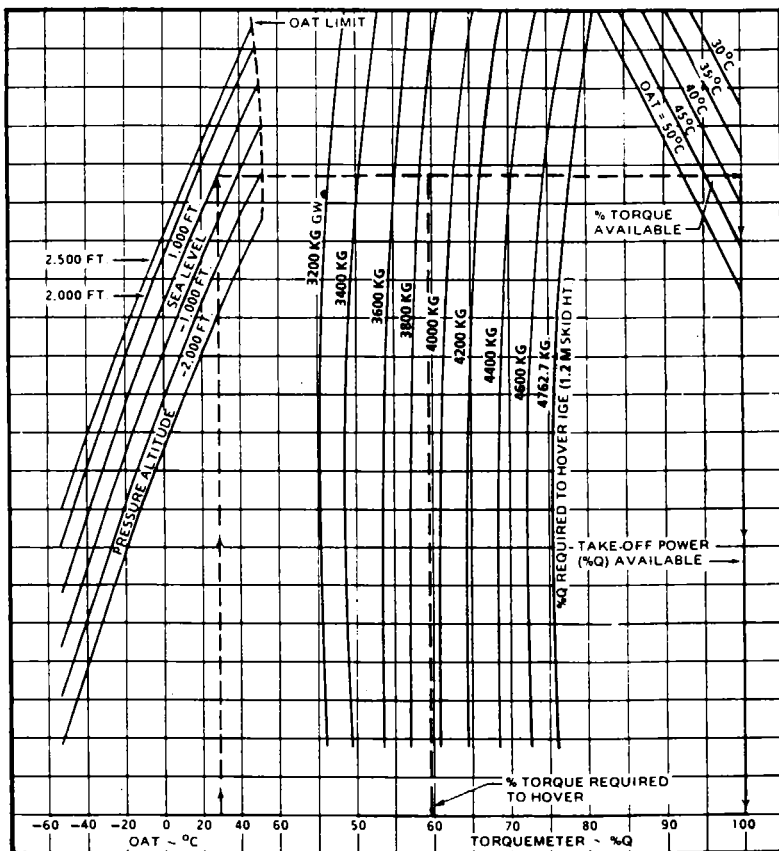


HOVER PERFORMANCE

IGE (1.2 M SKID HT.) POWER REQUIRED TO HOVER AND
TAKE-OFF POWER AVAILABLE

ENGINE RPM 100%
GENERATOR 150 AMPS

HEATER OFF
NO WIND



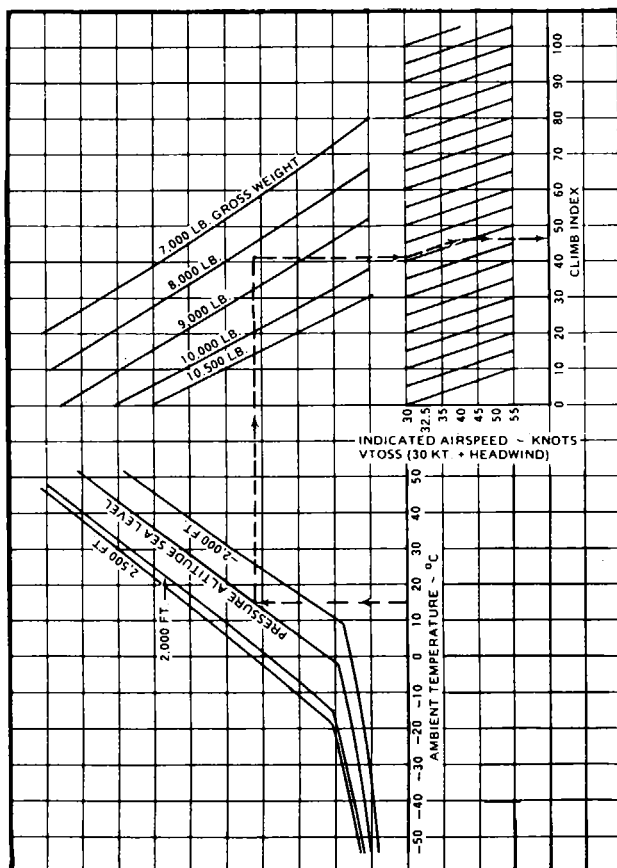
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PT6T-3B ENGINE TAKEOFF FLIGHT PATH (ENGLISH) CLIMB INDEX

2.5 MINUTE POWER
ENGINE RPM 97%
GENERATOR 150 AMPS

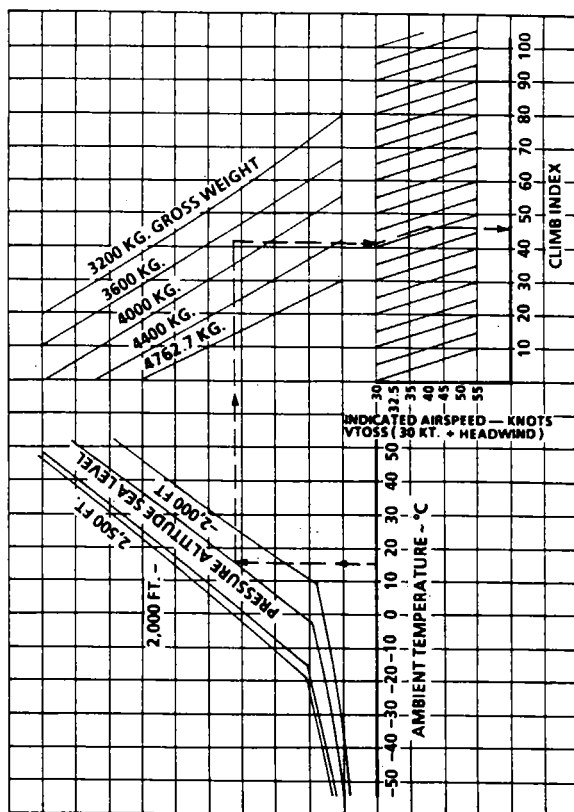
HEATER OFF
INOPERATIVE ENGINE SECURED
AIRSPEED = VTOSS (30 KT. IAS + HEADWIND)



**PT6T-3B ENGINE
TAKEOFF FLIGHT PATH
(METRIC)
CLIMB INDEX**

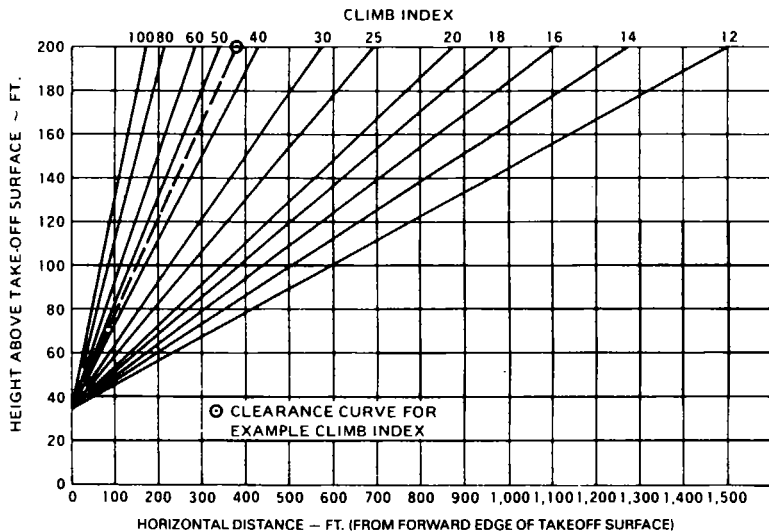
2.5 MIN. POWER
ENGINE RPM 97%
GENERATOR 150 AMPS

HEATER OFF
INOPERATIVE ENGINE SECURED
AIRSPEED = VTOSS (30 KT. IAS + HEADWIND)



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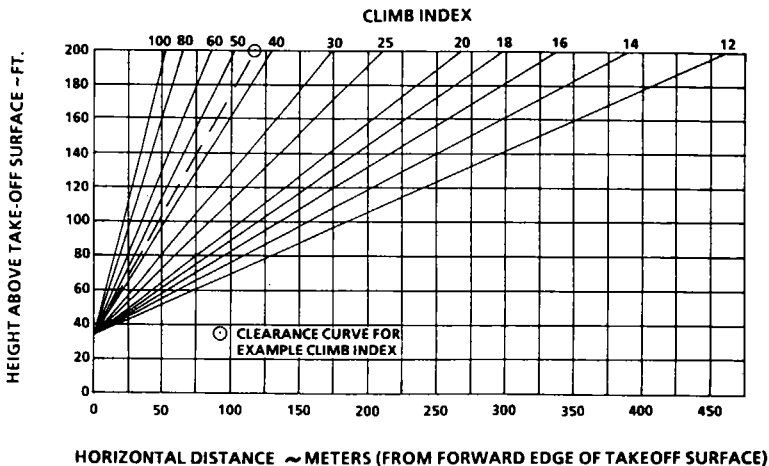
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**PT6T-3B ENGINE
TAKEOFF FLIGHT PATH
(ENGLISH)
OBSTACLE CLEARANCE****2.5 MINUTE POWER****ENGINE RPM 97%****GENERATOR 150 AMPS****HEATER OFF****INOPERATIVE ENGINE SECURED****AIRSPED VTOSS = 30 KT. IAS + HEADWIND**

**PT6T-3B ENGINE
TAKEOFF FLIGHT PATH
(METRIC)
OBSTACLE CLEARANCE**

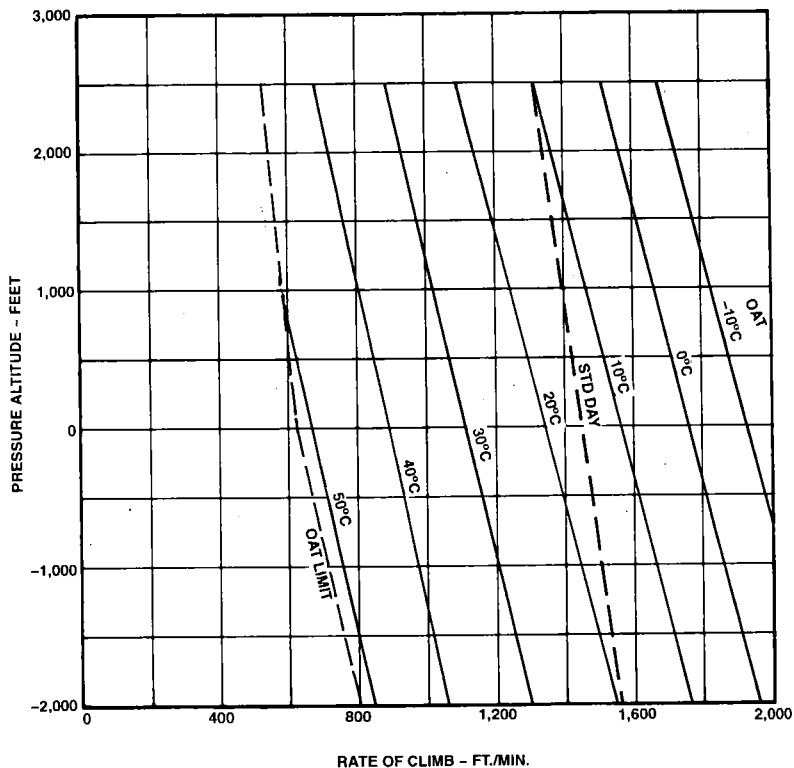
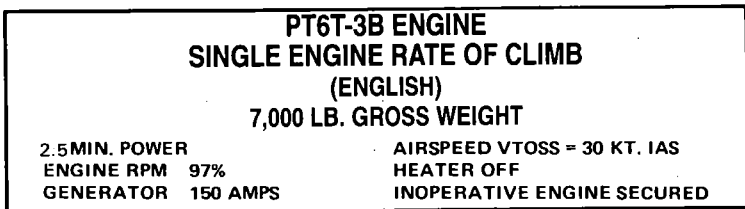
2.5 MIN. POWER
ENGINE RPM 97%
GENERATOR 150 AMPS

HEATER OFF
INOPERATIVE ENGINE SECURED
AIRSPEED VTOSS = 30 KT. IAS = HEADWIND



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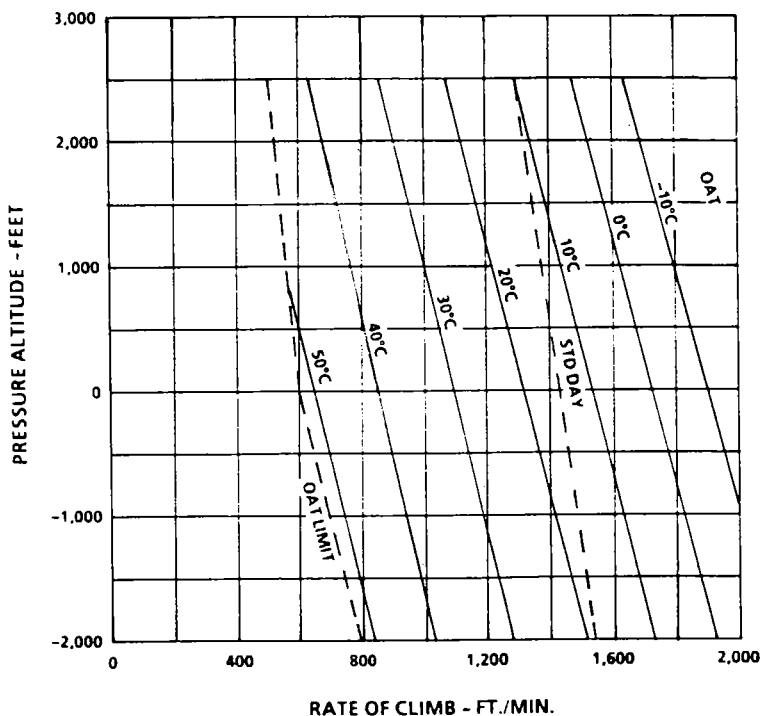
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**PT6T-3B ENGINE
SINGLE ENGINE RATE OF CLIMB
(METRIC)
3,200 KG. GROSS WEIGHT**

2.5 MIN. POWER
ENGINE RPM 97%
GENERATOR 150 AMPS

AIRSPEED VTOSS = 30 KT. IAS
HEATER OFF
INOPERATIVE ENGINE SECURED



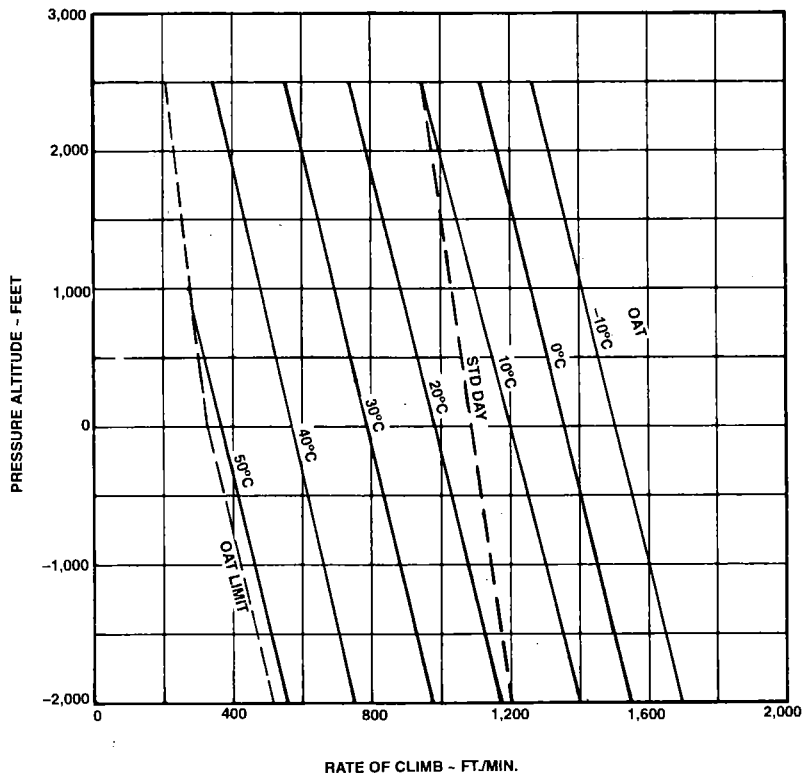
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PT6T-3B ENGINE
SINGLE ENGINE RATE OF CLIMB
(ENGLISH)
8,000 LB. GROSS WEIGHT

2.5 MIN. POWER
ENGINE RPM 97%
GENERATOR 150 AMPS

AIRSPED VTOSS = 30 KT. IAS
HEATER OFF
INOPERATIVE ENGINE SECURED

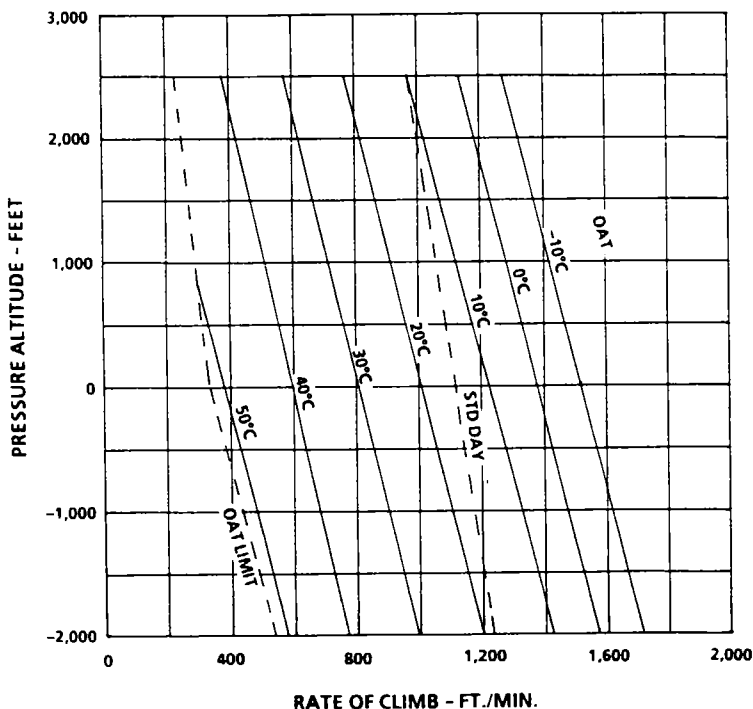


**PT6T-3B ENGINE
SINGLE ENGINE RATE OF CLIMB
(METRIC)**

3,600 KG. GROSS WEIGHT

2.5 MIN. POWER
ENGINE RPM 97%
GENERATOR 150 AMPS

AIRSPEED VTOSS = 30 KT. IAS
HEATER OFF
INOPERATIVE ENGINE SECURED



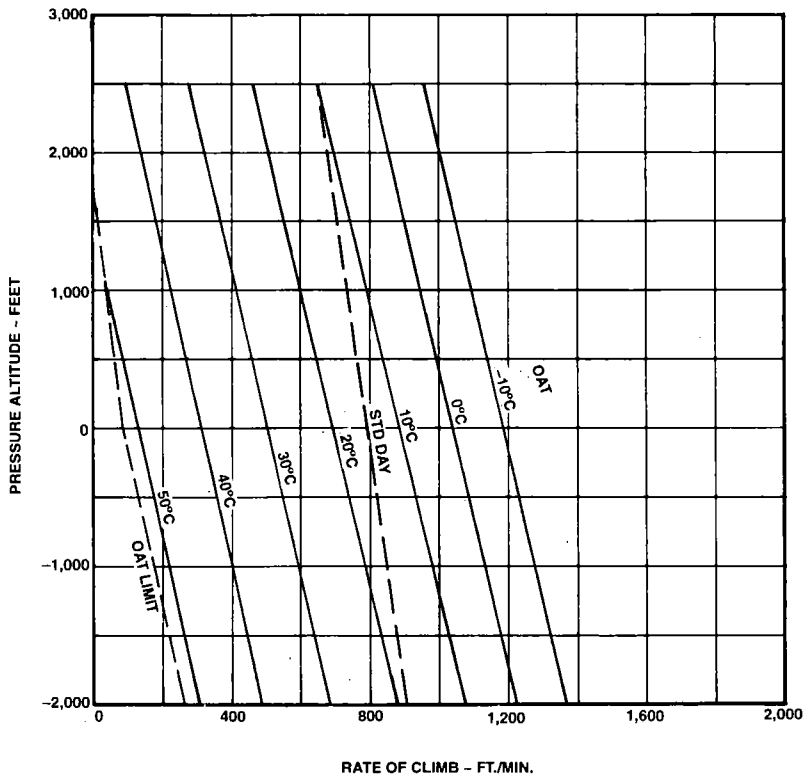
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PT6T-3B ENGINE
SINGLE ENGINE RATE OF CLIMB
(ENGLISH)
9,000 LB. GROSS WEIGHT

2.5 MIN. POWER
ENGINE RPM 97%
GENERATOR 150 AMPS

AIRSPED VTOSS = 30 KT. IAS
HEATER OFF
INOPERATIVE ENGINE SECURED

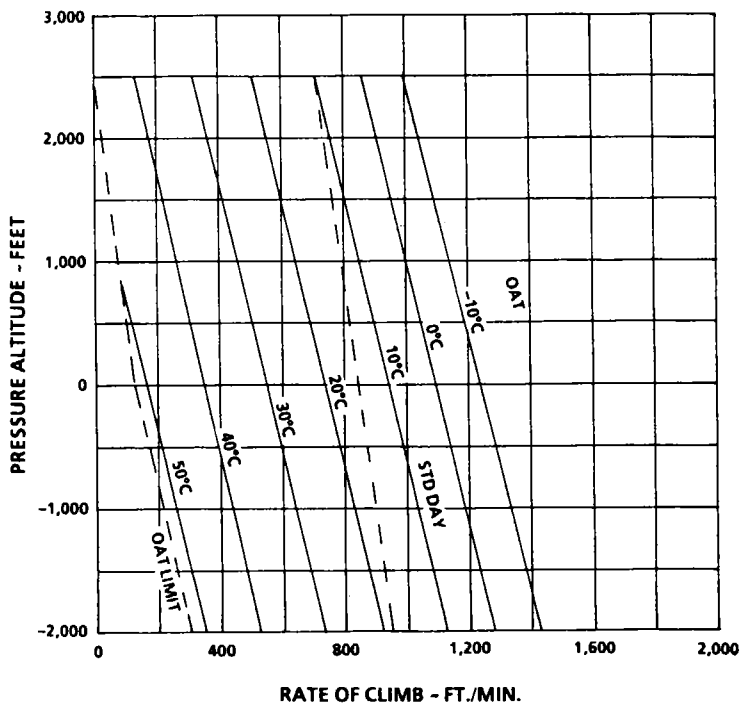


**PT6T-3B ENGINE
SINGLE ENGINE RATE OF CLIMB
(METRIC)**

4000 KG. GROSS WEIGHT

2.5 MIN. POWER
ENGINE RPM 97%
GENERATOR 150 AMPS

AIRSPED VTOSS = 30 KT. IAS
HEATER OFF
INOPERATIVE ENGINE SECURED



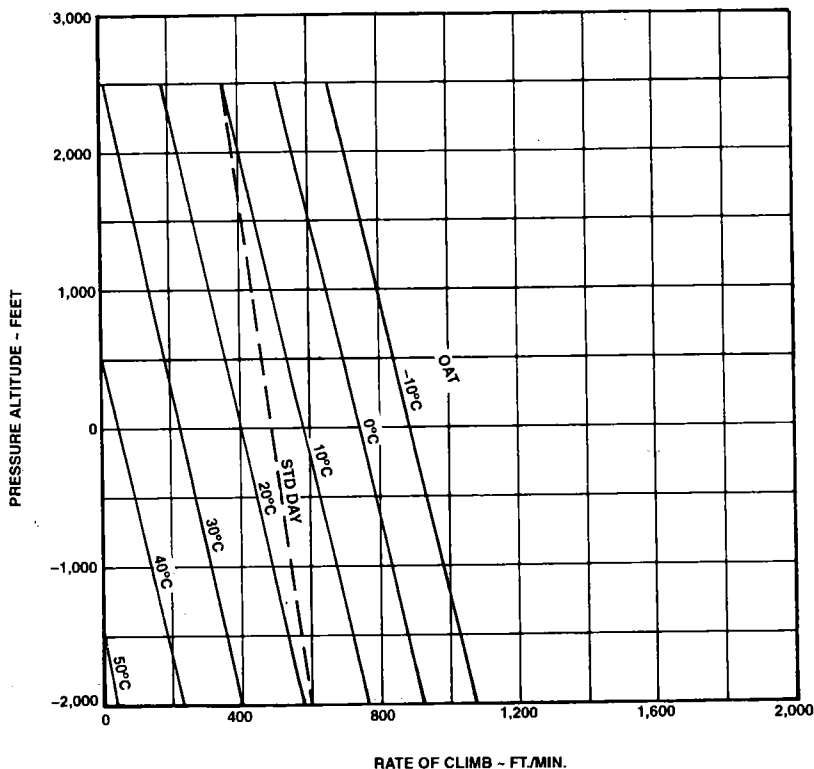
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PT6T-3B ENGINE
SINGLE ENGINE RATE OF CLIMB
(ENGLISH)
10,000 LB. GROSS WEIGHT

2.5 MIN. POWER
ENGINE RPM 97%
GENERATOR 150 AMPS

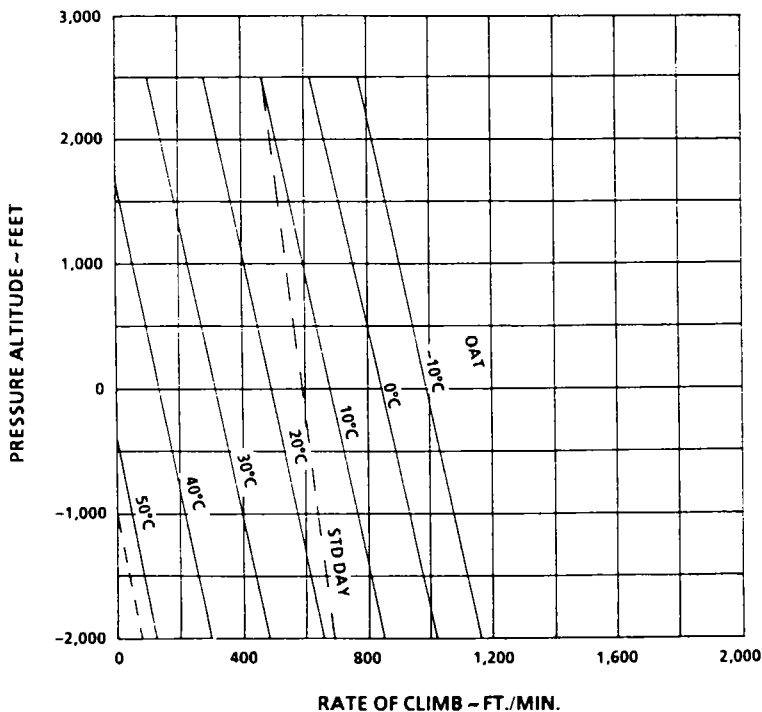
AIRSPED VTOSS = 30 KIAS
HEATER OFF
INOPERATIVE ENGINE SECURED



PT6T-3B ENGINE
SINGLE ENGINE RATE OF CLIMB
(METRIC)
4400 KG. GROSS WEIGHT

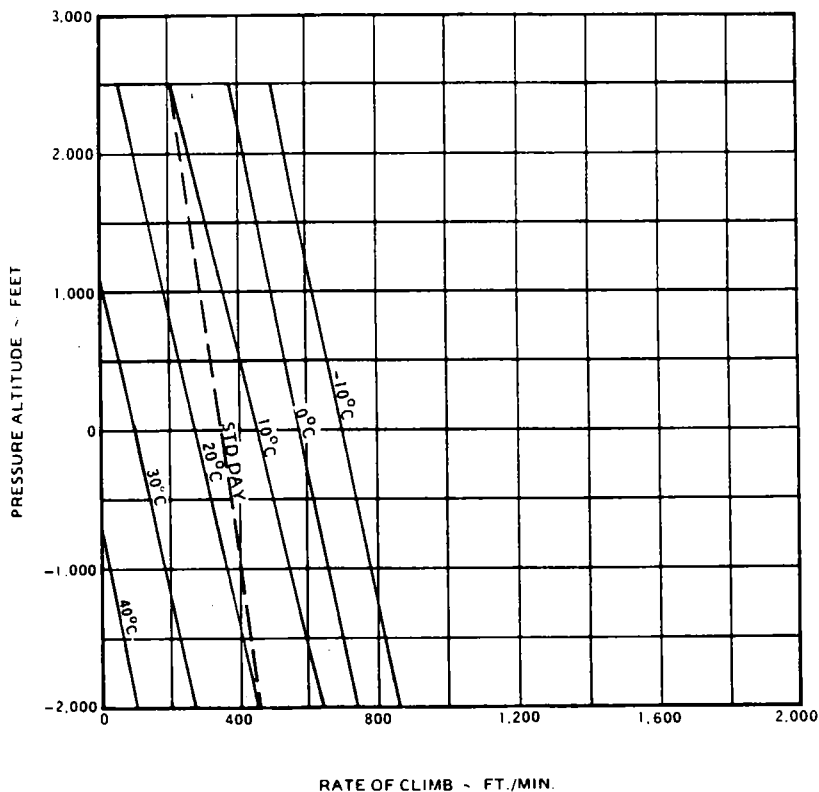
2.5 MIN. POWER
ENGINE RPM 97%
GENERATOR 150 AMPS

AIRSPEED VTOSS = 30 KIAS
HEATER OFF
INOPERATIVE ENGINE SECURED



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**PT6T-3B ENGINE
SINGLE ENGINE RATE OF CLIMB****10,500 LB. (4762.7 KG.) GROSS WEIGHT****2.5 MIN. POWER****AIRSPEED VTOSS = 30 KT. IAS****ENGINE RPM 97%****HEATER OFF****GENERATOR 150 AMPS****INOPERATIVE ENGINE SECURED**

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PT6T-3BF ENGINE TAKEOFF FLIGHT PATH

(ENGLISH)

CLIMB INDEX

HEATER OFF

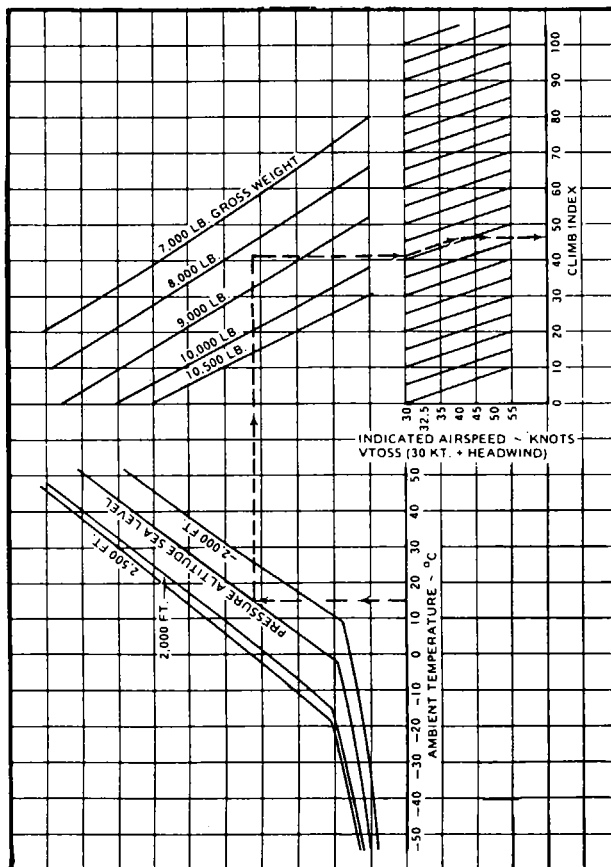
INOPERATIVE ENGINE SECURED

AIRSPED = VTOSS (30 KT. IAS + HEADWIND)

30 MINUTE POWER

ENGINE RPM 97%

GENERATOR 150 AMPS

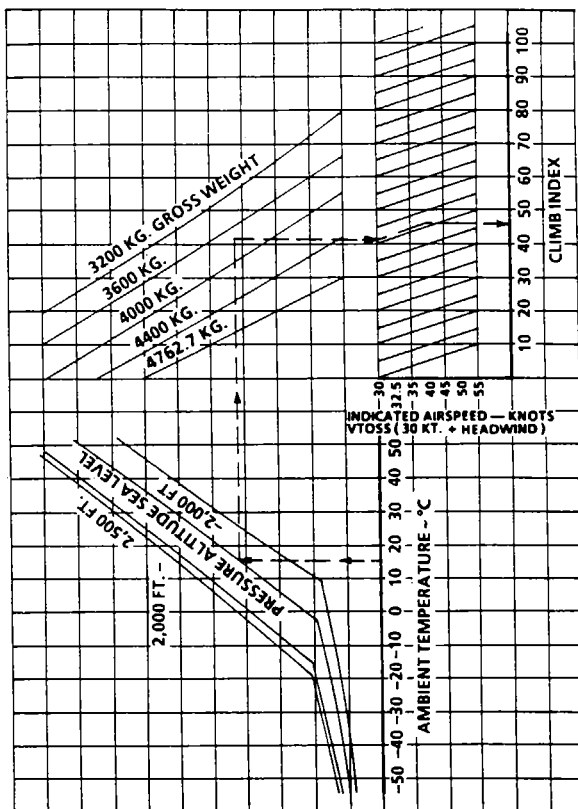


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**PT6T-3BF ENGINE
TAKEOFF FLIGHT PATH
(METRIC)
CLIMB INDEX**

30 MINUTE POWER
ENGINE RPM 97%
GENERATOR 150 AMPS

HEATER OFF
INOPERATIVE ENGINE SECURED
AIRSPEED = VTOSS (30 KT. IAS + HEADWIND)



412099-16

**PT6T-3BF ENGINE
TAKEOFF FLIGHT PATH
(ENGLISH)
OBSTACLE CLEARANCE**

30 MINUTE POWER

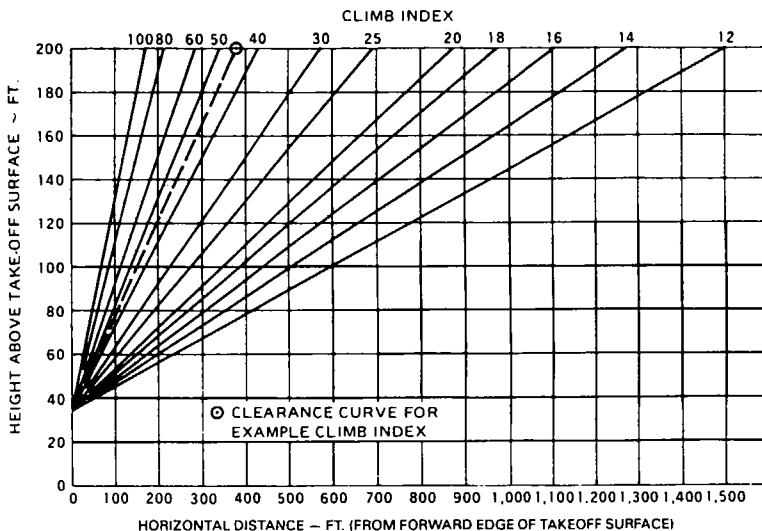
ENGINE RPM 97%

GENERATOR 150 AMPS

HEATER OFF

INOPERATIVE ENGINE SECURED

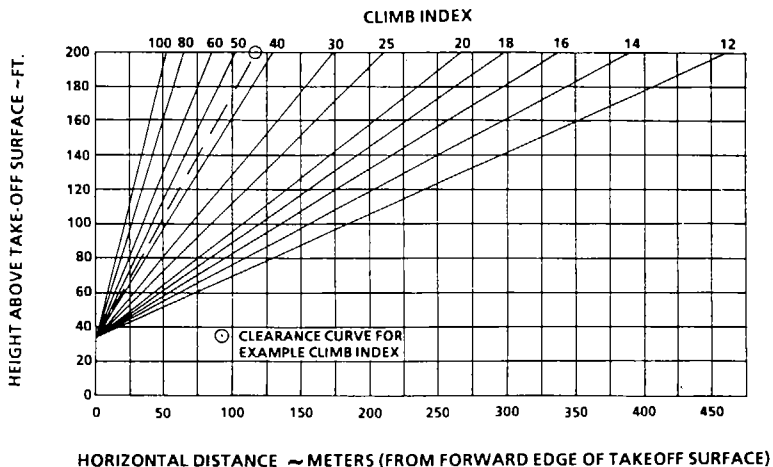
AIRSPED VTOSS = 30 KT. IAS + HEADWIND



BHT-412-FMS-22.2

**PT6T-3BF ENGINE
TAKEOFF FLIGHT PATH
(METRIC)
OBSTACLE CLEARANCE**

30 MINUTE POWER HEATER OFF
ENGINE RPM 97% INOPERATIVE ENGINE SECURED
GENERATOR 150 AMPS AIRSPEED VTOSS = 30 KT. IAS = HEADWIND

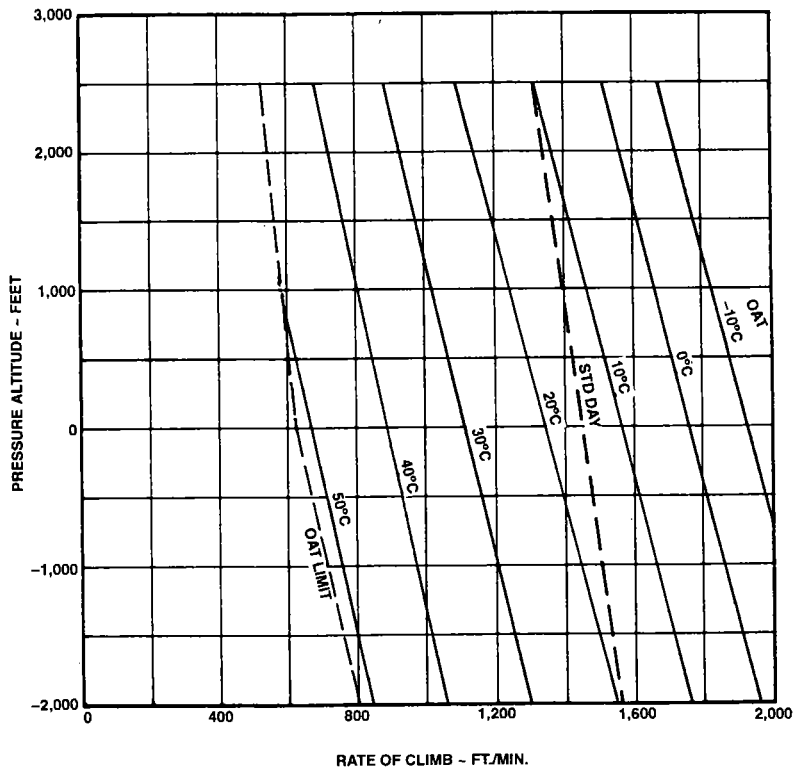


412099-17

PT6T-3BF ENGINE
SINGLE ENGINE RATE OF CLIMB
(ENGLISH)
7,000 LB. GROSS WEIGHT

30 MINUTE POWER
ENGINE RPM 97%
GENERATOR 150 AMPS

AIRSPEED VTOSS = 30 KT. IAS
HEATER OFF
INOPERATIVE ENGINE SECURED

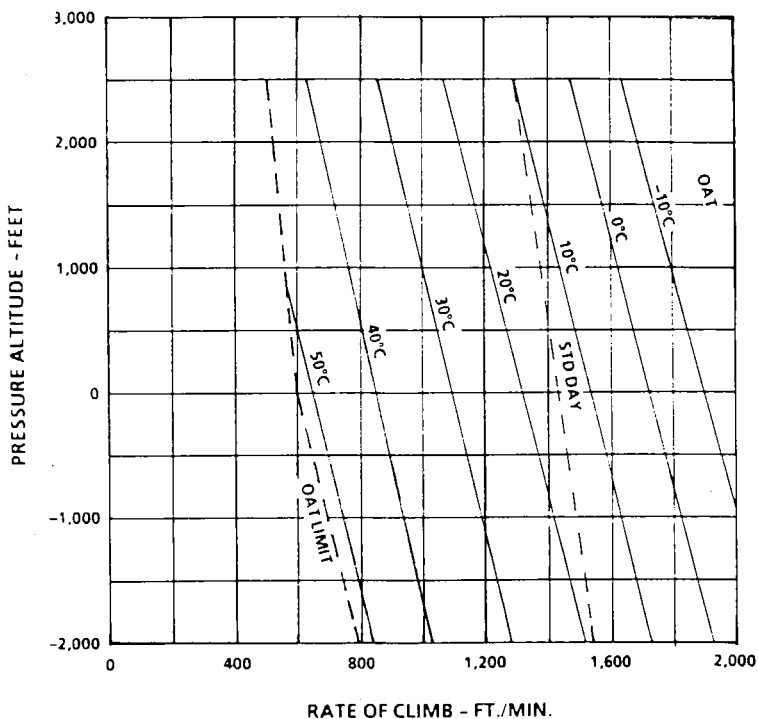


BHT-412-FMS-22.2

PT6T-3BF ENGINE
SINGLE ENGINE RATE OF CLIMB
(METRIC)
3,200 KG. GROSS WEIGHT

30 MINUTE POWER
ENGINE RPM 97%
GENERATOR 150 AMPS

AIR SPEED VTOSS = 30 KT. IAS
HEATER OFF
INOPERATIVE ENGINE SECURED

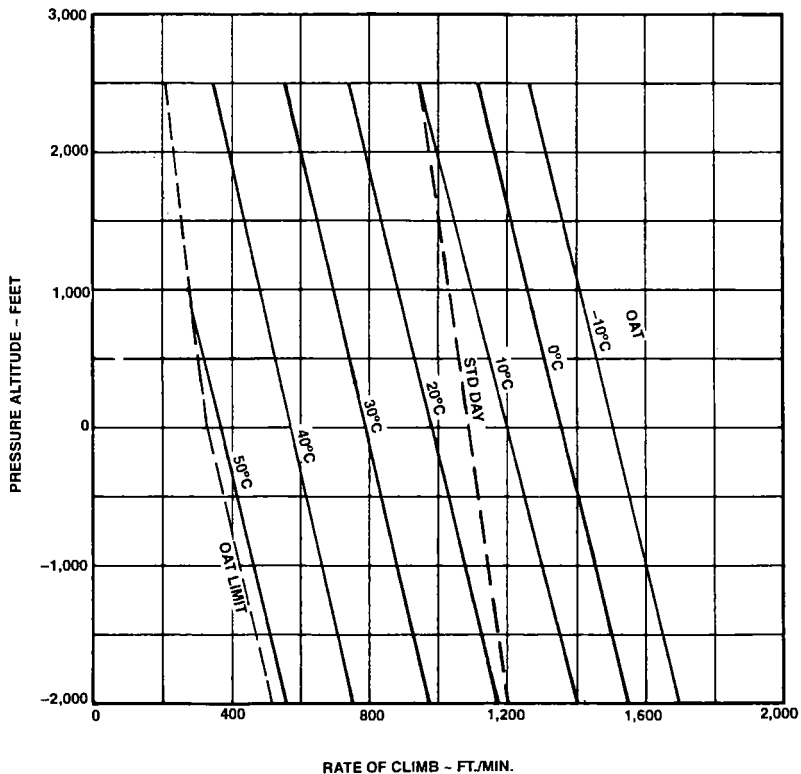


412099-18

**PT6T-3BF ENGINE
SINGLE ENGINE RATE OF CLIMB
(ENGLISH)
8,000 LB. GROSS WEIGHT**

30 MINUTE POWER
ENGINE RPM 97%
GENERATOR 150 AMPS

AIRSPEED VTOSS = 30 KT. IAS
HEATER OFF
INOPERATIVE ENGINE SECURED

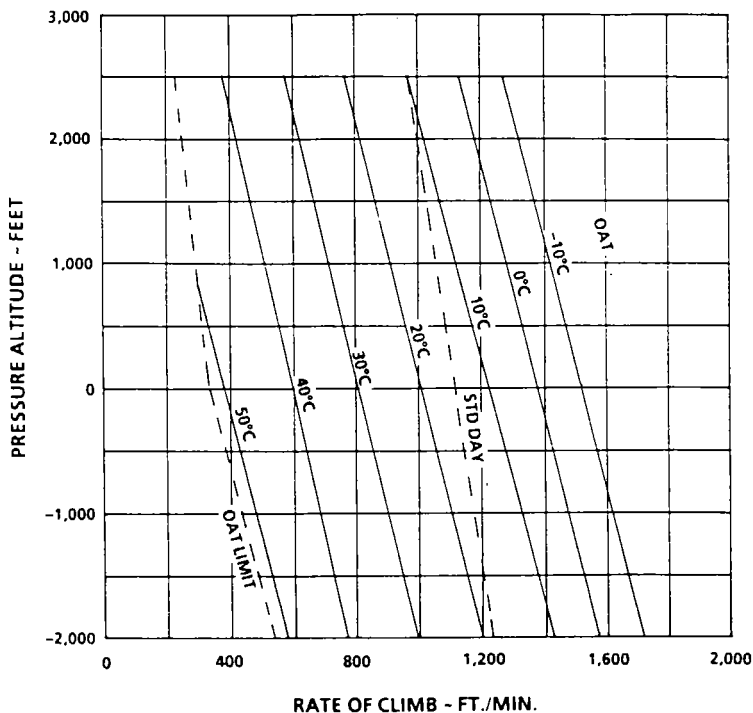


BHT-412-FMS-22.2

PT6T-3BF ENGINE
SINGLE ENGINE RATE OF CLIMB
(METRIC)
3,600 KG. GROSS WEIGHT

30 MINUTE POWER
ENGINE RPM 97%
GENERATOR 150 AMPS

AIRSPEED VTOSS = 30 KT. IAS
HEATER OFF
INOPERATIVE ENGINE SECURED

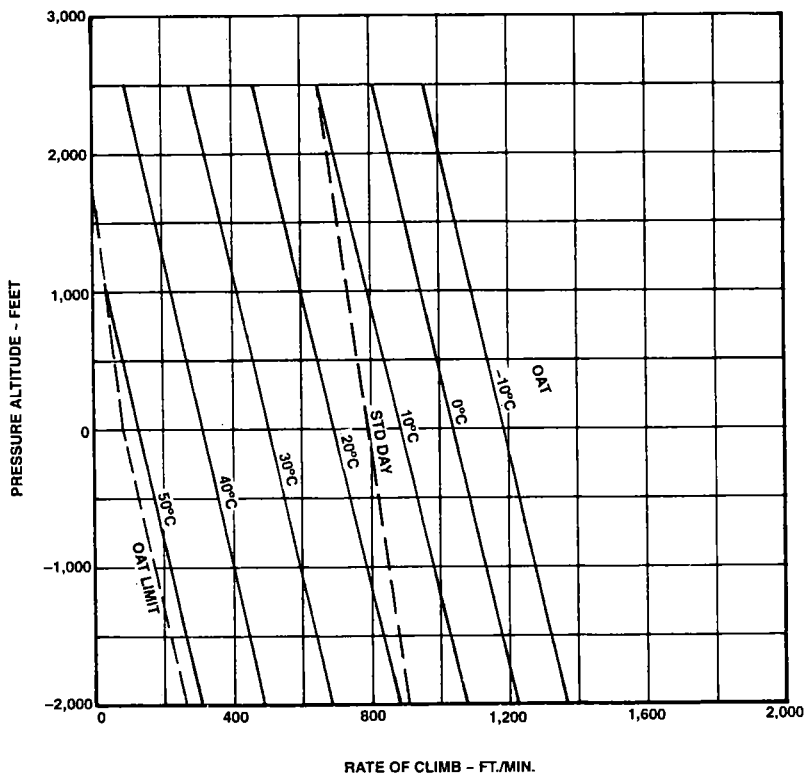


412099-19

PT6T-3BF ENGINE
SINGLE ENGINE RATE OF CLIMB
(ENGLISH)
9,000 LB. GROSS WEIGHT

30 MINUTE POWER
ENGINE RPM 97%
GENERATOR 150 AMPS

AIRSPED VTOSS = 30 KT. IAS
HEATER OFF
INOPERATIVE ENGINE SECURED

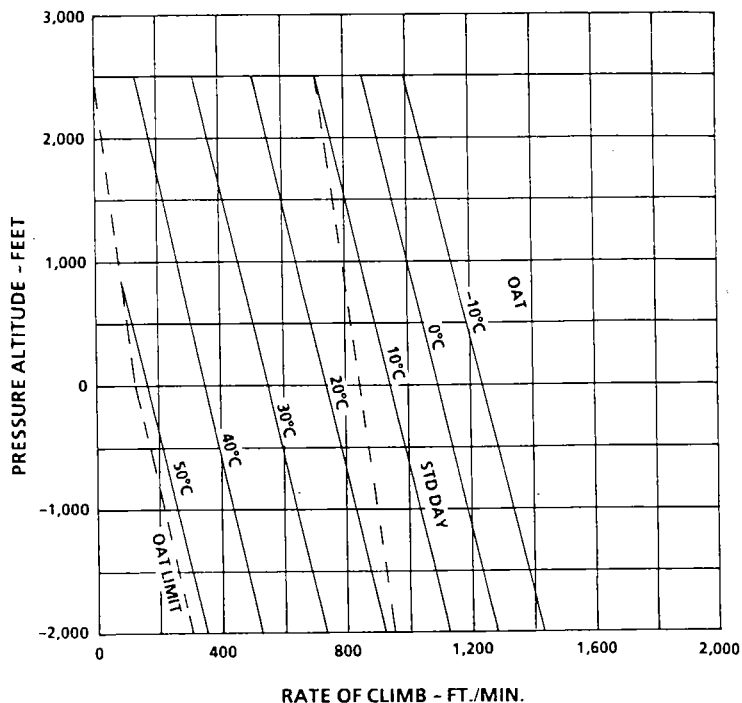


BHT-412-FMS-22.2

PT6T-3BF ENGINE
SINGLE ENGINE RATE OF CLIMB
(METRIC)
4000 KG. GROSS WEIGHT

30 MINUTE POWER
ENGINE RPM 97%
GENERATOR 150 AMPS

AIR SPEED VTOSS = 30 KT. IAS
HEATER OFF
INOPERATIVE ENGINE SECURED

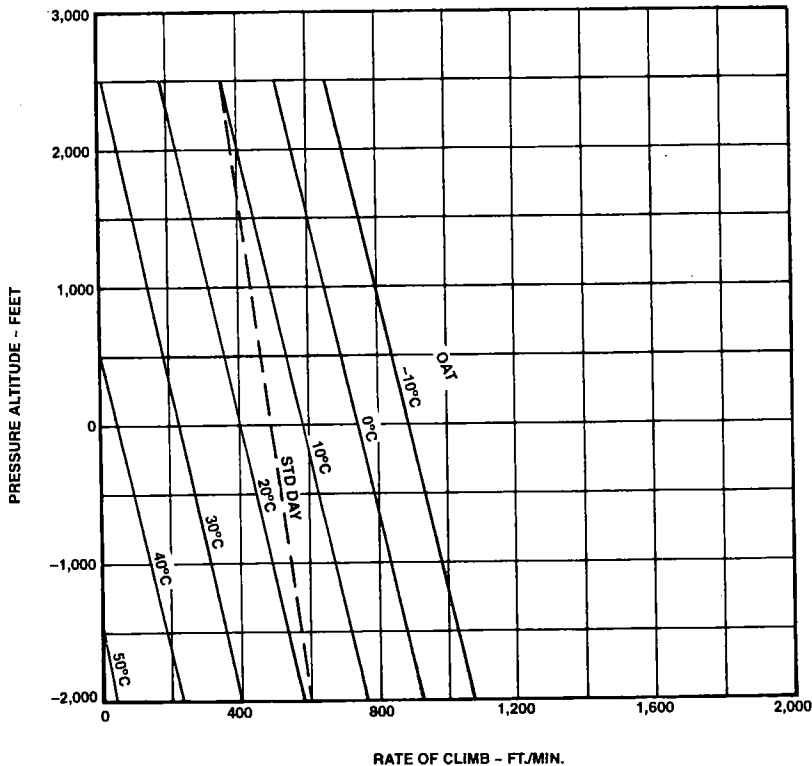


412099-20

PT6T-3BF ENGINE
SINGLE ENGINE RATE OF CLIMB
(ENGLISH)
10,000 LB. GROSS WEIGHT

30 MINUTE POWER
ENGINE RPM 97%
GENERATOR 150 AMPS

AIRSPEED VTOSS = 30 KIAS
HEATER OFF
INOPERATIVE ENGINE SECURED

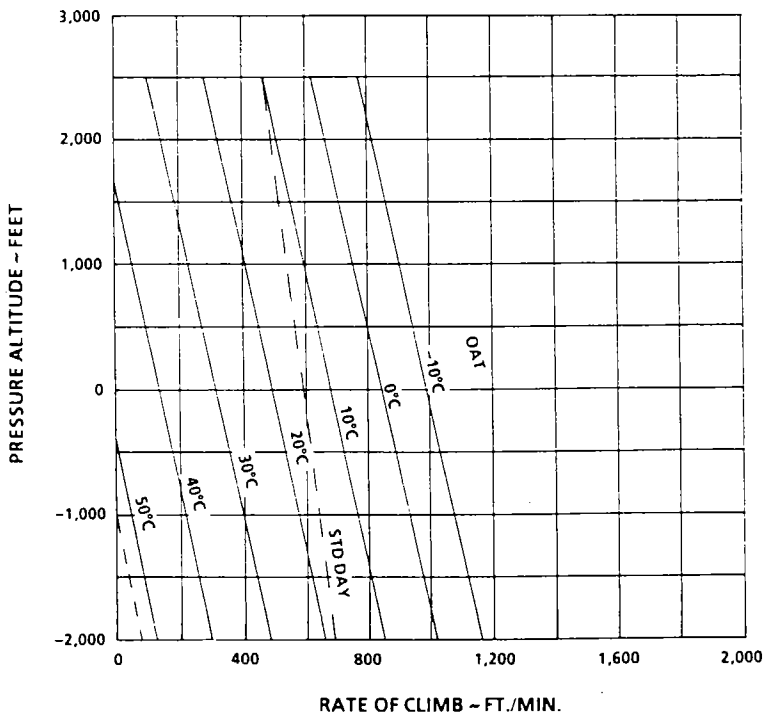


BHT-412-FMS-22.2

PT6T-3BF ENGINE
SINGLE ENGINE RATE OF CLIMB
(METRIC)
4400 KG. GROSS WEIGHT

30 MINUTE POWER
ENGINE RPM 97%
GENERATOR 150 AMPS

AIRSPED VTOSS = 30 KIAS
HEATER OFF
INOPERATIVE ENGINE SECURED



412099-21

**PT6T-3BF ENGINE
SINGLE ENGINE RATE OF CLIMB**

10,500 LB. (4762.7 KG.) GROSS WEIGHT

30 MINUTE POWER

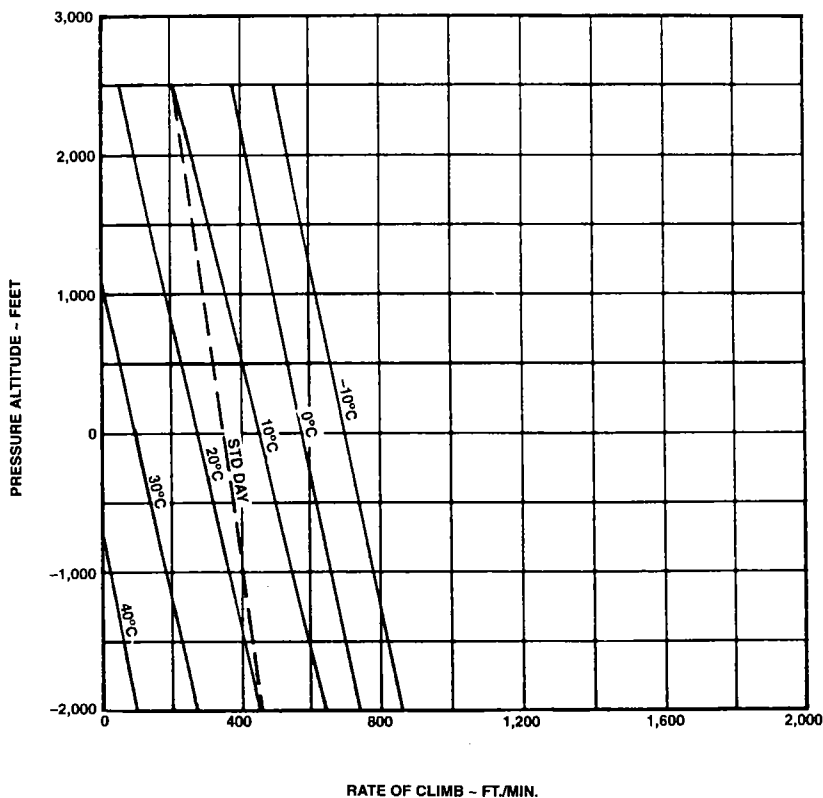
ENGINE RPM 97%

GENERATOR 150 AMPS

AIRSPEED VTOSS = 30 KT. IAS

HEATER OFF

INOPERATIVE ENGINE SECURED



412900-134



ROTORCRAFT FLIGHT MANUAL SUPPLEMENT

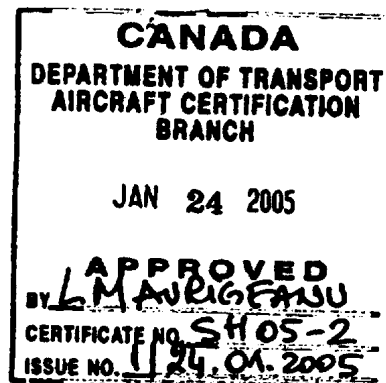
COLD WEATHER OPERATION WITH KEROSENE FUELS

CERTIFIED
24 JANUARY 2005

This supplement shall be attached to the Model 412 Flight Manual (BHT-412-FM-2) when conducting Cold Weather Operation with Kerosene Fuels.

Information contained herein supplements information in the basic Flight Manual. For Limitations, Procedures, and Performance Data not contained in this supplement, refer to the basic Flight Manual.

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24 JANUARY 2005

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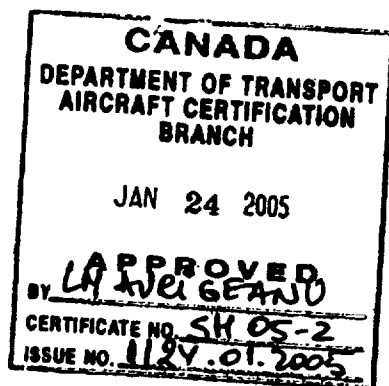
NOTE

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APPROVED



DATE JAN 24 2005

MANAGER

TRANSPORT CANADA
QUEBEC REGION
CIVIL AVIATION – AIRCRAFT CERTIFICATION

Section 1

LIMITATIONS

FUEL AND OIL

FUEL

Refer to Manufacturer's Data portion of this supplement for approved fuels list.

JET B OR JP-4

Fuel conforming to ASTM D-1655 Type B, MIL-T-5624 Grade JP-4, or NATO F-40 may be used at all ambient temperatures.

JET A, A-1, JP-5 OR JP-8 (KEROSENE TYPE FUELS)

1. Ambient Temperature Above -30°C (-22°F)

Fuel conforming to ASTM D-1655 Type A or A-1, MIL-T-5624 Grade JP-5, or NATO F-44 and MIL-T-83133 Grade JP-8, or NATO F-34 may be used without restriction.

2. Ambient Temperature Below -30°C (-22°F)

Operation with fuel conforming to ASTM D-1655 Type A is limited to ambient temperatures above -34°C (-29°F).

Operation with fuel conforming to ASTM D-1655 Type A-1, MIL-T-5624 Grade JP-5, or NATO F-44 and MIL-T-83133 Grade JP-8, or NATO F-34 is limited to ambient temperatures above -40°C (-40°F).

3. Engine Starting

Engine starting with fuel conforming to ASTM D-1655 Type A or A-1, MIL-T-5624 Grade JP-5, or NATO F-44 and MIL-T-83133 Grade JP-8, or NATO F-34 is limited to fuel temperatures above -30°C (-22°F).

Fuel temperature shall be measured by draining a quantity of fuel from the helicopter fuel tank and from the engine fuel inlet filter.

NOTE

Refer to Manufacturer's Data portion of this supplement for fuel temperature measurement procedure.

Section 2

NORMAL PROCEDURES

BEFORE EXTERIOR CHECK

Flight planning — Completed.

Gross weight and CG — Compute (refer to BHT-412-MD-2).

Publications — Checked.

Portable fire extinguishers — Condition and security.

Fuel — Measure fuel temperature as required.

NOTE

When OAT is below -30°C (-22°F), fuel temperature measurement procedure must be carried out for affected fuel types (refer to Limitations Section in this supplement). Fuel temperature measurement procedure is described in the Manufacturer's Data portion of this supplement.

Aft fuel sumps — Drain samples as follows:

FUEL TRANS switches — OFF.

BOOST PUMP switches — OFF.

ENGINE 1 and ENGINE 2 FUEL switches — OFF.

BAT BUS 1 switch — ON.

Aft fuel sump drain buttons (left and right) — Press.

NOTE

If aft sumps fail to drain, the sump valves may be operated manually.

Forward and middle fuel sumps — Drain samples as follows:

Press-to-drain valves — Press.

Fuel filters — Drain before first flight of day as follows:

BOOST PUMP switches — ON.

ENGINE 1 and ENGINE 2 FUEL switches — ON.

Fuel filter (left and right) — Drain samples.

ENGINE 1 and ENGINE 2 FUEL switches — OFF.

BOOST PUMP switches — OFF.

BAT BUS 1 switch — OFF.

Rotor tiedowns — Removed and secured.

Section 1

WEIGHT AND BALANCE

No change from basic manual.

Section 2

SYSTEMS DESCRIPTION

No change from basic manual.

Section 3

OPERATIONAL INFORMATION

No change from basic manual.

Section 4

HANDLING/SERVICING/MAINTENANCE

FUELS

Fuels conforming to the following commercial and military specifications are approved:

ASTM D-1655, Type A, A-1, or B

MIL-T-5624, Grade JP-4 or JP-5

NATO F-40 or F-44

Refer to Fuel Limitations in this supplement for ambient temperature limits.

The following fuel listing is provided for the convenience of the operator (Table 4-1 through Table 4-3). It shall be the responsibility of the operator and his fuel supplier to ensure that the fuel conforms to one of the approved specifications above.

Consult the engine manufacturer for alternate or emergency fuels.

FUEL SYSTEM SERVICING

Total capacity:

337.5 US gallons (1277.4 L).

333.7 US gallons (1262.8 L) for S/N 34001 — 34024.

Usable fuel:

330.5 US gallons (1251 L).

326.7 US gallons (1236.4 L) for S/N 34001 — 34024.

The fuel system is gravity serviced through a single filler port on the right side of aft fuselage. A grounding jack is provided below the fueling port.

NOTE

If fueling to a total of less than 1000 pounds (453.6 kg), open interconnect valve prior to fueling. Close interconnect valve prior to engine start.

Electrical/mechanical sump drain valves are located in each lower aft tank. Pushbutton switches for electrical operation of each drain valve are located on either side of the aft fuselage. To operate the drain valves, both FUEL switches must be in the OFF position and emergency dc bus 1 and essential dc bus 2 must be energized. Each lower aft tank also has a defueling valve. To drain the fuel, remove the plug and insert a standard fitting to open the spring-loaded poppet valve.

The lower forward and mid tanks have mechanical push-to-drain valves.

FUEL TEMPERATURE MEASUREMENT

Required apparatus:

- Measuring container with graduated scale
- Calibrated temperature meter with thermocouple probe suitable for measuring fuel at cold temperature

Procedure:

Perform the following prior to engine start:

1. Collect at least 250 cc fuel sample using the drain valve from either main feed fuel tank (left or right).
2. Measure fuel temperature immediately.
3. Record fuel temperature once thermocouple reading has stabilized.

If recorded fuel temperature is above -30°C (-22°F), repeat step 1 through step 3, but collecting fuel sample using the drain valve from either engine fuel inlet filter (left or right engine).

NOTE

Ensure container temperature is close to ambient and thermocouple is properly immersed in the fuel.

**Table 4-1. Fuels
COMMERCIAL TYPE A AND A-1**

FUEL VENDOR	ASTM D-1655, TYPE A PRODUCT NAME	ASTM D-1655, TYPE A-1 PRODUCT NAME
American Oil and Supply	American Jet Fuel Type A	American Jet Fuel Type A-1
ARCO (Atlantic Richfield)	Arcojet A	Arcojet A-1
Boron Oil	Jet A Kerosene	Jet A-1 Kerosene
British-American	B-A Jet Fuel JP-1	
British Petroleum	B.P. Jet A	B.P. A.T.K.
California-Texas		Caltex Jet A-1
Chevron	Chevron Jet A-50	Chevron Jet A-1
Cities Service	Citgo Turbine Type A	
Continental	Conoco Jet-50	Conoco Jet-60
Exxon Co. U.S.A.	Exxon Turbo Fuel A	Exxon Turbo Fuel A-1
Exxon International		Esso Turbo Fuel A-1
Gulf Oil	Gulf Jet A	Gulf Jet A-1
Mobil Oil	Mobil Jet A	Mobil Jet A-1
Phillips Petroleum	Philjet A-50	
Pure Oil	Purejet Turbine Fuel Type A	Purejet Turbine Fuel Type A-1
Shell Oil	AeroShell Turbine Fuel 640	AeroShell Turbine Fuel 650
Standard Oil of British Columbia	Chevron Jet Fuel A-50	Chevron Jet Fuel A-1
Standard Oil of California	Chevron Jet Fuel A-50	Chevron Jet Fuel A-1
Standard Oil of Indiana	American Jet Fuel Type A	American Jet Fuel Type A-1
Standard Oil of Kentucky	Standard Turbine Fuel A-50	Standard Turbine Fuel A-1
Standard Oil of New Jersey	Standard Jet A	Standard Jet A-1
Standard Oil of Ohio	Jet A Kerosene	Jet A-1 Kerosene
Standard Oil of Texas	Chevron Jet Fuel A-50	Chevron Jet Fuel A-1
Texaco	Texaco Avjet A	Texaco Avjet A-1
Union Oil	76 Turbine Fuel	

**Table 4-2. Fuels
COMMERCIAL TYPE B**

FUEL VENDOR	ASTM D-1655, TYPE B PRODUCT NAME
American Oil and Supply	American JP-4
ARCO (Atlantic Richfield)	Arcojet B
British-American	B-A Jet Fuel JP-4
British Petroleum	B.P. A.T.G.
California-Texas	Caltex Jet B
Chevron	Chevron Jet B
Continental	Conoco JP-4
Exxon Co. U.S.A.	Exxon Turbo Fuel 4
Exxon International	Esso Turbo Fuel 4
Gulf Oil	Gulf Jet B
Mobil Oil	Mobil Jet B
Phillips Petroleum	Philjet JP-4
Shell Oil	AeroShell Turbine Fuel JP-4
Standard Oil of California	Chevron Jet Fuel B
Standard Oil of Indiana	American JP-4
Standard Oil of Kentucky	Standard Turbine Fuel B
Standard Oil of New Jersey	Standard Jet B
Standard Oil of Texas	Chevron Jet Fuel B
Texaco	Texaco Avjet B
Union Oil	Union JP-4

**Table 4-3. Fuels
MILITARY**

COUNTRY	NATO F-34 (JP-8 TYPE)	NATO F-40 (JP-4 TYPE)	NATO F-44 (JP-5, JP-8 TYPE)
Belgium	BA-PF-7	BA-PF-2	3-GP-24
Canada		3-GP-22	3-GP-24
Denmark	D. Eng. R.D. 2453	MIL-T-5624, Grade JP-4	
France	AIR 3405	AIR 3407	AIR 3404
Germany		VTL 9130-006	VTL-9130-007 VTL-9130-010
Greece		MIL-T-5624, Grade JP-4	
Italy	AA-M-C.141	AER-M-C.142	AA-M-C.143
Netherlands	D. Eng. R.D. 2453	MIL-T-5624, Grade JP-4	D. Eng. R.D. 2498
Norway		MIL-T-5624, Grade JP-4	
Portugal	AIR 3405	MIL-T-5624, Grade JP-4	
Turkey		MIL-T-5624, Grade JP-4	
United Kingdom	D. Eng. R.D. 2453	D. Eng. R.D. 2454	D. Eng. R.D. 2498 D. Eng. R.D. 2452
United States	MIL-T-83133, Grade JP-8	MIL-T-5624, Grade JP-4	MIL-T-5624, Grade JP-5



Department of Transport

Supplemental Type Certificate

This approval is issued to:

Bell Helicopter Textron Canada Limited
12800, rue de l'Avenir
Mirabel, Quebec, J7J 1R4
Canada

Number: SH05-2

Issue No.: 1

Approval Date: January 24, 2005

Issue Date: January 24, 2005

Responsible Office:

Quebec

Aircraft/Engine Type or Model:

BELL 212, 412, 412 EP

Canadian Type Certificate or Equivalent:

H-86

Description of Type Design Change:

Cold weather use of Kerosene fuels for Bell 212 & 412 helicopters

**Installation/Operating Data,
Required Equipment and Limitations:**

Use of Kerosene fuels in cold weather will be permitted in accordance with Bell Helicopter Report No. 412-099-675, initial release, dated 3 January 2005, or later Transport Canada approved revision.

The applicable Rotorcraft Flight Manual Supplements are the following Bell Helicopter publications :

Model 212 :

- BHT-212-FMS-31, issue 0, dated 24 January 2005;

Model 412 :

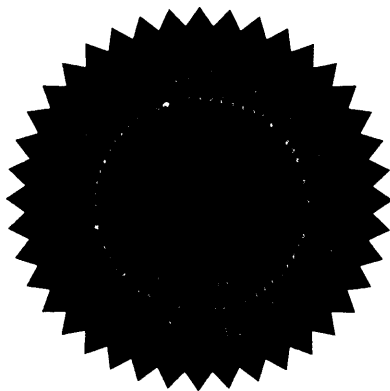
- BHT-412-FMS-23.1, issue 0, dated 24 January 2005;
- BHT-412-FMS-23.2, issue 0, dated 24 January 2005;
- BHT-412-FMS-23.3, issue 0, dated 24 January 2005;

Model 412EP :

- BHT-412-FMS-23.4, issue 0, dated 24 January 2005;

or later Transport Canada approved revision.

— End —



Conditions: This approval is only applicable to the type/model of aeronautical product specified therein. Prior to incorporating this modification, the installer shall establish that the interrelationship between this change and any other modification(s) incorporated will not adversely affect the airworthiness of the modified product.

Luize-Mihaela Avrigeanu
For Minister of Transport



ROTORCRAFT FLIGHT MANUAL

SUPPLEMENT SEAT CUSHION KIT 412-706-019

CERTIFIED
24 JULY 1987

This supplement shall be attached to Model 412 or 412EP Flight Manual when the 412-706-019 Seat Cushion Kit has been installed.

Information contained herein supplements information of basic Flight Manual. For Limitations, Procedures, and Performance Data not contained in this supplement, consult basic Flight Manual.

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REISSUE — 8 DECEMBER 1995

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APPROVED:


MANAGER

ROTORCRAFT CERTIFICATION OFFICE
FEDERAL AVIATION ADMINISTRATION
FT. WORTH, TX 76193-0170

GENERAL INFORMATION

INTRODUCTION

The Seat Cushion Kit is installed in conjunction with the Utility Passenger Seat Kit and provides increased comfort level for passengers.

Section 1

LIMITATIONS

WEIGHT/CG LIMITATIONS

Actual weight change shall be determined after Seat Cushion Kit is installed and ballast readjusted if necessary to return empty weight CG within allowable limits.

PLACARDS AND MARKINGS

DOORS MUST BE KEPT CLOSED DURING FLIGHT IF SEAT CUSHIONS INSTALLED

Located on inside of sliding passenger door.

Section 2

NORMAL PROCEDURES

BEFORE TAKEOFF

Passengers doors — Closed.



ROTORCRAFT FLIGHT MANUAL

33108 — 33213

36001 — 36019

AND

36020 — 36086

AND

36087 AND SUB

SUPPLEMENT FOR AUXILIARY FUEL OPERATIONS (412-706-009)

CERTIFIED
10 MARCH 1988

This supplement shall be attached to the 412 Flight Manual (BHT-412-FM-2, -3 or -4) when the 412-706-009 Auxiliary Fuel Kit has been installed.

The information contained herein supplements the information of the basic Flight Manual. For Limitations, Procedures, and Performance Data not contained in this supplement, consult the basic Flight Manual.

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REISSUE — 23 JUNE 1994

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FLIGHT MANUAL

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3-1/3-2	.0		
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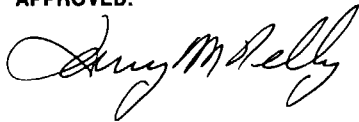
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APPROVED:



MANAGER

ROTORCRAFT CERTIFICATION OFFICE
FEDERAL AVIATION ADMINISTRATION
FT. WORTH, TX 76193-0170

INTRODUCTION

The Auxiliary Fuel Kit provides additional fuel capacity to extend the range of the helicopter. The kit consists of a left and right auxiliary fuel tank and the hardware and wiring necessary to complete the installation. The left or right auxiliary fuel tank may be removed as operational requirements dictate.

One fuel tank provides an additional 16.3 U.S. gallons (61.7 liters) of fuel. Both fuel tanks combined, provide additional 32.6 U.S. gallons (123.4 liters) of fuel.

Section 1

LIMITATIONS

WEIGHT/CG LIMITATIONS

NOTE

The contents of this supplement shall be used in conjunction with the basic Flight Manual for helicopters equipped with the 412-706-009 Auxiliary Fuel Tank Kit installed.

WEIGHT AND BALANCE

Actual weight change shall be determined after kit is installed and ballast readjusted, if necessary, to retain gross weight CG within allowable limits.

FUEL AND OIL LIMITATIONS

FUEL SYSTEM CAPACITIES

Basic system with left or right auxiliary tank:

Total usable fuel capacity is 346.7 U.S. gallons (1312.3 liters).

Basic system with both auxiliary tanks:

Total usable fuel capacity is 363.0 U.S. gallons (1374.1 liters).

Section 2

NORMAL PROCEDURES

IN-FLIGHT OPERATION

CAUTION

WHEN ONE CABIN MOUNTED AUXILIARY FUEL TANK ONLY IS USED, THE TANK INTERCONNECT SWITCH ON THE COCKPIT FUEL PANEL MUST BE PLACED IN THE OPEN POSITION WHEN THE FUEL QUANTITY INDICATION ON THE LOW SIDE WHICH DOES NOT HAVE AN AUXILIARY TANK, REDUCES TO APPROXIMATELY 500 LBS. THIS WILL ALLOW THE AUXILIARY FUEL TO BE SHARED BY BOTH ENGINE FEED TANKS AND PRECLUDE THE POSSIBILITY OF FUEL EXHAUSTION TO THE ENGINE BEING SUPPLIED BY THE SIDE WHICH DOES NOT HAVE AN AUXILIARY TANK. THE AUTOMATIC FEATURE OF THE TANK INTERCONNECT VALVE MAY NOT FUNCTION WITH ONLY ONE AUXILIARY TANK INSTALLED.

Section 3

EMERGENCY AND MALFUNCTION PROCEDURES

No change from basic manual.

Section 4

PERFORMANCE

No change from basic manual.

Section 1

MANUFACTURER'S DATA

WEIGHT AND BALANCE

AUXILIARY FUEL SYSTEM

AUXILIARY FUEL SYSTEM SERVICING

The auxiliary fuel tanks are interconnected with the basic fuel system to allow gravity flow of auxiliary fuel into main fuel cells as fuel is consumed. The auxiliary fuel system is serviced simultaneously with the basic fuel system through the single filler port located on the aft right side of the fuselage.

AUXILIARY FUEL LOADING TABLES

Total usable fuel capacity with 412-706-009 Auxiliary Fuel Kit (both tanks) installed is 363.0 U.S. gallons (1374.1 liters).

Total usable fuel capacity with one tank (left or right) installed is 346.7 U.S. gallons (1312.3 liters).

Fuel loading tables are presented for weight and balance computations in both English and Metric units. These tables shall be used in lieu of the tables for the basic fuel system when either or both auxiliary fuel tanks are installed. Weights and moments listed herein represent total fuel on board to include that contained in basic fuel cells. Refer to table 1-1 and 1-2 for English or 1-1M and 1-2M for Metric when both left and right auxiliary tanks are installed. Tables 1-3 and 1-4 for English or 1-3M and 1-4M for Metric apply to single auxiliary tank installed on left or right.

Table 1-1. Fuel loading with left and right auxiliary tanks (two 16.3 gal.) — longitudinal (English)

Longitudinal							
Jet B or JP-4 (6.5 Lb/U.S. Gallon)				Jet A, A1 or JP-5 (6.8 Lb/U.S. Gallon)			
Quantity (U.S. Gal.)	Weight (Pounds)	CG (Inches)	Moment (In-Lb)	Quantity (U.S. Gal)	Weight (Pounds)	CG (Inches)	Moment (In-Lb)
10	65	139.4	9061	10	68	139.4	9479
20	130	139.6	18148	20	136	139.6	18986
30	195	139.8	27261	30	204	139.8	28519
40	260	139.9	36374	40	272	139.9	38053
50	325	139.9	45468	50	340	139.9	47566
△ 58.3	379	139.9	53022	△ 58.3	397	139.9	55540
60	390	141.0	54990	60	408	141.0	57528
70	455	145.4	66157	70	476	145.4	69210
80	520	148.4	77168	80	544	148.4	80730
90	585	150.6	88101	90	612	150.6	92167
100	650	152.2	98930	100	680	152.2	103496
110	715	153.5	109753	110	748	153.5	114818
120	780	154.7	120666	120	816	154.7	126235
130	845	155.7	131567	130	884	155.7	137639
140	910	156.5	142415	140	952	156.5	148988
150	975	157.2	153270	150	1020	157.2	160344
160	1040	157.9	164216	160	1088	157.9	171795
170	1105	158.6	175253	170	1156	158.6	183342
* 172.6	1122	158.8	178174	* 172.6	1174	158.8	186431
180	1170	156.2	182754	180	1224	156.2	191189
190	1235	152.7	188585	190	1292	152.7	197288
200	1300	149.8	194740	200	1360	149.8	203728
* 205.8	1338	148.2	198292	* 205.8	1399	148.2	207332
210	1365	148.7	202976	210	1428	148.7	212344
220	1430	149.9	214357	220	1496	149.9	224250
230	1495	151.1	225895	230	1564	151.1	236320
240	1560	152.2	237432	240	1632	152.2	248390
250	1625	153.2	248950	250	1700	153.2	260440
260	1690	154.0	260260	260	1768	154.0	272272
270	1755	154.8	271674	270	1836	154.8	284213
□ 275.7	1792	155.2	278118	□ 275.7	1875	155.2	291000
280	1820	154.7	281554	280	1904	154.7	294549
290	1885	153.2	288782	290	1972	153.2	302110
300	1950	151.9	296205	300	2040	151.9	309876
310	2015	150.7	303661	310	2108	150.7	317676
320	2080	149.4	310752	320	2176	149.4	325094
* 327.7	2130	148.7	316731	* 327.7	2228	148.7	331304
330	2145	149.0	319605	330	2244	149.0	334356
340	2210	149.7	330837	340	2312	149.7	346106
350	2275	150.4	342160	350	2380	150.4	357952
360	2340	151.2	353808	360	2448	151.2	370138
■ 363.0	2360	151.3	357068	■ 363.0	2469	151.3	373560

NOTE: All data above represents usable fuel (basic and auxiliary) based on nominal density at 15°C (59°F).

* Quantity at which CG of fuel changes direction.

△ Quantity resulting in maximum forward CG of helicopter (at any weight).

□ Quantity resulting in maximum aft CG of helicopter (weight empty 6520 lb or less).

■ Quantity resulting in maximum aft CG of helicopter (weight empty more than 6520 lb).

Table 1-1M. Fuel loading with left and right auxiliary tanks (two 61.7 liter) — longitudinal (Metric)

Longitudinal							
Jet B or JP-4 (0.779 kg/liter)				Jet A, A1 or JP-5 (0.815 kg/liter)			
Quantity (liters)	Weight (kg)	CG (mm)	Moment (kg-mm)	Quantity (liters)	Weight (kg)	CG (mm)	Moment (kg-mm)
40	31.2	3541	110479	40	32.6	3541	115437
80	62.3	3547	220978	80	65.2	3547	231264
120	93.5	3551	332019	120	97.8	3551	347288
160	124.6	3552	442579	160	130.4	3552	463181
200	155.8	3552	553402	200	163.0	3552	578976
△ 220.7	171.9	3553	610761	△ 220.7	179.9	3553	639185
240	187.0	3632	679184	240	195.6	3632	710419
280	218.1	3726	812641	280	228.2	3726	850273
320	249.3	3795	946094	320	260.8	3795	989736
360	280.4	3848	1078979	360	293.4	3848	1129003
400	311.6	3886	1210878	400	326.0	3886	1266836
440	342.8	3917	1342748	440	358.6	3917	1404636
480	373.9	3947	1475783	480	391.2	3947	1544066
520	405.1	3970	1608247	520	423.8	3970	1682486
560	436.2	3988	1739566	560	456.4	3988	1820123
600	467.4	4008	1873339	600	489.0	4008	1959912
640	498.6	4026	2007364	640	521.6	4026	2099962
* 653.2	508.8	4033	2051990	* 653.2	532.3	4033	2146766
680	529.7	3973	2104498	680	554.2	3973	2201837
720	560.9	3879	2175731	720	586.8	3879	2276197
760	592.0	3800	2249600	760	619.4	3800	2353720
* 778.8	606.7	3765	2284226	* 778.8	634.6	3765	2389269
800	623.2	3785	2358812	800	652.0	3785	2467820
840	654.4	3815	2496536	840	684.6	3815	2611749
880	685.5	3843	2634377	880	717.2	3843	2756200
920	716.7	3874	2776496	920	749.8	3874	2904725
960	747.8	3899	2915672	960	782.4	3899	3050578
1000	779.0	3922	3055238	1000	815.0	3922	3196430
1040	810.2	3941	3192998	1040	847.6	3941	3340392
□ 1043.7	813.0	3942	3204846	□ 1043.7	850.5	3942	3352671
1080	841.3	3907	3286959	1080	880.2	3907	3438941
1120	872.5	3871	3377448	1120	912.8	3871	3533449
1160	903.6	3838	3468017	1160	945.4	3838	3628445
1200	934.8	3807	3558784	1200	978.0	3807	3723246
* 1240.2	966.1	3776	3647994	* 1240.2	1010.7	3776	3816403
1280	997.1	3800	3788980	1280	1043.2	3800	3964160
1320	1028.3	3820	3928106	1320	1075.8	3820	4109556
1360	1059.4	3838	4065977	1360	1108.4	3838	4254039
■ 1374.1	1070.4	3843	4113547	■ 1374.1	1119.9	3843	4303776

NOTE: All data above represents usable fuel (basic and auxiliary) based on nominal density at 15°C (59°F).

* Quantity at which CG of fuel changes direction.

△ Quantity resulting in maximum forward CG of helicopter (at any weight).

□ Quantity resulting in maximum aft CG of helicopter (weight empty 2957 kg or less).

■ Quantity resulting in maximum aft CG of helicopter (weight empty more than 2957 kg).

Table 1-2. Fuel loading with left and right auxiliary tanks (two 16.3 gal.) — lateral (English)

Lateral							
Jet B or JP-4 (6.5 Lb/U.S. Gallon)				Jet A, A1 or JP-5 (6.8 Lb/U.S. Gallon)			
Quantity (U.S. Gal.)	Weight (Pounds)	CG (Inches)	Moment (In-Lb)	Quantity (U.S. Gal.)	Weight (Pounds)	CG (Inches)	Moment (In-Lb)
10	65	0	0	10	68	0	0
20	130	0	0	20	136	0	0
30	195	0	0	30	204	0	0
40	260	0	0	40	272	0	0
50	325	0	0	50	340	0	0
* 54.6	355	0	0	* 54.6	371	0	0
60	390	-0.03	-12	60	408	-0.03	-12
70	455	-0.06	-27	70	476	-0.06	-29
* 80.0	520	-0.05	-26	* 80.0	544	-0.05	-27
90	585	-0.04	-23	90	612	-0.04	-24
100	650	-0.04	-26	100	680	-0.04	-27
110	715	-0.03	-21	110	748	-0.03	-22
120	780	-0.03	-23	120	816	-0.03	-24
130	845	-0.03	-25	130	884	-0.03	-27
140	910	-0.03	-27	140	952	-0.03	-29
150	975	-0.03	-29	150	1020	-0.03	-31
160	1040	-0.02	-21	160	1088	-0.02	-22
170	1105	-0.02	-22	170	1156	-0.02	-23
* 172.6	1122	-0.02	-22	* 172.6	1174	-0.02	-23
180	1170	-0.03	-35	180	1224	-0.03	-37
190	1235	-0.44	-543	190	1292	-0.44	-568
200	1300	-0.55	-715	200	1360	-0.55	-748
* 205.8	1337	-0.60	-802	* 205.8	1399	-0.60	-839
▶ 210	1365	-0.59	-805	▶ 210	1428	-0.59	-843
220	1430	-0.56	-801	220	1496	-0.56	-838
230	1495	-0.54	-807	230	1564	-0.54	-845
240	1560	-0.52	-811	240	1632	-0.52	-849
250	1625	-0.50	-813	250	1700	-0.50	-850
260	1690	-0.48	-811	260	1768	-0.48	-849
270	1755	-0.46	-807	270	1836	-0.46	-845
280	1820	-0.44	-801	280	1904	-0.44	-838
290	1885	-0.43	-811	290	1972	-0.43	-848
300	1950	-0.41	-800	300	2040	-0.41	-836
310	2015	-0.40	-806	310	2108	-0.40	-843
320	2080	-0.39	-811	320	2176	-0.39	-849
330	2145	-0.38	-815	330	2244	-0.38	-853
340	2210	-0.36	-796	340	2312	-0.36	-832
350	2275	-0.35	-796	350	2380	-0.35	-833
360	2340	-0.34	-796	360	2448	-0.34	-832
363.0	2360	-0.34	-802	363.0	2469	-0.34	-839

NOTE: All data above represents usable fuel (basic and auxiliary) based on nominal density at 15°C (59°F).

* Quantity at which CG of fuel changes direction.

▶ Quantity resulting in maximum lateral CG of helicopter.

Table 1-2M. Fuel loading with left and right auxiliary tanks (two 61.7 liter) — lateral (Metric)

Lateral							
Jet B or JP-4 (0.779 kg/liter)				Jet A, A1 or JP-5 (0.815 kg/liter)			
Quantity (liters)	Weight (kg)	CG (mm)	Moment (kg-mm)	Quantity (liters)	Weight (kg)	CG (mm)	Moment (kg-mm)
40	31.2	0	0	40	32.6	0	0
80	62.3	0	0	80	65.2	0	0
120	93.5	0	0	120	97.8	0	0
160	124.6	0	0	160	130.4	0	0
200	155.8	0	0	200	163.0	0	0
* 206.6	161.0	0	0	* 206.6	168.4	0	0
240	187.0	-1	-187	240	195.6	-1	-196
280	218.1	-1	-218	280	228.2	-1	-228
* 302.7	235.8	-1	-236	* 302.7	246.7	-1	-247
320	249.3	-1	-249	320	260.8	-1	-261
360	280.4	-1	-280	360	293.4	-1	-293
400	311.6	-1	-312	400	326.0	-1	-326
440	342.8	-1	-343	440	358.6	-1	-359
480	373.9	-1	-374	480	391.2	-1	-391
520	405.1	-1	-405	520	423.8	-1	-424
560	436.2	-1	-436	560	456.4	-1	-456
600	467.4	-1	-467	600	489.0	-1	-489
640	498.6	-1	-499	640	521.6	-1	-522
* 653.2	508.8	-1	-509	* 653.2	532.3	-1	-532
680	529.7	-8	-4238	680	554.2	-8	-4434
720	560.9	-11	-6170	720	586.8	-11	-6455
760	592.0	-14	-8288	760	619.4	-14	-8672
* 778.6	606.5	-15	-9098	* 778.6	634.5	-15	-9518
▶ 800	623.2	-15	-9348	▶ 800	652.0	-15	-9780
840	654.4	-14	-9162	840	684.6	-14	-9584
880	685.5	-13	-8912	880	717.2	-13	-9324
920	716.7	-13	-9317	920	749.8	-13	-9747
960	747.8	-12	-8974	960	782.4	-12	-9389
1000	779.0	-12	-9348	1000	815.0	-12	-9780
1040	810.2	-11	-8912	1040	847.6	-11	-9324
1080	841.3	-11	-9254	1080	880.2	-11	-9682
1120	872.5	-11	-9598	1120	912.8	-11	-10041
1160	903.6	-10	-9036	1160	945.4	-10	-9454
1200	934.8	-10	-9348	1200	978.0	-10	-9780
1240	966.0	-10	-9660	1240	1010.6	-10	-10106
1280	997.1	-9	-8974	1280	1043.2	-9	-9389
1320	1028.3	-9	-9255	1320	1075.8	-9	-9682
1360	1059.4	-9	-9535	1360	1108.1	-9	-9973
1374.1	1070.4	-9	-9634	1374.1	1119.9	-9	-10079

NOTE: All data above represents usable fuel (basic and auxiliary) based on nominal density at 15°C (59°F).

* Quantity at which CG of fuel changes direction.

▶ Quantity resulting in maximum lateral CG of helicopter.

Table 1-3. Fuel loading with left or right auxiliary tanks (one 16.3 gal.) — longitudinal (English)

Longitudinal							
Jet B or JP-4 (6.5 Lb/U.S. Gallon)				Jet A, A1 or JP-5 (6.8 Lb/U.S. Gallon)			
Quantity (U.S. Gal.)	Weight (Pounds)	CG (Inches)	Moment (In-Lb)	Quantity (U.S. Gal)	Weight (Pounds)	CG (Inches)	Moment (In-Lb)
10	65	139.4	9061	10	68	139.4	9479
20	130	139.6	18148	20	136	139.6	18986
30	195	139.8	27261	30	204	139.8	28519
40	260	139.9	36374	40	272	139.9	38053
50	325	139.9	45468	50	340	139.9	47566
△ 58.3	379	139.9	53022	△ 58.3	397	139.9	55540
60	390	140.9	54951	60	408	140.9	57487
70	455	145.8	66339	70	476	145.8	69401
80	520	149.0	77480	80	544	149.0	81056
90	585	151.5	88628	90	612	151.5	92718
100	650	153.2	99580	100	680	153.2	104176
110	715	154.8	110682	110	748	154.8	115790
120	780	156.0	121680	120	816	156.0	127296
130	845	157.0	132665	130	884	157.0	138788
140	910	158.2	143962	140	952	158.2	150606
150	975	159.2	155220	150	1020	159.2	162384
* 156.3	1016	159.7	162255	* 156.3	1063	159.7	169761
160	1040	158.0	164320	160	1088	158.0	171904
170	1105	154.4	170612	170	1156	154.4	178486
180	1170	150.9	176553	180	1224	150.9	184702
* 189.4	1231	148.1	182311	* 189.4	1288	148.1	190753
200	1300	149.6	194480	200	1360	149.6	203456
210	1365	150.8	205842	210	1428	150.8	215342
220	1430	151.9	217217	220	1496	151.9	227242
230	1495	152.9	228586	230	1564	152.9	239136
240	1560	153.9	240084	240	1632	153.9	251165
250	1625	154.9	251713	250	1700	154.9	263330
□ 259.4	1686	155.5	262173	□ 259.4	1764	155.5	274302
270	1755	154.1	270446	270	1836	154.1	282928
280	1820	152.6	277732	280	1904	152.6	290550
290	1885	151.2	285012	290	1972	151.2	298166
300	1950	150.0	292500	300	2040	150.0	306000
* 311.4	2024	148.6	300766	* 311.4	2117	148.6	314586
320	2080	149.3	310544	320	2176	149.3	324877
330	2145	150.1	321965	330	2244	150.1	336824
340	2210	150.9	333489	340	2312	150.9	348881
■ 346.7	2254	151.4	341256	■ 346.7	2358	151.4	357001

NOTE: All data above represents usable fuel (basic and auxiliary) based on nominal density at 15°C (59°F).

* Quantity at which CG of fuel changes direction.

△ Quantity resulting in maximum forward CG of helicopter (at any weight).

□ Quantity resulting in maximum aft CG of helicopter (weight empty 6630 lb or less).

■ Quantity resulting in maximum aft CG of helicopter (weight empty more than 6630 lb).

Table 1-3M. Fuel loading with left or right auxiliary tanks (one 61.7 liter) — longitudinal (Metric)

Longitudinal							
Jet B or JP-4 (0.779 kg/liter)				Jet A, A1 or JP-5 (0.815 kg/liter)			
Quantity (liters)	Weight (kg)	CG (mm)	Moment (kg-mm)	Quantity (liters)	Weight (kg)	CG (mm)	Moment (kg-mm)
40	31.2	3541	110479	40	32.6	3541	115437
80	62.3	3547	220978	80	65.2	3547	231264
120	93.5	3551	332019	120	97.8	3551	347288
160	124.6	3552	442579	160	130.4	3552	463181
200	155.8	3552	553402	200	163.0	3552	578976
△ 220.7	171.9	3553	610761	△ 220.7	179.9	3553	639185
240	187.0	3637	680119	240	195.6	3637	711397
280	218.1	3739	815476	280	228.2	3739	853240
320	249.3	3815	951080	320	260.8	3815	994952
360	280.4	3871	1085428	360	293.4	3871	1135751
400	311.6	3917	1220537	400	326.0	3917	1276942
440	342.8	3952	1354746	440	358.6	3952	1417187
480	373.9	3983	1489244	480	391.2	3983	1558150
520	405.1	4013	1625666	520	423.8	4013	1700709
560	436.2	4036	1760503	560	456.4	4036	1842030
* 591.5	460.8	4057	1869466	* 591.5	482.1	4057	1955880
600	467.4	4046	1891100	600	489.0	4046	1978494
640	498.6	3927	1958002	640	521.6	3927	2048323
680	529.7	3838	2032989	680	554.2	3838	2127020
* 717.1	558.6	3762	2101453	* 717.1	584.4	3762	2198513
720	560.9	3764	2111228	720	586.8	3764	2208715
760	592.0	3800	2249600	760	619.4	3800	2353720
800	623.2	3833	2388726	800	652.0	3833	2499116
840	654.4	3861	2526638	840	684.6	3861	2643241
880	685.5	3889	2665910	880	717.2	3889	2789191
920	716.7	3917	2807314	920	749.8	3917	2936967
960	747.8	3940	2946332	960	782.4	3940	3082656
□ 982.0	765.0	3950	3021750	□ 982.0	800.3	3950	3161185
1000	779.0	3934	3064586	1000	815.0	3934	3206210
1040	810.2	3894	3154919	1040	847.6	3894	3300554
1080	841.3	3856	3244053	1080	880.2	3856	3394051
1120	872.5	3823	3335668	1120	912.8	3823	3489634
1160	903.6	3790	3424644	1160	945.4	3790	3583066
* 1178.5	918.1	3775	3465828	* 1178.5	960.5	3775	3625888
1200	934.8	3787	3540088	1200	978.0	3787	3703686
1240	966.0	3807	3677562	1240	1010.6	3807	3847354
1280	997.1	3830	3818893	1280	1043.2	3830	3995456
■ 1312.3	1022.3	3845	3930744	■ 1312.3	1069.5	3845	4112228

NOTE: All data above represents usable fuel (basic and auxiliary) based on nominal density at 15°C (59°F).

* Quantity at which CG of fuel changes direction.

△ Quantity resulting in maximum forward CG of helicopter (at any weight).

□ Quantity resulting in maximum aft CG of helicopter (weight empty 3007 kg or less).

■ Quantity resulting in maximum aft CG of helicopter (weight empty more than 3007 kg).

Table 1-4. Fuel loading with left or right auxiliary tanks (one 16.3 gal.) — lateral (English) (Sheet 1 of 2)

Lateral					
Jet B or JP-4 (6.5 Lb/U.S. Gallon)					
Quantity (U.S. Gal.)	Weight (Pounds)	Lateral (16.3 Left)		Lateral (16.3 Right)	
		CG (Inches)	Moment (In-Lb)	CG (Inches)	Moment (In-Lb)
10	65	0	0	0	0
20	130	0	0	0	0
30	195	0	0	0	0
40	260	0	0	0	0
50	325	0	0	0	0
* 54.6	355	0	0	0	0
*** 60	390	-0.04	-16	-0.05	-20
70	455	-0.43	-196	0.38	173
80	520	-0.84	-437	0.80	416
90	585	-1.20	-702	1.15	673
100	650	-1.58	-1027	1.50	975
110	715	-1.82	-1301	1.77	1266
120	780	-2.07	-1615	2.01	1568
130	845	-2.27	-1918	2.22	1876
140	910	-2.43	-2211	2.39	2175
150	975	-2.52	-2457	2.49	2428
* ► 152.6	992	-2.53	-2510	2.49	2470
** 156.3	1016	-2.48	-2520		
** 157.8	1026	-2.67	-2739		
160	1040	-2.69	-2798	2.13	2215
170	1105	-2.69	-2972	1.83	2022
180	1170	-2.69	-3147	1.57	1837
** 182.8	1188	-2.68	-3184		
+ 190	1235	-2.67	-3297	1.37	1692
200	1300	-2.54	-3302	1.29	1677
210	1365	-2.42	-3303	1.23	1679
220	1430	-2.30	-3289	1.18	1687
230	1495	-2.20	-3289	1.13	1689
240	1560	-2.11	-3292	1.08	1685
250	1625	-2.03	-3299	1.04	1690
260	1690	-1.95	-3296	1.00	1690
270	1755	-1.88	-3299	0.96	1685
280	1820	-1.81	-3294	0.92	1674
290	1885	-1.75	-3299	0.89	1678
300	1950	-1.69	-3296	0.86	1677
310	2015	-1.64	-3305	0.84	1693
320	2080	-1.58	-3286	0.81	1685
330	2145	-1.54	-3303	0.78	1673
340	2210	-1.49	-3293	0.76	1680
346.7	2254	-1.46	-3291	0.75	1691

* Quantity at which CG of fuel changes direction.

** Quantity at which CG of fuel changes direction for left auxiliary tank only.

*** Quantity at which CG of fuel changes direction for right auxiliary tank only.

+ Quantity resulting in maximum lateral CG of helicopter for left auxiliary tank.

► Quantity resulting in maximum lateral CG of helicopter for right auxiliary tank.

Table 1-4. Fuel loading with left or right auxiliary tanks (one 16.3 gal.) — lateral (English) (Sheet 2)

Lateral					
Jet A, A1 or JP-5 (6.8 Lb/U.S. Gallon)					
Quantity (U.S. Gal.)	Weight (Pounds)	Lateral (16.3 Left)		Lateral (16.3 Right)	
		CG (Inches)	Moment (In-Lb)	CG (Inches)	Moment (In-Lb)
10	68	0	0	0	0
20	136	0	0	0	0
30	204	0	0	0	0
40	272	0	0	0	0
50	340	0	0	0	0
* 54.6	371	0	0	0	0
*** 60	408	-0.04	-16	-0.05	-20
70	476	-0.43	-205	0.38	181
80	544	-0.84	-457	0.80	435
90	612	-1.20	-734	1.15	704
100	680	-1.58	-1074	1.50	1020
110	748	-1.82	-1361	1.77	1324
120	816	-2.07	-1689	2.01	1640
130	884	-2.27	-2007	2.22	1962
140	952	-2.43	-2313	2.39	2275
150	1020	-2.52	-2570	2.49	2540
* ► 152.6	1038	-2.53	-2626	2.49	2585
** 156.3	1063	-2.48	-2636		
** 157.8	1073	-2.67	-2865		
160	1088	-2.69	-2927	2.13	2317
170	1156	-2.69	-3110	1.83	2115
180	1224	-2.69	-3293	1.57	1922
** 182.8	1243	-2.68	-3331		
+ 190	1292	-2.67	-3450	1.37	1770
200	1360	-2.54	-3454	1.29	1754
210	1428	-2.42	-3456	1.23	1756
220	1496	-2.30	-3441	1.18	1765
230	1564	-2.20	-3441	1.13	1767
240	1632	-2.11	-3444	1.08	1763
250	1700	-2.03	-3451	1.04	1768
260	1768	-1.95	-3448	1.00	1768
270	1836	-1.88	-3452	0.96	1763
280	1904	-1.81	-3446	0.92	1752
290	1972	-1.75	-3451	0.89	1755
300	2040	-1.69	-3448	0.86	1754
310	2108	-1.64	-3457	0.84	1771
320	2176	-1.58	-3438	0.81	1763
330	2244	-1.54	-3456	0.78	1750
340	2312	-1.49	-3445	0.76	1757
346.7	2358	-1.46	-3443	0.75	1769

* Quantity at which CG of fuel changes direction.

** Quantity at which CG of fuel changes direction for left auxiliary tank only.

*** Quantity at which CG of fuel changes direction for right auxiliary tank only.

+ Quantity resulting in maximum lateral CG of helicopter for left auxiliary tank.

► Quantity resulting in maximum lateral CG of helicopter for right auxiliary tank.

Table 1-4M. Fuel loading with left or right auxiliary tanks (one 61.7 liter) – lateral (Metric) (Sheet 1 of 2)

Lateral					
Jet B or JP-4 (0.779 kg/liter)					
Quantity (liters)	Weight (kg)	Lateral (61.7 Left)		Lateral (61.7 Right)	
		CG (mm)	Moment (kg-mm)	CG (mm)	Moment (kg-mm)
40	31.2	0	0	0	0
80	62.3	0	0	0	0
120	93.5	0	0	0	0
160	124.6	0	0	0	0
200	155.8	0	0	0	0
* 206.6	160.9	0	0	0	0
240	187.0	-4	-748	2	374
280	218.1	-15	-3272	14	3053
320	249.3	-26	-6482	25	6233
360	280.4	-35	-9814	34	9534
400	311.6	-43	-13399	42	13087
440	342.8	-50	-17140	49	16797
480	373.9	-56	-20938	55	20565
520	405.1	-61	-24711	60	24306
560	436.2	-64	-27917	63	27481
* ► 577.6	450.0	-64	-28800	63	28350
** 591.5	460.8	-63	-29030		
** 597.1	465.1	-68	-31627		
600	467.4	-68	-31783	55	25707
640	498.6	-68	-33905	47	23434
680	529.7	-68	-36020	40	21188
** 691.8	538.9	-68	-36645		
+ 720	560.9	-68	-38141	35	19632
760	592.0	-64	-37888	33	19536
800	623.2	-61	-38015	31	19319
840	654.4	-58	-37955	30	19632
880	685.5	-55	-37703	28	19194
920	716.7	-53	-37985	27	19351
960	747.8	-51	-38138	26	19443
1000	779.0	-49	-38171	25	19475
1040	810.2	-47	-38079	24	19445
1080	841.3	-45	-37859	23	19350
1120	872.5	-43	-37518	22	19195
1160	903.6	-42	-37951	21	18976
1200	934.8	-41	-38327	21	19631
1240	966.0	-39	-37674	20	19320
1280	977.1	-38	-37890	19	18945
1312.3	1022.3	-37	-37825	19	19424

* Quantity at which CG of fuel changes direction.

** Quantity at which CG of fuel changes direction for left auxiliary tank only.

+ Quantity resulting in maximum lateral CG of helicopter for left auxiliary tank.

► Quantity resulting in maximum lateral CG of helicopter for right auxiliary tank.

Table 1-4M. Fuel loading with left or right auxiliary tanks (one 61.7 liter) — lateral (Metric) (Sheet 2)

Lateral					
Jet A, A1 or JP-5 (0.815 kg/liter)					
Quantity (liters)	Weight (kg)	Lateral (61.7 Left)		Lateral (61.7 Right)	
		CG (mm)	Moment (kg-mm)	CG (mm)	Moment (kg-mm)
40	32.6	0	0	0	0
80	65.2	0	0	0	0
120	97.8	0	0	0	0
160	130.4	0	0	0	0
200	163.0	0	0	0	0
* 206.6	168.4	0	0	0	0
240	195.6	-4	-782	2	391
280	228.2	-15	-3423	14	3195
320	260.8	-26	-6781	25	6520
360	293.4	-35	-10269	34	9976
400	326.0	-43	-14018	42	13692
440	358.6	-50	-17930	49	17571
480	391.2	-56	-21907	55	21516
520	423.8	-61	-25852	60	25428
560	456.4	-64	-29210	63	28753
* ► 577.6	470.7	-64	-30125	63	29654
** 591.5	482.1	-63	-30372		
** 597.1	486.6	-68	-33089		
600	489.0	-68	-33252	55	26895
640	521.6	-68	-35469	47	24515
680	554.2	-68	-37686	40	22168
** 691.8	563.8	-68	-38338		
+ 720	586.8	-68	-39902	35	20538
760	619.4	-64	-39642	33	20440
800	652.0	-61	-39772	31	20212
840	684.6	-58	-39707	30	20538
880	717.2	-55	-39446	28	20082
920	749.8	-53	-39739	27	20245
960	782.4	-51	-39902	26	20342
1000	815.0	-49	-39935	25	20375
1040	847.6	-47	-39837	24	20342
1080	880.2	-45	-39609	23	20245
1120	912.8	-43	-39250	22	20082
1160	945.4	-42	-39707	21	19853
1200	978.0	-41	-40098	21	20538
1240	1010.6	-39	-39413	20	20212
1280	1043.2	-38	-39642	19	19821
1312.3	1069.5	-37	-39572	19	20321

- * Quantity at which CG of fuel changes direction.
- ** Quantity at which CG of fuel changes direction for left auxiliary tank only.
- + Quantity resulting in maximum lateral CG of helicopter for left auxiliary tank.
- Quantity resulting in maximum lateral CG of helicopter for right auxiliary tank.

Bell **412/412EP**
MODELS

ROTORCRAFT FLIGHT MANUAL

SUPPLEMENT TWO-SPEED INTERNAL HOIST

(412-899-223)

OR

(214-706-003)

CERTIFIED

SEPTEMBER 19, 1988

This supplement shall be attached to the Model 412 & 412EP Flight Manual when the 412-899-223 or 214-706-003 Internal Hoist has been installed.

Information contained herein supplements information of basic Flight Manual. For Limitations, Procedures, and Performance Data not contained in this supplement, consult basic Flight Manual.

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FLIGHT MANUAL		35/36	0
Title — NP	0	MANUFACTURER'S DATA	
A — B	0	37 — 38	0
i/ii	0		
1 — 34	0		

NOTE

Revised text is indicated by a black vertical line. Insert latest revision pages; dispose of superseded pages.

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APPROVED:

H Whitlock

for MANAGER

ROTORCRAFT CERTIFICATION OFFICE
FEDERAL AVIATION ADMINISTRATION
FT. WORTH, TX 76193-0170

GENERAL INFORMATION

The Two Speed Internal Hoist enables cargo and emergency rescue operations in areas where landing cannot be accomplished. The hoist can raise or lower loads up to 600 pounds (272 kilograms). The hoist contains 250 usable feet (76.2 meters) cable. Each of the four cabin mounting locations allows the hoist to be extended 90 degrees outboard. The hoist provides two extend/retrieve speeds (HIGH and LOW). With LOW speed selected, a

continuously variable speed range from zero to 125 feet/minute (38.1 meters/minute) is available. With HIGH speed selected, a continuously variable speed range from zero to 250 feet/minute (76.2 meters/minute) is available. An electrically actuated cable cutting device allows either the pilot or hoist operator to sever the cable if necessary. A manually actuated cutting device is provided for use in the event of an electrical failure.

Section 1

LIMITATIONS

1-3. TYPES OF OPERATION

Hoist operations shall be conducted under appropriate operating rules for external loads.

Passenger operations with hoist installed are approved if hoist is stowed and electrical system is deactivated.

Hoist operations are prohibited during instrument meteorological conditions.

1-4. FLIGHT CREW

A crewmember wearing an approved safety harness in passenger compartment is required during all phases of hoist operations. Crewmember shall wear protective gloves for guiding cable during operation. The hoist operator shall be familiar with hoist operating procedures and limitations.

1-5. CONFIGURATION

1-5-A. REQUIRED EQUIPMENT

Hoist cable antichafing guard shall be installed on standard or high skid landing gear (with or without floats) on same side of helicopter as hoist.

1-5-B. OPTIONAL EQUIPMENT

Fixed passenger step shall not be installed concurrently with internal hoist.

Retractable passenger steps shall be stowed during hoist operations.

Hoist operation with flight director in coupled mode is prohibited.

Hoisting or lowering an empty litter in open position (except Stokes litter) is prohibited.

Refer to appropriate Flight Manual Supplement(s) for additional limitations, procedures, and performance data.

1-6. WEIGHT AND CENTER OF GRAVITY

Actual weight change shall be determined after hoist is installed and ballast readjusted, if necessary.

For maximum gross weight, including hoist load, refer to applicable Flight Manual or BHT-412-FM-19.1 when Increased Gross Weight and Takeoff Horsepower kit is installed.

Maximum hoist load is 600 pounds (272 kilograms). This is a structural limitation only and does not ensure that longitudinal or lateral CG will remain within approved limits. Maximum allowable hoist load varies with gross weight, center of gravity, and hoist location. Refer to appropriate Hoist Loading Schedule.

NOTE

The center of gravity of hoist load in forward position is F.S. 82 (2083 mm) and B.L. 60 (1524 mm). The center of gravity of hoist load in aft position is F.S. 131 (3327 mm) and B.L. 64.4 (1636 mm).

For Longitudinal vs. Lateral CG limits with internal hoist refer to Internal hoist CG envelope figure 1-1.

1-7. AIRSPPEED

VNE with asymmetrical door configuration is 20 KIAS.

VNE with hinged panels locked open and cargo doors open is 20 KIAS.

VNE with hinged panels removed and cargo doors removed or secured open is 60 KIAS.

1-23. HOIST SPEED

HIGH speed — Limited to hoist loads of 300 pounds (136 kilograms) or less.

LOW speed — Limits of basic hoist (600 pounds., 272 kilograms).

1-24. HOIST DUTY CYCLE LIMITATIONS

The hoist is approved for continuous operation with loads not to exceed 600 pounds (272 kilograms).

1-25. ALLOWABLE HOIST LOAD

Select hoist loading schedules (figures 1-2 through 1-5) appropriate for position in which hoist is installed.

NOTE

Hoist loading schedules are based on most adverse loading combinations of pilot, copilot, and hoist operator, each weighing 170 or 200 pounds (77.1 or 90.7 kilograms), and on a weight empty CG of 0.3 inches (7.3 mm) to right

of centerline prior to adding hoist. If lateral CG is appreciably different or crewmember weights are out of this range, allowable hoist load shall be computed. For computation, assume hoist operator in forward position to be located at F.S. 87 (2210 mm) and B.L. 40 (1016mm), and in aft position F.S. 125 (3175mm) and B.L. 40 (1016mm).

1-25-A. LEFT HOIST INSTALLATIONS

Enter appropriate schedule, figures 1-2 through 1-5 at gross weight of helicopter prior to hoisting. Proceed vertically to intersect with diagonal line representing number of crewmembers on board, top of schedule, or right cutoff line. Proceed horizontally to left to read maximum allowable hoist load. Intersecting with right cutoff line gives maximum load which does not cause helicopter to exceed gross weight limitations.

Using Weight empty chart, Section 5 and left hoist loading schedules ensures that both longitudinal and lateral limits are not exceeded during first hoist operation. However, for subsequent hoisting, additional precautions must be taken to avoid exceeding forward longitudinal limits.

1-25-A-1. LEFT FORWARD HOIST LOCATION

To continue using maximum allowable hoist capability: (Refer to figure 1-2 through 1-5)

- a. put hoisted load (people or cargo) along side of island, or
- b. when hoisted load is put immediately forward of island, reduce maximum hoist load to 300 pounds.

WARNING

DO NOT PUT HOISTED LOAD IN FORWARD AREA OF PASSENGER COMPARTMENT UNLESS MAXIMUM HOIST LOADS ARE COMPUTED FOR THAT CONFIGURATION.

1-25-A-2. LEFT AFT HOIST LOCATION

To continue using maximum allowable hoist capability: (Refer to figure 1-2 through 1-5)

- a. put hoisted load along island or immediately forward of island, or
- b. ensure empty weight CG is within Area A. Refer to Weight empty chart, Section 5.

1-25-B. RIGHT HOIST INSTALLATIONS — NORMAL OPERATIONS

Right lateral limit for hoist operations varies with longitudinal center of gravity of the helicopter. The loading schedules have been modified to account for this variation.

- a. Starting with appropriate schedule for number of crewmembers on board, enter at gross weight of helicopter prior to hoisting.
- b. Proceed vertically to intersect with diagonal line representing helicopter center of gravity prior to hoisting, top of schedule, or right cutoff line.
- c. Proceed horizontally to left to read maximum allowable hoist load.

When helicopter center of gravity is between STA. lines, interpolate to determine CG.

Intersecting right cutoff line gives maximum load which does not cause helicopter to exceed gross weight limitations or forward longitudinal limits.

For multiple hoists during a single flight, after each hoist operation enter appropriate schedule at revised gross weight and proceed to new center of gravity to determine maximum allowable hoist load.

EXAMPLE 1: NORMAL

Determine Hoist Load when hoist is in R/H FWD POSITION and crew consist of Pilot, Copilot and Hoist Operator.

GIVEN:

Gross Weight — 9,500 lbs.

CG — STA. 135.5 before hoisting

From appropriate 11,600 lb. GW schedule obtain hoist load as follows:

Enter gross weight at 9,500 lbs.

Proceed up GW line to interpolated STA. 135.5

Proceed left to read hoist load of 210 lbs. Point (A).

EXAMPLE 2: NORMAL

Determine Hoist Load when hoist is in R/H FWD POSITION and crew consist of Pilot, Copilot and Hoist Operator.

GIVEN:

Gross Weight — 9,500 lbs.

CG — STA. 138.5 before hoisting

From appropriate 11,600 lb. GW schedule obtain hoist load as follows:

Enter gross weight at 9,500 lbs.

Proceed up GW line to STA. 138.5

Proceed left to read hoist load of 550 lbs. Point (B).

1-25-C. RIGHT HOIST INSTALLATION - PENALTY REGION OPERATION

The dashed line on schedules represents longitudinal center of gravity prior to hoisting which will result in a gross weight center of gravity at Sta. 135.2 and B.L. 4.5 during hoist operations with maximum hoist loads derived using this line. This center of gravity is the corner of but not in Penalty Region shown in Limitations.

Hoist loads derived for Normal Operations may be increased when GW/CG combinations are forward of those represented by dashed line. Loads may be increased up to but not greater than those defined by dashed line. However, this procedure will result in operations within Penalty Region. Refer to Section 1, Internal Hoist CG Envelope, for Penalty Region.

EXAMPLE 3: PENALTY REGION

Determine Hoist Load when hoist is in R/H FWD POSITION and crew consist of Pilot, Copilot and Hoist Operator.

GIVEN:

Gross Weight — 9,500 lbs.

CG — STA. 135.5 before hoisting

From appropriate 11,600 lb. GW schedule obtain hoist load as previously determined in Example 1 the maximum hoist load for normal operations is 210 lbs. Point (A).

To increase hoist load to maximum for condition without exceeding GW/CG limits, proceed up to dashed line and read left to find 435 lbs. Point (C).

The Penalty Region is any load greater than Point (A) up to maximum load at Point (C).

For GW vs. CG combinations aft of the CG represented by the dashed line (see Example 2), there is no Penalty Region.

1-26. WEIGHT EMPTY CHART

The Weight empty chart for internal hoisting operations is shown in Section 5. Refer to the maintenance manual for additional information.

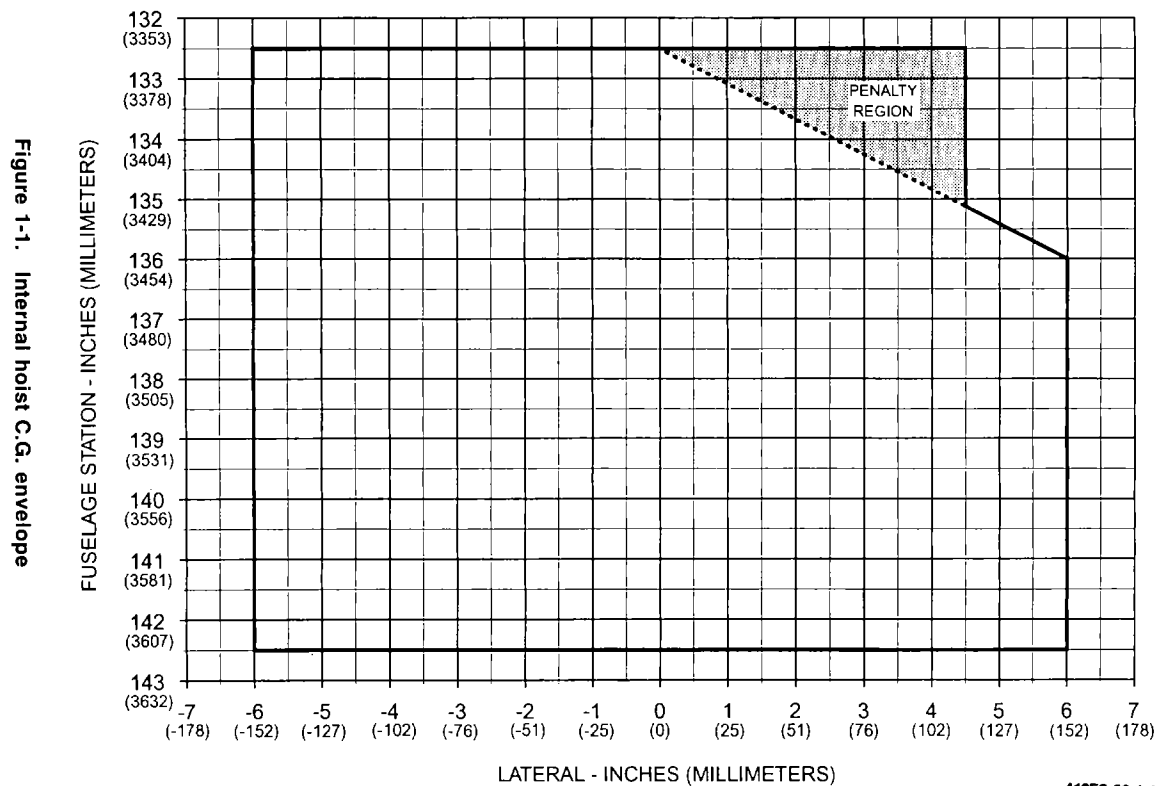
NOTE

Allowable hoist load must be computed when weight empty is not within specified guidelines, shown in Section 5.

NOTE

Allowable hoist loads must be computed when AUX Fuel kits are installed.

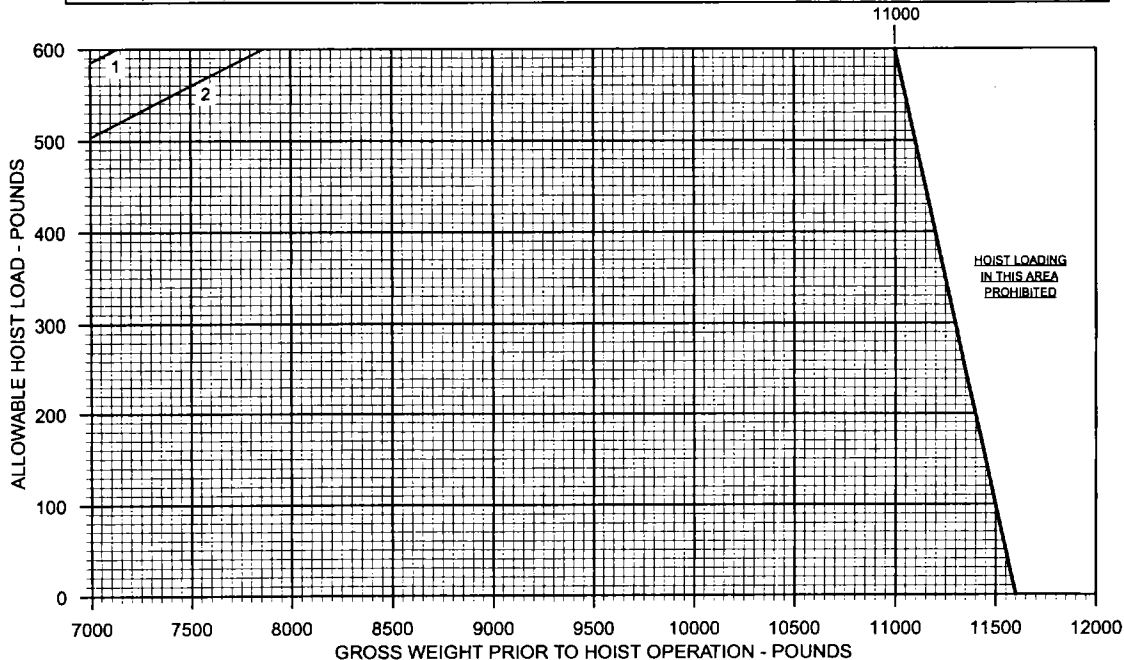
Longitudinal/Lateral C.G. Envelope for Internal Hoist Operations



412FS-26-1-1

L/H FWD POSITION

- 1 PILOT AND HOIST OPERATOR - 600 LB HOIST LOAD IS ALLOWED AT GROSS WEIGHTS FROM 7133 TO 11000 LBS.
 2 PILOT, COPILOT, AND HOIST OPERATOR - 600 LB HOIST LOAD IS ALLOWED AT GROSS WEIGHTS FROM 7866 TO 11000 LBS.

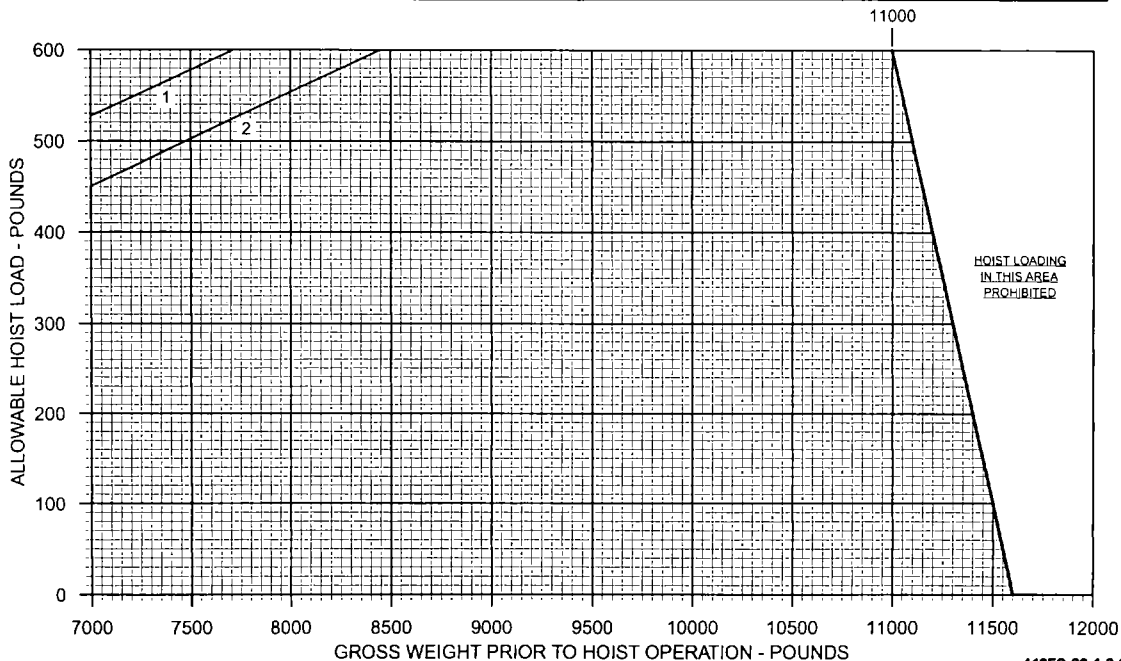


412FS-26-1-2-1

Figure 1-2. Hoist loading schedules 11,600 lb. GW (English) (Sheet 1 of 6)

L/H AFT POSITION

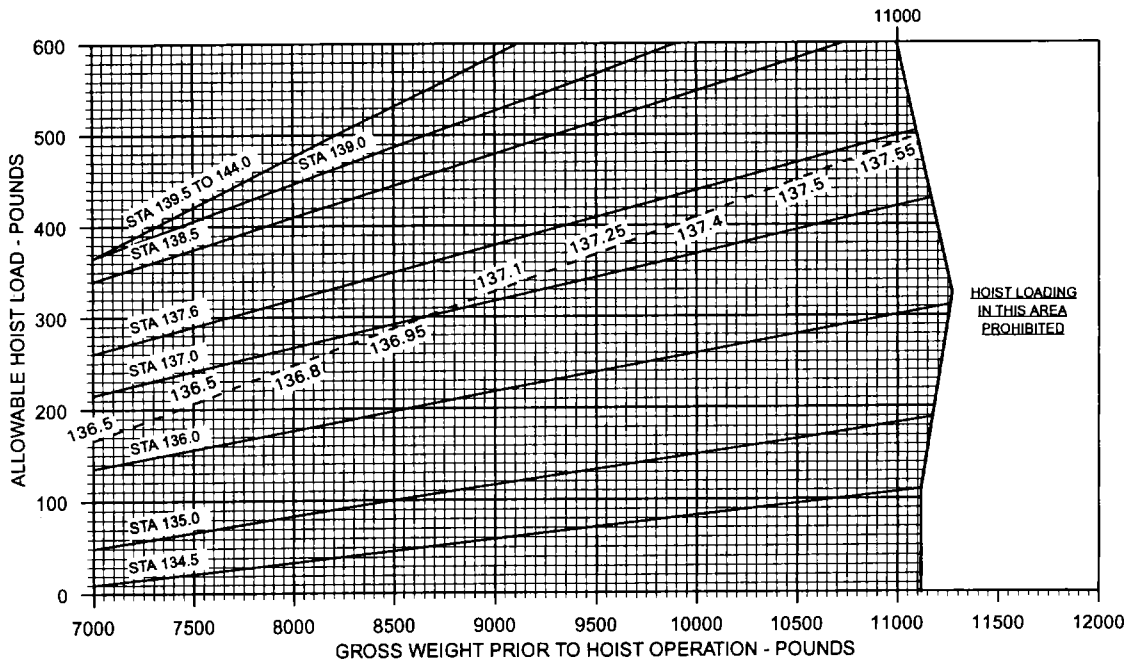
- 1 PILOT AND HOIST OPERATOR - 600 LB HOIST LOAD IS ALLOWED AT GROSS WEIGHTS FROM 7133 TO 11000 LBS.
- 2 PILOT, COPILOT, AND HOIST OPERATOR - 600 LB HOIST LOAD IS ALLOWED AT GROSS WEIGHTS FROM 7866 TO 11000 LBS.



412FS-26-1-2-2

Figure 1-2. Hoist loading schedules 11,600 lb. GW (English) (Sheet 2 of 6)

R/H FWD POSITION - PILOT AND HOIST OPERATOR

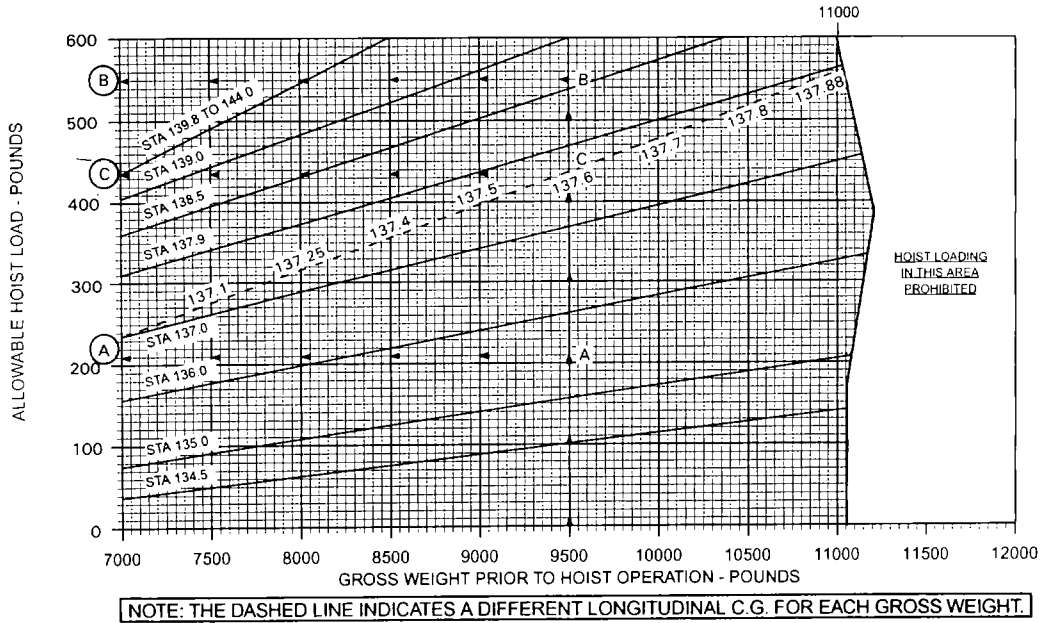


NOTE: THE DASHED LINE INDICATES A DIFFERENT LONGITUDINAL C.G. FOR EACH GROSS WEIGHT.

412F8-26-1-2-3

Figure 1-2. Hoist loading schedules 11,600 lb. GW (English) (Sheet 3 of 6)

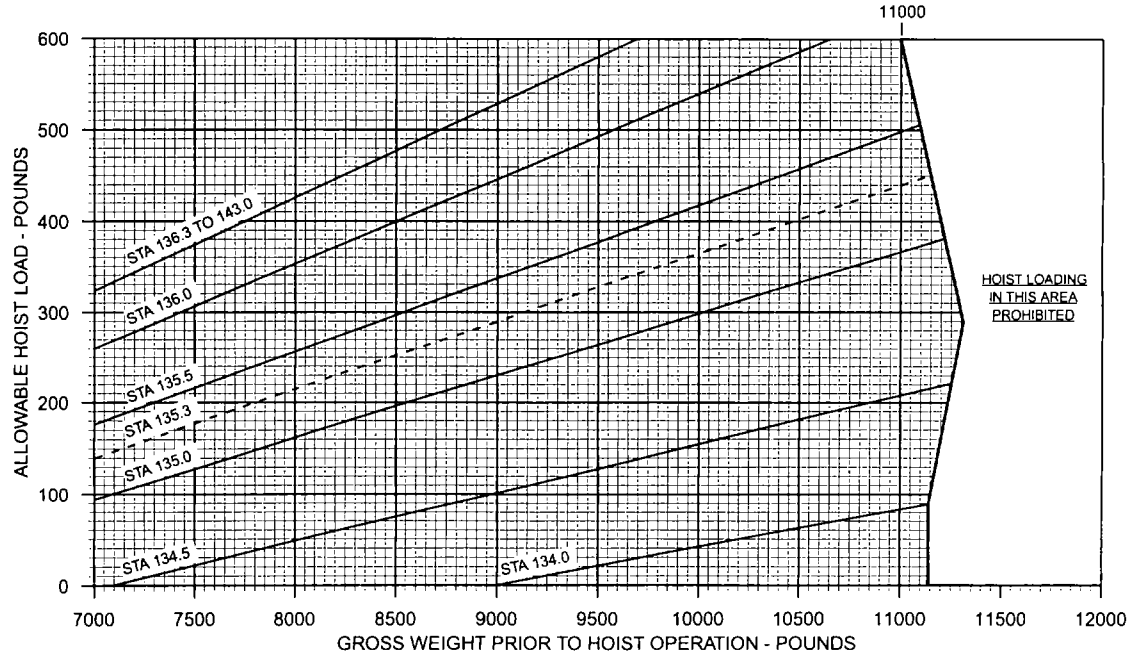
R/H FWD POSITION - PILOT, COPILOT AND HOIST OPERATOR



412FS-26-1-2-4

Figure 1-2. Hoist loading schedules 11,600 lb. GW (English) (Sheet 4 of 6)

R/H AFT POSITION - PILOT AND HOIST OPERATOR

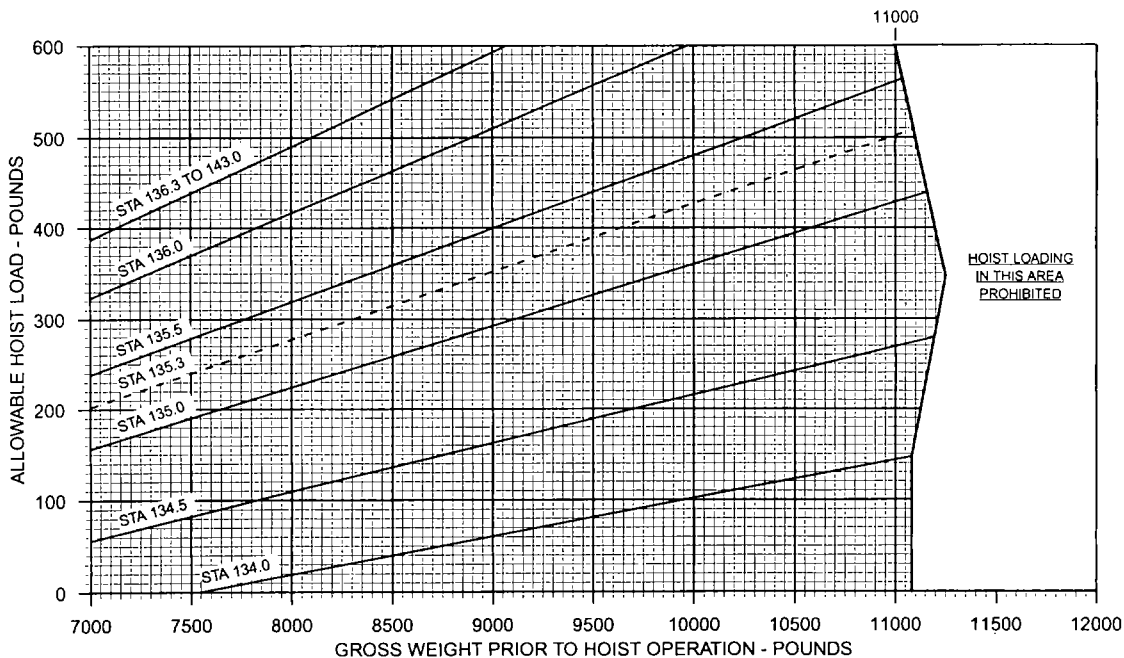


NOTE: THE DASHED LINE IS A CONSTANT LONGITUDINAL C.G. FOR ALL GROSS WEIGHTS.

412FS-26-1-2-5

Figure 1-2. Hoist loading schedules 11,600 lb. GW (English) (Sheet 5 of 6)

R/H AFT POSITION - PILOT, COPILOT AND HOIST OPERATOR



NOTE: THE DASHED LINE IS A CONSTANT LONGITUDINAL C.G. FOR ALL GROSS WEIGHTS.

412FS-26-1-2-4

Figure 1-2. Hoist loading schedules 11,600 lb. GW (English) (Sheet 6 of 6)

L/H FWD POSITION

- 1 PILOT AND HOIST OPERATOR - 600 LB HOIST LOAD IS ALLOWED AT GROSS WEIGHTS FROM 3233.8 TO 4989.5 KGS.
 2 PILOT, COPILOT, AND HOIST OPERATOR - 600 LB HOIST LOAD IS ALLOWED AT GROSS WEIGHTS FROM 3566.5 TO 4989.5 KGS.

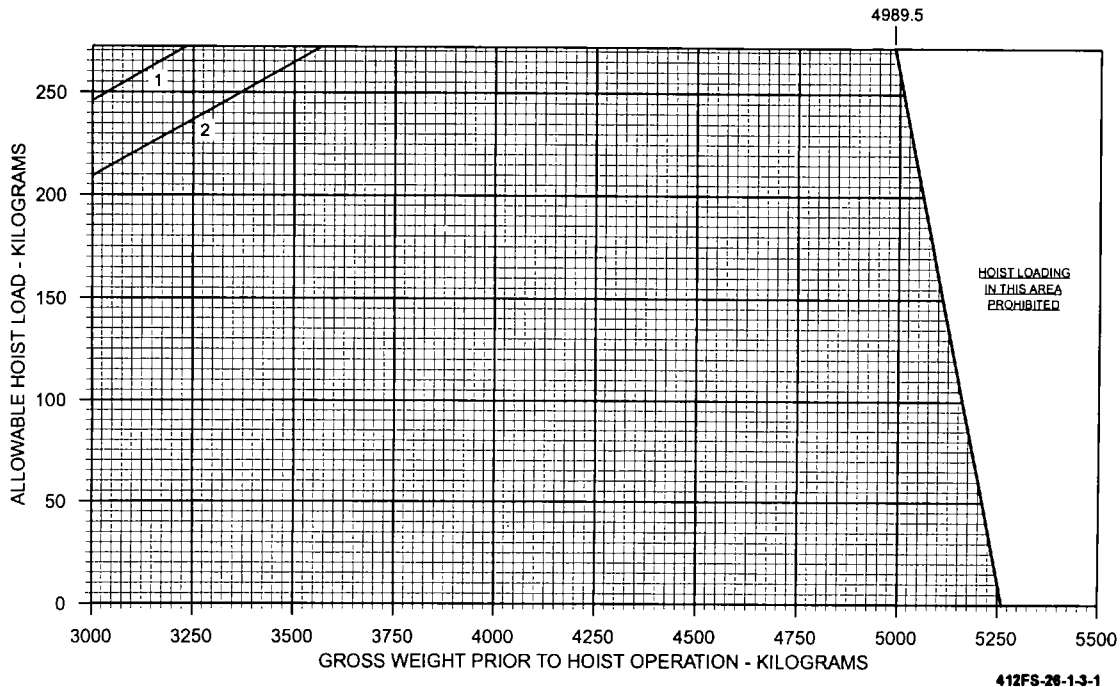
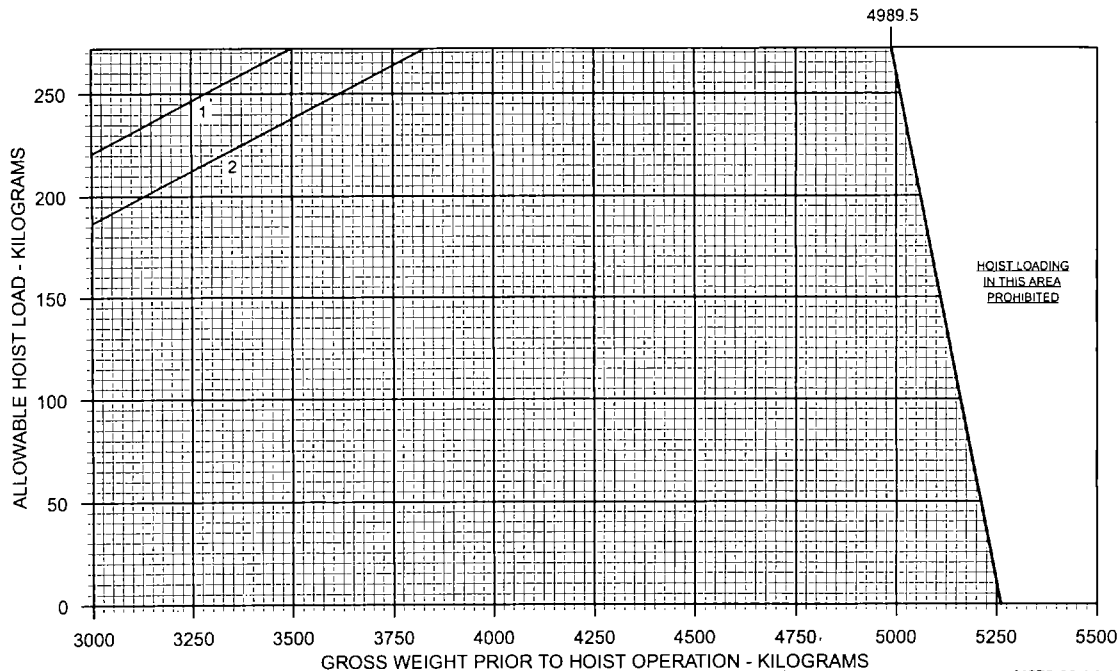


Figure 1-3. Hoist loading schedules 5261 kg. GW (Metric) (Sheet 1 of 6)

L/H AFT POSITION

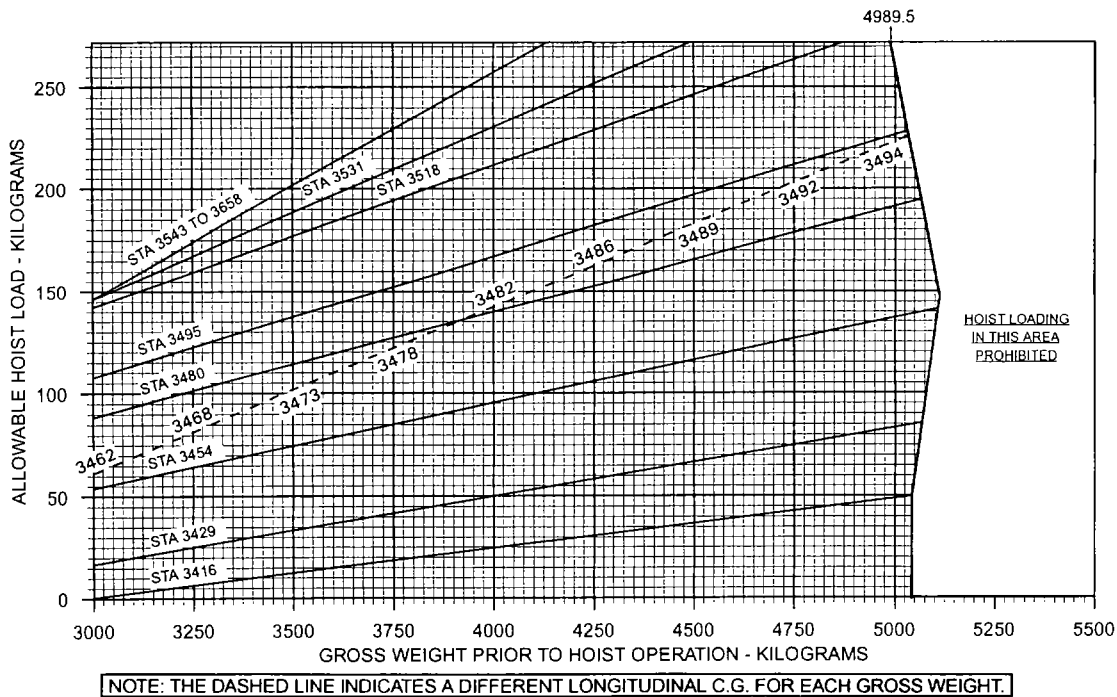
- 1 PILOT AND HOIST OPERATOR - 600 LB HOIST LOAD IS ALLOWED AT GROSS WEIGHTS FROM 3498.2 TO 4989.5 KGS.
 2 PILOT, COPILOT, AND HOIST OPERATOR - 600 LB HOIST LOAD IS ALLOWED AT GROSS WEIGHTS FROM 3830.8 TO 4989.5 KGS.



412FS-26-1-3-2

Figure 1-3. Hoist loading schedules 5261 kg. GW (Metric) (Sheet 2 of 6)

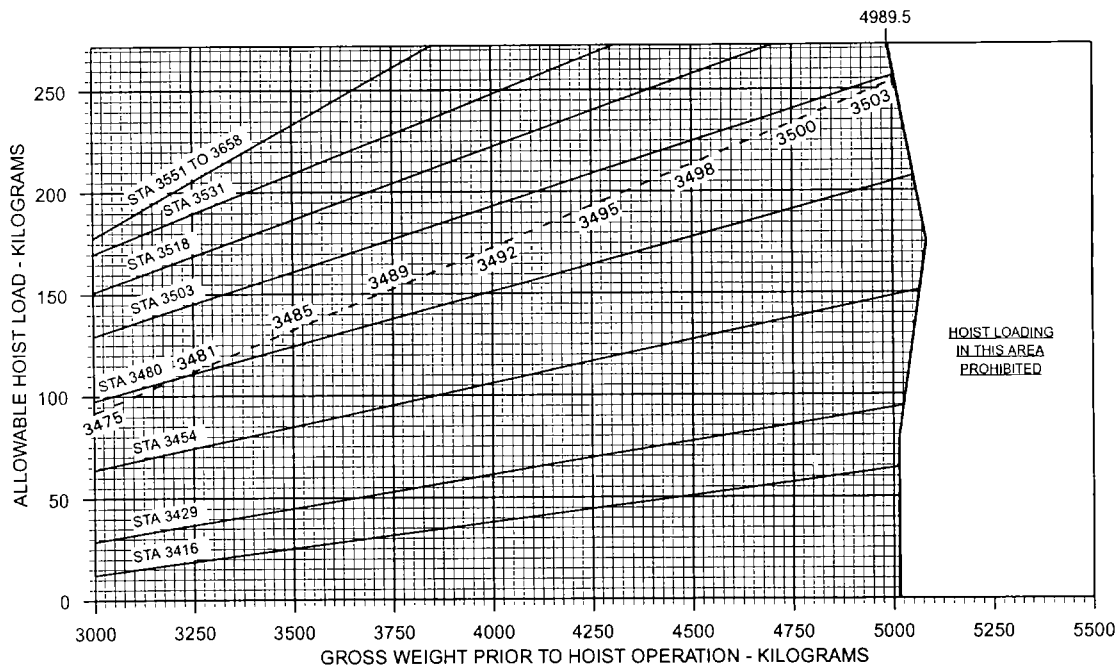
R/H FWD POSITION - PILOT AND HOIST OPERATOR



412FS-26-1-3-3

Figure 1-3. Hoist loading schedules 5261 kg. GW (Metric) (Sheet 3 of 6)

R/H FWD POSITION - PILOT, COPILOT AND HOIST OPERATOR

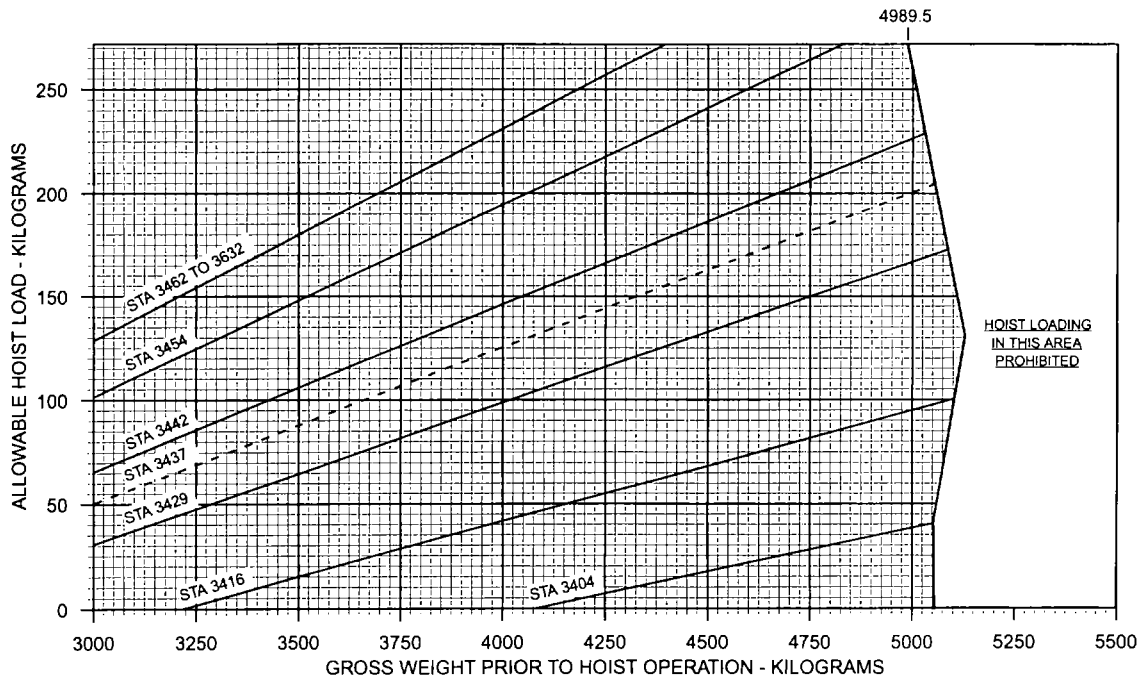


NOTE: THE DASHED LINE INDICATES A DIFFERENT LONGITUDINAL C.G. FOR EACH GROSS WEIGHT.

412FS-26-1-3-4

Figure 1-3. Hoist loading schedules 5261 kg. GW (Metric) (Sheet 4 of 6)

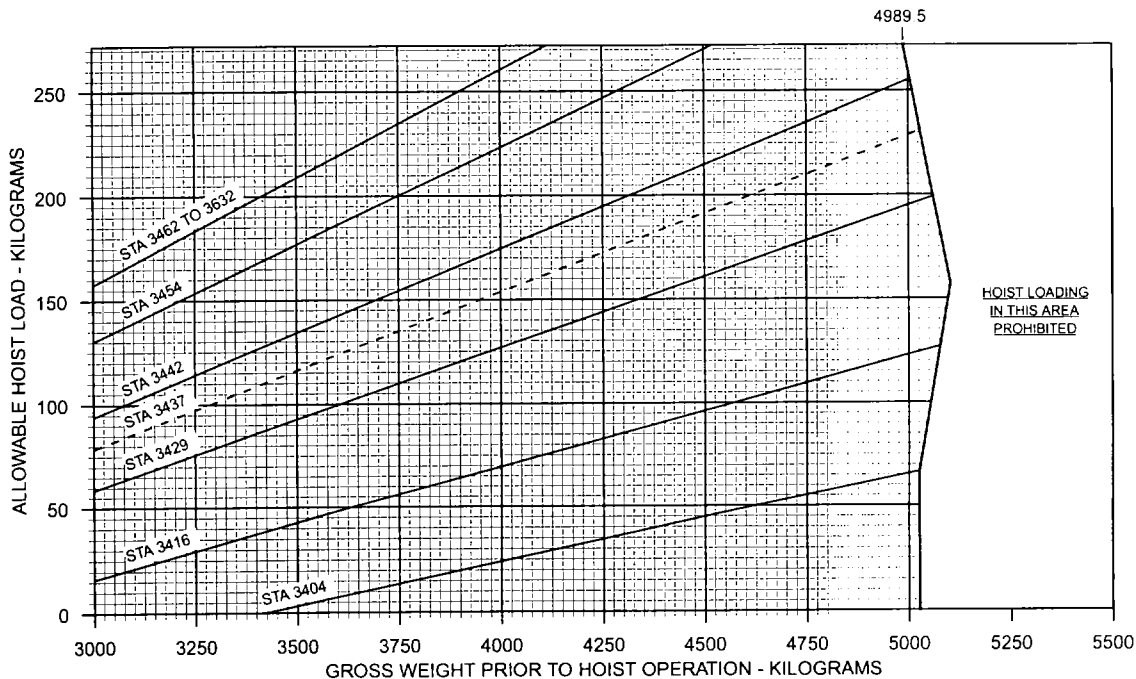
R/H AFT POSITION - PILOT AND HOIST OPERATOR



412FS-26-1-3-6

Figure 1-3. Hoist loading schedules 5261 kg. GW (Metric) (Sheet 5 of 6)

R/H AFT POSITION - PILOT, COPILOT AND HOIST OPERATOR



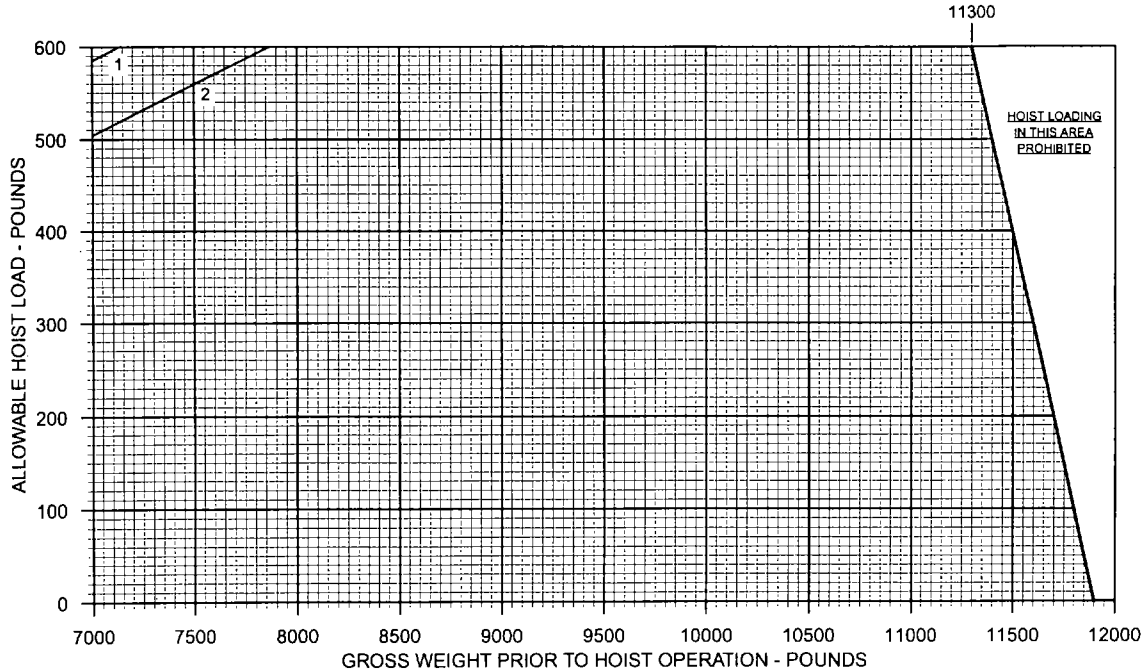
NOTE: THE DASHED LINE IS A CONSTANT LONGITUDINAL C.G. FOR ALL GROSS WEIGHTS.

412FS-26-1-3-8

Figure 1-3. Hoist loading schedules 5261 kg. GW (Metric) (Sheet 6 of 6)

L/H FWD POSITION

- 1 PILOT AND HOIST OPERATOR - 600 LB HOIST LOAD IS ALLOWED AT GROSS WEIGHTS FROM 7133 TO 11300 LBS.
 2 PILOT, COPILOT, AND HOIST OPERATOR - 600 LB HOIST LOAD IS ALLOWED AT GROSS WEIGHTS FROM 7866 TO 11300 LBS.

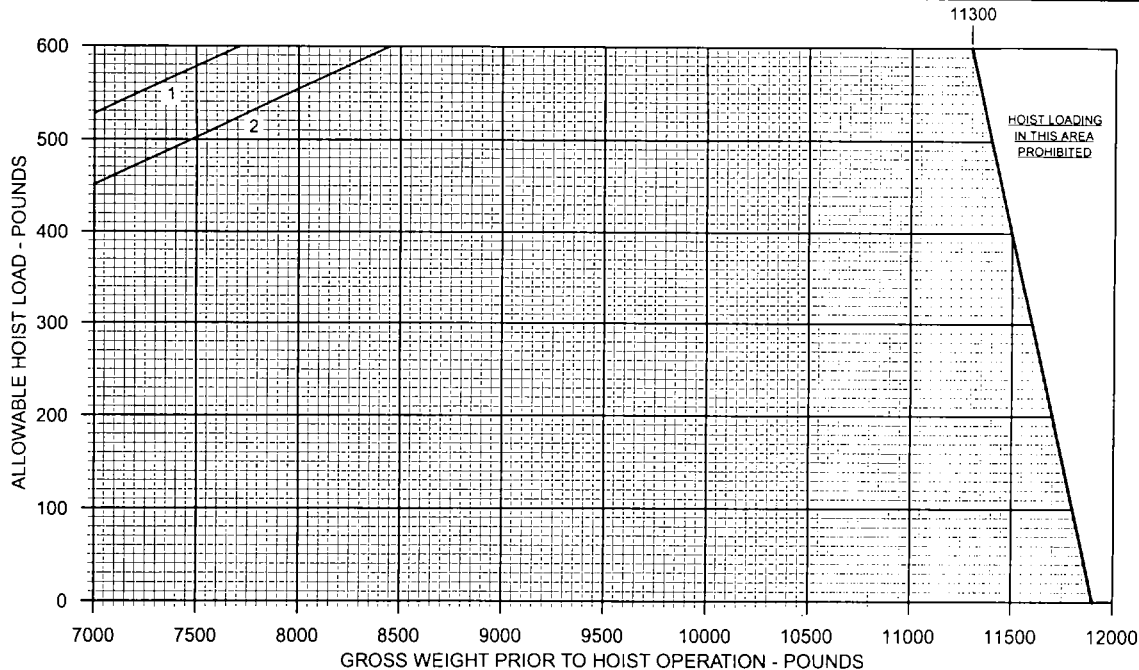


412FS-26-1-4-1

Figure 1-4. Hoist loading schedules 11,900 lb. GW (English) (Sheet 1 of 6)

L/H AFT POSITION

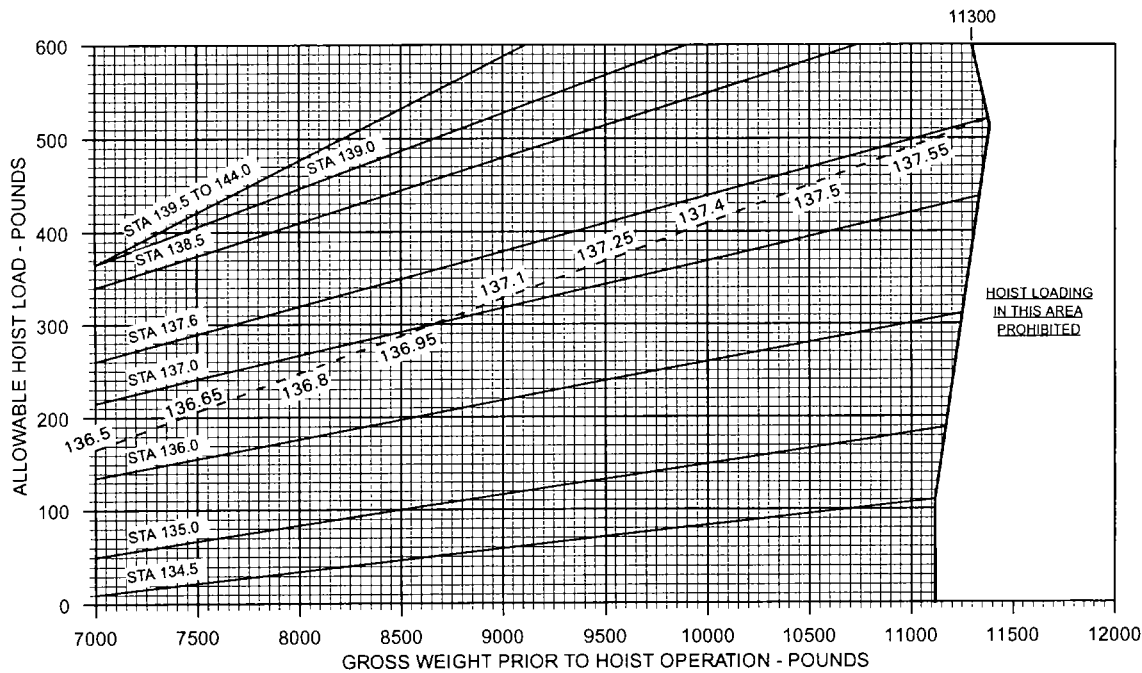
- 1 PILOT AND HOIST OPERATOR - 600 LB HOIST LOAD IS ALLOWED AT GROSS WEIGHTS FROM 7715 TO 11300 LBS.
- 2 PILOT, COPILOT, AND HOIST OPERATOR - 600 LB HOIST LOAD IS ALLOWED AT GROSS WEIGHTS FROM 8449 TO 11300 LBS.



412FS-26-1-4-2

Figure 1-4. Hoist loading schedules 11,900 lb. GW (English) (Sheet 2 of 6)

R/H FWD POSITION - PILOT AND HOIST OPERATOR

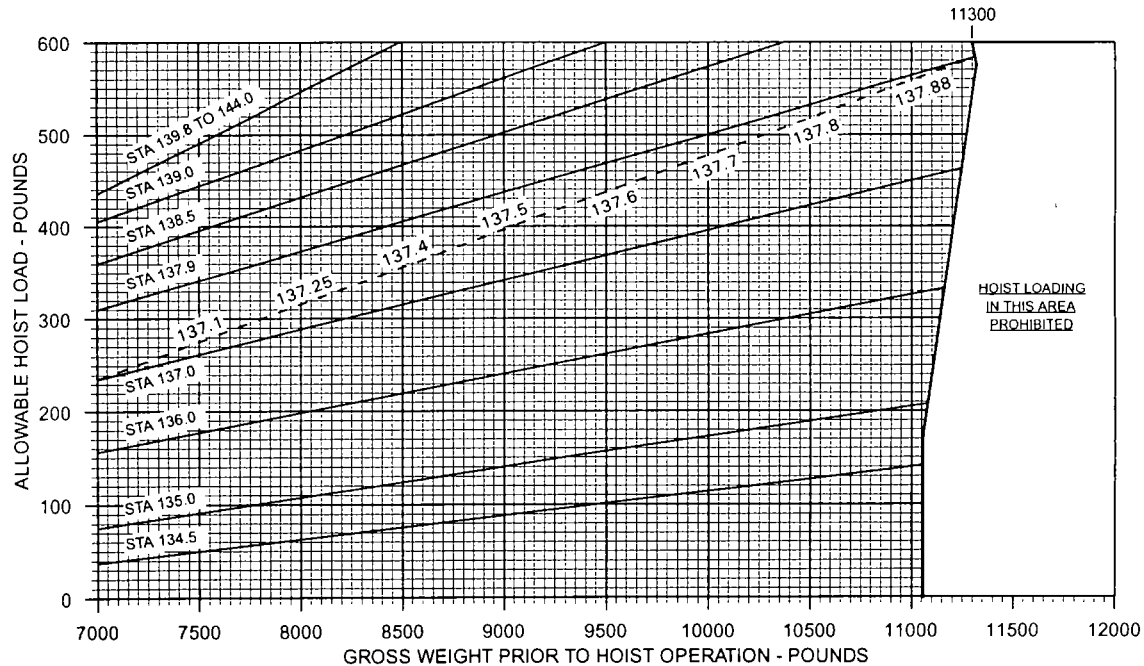


NOTE: THE DASHED LINE INDICATES A DIFFERENT LONGITUDINAL C.G. FOR EACH GROSS WEIGHT.

412FS-26-14-3

Figure 1-4. Hoist loading schedules 11,900 lb. GW (English) (Sheet 3 of 6)

R/H FWD POSITION - PILOT, COPILOT AND HOIST OPERATOR

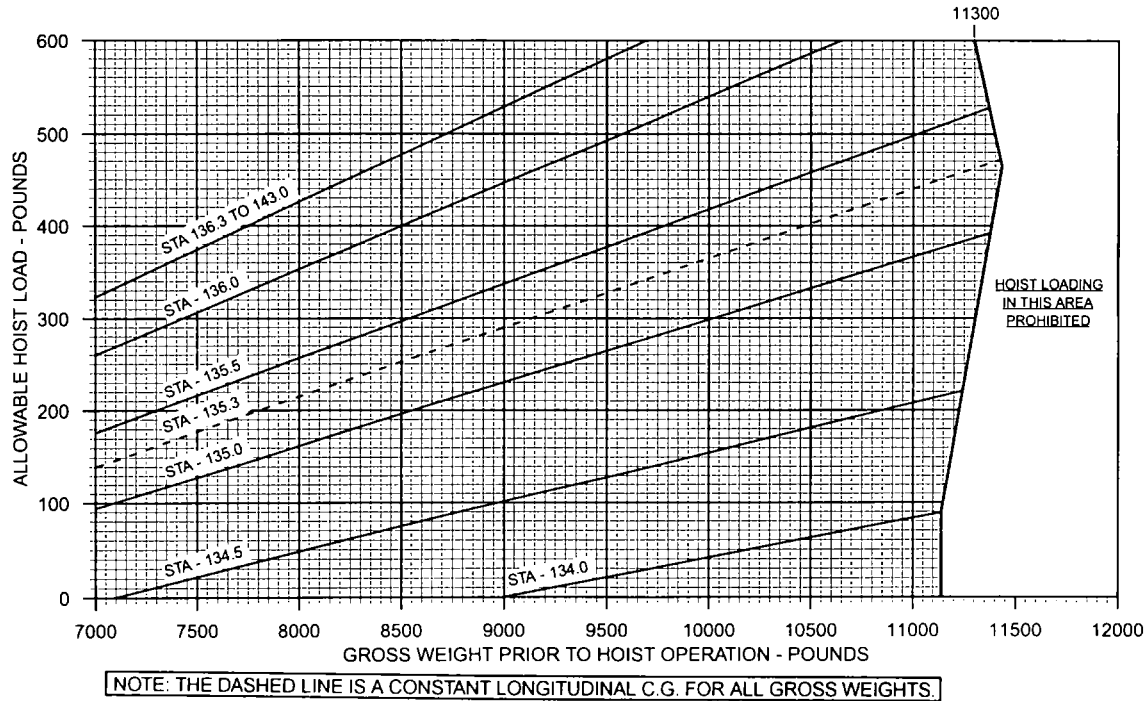


NOTE: THE DASHED LINE INDICATES A DIFFERENT LONGITUDINAL C.G. FOR EACH GROSS WEIGHT.

412FS-26-1-4-4

Figure 1-4. Hoist loading schedules 11,900 lb. GW (English) (Sheet 4 of 6)

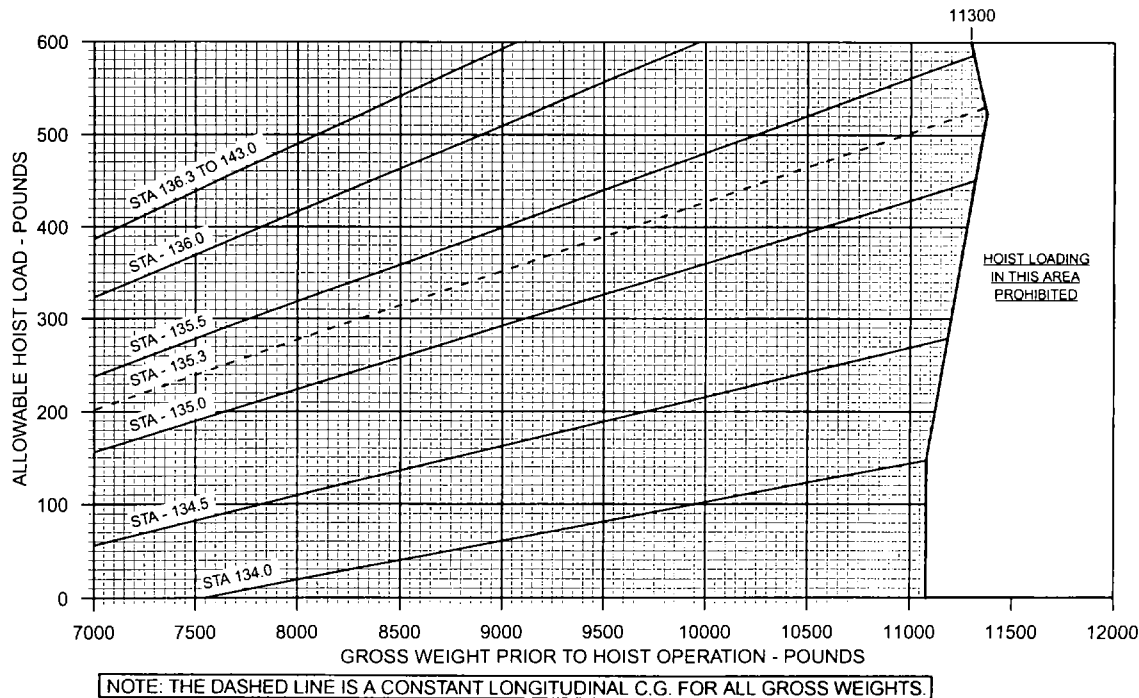
R/H AFT POSITION - PILOT AND HOIST OPERATOR



412FS-26-1-4-8

Figure 1-4. Hoist loading schedules 11,900 lb. GW (English) (Sheet 5 of 6)

R/H AFT POSITION - PILOT, COPILOT AND HOIST OPERATOR

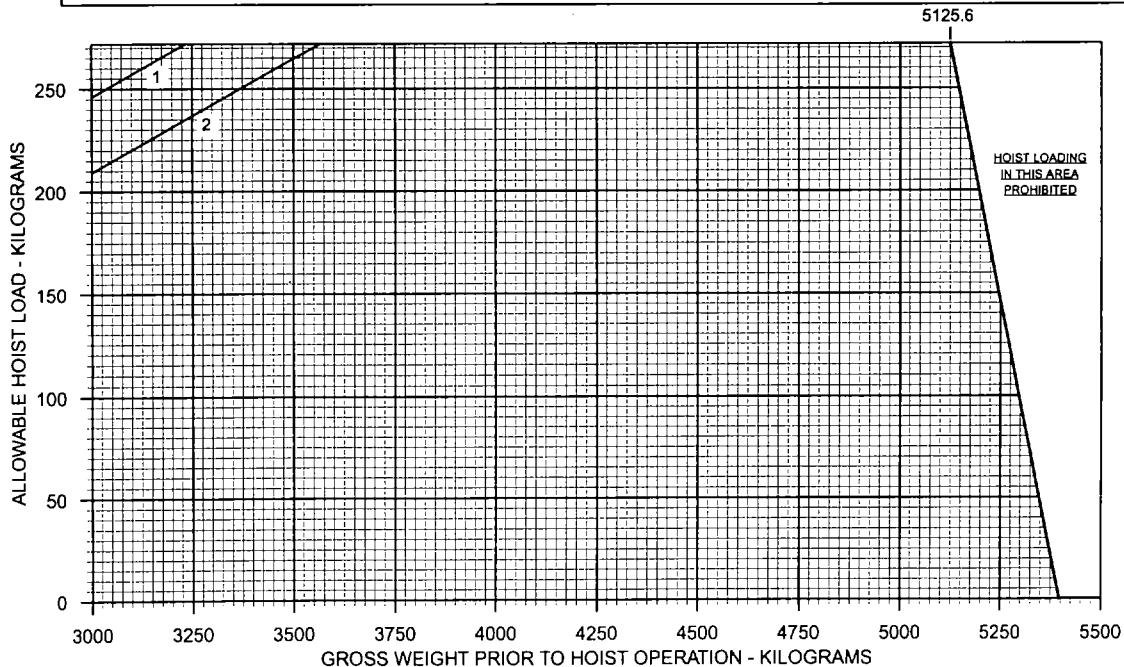


412FS-26-1-4-6

Figure 1-4. Hoist loading schedules 11,900 lb. GW (English) (Sheet 6 of 6)

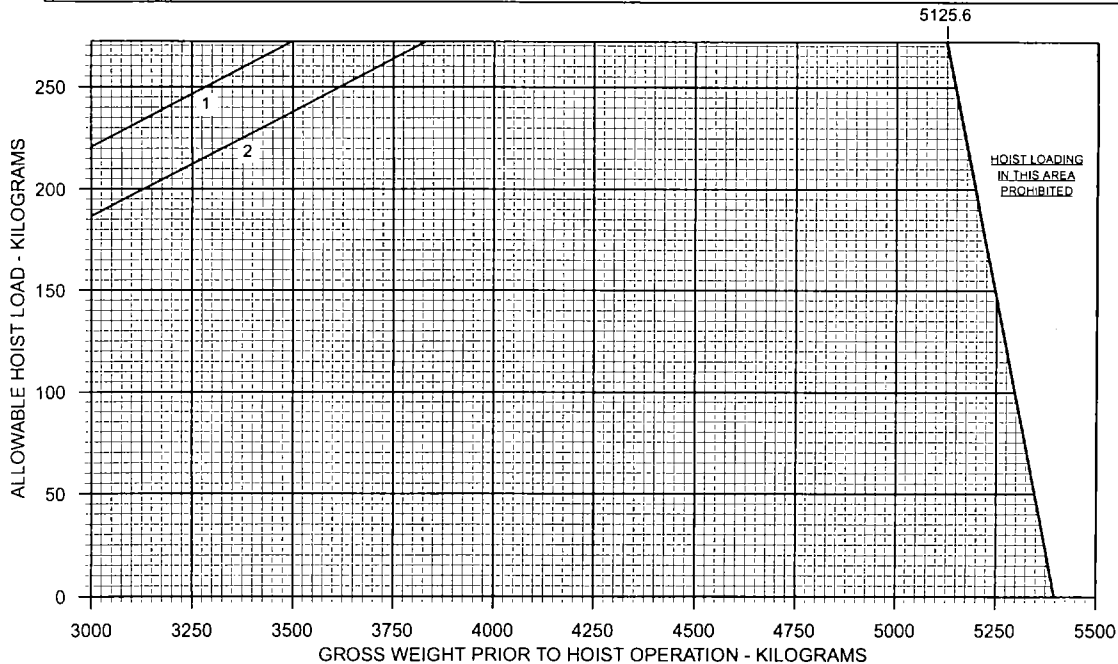
L/H FWD POSITION

- 1 PILOT AND HOIST OPERATOR - 272 KG HOIST LOAD IS ALLOWED AT GROSS WEIGHTS FROM 3233.8 TO 5125.6 KGS.
 2 PILOT, COPILOT, AND HOIST OPERATOR - 272 KG HOIST LOAD IS ALLOWED AT GROSS WEIGHTS FROM 3566.5 TO 5125.6 KGS.



L/H AFT POSITION

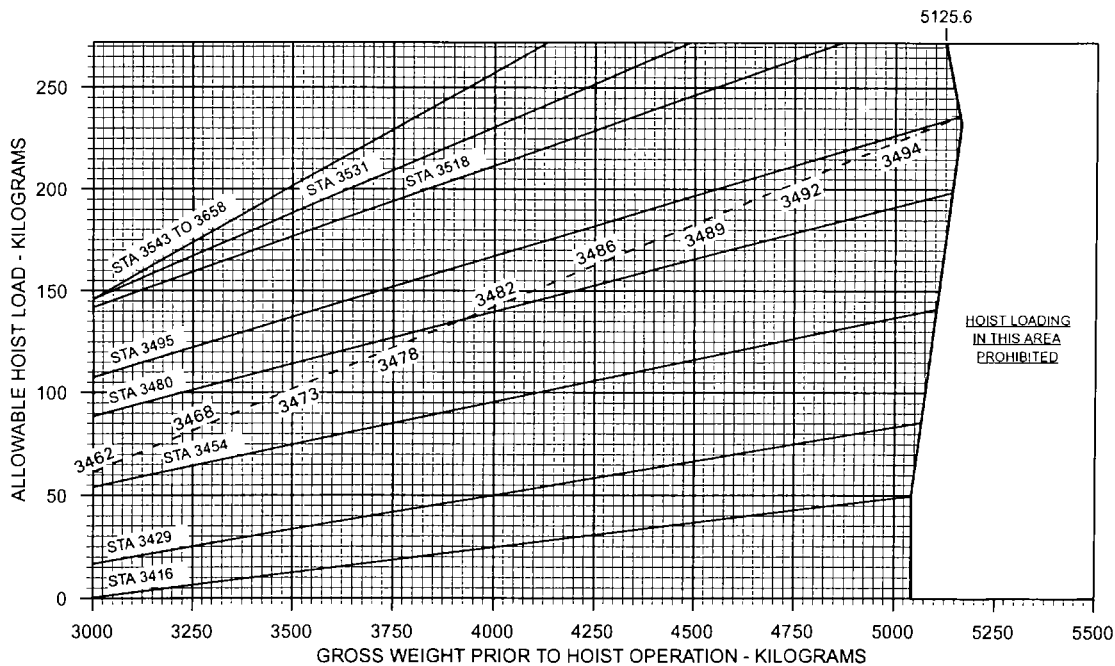
- 1 PILOT AND HOIST OPERATOR - 272 KG HOIST LOAD IS ALLOWED AT GROSS WEIGHTS FROM 3498.2 TO 5125.6 KGS.
- 2 PILOT, COPILOT, AND HOIST OPERATOR - 272 KG HOIST LOAD IS ALLOWED AT GROSS WEIGHTS FROM 3830.8 TO 5125.6 KGS.



412FS-26-1-4-2

Figure 1-5. Hoist loading schedules 5397 kg. GW (Metric) (Sheet 2 of 6)

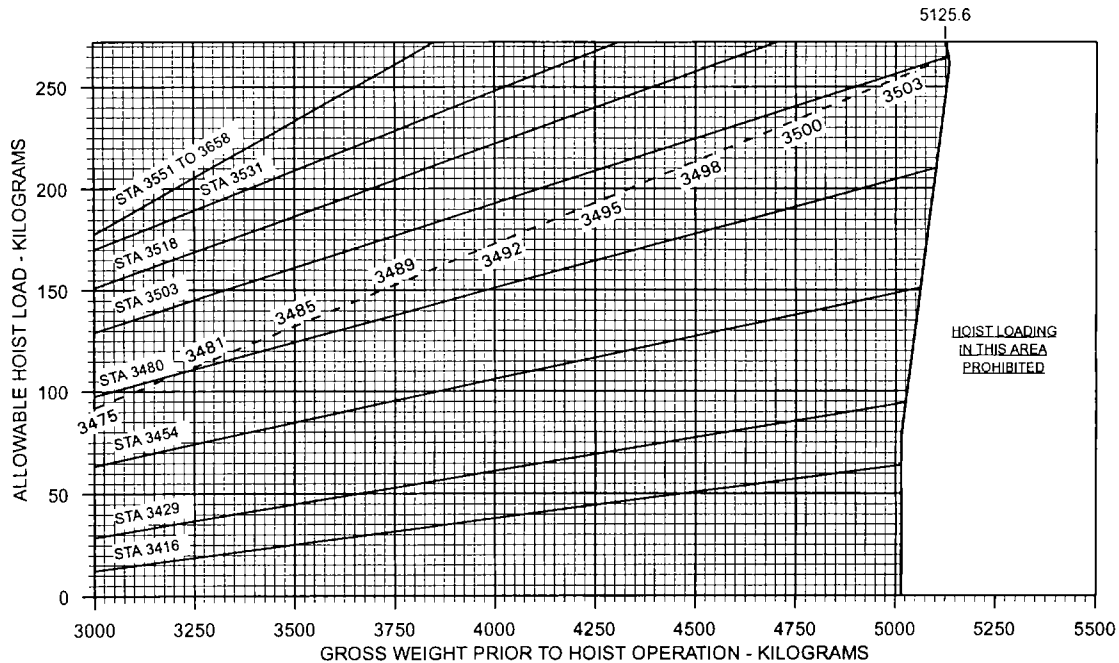
R/H FWD POSITION - PILOT AND HOIST OPERATOR



412FS-26-1-6-3

Figure 1-5. Hoist loading schedules 5397 kg. GW (Metric) (Sheet 3 of 6)

R/H FWD POSITION - PILOT, COPILOT AND HOIST OPERATOR

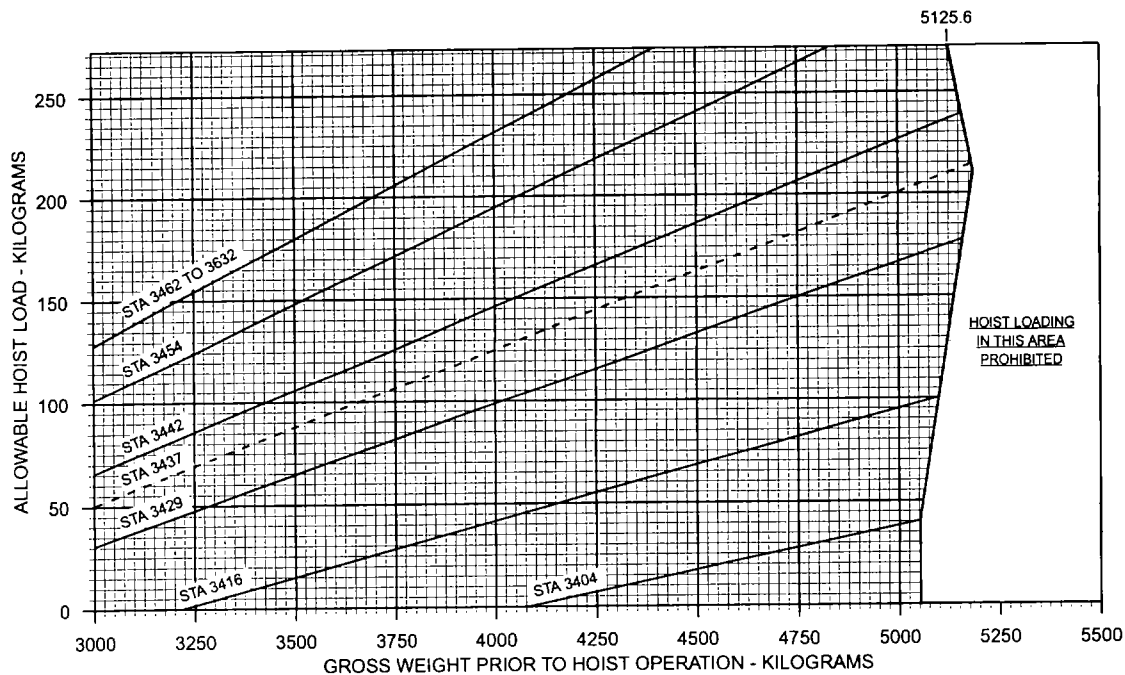


NOTE: THE DASHED LINE INDICATES A DIFFERENT LONGITUDINAL C.G. FOR EACH GROSS WEIGHT.

412FS-26-1-5-4

Figure 1-5. Hoist loading schedules 5397 kg. GW (Metric) (Sheet 4 of 6)

R/H AFT POSITION - PILOT AND HOIST OPERATOR

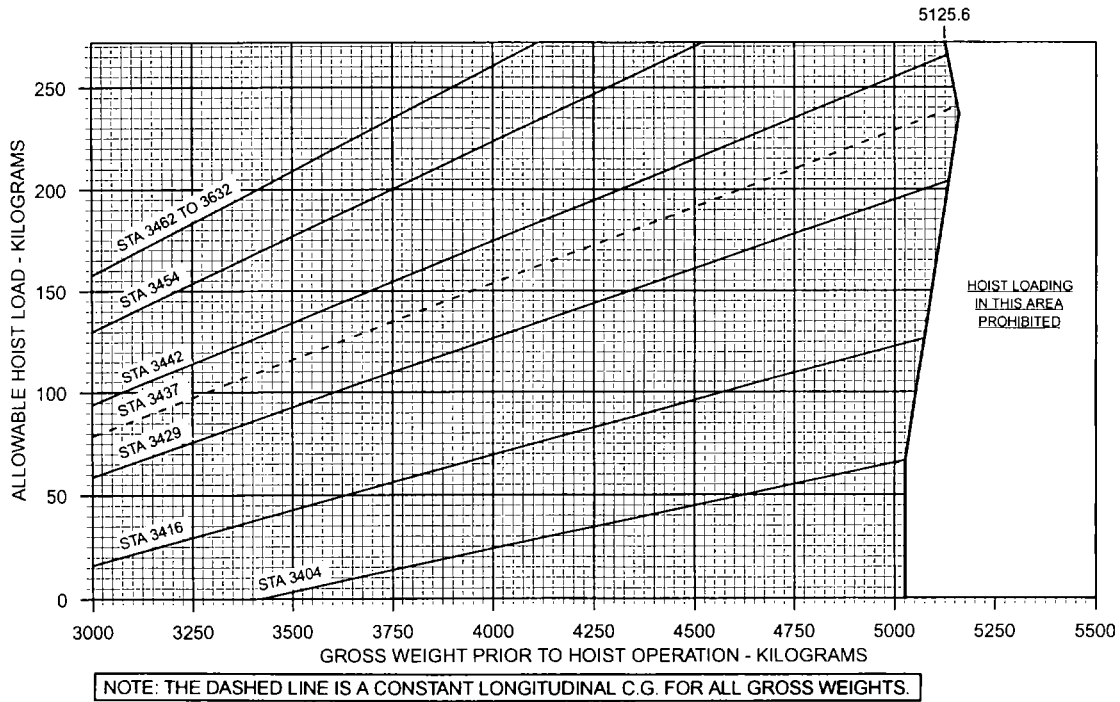


412FS-26-1-B-5

Figure 1-5. Hoist loading schedules 5397 kg. GW (Metric) (Sheet 5 of 6)

R/H AFT POSITION - PILOT, COPILOT AND HOIST OPERATOR

Figure 1-5. Hoist loading schedules 5397 kg. GW (Metric) (Sheet 6 of 6)



412FS-26-1-5-6

Section 2

NORMAL PROCEDURES

2-2. FLIGHT PLANNING

WARNING

HOIST LOAD CAN CAUSE LONGITUDINAL OR LATERAL CG LIMITS TO BE EXCEEDED. GROSS WEIGHT AND CENTER OF GRAVITY SHALL BE COMPUTED TO ASSURE LOADING WITHIN APPROVED LIMITS.

CAUTION

IF ADDITIONAL LOADS ARE CARRIED DURING HOISTING OPERATIONS, LOADS SHOULD BE PLACED ON SIDE OF HELICOPTER OPPOSITE HOIST POSITION.

Gross weight and CG — Compute with and without hoist load.

2-4. INTERIOR AND PRESTART CHECK

2-4-A. HOIST INSTALLATION CHECK

NOTE

If pilot plans to operate hoist with other crewmember in passenger compartment, hoist shall be installed in forward right position.

Hoist — Installed in desired position; check roof and floor stud adapters and locking collars properly secured.

Boom actuator — Installed in proper position; all fittings secured.

AIRCRAFT POSITION switch (on hoist control box, figure 2-1) — Set in proper position.

Hook — Rotates freely on cable.

Cable — Check proper routing through guide rollers, pulleys, and drums.

Gearbox oil levels — Check sight glasses.

Hoist operators pendant — Installed; connectors secured.

Electrical power cables — Condition; connectors secured.

WARNING

ACTUATION OF CABLE CUT SWITCH ON PEDESTAL CAN CUT CABLE REGARDLESS OF HOIST PWR SWITCH POSITION. ACTUATION OF CABLE CUT SWITCH ON HOIST CAN CUT CABLE, EVEN IF CABLE CUT CIRCUIT BREAKER IS OUT.

CABLE CUT switches (pedestal and hoist) — Off; covers safetied.

Safety vests, tether straps, hoisting slings, and litters — Condition; secured or stored.

2-4-B. HOIST OPERATION CHECK

Cargo doors and hinged panels — Secured open or removed.

HOIST PWR. CONT and CABLE CUT circuit breakers — In.

BATTERY switches — ON (or connect external power).

NON ESNTL BUS switch — MANUAL.

ICS — Check intercom between pilot and hoist operator using hoist pendant ICS trigger and HOT MIC switch (right ICS box only).

HOIST PWR switch — ON. Check that blue HOIST POWER light on hoist control box and amber CAUTION light on hoist pendant illuminates.

Hoist pendant CAUTION and OVER TEMP indicators — Pres to test.

HOIST UP/DOWN, BOOM IN/OUT, and SPEED HIGH/LOW switches (pilot and operator) — Actuate to check all hoist functions for proper operation. Check that pilot HOIST switch overrides pendant HOIST switch.

Hoist OVERTEMP warning lights — Press to test.



MAINTAIN TENSION ON HOIST CABLE WHILE REELING IN AND OUT TO PREVENT SLACK.

HOIST and BOOM switches (pilot and operator) — Actuate to check all hoist functions for proper operation. Check that pilot HOIST switch overrides operator pendant HOIST switch.

Hoist cable — Check for corrosion, kinks, flat spots, fraying, or broken strands.

Up limit switch actuator - Raise while hoist is reeling in and check hoist motor stops; then release and check hoist resumes operation.

Reduce hoist speed as cable approaches up limit. Check that hoist stops when hook reaches up limit without excess tension on cable.

Hoist - Stowed for flight; hook restraint secured.

HOIST PWR switch - OFF.

NON ESNTL BUS switch - NORMAL.

BATTERY switches - OFF.

NOTE

Ground crewmember should be instructed to discharge helicopter static electricity before attaching load to hoist when possible.

2-6. SYSTEMS CHECK

Cargo doors and hinged panels - Secured open or removed.

CABLE CUT switches (pedestal and hoist) - Off; covers safetied.

HOIST PWR, CONT, and CABLE CUT circuit breakers - In.

2-6-A. BEFORE TAKEOFF

Safety vests and straps - On and secured to helicopter.

Gloves - On.

STEP switch (if installed) - STOW.

2-9. IN-FLIGHT OPERATIONS

Maximum hoist load shall be determined prior to each hoist operation.

NOTE

The Height-Velocity Diagram is not a limitation for internal hoist operations under an appropriate operating certificate.

HOIST PWR switch - ON.

WARNING

HOIST OPERATOR SHALL BE SECURED TO HELICOPTER WITH AN APPROVED SAFETY HARNESS DURING HOIST OPERATIONS.

Establish hover over hoist operation area.

Hoist hook restraint - Removed.

SPEED switch — As desired (refer to limitations).

BOOM switch (or pilots HOIST switch) - OUT.

NOTE

Each hoist operation performed is defined as reeling hoist cable out and then in while hovering with any weight on hoist, regardless of whether the hoist was used for training or an actual rescue.

The pilot must record each operation in the penalty CG region. For each hoist operation performed within penalty CG region, four (4) additional hours of usage must be logged against the main rotor yoke, mast and lower cone seat.

HOIST switch - DOWN.

Discharge static electricity when possible, and connect hook to load, observing allowable hoist load.

NOTE

As hook nears the up or down limits, hoist speed automatically slows.

HOIST switch - UP.

CAUTION

USE CARE TO PREVENT CABLE, HOOK, AND LOAD FROM FOULING ON FUSELAGE OR LANDING GEAR.

Maintain zero ground speed until load is clear of obstructions.

BOOM switch - IN to swing hoist boom and load into cabin, if possible.

Takeoff into wind, if possible, allowing adequate hoist load clearance over obstacles if load is not internal.

CAUTION

AIRSPEED WITH EXTERNAL LOAD IS LIMITED BY CONTROLLABILITY. CAUTION SHOULD BE EXERCISED WHEN CARRYING AN EXTERNAL LOAD. HANDLING CHARACTERISTICS MAY BE AFFECTED BY THE SIZE, WEIGHT, AND SHAPE OF LOAD.

Airspeed - As required for adequate controllability, not to exceed limits for hoist operations (20 or 60 KIAS, as applicable).

2-13. LITTER HOISTING

When emergency transportation of a patient by litter is essential, every effort should be made to land the helicopter for litter loading. Litter hoisting can be hazardous and should be accomplished only when a landing is not feasible and the condition of the patient precludes the use of the personnel hoisting sling.

In addition to all other procedures contained herein, the following shall apply to litter hoisting operations.

2-13-A. EMPTY LITTER**WARNING**

HOISTING OR LOWERING AN EMPTY LITTER IN OPEN POSITION IS PROHIBITED. AN EMPTY LITTER CAN OSCILLATE UNCONTROLLABLY IN ROTOR WASH AND FLY UPWARD, STRIKING FUSELAGE OR TAIL ROTOR.

Prior to hoisting or lowering an empty litter, litter shall be closed and secured with straps. Litter should be suspended in a near-vertical position and sling straps should be drawn tight.

2-13-B. LOADED LITTER**WARNING**

LITTER PATIENT SHALL BE SECURED TO LITTER WITH

SAFETY STRAPS.

HOIST HOOK CATCH SHALL BE SECURED WITH SAFETY PIN PRIOR TO HOISTING LITTER PATIENT.

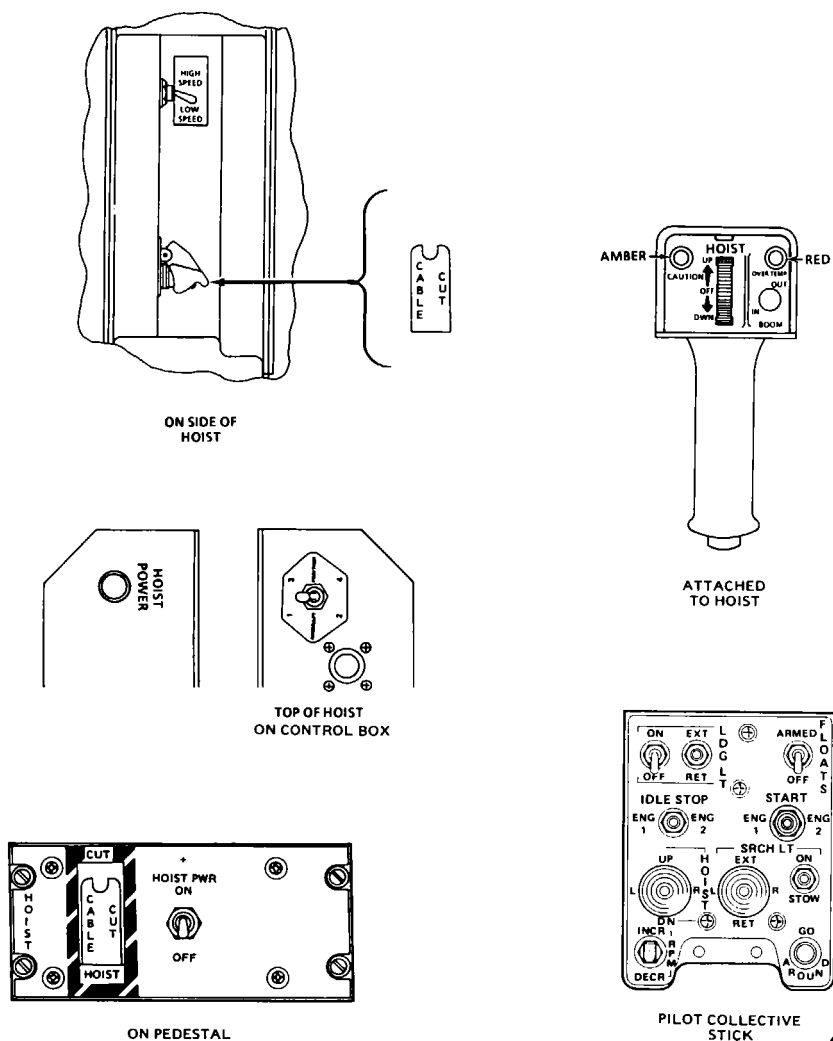
Litter sling straps should be adjusted so that litter is 24 to 28 inches (61 to 71 centimeters) below hoist hook.

NOTE

If litter is suspended too far below hook, litter cannot be loaded in helicopter with hoist hook at up limit.

CAUTION

A LOADED LITTER CAN ROTATE ABOUT CABLE DURING HOISTING. HOIST OPERATOR MAY HAVE TO GRASP LITTER SLING STRAPS TO CONTROL ROTATION AS LITTER APPROACHES LANDING GEAR.



412FS-26-2-1

Figure 2-1. Internal hoist controls

Section 3

EMERGENCY/MALFUNCTION PROCEDURES

3-15. HOIST LOAD JETTISON

To jettison hoist load in an emergency, actuate CABLE CUT switch (located on pedestal or hoist).

In the event of failure of CABLE CUT switch, sever cable with manual cable cutter (stowed in pouch on hoist).

3-15-A. HOIST OVERTEMP WARNING LIGHT

In the event that the OVERTEMP warning light (located on pendant) illuminates, continue present operation until hoist cable is reeled in. Leave HOIST PWR switch ON (for cooling fan operation). When OVERTEMP light extinguishes, hoisting may be resumed as desired.

Section 4

PERFORMANCE

No change from basic manual.

Section 5

WEIGHT AND BALANCE

5-11. WEIGHT EMPTY CHART

The Weight empty chart for internal hoisting operations is shown in figure 5-1. Refer to the maintenance manual for additional information.

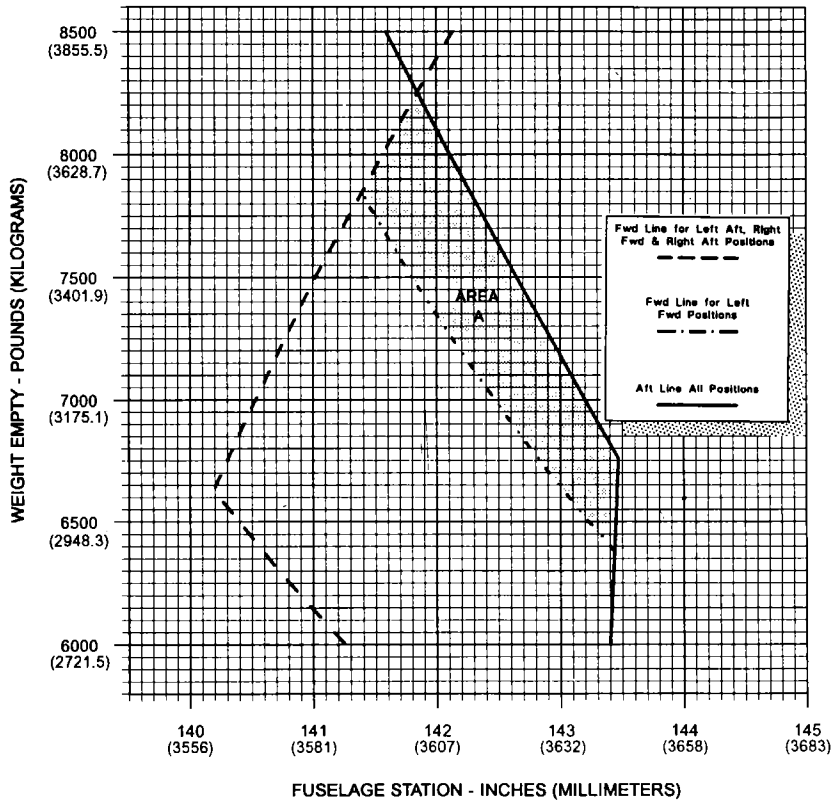
NOTE

Allowable hoist load must be computed when weight empty is not within specified guidelines.

NOTE

Allowable hoist load must be computed when AUX Fuel kits are installed.

412 Weight Empty Chart for Internal Hoist Operations



412FMS-26-8-1

Figure 5-1. Weight empty chart



ROTORCRAFT FLIGHT MANUAL

SUPPLEMENT FOR LITTER KIT OPERATION

(205-706-047)

CERTIFIED
14 OCTOBER 1988

This supplement shall be attached to the Models 412 and 412 EP Flight Manual when the Litter Kit 205-706-047 has been installed.

The information contained herein supplements the information of the basic Flight Manual. For Limitations, Procedures, and Performance Data not contained in this supplement, consult the basic Flight Manual.

Bell Helicopter **TEXTRON**

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REISSUE — 23 JUNE 1994

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BHT-412-FMS-27

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FLIGHT MANUAL

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4-1/4-2	0		

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MANAGER

ROTORCRAFT CERTIFICATION OFFICE
FEDERAL AVIATION ADMINISTRATION
FT. WORTH, TX 76193-0170

NOTE: Revised text is indicated by a black vertical line.
Insert latest revision pages; dispose of superseded pages.

INTRODUCTION

The Litter Kit provides three litters and the provisions for installing up to three litters in the helicopter. A cabin attendant seat is also included in the kit.

Section 1

LIMITATIONS

WEIGHT/CG LIMITATIONS

Actual weight change shall be determined after the litter(s) are installed and ballast readjusted if necessary, to return empty weight CG within allowable limits.

MINIMUM FLIGHT CREW

The minimum flight crew for litter operations shall consist of a pilot and a second crewmember or cabin attendant, both of whom shall be trained in and capable of assisting in litter patient emergency evacuation procedures.

Section 2

NORMAL PROCEDURES

LITTER LOADING

Secure patients to litters, then load litters aboard the helicopter in sequence from top to bottom. When only two patients are carried, they should occupy the top and center litter positions. When only one patient is carried, the center litter should be used.

opened. Refer to Section 3. Emergency Procedures for unloading procedures when cabin doors cannot be opened.

Open cabin door and unload litters and patients from the helicopter in sequence from bottom to top.

LITTER UNLOADING

Litters to be handled by one person inside cabin and one person outside cabin.

NOTE

Normal unloading procedures apply when either passenger door can be

Section 3

EMERGENCY AND MALFUNCTION PROCEDURES

UNLOADING THROUGH EMERGENCY EXITS

NOTE

In the event that cabin doors can not be opened, litter patients shall be unloaded through emergency pop-out windows. After all litter patients have been removed, ambulatory patients may then exit.

Remove emergency pop-out window by pushing at corners as marked.

Unstrap patient on center litter and remove patient through window opening.

Disconnect top litter at end near open window and lower end to rest on center litter. Remove patient retention straps and slide patient down litter and out through window opening.

Raise top and center litter ends near open window and engage center litter in brackets for top litter. Disconnect bottom litter. Raise bottom litter at end near open window and rest handles on the lower surface of the window opening. Unstrap patient and slide patient up litter and through window opening.

Section 4

PERFORMANCE

No change from basic Flight Manual.

Section 1

WEIGHT AND BALANCE

TABLE OF MOMENTS (IN-LB)		TABLE OF MOMENTS <u>(kg • mm)</u> 100	
Weight (Pounds)	LITTER PATIENT	Weight (K.G.)	LITTER PATIENT
	Loaded Laterally F.S. 117		Loaded Laterally 2972 mm
100	11700	50	1486.0
110	12870	55	1634.6
120	14040	60	1783.2
130	15210	65	1931.8
140	16380	70	2080.4
150	17550	75	2229.0
160	18720	77.1	2291.4
170	19890	80	2377.6
180	21060	85	2526.2
190	22230	90	2674.8
200	23400	95	2823.4
210	24570	100	2972.0
220	25740	105	3120.6
		110	3269.2



**ROTORCRAFT
FLIGHT MANUAL**

33108 — 33213

36001 — 36019

AND

33214 — 33999

36020 AND SUB

**SUPPLEMENT FOR
DUAL BATTERY INSTALLATION**

(412-899-225)

**CERTIFIED
APRIL 5, 1989**

This supplement shall be attached to the Model 412 Flight Manual (BHT-412-FM-2 or -3) when the 412-899-225 Dual Battery Installation has been installed.

The information contained herein supplements the information of the basic Flight Manual. For Limitations, Procedures, and Performance Data not contained in this supplement, consult the basic Flight Manual.

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APRIL 5, 1989

REISSUED — 8 OCTOBER 1991

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FLIGHT MANUAL

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FT. WORTH, TEXAS 76193-0170

NOTE

Revised text is indicated by a black vertical line. Insert latest revision pages; dispose of superseded pages.

INTRODUCTION

The dual battery system consists of two 25.2 volt, 40 ampere-hour batteries located in the nose compartment. Both batteries are connected to BAT BUS NO. 1 in parallel. BATT 1 HOT and BATT 2 HOT warnings are located on the caution segment panel. Battery switches are located on the instrument panel and are labeled BATT 1 and BATT 2. There are two circuit breakers, labeled BAT 2 CB1 and BAT 2 CB2, located on the nose compartment floor, on the right side of the helicopter.

Section 1

LIMITATIONS

WEIGHT/CG LIMITATIONS

Weight change shall be determined after the dual batteries have been installed, and ballast shall be readjusted (if necessary) to return empty weight CG to within allowable limits.

ELECTRICAL LIMITATIONS

BATTERY LIMITATIONS

Maximum battery case temperature is 54.5 °C (130 °F), as indicated by illumination of BATT 1 HOT or BATT 2 HOT segment lights.

WARNING

BATTERY SHALL NOT BE USED FOR ENGINE START AFTER ILLUMINATION OF BATT 1 HOT OR BATT 2 HOT SEGMENT LIGHT. BATTERY SHALL BE REMOVED AND SERVICED IN ACCORDANCE WITH MANUFACTURER'S INSTRUCTIONS.

Section 2

NORMAL PROCEDURES

BEFORE EXTERIOR CHECK

Flight planning — Completed.

Gross weight and CG — Compute (refer to Weight and Balance section in Manufacturer's Data BHT-412-MD).

Publications — Checked.

Portable fire extinguishers — Installed and secured.

Aft fuel sumps — Drain samples as follows:

FUEL TRANS switches — OFF.

BOOST PUMP switches — OFF.

FUEL switches — OFF.

BATT 1 switch — ON.

BATTERY BUS 1 switch — ON.

Aft fuel sump drain buttons (left and right) — Depress.

NOTE

If aft sumps fail to drain, the sump valves may be operated manually.

Forward and middle fuel sumps — Drain samples as follows:

Press-to-drain valves — Press.

Fuel filters — Drain before first flight of day as follows:

BOOST PUMP switches — ON.

FUEL switches — ON.

Fuel filter (left and right) — Drain samples.

FUEL switches — OFF.

BOOST PUMP switches — OFF.

BATT 1 switch — OFF.

BATTERY BUS 1 switch — AUTO TRIP OFF.

EXTERIOR CHECK

1. FUSELAGE — FRONT

Nose compartment — Condition; batteries connected; door secured.

NOTE

Ballast is not allowed beneath batteries.

PRESTART CHECK

NOTE

BATT 1 switch and BATT 2 switch will be ON during GROUND POWER START.

NOTE

Both batteries, or either single battery may be used for engine start when properly charged. This procedure shows a dual battery start.

BATT 1 switch — ON.

BATT 2 switch — ON.

BATTERY BUS 1 and BUS 2 switches — ON; check BATTERY caution light illuminates.

ENGINE STARTING

ENGINE 1 START.

N1 RPM — Check 71% minimum. BATT 2 switch — OFF.

NOTE

After start, in order to avoid excessive generator drive loads, charge only one battery at a time.

GEN 1 switch — ON; check ammeter load increases.

BATT 2 switch — ON when ammeter load drops below 200 amps.

NOTE

Before attempting generator assisted start on second engine, it is recommended that the battery be charged until ammeter load drops below 150 amps.

AMPS 1 indicator — Check at or below 150 amps.

ENGINE SHUTDOWN

Lighting and miscellaneous switches — OFF.

BATT 1 and BATT 2 switches — OFF.

NOTE

BATTERY BUS 2 switch will trip OFF automatically when BATT 1 and BATT 2 switches are turned OFF.

Collective down lock — Secured as desired.

Section 3

EMERGENCY AND MALFUNCTION PROCEDURES

Table 3-1. Warning lights

PANEL WORDING	FAULT CONDITION	CORRECTIVE ACTION
BATT 1 HOT or BATT 2 HOT	Battery overheating.	Affected battery switch — OFF. Land as soon as practical.

WARNING

BATTERY SHALL NOT BE USED FOR ENGINE START AFTER ILLUMINATION OF BATT 1 HOT OR BATT 2 HOT LIGHT. BATTERY SHALL BE REMOVED AND SERVICED IN ACCORDANCE WITH MANUFACTURERS INSTRUCTIONS PRIOR TO RETURN TO SERVICE.

Table 3-2. Caution lights

PANEL WORDING	FAULT CONDITION	CORRECTIVE ACTION
BATTERY	BATTERY BUS 1 and BUS 2 switches/relays in the same position.	Turn BATT BUS 1 switch ON and BATT BUS 2 switch OFF. If light remains on, reverse switches.

Section 4

PERFORMANCE

No change from basic manual

Section 6

CATEGORY A OPERATIONS

EMERGENCY PROCEDURES

Table 6-1. Warning lights — Takeoff prior to CDP

PANEL WORDING	FAULT CONDITION	CORRECTIVE ACTION
BATT 1 HOT or BATT 2 HOT	Battery overheating.	Land immediately. Affected battery switch — OFF.

WARNING

BATTERY SHALL NOT BE USED FOR ENGINE START AFTER ILLUMINATION OF BATT 1 HOT OR BATT 2 HOT LIGHT. BATTERY SHALL BE REMOVED AND SERVICED IN ACCORDANCE WITH MANUFACTURER'S INSTRUCTIONS PRIOR TO RETURN TO SERVICE.

EMERGENCY PROCEDURES**Table 6-2. Warning lights — Takeoff after CDP**

PANEL WORDING	FAULT CONDITION	CORRECTIVE ACTION
BATT 1 HOT or BATT 2 HOT	Battery overheating.	Accelerate to V_{ross}. Affected battery switch — OFF. Land as soon as practical.

WARNING

BATTERY SHALL NOT BE USED FOR ENGINE START AFTER ILLUMINATION OF BATT 1 HOT OR BATT 2 HOT LIGHT. BATTERY SHALL BE REMOVED AND SERVICED IN ACCORDANCE WITH MANUFACTURER'S INSTRUCTIONS PRIOR TO RETURN TO SERVICE.

EMERGENCY PROCEDURES

Table 6-3. Warning lights — Landing prior to LDP

PANEL WORDING	FAULT CONDITION	CORRECTIVE ACTION
BATT 1 HOT or BATT 2	Battery overheating.	Affected battery switch — OFF. Land as soon as practical.

WARNING

BATTERY SHALL NOT BE USED FOR ENGINE START AFTER ILLUMINATION OF BATT 1 HOT OR BATT 2 HOT LIGHT. BATTERY SHALL BE REMOVED AND SERVICED IN ACCORDANCE WITH MANUFACTURER'S INSTRUCTIONS PRIOR TO RETURN TO SERVICE.

EMERGENCY PROCEDURES

Table 6-4. Warning lights — Landing after LDP

PANEL WORDING	FAULT CONDITION	CORRECTIVE ACTION
BATT 1 HOT or BATT 2 HOT	Battery overheating.	Land immediately. Affected battery switch — OFF.
		WARNING BATTERY SHALL NOT BE USED FOR ENGINE START AFTER ILLUMINATION OF BATT 1 HOT OR BATT 2 HOT LIGHT. BATTERY SHALL BE REMOVED AND SERVICED IN ACCORDANCE WITH MANUFACTURER'S INSTRUCTIONS PRIOR TO RETURN TO SERVICE.

Section 1

MANUFACTURER'S DATA

WEIGHT AND BALANCE

No change from basic manual

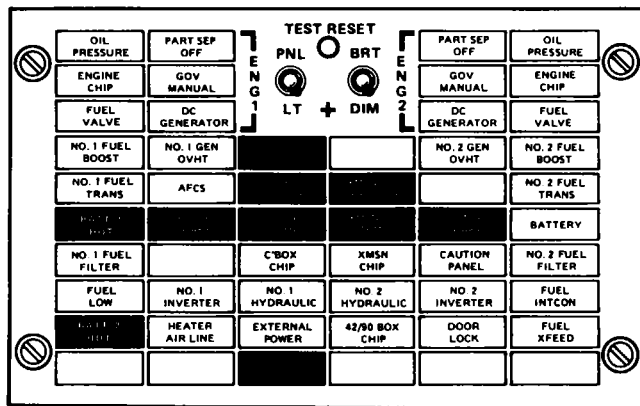
Section 2

MANUFACTURER'S DATA

SYSTEMS DESCRIPTION

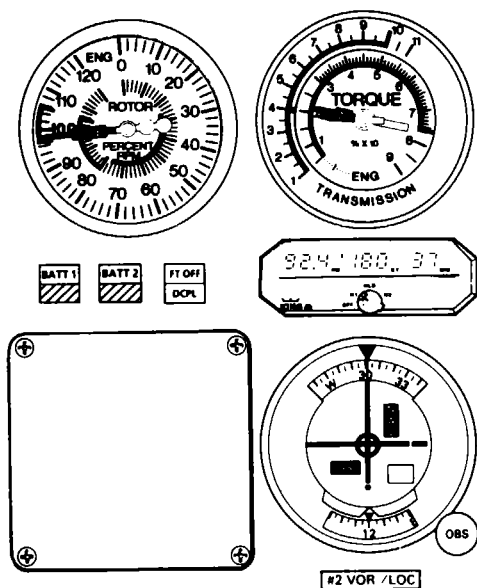
INSTRUMENT PANEL AND CONSOLES

Incorporation of Dual Battery Installation will alter the instrument panel and dc power system as shown in figures 2-1 thru 2-3.



412075-354

Figure 2-1. Instrument panel



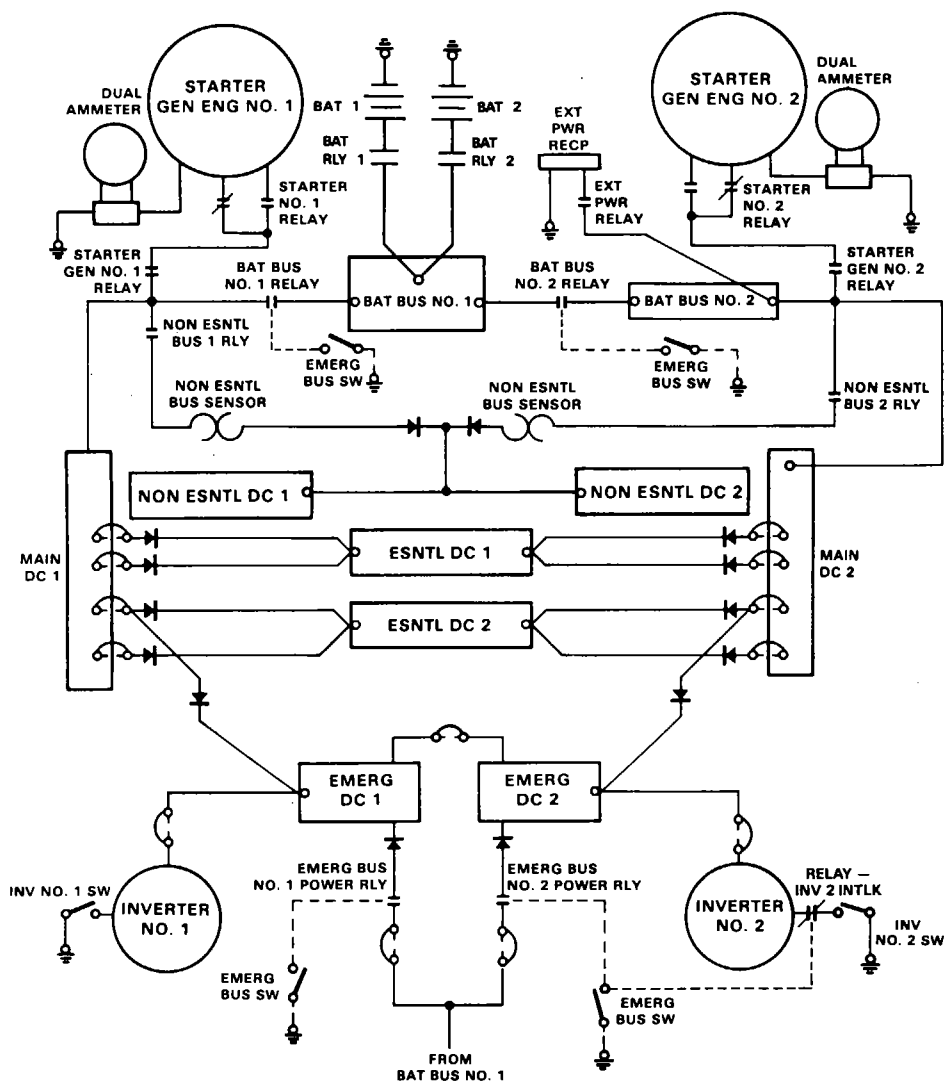
PILOT PANEL

412075-353

Figure 2-2. Instrument panel



Figure 2-3. DC electrical system (Sheet 1 of 2)



**EFFECTIVE S/N 33108 THRU 33139
S/N 33150 THRU 33154**

Figure 2-3. DC electrical system (Sheet 2)



ROTORCRAFT FLIGHT MANUAL

33108 — 33213

36001 — 36019

AND

33214 — 33999

36020 AND SUB

SUPPLEMENT FOR REMOVAL OF UPPER AFT CENTER FUEL CELL

(412-899-227)

CERTIFIED
23 MAY 1989

This supplement shall be attached to the Model 412 Flight Manual (BHT-412-FM-2 or -3) when the upper aft center fuel cell has been removed.

The information contained herein supplements the information of the basic Flight Manual. For Limitations, Procedures, and Performance Data not contained in this supplement, consult the basic Flight Manual.

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23 MAY 1989

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FLIGHT MANUAL

MANUFACTURER'S DATA

PAGE	REVISION NO.	PAGE	REVISION NO.
Title	.0	3 - 16	.0
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A/B	.0	19/20	.0
1 - 2	.0		

APPROVED:



MANAGER

ROTORCRAFT CERTIFICATION OFFICE
FEDERAL AVIATION ADMINISTRATION
FT. WORTH, TEXAS 76193-0170

NOTE

Revised text is indicated by a black vertical line. Insert latest revision pages; dispose of superseded pages.

Section 1

LIMITATIONS

OPTIONAL EQUIPMENT

Auxiliary fuel kits shall not be installed when the aft upper center fuel cell is removed.

WEIGHT CG LIMITATIONS

Actual weight change shall be determined after the upper aft center fuel cell is removed and balance readjusted, if necessary, to retain empty weight CG within limits.

FUEL SYS CAP
1736 LBS
AUX FUEL KITS
NOT ALLOWED

412899-61

Figure 1. Placards and decals

Section 2

NORMAL PROCEDURES

No change to basic manual.

Section 3

EMERGENCY AND MALFUNCTION PROCEDURES

No change to basic manual.

Section 4

PERFORMANCE

No change to basic manual.

Section 1

MANUFACTURER'S DATA

WEIGHT AND BALANCE

WEIGHT EMPTY CENTER OF GRAVITY

With the upper aft center fuel cell removed, the total net weight and C.G. changes are depicted in Table 1.

Table 1. Weight and center of gravity changes

WEIGHT CHANGE (LBS)		LONGITUDINAL	LATERAL	VERTICAL
-28.9	ARM (IN) MOMENT	190.2 -5500.9	0.1 -2.1	35.3 -1020.6

FUEL LOADING

Due to the fuel flow sequencing between the tanks, the fuel loading CG will vary between fuselage station 139.4 and 153.9.

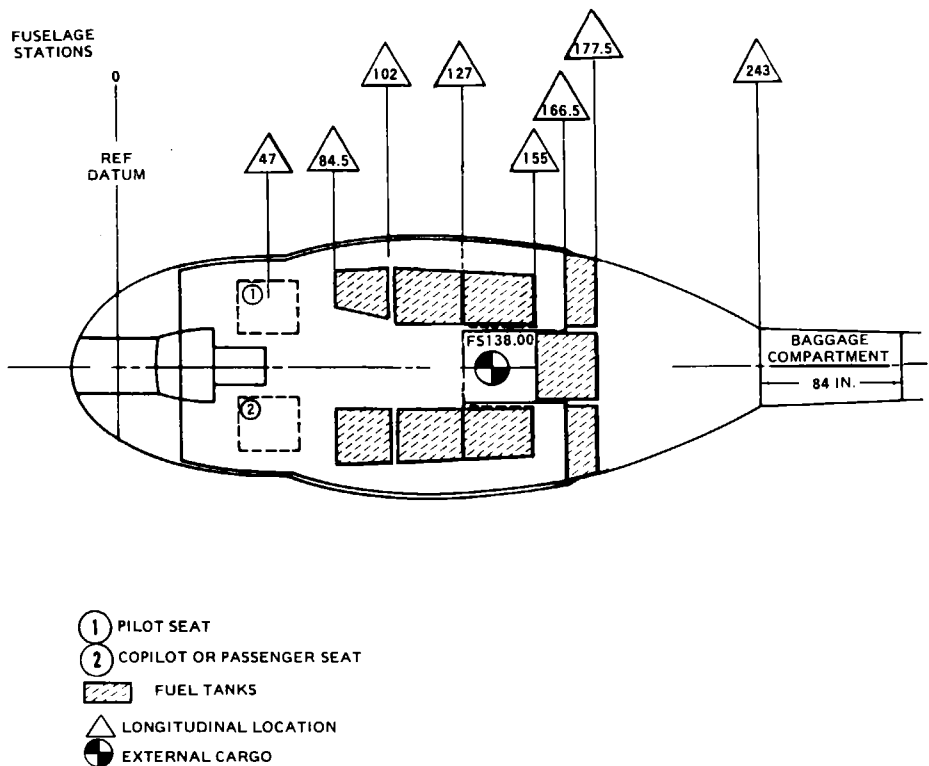
Critical fuel CG's are shown on fuel tables 1, 2, 2M and 3, 3M.

With normal crew and passenger loading, gross weight CG will remain within limits at any fuel quantity.

Figure 2 depicts fuel tank locations by station number.

NOTE

Station 0 (reference datum) is located 20 inches (508 millimeters) aft of the most forward point of the cabin nose.



412061-97

Figure 2. Internal fuel tank station location

Table 2. Fuel loading table (English)

USABLE FUEL LOADING TABLE, UPPER AFT CENTER TANK REMOVED Horizontal (English)							
Jet B, JP-4 (6.5 Lbs/Gal)				Jet A, A-1, JP-5 (6.8 Lbs/Gal)			
U.S. Gal.	Weight (Lb)	CG (In)	Moment (In•Lb)	U.S. Gal.	Weight (Lb)	CG (In)	Moment (In•Lb)
10	65	139.4	9061	10	68	139.4	9479
20	130	139.6	18148	20	136	139.6	18986
30	195	139.8	27261	30	204	139.8	28519
40	260	139.9	36374	40	272	139.9	38053
50	325	139.9	45468	50	340	139.9	47566
58.3	379	139.9	53022	58.3	397	139.9	55540
60	390	140.6	54834	60	408	140.6	57365
70	455	144.7	65839	70	476	144.7	68877
80	520	147.8	76856	80	544	147.8	80403
90	585	150.1	87809	90	612	150.1	91861
100	650	152.0	98800	100	680	152.0	103360
110	715	153.6	109824	110	748	153.6	114893
**112	728	153.9	112039	**112	762	153.9	117272
120	780	150.0	117000	120	816	150.0	122400
130	845	145.5	122948	130	884	145.5	128622
140	910	141.8	129038	140	952	141.8	134994
145.2	944	140.1	132254	145.2	987	140.1	138279
150	975	141.0	137475	150	1020	141.0	143820
160	1040	142.8	148512	160	1088	142.8	155366
170	1105	144.3	159452	170	1156	144.3	166811
180	1170	145.7	170469	180	1224	145.7	178337
190	1235	146.9	181422	190	1292	146.9	189795
***192.1	1249	147.2	183853	***192.1	1306	147.2	192243
200	1300	145.8	189540	200	1360	145.8	198288
210	1365	144.4	197106	210	1428	144.4	206203
220	1430	143.0	204490	220	1496	143.0	213928
230	1495	141.7	211842	230	1564	141.7	221619
240	1560	140.6	219336	240	1632	140.6	229459
*244.1	1587	140.1	222339	*244.1	1660	140.1	232566
250	1625	140.8	228800	250	1700	140.8	239360
260	1690	141.9	239811	260	1768	141.9	250879
267	1736	142.6	247554	267	1816	142.6	258962

Most forward C.G. condition at weight empties under 6260 pounds has no fuel.

*Most critical fuel amount for most forward C.G. condition at weight empties 6260 pounds or greater.

**Most critical fuel amount for most aft C.G. condition at weight empties up to 7600 pounds.

***Most critical fuel amount for most aft C.G. condition at weight empties 7600 pounds or greater.

Weights given are nominal weights at 15°C.

NOTE

This table is invalid with auxiliary fuel tank(s) installed.

Table 2M. Fuel loading table (Metric)

USABLE FUEL LOADING TABLE, UPPER AFT CENTER TANK REMOVED Horizontal (Metric)							
Jet B, JP-4 (0.779 kg/L.)				Jet A, A-1, JP-5 (0.815 kg/L.)			
Liters	Weight (kg)	CG (mm)	Moment (kg*mm)	Liters	Weight (kg)	CG (mm)	Moment (kg*mm)
40	31.2	3541	110479	40	32.6	3541	115437
80	62.3	3547	220978	80	65.2	3547	231264
120	93.5	3551	332019	120	97.8	3551	347288
160	124.6	3552	442579	160	130.4	3552	463181
200	155.8	3552	553402	200	163.0	3552	578976
220.7	172.0	3553	611116	220.7	179.9	3553	639185
240	187.0	3617	676379	240	195.6	3617	707485
280	218.1	3708	808715	280	228.2	3708	846166
320	249.3	3785	943601	320	260.8	3785	987128
360	280.4	3838	1076175	360	293.4	3838	1126069
400	311.6	3886	1210878	400	326.0	3886	1266836
**424.1	330.4	3909	1291534	**424.1	345.6	3909	1350950
440	342.8	3856	1321837	440	358.6	3856	1382762
480	373.9	3731	1395021	480	391.2	3731	1459567
520	405.1	3625	1468488	520	423.8	3625	1536275
549.6	428.1	3558	1523180	549.6	447.9	3558	1593628
560	436.2	3569	1556798	560	456.4	3569	1628892
600	467.4	3620	1691988	600	489.0	3620	1770180
640	498.6	3663	1826372	640	521.6	3663	1910621
680	529.7	3698	1958831	680	554.2	3698	2049432
720	560.9	3734	2094401	720	586.8	3734	2191111
***727.2	566.5	3738	2117577	***727.2	592.7	3738	2215513
760	592.0	3701	2190992	760	619.4	3701	2292399
800	623.2	3663	2282782	800	652.0	3663	2388276
840	654.4	3625	2372200	840	684.6	3625	2481675
880	685.5	3592	2462316	880	717.2	3592	2576182
920	716.7	3561	2552169	920	749.8	3561	2670038
*923.7	719.6	3559	2561056	*923.7	752.8	3559	2679215
960	747.8	3586	2681611	960	782.4	3586	2805686
1000	779.0	3614	2815306	1000	815.0	3614	2945410
1010.7	787.3	3622	2851601	1010.7	823.7	3622	2983441

Most forward C.G. condition at weight empties under 2839 kilograms has no fuel.

*Most critical fuel amount for most forward C.G. condition at weight empties 2839 kilograms or greater.

**Most critical fuel amount for most aft C.G. condition at weight empties up to 3447 kilograms.

***Most critical fuel amount for most aft C.G. condition at weight empties 3447 kilograms or greater.

Weights given are nominal weights at 15°C.

NOTE

This table is invalid with auxiliary fuel tank(s) installed.

Table 3. Fuel loading table — lateral (English)

USABLE FUEL LOADING TABLE, UPPER AFT CENTER TANK REMOVED Lateral (English)							
Jet B, JP-4 (6.5 Lbs/Gal)				Jet A, A-1, JP-5 (6.8 Lbs/Gal)			
U.S. Gal.	Weight (Lb)	CG (In)	Moment (In•Lb)	U.S. Gal.	Weight (Lb)	CG (In)	Moment (In•Lb)
10	65	0	0	10	68	0	0
20	130	0	0	20	136	0	0
30	195	0	0	30	204	0	0
40	260	0	0	40	272	0	0
50	325	0	0	50	340	0	0
58.3	379	0	0	58.3	397	0	0
60	390	0	0	60	408	0	0
70	455	0	0	70	476	0	0
80	520	0	0	80	544	0	0
90	585	0	0	90	612	0	0
100	650	0	0	100	680	0	0
110	715	0	0	110	748	0	0
112	728	0	0	112	762	0	0
120	780	-0.46	-359	120	816	-0.46	-375
130	845	-0.63	-532	130	884	-0.63	-557
140	910	-0.77	-701	140	952	-0.77	-733
*145.2	944	-0.83	-784	*145.2	987	-0.83	-819
150	975	-0.80	-780	150	1020	-0.80	-816
160	1040	-0.75	-780	160	1088	-0.75	-816
170	1105	-0.71	-785	170	1156	-0.71	-821
180	1170	-0.67	-784	180	1224	-0.67	-820
190	1235	-0.63	-778	190	1292	-0.63	-814
200	1300	-0.60	-780	200	1360	-0.60	-816
210	1365	-0.57	-778	210	1428	-0.57	-814
220	1430	-0.54	-772	220	1496	-0.54	-808
230	1495	-0.52	-777	230	1564	-0.52	-813
240	1560	-0.50	-780	240	1632	-0.50	-816
250	1625	-0.48	-780	250	1700	-0.48	-816
260	1690	-0.46	-777	260	1768	-0.46	-813
267	1736	-0.45	-781	267	1816	-0.45	-817

*Most critical fuel amount for most lateral C.G. condition.

Weights given are nominal weights at 15°C.

NOTE

This table is invalid with auxiliary fuel tank(s) installed.

Table 3M. Fuel loading table — lateral (Metric)

USABLE FUEL LOADING TABLE, UPPER AFT CENTER TANK REMOVED Lateral (Metric)							
Jet B, JP-4 (0.779 kg/L.)				Jet A, A-1, JP-5 (0.815 kg/L.)			
Liter	Weight (kg)	CG (mm)	Moment (kg*mm)	Liter	Weight (kg)	CG (mm)	Moment (kg*mm)
40	31.2	0	0	40	32.6	0	0
80	62.3	0	0	80	65.2	0	0
120	93.5	0	0	120	97.8	0	0
160	124.6	0	0	160	130.4	0	0
200	155.8	0	0	200	163.0	0	0
240	187.0	0	0	240	195.6	0	0
280	213.1	0	0	280	228.2	0	0
320	249.3	0	0	320	260.8	0	0
360	280.4	0	0	360	293.4	0	0
400	311.6	0	0	400	326.0	0	0
424.1	330.4	0	0	424.1	345.6	0	0
440	342.8	-10	-3428	440	358.6	-10	-3586
480	373.9	-15	-5609	480	391.2	-15	-5868
520	405.1	-19	-7697	520	423.8	-19	-8052
*549.6	428.1	-21	-8990	*549.6	447.9	-21	-9406
560	436.2	-21	-9160	560	456.4	-21	-9584
600	467.4	-19	-8881	600	489.0	-19	-9291
640	498.6	-18	-8975	640	521.6	-18	-9389
680	529.7	-17	-9005	680	554.2	-17	-9421
720	560.9	-16	-8974	720	586.8	-16	-9389
760	592.0	-15	-8880	760	619.4	-15	-9291
800	623.2	-14	-8725	800	652.0	-14	-9128
840	654.4	-14	-9162	840	684.6	-14	-9584
880	685.5	-13	-8912	880	717.2	-13	-9324
920	716.7	-12	-8600	920	749.8	-12	-8998
960	747.8	-12	-8974	960	782.4	-12	-9389
1000	779.0	-11	-8569	1000	815.0	-11	-8965
1010.7	787.3	-11	-8660	1010.7	823.7	-11	-9061

*Most critical fuel amount for most lateral C.G. condition.

Weights given are nominal weights at 15°C.

NOTE

This table is invalid with auxiliary fuel tank(s) installed.

SAMPLE LOADING PROBLEM **(ENGLISH)**

The helicopter is chartered to transport nine passengers and 180 pounds of baggage on a trip that will require approximately 220 U.S. gallons of fuel one way. The helicopter will be refueled and the 190-pound pilot will return alone. Determine extreme CG conditions for both flights.

OUTBOUND FLIGHT

	<u>LONGITUDINAL</u>			<u>LATERAL</u>	
	<u>WEIGHT</u> <u>(lbs.)</u>	<u>CG</u> <u>(in.)</u>	<u>MOMENT</u> <u>($\frac{\text{lbs}\cdot\text{in}}{100}$)</u>	<u>CG</u> <u>(in.)</u>	<u>MOMENT</u> <u>($\frac{\text{lbs}\cdot\text{in}}{100}$)</u>
Weight Empty	7265	142.3	1034116	-0.2	-1631
+Oil	25		4146	0	0
+Pilot	190		8930	+22.0	+4180
+Passengers, (5 man seat)	850		99450	0	0
+Passengers, (4 man seat)	680		59160	0	0
+Baggage	180		46980	0	0
Basic Operating Weight + Payload	9190	136.3	1252782	+0.3	+2549
+Takeoff Fuel (267 U.S. Gallons)	1816	142.6	258962	-0.45	-817
Takeoff Condition	11006	137.4	1511744	+0.2	+1732
Basic Operating Weight + Payload	9190	136.3	1252782	+0.3	+2549
+Critical Forward Fuel (244.1 U.S. Gallons)	1660	140.1	232566	-0.49	-813
Most Forward Condition	10850	136.9	1485348	+0.2	+1736
Basic Operating Weight + Payload	9190	136.3	1252782	+0.3	+2549
+Landing Fuel (47 U.S. Gallons)	320	139.9	44768	0	0
Landing Condition	9510	136.4	1297550	+0.3	+2549

RETURN FLIGHT

LONGITUDINALLATERAL

	<u>WEIGHT</u> <u>(lbs.)</u>	<u>CG</u> <u>(in.)</u>	<u>MOMENT</u> <u>($\frac{\text{lbs} \cdot \text{in}}{100}$)</u>	<u>CG</u> <u>(in.)</u>	<u>MOMENT</u> <u>($\frac{\text{lbs} \cdot \text{in}}{100}$)</u>
Weight Empty	7265	142.3	1034116	-0.2	-1631
+Oil	25		4146	0	0
+Pilot	190		8930	+22.0	+4180
Basic Operating Weight	7480	140.0	1047192	+0.3	+2549
+Takeoff Fuel (267 U.S. Gallons)	1816	142.6	258962	-0.45	-817
Takeoff Condition	9296	140.5	1306154	+0.2	+1732
Basic Operating Weight	7480	140.0	1047192	+0.3	+2549
+Critical Forward Fuel (244.1 U.S. Gallons)	1660	140.1	232566	-0.49	-813
Most Forward Condition	9140	140.0	1279758	+0.2	+1736
Basic Operating Weight	7480	140.0	1047192	+0.3	+2549
+Critical Aft Fuel (112 U.S. Gallons)	762	153.9	117272	0	0
Most Aft Condition	8242	141.3	1164464	+0.3	+2549
Basic Operating Weight	7480	140.0	1047192	+0.3	+2549
+Landing Fuel (47 U.S. Gallons)	320	139.9	44768	0	0
Landing Condition	7800	140.0	1091960	+0.3	+2549

SAMPLE LOADING PROBLEM (METRIC)

The helicopter is chartered to transport nine passengers and 80 kg of baggage for a trip that will require approximately 830 liters of fuel one way. The helicopter will be refueled and the 90 kg pilot will return alone. Determine extreme CG conditions for both flights.

OUTBOUND FLIGHT

	<u>LONGITUDINAL</u>			<u>LATERAL</u>	
	WEIGHT (kg)	CG (mm)	MOMENT $\left(\frac{\text{kg} \cdot \text{mm}}{100}\right)$	CG (mm)	MOMENT $\left(\frac{\text{kg} \cdot \text{mm}}{100}\right)$
Weight Empty	3295.3	3614	119092.1	-6	-198
+Oil	11.3		485.3	0	0
+Pilot	90.0		1074.6	+559	+503
+Passengers, (5 man seat)	375.0		11145.0	0	0
+Passengers, (4 man seat)	300.0		6630.0	0	0
+Baggage	80.0		5296.0	0	0
Basic Operating Weight + Payload	4151.6	3462	143723.0	+7	+305
+Takeoff Fuel (1010.7 Liters)	823.7	3622	29834.4	-11	-91
Takeoff Condition	4975.3	3488	173557.4	+4	+214
Basic Operating Weight + Payload	4151.6	3462	143723.0	+7	+305
+Forward Fuel (923.7 Liters)	752.8	3559	26792.2	-12	-90
Most Forward Condition	4904.4	3477	170515.2	+4	+215
Basic Operating Weight + Payload	4151.6	3462	143723.0	+7	+305
+Landing Fuel (180.7 Liters)	147.3	3552	5232.1	0	0
Landing Condition	4298.8	3465	148955.1	+7	+305

RETURN FLIGHT

	LONGITUDINAL			LATERAL	
	WEIGHT (kg)	CG (mm)	MOMENT ($\frac{\text{kg} \cdot \text{mm}}{100}$)	CG (mm)	MOMENT ($\frac{\text{kg} \cdot \text{mm}}{100}$)
Weight Empty	3295.3	3614	119092.1	-6	-198
+Oil	11.3		485.3	0	0
+Pilot	90.0		1074.6	+559	+503
Basic Operating Weight	3396.6	3552	120652.0	+9	+305
+Takeoff Fuel (1010.7 Liters)	823.7	3622	29834.4	-11	-91
Takeoff Condition	4220.3	3566	150486.4	+5	+214
Basic Operating Weight	3396.6	3552	120652.0	+9	+305
+Forward Fuel (923.7 Liters)	752.8	3559	26792.2	-12	-90
Most Forward Condition	4149.4	3553	147444.2	+5	+215
Basic Operating Weight	3396.6	3552	120652.0	+9	+305
+Critical Aft Fuel (424.1 Liters)	345.6	3909	13509.5	0	0
Most Aft Condition	3742.2	3585	134161.5	+8	+305
Basic Operating Weight	3396.6	3552	120652.0	+9	+305
+Landing Fuel (180.7 Liters)	147.3	3552	5232.1	0	0
Landing Condition	3543.9	3552	125884.1	+9	+305

Section 2

MANUFACTURER'S DATA

SYSTEMS DESCRIPTION

INSTRUMENT PANEL AND CONSOLES

When the upper aft center fuel cell is removed, a 1736 lb fuel capacity placard is mounted on the center section of the instrument panel. See figure 3.

FUEL SYSTEM

DESCRIPTION — MECHANICAL

The fuel system (figure 4) is comprised of 9 crash resistant fuel cells. Six of the cells are located below the cabin floor and three are located aft of the cabin and above the level of the underfloor

cells. Refer to figure 5 for fuel burn sequence. Partial cell dividers in two of the aft cells and the system interconnect valve provide 52.5 gallons (198.7 liters) isolated fuel supply for each engine.

FUEL TRANSFER AND FILLING

Each lower fuel cell is joined with its opposite (left and right), and with the upper cell interconnect system.

FUEL QUANTITY SYSTEM

The DIGITS TEST button is functionally inoperative.

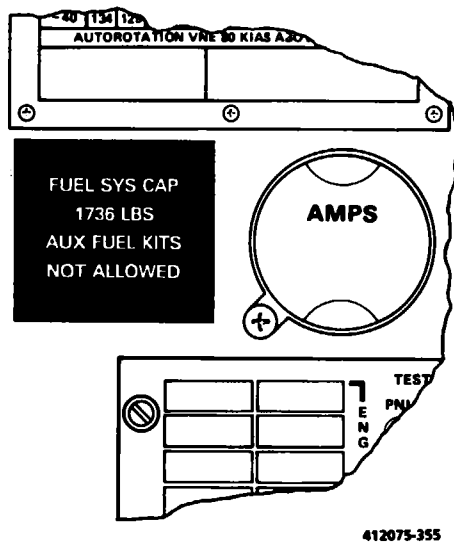


Figure 3. Instrument panel

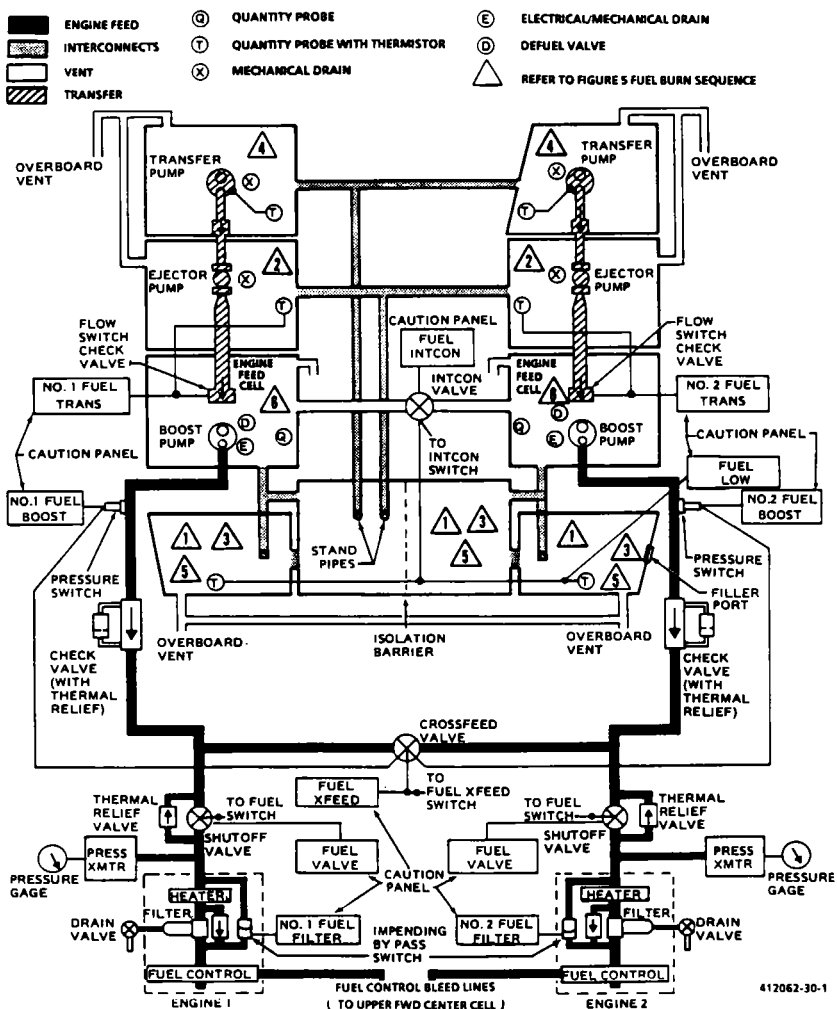
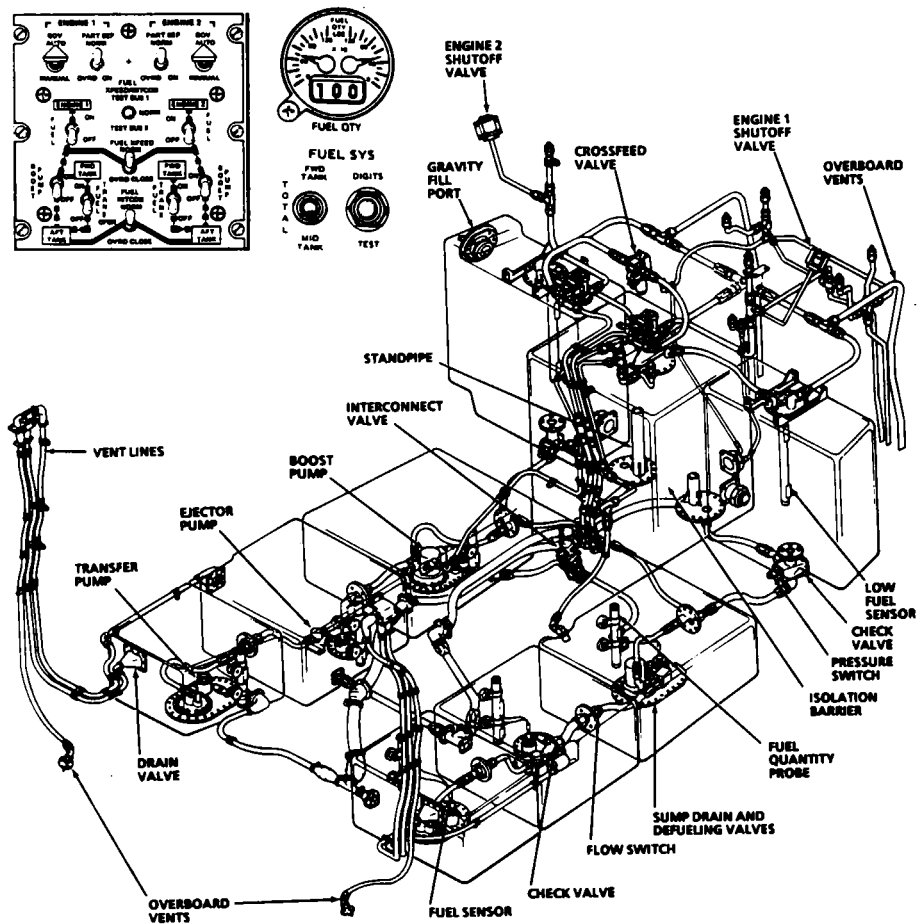
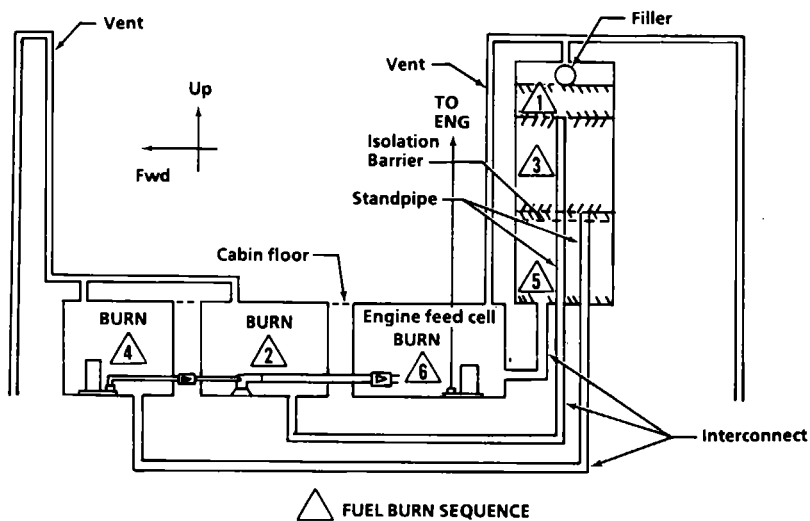


Figure 4. Fuel system schematic (Sheet 1 of 2)



412062-30-2

Figure 4. Fuel system schematic (Sheet 2)



412062-31

Figure 5. Fuel burn sequence

Section 4

MANUFACTURER'S DATA

HANDLING/SERVICING/MAINTENANCE

SERVICING

FUEL SYSTEM SERVICING

Total capacity:

274.7 U.S gallons (1037.0 liters).

Usable fuel:

267.7 U.S. gallons (1010.7 liters).

BELL MODEL 412

ROTORCRAFT FLIGHT MANUAL

SUPPLEMENT FOR CATEGORY B OPERATIONS WITH APPROVED CONFIGURATION OF NINE OR LESS PASSENGER SEATS

SUPPLEMENTAL TYPE CERTIFICATE NO. SH7727SW

CERTIFIED
FEBRUARY 8, 1990

This supplement shall be attached to the Model 412 Flight Manual when helicopter is equipped with an approved nine or less passenger seat configuration.

The information contained herein supplements the information of the basic Flight Manual. For Limitations, Procedures, and Performance Data not contained in this supplement, consult the basic Flight Manual.

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FEBRUARY 8, 1990

LOG OF REVISIONS

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Page	Revision No.	Page	Revision No.
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FT. WORTH, TX 76193-0170NOTE: Revised text is indicated by a black vertical line.
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to the restriction on the title page of this document.

INTRODUCTION:

This supplement removes the Height-velocity diagram as a limitation when helicopter is equipped with an approved configuration of nine or less passenger seats.

Section 1

LIMITATIONS

TYPE OF OPERATION

Flight shall be conducted in accordance with Category B operations and an approved configuration of nine or less passenger seats.

WEIGHT/CG LIMITATIONS

Actual weight change shall be determined after approved seating is installed, and ballast shall be readjusted (if necessary) to return empty weight CG to within allowable limits.

ALTITUDE LIMITATIONS

Maximum altitude for takeoff and landing is 9000 feet density altitude.

Section 2

NORMAL PROCEDURES

TAKEOFF AND LANDING

NOTE

The Height-velocity diagram does not represent a limitation.

Refer to Performance Data, Section 4.

Section 3

EMERGENCY AND MALFUNCTION PROCEDURES

No change from basic Flight Manual.

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Section 4

PERFORMANCE

HEIGHT-VELOCITY ENVELOPE

Operation in height-velocity envelope is critical in the event of a single engine failure during takeoff, landing, or other operation near the surface (figure 4-1). The AVOID area of the Height-velocity diagram defines the combinations of airspeed and height above ground from which a safe single engine landing on a smooth, level, firm surface cannot be assured.

The Height-velocity diagram is valid only when the Weight-Altitude-Temperature limitations are not exceeded (refer to basic Flight Manual). The diagram does not define the conditions which assure continued flight following an engine failure nor the conditions from which a safe power-off landing can be made.

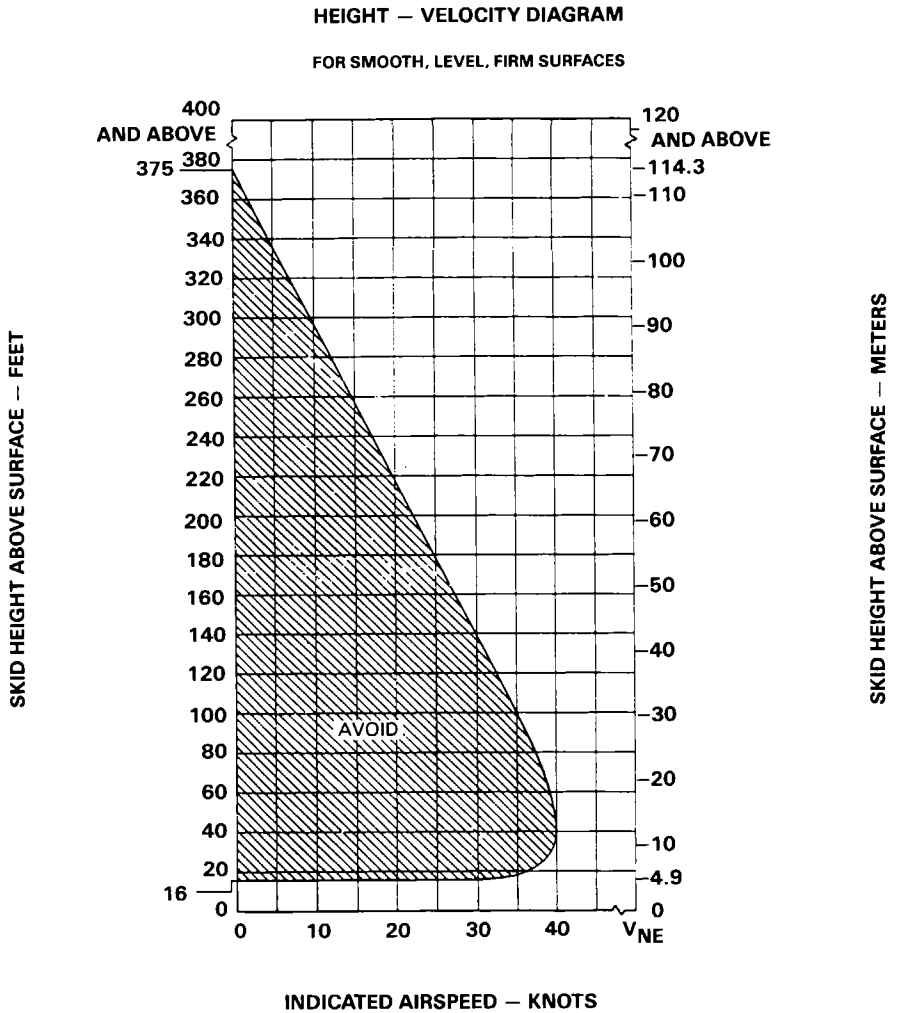


Figure 4-1. Height-velocity diagram (OEI)

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BELL MODEL 412

ROTORCRAFT FLIGHT MANUAL

SUPPLEMENT FOR LORAN C NAVIGATION SYSTEM (KING KLN-88) 412-899-231

**CERTIFIED
22 JUNE 1990**

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This supplement shall be attached to the Model 412 Flight Manual when the Loran C Navigation System (King KLN-88) has been installed.

The information contained herein supplements the information of the basic Flight Manual. For Limitations, Procedures, and Performance Data not contained in this supplement, consult the basic Flight Manual

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22 JUNE 1990

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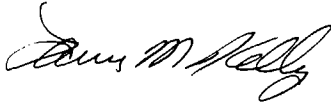
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FLIGHT MANUAL

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INTRODUCTION

The Loran C Navigation System is a navigation aid for use in the North American geographic area as defined in King KLN-88 Pilot Guide.

Visual navigation data, when selected, is presented on the pilot HSI in the form of L/R steering, bearing-to-waypoint, and "TO" indications. L/R steering and "TO" indications are also duplicated on the copilot HSI.

The system consists of a combined Loran C receiver and navigational computer, an antenna, a four-way annunciator, switching circuitry and associated wiring.

Section 1

LIMITATIONS

TYPE OF OPERATIONS

The Loran C system, as installed in this helicopter, is certified for operation in day or night VFR non-icing conditions.

OPERATIONAL LIMITATIONS

A KLN-88 Pilot Guide (King P/N 006-08458-0000, Operation Revision Status ORS 01) dated August 1989 or later revision, shall be accessible by the flight crew at all times during flight.

The Loran C Navigation shall be operated in accordance with the manufacturers instructions with the following exceptions:

This Loran C cannot be coupled to the flight director or helipilot.

There is no fuel management data available in this installation.

It is the responsibility of the pilot to verify that any navigation or communications data used is correct.

PLACARD AND DECALS

**LORAN C APPROVED
FOR VFR ONLY**

(located on instrument panel)

**LATERAL MODES
EXCEPT HDG & GA
ARE INOP WHEN
LRN IS SELECTED**

(located on instrument panel)

WEIGHT/CG LIMITATIONS

Actual weight change shall be determined after the Loran C is installed and ballast readjusted as required to return empty weight CG to within allowable limits.

Section 2

NORMAL PROCEDURES

EXTERIOR CHECK

7. CABIN TOP

Loran C antenna — Condition and security.

PRESTART CHECK

LORAN PWR and FAN circuit breakers — In.

Loran C unit — Verify off.

BEFORE TAKEOFF

Loran C unit — Turn on. Verify operational revision status on initial display page is identical to that of available KLN-88 Pilot Guide.

Pilot HSI CRS pointer — Align to desired course shown on Loran display.

NAV/LRN switch annunciator — Press; verify LRN segment illuminated and NAV segment extinguished.

Pilot and copilot HSI deviation bars — Verify centered and "TO" indication displayed.

Pilot HSI bearing pointer — Verify bearing to waypoint displayed.

NOTE

For additional normal procedures, except fuel management, refer to KLN-88 Pilot Guide.

Section 3

EMERGENCY AND MALFUNCTION PROCEDURES

NO CHANGE

Section 4

PERFORMANCE

NO CHANGE

***BELL* MODEL 412**

ROTORCRAFT FLIGHT MANUAL

33108 THROUGH 33213
AND
36001 THROUGH 36019

SUPPLEMENT FOR IMPROVED HOVER PERFORMANCE WITH PT6T-3BE ENGINES AND 5-MINUTE TAKEOFF POWER RATING (412-570-001-103)

**CERTIFIED
OCTOBER 12, 1990**

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This supplement shall be attached to the Model 412 Flight Manual when the Improved Hover Performance Modification (412-570-001-103) has been installed.

The information contained herein supplements the information of the basic Flight Manual. For Limitations, Procedures, and Performance Data not contained in this supplement, consult the basic Flight Manual

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FLIGHT MANUAL

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Section 1

LIMITATIONS

WEIGHT/CG LIMITATIONS

Actual weight change shall be determined after components are installed and ballast readjusted, if necessary, to return empty weight CG to within allowable limits.

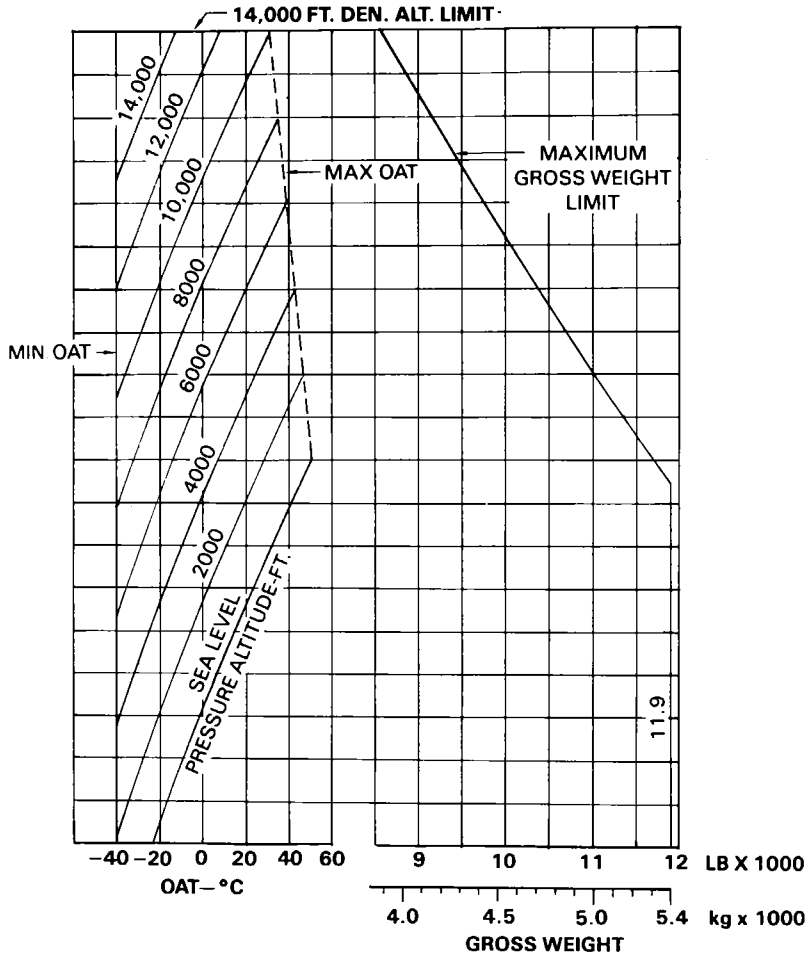
Refer to Weight-Altitude-Temperature Limitations chart for maximum allowable weight for takeoff, landing, and IGE hover operation.

AIRSPPEED LIMITATIONS

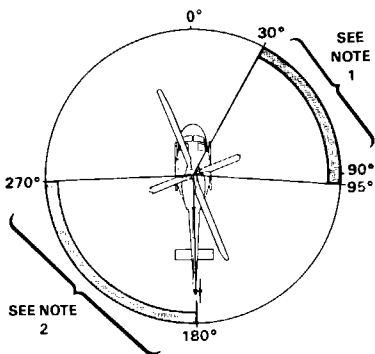
Airspeed shall not exceed 105 KIAS (or placarded V_{NE} , if less) when operating above 81% mast torque.

Refer to Maximum Speed-Sideward and Rearward Flight, Crosswind and Tailwind At A Hover chart.

NOTE: ALLOWABLE GROSS WEIGHTS OBTAINED FROM THIS CHART MAY EXCEED CONTINUOUS HOVER CAPABILITY UNDER CERTAIN AMBIENT CONDITIONS. REFER TO HOVER CEILING CHARTS IN SECTION 4.



Weight-altitude-temperature limitations for takeoff, landing, and in-ground-effect maneuvers



OGE CRITICAL RELATIVE WIND AZIMUTH

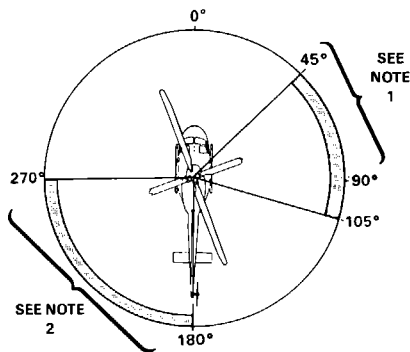
NOTES:

1. Pedal critical wind azimuth-hovering with the relative wind within these azimuth angles can result in the following:

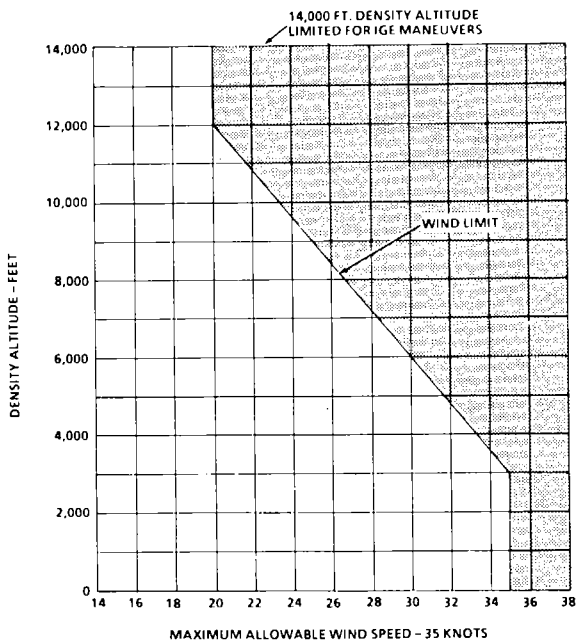
a. Inability to maintain heading due to large left pedal requirements for certain wind velocities.

b. Reduction of available left pedal control with a directional AFCS hardover.

2. Longitudinal cyclic critical wind azimuth — aft cyclic may be limited with longitudinal AFCS hardover.



IGE CRITICAL RELATIVE WIND AZIMUTH



Maximum speed — sideward and rearward flight, crosswind and tailwind at a hover

POWER PLANT LIMITATIONS

Pratt and Whitney Aircraft of Canada, Ltd.
PT6T-3BE.

POWER TURBINE RPM (N_{II}) LIMITS

Minimum in cruise	97%
Minimum for hover, takeoff, and climb	100%
Maximum continuous	100%

MAST TORQUE LIMITS**TWIN ENGINE OPERATION**

Maximum continuous	81%
Takeoff range (5 minutes maximum)	81 to 100%

WARNING

TAKEOFF POWER SHALL NOT BE
USED ABOVE 105 KIAS.

Maximum	100%
---------	------

CAUTION

WHEN OPERATING NEAR THE
MAXIMUM MAST TORQUE LIMIT,
INADVERTANT OVERTORQUE MAY
OCCUR DURING MANEUVERING
FLIGHT CONDITIONS INVOLVING
TURNS AND/OR NOSE DOWN

ATTITUDE CHANGES. DECREASE
POWER TO 90% MAST TORQUE
PRIOR TO MANEUVERING
HELICOPTER.

Intentional use of mast torque over 100% is
prohibited.

TRANSMISSION TORQUE LIMITS

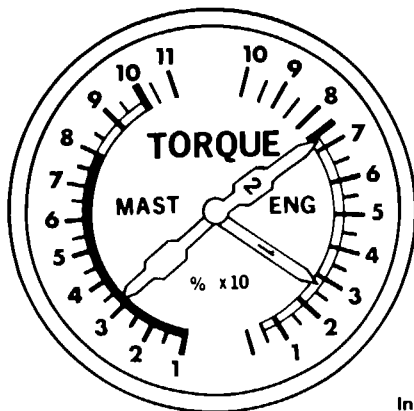
Deleted. See MAST TORQUE LIMITS.

ROTOR LIMITATIONS**ROTOR RPM (N_R) LIMITS — POWER ON**

Minimum	97%
Continuous operation	97 to 100%
Maximum continuous	100%
Operation with mast torque at or below 32%	100 to 104.5%
Maximum with mast torque at or below 32%	104.5%

FUEL AND OIL LIMITATIONS**TRANSMISSION, INTERMEDIATE AND TAIL
ROTOR GEARBOX OIL**

Turbine Oil 555 is the only approved oil for use in
the transmission and gearboxes.



Instrument markings

TRIPLE TORQUE INDICATOR**MAST TORQUE**

GREEN	10 to 81%	Continuous operation
YELLOW	81 to 100%	5 minute takeoff range
RED	100%	Maximum

ENGINE

GREEN	5 to 58.9%	Continuous OEI operation
YELLOW	58.9 to 73.2%	30 minute OEI range
RED	73.2%	Maximum OEI

Section 2

NORMAL PROCEDURES

EXTERIOR CHECK

FUSELAGE — AFT LEFT SIDE

Check OVER TORQ warning flag (cat's eye) for indication of overtorque.

PRESTART CHECK

OVER TORQ caution light — Press. Check light illuminates and MAST TORQUE indicator reads $105 \pm 1\%$.

CAUTION

IF MAST TORQUE INDICATOR INDICATES AN ERROR GREATER THAN $\pm 1\%$ FROM THE 105% POSITION, THE MAST TORQUE SYSTEM IS UNRELIABLE. MAINTENANCE ACTION IS REQUIRED.

BEFORE TAKEOFF

Throttles — Full open. Adjust frictions.

RPM switch — Minimum beep (DECR for 4-5 seconds).

RPM switch — Minimum trim (-2 for 4-5 seconds).

N_R — Check 95% or greater.

RPM switch — Adjust to obtain matching torque at 100% N_R .

Flight instruments — Check operation and set.

TAKEOFF

Area — Clear.

NOTE

As collective is increased, it may be necessary to rematch engine torques prior to reaching hover.

RPM switch — Adjust to obtain matching torque or ITT, as required, and 100% N_R .

Hover power — Check torque required to hover at four feet skid height.

Section 3

EMERGENCY AND MALFUNCTION PROCEDURES

EMERGENCY PROCEDURES

Caution lights

PANEL WORDING	FAULT CONDITION	CORRECTIVE ACTION
OVER TORQ	Mast torque exceeds 100.5%	Reduce power or severity of maneuver. Land as soon as practical.

Section 4

PERFORMANCE

INTRODUCTION

The performance data presented herein are derived from the engine manufacturer's specification power for the engine less installation losses when used with the 412-570-001-103 Improved Hover Performance modification. These data are applicable to the basic helicopter without any optional equipment which would appreciably affect lift, drag, or power available.

HOVER CEILING OGE.

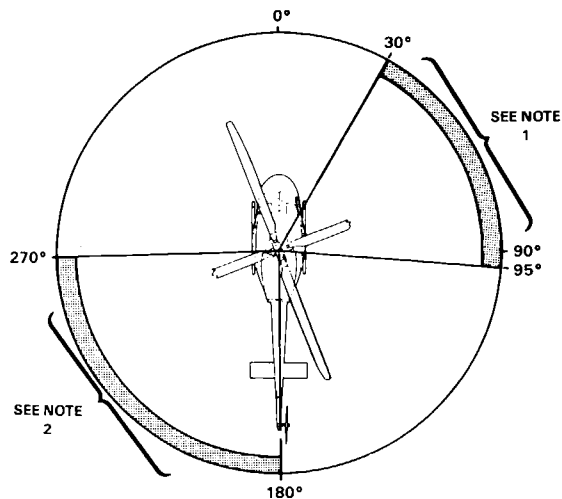
AREA A (unshaded area) as shown on the hover ceiling charts presents hover performance for which satisfactory cyclic and directional control

have been demonstrated in relative winds of 35 knots from any direction at or below 3000 feet HD. Improved control margins will be realized by avoiding winds in the critical relative wind azimuth areas.

AREA B (shaded area) as shown on hover ceiling charts presents additional hover performance which can be realized in calm winds or winds outside the critical relative wind azimuth areas.

NOTE

Tail rotor or cyclic control margin may preclude operation in AREA B of the hover ceiling charts when the relative wind is in the respective critical wind azimuth area.



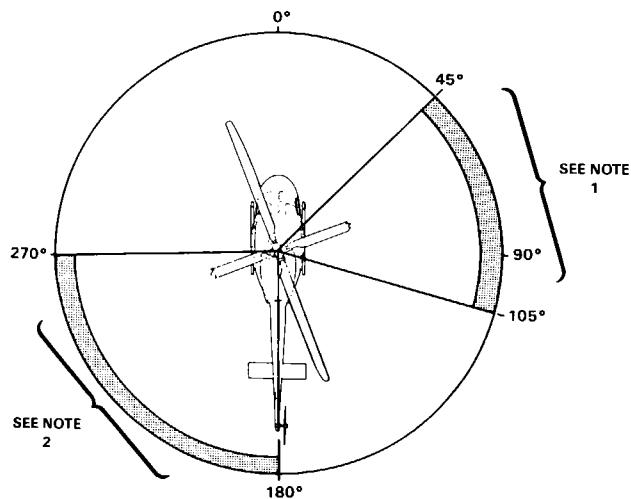
NOTES:

1. Pedal critical wind azimuth-hovering with the relative wind within these azimuth angles can result in the following:

- a. Inability to maintain heading due to large left pedal requirements for certain wind velocities.
- b. Reduction of available left pedal control with a directional AFCS hardover.

2. Longitudinal cyclic critical wind azimuth – aft cyclic may be limited with longitudinal AFCS hardover.

OGE CRITICAL RELATIVE WIND AZIMUTH



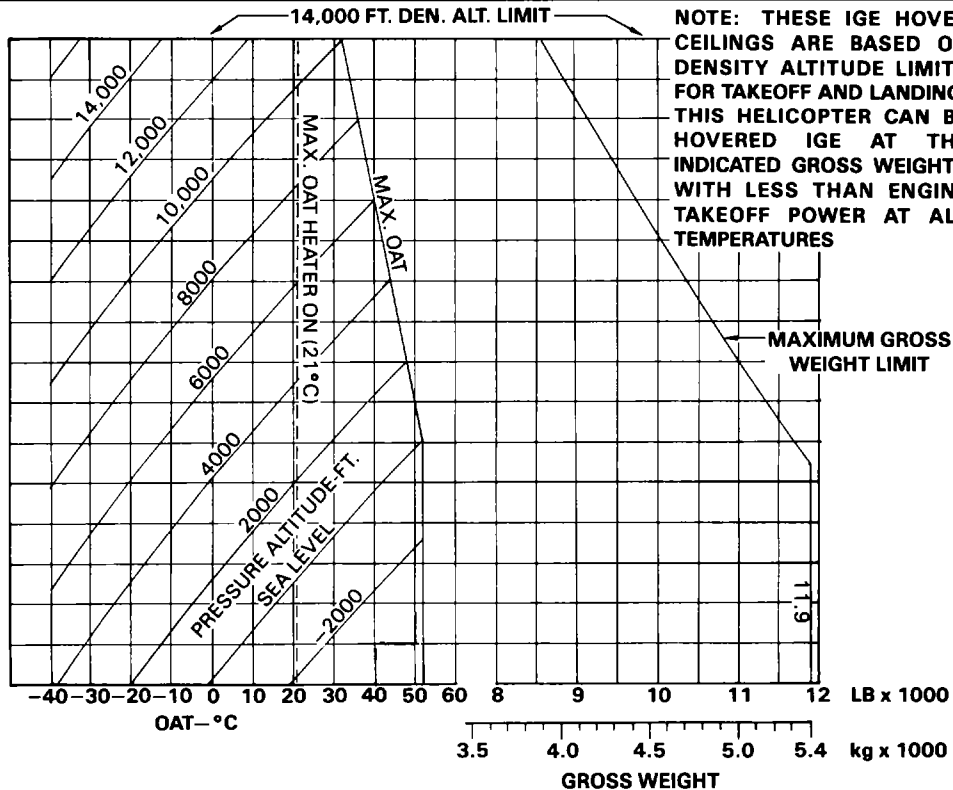
IGE CRITICAL RELATIVE WIND AZIMUTH

Critical relative wind azimuths

HOVER CEILING IN GROUND EFFECT

POWER: SEE NOTE BELOW
ENGINE RPM 100%
GENERATOR 150 AMPS (EA.)

SKID HEIGHT 4 FEET
HEATER ON OR OFF
-40 TO 52°C



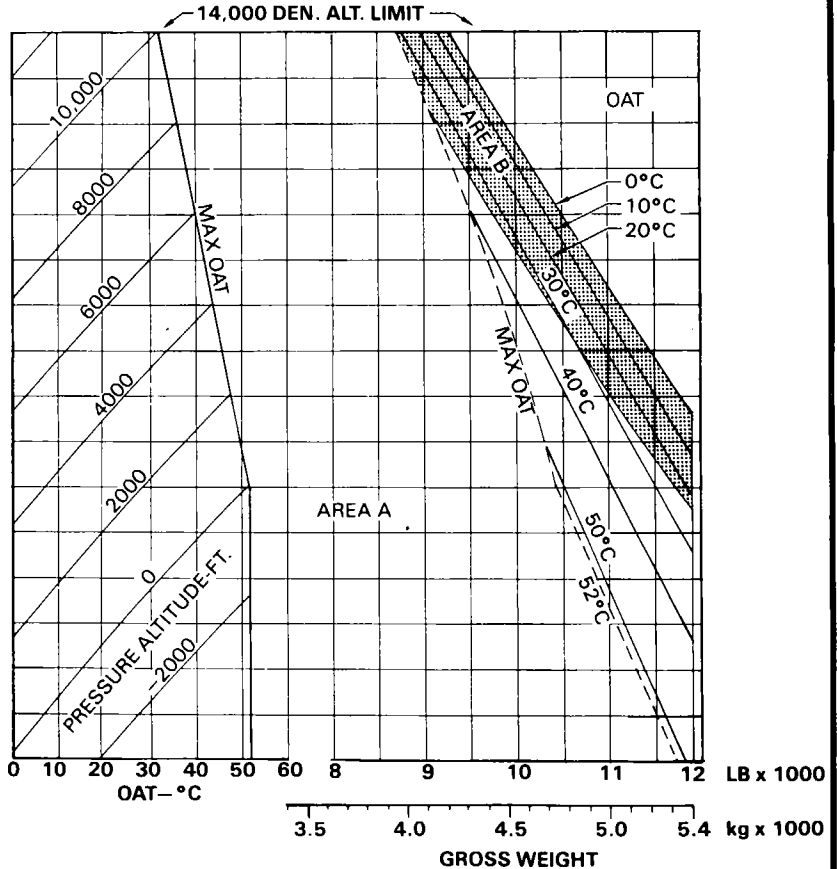
Hover ceiling (Sheet 1 of 10)

HOVER CEILING
OUT OF GROUND EFFECT

ENGINE TAKEOFF POWER
ENGINE RPM 100%
GENERATOR 150 AMPS (EA.)

SKID HEIGHT 60 FT.
HEATER OFF
0 TO 52°C

CAUTION: OGE HOVER OPERATION MAY RESULT IN VIOLATION OF H-V LIMITATIONS.

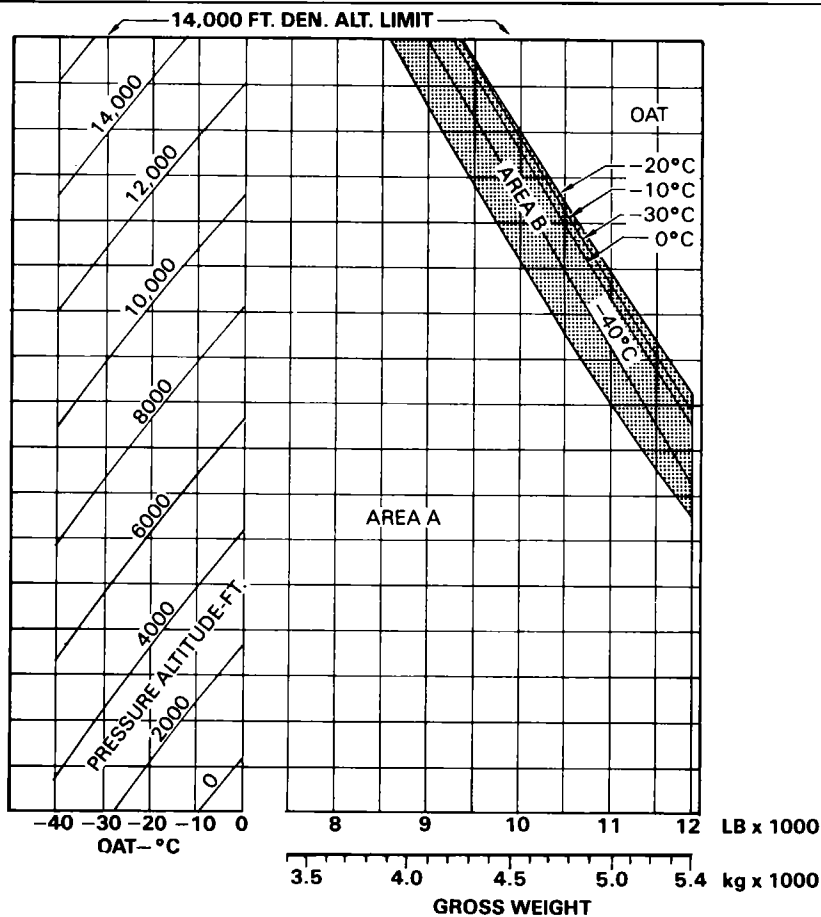


HOVER CEILING OUT OF GROUND EFFECT

ENGINE TAKEOFF POWER
ENGINE RPM 100%
GENERATOR 150 AMPS (EA.)

SKID HEIGHT 60 FT.
HEATER OFF
-40 TO 0°C

CAUTION: OGE HOVER OPERATION MAY RESULT IN VIOLATION OF H-V LIMITATIONS.



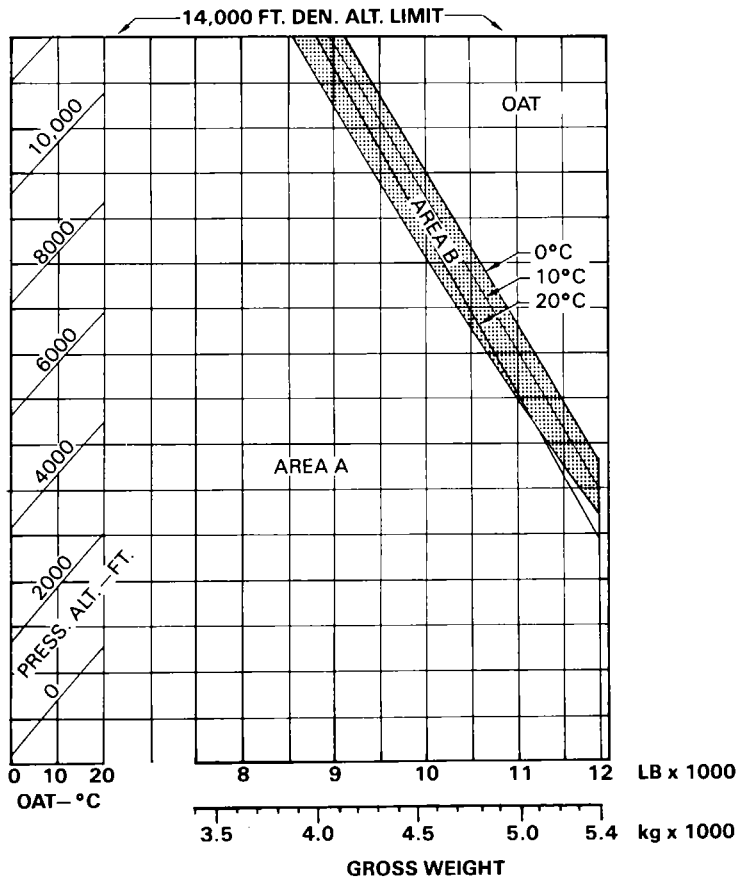
Hover ceiling (Sheet 3 of 10)

HOVER CEILING
OUT OF GROUND EFFECT

ENGINE TAKEOFF POWER
ENGINE RPM 100%
GENERATOR 150 AMPS (EA.)

SKID HEIGHT 60 FT.
HEATER ON
0 TO 20°C

CAUTION: OGE HOVER OPERATION MAY RESULT IN VIOLATION OF H-V LIMITATIONS.

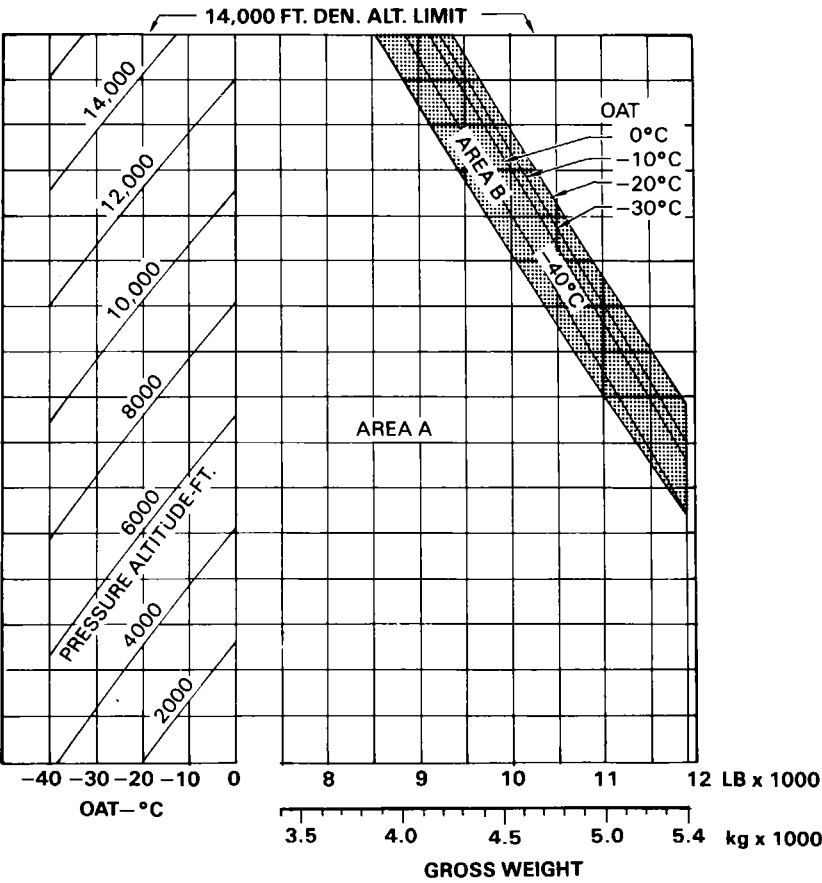


HOVER CEILING
OUT OF GROUND EFFECT

ENGINE TAKEOFF POWER
ENGINE RPM 100%
GENERATOR 150 AMPS (EA.)

SKID HEIGHT 60 FT.
HEATER ON
-40 TO 0°C

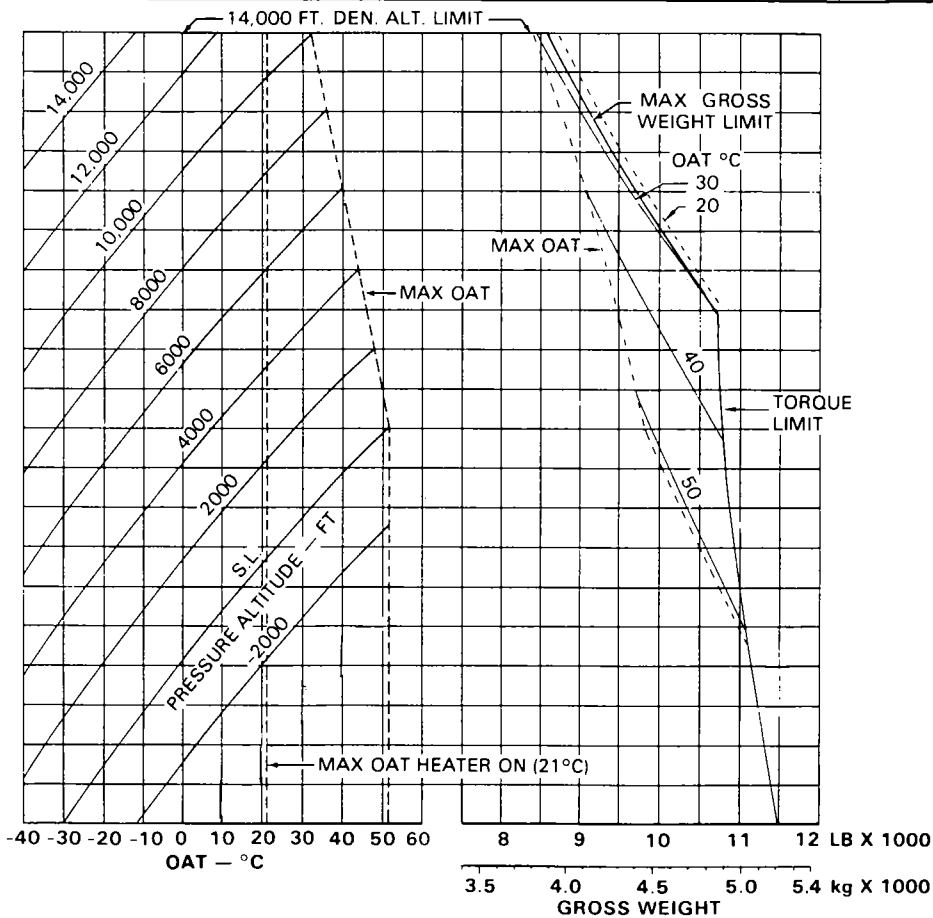
CAUTION: OGE HOVER OPERATION MAY RESULT IN VIOLATION OF H-V LIMITATIONS.



HOVER CEILING IN GROUND EFFECT

MAXIMUM CONTINUOUS POWER
ENGINE RPM 100%
GENERATOR 150 AMPS (EA.)

SKID HEIGHT 4 FEET
HEATER ON OR OFF
-40° TO 52°C



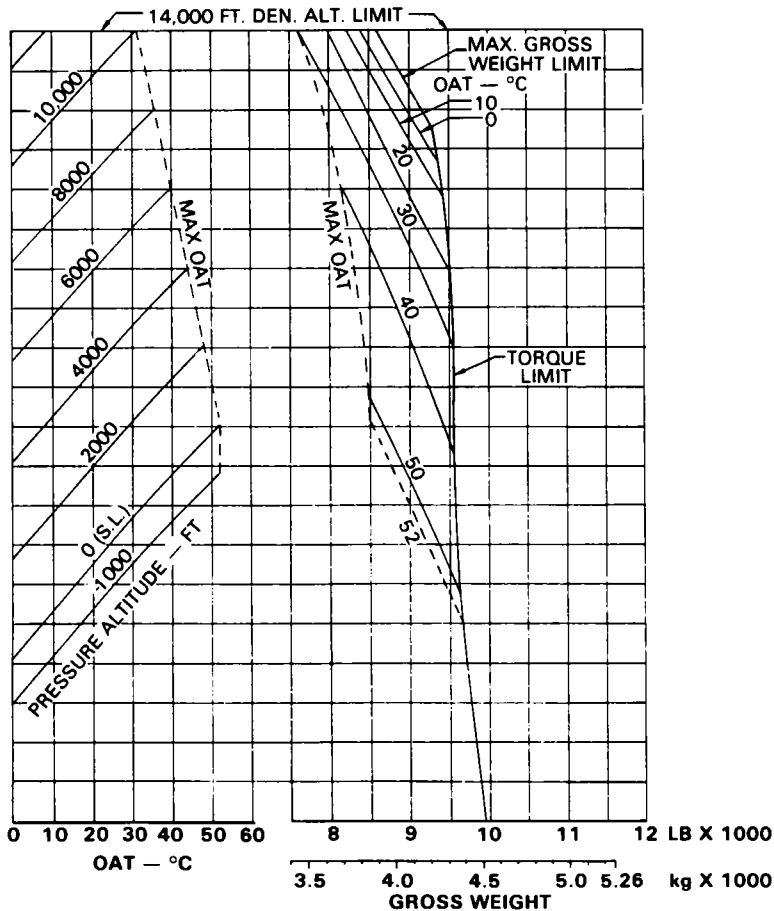
Hover ceiling (Sheet 6 of 10)

HOVER CEILING OUT OF GROUND EFFECT

MAXIMUM CONTINUOUS POWER
ENGINE RPM 100%
GENERATOR 150 AMPS (EA.)

SKID HEIGHT 60 FT.
HEATER OFF
0 TO 52°C

CAUTION: OGE HOVER OPERATION MAY RESULT IN VIOLATION OF H-V LIMITATIONS.



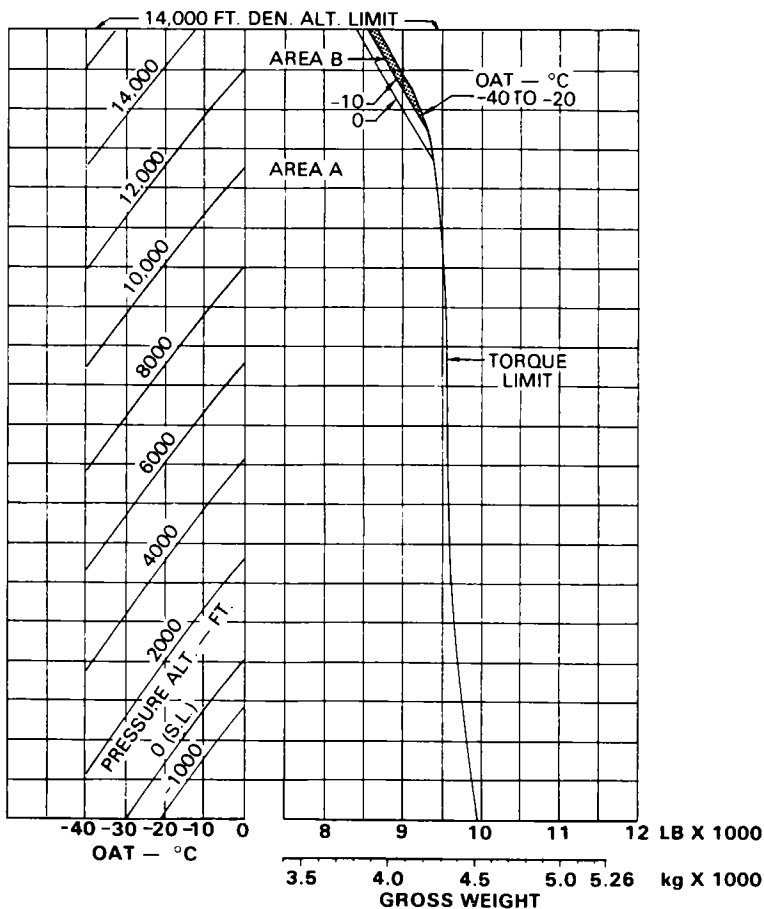
Hover ceiling (Sheet 7 of 10)

HOVER CEILING OUT OF GROUND EFFECT

MAXIMUM CONTINUOUS POWER
ENGINE RPM 100%
GENERATOR 150 AMPS (EA.)

SKID HEIGHT 60 FT.
HEATER OFF
-40 TO 0°C

CAUTION: OGE HOVER OPERATION MAY RESULT IN VIOLATION OF H-V LIMITATIONS.

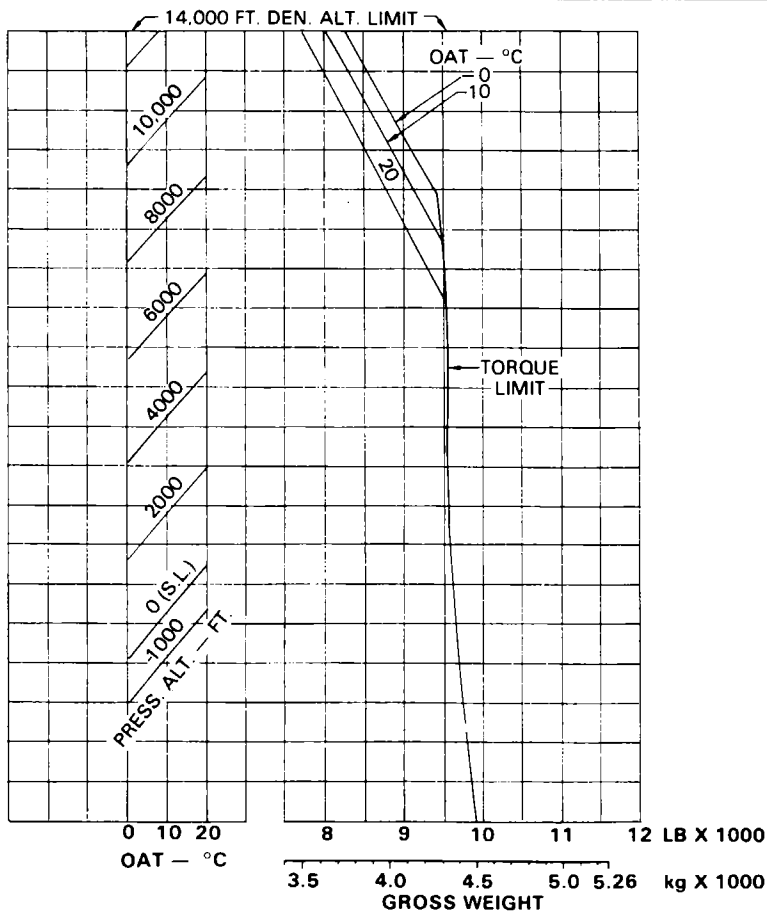


HOVER CEILING OUT OF GROUND EFFECT

MAXIMUM CONTINUOUS POWER
 ENGINE RPM 100%
 GENERATOR 150 AMPS (EA.)

SKID HEIGHT 60 FT.
 HEATER ON
 0 TO 20°C

CAUTION: OGE HOVER OPERATION MAY RESULT IN VIOLATION OF H-V LIMITATIONS.



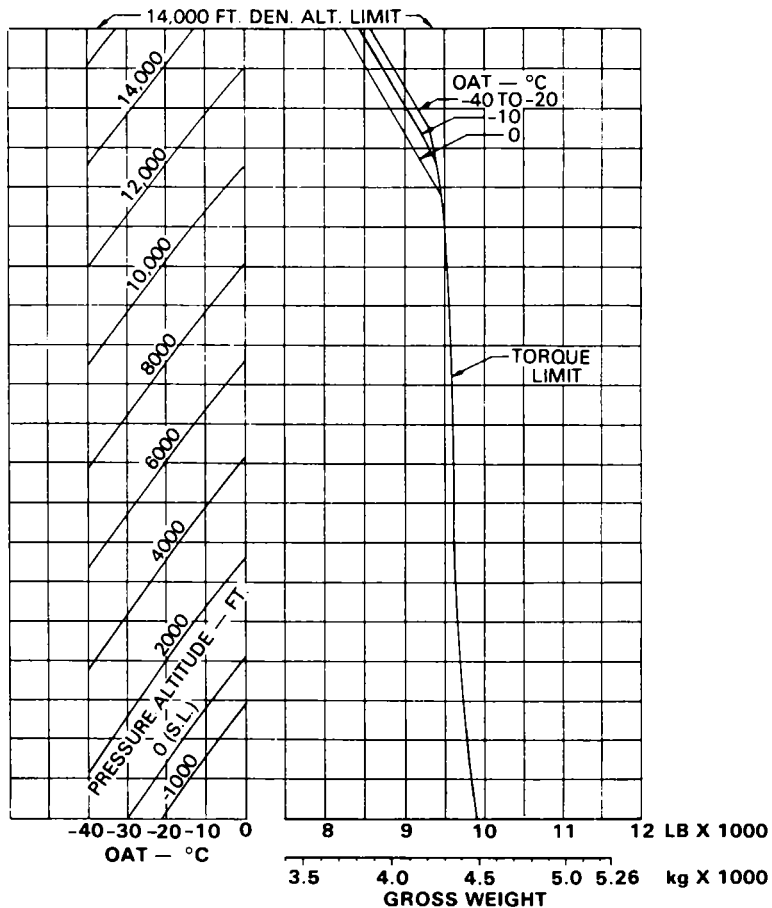
Hover ceiling (Sheet 9 of 10)

HOVER CEILING OUT OF GROUND EFFECT

MAXIMUM CONTINUOUS POWER
ENGINE RPM 100%
GENERATOR 150 AMPS (EA.)

SKID HEIGHT 60 FT.
HEATER ON
-40 TO 0°C

CAUTION: OGE HOVER OPERATION MAY RESULT IN VIOLATION OF H-V LIMITATIONS.

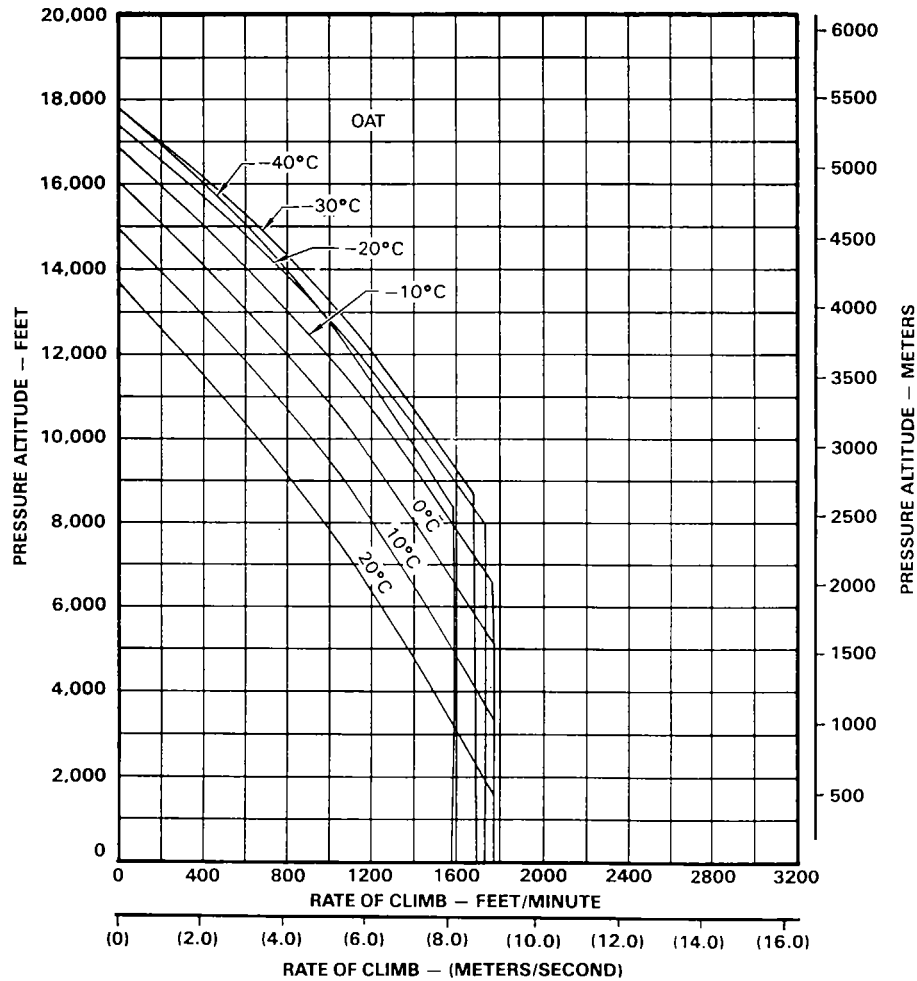


TWIN ENGINE RATE OF CLIMB
GROSS WEIGHT 11,900 LB (5398 kg)

TAKEOFF POWER
ENGINE RPM 100%
GENERATOR 150 AMPS (EA.)

70 KIAS
HEATER ON

- WITH ALL DOORS OPEN OR REMOVED:
- 1. CLIMB SPEED IS 60 KIAS
 - 2. RATE OF CLIMB WILL DECREASE 275 FT./MIN.



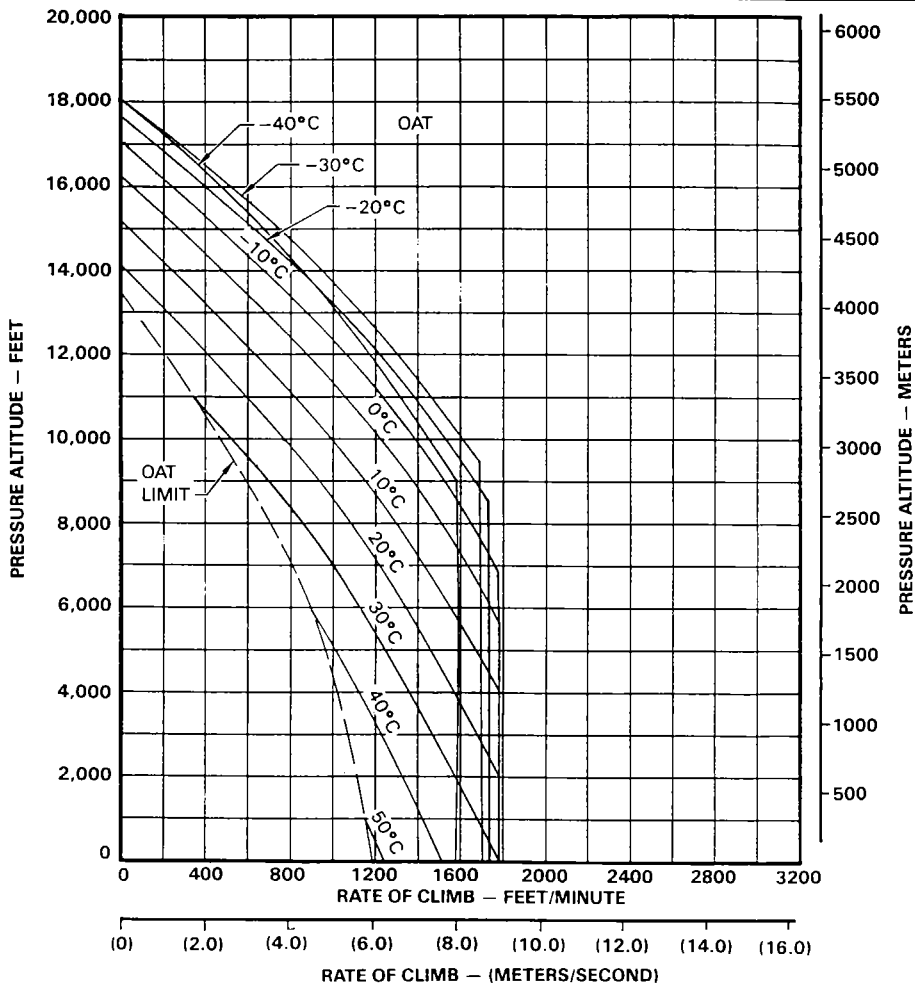
Twin engine rate of climb (Sheet 1 of 24)

TWIN ENGINE RATE OF CLIMB
GROSS WEIGHT 11,900 LB (5398 kg)

TAKEOFF POWER
ENGINE RPM 100%
GENERATOR 150 AMPS (EA.)

70 KIAS
HEATER OFF

- WITH ALL DOORS OPEN OR REMOVED:
1. CLIMB SPEED IS 60 KIAS
 2. RATE OF CLIMB WILL DECREASE 275 FT./MIN.



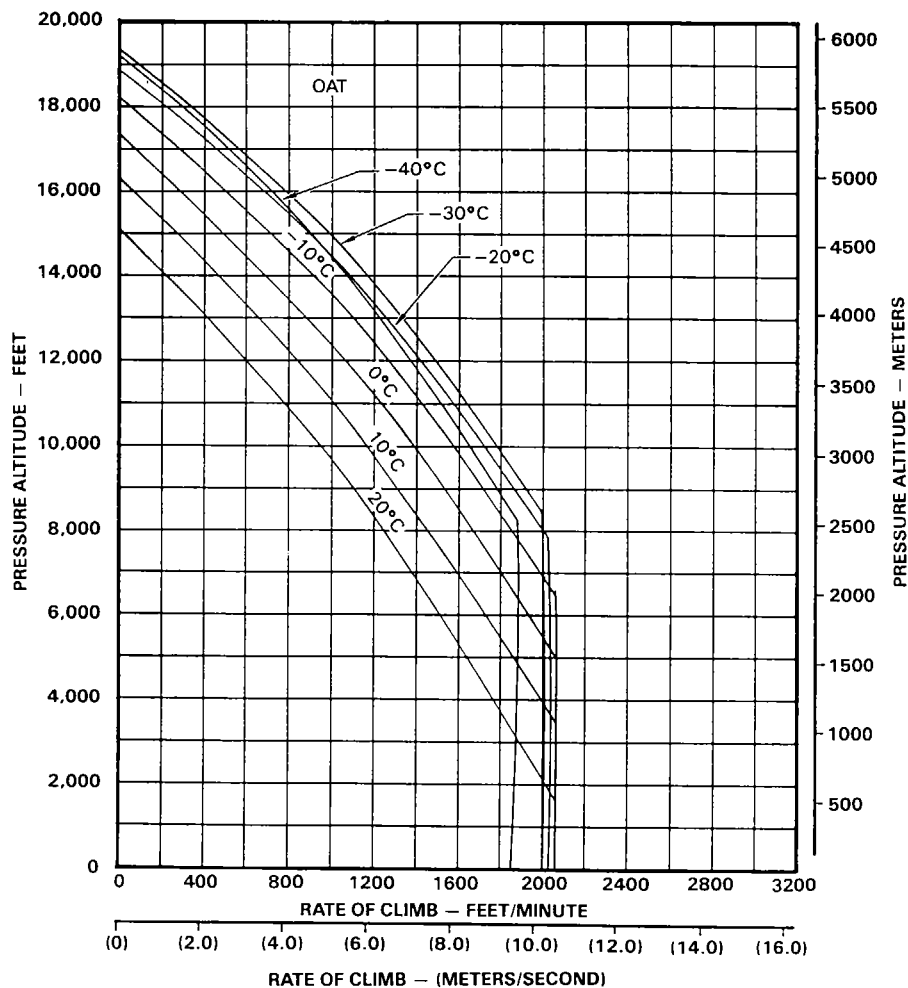
Twin engine rate of climb (Sheet 2 of 24)

TWIN ENGINE RATE OF CLIMB GROSS WEIGHT 11,000 LB (4990 kg)

TAKEOFF POWER
ENGINE RPM 100%
GENERATOR 150 AMPS (EA.)

70 KIAS
HEATER ON

WITH ALL DOORS OPEN OR REMOVED: 1. CLIMB SPEED IS 60 KIAS
2. RATE OF CLIMB WILL DECREASE 275 FT./MIN.



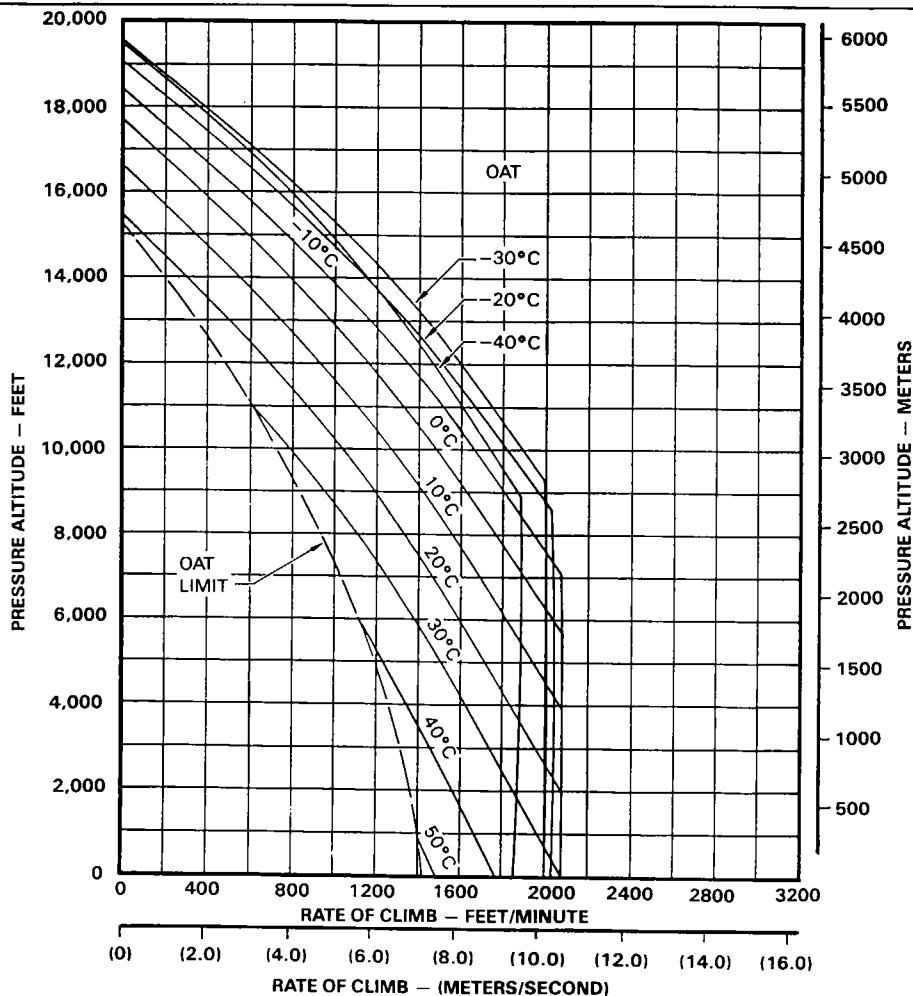
Twin engine rate of climb (Sheet 3 of 24)

TWIN ENGINE RATE OF CLIMB
GROSS WEIGHT 11,000 LB (4990 kg)

TAKEOFF POWER
ENGINE RPM 100%
GENERATOR 150 AMPS (EA.)

70 KIAS
HEATER OFF

- WITH ALL DOORS OPEN OR REMOVED:
1. CLIMB SPEED IS 60 KIAS
 2. RATE OF CLIMB WILL DECREASE 275 FT./MIN.



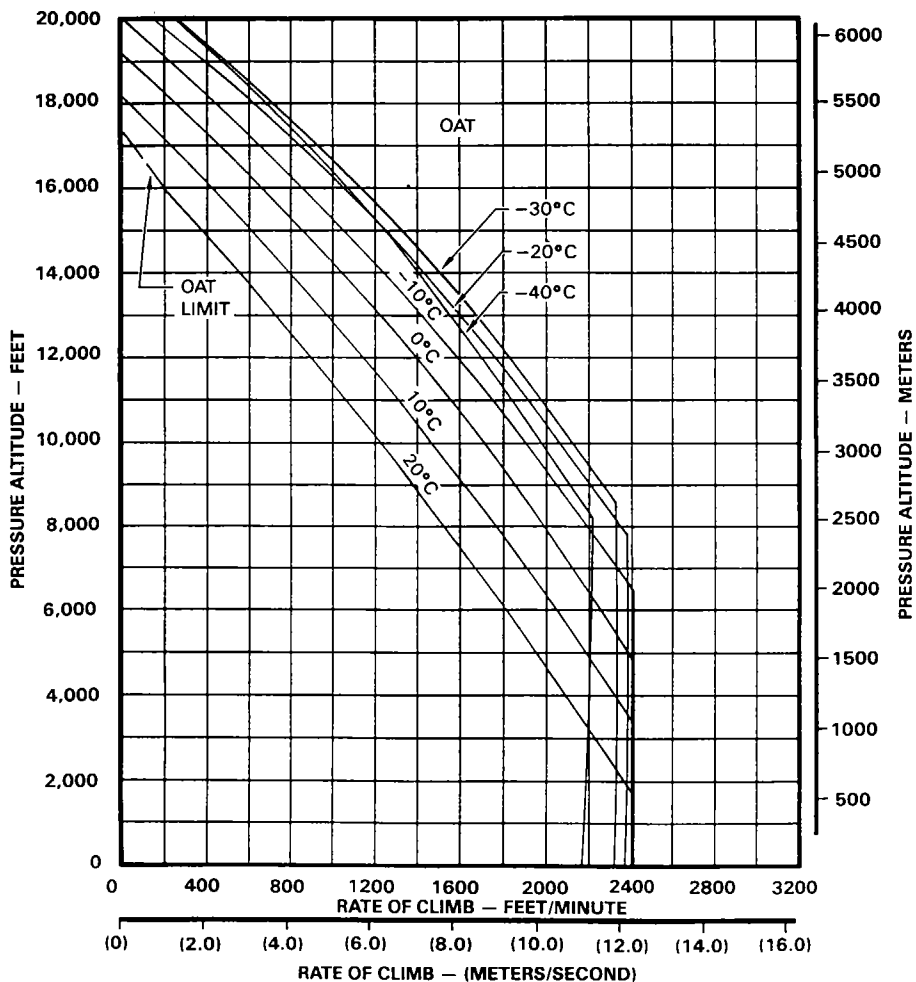
Twin engine rate of climb (Sheet 4 of 24)

TWIN ENGINE RATE OF CLIMB GROSS WEIGHT 10,000 LB (4536 kg)

TAKEOFF POWER
ENGINE RPM 100%
GENERATOR 150 AMPS (EA.)

70 KIAS
HEATER ON

- WITH ALL DOORS OPEN OR REMOVED: 1. CLIMB SPEED IS 60 KIAS
2. RATE OF CLIMB WILL DECREASE 275 FT./MIN.



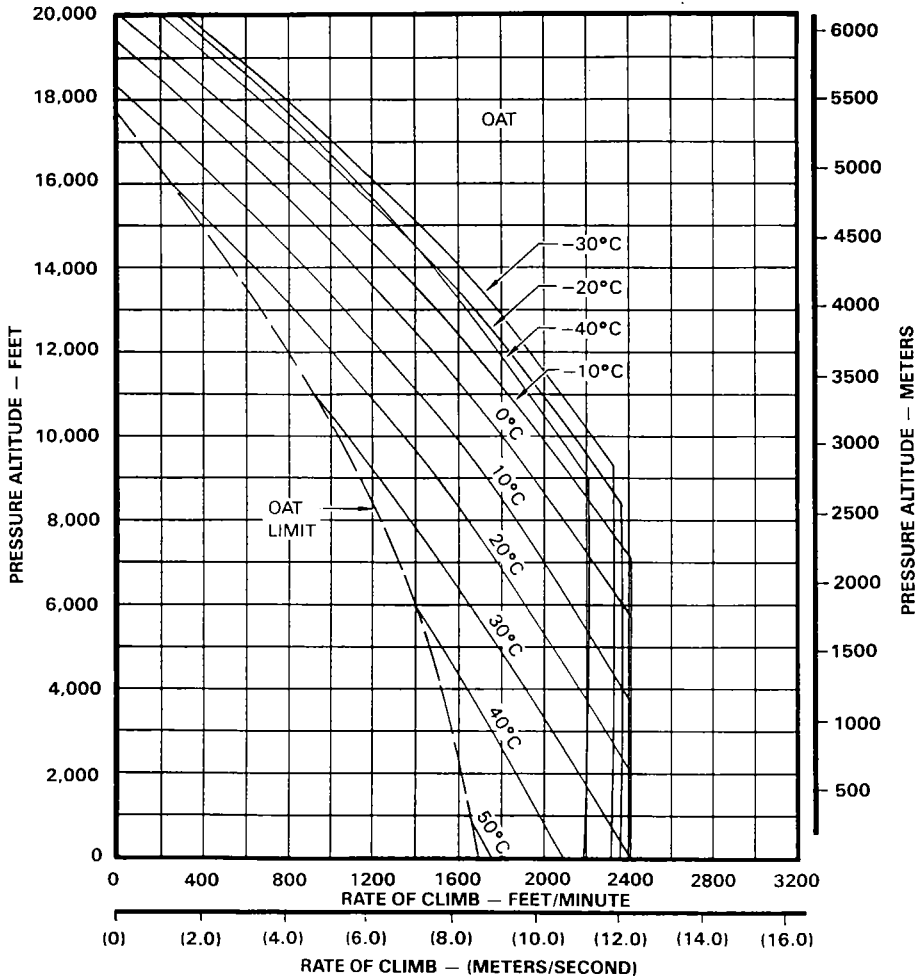
Twin engine rate of climb (Sheet 5 of 24)

TWIN ENGINE RATE OF CLIMB
GROSS WEIGHT 10,000 LB (4536 kg)

TAKEOFF POWER
ENGINE RPM 100%
GENERATOR 150 AMPS (EA.)

70 KIAS
HEATER OFF

WITH ALL DOORS OPEN OR REMOVED: 1. CLIMB SPEED IS 60 KIAS
2. RATE OF CLIMB WILL DECREASE 275 FT./MIN.



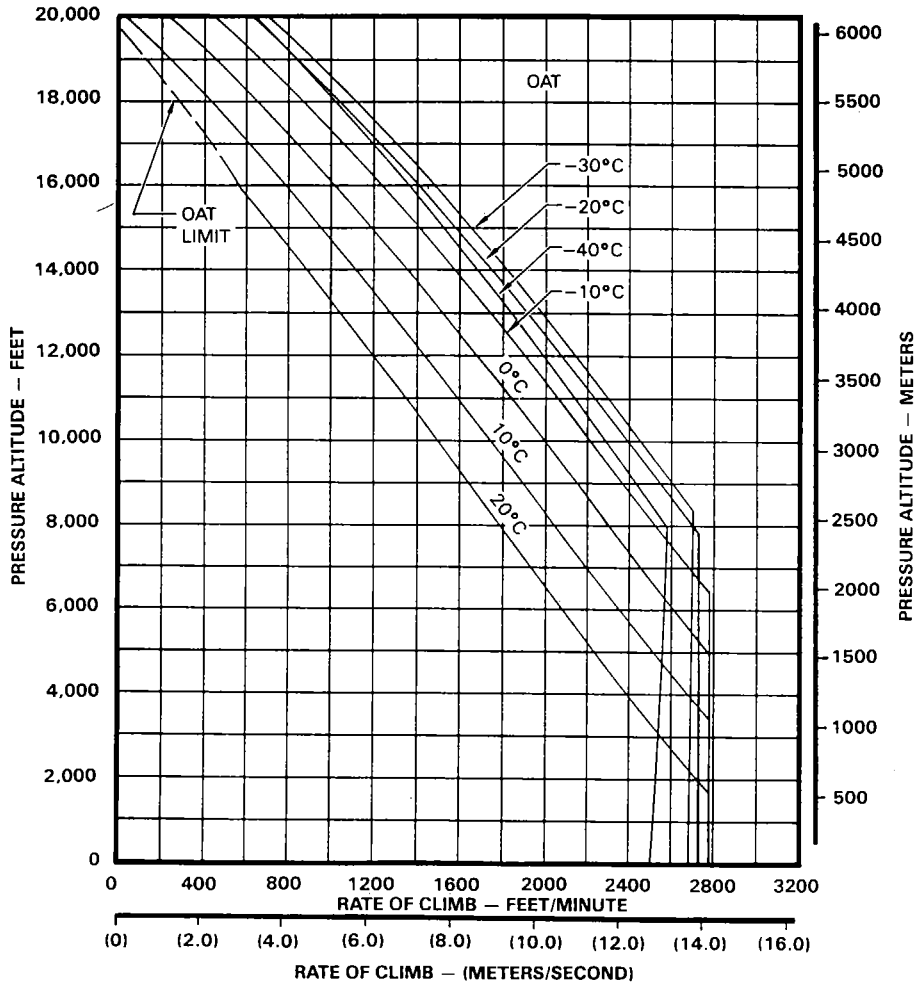
Twin engine rate of climb (Sheet 6 of 24)

TWIN ENGINE RATE OF CLIMB GROSS WEIGHT 9000 LB (4082 kg)

TAKEOFF POWER
ENGINE RPM 100%
GENERATOR 150 AMPS (EA.)

70 KIAS
HEATER ON

- WITH ALL DOORS OPEN OR REMOVED: 1. CLIMB SPEED IS 60 KIAS
2. RATE OF CLIMB WILL DECREASE 275 FT./MIN.



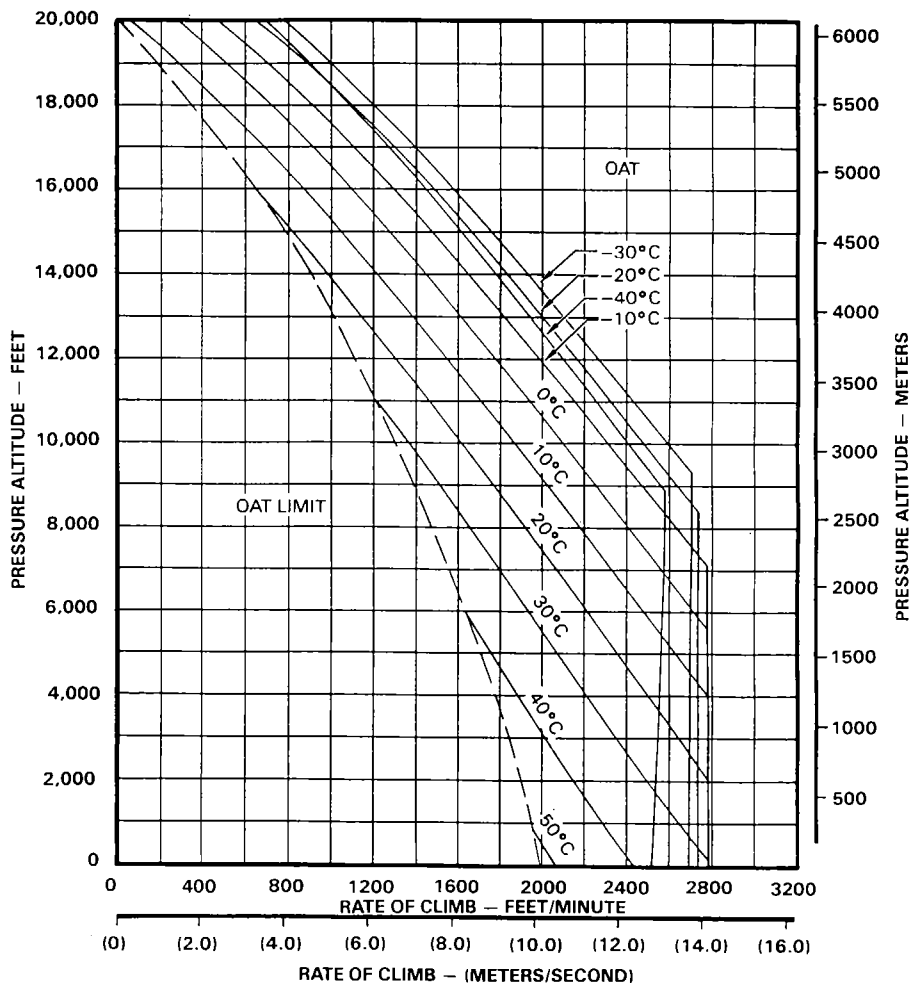
Twin engine rate of climb (Sheet 7 of 24)

TWIN ENGINE RATE OF CLIMB
GROSS WEIGHT 9000 LB (4082 kg)

TAKEOFF POWER
ENGINE RPM 100%
GENERATOR 150 AMPS (EA.)

70 KIAS
HEATER OFF

WITH ALL DOORS OPEN OR REMOVED: 1. CLIMB SPEED IS 60 KIAS
2. RATE OF CLIMB WILL DECREASE 275 FT./MIN.



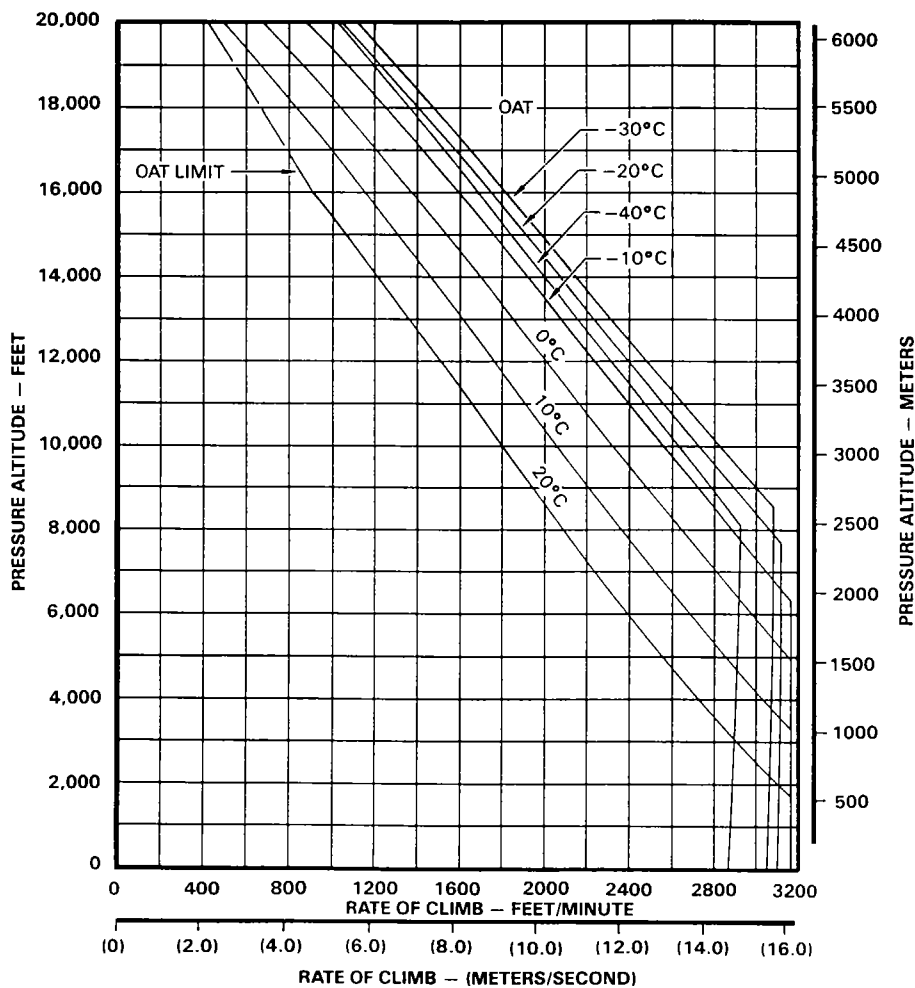
Twin engine rate of climb (Sheet 8 of 24)

TWIN ENGINE RATE OF CLIMB GROSS WEIGHT 8000 LB (3629 kg)

TAKEOFF POWER
ENGINE RPM 100%
GENERATOR 150 AMPS (EA.)

70 KIAS
HEATER ON

WITH ALL DOORS OPEN OR REMOVED: 1. CLIMB SPEED IS 60 KIAS
2. RATE OF CLIMB WILL DECREASE 275 FT./MIN.



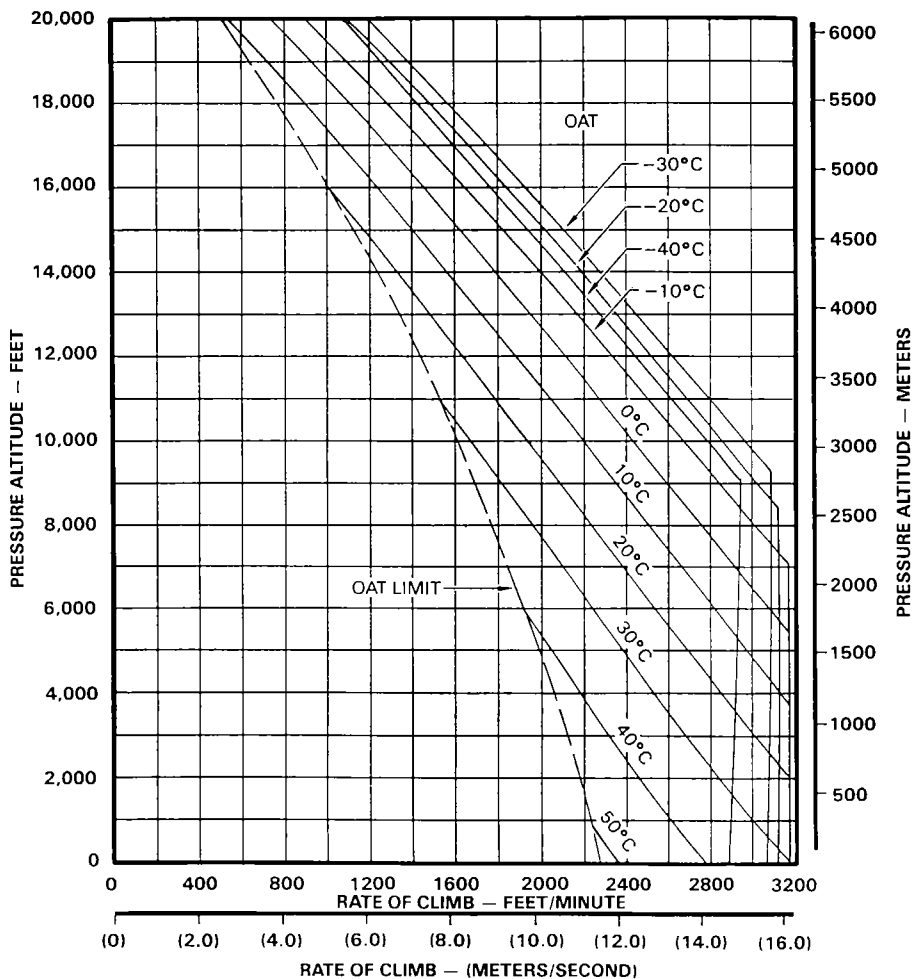
Twin engine rate of climb (Sheet 9 of 24)

TWIN ENGINE RATE OF CLIMB
GROSS WEIGHT 8000 LB (3629 kg)

TAKEOFF POWER
ENGINE RPM 100%
GENERATOR 150 AMPS (EA.)

70 KIAS
HEATER OFF

WITH ALL DOORS OPEN OR REMOVED: 1. CLIMB SPEED IS 60 KIAS
2. RATE OF CLIMB WILL DECREASE 275 FT./MIN.



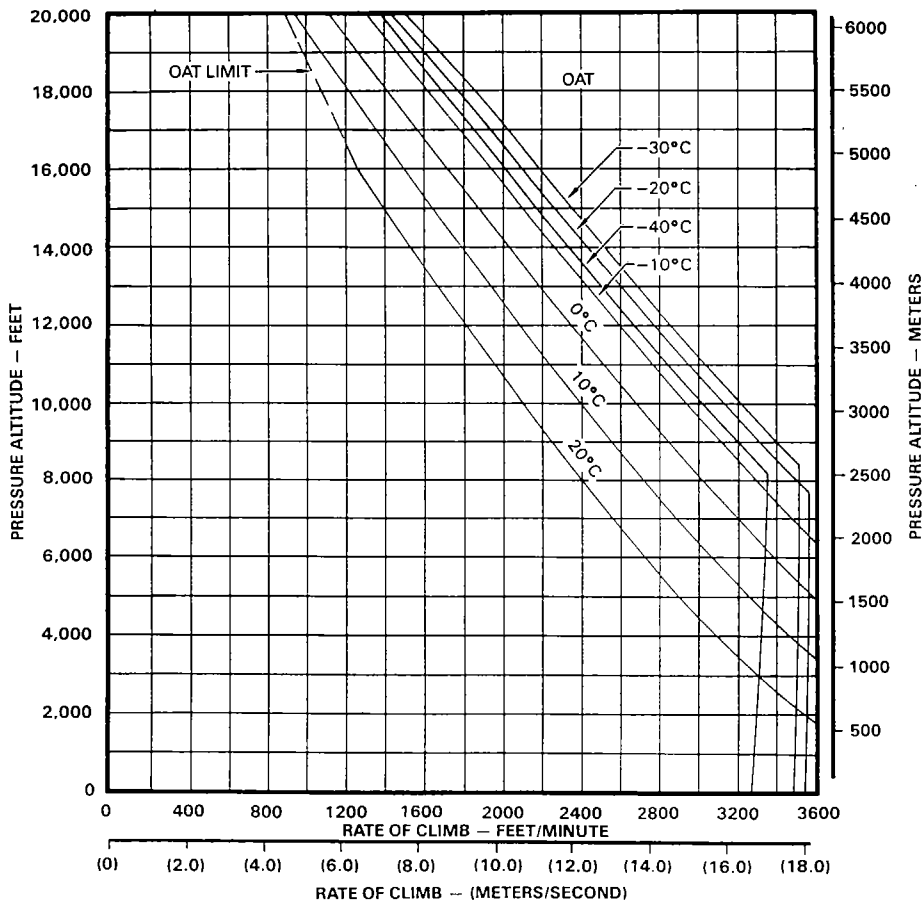
Twin engine rate of climb (Sheet 10 of 24)

TWIN ENGINE RATE OF CLIMB GROSS WEIGHT 7000 LB (3175 kg)

TAKEOFF POWER
ENGINE RPM 100%
GENERATOR 150 AMPS (EA.)

70 KIAS
HEATER ON

- WITH ALL DOORS OPEN OR REMOVED: 1. CLIMB SPEED IS 60 KIAS
2. RATE OF CLIMB WILL DECREASE 275 FT./MIN.



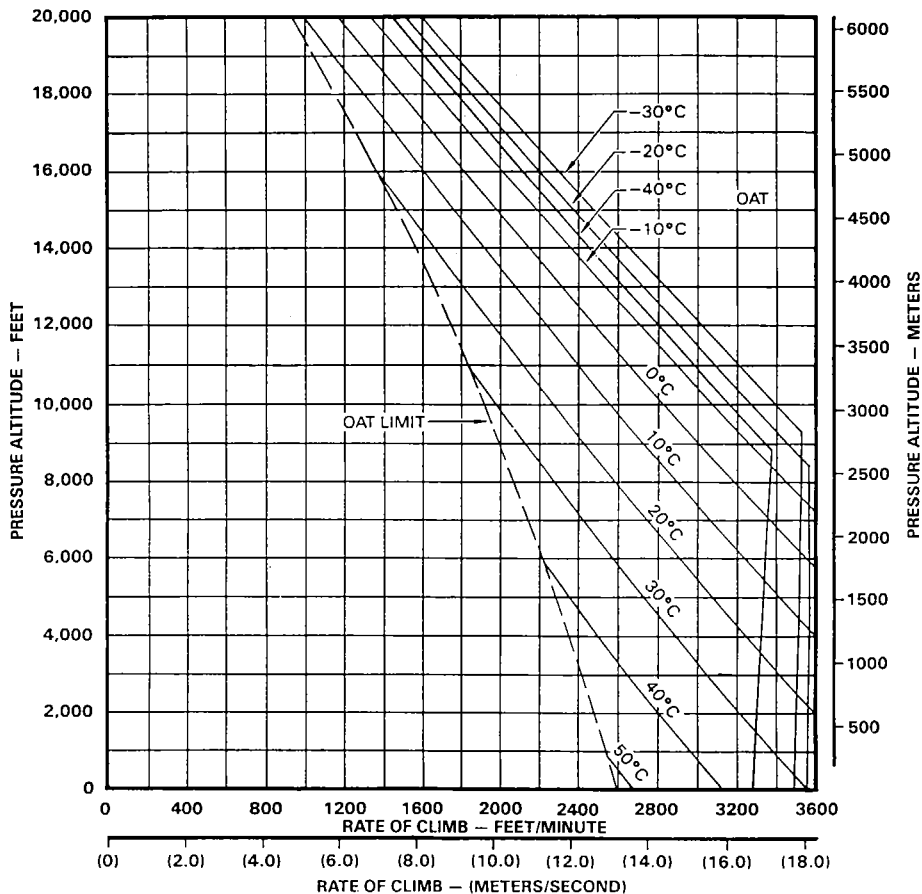
Twin engine rate of climb (Sheet 11 of 24)

TWIN ENGINE RATE OF CLIMB
GROSS WEIGHT 7000 LB (3175 kg)

TAKEOFF POWER
ENGINE RPM 100%
GENERATOR 150 AMPS (EA.)

70 KIAS
HEATER OFF

WITH ALL DOORS OPEN OR REMOVED: 1. CLIMB SPEED IS 60 KIAS
2. RATE OF CLIMB WILL DECREASE 275 FT./MIN.



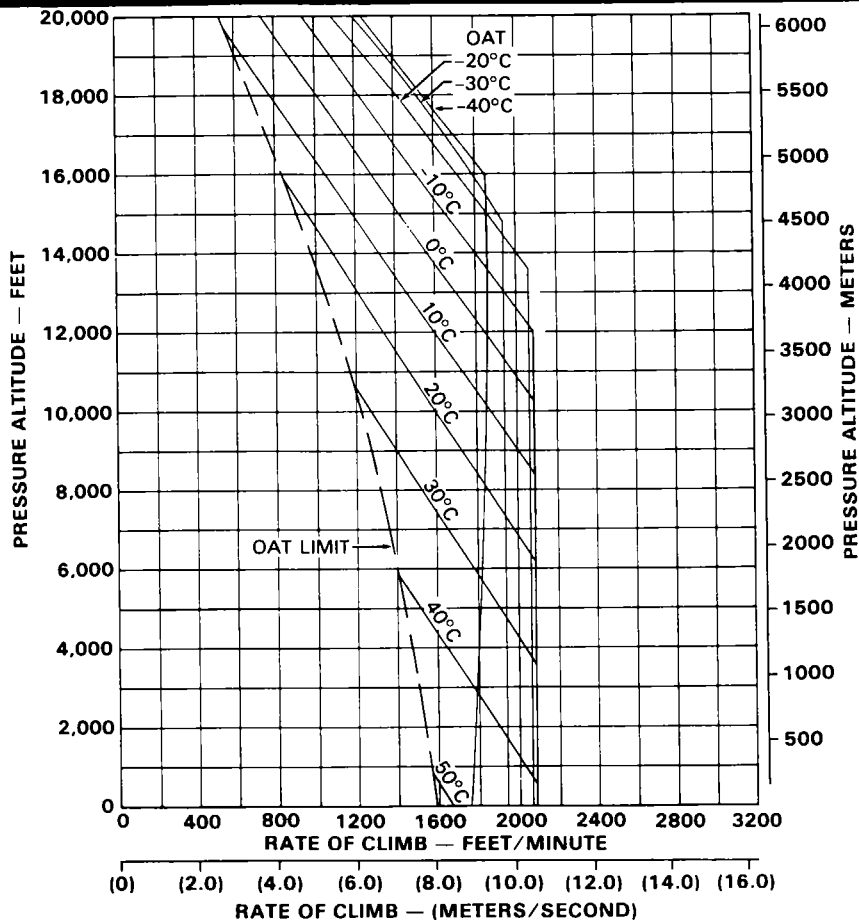
TWIN ENGINE RATE OF CLIMB

GROSS WEIGHT 7000 LB (3175 kg)

MAXIMUM CONTINUOUS POWER
ENGINE RPM 100%
GENERATOR 150 AMPS (EA.)

70 KIAS
HEATER OFF

WITH ALL DOORS OPEN OR REMOVED: 1. CLIMB SPEED IS 60 KIAS
2. RATE OF CLIMB WILL DECREASE
275 FT./MIN.



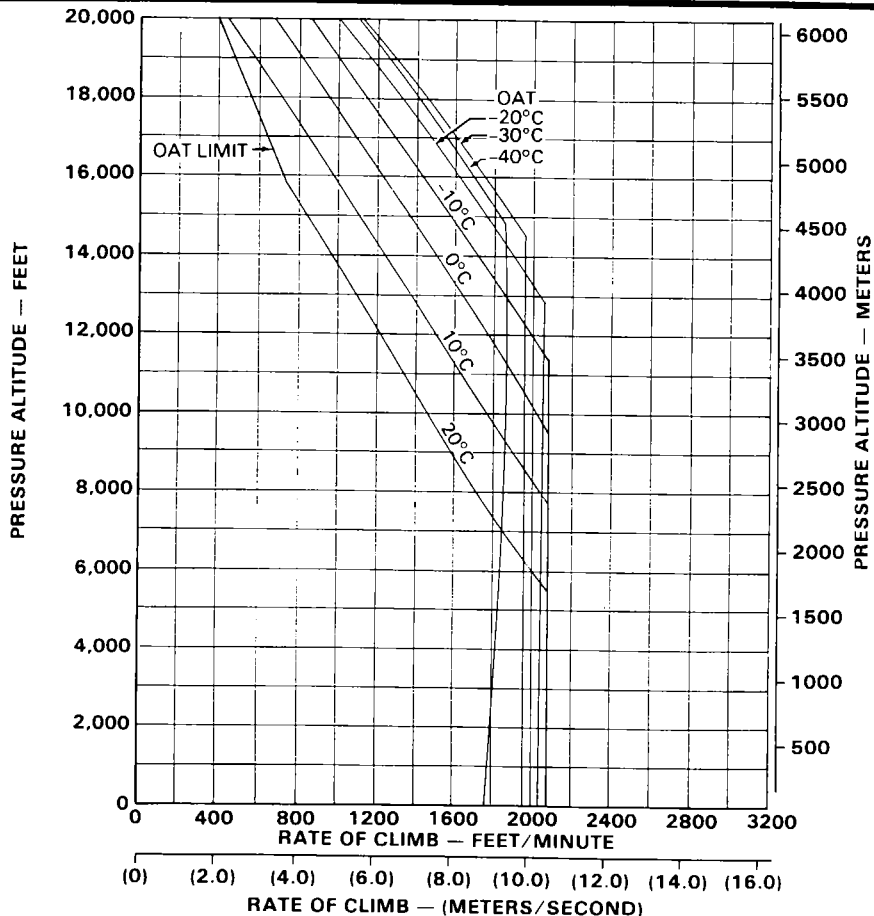
Twin engine rate of climb (Sheet 13 of 24)

TWIN ENGINE RATE OF CLIMB
GROSS WEIGHT 7000 LB (3175 kg)

MAXIMUM CONTINUOUS POWER
ENGINE RPM 100%
GENERATOR 150 AMPS (EA.)

70 KIAS
HEATER ON

WITH ALL DOORS OPEN OR REMOVED: 1. CLIMB SPEED IS 60 KIAS
2. RATE OF CLIMB WILL DECREASE
275 FT /MIN.

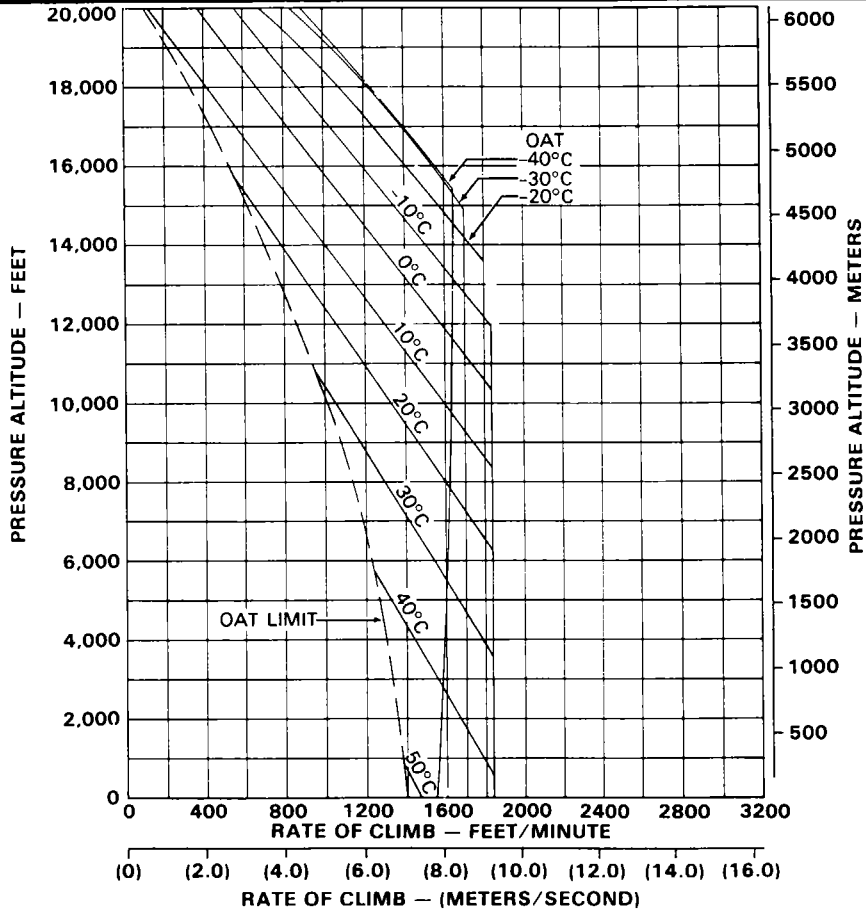


TWIN ENGINE RATE OF CLIMB GROSS WEIGHT 8000 LB (3629 kg)

MAXIMUM CONTINUOUS POWER
ENGINE RPM 100%
GENERATOR 150 AMPS (EA.)

70 KIAS
HEATER OFF

WITH ALL DOORS OPEN OR REMOVED: 1. CLIMB SPEED IS 60 KIAS
2. RATE OF CLIMB WILL DECREASE
275 FT / MIN.



TWIN ENGINE RATE OF CLIMB

GROSS WEIGHT 8000 LB (3629 kg)

MAXIMUM CONTINUOUS POWER

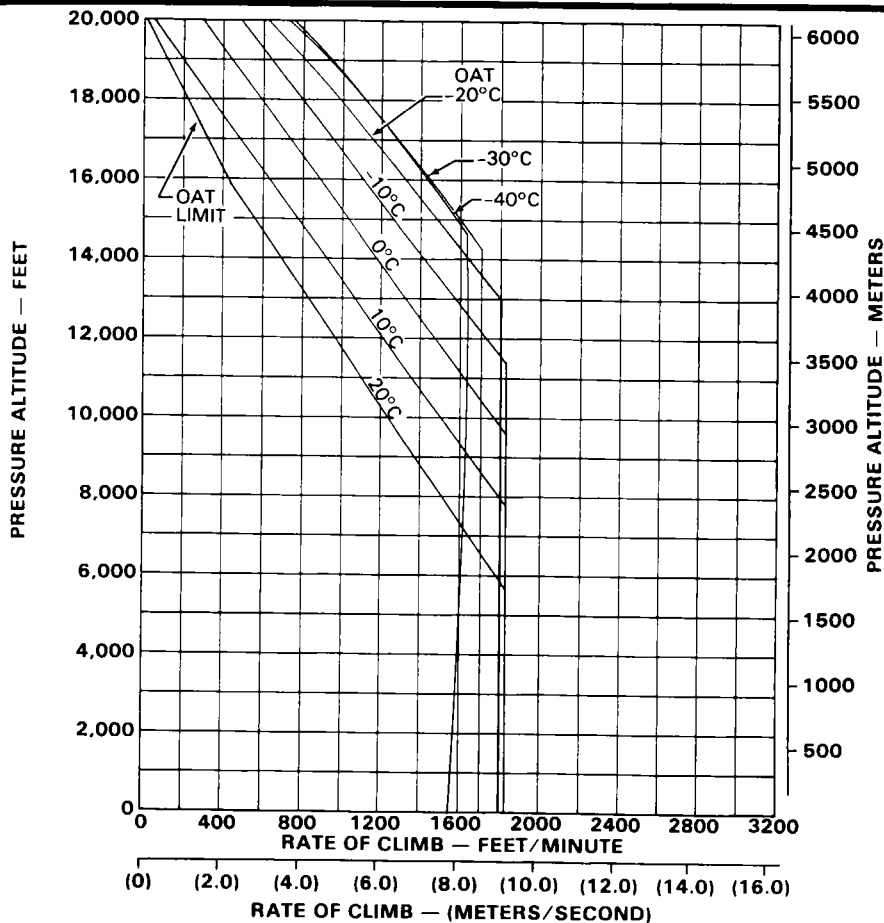
ENGINE RPM 100%

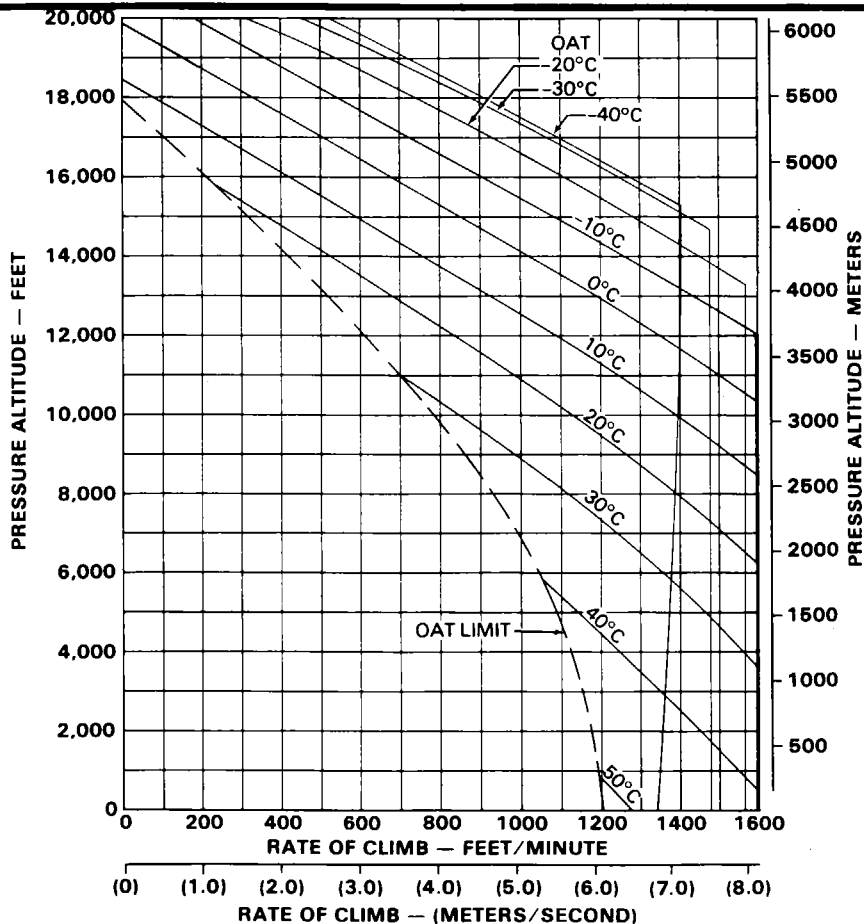
GENERATOR 150 AMPS (EA.)

70 KIAS

HEATER ON

- WITH ALL DOORS OPEN OR REMOVED:
1. CLIMB SPEED IS 60 KIAS
 2. RATE OF CLIMB WILL DECREASE 275 FT./MIN.



TWIN ENGINE RATE OF CLIMB
GROSS WEIGHT 9000 LB (4082 kg)MAXIMUM CONTINUOUS POWER
ENGINE RPM 100%
GENERATOR 150 AMPS (EA.)70 KIAS
HEATER OFFWITH ALL DOORS OPEN OR REMOVED: 1. CLIMB SPEED IS 60 KIAS
2. RATE OF CLIMB WILL DECREASE
275 FT./MIN.

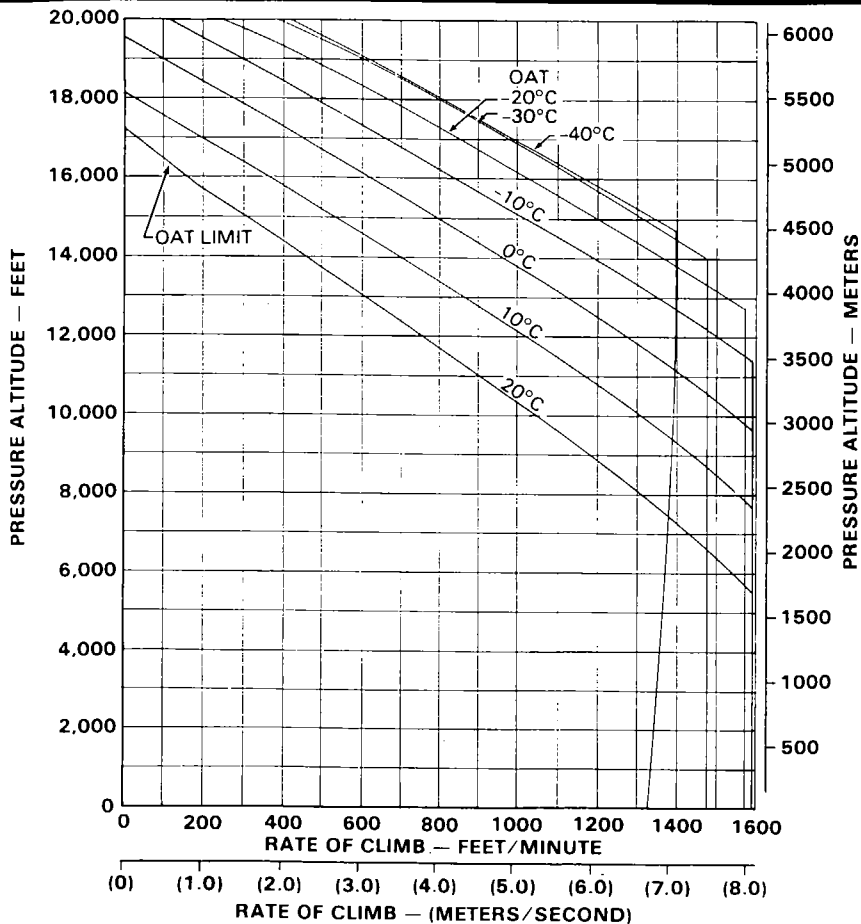
Twin engine rate of climb (Sheet 17 of 24)

TWIN ENGINE RATE OF CLIMB
GROSS WEIGHT 9000 LB (4082 kg)

MAXIMUM CONTINUOUS POWER
ENGINE RPM 100%
GENERATOR 150 AMPS (EA.)

70 KIAS
HEATER ON

WITH ALL DOORS OPEN OR REMOVED: 1. CLIMB SPEED IS 60 KIAS
2. RATE OF CLIMB WILL DECREASE
275 FT / MIN.



Twin engine rate of climb (Sheet 18 of 24)

TWIN ENGINE RATE OF CLIMB

GROSS WEIGHT 10,000 LB (4536 kg)

MAXIMUM CONTINUOUS POWER

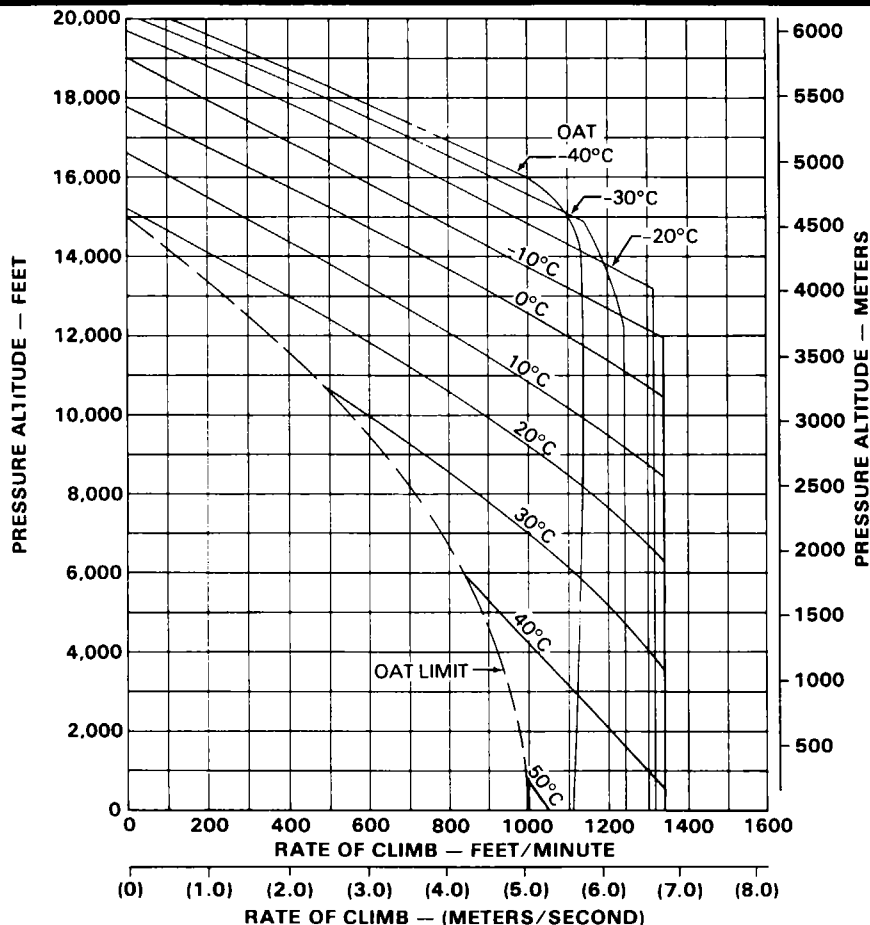
ENGINE RPM 100%

GENERATOR 150 AMPS (EA.)

70 KIAS

HEATER OFF

- WITH ALL DOORS OPEN OR REMOVED:
1. CLIMB SPEED IS 60 KIAS
 2. RATE OF CLIMB WILL DECREASE 275 FT./MIN.



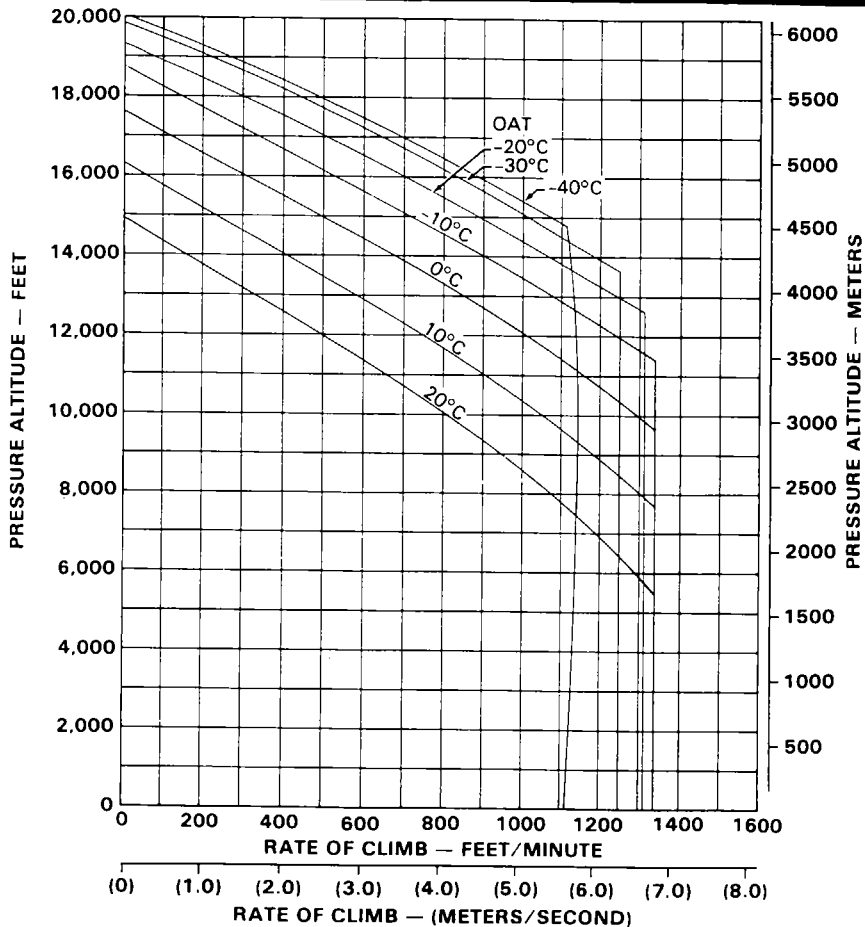
Twin engine rate of climb (Sheet 19 of 24)

TWIN ENGINE RATE OF CLIMB
GROSS WEIGHT 10,000 LB (4536 kg)

MAXIMUM CONTINUOUS POWER
ENGINE RPM 100%
GENERATOR 150 AMPS (EA.)

70 KIAS
HEATER ON

- WITH ALL DOORS OPEN OR REMOVED:
1. CLIMB SPEED IS 60 KIAS
 2. RATE OF CLIMB WILL DECREASE 275 FT / MIN.



Twin engine rate of climb (Sheet 20 of 24)

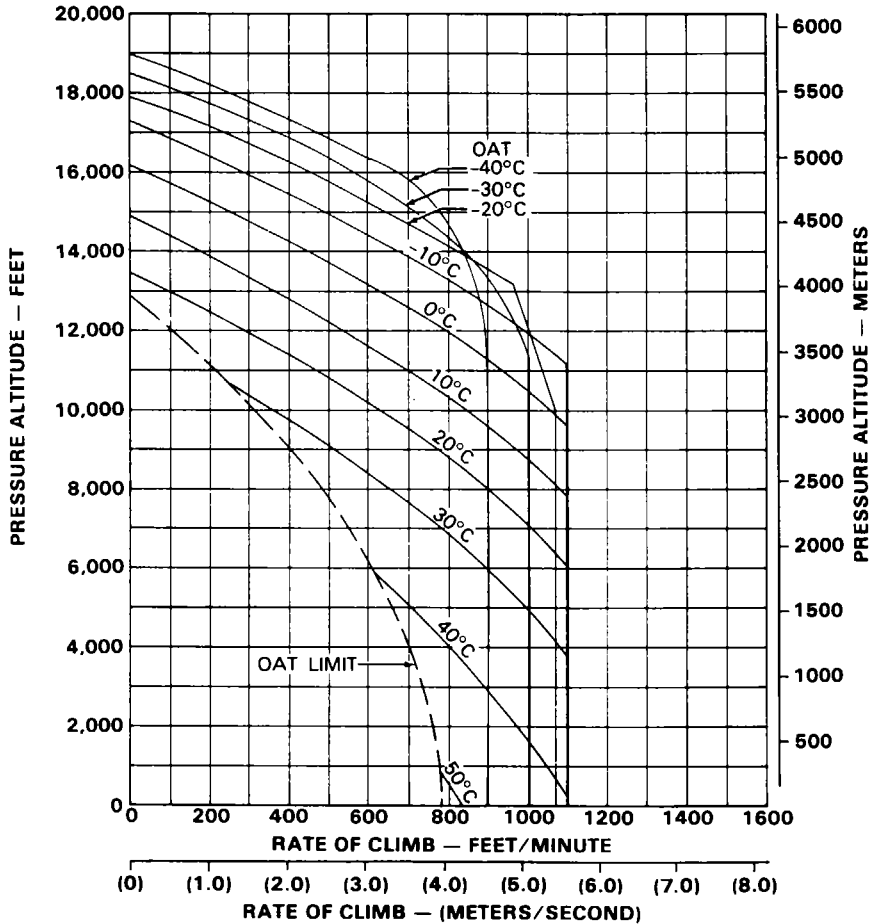
TWIN ENGINE RATE OF CLIMB

GROSS WEIGHT 11,000 LB (4990 kg)

MAXIMUM CONTINUOUS POWER
ENGINE RPM 100%
GENERATOR 150 AMPS (EA.)

70 KIAS
HEATER OFF

WITH ALL DOORS OPEN OR REMOVED: 1. CLIMB SPEED IS 60 KIAS
2. RATE OF CLIMB WILL DECREASE
275 FT./MIN.



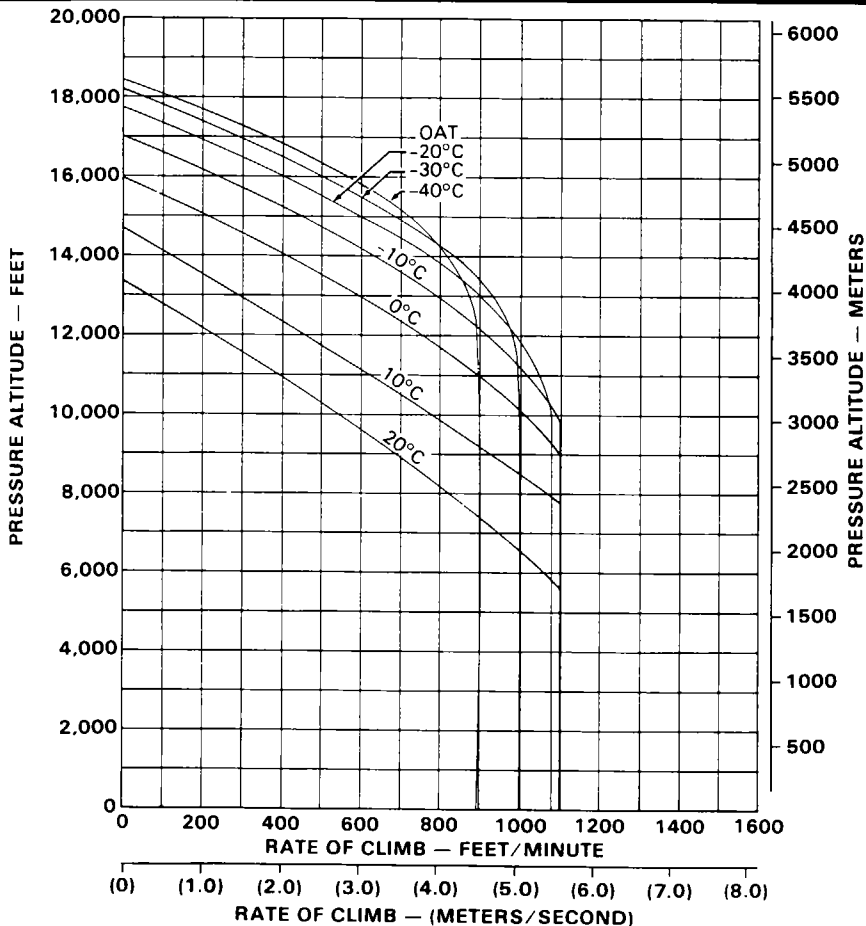
Twin engine rate of climb (Sheet 21 of 24)

TWIN ENGINE RATE OF CLIMB
GROSS WEIGHT 11,000 LB (4990 kg)

MAXIMUM CONTINUOUS POWER
ENGINE RPM 100%
GENERATOR 150 AMPS (EA.)

70 KIAS
HEATER ON

WITH ALL DOORS OPEN OR REMOVED: 1. CLIMB SPEED IS 60 KIAS
2. RATE OF CLIMB WILL DECREASE
275 FT / MIN.



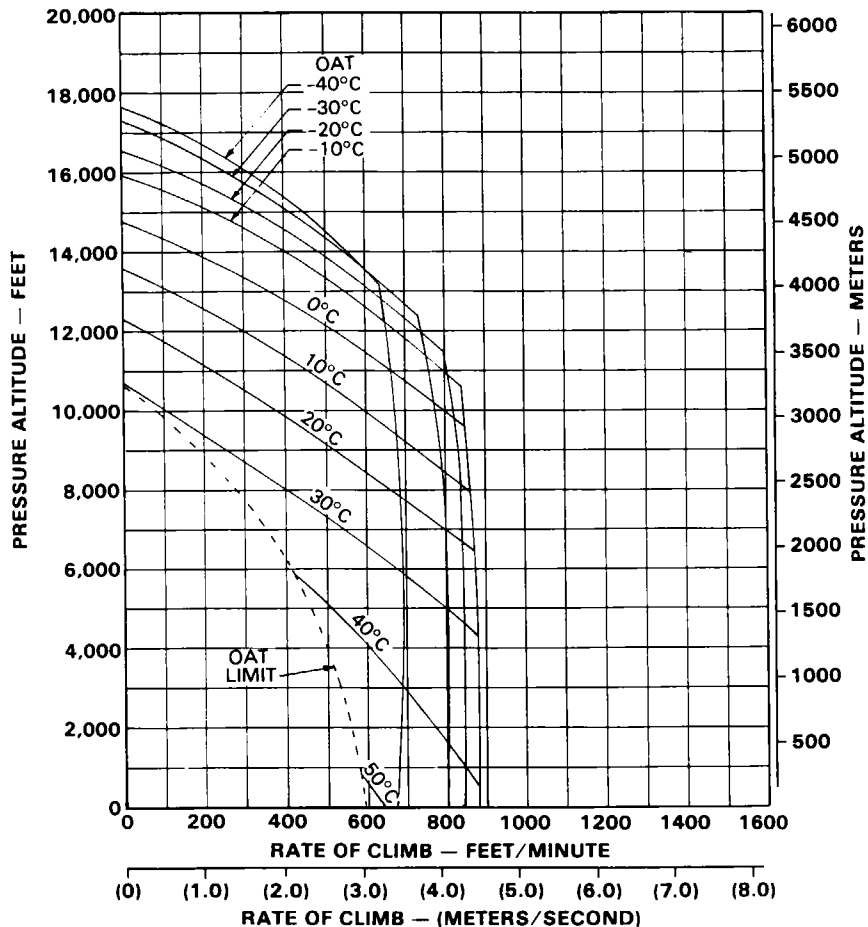
Twin engine rate of climb (Sheet 22 of 24)

TWIN ENGINE RATE OF CLIMB GROSS WEIGHT 11,900 LB (5398 kg)

MAXIMUM CONTINUOUS POWER
ENGINE RPM 100%
GENERATOR 150 AMPS (EA.)

70 KIAS
HEATER OFF

WITH ALL DOORS OPEN OR REMOVED: 1. CLIMB SPEED IS 60 KIAS
2. RATE OF CLIMB WILL DECREASE
275 FT./MIN.



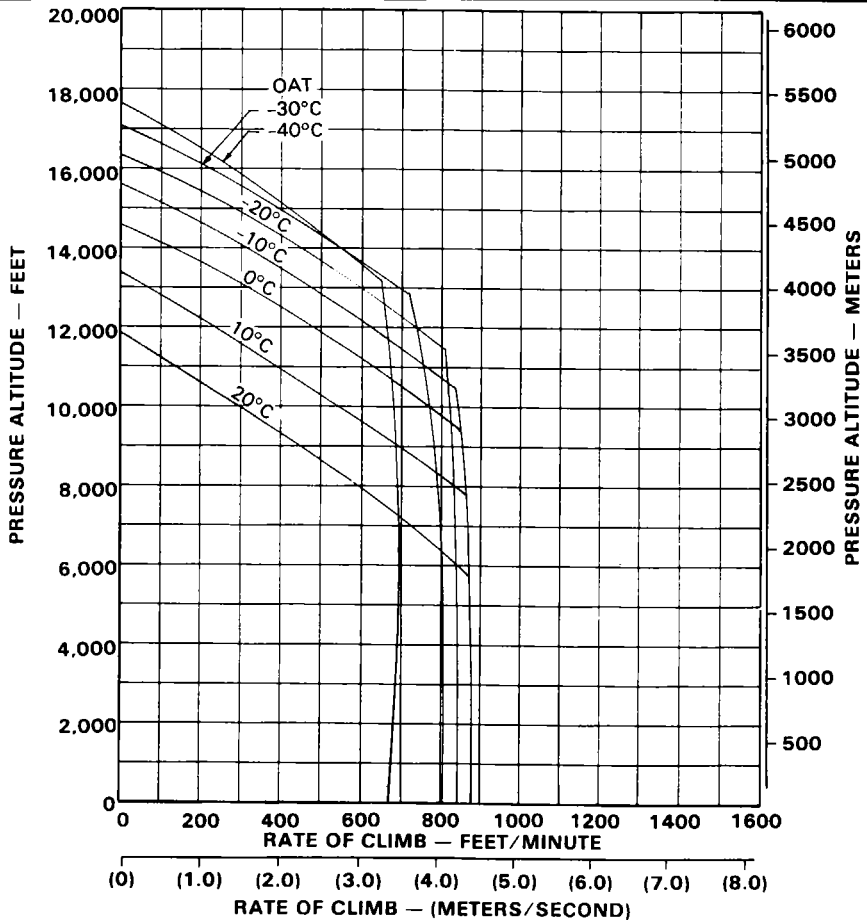
Twin engine rate of climb (Sheet 23 of 24)

TWIN ENGINE RATE OF CLIMB
GROSS WEIGHT 11,900 LB (5398 kg)

MAXIMUM CONTINUOUS POWER
ENGINE RPM 100%
GENERATOR 150 AMPS (EA.)

70 KIAS
HEATER ON

WITH ALL DOORS OPEN OR REMOVED: 1. CLIMB SPEED IS 60 KIAS
2. RATE OF CLIMB WILL DECREASE
275 FT./MIN.



Twin engine rate of climb (Sheet 24 of 24)

Section 1

MANUFACTURER'S DATA

WEIGHT AND BALANCE

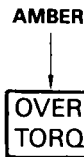
No change from basic Flight Manual

Section 2

MANUFACTURER'S DATA

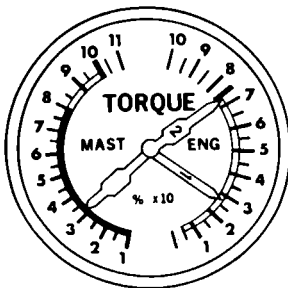
SYSTEMS DESCRIPTION

INSTRUMENT PANEL AND OVERTORQUE WARNING SYSTEM CONSOLES



(located on instrument panel)

An OVER TRQ warning flag (cat's eye) is located in the lower left aft avionics compartment. If mast torque exceeds 108%, the warning flag will be tripped showing alternating black and white sections.



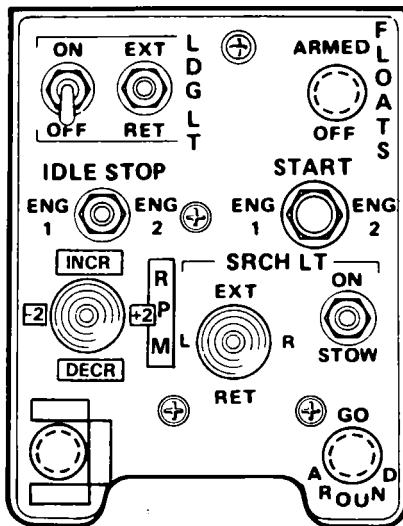
(Replaces TRANSMISSION TORQUE gage)

POWERPLANT

The pilot RPM switch is mounted on the collective switchbox. The pilot switch is a five-position momentary-on type switch. The INCR position increases engine RPM, and the DECR position decreases engine RPM. The INCR/DECR positions control the governors on both engines simultaneously. Regulated engine RPM may be adjusted in flight through the operating range of 97 to 101.5 (± 0.5) % by moving the switch.

The RPM +2/-2 switch increases or decreases engine No. 2 RPM to provide torque or ITT matching. Engine 2 trim range is 2.0 — 2.5% N_{II}. Engine 2 governor should be at least 95% minimum trim — minimum beep.

The copilot does not have trim capability. For location of RPM switch, refer to collective control panel.



Collective control panel.

Section 3

MANUFACTURER'S DATA

OPERATIONAL INFORMATION

No change from basic Flight Manual

Section 4

MANUFACTURER'S DATA

HANDLING/SERVICING/MAINTENANCE

No change from basic Flight Manual

Bell **412** MODEL

ROTORCRAFT FLIGHT MANUAL

SUPPLEMENT CATEGORY B OPERATIONS WHEN CONFIGURED WITH NINE OR LESS PASSENGER SEATS

33108 — 33213

34001 — 34024

36001 — 36019

AND

33001 — 33107

**WHEN 412-075-008-111
TORQUEMETER IS INSTALLED
(BHT-412-FMS-19.1)
CERTIFIED
10 APRIL 1991**

This supplement shall be attached to the Model 412 Flight Manual when helicopter is configured with nine or less passengers seat configuration.

Information contained herein supplements information of basic Flight Manual. For Limitations, Procedures, and Performance Data not contained in this supplement, or other applicable supplements, consult basic Flight Manual.

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Reissue	0	10 MAY 96	Reissue	0	19 MAR 03

LOG OF PAGES

PAGE	REVISION NO.	PAGE	REVISION NO.
FLIGHT MANUAL		i/ii	0
Title	0	1 — 12	0
NP	0		
A — B	0		

NOTE

Revised text is indicated by a black vertical line. Insert latest revision pages; dispose of superseded pages.

LOG OF FAA APPROVED REVISIONS

Original 0 10 APR 91
Reissue..... 0 10 MAY 96

Revision 1 23 APR 98
Reissue 0 19 MAR 03

APPROVED:

DATE: MAR 19 2003



MANAGER

ROTORCRAFT CERTIFICATION OFFICE
FEDERAL AVIATION ADMINISTRATION
FT. WORTH, TX 76193-0170

GENERAL INFORMATION

This supplement removes the Height-velocity diagram as a limitation and increases the density altitude limit for takeoff, landing, and in-ground-effect maneuvers when helicopter is configured with nine or less passenger seats.

Section 1

LIMITATIONS

1-3. TYPES OF OPERATION

Flights may be conducted in accordance with this supplement only when the helicopter is configured with nine or less passenger seats.

1-6. WEIGHT AND CENTER OF GRAVITY

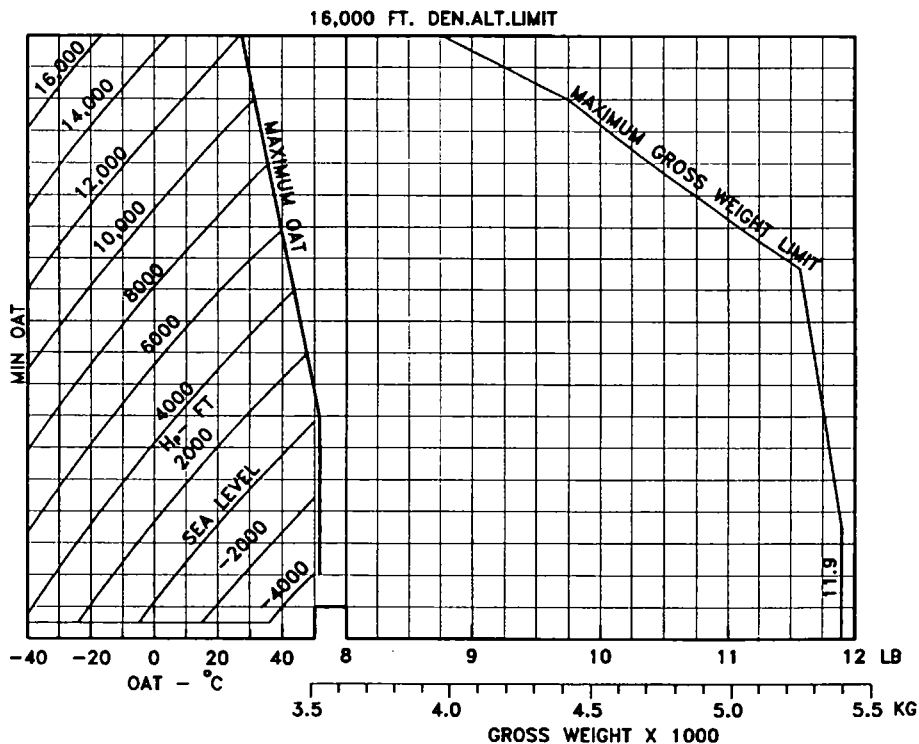
Maximum gross weight for takeoff, landing, and in-ground-effect maneuvers is 11,900 pounds (5,398 kilograms) or as shown in Hover ceiling in ground effect (takeoff power) chart, refer to SECTION 4, whichever is less.

Actual weight change shall be determined after seating is installed, and ballast shall be adjusted (if necessary) to return empty weight CG to within allowable limits.

1-8. ALTITUDE

Maximum density altitude for takeoff, landing, and in-ground-effect maneuvers is 16,000 feet. Refer to Weight-Altitude-Temperature Limitations chart (Figure 1-1).

NOTE: ALLOWABLE GROSS WEIGHTS OBTAINED FROM THIS CHART MAY EXCEED HOVER CAPABILITY UNDER CERTAIN AMBIENT CONDITIONS. REFER TO HOVER CEILING CHARTS IN SECTION 4.



412FMS/001

Figure 1-1. Weight-altitude-temperature limitations for takeoff, landing, and in-ground-effect maneuvers

Section 2

NORMAL PROCEDURES

2-2. FLIGHT PLANNING

Refer to Performance Data, Section 4.

NOTE

The Height-velocity diagram does not represent a limitation.

Section 3

EMERGENCY/MALFUNCTION PROCEDURES

No change from basic manual.

Section 4

PERFORMANCE

4-4. HEIGHT - VELOCITY ENVELOPE

Operation in height-velocity envelope is critical in the event of a single engine failure during takeoff, landing, or other operation near the surface (Figure 4-1). The AVOID area of the Height-velocity diagram defines the combinations of airspeed and height above ground from which a safe single engine landing on a smooth, level, firm surface cannot be assured.

The Height-velocity diagram is valid only when the Hover Ceiling Out of Ground Effect performance envelope is not exceeded (refer to Figure 4-2). The Hover Ceiling In Ground Effect performance chart (refer to Figure 4-3) does not define the conditions which assure continued flight following an engine failure nor the conditions from which a safe power-off landing can be made.

4-5. HOVER CEILING

The Hover Ceiling In Ground Effect charts (Figures 4-3 and 4-4) provide the maximum allowable gross weights for takeoff, landing, and IGE maneuvers at all pressure altitude and outside air temperature conditions with heater on or off. Conversely, the hover ceiling altitude can be determined for any given gross weight.

Adequate cyclic and directional control are available at the gross weights allowed by the Hover Ceiling IGE charts in winds up to 35 knots from any direction at or below 3,000 feet HD (refer to Basic Flight Manual). Above 3,000 feet HD, improved control margins will be realized by avoiding winds in the critical wind azimuth area (Figure 4-5).

4-6. TAKEOFF DISTANCE

The Takeoff Distance charts (Figure 4-6) provide takeoff distances required to clear a 50-foot or 15-meter obstacle in a zero wind condition. Takeoff is initiated from a hover at 4 feet (1.2 meters) skid height with climbout of 45 knots.

NOTE

Downwind takeoffs are not recommended because the published takeoff distance performance cannot be achieved.

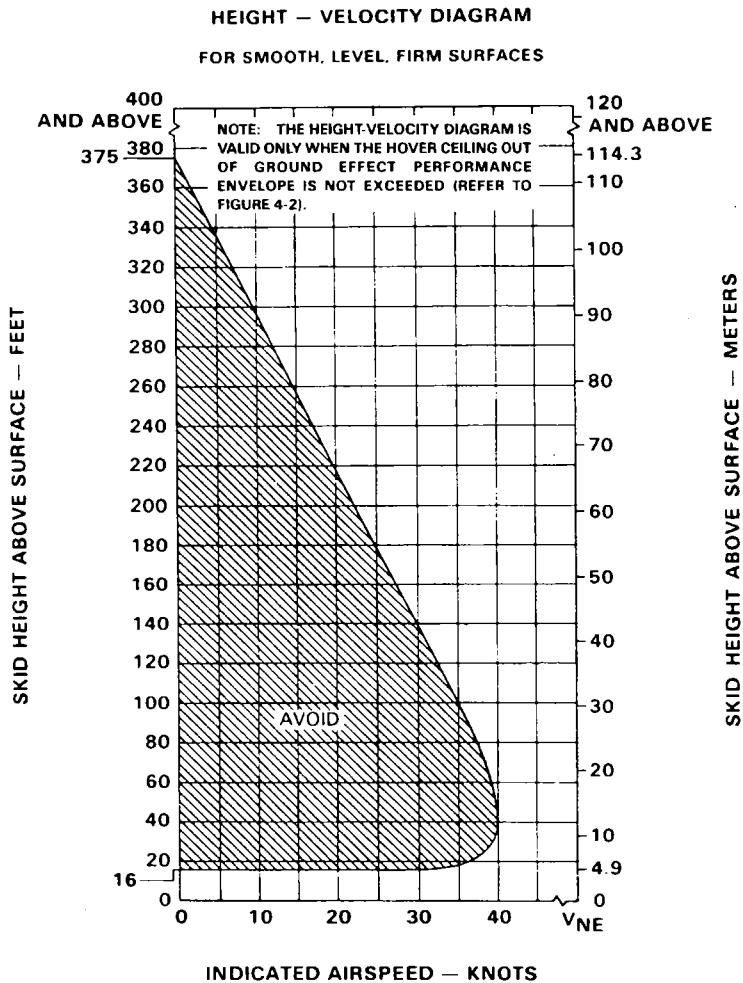


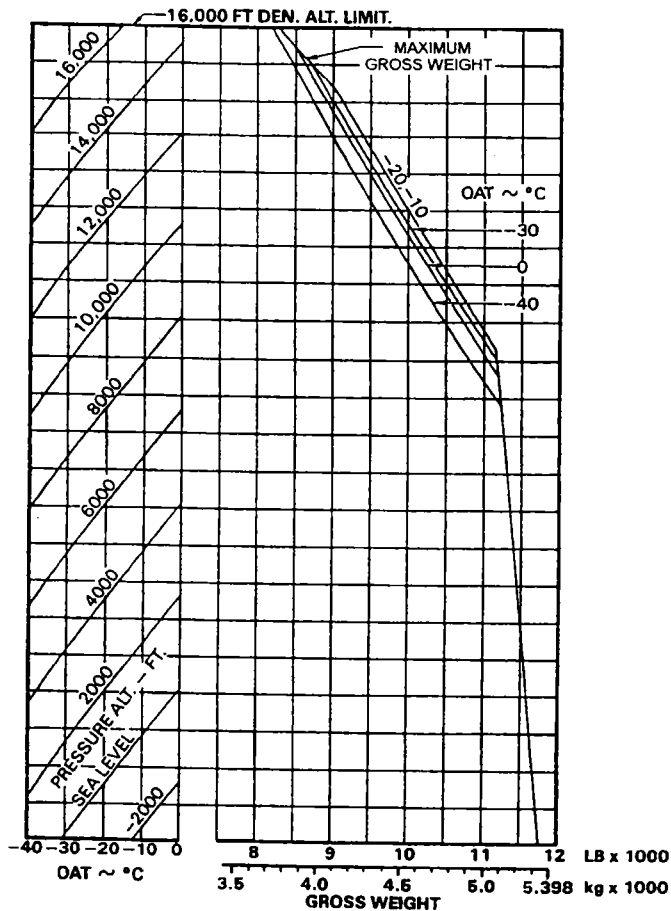
Figure 4-1. Height-velocity diagram (OEI)

HOVER CEILING OUT OF GROUND EFFECT

TAKEOFF POWER
ENGINE RPM 100%
GENERATOR 150 AMPS (EA.)

SKID HEIGHT 60 FEET
HEATER ON OR OFF
-40°C TO 0°C

NOTE: THESE DATA VALID FOR ZERO WIND OUTSIDE OF THE CRITICAL WIND AZIMUTH AREA (REFER TO FIGURE 4-5)



412FMS003

Figure 4-2. Hover ceiling out of ground effect (Sheet 1 of 2)

HOVER CEILING OUT OF GROUND EFFECT

TAKEOFF POWER
ENGINE RPM 100%
GENERATOR 150 AMPS (EA.)

SKID HEIGHT 60 FEET
HEATER ON OR OFF
0°C TO 52°C

NOTE: THESE DATA VALID FOR ZERO WIND OUTSIDE OF THE CRITICAL WIND AZIMUTH AREA (REFER TO FIGURE 4-5)

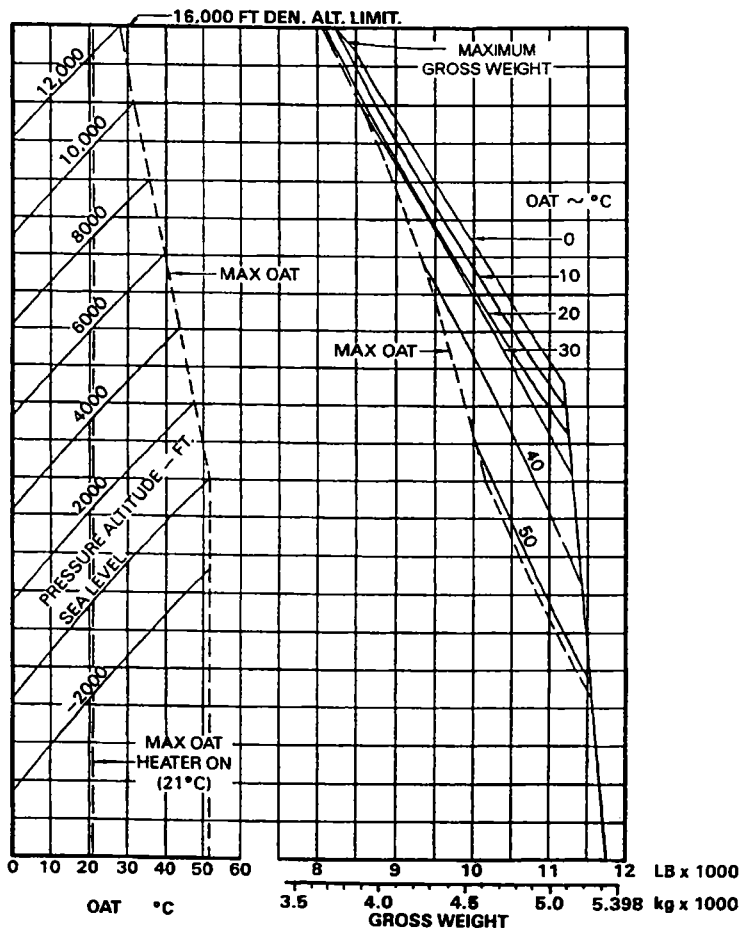


Figure 4-2. Hover ceiling out of ground effect (Sheet 2 of 2)

HOVER CEILING IN GROUND EFFECT MAXIMUM WEIGHT FOR TAKEOFF, LANDING, AND IN-GROUND-EFFECT MANEUVERS

TAKEOFF POWER
ENGINE RPM 100%
GENERATOR 150 AMPS (EA.)

SKID HEIGHT 4 FEET
HEATER ON OR OFF
-40°C TO 52°C

NOTE: THESE DATA VALID FOR ZERO WIND OUTSIDE OF THE CRITICAL WIND AZIMUTH AREA (REFER TO FIGURE 4-5)

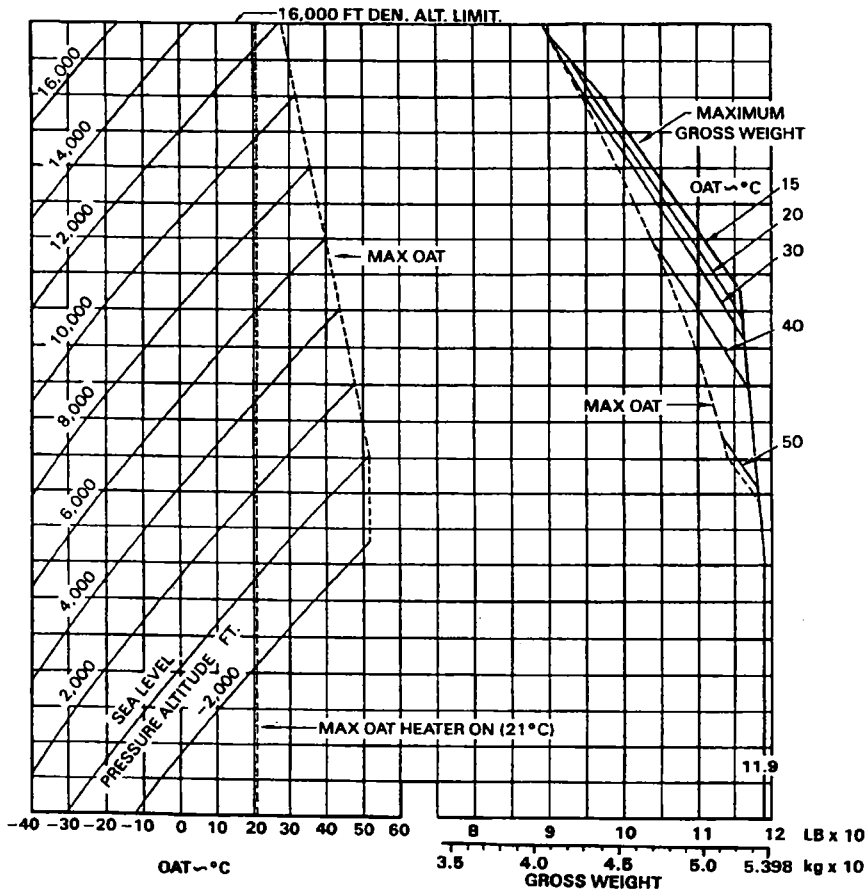


Figure 4-3. Hover ceiling in ground effect (takeoff power)

HOVER CEILING IN GROUND EFFECT MAXIMUM WEIGHT FOR TAKEOFF, LANDING, AND IN-GROUND-EFFECT MANEUVERS

MAXIMUM CONTINUOUS POWER
ENGINE RPM 100%
GENERATOR 150 AMPS (EA.)

SKID HEIGHT 4 FEET
HEATER ON OR OFF
-40°C TO 52°C

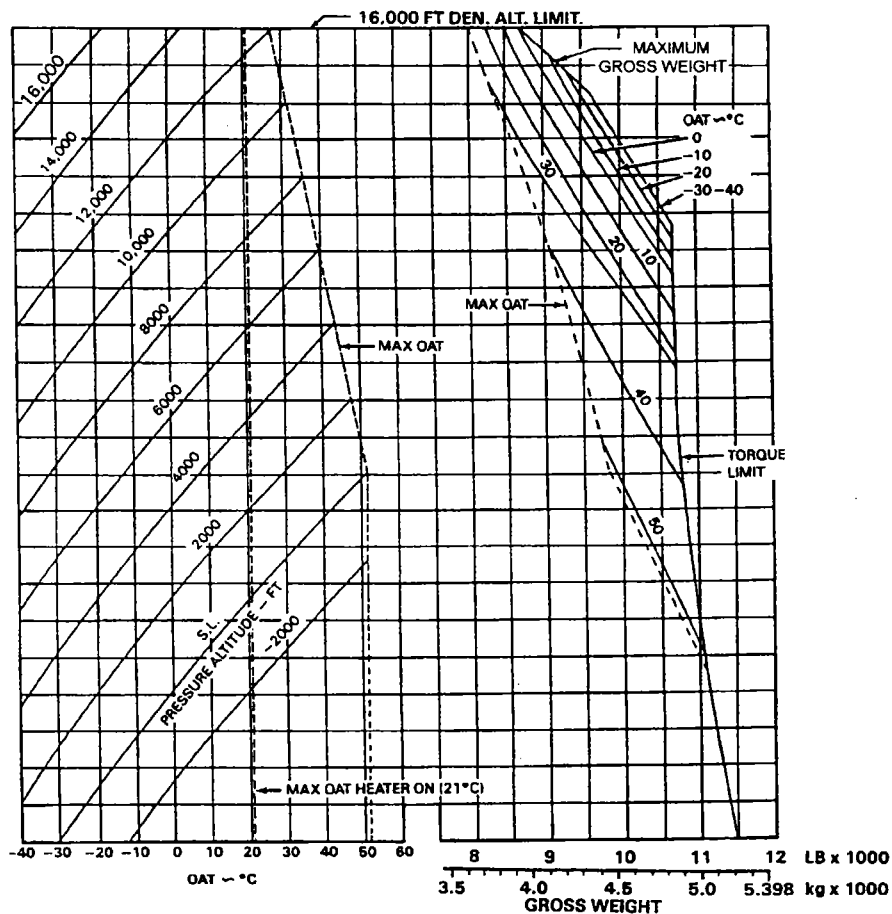
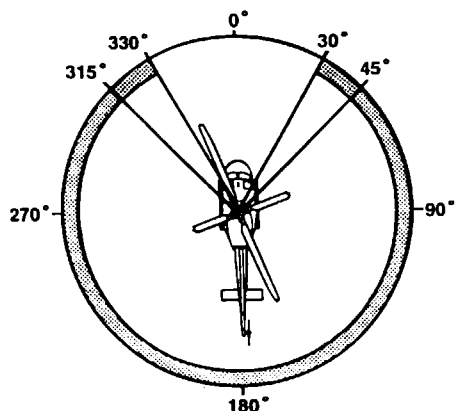
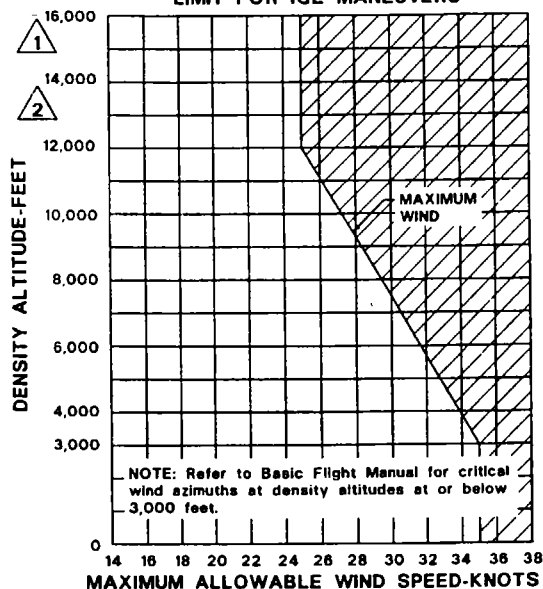


Figure 4-4. Hover ceiling in ground effect (maximum continuous power)

CRITICAL RELATIVE WIND AZIMUTH

18,000 FT. H_D

LIMIT FOR IGE MANEUVERS



Critical wind azimuth — hovering with the relative wind azimuth angles in shaded area  can result in the following:

a. Inability to maintain heading due to large left pedal requirements for certain wind velocities.

b. Reduction of available left pedal control with a directional AFCS hardover.

c. Aft cyclic may be limited with longitudinal AFCS hardover.

d. Hover performance is valid for all headings in calm wind.

① For H_D from 14,000 to 16,000 ft winds up to $\pm 30^\circ$ off nose for hover performance to be valid.

② For H_D below 14,000 ft winds up to $\pm 45^\circ$ off nose for hover performance to be valid.

Figure 4-5. Critical relative wind azimuth

TAKEOFF DISTANCE OVER 50 FOOT OBSTACLE

HOVER POWER +15% TORQUE (not to exceed limits)
ENGINE RPM 100%
GENERATOR 150 AMPS (EA.)

INITIATED FROM 4FT. SKID HEIGHT
 $V_{TOCS} = 45 \text{ KIAS}$
HEATER ON OR OFF

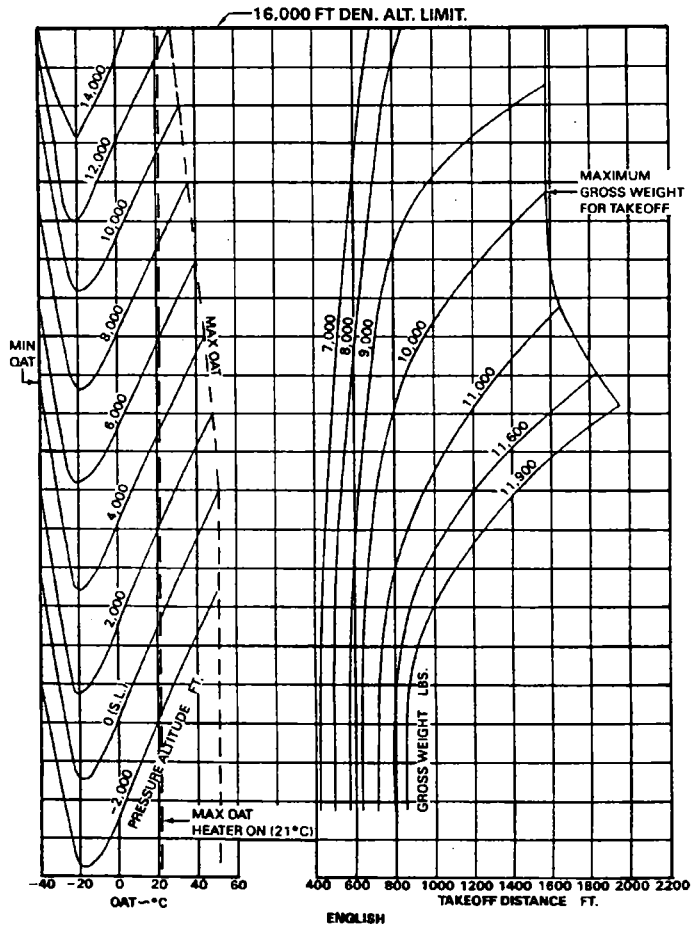
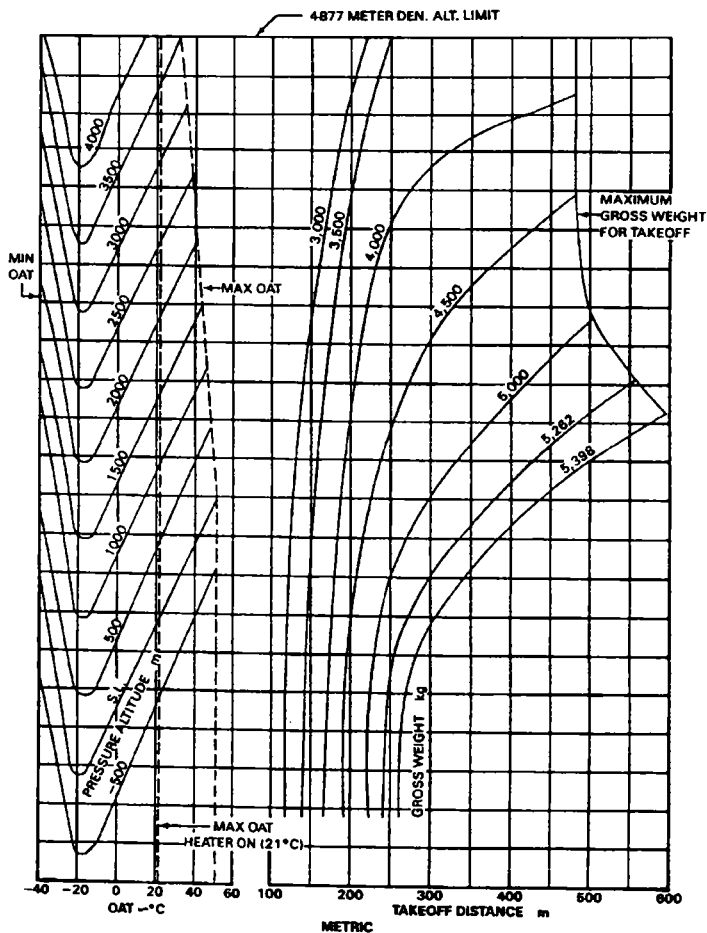


Figure 4-6. Takeoff distance (English)

TAKEOFF DISTANCE OVER 15 METER OBSTACLE

HOVER POWER +15% TORQUE (not to exceed limits)
ENGINE RPM 100%
GENERATOR 150 AMPS (EA.)

INITIATED FROM 1.2 METER SKID HEIGHT
 $V_{TOCS} = 45$ KIAS
HEATER ON OR OFF



412FMS009

Figure 4-7. Takeoff distance (Metric)

Bell **MODELS 412/412EP**

ROTORCRAFT FLIGHT MANUAL

SUPPLEMENT FOR INCREASED GENERATOR CAPACITY KIT (412-706-026)

CERTIFIED
29 OCTOBER 1992

This supplement shall be attached to the Models 412 and 412EP Flight Manuals when the Increased Generator Capacity Kit is installed.

The information contained herein supplements the information of the basic Flight Manual. For Limitations, Procedures, and Performance Data not contained in this supplement, consult the basic Flight Manual

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REISSUE — 5 OCTOBER 1994

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INTRODUCTION

The Increased Generator Capacity Kit increases the amperage limit from 150 to 200 amps on each generator. The incremental performance losses for the additional 50 amps each is presented in Section 4.

Section 1

LIMITATIONS

WEIGHT/CG LIMITATIONS

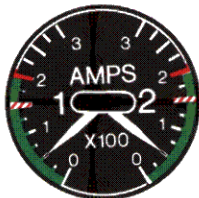
Actual weight change shall be determined after kit is installed and ballast adjusted (if necessary) to return empty weight CG within allowable limits.

CAUTION

GENERATOR LIMITATIONS

Continuous operation	0 to 200 amps
Maximum for operation above 15,000 feet density altitude	150 amps
Maximum	200 amps

DURING SINGLE GENERATOR OPERATION, ELECTRICAL LOADS SHALL BE REDUCED BEFORE RESTORING POWER TO NONESSENTIAL BUS TO ENSURE GENERATOR LOAD LIMIT IS NOT EXCEEDED.



AMMETER



0 to 200 AMPS
150 AMPS
200 AMPS

Continuous operation
Maximum for operation
above 15,000 ft H_D
Maximum

INSTRUMENT MARKINGS

Section 2

NORMAL PROCEDURES

No change from basic Flight Manual.

Section 3

EMERGENCY AND MALFUNCTION PROCEDURES

ELECTRICAL POWER FAILURES

DC POWER FAILURE

INDICATIONS:

DC GENERATOR caution light illuminates.

All lighting and avionics on nonessential buses inoperative.

PROCEDURE:

GEN FIELD and GEN RESET circuit breakers — Check in.

GEN switch (affected generator) — RESET, then ON.

If generator remains inoperative, proceed as follows:

GEN switch (affected generator) — OFF.

If No. 2 Generator failed:

BATTERY BUS 2 switch — OFF;

BATTERY BUS 1 switch — ON.

If nonessential bus power is required, proceed as follows:

Switch off all unnecessary equipment.

CAUTION

DO NOT SET NON-ESNTL BUS SWITCH TO MANUAL BEFORE TURNING OFF UNNECESSARY EQUIPMENT TO ENSURE GENERATOR LOAD LIMIT IS NOT EXCEEDED.

NON-ESNTL BUS switch — MANUAL.

DC AMPS — Monitor.

Equipment switches — As desired/off as necessary to maintain generator load below maximum limit.

NOTE

During single engine operation, avoid generator load above 150 amps to attain climb performance presented in basic Flight Manual.

Section 4

PERFORMANCE

PERFORMANCE VARIATIONS

Performance variation charts are provided to determine hover and climb performance decrements due to the additional power requirements for the generators when operating at 200 amps each.

The charts are organized into three performance sections according to helicopter configuration and respective flight manual and supplements to which the charts apply.

PERFORMANCE SECTION APPLICATION

PERFORMANCE SECTION	HELICOPTER SERIAL NUMBERS	EQUIPMENT REQUIRED	FLIGHT MANUAL/ SUPPLEMENT
Section 4A	33001-33107	None	*BHT-412-FM-1
Section 4B	33108-33213 36001-36019	None	*BHT-412-FM-2
	33001-33107	Increased Gross Weight and Takeoff Horsepower (412-075-008-111)	BHT-412-FMS-19.1
Section 4C	36020 — 36086 AND 36087 AND SUB	None	*BHT-412-FM-3
			*BHT-412-FM-4
	33108-33213 36001-36019	Improved Hover Performance Modification (412-570-001-103) or Increased Maximum Continuous Power Kit (412-706-029)	BHT-412-FMS-34.2
			BHT-412-FMS-41
	33001-33107	Increased Maximum Continuous Power Kit (412-706-029)	BHT-412-FMS-41

* Basic Flight Manual or appropriate optional equipment supplement.

Section 4A

BHT-412-FM-1

PERFORMANCE

TWIN ENGINE HOVER AND RATE OF CLIMB DECREASE DUE TO 200 AMP GENERATOR LOADS.

Enter appropriate chart with pressure altitude and OAT to determine whether or not performance reduction is required. If applicable, decrease performance data in basic flight manual or appropriate optional equipment supplement as indicated on chart (rate of climb reduction of 30 feet per minute or hover gross weight reduction of 50 pounds out of ground effect or 60 pounds in ground effect).

HOVER PERFORMANCE VARIATION FOR BHT-412-FM-1

TAKEOFF POWER
GENERATOR 200 AMPS (EACH)

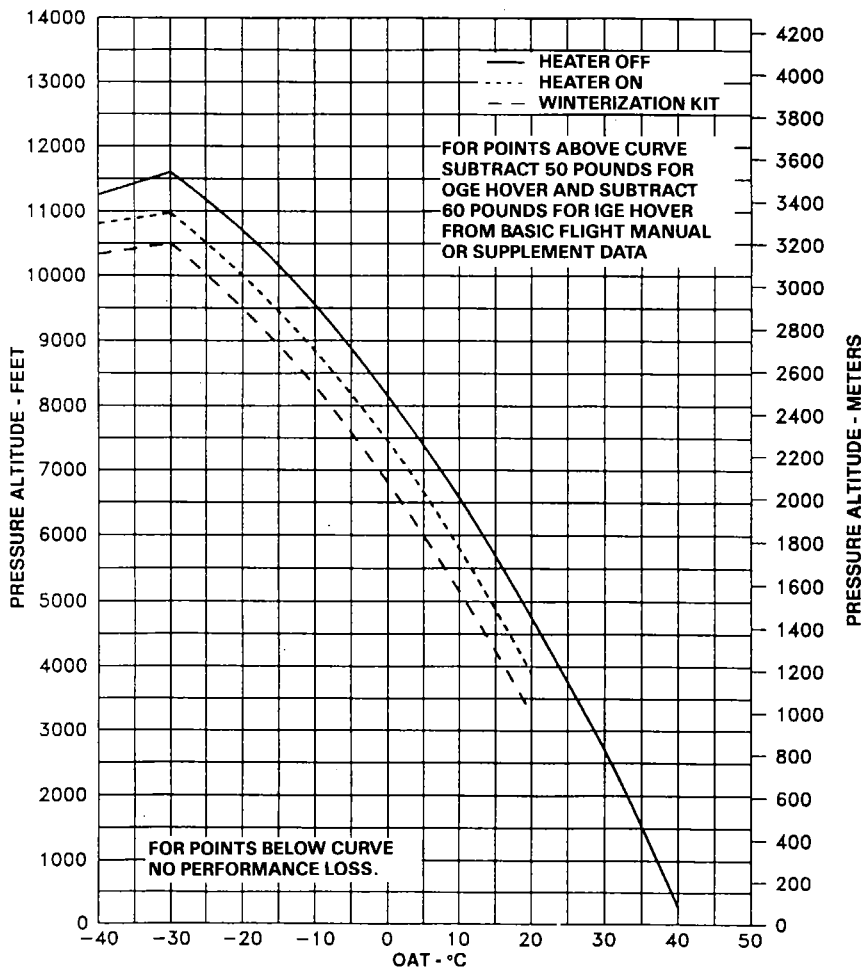
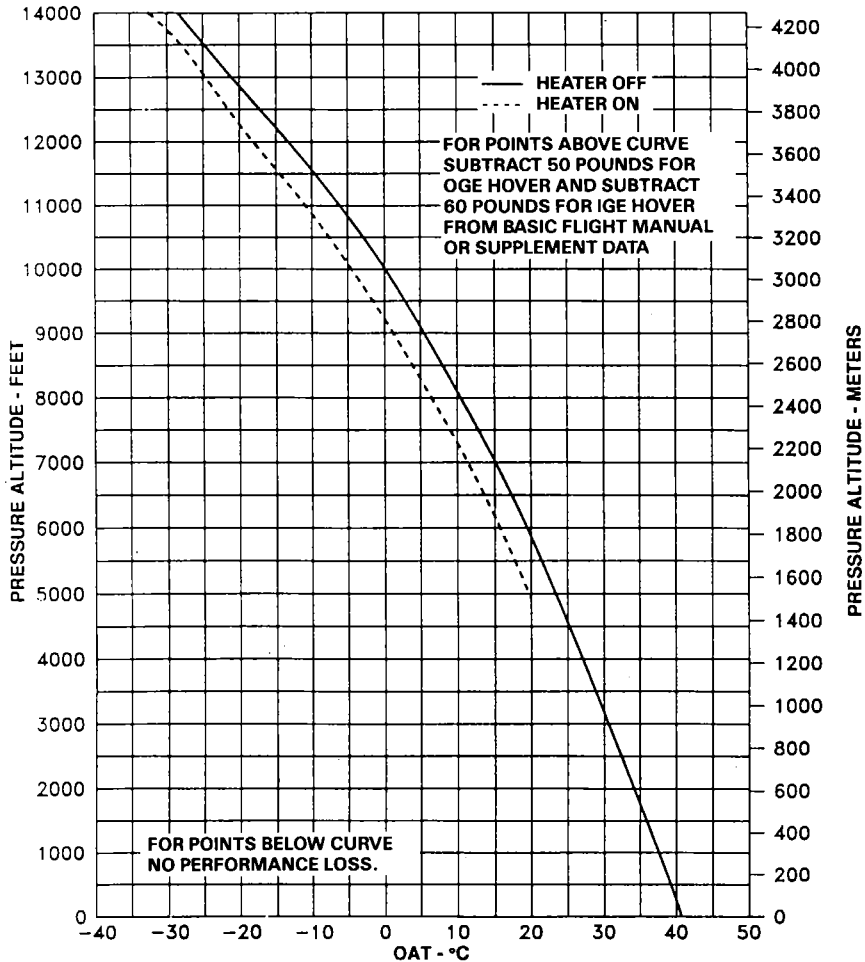


Figure 4A-1. Hover performance variation — takeoff power

**HOVER PERFORMANCE VARIATION
FOR BHT-412-FM-1****MAXIMUM CONTINUOUS POWER
GENERATOR 200 AMPS (EACH)****Figure 4A-2. Hover performance variation — maximum continuous power**

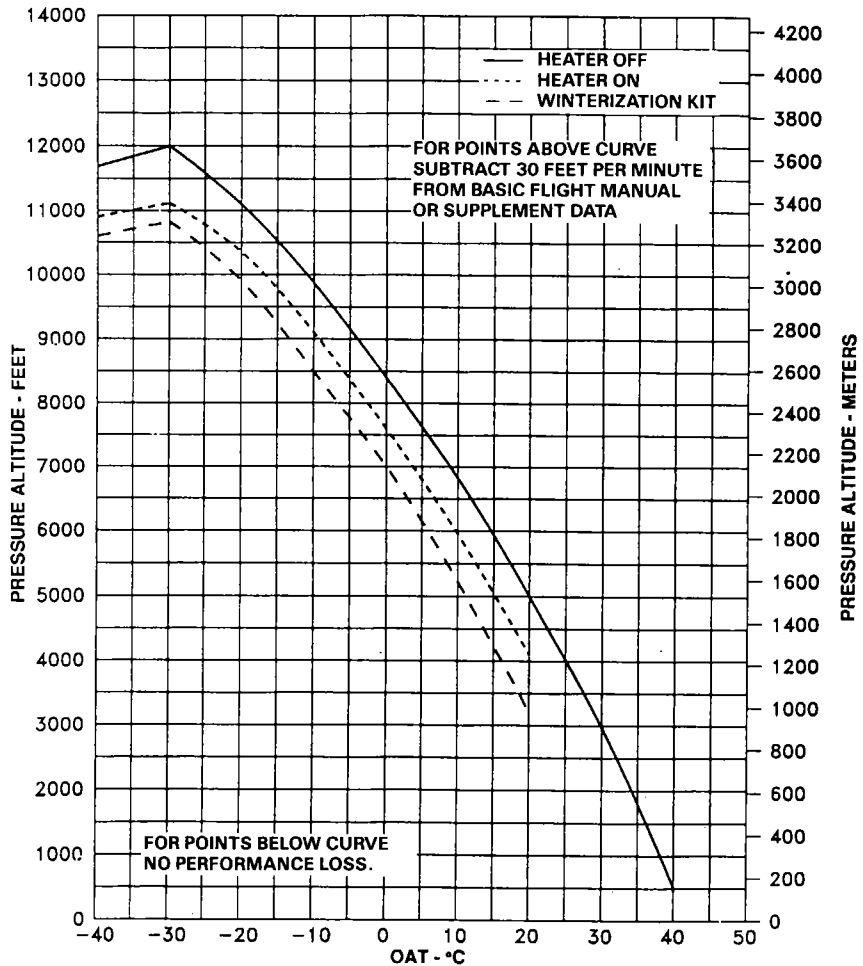
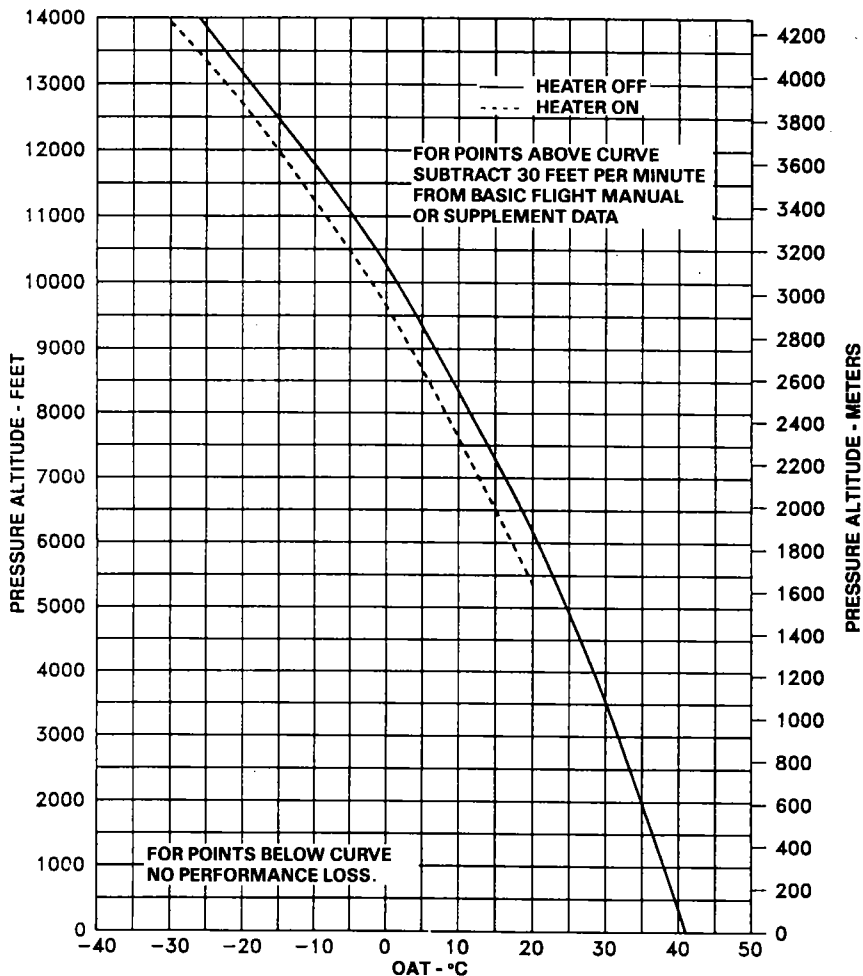
**CLIMB PERFORMANCE VARIATION
FOR BHT-412-FM-1****TAKEOFF POWER
GENERATOR 200 AMPS (EACH)**

Figure 4A-3. Climb performance variation — takeoff power

**CLIMB PERFORMANCE VARIATION
FOR BHT-412-FM-1****MAXIMUM CONTINUOUS POWER
GENERATOR 200 AMPS (EACH)****Figure 4A-4. Climb performance variation — maximum continuous power**

Section 4B

BHT-412-FM-2
BHT-412-FMS-19.1

PERFORMANCE

TWIN ENGINE HOVER AND RATE OF CLIMB DECREASE DUE TO 200 AMP GENERATOR LOADS.

Enter appropriate chart with pressure altitude and OAT to determine whether or not performance reduction is required. If applicable, decrease performance data in basic flight manual or appropriate optional equipment supplement as indicated on chart (rate of climb reduction of 30 feet per minute or hover gross weight reduction of 50 pounds out of ground effect or 60 pounds in ground effect).

**HOVER PERFORMANCE VARIATION
FOR BHT-412-FM-2 AND BHT-412-FMS 19.1**

**TAKEOFF POWER
GENERATOR 200 AMPS (EACH)**

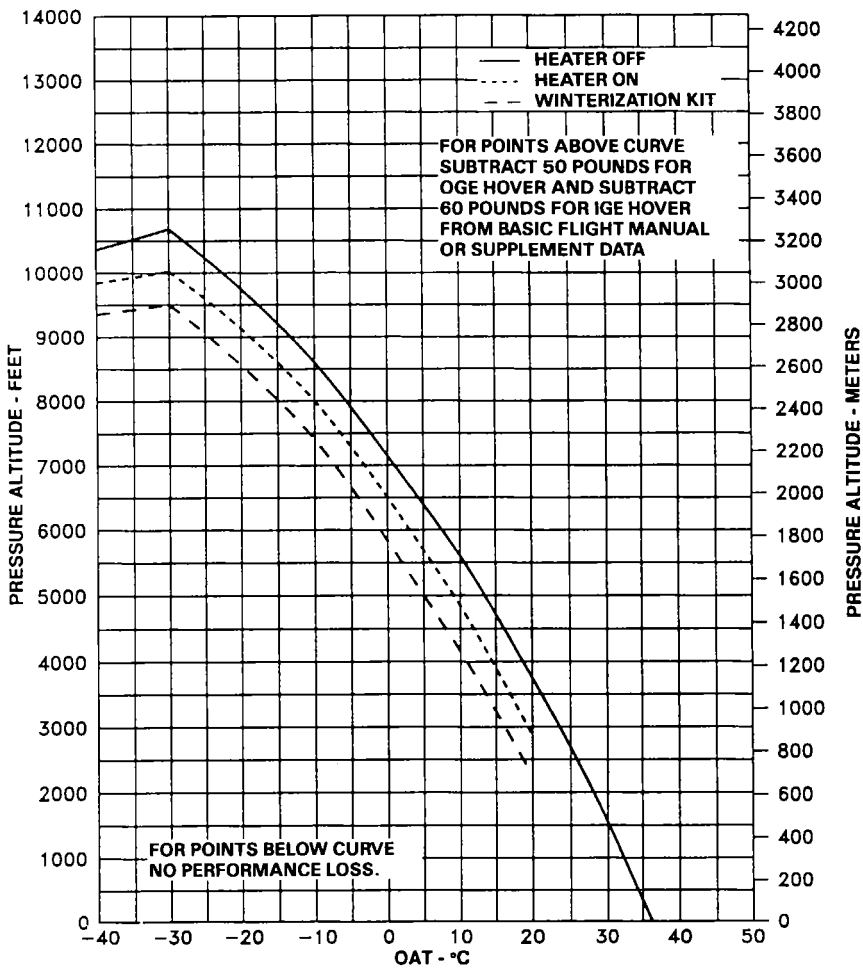


Figure 4B-1. Hover performance variation — takeoff power

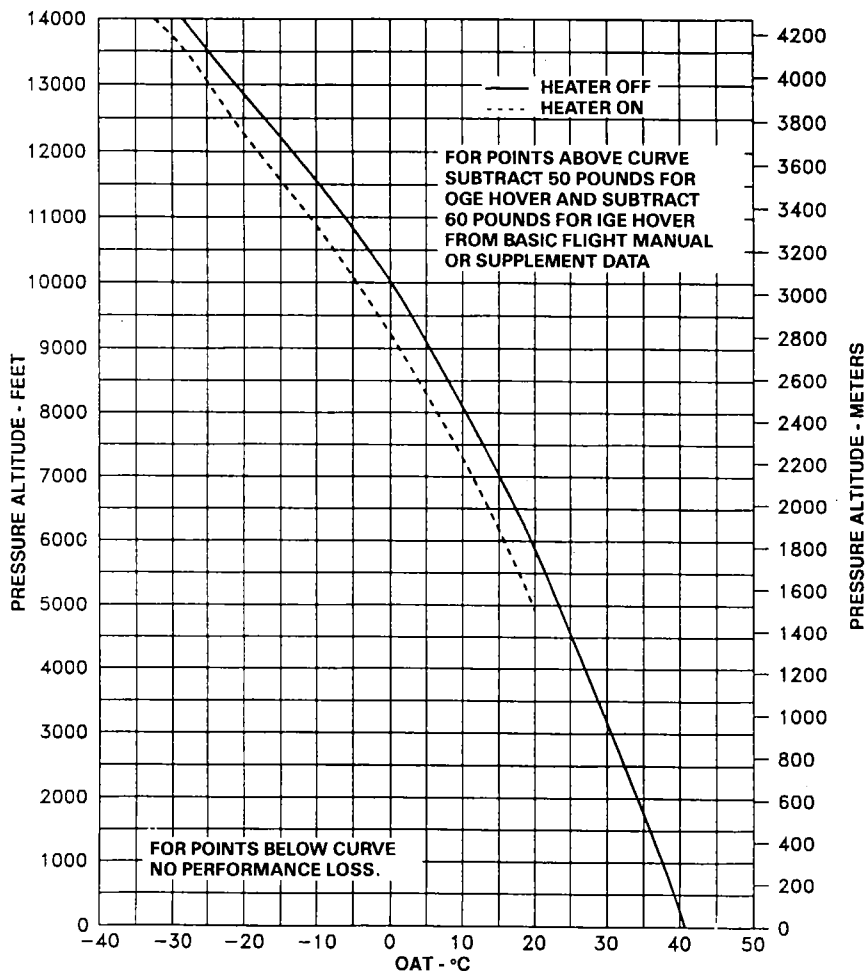
**HOVER PERFORMANCE VARIATION
FOR BHT-412-FM-2 AND BHT-412-FMS-19.1****MAXIMUM CONTINUOUS POWER
GENERATOR 200 AMPS (EACH)**

Figure 4B-2. Hover performance variation — maximum continuous power

**CLIMB PERFORMANCE VARIATION
FOR BHT-412-FM-2 AND BHT-412-FMS-19.1**

**TAKEOFF POWER
GENERATOR 200 AMPS (EACH)**

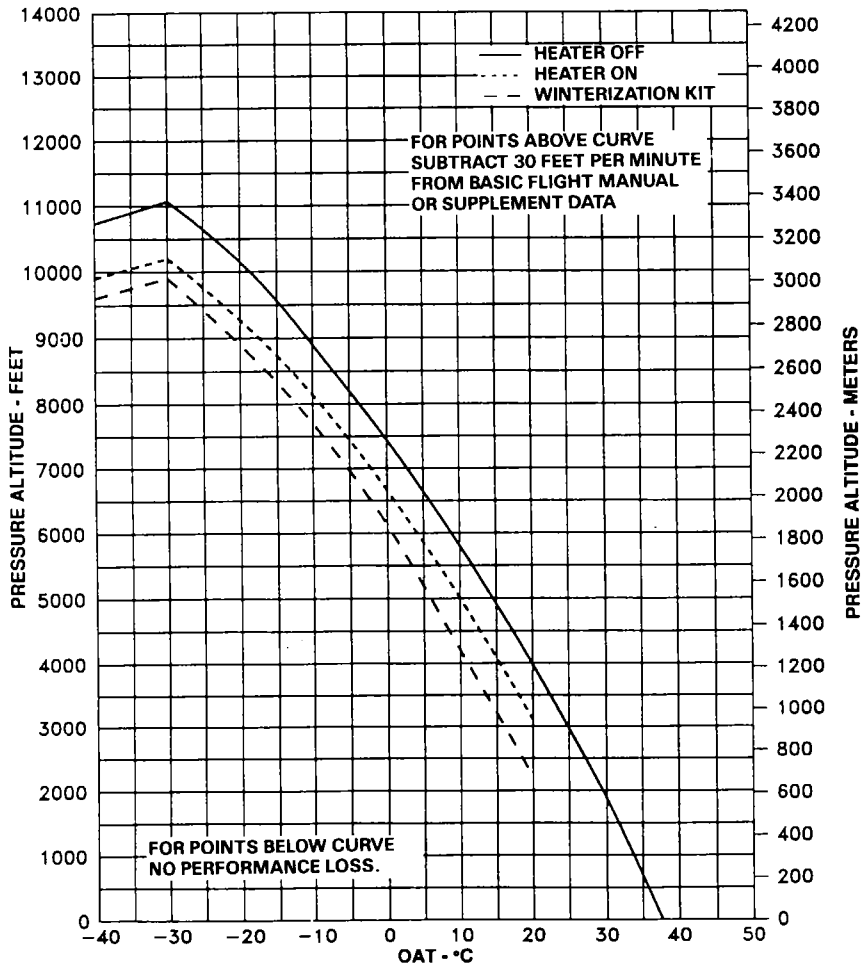


Figure 4B-3. Climb performance variation — takeoff power

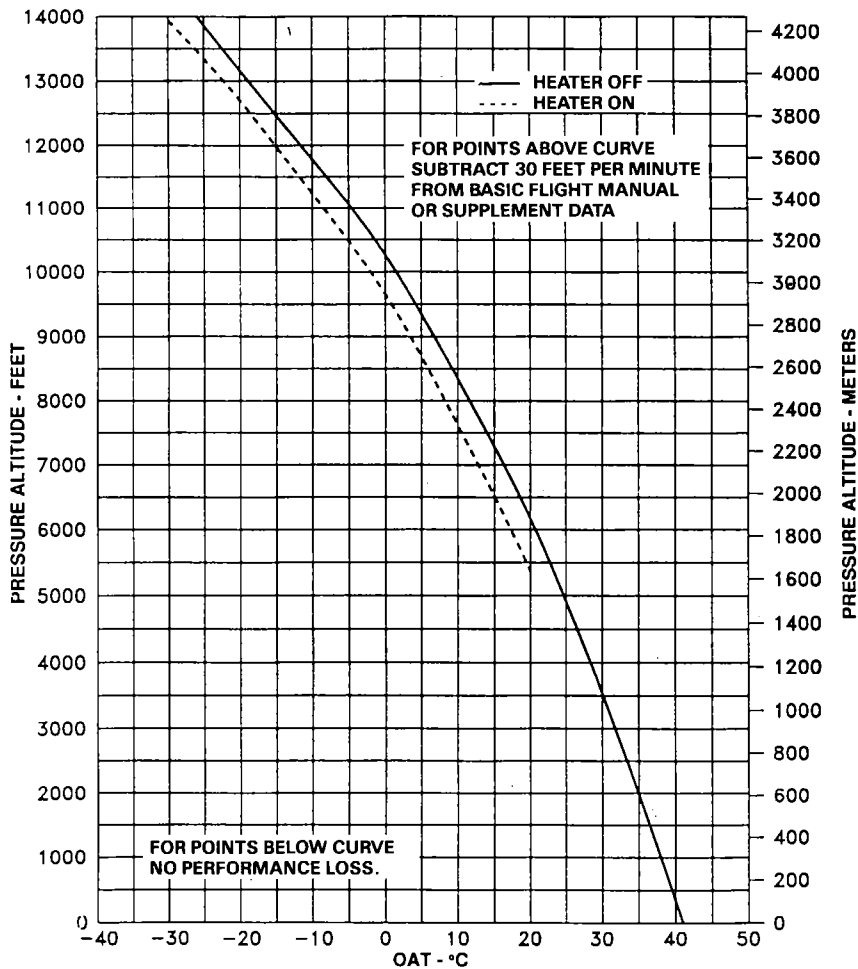
**CLIMB PERFORMANCE VARIATION
FOR BHT-412-FM-2 AND BHT-412-FMS-19.1****MAXIMUM CONTINUOUS POWER
GENERATOR 200 AMPS (EACH)**

Figure 4B-4. Climb performance variation — maximum continuous power

Section 4C

BHT-412-FM-3
BHT-412-FM-4
BHT-412-FMS-34.2
BHT-412-FMS-41

PERFORMANCE**TWIN ENGINE HOVER AND RATE OF CLIMB DECREASE DUE TO 200 AMP GENERATOR LOADS.**

Enter appropriate chart with pressure altitude and OAT to determine whether or not performance reduction is required. If applicable, decrease performance data in basic flight manual or appropriate optional equipment supplement as indicated on chart (rate of climb reduction of 30 feet per minute or hover gross weight reduction of 50 pounds out of ground effect or 60 pounds in ground effect).

HOVER PERFORMANCE VARIATION
FOR BHT-412-FM-3, BHT-412-FM-4, BHT-412-FMS-34.2, AND BHT-412-FMS-41

TAKEOFF POWER
GENERATOR 200 AMPS (EACH)

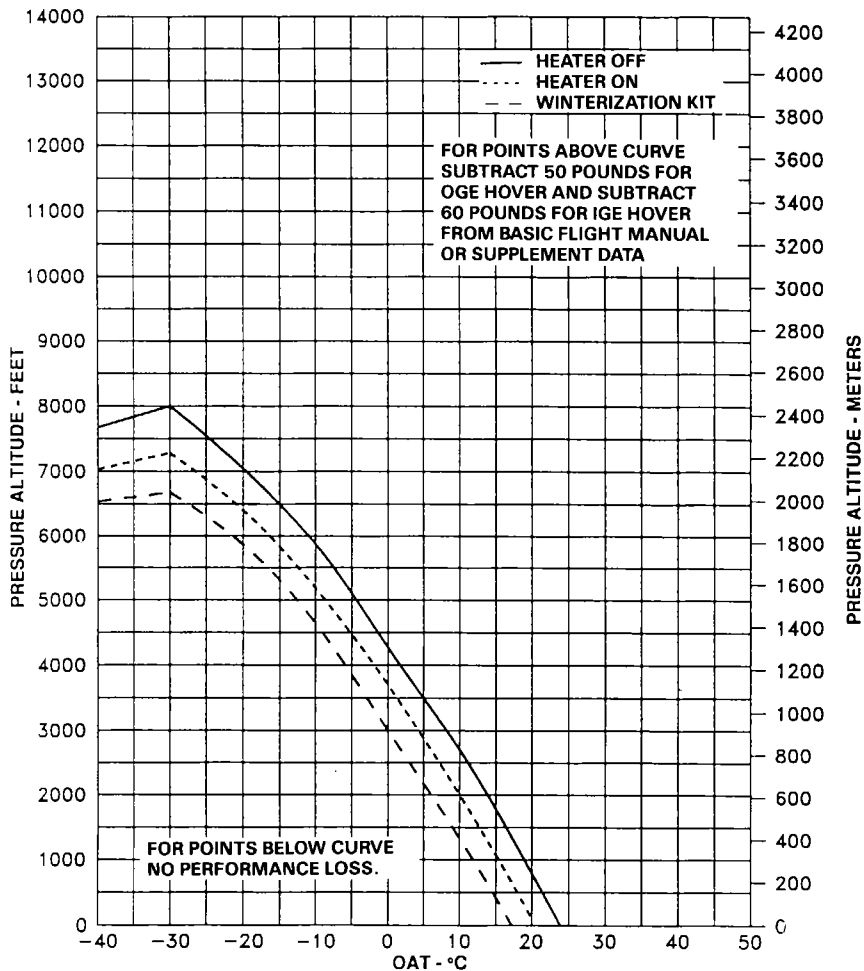


Figure 4C-1. Hover performance variation — takeoff power

HOVER PERFORMANCE VARIATION
FOR BHT-412-FM-3, BHT-412-FM-4, BHT-412-FMS-34.2, AND BHT-412-FMS-41

**MAXIMUM CONTINUOUS POWER
GENERATOR 200 AMPS (EACH)**

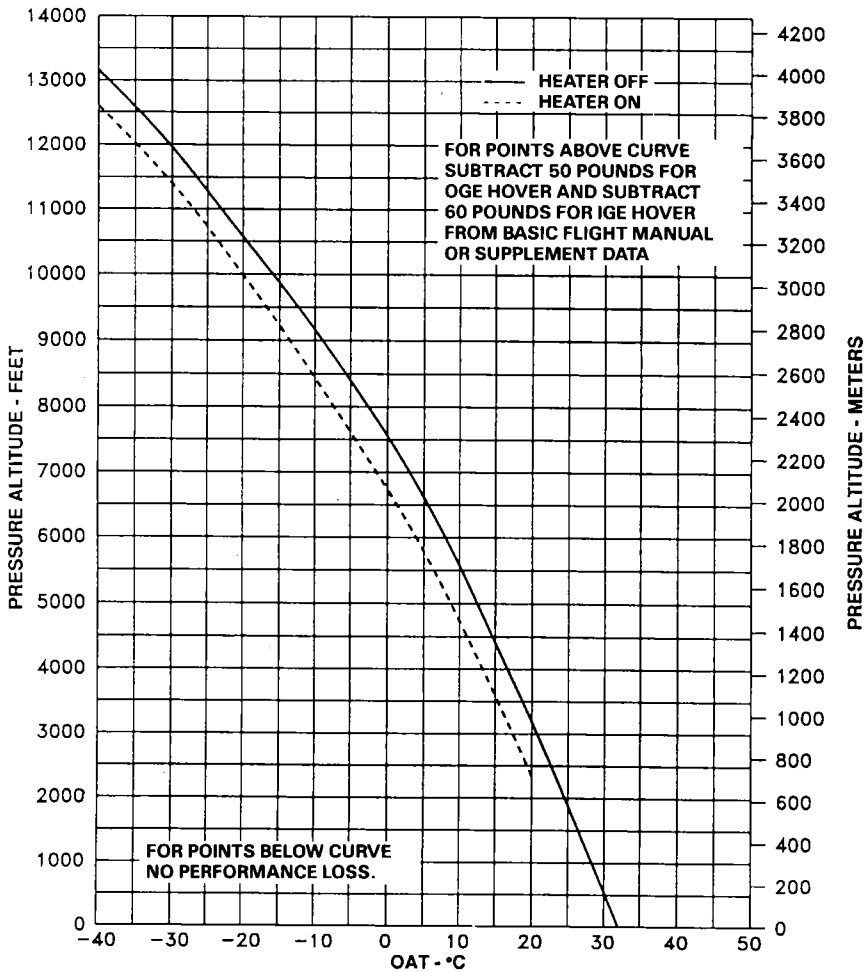


Figure 4C-2. Hover performance variation — maximum continuous power

CLIMB PERFORMANCE VARIATION
FOR BHT-412-FM-3, BHT-412-FM-4, BHT-412-FMS-34.2, AND BHT-412-FMS-41

TAKEOFF POWER
GENERATOR 200 AMPS (EACH)

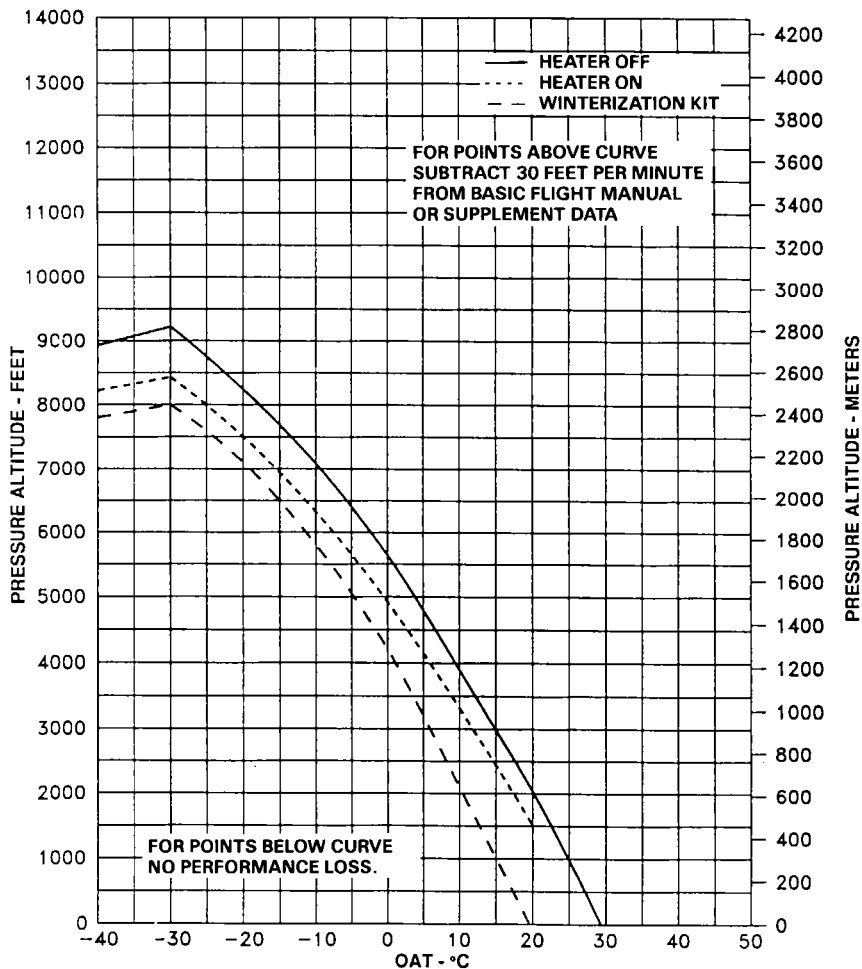


Figure 4C-3. Climb performance variation — takeoff power

CLIMB PERFORMANCE VARIATION
FOR BHT-412-FM-3, BHT-412-FM-4, BHT-412-FMS-34.2, AND BHT-412-FMS-41

MAXIMUM CONTINUOUS POWER
GENERATOR 200 AMPS (EACH)

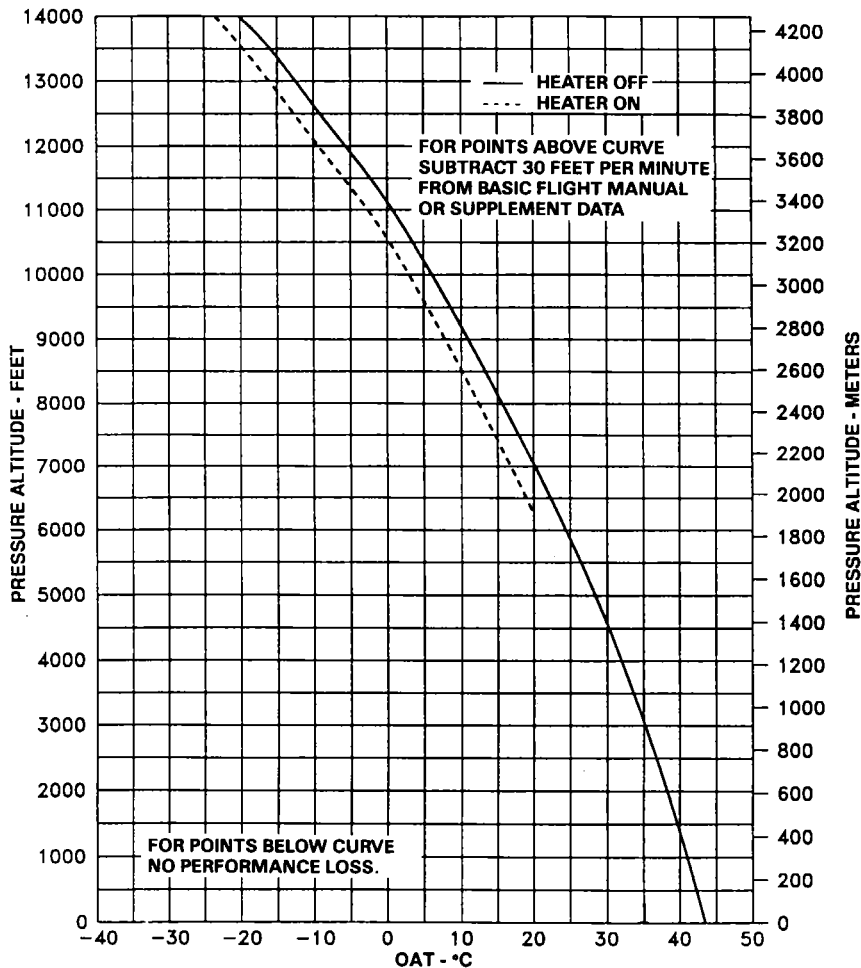


Figure 4C-4. Climb performance variation — maximum continuous power

Bell
MODEL **412**

ROTORCRAFT FLIGHT MANUAL

SUPPLEMENT

FOLDING STEP 412-899-287

CERTIFIED
25 OCTOBER 1993

This supplement shall be attached to the Model 412 Flight Manual when the 412-899-287 folding step has been installed.

Information contained herein supplements information of basic Flight Manual. For Limitations, Procedures, and Performance Data not contained in this supplement, consult basic Flight Manual.

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for MANAGER

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Section 1

LIMITATIONS

OPERATING LIMITATIONS

If necessary, to return empty weight CG within allowable limits.

The 412-706-004 Emergency float kit shall not be installed in conjunction with the folding step.

WEIGHT — CG LIMITATIONS

Actual weight change shall be determined after kit is installed and ballast readjusted,

Section 2

NORMAL PROCEDURES

NOTE

After passenger loading/
unloading, stow step (up).

6. FUSELAGE — CABIN RIGHT SIDE

Folding step — Stowed (up).

EXTERIOR CHECK

2. FUSELAGE — CABIN LEFT SIDE

Folding step — Stowed (up).

Section 3

EMERGENCY AND MALFUNCTION PROCEDURES

No change from basic Flight Manual.

Section 4

PERFORMANCE

No change from basic Flight Manual.



ROTORCRAFT FLIGHT MANUAL

SUPPLEMENT ENGINE NO. 2 GOVERNOR TRIM SWITCH

TB 412-93-118

33001 — 33213

36001 — 36019

CERTIFIED
28 JULY 1994

This supplement shall be attached to the Bell Helicopter Model 412 Flight Manual (BHT-412-FM-1 and BHT-412-FM-2) when engine #2 governor trim switch has been installed per TB 412-93-118.

Information contained herein supplements information of basic Flight Manual. For Limitations, Procedures, and Performance Data not contained in this supplement, consult basic Flight Manual.

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Section 1

LIMITATIONS

No change from basic manual.

Section 2

NORMAL PROCEDURES

BEFORE TAKEOFF

Throttles — Full open. Adjust frictions.

RPM INCR/DECR switch — Minimum beep (DECR for 4-5 seconds).

RPM INCR/DECR switch — Minimum trim (-2 for 4-5 seconds).

N_R — Check 95% or greater.

RPM INCR/DECR switch — Adjust to obtain matching torque at 100% N_R .

Flight instruments — Check operation and set.

TAKEOFF

Area — Clear.

NOTE

As collective is increased, it may be necessary to rematch engine torque prior to reaching hover.

RPM INCR/DECR switch — Adjust to obtain matching torque or ITT, as required, and 100% N_R .

Hover power — Check torque required to hover at four feet skid height.

Section 3

EMERGENCY PROCEDURES

No change from basic manual.

Section 4

MALFUNCTION PROCEDURES

No change from basic manual.

Section 5

OPTIONAL EQUIPMENT SUPPLEMENTS

No change from basic manual.

Section 6

CATEGORY A OPERATIONS

No change from basic manual.

Section 1

MANUFACTURER'S DATA

Weight and Balance

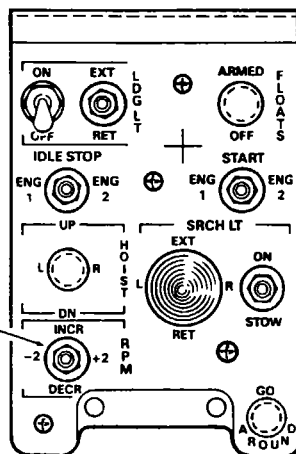
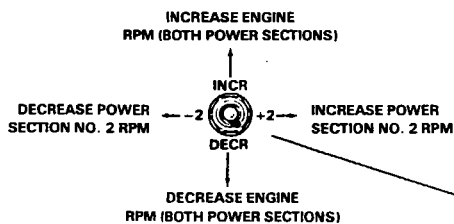
No change from basic manual.

Section 2

MANUFACTURER'S DATA

SYSTEM DESCRIPTION

The +2/-2 switch allows the pilot to match engine performance and improve total engine power available.



412-FMS-48.1 48.2-2-1

Figure 2-1. Engine RPM INCR/DECR switch.

Bell
MODELS **412/412EP**

ROTORCRAFT FLIGHT MANUAL

SUPPLEMENT SELF SEALING FUEL CELLS

412-899-175

S/N 33108 — 33213

AND

S/N 36001 — 36019

S/N 36020 — 36086

S/N 36087 AND SUB

CERTIFIED

19 SEPTEMBER 1997

This supplement shall be attached to Model 412 or 412EP Flight Manual when SELF SEALING FUEL CELLS are installed.

Information contained herein supplements information of basic Flight Manual. For Limitations, Procedures, and Performance Data not contained in this supplement, consult basic Flight Manual.

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NOTE

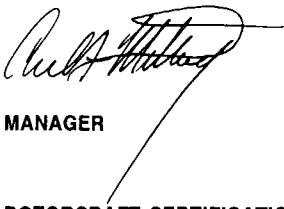
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GENERAL INFORMATION

The customizing-lower self sealing fuel cell installation minimizes the loss of fuel in the event of puncture damage by small objects.

Due to the increased wall thickness of the cell, the total and useable fuel capacities are less than those of the basic helicopter.

Section 1

LIMITATIONS

1-6. WEIGHT AND CENTER OF GRAVITY

necessary, to retain gross weight CG within allowable limits.

Actual weight change shall be determined after installation and ballast readjusted, if

Section 2

NORMAL PROCEDURES

No change from basic Flight Manual.

Section 3

EMERGENCY/MALFUNCTION PROCEDURES

No change from basic Flight Manual.

Section 4

PERFORMANCE

No change from basic Flight Manual.

Section 5

WEIGHT AND BALANCE

5-7. FUEL LOADING

Fuel loading tables lists usable fuel quantities in 10 gallon (40 liter) increments, with weights and moments in both english and metric units for balance computation. Critical fuel loading for computing most forward and aft CGs are denoted.

5-7-A. BASIC SYSTEM — SELF SEALING FUEL CELL.

Total capacity: 328.3 U.S. gallons (1242.2 liters).

Usable fuel: 321.3 U.S. gallons (1215.7 liters).

- Tables 5-1 and 5-2 - Provides longitudinal CG data for approved fuels.
- Tables 5-3 and 5-4 - Provides lateral CG data for approved fuels.

5-7-B. BASIC SYSTEM WITH ONE LONG RANGE AUXILIARY FUEL TANK (LEFT OR RIGHT).

Refer to BHT-412-FMS-17.2/17.3/17.4.

Total capacity: 410.0 U.S. gallons (1551.7 liters).

Usable fuel: 403.0 U.S. gallons (1525.7 liters).

- Table 5-5 - Provides longitudinal CG data for approved fuels.
- Table 5-6 - Provides lateral CG data for left side installation.

- Table 5-7 - Provides lateral CG data for right side installation.

5-7-C. BASIC SYSTEM WITH BOTH LONG RANGE AUXILIARY FUEL TANKS.

Refer to BHT-412-FMS-17.2/17.3/17.4.

Total capacity: 491.6 U.S. gallons (1860.9 liters).

Usable fuel: 484.6 U.S. gallons (1834.4 liters).

- Table 5-8 - Provides longitudinal CG data for approved fuels.
- Table 5-9 - Provides lateral CG data for approved fuels.

5-7-D. BASIC SYSTEM WITH ONE SEAT TYPE AUXILIARY FUEL TANK (LEFT OR RIGHT).

Refer to BHT-412-FMS-25.2/25.3/25.4.

Total capacity: 344.6 U.S. gallons (1304.2 liters).

Usable fuel: 337.6 U.S. gallons (1277.7 liters).

- Table 5-10 - Provides longitudinal CG data for approved fuels.
- Table 5-11 - Provides lateral CG data for left side installation.
- Table 5-12 - Provides lateral CG data for right side installation.

**5-7-E. BASIC SYSTEM WITH BOTH
SEAT TYPE AUXILIARY TANKS.**

Refer to BHT-412-FMS-25.2/25.3/25.4.

Total capacity: 360.9 U.S. gallons (1365.9
liters).

Usable fuel: 353.9 U.S. gallons (1339.4
liters).

- Table 5-13 - Provides longitudinal CG data for approved fuels.
- Table 5-14 - Provides lateral CG data for approved fuels.

Table 5-1. Usable fuel loading table self-sealing tanks (English)
Jet A, A-1, JP-5, JP-8 (6.8 Lbs/Gal)

U.S. GALLONS	WEIGHT (lb)	LONG CG (in.)	MOMENT (in-lb)
10	68	139.5	9486
20	136	139.7	18999
30	204	139.8	28519
40	272	139.9	38053
50	340	139.9	47566
* 54.6	371	139.9	51903
60	408	143.6	58589
70	476	148.0	70448
80	544	151.4	82362
90	612	154.1	94309
100	680	156.4	106352
110	748	158.2	118334
120	816	159.6	130234
130	884	160.8	142147
136.3	927	161.5	149711
140	952	159.6	151939
150	1020	155.0	158100
160	1088	151.0	164288
167.3	1138	148.8	169334
170	1156	149.2	172475
180	1224	150.7	184457
190	1292	152.0	196384
200	1360	153.2	208352
210	1428	154.4	220483
220	1496	155.3	232329
230	1564	156.0	243984
** 237.3	1614	156.7	252914
240	1632	156.2	254918
250	1700	154.8	263160
260	1768	153.0	270504
270	1836	151.6	278338
280	1904	150.2	285981
285.9	1944	149.5	290628
290	1972	149.8	295406
300	2040	150.7	307428
310	2108	151.5	319362
320	2176	152.3	331405
*** 321.3	2185	152.4	332994

* Most critical amount for most forward C.G. condition at weight empty is 6580 pounds or greater.

** Most critical fuel amount for most aft C.G. condition at weight empty is up to 6450 pounds.

*** Most critical fuel amount for most aft C.G. condition at weight empty is 6450 pounds or greater.
Weights given are nominal weights at 15 C.

NOTE

This table is invalid with auxiliary fuel tank(s) installed.

412-FMS-63-5-1-1

Table 5-1M. Usable fuel loading table self-sealing tanks (Metric)
Jet A, JP-5, JP-8 (0.815 kg/l)

LITERS	WEIGHT (kg)	LONG CG (mm)	MOMENT (kg-mm/100)
40	32.6	3551	1158
80	65.2	3551	2315
120	97.8	3551	3473
160	130.4	3553	4633
200	163.0	3553	5791
* 206.7	168.5	3553	5987
240	195.6	3691	7220
280	228.2	3795	8660
320	260.8	3876	10109
360	293.4	3945	11575
400	326.0	3998	13033
440	358.6	4041	14491
480	391.2	4074	15937
515.7	420.3	4102	17241
520	423.8	4087	17321
560	456.4	3962	18083
600	489.0	3851	18831
633.2	516.1	3780	19509
640	521.6	3787	19753
680	554.2	3825	21198
720	586.8	3863	22668
760	619.4	3894	24119
800	652.0	3922	25571
840	684.6	3950	27042
880	717.2	3973	28494
** 898.2	732.0	3980	29134
920	749.8	3955	29655
960	782.4	3912	30607
1000	815.0	3871	31549
1040	847.6	3835	32505
1080	880.2	3797	33421
1082.1	881.9	3797	33486
1120	912.8	3820	34869
1160	945.4	3840	36303
1200	978.0	3861	37761
*** 1215.7	990.8	3871	38354

* Most critical amount for most forward C.G. condition at weight empty is 2984 kilograms or greater.

** Most critical fuel amount for most aft C.G. condition weight empty is up to 2925 kilograms.

*** Most critical fuel amount for most aft C.G. condition at weight empty is 2925 kilograms or greater.
Weights given are nominal weights at 15 C.

NOTE

This table is invalid with auxiliary fuel tank(s) installed.

412-FMS-63-5-1-2

Table 5-2. Usable fuel loading table self-sealing tanks (English)
Jet B, JP-4 (6.5 Lbs/Gal)

U.S. GALLONS	WEIGHT (lb)	LONG CG (in.)	MOMENT (in-lb)
10	65	139.5	9068
20	130	139.7	18161
30	195	139.8	27261
40	260	139.9	36374
50	325	139.9	45468
54.6	355	139.9	49665
60	390	143.6	56004
70	455	148.0	67340
80	520	151.4	78728
90	585	154.1	90149
100	650	156.4	101660
110	715	158.2	113113
120	780	159.6	124488
130	845	160.8	135876
136.3	886	161.5	143089
140	910	159.6	145236
150	975	155.0	151125
160	1040	151.0	157040
167.3	1087	148.8	161746
170	1105	149.2	164866
180	1170	150.7	176319
190	1235	152.0	187720
200	1300	153.2	199160
210	1365	154.4	210756
220	1430	155.3	222079
230	1495	156.0	233220
237.3	1542	156.7	241631
240	1560	156.2	243672
250	1625	154.8	251550
260	1690	153.0	258570
270	1755	151.6	266058
280	1820	150.2	273364
285.9	1858	149.5	277771
290	1885	149.8	282373
300	1950	150.7	293865
310	2015	151.5	305273
320	2080	152.3	316784
321.3	2088	152.4	318211

* Most critical amount for most forward C.G. condition at weight empty is 6580 pounds or greater.

** Most critical fuel amount for most aft C.G. condition at weight empty is up to 6450 pounds.

*** Most critical fuel amount for most aft C.G. condition at weight empty is 6450 pounds or greater.

Weights given are nominal weights at 15 C.

NOTE

This table is invalid with auxiliary fuel tank(s) installed.

412-FMS-63-5-2-1

Table 5-2M. Usable fuel loading table self-sealing tanks (Metric)
Jet B, JP-4 (0.779 kg/l)

LITERS	WEIGHT (kg)	LONG CG (mm)	MOMENT (kg-mm/100)
40	31.2	3551	1108
80	62.3	3551	2212
120	93.5	3551	3320
160	124.6	3553	4427
200	155.8	3553	5536
* 206.7	161.0	3553	5720
240	187.0	3691	6902
280	218.1	3795	8277
320	249.3	3876	9663
360	280.4	3945	11062
400	311.6	3998	12458
440	342.8	4041	13853
480	373.9	4074	15233
515.7	401.7	4102	16478
520	405.1	4087	16556
560	436.2	3962	17282
600	467.4	3851	18000
633.2	493.3	3780	18647
640	498.6	3787	18882
680	529.7	3825	20261
720	560.9	3863	21668
760	592.0	3894	23052
800	623.2	3922	24442
840	654.4	3950	25849
880	685.5	3973	27235
** 898.2	699.7	3980	27848
920	716.7	3955	28345
960	747.8	3912	29254
1000	779.0	3871	30155
1040	810.2	3835	31071
1080	841.3	3797	31944
1082.1	843.0	3797	32009
1120	872.5	3820	33330
1160	903.6	3840	34698
1200	934.8	3861	36093
*** 1215.7	947.0	3871	36658

* Most critical amount for most forward C.G. condition at weight empty is 2984 kilograms or greater.

** Most critical fuel amount for most aft C.G. condition at weight empty is up to 2925 kilograms.

*** Most critical fuel amount for most aft C.G. condition at weight empty is 2925 kilograms or greater.
Weights given are nominal weights at 15 C.

NOTE

This table is invalid with auxiliary fuel tank(s) installed.

412-FMS-63-5-2-2

Table 5-3. Lateral usable fuel loading table self-sealing tanks (English)
Jet A, A-1, JP-5, JP-8 (6.8 Lbs/Gal)

U.S. GALLON	WEIGHT (lbs.)	LAT CG (in.)	MOMENT (in-lb)
10	68	0	0
20	136	0	0
30	204	0	0
40	272	0	0
50	340	0	0
54.6	371	0	0
60	408	-0.06	-24
70	476	-0.05	-24
80	544	-0.05	-27
90	612	-0.04	-24
100	680	-0.04	-27
110	748	-0.04	-30
120	816	-0.03	-24
130	884	-0.03	-27
136.3	927	-0.03	-28
140	952	-0.31	-295
150	1020	-0.50	-510
160	1088	-0.63	-685
167.3	1138	-0.70	-796
170	1156	-0.69	-798
180	1224	-0.65	-796
190	1292	-0.61	-788
200	1360	-0.58	-789
210	1428	-0.55	-785
220	1496	-0.53	-793
230	1564	-0.51	-798
240	1632	-0.49	-800
250	1700	-0.47	-799
260	1768	-0.45	-796
270	1836	-0.43	-789
280	1904	-0.41	-781
290	1972	-0.40	-789
300	2040	-0.39	-796
310	2108	-0.37	-780
320	2176	-0.36	-783
321.3	2185	-0.36	-787

* Most critical fuel amount for left side most lateral C.G. condition.
 Weights given are nominal weights at 15 C.

NOTE

This table is invalid with auxiliary fuel tank(s) installed.

412-FMS-63-5-3-1

Table 5-3M. Lateral usable fuel loading table self-sealing tanks (Metric)
Jet A, A-1, JP-5, JP-8 (0.815 kg/l)

LITERS	WEIGHT (kg)	LAT CG (mm)	MOMENT (kg-mm)
40	32.6	0	0
80	65.2	0	0
120	97.8	0	0
160	130.4	0	0
200	163.0	0	0
206.6	168.4	0	0
240	195.6	-2	-391
280	228.2	-1	-228
320	260.8	-1	-261
360	293.4	-1	-293
400	326.0	-1	-326
440	358.6	-1	-359
480	391.2	-1	-391
515.6	420.2	-1	-420
520	423.8	-6	-2543
560	456.4	-12	-5477
600	489.0	-15	-7335
633.0	515.9	-18	-9286
640	521.6	-18	-9389
680	554.2	-17	-9421
720	586.8	-15	-8802
760	619.4	-15	-9291
800	652.0	-14	-9128
840	684.6	-13	-8900
880	717.2	-13	-9324
920	749.8	-12	-8998
960	782.4	-12	-9389
1000	815.0	-11	-8965
1040	847.6	-11	-9324
1080	880.2	-10	-8802
1120	912.8	-10	-9128
1160	945.4	-10	-9454
1200	978.0	-9	-8802
1215.7	990.8	-9	-8917

* Most critical fuel amount for left side most lateral C.G. condition.
 Weights given are nominal weights at 15 C.

NOTE

This table is invalid with auxiliary fuel tank(s) installed.

412-FMS-63-5-3-2

Table 5-4. Lateral usable fuel loading table self-sealing tanks (English)
Jet B, JP-4 (6.5 Lbs/Gal)

U.S. GALLON	WEIGHT (lbs.)	LAT CG (in.)	MOMENT (in-lb)
10	65	0	0
20	130	0	0
30	195	0	0
40	260	0	0
50	325	0	0
54.6	355	0	0
60	390	-0.06	-23
70	455	-0.05	-23
80	520	-0.05	-26
90	585	-0.04	-23
100	650	-0.04	-26
110	715	-0.04	-29
120	780	-0.03	-23
130	845	-0.03	-25
136.3	886	-0.03	-27
140	910	-0.31	-282
150	975	-0.50	-488
160	1040	-0.63	-655
167.3	1087	-0.70	-761
170	1105	-0.69	-762
180	1170	-0.65	-761
190	1235	-0.61	-753
200	1300	-0.58	-754
210	1365	-0.55	-751
220	1430	-0.53	-758
230	1495	-0.51	-762
240	1560	-0.49	-764
250	1625	-0.47	-764
260	1690	-0.45	-761
270	1755	-0.43	-755
280	1820	-0.41	-746
290	1885	-0.40	-754
300	1950	-0.39	-761
310	2015	-0.37	-746
320	2080	-0.36	-749
321.3	2088	-0.36	-752

* Most critical fuel amount for left side most lateral C.G. condition.
 Weights given are nominal weights at 15 C.

NOTE

This table is invalid with auxiliary fuel tank(s) installed.

412-FMS-63-5-4-1

Table 5-4M. Lateral usable fuel loading table self-sealing tanks (Metric)
Jet B, JP-4 (0.779 kg/l)

LITERS	WEIGHT (kg)	LAT CG (mm)	MOMENT (kg-mm)
40	31.2	0	0
80	62.3	0	0
120	93.5	0	0
160	124.6	0	0
200	155.8	0	0
206.6	160.9	0	0
240	187.0	-2	-374
280	218.1	-1	-218
320	249.3	-1	-249
360	280.4	-1	-280
400	311.6	-1	-312
440	342.8	-1	-343
480	373.9	-1	-374
515.6	401.7	-1	-402
520	405.1	-6	-2430
560	436.2	-12	-5235
600	467.4	-15	-7011
633.0	493.1	-18	-8876 *
640	498.6	-18	-8974
680	529.7	-17	-9005
720	560.9	-15	-8413
760	592.0	-15	-8881
800	623.2	-14	-8725
840	654.4	-13	-8507
880	685.5	-13	-8912
920	716.7	-12	-8600
960	747.8	-12	-8974
1000	779.0	-11	-8569
1040	810.2	-11	-8912
1080	841.3	-10	-8413
1120	872.5	-10	-8725
1160	903.6	-10	-9036
1200	934.8	-9	-8413
1215.7	947.0	-9	-8523

* Most critical fuel amount for left side most lateral C.G. condition.
Weights given are nominal weights at 15 C.

NOTE

This table is invalid with auxiliary fuel tank(s) installed.

412-FMS-63-5-4-2

Table 5-5. USABLE FUEL LOADING TABLE WITH SELF-SEALING TANKS AND 81.7 GAL AUX FUEL
(English)

Jet A, A-1, JP-5, JP-8 (6.8 Lbs/Gal)

U.S. GALLON	WEIGHT (lbs.)	LONG CG (in.)	MOMENT (in-lb)
10	68	139.5	9486
20	136	139.7	18999
30	204	139.8	28519
40	272	139.9	38053
50	340	139.9	47566
* 54.6	371.3	139.9	51942
60	408	143.6	58589
70	476	147.4	70162
80	544	149.8	81491
90	612	151.8	92902
100	680	153.3	104244
110	748	154.5	115566
120	816	155.6	126970
130	884	156.4	138258
140	952	157.2	149654
150	1020	157.9	161058
160	1088	158.5	172448
170	1156	159.0	183804
170.2	1157.4	159.0	184020
180	1224	155.4	190210
190	1292	152.0	196384
200	1360	149.2	202912
201.2	1368.2	148.9	203719
210	1428	149.6	213629
220	1496	150.4	224998
230	1564	151.1	236320
240	1632	151.8	247738
250	1700	152.5	259250
260	1768	153.0	270504
270	1836	153.6	282010
280	1904	154.0	293216
290	1972	154.5	304674
300	2040	154.9	315996
** 303.0	2060.4	155.0	319362
310	2108	154.1	324843
320	2176	152.9	332710
330	2244	151.7	340415
340	2312	150.6	348187
350	2380	149.5	355810
351.6	2390.9	149.4	357197
360	2448	149.8	366710
370	2516	150.3	376155
380	2584	150.8	386667
390	2652	151.2	400982
400	2720	151.6	412352
*** 403.0	2740.4	151.7	415719

Jet B, JP-4 (6.5 Lbs/Gal)

U.S. GALLON	WEIGHT (lbs.)	LONG CG (in.)	MOMENT (in-lb)
10	65	139.5	9068
20	130	139.7	18161
30	195	139.8	27261
40	260	139.9	36374
50	325	139.9	45468
* 54.6	354.9	139.9	49651
60	390	143.6	56004
70	455	147.4	67067
80	520	149.8	77896
90	585	151.8	88803
100	650	153.3	99645
110	715	154.5	110468
120	780	155.6	121368
130	845	156.4	132158
140	910	157.2	143052
150	975	157.9	153953
160	1040	158.5	164840
170	1105	159.0	175695
170.2	1106.3	159.0	175902
180	1170	155.4	181818
190	1235	152.0	187720
200	1300	149.2	193960
201.2	1307.8	148.9	194731
210	1365	149.6	204204
220	1430	150.4	215072
230	1495	151.1	225895
240	1560	151.8	236808
250	1625	152.5	247813
260	1690	153.0	258570
270	1755	153.6	269568
280	1820	154.0	280280
290	1885	154.5	291233
300	1950	154.9	302055
** 303.0	1969.5	155.0	305273
310	2015	154.1	310512
320	2080	152.9	318032
330	2145	151.7	325397
340	2210	150.6	332826
350	2275	149.5	340113
351.6	2285.4	149.4	341439
360	2340	149.8	350532
370	2405	150.3	361472
380	2470	150.8	372476
390	2535	151.2	383292
400	2600	151.6	394160
*** 403.0	2619.5	151.7	397378

Most critical amount for most forward C.G. condition at a weight empty below 6750 pounds has no fuel.

* Most critical amount for most forward C.G. condition at a weight empty of 6750 pounds or greater.

** Most critical fuel amount for most aft C.G. condition at a weight empty below 6100 pounds.

*** Most critical fuel amount for most aft C.G. condition at a weight empty of 6100 pounds or greater.

Weights given are nominal weights at 15 C.

Table 5-5M. USABLE FUEL LOADING TABLE WITH SELF-SEALING TANKS AND 309 LITERS AUX FUEL
(Metric)

Jet A, A-1, JP-5, JP-8 (.815kg/l)

Jet B, JP-4 (.779 kg/l)

LITERS	WEIGHT (kg)	LONG CG (mm)	MOMENT (kg-mm)
40	32.6	3543	115502
80	65.2	3548	231330
120	97.8	3551	347288
160	130.4	3553	463311
200	163.0	3553	579139
* 206.7	168.5	3553	598540
240	195.6	3691	721960
280	228.2	3772	860770
320	260.8	3830	998864
360	293.4	3876	1137218
400	326.0	3912	1275312
440	358.6	3942	1413601
480	391.2	3967	1551890
520	423.8	3988	1690114
560	456.4	4008	1829251
600	489.0	4023	1967247
640	521.6	4036	2105178
** 644.2	525.0	4038	2120043
680	554.2	3950	2189090
720	586.8	3861	2265635
760	619.4	3785	2344429
761.7	620.8	3782	2347811
800	652.0	3800	2477600
840	684.8	3823	2617226
880	717.2	3843	2756200
920	749.8	3861	2894978
960	782.4	3879	3034930
1000	815.0	3875	3158125
1040	847.6	3906	3310726
1080	880.2	3919	3449504
1120	912.8	3931	3588217
1146.7	934.6	3938	3680299
1160	945.4	3926	3711640
1200	978.0	3879	3793662
1240	1010.6	3872	3913043
1280	1043.2	3830	3995456
1320	1075.8	3801	4089116
1330.6	1084.4	3794	4114362
1360	1108.4	3805	4217462
1400	1141.0	3818	4356338
1440	1173.6	3830	4494888
1480	1206.2	3840	4631808
1520	1238.8	3851	4770619
*** 1525.2	1243.0	3852	4788182

LITERS	WEIGHT (kg)	LONG CG (mm)	MOMENT (kg-mm)
40	31.2	3543	110400
80	62.3	3548	221111
120	93.5	3551	331947
160	124.6	3553	442846
200	155.8	3553	553557
* 206.7	161.0	3553	572102
240	187.0	3691	690069
280	218.1	3772	822749
320	249.3	3830	954742
360	280.4	3876	1086985
400	311.6	3912	1218979
440	342.8	3942	1351160
480	373.9	3967	1483341
520	405.1	3988	1615459
560	436.2	4008	1748450
600	467.4	4023	1880350
640	498.6	4036	2012188
** 644.2	501.8	4038	2026397
680	529.7	3950	2092394
720	560.9	3861	2165558
760	592.0	3785	2240871
761.7	593.4	3782	2244104
800	623.2	3800	2368160
840	654.4	3823	2501618
880	685.5	3843	2634453
920	716.7	3861	2767101
960	747.8	3879	2900871
1000	779.0	3875	3018625
1040	810.2	3906	3164485
1080	841.3	3919	3297133
1120	872.5	3931	3429719
1146.7	893.3	3938	3517734
1160	903.6	3926	3547691
1200	934.8	3879	3626089
1240	966.0	3872	3740197
1280	997.1	3830	3818970
1320	1028.3	3801	3908492
1330.6	1036.5	3794	3932823
1360	1059.4	3805	4031169
1400	1090.6	3818	4163911
1440	1121.8	3830	4296341
1480	1152.9	3840	4427213
1520	1184.1	3851	4559892
*** 1525.2	1188.1	3852	4576680

Most critical amount for most forward C.G. condition at a weight empty below 3062 kilograms has no f

* Most critical amount for most forward C.G. condition at a weight empty of 3062 kilograms or greater.

** Most critical fuel amount for most aft C.G. condition at a weight empty below 2767 kilograms.

*** Most critical fuel amount for most aft C.G. condition at a weight empty of 2767 kilograms or greater.

Weights given are nominal weights at 15 C.

412-FMS-63-5-5-2

Table 5-6. USABLE FUEL LOADING TABLE WITH SELF-SEALING TANKS AND 81.7 GAL AUX FUEL (LH)
(ENGLISH)

Jet A, A-1, JP-5, JP-8 (6.8 Lbs/Gal)

U.S. GALLON	WEIGHT (lbs.)	LAT CG (in.)	MOMENT (in-lb)
10	68	0	0
20	136	0	0
30	204	0	0
40	272	0	0
50	340	0	0
54.6	371.3	0	0
60	408	-0.13	-53
70	476	-0.72	-343
80	544	-1.78	-968
90	612	-2.63	-1610
100	680	-3.30	-2244
110	748	-3.84	-2872
120	816	-4.29	-3501
130	884	-4.67	-4128
140	952	-5.00	-4760
150	1020	-5.27	-5375
160	1088	-5.50	-5984
170	1156	-5.71	-6601
170.2	1157.4	-5.71	-6609
180	1224	-5.72	-7001
190	1292	-5.56	-7184
200	1360	-5.41	-7358
201.2	1368.2	-5.39	-7374
210	1428	-5.54	-7911
220	1496	-5.69	-8512
230	1564	-5.84	-9134
240	1632	-5.97	-9743
250	1700	-6.08	-10336
260	1768	-6.19	-10944
270	1836	-6.29	-11548
280	1904	-6.39	-12167
290	1972	-6.48	-12779
300	2040	-6.56	-13382
303.0	2060.4	-6.58	-13557
310	2108	-6.43	-13554
320	2176	-6.23	-13556
330	2244	-6.04	-13554
340	2312	-5.86	-13548
350	2380	-5.70	-13566
351.6	2390.9	-5.67	-13556
360	2448	-5.75	-14076
370	2516	-5.84	-14693
380	2584	-5.92	-15297
390	2652	-6.00	-15912
400	2720	-6.07	-16510
* 403.0	2740.4	-6.09	-16689

Jet B, JP-4 (6.5 Lbs/Gal)

U.S. GALLON	WEIGHT (lbs.)	LAT CG (in.)	MOMENT (in-lb)
10	65	0	0
20	130	0	0
30	195	0	0
40	260	0	0
50	325	0	0
54.6	354.9	0	0
60	390	-0.13	-51
70	455	-0.72	-328
80	520	-1.78	-926
90	585	-2.63	-1539
100	650	-3.30	-2145
110	715	-3.84	-2746
120	780	-4.29	-3346
130	845	-4.67	-3946
140	910	-5.00	-4550
150	975	-5.27	-5138
160	1040	-5.50	-5720
170	1105	-5.71	-6310
170.2	1106.3	-5.71	-6317
180	1170	-5.72	-6692
190	1235	-5.56	-6867
200	1300	-5.41	-7033
201.2	1307.8	-5.39	-7049
210	1365	-5.54	-7562
220	1430	-5.69	-8137
230	1495	-5.84	-8731
240	1560	-5.97	-9313
250	1625	-6.08	-9880
260	1690	-6.19	-10461
270	1755	-6.29	-11039
280	1820	-6.39	-11630
290	1885	-6.48	-12215
300	1950	-6.56	-12792
303.0	1969.5	-6.58	-12959
310	2015	-6.43	-12956
320	2080	-6.23	-12958
330	2145	-6.04	-12956
340	2210	-5.86	-12951
350	2275	-5.70	-12968
351.6	2285.4	-5.67	-12958
360	2340	-5.75	-13455
370	2405	-5.84	-14045
380	2470	-5.92	-14622
390	2535	-6.00	-15210
400	2600	-6.07	-15782
* 403.0	2619.5	-6.09	-15953

* Most critical fuel amount for left side most lateral C.G. condition.

Weights given are nominal weights at 15 C.

412-FMS-63-5-6-1

Table 5-6M. USABLE FUEL LOADING TABLE WITH SELF-SEALING TANKS AND 309 LITERS AUX FUEL (LH)
(Metric)

Jet A, A-1, JP-5, JP-8 (.815kg/l)

	WEIGHT	LAT CG	MOMENT
LITERS	(kg)	(mm)	(kg-mm)

40	32.6	0	0
80	65.2	0	0
120	97.8	0	0
160	130.4	0	0
200	163.0	0	0
206.7	168.5	0	0
240	195.6	-6	-1174
280	228.2	-28	-6390
320	260.8	-56	-14605
360	293.4	-76	-22298
400	326.0	-92	-29992
440	358.6	-105	-37653
480	391.2	-116	-45379
520	423.8	-126	-53399
560	456.4	-132	-60245
600	489.0	-139	-67971
640	521.6	-145	-75632
644.2	525.0	-145	-76128
680	554.2	-145	-80359
720	586.8	-141	-82739
760	619.4	-137	-84858
761.7	620.8	-137	-85048
800	652.0	-141	-91932
840	684.6	-145	-99267
880	717.2	-149	-106863
920	749.8	-152	-113970
960	782.4	-156	-122054
1000	815.0	-158	-128770
1040	847.6	-161	-136464
1080	880.2	-164	-144353
1120	912.8	-166	-151525
1146.7	934.6	-167	-156072
1160	945.4	-165	-155991
1200	978.0	-160	-156480
1240	1010.6	-155	-156643
1280	1043.2	-150	-156480
1320	1075.8	-145	-155991
1330.6	1084.4	-144	-156159
1360	1108.4	-146	-161826
1400	1141.0	-148	-168868
1440	1173.6	-150	-176040
1480	1206.2	-152	-183342
1520	1238.8	-154	-190775
* 1525.2	1243.0	-155	-192671

Jet B, JP-4 (.779 kg/l)

	WEIGHT	LAT CG	MOMENT
LITERS	(kg)	(mm)	(kg-mm)

40	31.2	0	0
80	62.3	0	0
120	93.5	0	0
160	124.6	0	0
200	155.8	0	0
206.7	161.0	0	0
240	187.0	-5	-935
280	218.1	-25	-5453
320	249.3	-55	-13710
360	280.4	-75	-21033
400	311.6	-91	-28356
440	342.8	-105	-35990
480	373.9	-116	-43375
520	405.1	-126	-51040
560	436.2	-133	-58020
600	467.4	-140	-65436
640	498.6	-147	-73288
644.2	501.8	-145	-72766
680	529.7	-145	-76809
720	560.9	-141	-79084
760	592.0	-137	-81109
761.7	593.4	-137	-81291
800	623.2	-142	-88494
840	654.4	-146	-95537
880	685.5	-150	-102828
920	716.7	-152	-108935
960	747.8	-155	-115915
1000	779.0	-158	-123082
1040	810.2	-161	-130436
1080	841.3	-164	-137976
1120	872.5	-166	-144832
1146.7	893.3	-167	-149178
1160	903.6	-165	-149101
1200	934.8	-158	-147698
1240	966.0	-156	-150690
1280	997.1	-150	-149568
1320	1028.3	-145	-149101
1330.6	1036.5	-144	-149261
1360	1059.4	-146	-154678
1400	1090.6	-149	-162499
1440	1121.8	-151	-169386
1480	1152.9	-153	-176397
1520	1184.1	-154	-182348
* 1525.2	1188.1	-155	-184160

* Most critical fuel amount for left side most lateral C.G. condition.
Weights given are nominal weights at 15 C.

412-FMS-63-5-6-2

Table 5-7. USABLE FUEL LOADING TABLE WITH SELF-SEALING TANKS AND 81.7 GAL AUX FUEL (RH)
(English)

Jet A, A-1, JP-5, JP-8 (6.8 Lbs/Gal)

U.S. GALLON	WEIGHT (lbs.)	LAT CG (in.)	MOMENT (in-lb)
10	68	0	0
20	136	0	0
30	204	0	0
40	272	0	0
50	340	0	0
54.6	371.3	0	0
60	408	0.01	4
70	476	0.62	295
80	544	1.68	914
90	612	2.55	1561
100	680	3.23	2196
110	748	3.77	2820
120	816	4.23	3452
130	884	4.61	4075
140	952	4.95	4712
150	1020	5.22	5324
160	1088	5.46	5940
170	1156	5.66	6543
170.2	1157.4	5.66	6551
180	1224	5.03	6157
190	1292	4.63	5982
200	1360	4.42	6011
201.2	1368.2	4.23	5787
210	1428	4.42	6312
220	1496	4.63	6926
230	1564	4.83	7554
240	1632	4.99	8144
250	1700	5.14	8738
260	1768	5.30	9370
270	1836	5.43	9969
280	1904	5.56	10586
290	1972	5.68	11201
300	2040	5.79	11812
303.0	2060.4	5.82	11992
310	2108	5.69	11995
320	2176	5.51	11990
330	2244	5.34	11983
340	2312	5.19	11999
350	2380	5.04	11995
351.6	2390.9	5.01	11978
360	2448	5.11	12509
370	2516	5.21	13108
380	2584	5.31	13721
390	2652	5.40	14321
400	2720	5.48	14906
* 403.0	2740.4	5.51	15100

Jet B, JP-4 (6.5 Lbs/Gal)

U.S. GALLON	WEIGHT (lbs.)	LAT CG (in.)	MOMENT (in-lb)
10	65	0	0
20	130	0	0
30	195	0	0
40	260	0	0
50	325	0	0
54.6	354.9	0	0
60	390	0.01	4
70	455	0.62	282
80	520	1.68	874
90	585	2.55	1492
100	650	3.23	2100
110	715	3.77	2696
120	780	4.23	3299
130	845	4.61	3895
140	910	4.95	4505
150	975	5.22	5090
160	1040	5.46	5678
170	1105	5.66	6254
170.2	1106.3	5.66	6262
180	1170	5.03	5885
190	1235	4.63	5718
200	1300	4.42	5746
201.2	1307.8	4.23	5532
210	1365	4.42	6033
220	1430	4.63	6621
230	1495	4.83	7221
240	1560	4.99	7784
250	1625	5.14	8353
260	1690	5.30	8957
270	1755	5.43	9530
280	1820	5.56	10119
290	1885	5.68	10707
300	1950	5.79	11291
303.0	1969.5	5.82	11462
310	2015	5.69	11465
320	2080	5.51	11461
330	2145	5.34	11454
340	2210	5.19	11470
350	2275	5.04	11466
351.6	2285.4	5.01	11450
360	2340	5.11	11957
370	2405	5.21	12530
380	2470	5.31	13116
390	2535	5.40	13689
400	2600	5.48	14248
* 403.0	2619.5	5.51	14433

* Most critical fuel amount for right side most lateral C.G. condition.
Weights given are nominal weights at 15 C.

412-FMS-63-5-7-1

Table 5-7M. USABLE FUEL LOADING TABLE WITH SELF-SEALING TANKS AND 309 LITERS AUX FUEL (RH)
(Metric)

Jet A, A-1, JP-5, JP-8 (.815kg/l)

LITERS	WEIGHT (kg)	LAT CG (mm)	MOMENT (kg-mm)
40	32.6	0	0
80	65.2	0	0
120	97.8	0	0
160	130.4	0	0
200	163.0	0	0
206.7	168.5	0	0
240	195.6	3	497
280	228.2	25	5705
320	260.8	53	13822
360	293.4	74	21712
400	326.0	90	29340
440	358.6	103	36936
480	391.2	114	44597
520	423.8	123	52127
560	456.4	131	59788
600	489.0	138	67482
640	521.6	143	74589
644.2	525.0	144	75603
680	554.2	128	70938
720	586.8	117	68656
760	619.4	108	66895
761.7	620.8	107	66424
800	652.0	113	73676
840	684.6	119	81467
880	717.2	124	88933
920	749.8	128	95974
960	782.4	132	103277
1000	815.0	136	110840
1040	847.6	140	118664
1080	880.2	143	125869
1120	912.8	146	133269
1146.7	934.6	148	138315
1160	945.4	146	138028
1200	978.0	139	135942
1240	1010.6	141	142495
1280	1043.2	137	142918
1320	1075.8	128	137702
1330.6	1084.4	127	137724
1360	1108.4	129	142984
1400	1141.0	132	150612
1440	1173.6	135	158436
1480	1206.2	137	165249
1520	1238.8	140	173432
* 1525.2	1243.0	140	174025

Jet B, JP-4 (.779 kg/l)

LITERS	WEIGHT (kg)	LAT CG (mm)	MOMENT (kg-mm)
40	31.2	0	0
80	62.3	0	0
120	93.5	0	0
160	124.6	0	0
200	155.8	0	0
206.7	161.0	0	0
240	187.0	3	475
280	218.1	25	5453
320	249.3	53	13212
360	280.4	74	20753
400	311.6	90	28044
440	342.8	103	35304
480	373.9	114	42627
520	405.1	123	49825
560	436.2	131	57147
600	467.4	138	64501
640	498.6	143	71294
644.2	501.8	144	72264
680	529.7	128	67804
720	560.9	117	65623
760	592.0	108	63940
761.7	593.4	107	63490
800	623.2	113	70422
840	654.4	119	77869
880	685.5	124	85004
920	716.7	128	91735
960	747.8	132	98715
1000	779.0	136	105944
1040	810.2	140	113422
1080	841.3	143	120309
1120	872.5	146	127382
1146.7	893.3	148	132205
1160	903.6	146	131931
1200	934.8	139	129937
1240	966.0	141	136200
1280	997.1	137	136605
1320	1028.3	128	131620
1330.6	1036.5	127	131640
1360	1059.4	129	136668
1400	1090.6	132	143959
1440	1121.8	135	151438
1480	1152.9	137	157950
1520	1184.1	140	165771
* 1525.2	1188.1	140	166338

* Most critical fuel amount for right side most lateral C.G. condition.
Weights given are nominal weights at 15 C.

Table 5-8. USABLE FUEL LOADING TABLE WITH SELF-SEALING TANKS AND 163.4 GAL AUX FUEL
(ENGLISH)

Jet A, A-1, JP-5, JP-8 (6.8 Lbs/Gal)

U.S. GALLONS	WEIGHT (lbs.)	LONG CG (in.)	MOMENT (in-lb)
10	68	139.5	9486
20	136	139.7	18999
30	204	139.8	28519
40	272	139.9	38053
50	340	139.9	47566
54.6	371.3	139.9	51942
60	408	143.6	58589
70	476	147.3	70115
80	544	149.2	81165
90	612	150.7	92228
100	680	151.8	103224
110	748	152.8	114294
120	816	153.6	125338
130	884	154.3	136401
140	952	154.8	147370
150	1020	155.4	158508
160	1088	155.8	169510
170	1156	156.2	180567
180	1224	156.6	191678
190	1292	157.0	202844
200	1360	157.2	213792
204.1	1387.9	157.3	218314
210	1428	155.5	222054
220	1496	152.7	228439
230	1564	150.1	234756
235.2	1599.4	148.9	238145
240	1632	149.2	243494
250	1700	149.7	254490
260	1768	150.2	265554
270	1836	150.6	276502
280	1904	151.1	287694
290	1972	151.5	298758
300	2040	151.9	309876
310	2108	152.3	321048
320	2176	152.6	332058
330	2244	152.9	343108
340	2312	153.2	354198
350	2380	153.5	365330
360	2448	153.7	376258
368.6	2506.5	153.9	385747
370	2516	153.8	388961
380	2584	152.8	394835
390	2652	151.8	402574
400	2720	150.8	410176
410	2788	149.9	417921
417.2	2837.0	149.3	423558
420	2856	149.4	426686
430	2924	149.6	437430
440	2992	149.9	448501
450	3060	150.2	459612
460	3128	150.5	470764
470	3196	150.8	481957
480	3264	151.0	492864
** 484.6	3295.3	151.2	498246

Jet B, JP-4 (6.5 Lbs/Gal)

U.S. GALLONS	WEIGHT (lbs.)	LONG CG (in.)	MOMENT (in-lb)
10	65	139.5	9068
20	130	139.7	18161
30	195	139.8	27261
40	260	139.9	36374
50	325	139.9	45468
54.6	354.9	139.9	49651
60	390	143.6	56004
70	455	147.3	67022
80	520	149.2	77584
90	585	150.7	88160
100	650	151.8	98670
110	715	152.8	109254
120	780	153.6	119808
130	845	154.3	130384
140	910	154.8	140868
150	975	155.4	151515
160	1040	155.8	162032
170	1105	156.2	172601
180	1170	156.6	183222
190	1235	157.0	193895
200	1300	157.2	204360
204.1	1326.7	157.3	208682
210	1365	155.5	212258
220	1430	152.7	218361
230	1495	150.1	224400
235.2	1528.8	148.9	227638
240	1560	149.2	232752
250	1625	149.7	243263
260	1690	150.2	253838
270	1755	150.6	264303
280	1820	151.1	275002
290	1885	151.5	285578
300	1950	151.9	296205
310	2015	152.3	306885
320	2080	152.6	317408
330	2145	152.9	327971
340	2210	153.2	338572
350	2275	153.5	349213
360	2340	153.7	359568
368.6	2395.9	153.9	368729
370	2405	153.8	369889
380	2470	152.8	377416
390	2535	151.8	384813
400	2600	150.8	392080
410	2665	149.9	399484
417.2	2711.8	149.3	404872
420	2730	149.4	407862
430	2795	149.6	418132
440	2860	149.9	428714
450	2925	150.2	439335
460	2990	150.5	449995
470	3055	150.8	460694
480	3120	151.0	471120
** 484.6	3149.9	151.2	476265

Most critical amount for most forward C.G. condition at a weight empty below 6920 pounds has no fuel.

• Most critical amount for most forward C.G. condition at a weight empty of 6920 pounds or greater.

•• Most critical fuel amount for most aft C.G. condition.

Weights given are nominal weights at 15 C.

Table 5-8M. USABLE FUEL LOADING TABLE WITH SELF-SEALING TANKS AND 618 LITERS AUX FUEL

(Metric)

Jet A, A-1, JP-5, JP-8 (.815kg/l)

Jet B, JP-4 (.779 kg/l)

LITERS	WEIGHT (kg)	LONG CG (mm)	MOMENT (kg-mm)
40	32.6	3543	115502
80	65.2	3548	231330
120	97.8	3551	347288
160	130.4	3553	463311
200	163.0	3553	579139
206.7	168.5	3553	598540
240	195.6	3691	721960
280	228.2	3764	858945
320	260.8	3810	993648
360	293.4	3843	1127536
400	326.0	3866	1260316
440	358.6	3894	1396388
480	391.2	3917	1532330
520	423.8	3929	1665110
560	456.4	3945	1800498
600	489.0	3957	1934973
640	521.6	3967	2069187
680	554.2	3978	2204608
720	586.8	3988	2340158
760	619.4	3993	2473264
772.7	629.8	3996	2516483
800	652.0	3952	2576704
840	684.6	3866	2646664
880	717.2	3797	2723208
890.2	725.5	3782	2743890
920	749.8	3792	2843242
960	782.4	3807	2978597
1000	815.0	3820	3113300
1040	847.6	3810	3229356
1080	880.2	3846	3385249
1120	912.8	3856	3519757
1160	945.4	3864	3653026
1200	978.0	3874	3788772
1240	1010.6	3882	3923149
1280	1043.2	3890	4058048
1320	1075.8	3897	4192393
1360	1108.4	3904	4327194
1395.2	1137.1	3910	4446014
1400	1141.0	3906	4456746
1440	1173.6	3879	4552394
1480	1206.2	3852	4646282
1520	1238.8	3827	4740888
1560	1271.4	3803	4835134
1579.2	1287.0	3792	4880486
1600	1304.0	3795	4948680
1640	1336.6	3802	5081753
1680	1369.2	3813	5220760
1720	1401.8	3820	5354876
1760	1434.4	3825	5486580
1800	1467.0	3833	5623011
** 1834.4	1495.0	3840	5740938

LITERS	WEIGHT (kg)	LONG CG (mm)	MOMENT (kg-mm)
40	31.2	3543	110400
80	62.3	3548	221111
120	93.5	3551	331947
160	124.6	3553	442846
200	155.8	3553	553557
206.7	161.0	3553	572102
240	187.0	3691	690069
280	218.1	3764	821004
320	249.3	3810	949757
360	280.4	3843	1077731
400	311.6	3866	1204646
440	342.8	3894	1334707
480	373.9	3917	1464645
520	405.1	3929	1591559
560	436.2	3945	1720967
600	467.4	3957	1849502
640	498.6	3967	1977788
680	529.7	3978	2107226
720	560.9	3988	2236789
760	592.0	3993	2364016
772.7	601.9	3996	2405325
800	623.2	3952	2462886
840	654.4	3866	2529756
880	685.5	3797	2602919
890.2	693.5	3782	2622688
920	716.7	3792	2717651
960	747.8	3807	2847027
1000	779.0	3820	2975780
1040	810.2	3810	3086710
1080	841.3	3846	3235717
1120	872.5	3856	3364283
1160	903.6	3864	3491865
1200	934.8	3874	3621415
1240	966.0	3882	3749857
1280	997.1	3890	3878797
1320	1028.3	3897	4007207
1360	1059.4	3904	4136054
1395.2	1086.9	3910	4249626
1400	1090.6	3906	4259884
1440	1121.8	3879	4351307
1480	1152.9	3852	4441048
1520	1184.1	3827	4531474
1560	1215.2	3803	4621558
1579.2	1230.2	3792	4664906
1600	1246.4	3795	4730088
1640	1277.6	3802	4857283
1680	1308.7	3813	4990149
1720	1339.9	3820	5118342
1760	1371.0	3825	5244228
1800	1402.2	3833	5374633
** 1834.4	1429.0	3840	5487351

Most critical amount for most forward C.G. condition at a weight empty below 3139 kilograms has n

* Most critical amount for most forward C.G. condition at a weight empty of 3062 kilograms or greater

** Most critical fuel amount for most aft C.G. condition

Weights given are nominal weights at 15 C.

Table 5-9. USABLE FUEL LOADING TABLE WITH SELF-SEALING TANKS AND 163.4 GAL AUX FUEL
(English)

Jet A, A-1, JP-5, JP-8 (6.8 Lbs/Gal)

Jet B, JP-4 (6.5 Lbs/Gal)

U.S. GALLONS	WEIGHT (lbs.)	LAT CG (in.)	MOMENT (in-lb)
10	68	0	0
20	136	0	0
30	204	0	0
40	272	0	0
50	340	0	0
54.6	371.3	0	0
60	408	-0.08	-33
70	476	-0.04	-19
80	544	-0.04	-22
90	612	-0.04	-24
100	680	-0.04	-27
110	748	-0.03	-22
120	816	-0.03	-24
130	884	-0.03	-27
140	952	-0.03	-29
150	1020	-0.02	-20
160	1088	-0.02	-22
170	1156	-0.02	-23
180	1224	-0.02	-24
190	1292	-0.02	-26
200	1360	-0.02	-27
204.1	1387.9	-0.02	-28
210	1428	-0.25	-357
220	1496	-0.35	-524
230	1564	-0.45	-704
235.2	1599.4	-0.50	-800
240	1632	-0.48	-783
250	1700	-0.46	-782
260	1768	-0.44	-778
270	1836	-0.43	-789
280	1904	-0.42	-800
290	1972	-0.40	-789
300	2040	-0.39	-796
310	2108	-0.37	-780
320	2176	-0.37	-805
330	2244	-0.36	-808
340	2312	-0.34	-786
350	2380	-0.33	-785
360	2448	-0.32	-783
370	2516	-0.32	-805
380	2584	-0.31	-801
390	2652	-0.30	-796
400	2720	-0.30	-816
410	2788	-0.29	-809
420	2856	-0.28	-800
430	2924	-0.27	-789
440	2992	-0.26	-778
450	3060	-0.26	-796
460	3128	-0.25	-782
470	3196	-0.25	-799
480	3264	-0.24	-783
484.6	3295.3	-0.24	-791

U.S. GALLONS	WEIGHT (lbs.)	LAT CG (in.)	MOMENT (in-lb)
10	65	0	0
20	130	0	0
30	195	0	0
40	260	0	0
50	325	0	0
54.6	354.9	0	0
60	390	-0.08	-31
70	455	-0.04	-18
80	520	-0.04	-21
90	585	-0.04	-23
100	650	-0.04	-26
110	715	-0.03	-21
120	780	-0.03	-23
130	845	-0.03	-25
140	910	-0.03	-27
150	975	-0.02	-20
160	1040	-0.02	-21
170	1105	-0.02	-22
180	1170	-0.02	-23
190	1235	-0.02	-25
200	1300	-0.02	-26
204.1	1326.7	-0.02	-27
210	1365	-0.25	-341
220	1430	-0.35	-501
230	1495	-0.45	-673
235.2	1528.8	-0.50	-764
240	1560	-0.48	-749
250	1625	-0.46	-748
260	1690	-0.44	-744
270	1755	-0.43	-755
280	1820	-0.42	-764
290	1885	-0.40	-754
300	1950	-0.39	-761
310	2015	-0.37	-746
320	2080	-0.37	-770
330	2145	-0.36	-772
340	2210	-0.34	-751
350	2275	-0.33	-751
360	2340	-0.32	-749
370	2405	-0.32	-770
380	2470	-0.31	-766
390	2535	-0.30	-761
400	2600	-0.30	-780
410	2665	-0.29	-773
420	2730	-0.28	-764
430	2795	-0.27	-755
440	2860	-0.26	-744
450	2925	-0.26	-761
460	2990	-0.25	-748
470	3055	-0.25	-764
480	3120	-0.24	-749
484.6	3149.9	-0.24	-756

• Most critical fuel amount for left side most lateral C.G. condition.

Weights given are nominal weights at 15 C.

Table 5-9M. USABLE FUEL LOADING TABLE WITH SELF-SEALING TANKS AND 618 LITERS AUX FUEL

(Metric)

Jet A, A-1, JP-5, JP-8 (.815kg/l)

LITERS	WEIGHT (kg)	LAT CG (mm)	MOMENT (kg-mm)
40	32.6	0	0
80	65.2	0	0
120	97.8	0	0
160	130.4	0	0
200	163.0	0	0
206.7	168.5	0	0
240	195.6	-1	-196
280	228.2	-1	-228
320	260.8	-1	-261
360	293.4	-1	-293
400	326.0	-1	-326
440	358.6	-1	-359
480	391.2	-1	-391
520	423.8	-1	-424
560	456.4	-1	-456
600	489.0	-1	-489
640	521.6	-1	-522
680	554.2	-1	-554
720	586.8	-1	-587
760	619.4	-1	-619
772.7	629.8	-1	-630
800	652.0	-7	-4564
840	684.6	-9	-6161
880	717.2	-12	-8606
890.2	725.5	-13	-9432
920	749.8	-12	-8998
960	782.4	-12	-9389
1000	815.0	-11	-8965
1040	847.6	-11	-9324
1080	880.2	-10	-8802
1120	912.8	-10	-9128
1160	945.4	-10	-9454
1200	978.0	-9	-8802
1240	1010.6	-9	-9095
1280	1043.2	-9	-9389
1320	1075.8	-8	-8606
1360	1108.4	-8	-8867
1395.2	1137.1	-8	-9097
1400	1141.0	-8	-9128
1440	1173.6	-8	-9389
1480	1206.2	-8	-9650
1520	1238.8	-7	-8672
1560	1271.4	-7	-8900
1579.2	1287.0	-7	-9009
1600	1304.0	-7	-9128
1640	1336.6	-7	-9356
1680	1369.2	-7	-9584
1720	1401.8	-6	-8411
1760	1434.4	-6	-8606
1800	1467.0	-6	-8802
1834.4	1495.0	-6	-8970

LITERS	WEIGHT (kg)	LAT CG (mm)	MOMENT (kg-mm)
40	31.2	0	0
80	62.3	0	0
120	93.5	0	0
160	124.6	0	0
200	155.8	0	0
206.7	161.0	0	0
240	187.0	-1	-187
280	218.1	-1	-218
320	249.3	-1	-249
360	280.4	-1	-280
400	311.6	-1	-312
440	342.8	-1	-343
480	373.9	-1	-374
520	405.1	-1	-405
560	436.2	-1	-436
600	467.4	-1	-467
640	498.6	-1	-499
680	529.7	-1	-530
720	560.9	-1	-561
760	592.0	-1	-592
772.7	601.9	-1	-602
800	623.2	-7	-4362
840	654.4	-9	-5889
880	685.5	-12	-8226
890.2	693.5	-13	-9015
920	716.7	-12	-8600
960	747.8	-12	-8974
1000	779.0	-11	-8569
1040	810.2	-11	-8912
1080	841.3	-10	-8413
1120	872.5	-10	-8725
1160	903.6	-10	-9036
1200	934.8	-9	-8413
1240	966.0	-9	-8694
1280	997.1	-9	-8974
1320	1028.3	-8	-8226
1360	1059.4	-8	-8476
1395.2	1086.9	-8	-8695
1400	1090.6	-8	-8725
1440	1121.8	-8	-8974
1480	1152.9	-8	-9223
1520	1184.1	-7	-8289
1560	1215.2	-7	-8507
1579.2	1230.2	-7	-8611
1600	1246.4	-7	-8725
1640	1277.6	-7	-8943
1680	1308.7	-7	-9161
1720	1339.9	-6	-8039
1760	1371.0	-6	-8226
1800	1402.2	-6	-8413
1834.4	1429.0	-6	-8574

* Weights given are nominal weights at 15 C.
Weights given are nominal weights at 15 C.

412-FMS-63-5-9-2

Table 5-10. USABLE FUEL LOADING TABLE WITH SELF-SEALING TANKS AND 16.3 GAL AUX FUEL
(English)

Jet A, A-1, JP-5, JP-8 (6.8 Lbs/Gal)

Jet B, JP-4 (6.5 Lbs/Gal)

U.S. GALLON	WEIGHT (lbs.)	LONG CG (in.)	MOMENT (in-lb)
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10	68	139.5	9486
20	136	139.7	18999
30	204	139.8	28519
40	272	139.9	38053
50	340	139.9	47566
* 54.6	371.3	139.9	51942
60	408	143.2	58426
70	476	147.4	70162
80	544	150.5	81872
90	612	152.7	93452
100	680	154.5	105060
110	748	155.9	116613
120	816	157.1	128194
130	884	158.2	139849
140	952	159.1	151463
150	1020	159.9	163098
152.6	1037.7	160.2	166236
160	1088	157.1	170925
170	1156	153.3	177215
180	1224	150.0	183600
183.6	1248.5	148.9	185899
190	1292	149.8	193542
200	1360	151.1	205496
210	1428	152.3	217484
220	1496	153.3	229337
230	1564	154.3	241325
240	1632	155.2	253286
250	1700	156.0	265200
** 253.6	1724.5	156.3	269536
260	1768	155.2	274394
270	1836	153.7	282193
280	1904	152.3	289979
290	1972	151.0	297772
300	2040	149.8	305592
302.2	2055.0	149.5	307217
310	2108	150.2	316622
320	2176	150.9	328358
330	2244	151.7	340415
*** 337.6	2295.7	152.2	349402

U.S. GALLON	WEIGHT (lbs.)	LONG CG (in.)	MOMENT (in-lb)
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10	65	139.5	9068
20	130	139.7	18161
30	195	139.8	27261
40	260	139.9	36374
50	325	139.9	45468
* 54.6	354.9	139.9	49651
60	390	143.2	55848
70	455	147.4	67067
80	520	150.5	78260
90	585	152.7	89330
100	650	154.5	100425
110	715	155.9	111469
120	780	157.1	122538
130	845	158.2	133679
140	910	159.1	144781
150	975	159.9	155903
152.6	991.9	160.2	158902
160	1040	157.1	163384
170	1105	153.3	169397
180	1170	150.0	175500
183.6	1193.4	148.9	177697
190	1235	149.8	185003
200	1300	151.1	196430
210	1365	152.3	207890
220	1430	153.3	219219
230	1495	154.3	230679
240	1560	155.2	242112
250	1625	156.0	253500
** 253.6	1648.4	156.3	257645
260	1690	155.2	262288
270	1755	153.7	269744
280	1820	152.3	277186
290	1885	151.0	284635
300	1950	149.8	292110
302.2	1964.3	149.5	293663
310	2015	150.2	302653
320	2080	150.9	313872
330	2145	151.7	325397
*** 337.6	2194.4	152.2	333988

Most critical amount for most forward C.G. condition at a weight empty below 6580 pounds has no fuel.

* Most critical amount for most forward C.G. condition at a weight empty of 6580 pounds or greater.

** Most critical fuel amount for most aft C.G. condition at a weight empty below 6650 pounds.

*** Most critical fuel amount for most aft C.G. condition at a weight empty of 6650 pounds or greater.

Weights given are nominal weights at 15 C.

Table 5-10M. USABLE FUEL LOADING TABLE WITH SELF-SEALING TANKS AND 62 LITERS AUX FUEL
(Metric)

Jet A, A-1, JP-5, JP-8 (.815kg/l)

Jet B, JP-4 (.779 kg/l)

LITERS	WEIGHT (kg)	LONG CG (mm)	MOMENT (kg-mm)
40	32.6	3543	115502
80	65.2	3548	231330
120	97.8	3551	347288
160	130.4	3553	463311
200	163.0	3553	579139
* 206.7	168.5	3553	598540
240	195.6	3679	719612
280	228.2	3777	861911
320	260.8	3850	1004080
360	293.4	3904	1145434
400	326.0	3946	1286396
440	358.6	3981	1427587
480	391.2	4010	1568712
520	423.8	4034	1709609
560	456.4	4057	1851615
** 577.4	470.6	4069	1914794
600	489.0	4005	1958445
640	521.6	3903	2035805
680	554.2	3813	2113165
694.9	566.3	3782	2141911
720	586.8	3806	2233361
760	619.4	3840	2378496
800	652.0	3871	2523892
840	684.6	3900	2669940
880	717.2	3925	2815010
920	749.8	3948	2960210
959.8	782.2	3970	3105481
960	782.4	3969	3105346
1000	815.0	3927	3200505
1040	847.6	3887	3294621
1080	880.2	3850	3388770
1120	912.8	3816	3483245
1143.8	932.2	3797	3539552
1160	945.4	3806	3598192
1200	978.0	3828	3743784
1240	1010.6	3846	3886768
*** 1277.7	1041.3	3867	4026806

LITERS	WEIGHT (kg)	LONG CG (mm)	MOMENT (kg-mm)
40	31.2	3543	110400
80	62.3	3548	221111
120	93.5	3551	331947
160	124.6	3553	442846
200	155.8	3553	553557
* 206.7	161.0	3553	572102
240	187.0	3679	687826
280	218.1	3777	823839
320	249.3	3850	959728
360	280.4	3904	1094838
400	311.6	3946	1229574
440	342.8	3981	1364528
480	373.9	4010	1499419
520	405.1	4034	1634093
560	436.2	4057	1769826
** 577.4	449.8	4069	1830214
600	467.4	4005	1871937
640	498.6	3903	1945880
680	529.7	3813	2019822
694.9	541.3	3782	2047299
720	560.9	3806	2134709
760	592.0	3840	2273434
800	623.2	3871	2412407
840	654.4	3900	2552004
880	685.5	3925	2690666
920	716.7	3948	2829453
959.8	747.7	3970	2968306
960	747.8	3969	2968177
1000	779.0	3927	3059133
1040	810.2	3887	3149092
1080	841.3	3850	3239082
1120	872.5	3816	3329384
1143.8	891.0	3797	3383204
1160	903.6	3806	3439254
1200	934.8	3828	3578414
1240	966.0	3846	3715082
*** 1277.7	995.3	3867	3848935

Most critical amount for most forward C.G. condition at a weight empty below 2984 kilograms has no fuel.

* Most critical amount for most forward C.G. condition at a weight empty of 2984 kilograms or greater.

** Most critical fuel amount for most aft C.G. condition at a weight empty below 3016 kilograms.

*** Most critical fuel amount for most aft C.G. condition at a weight empty of 3016 kilograms or greater.

Weights given are nominal weights at 15 C.

Table 5-11. USABLE FUEL LOADING TABLE WITH SELF-SEALING TANKS AND 16.3 GAL AUX FUEL (LH)
(English)

Jet A, A-1, JP-5, JP-8 (6.8 Lbs/Gal)

U.S. GALLON	WEIGHT (lbs.)	LAT CG (in.)	MOMENT (in-lb)
----------------	------------------	-----------------	-------------------

10	68	0	0
20	136	0	0
30	204	0	0
40	272	0	0
50	340	0	0
54.6	371.3	0	0
60	408	-0.17	-69
70	476	-0.69	-328
80	544	-1.02	-555
90	612	-1.42	-869
100	680	-1.75	-1190
110	748	-2.00	-1496
120	816	-2.22	-1812
130	884	-2.40	-2122
140	952	-2.55	-2428
150	1020	-2.58	-2632
152.6	1037.7	-2.54	-2636
160	1088	-2.74	-2981
170	1156	-2.74	-3167
180	1224	-2.73	-3342
183.6	1248.5	-2.72	-3396
190	1292	-2.63	-3398
200	1360	-2.49	-3386
210	1428	-2.37	-3384
220	1496	-2.27	-3396
230	1564	-2.20	-3441
240	1632	-2.08	-3395
250	1700	-2.00	-3400
253.6	1724.5	-1.97	-3397
260	1768	-1.92	-3395
270	1836	-1.85	-3397
280	1904	-1.78	-3389
290	1972	-1.72	-3392
300	2040	-1.67	-3407
302.2	2055.0	-1.65	-3391
310	2108	-1.61	-3394
320	2176	-1.57	-3416
330	2244	-1.52	-3411
337.6	2295.7	-1.48	-3398

Jet B, JP-4 (6.5 Lbs/Gal)

U.S. GALLON	WEIGHT (lbs.)	LAT CG (in.)	MOMENT (in-lb)
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10	65	0	0
20	130	0	0
30	195	0	0
40	260	0	0
50	325	0	0
54.6	354.9	0	0
60	390	-0.17	-66
70	455	-0.69	-314
80	520	-1.02	-530
90	585	-1.42	-831
100	650	-1.75	-1138
110	715	-2.00	-1430
120	780	-2.22	-1732
130	845	-2.40	-2028
140	910	-2.55	-2321
150	975	-2.58	-2516
152.6	991.9	-2.54	-2519
160	1040	-2.74	-2850
170	1105	-2.74	-3028
180	1170	-2.73	-3194
183.6	1193.4	-2.72	-3246
190	1235	-2.63	-3248
200	1300	-2.49	-3237
210	1365	-2.37	-3235
220	1430	-2.27	-3246
230	1495	-2.20	-3289
240	1560	-2.08	-3245
250	1625	-2.00	-3250
253.6	1648.4	-1.97	-3247
260	1690	-1.92	-3245
270	1755	-1.85	-3247
280	1820	-1.78	-3240
290	1885	-1.72	-3242
300	1950	-1.67	-3257
302.2	1964.3	-1.65	-3241
310	2015	-1.61	-3244
320	2080	-1.57	-3266
330	2145	-1.52	-3260
337.6	2194.4	-1.48	-3248

* Most critical fuel amount for left side most lateral C.G. condition.
Weights given are nominal weights at 15 C.

Table 5-11M. USABLE FUEL LOADING TABLE WITH SELF-SEALING TANKS AND 62 LITERS AUX FUEL (LH)
(Metric)

Jet A, A-1, JP-5, JP-8 (.815kg/l)

LITERS	WEIGHT (kg)	LAT CG (mm)	MOMENT (kg-mm)
40	32.6	0	0
80	65.2	0	0
120	97.8	0	0
160	130.4	0	0
200	163.0	0	0
206.7	168.5	0	0
240	195.6	-8	-1565
280	228.2	-22	-5020
320	260.8	-31	-8085
360	293.4	-41	-12029
400	326.0	-48	-15648
440	358.6	-54	-19364
480	391.2	-60	-23472
520	423.8	-64	-27123
560	456.4	-66	-30122
577.4	470.6	-64	-30117
600	489.0	-70	-34230
640	521.6	-70	-36512
680	554.2	-69	-38240
694.9	566.3	-69	-39078
720	586.8	-67	-39316
760	619.4	-63	-39022
800	652.0	-60	-39120
840	684.6	-57	-39022
880	717.2	-54	-38729
920	749.8	-52	-38990
959.8	782.2	-50	-39112
960	782.4	-50	-39120
1000	815.0	-48	-39120
1040	847.6	-46	-38990
1080	880.2	-44	-38729
1120	912.8	-43	-39250
1143.8	932.2	-42	-39152
1160	945.4	-41	-38761
1200	978.0	-40	-39120
1240	1010.6	-39	-39413
1277.7	1041.3	-38	-39570

Jet B, JP-4 (.779 kg/l)

LITERS	WEIGHT (kg)	LAT CG (mm)	MOMENT (kg-mm)
40	31.2	0	0
80	62.3	0	0
120	93.5	0	0
160	124.6	0	0
200	155.8	0	0
206.7	161.0	0	0
240	187.0	-8	-1496
280	218.1	-22	-4799
320	249.3	-31	-7728
360	280.4	-41	-11498
400	311.6	-48	-14957
440	342.8	-54	-18509
480	373.9	-60	-22435
520	405.1	-64	-25925
560	436.2	-66	-28792
577.4	449.8	-64	-28787
600	467.4	-70	-32718
640	498.6	-70	-34899
680	529.7	-69	-36551
694.9	541.3	-69	-37352
720	560.9	-67	-37579
760	592.0	-63	-37299
800	623.2	-60	-37392
840	654.4	-57	-37299
880	685.5	-54	-37018
920	716.7	-52	-37267
959.8	747.7	-50	-37384
960	747.8	-50	-37392
1000	779.0	-48	-37392
1040	810.2	-46	-37267
1080	841.3	-44	-37018
1120	872.5	-43	-37517
1143.8	891.0	-42	-37423
1160	903.6	-41	-37049
1200	934.8	-40	-37392
1240	966.0	-39	-37672
1277.7	995.3	-38	-37822

* Most critical fuel amount for left side most lateral C.G. condition.
Weights given are nominal weights at 15 C.

412-FMS-63-5-11-2

Table 5-12. USABLE FUEL LOADING TABLE WITH SELF-SEALING TANKS AND 16.3 GAL AUX FUEL (RH)
(English)

Jet A, A-1, JP-5, JP-8 (6.8 Lbs/Gal)

U.S. GALLON	WEIGHT (lbs.)	LAT CG (in.)	MOMENT (in-lb)
10	68	0	0
20	136	0	0
30	204	0	0
40	272	0	0
50	340	0	0
54.6	371.3	0	0
60	408	0.06	24
70	476	0.60	286
80	544	0.93	506
90	612	1.34	820
100	680	1.67	1136
110	748	1.93	1444
120	816	2.16	1763
130	884	2.35	2077
140	952	2.50	2380
150	1020	2.53	2581
152.6	1037.7	2.49	2584
160	1088	2.05	2230
170	1156	1.77	2046
180	1224	1.53	1873
183.6	1248.5	1.45	1810
190	1292	1.40	1809
200	1360	1.33	1809
210	1428	1.27	1814
220	1496	1.21	1810
230	1564	1.16	1814
240	1632	1.11	1812
250	1700	1.07	1819
253.6	1724.5	1.05	1811
260	1768	1.02	1803
270	1836	0.99	1818
280	1904	0.95	1809
290	1972	0.92	1814
300	2040	0.89	1816
302.2	2055.0	0.88	1808
310	2108	0.86	1813
320	2176	0.83	1806
330	2244	0.80	1795
337.6	2295.7	0.79	1814

Jet B, JP-4 (6.5 Lbs/Gal)

U.S. GALLON	WEIGHT (lbs.)	LAT CG (in.)	MOMENT (in-lb)
10	65	0	0
20	130	0	0
30	195	0	0
40	260	0	0
50	325	0	0
54.6	354.9	0	0
60	390	0.06	23
70	455	0.60	273
80	520	0.93	484
90	585	1.34	784
100	650	1.67	1086
110	715	1.93	1380
120	780	2.16	1685
130	845	2.35	1986
140	910	2.50	2275
150	975	2.53	2467
152.6	991.9	2.49	2470
160	1040	2.05	2132
170	1105	1.77	1956
180	1170	1.53	1790
183.6	1193.4	1.45	1730
190	1235	1.40	1729
200	1300	1.33	1729
210	1365	1.27	1734
220	1430	1.21	1730
230	1495	1.16	1734
240	1560	1.11	1732
250	1625	1.07	1739
253.6	1648.4	1.05	1731
260	1690	1.02	1724
270	1755	0.99	1737
280	1820	0.95	1729
290	1885	0.92	1734
300	1950	0.89	1736
302.2	1964.3	0.88	1729
310	2015	0.86	1733
320	2080	0.83	1726
330	2145	0.80	1716
337.6	2194.4	0.79	1734

* Most critical fuel amount for right side most lateral C.G. condition.
Weights given are nominal weights at 15 C.

412-FMS-63-5-12-1

Table 5-12M. USABLE FUEL LOADING TABLE WITH SELF-SEALING TANKS AND 62 LITERS AUX FUEL (RH)
(Metric)

Jet A, A-1, JP-5, JP-8 (.815kg/l)

Jet B, JP-4 (.779 kg/l)

LITERS	WEIGHT (kg)	LAT CG (mm)	MOMENT (kg-mm)
40	32.6	0	0
80	65.2	0	0
120	97.8	0	0
160	130.4	0	0
200	163.0	0	0
206.7	168.5	0	0
240	195.6	5	978
280	228.2	19	4336
320	260.8	29	7563
360	293.4	36	10562
400	326.0	46	14996
440	358.6	53	19006
480	391.2	58	22690
520	423.8	62	26276
560	456.4	65	29666
577.4	470.6	63	29647
600	489.0	53	25917
640	521.6	46	23994
680	554.2	39	21614
694.9	566.3	37	20955
720	586.8	36	21125
760	619.4	34	21060
800	652.0	32	20864
840	684.6	31	21223
880	717.2	29	20799
920	749.8	28	20994
959.8	782.2	27	21120
960	782.4	27	21125
1000	815.0	26	21190
1040	847.6	25	21190
1080	880.2	24	21125
1120	912.8	23	20994
1143.8	932.2	22	20508
1160	945.4	22	20799
1200	978.0	21	20538
1240	1010.6	21	21223
1277.7	1041.3	20	20827

LITERS	WEIGHT (kg)	LAT CG (mm)	MOMENT (kg-mm)
40	31.2	0	0
80	62.3	0	0
120	93.5	0	0
160	124.6	0	0
200	155.8	0	0
206.7	161.0	0	0
240	187.0	5	935
280	218.1	19	4144
320	249.3	29	7229
360	280.4	36	10096
400	311.6	46	14334
440	342.8	53	18166
480	373.9	58	21687
520	405.1	62	25115
560	436.2	65	28356
577.4	449.8	63	28337
600	467.4	53	24772
640	498.6	46	22934
680	529.7	39	20659
694.9	541.3	37	20029
720	560.9	36	20192
760	592.0	34	20129
800	623.2	32	19942
840	654.4	31	20285
880	685.5	29	19880
920	716.7	28	20067
959.8	747.7	27	20187
960	747.8	27	20192
1000	779.0	26	20254
1040	810.2	25	20254
1080	841.3	24	20192
1120	872.5	23	20067
1143.8	891.0	22	19602
1160	903.6	22	19880
1200	934.8	21	19631
1240	966.0	21	20285
1277.7	995.3	20	19907

* Most critical fuel amount for right side most lateral C.G. condition.
Weights given are nominal weights at 15 C.

412-FMS-63-5-12-2

Table 5-13. USABLE FUEL LOADING TABLE WITH SELF-SEALING TANKS AND 32.6 GAL AUX FUEL
(English)

Jet A, A-1, JP-5, JP-8 (6.8 Lbs/Gal)

U.S. GALLON	WEIGHT (lbs.)	LONG CG (in.)	MOMENT (in-lb)
----------------	------------------	------------------	-------------------

10	68	139.5	9486
20	136	139.7	18999
30	204	139.8	28519
40	272	139.9	38053
50	340	139.9	47566
* 54.6	371.3	139.9	51942
60	408	143.1	58385
70	476	147.0	69972
80	544	149.8	81491
90	612	151.8	92902
100	680	153.3	104244
110	748	154.5	115566
120	816	155.6	126970
130	884	156.4	138258
140	952	157.2	149654
150	1020	157.9	161058
160	1088	158.5	172448
168.9	1148.5	159.2	182844
170	1156	158.7	183457
180	1224	155.1	189842
190	1292	151.9	196255
199.9	1359.3	149.0	202539
200	1360	149.0	202640
210	1428	150.2	214486
220	1496	151.4	226494
230	1564	152.5	238510
240	1632	153.4	250349
250	1700	154.3	262310
260	1768	155.1	274217
** 269.9	1835.3	155.9	286126
270	1836	155.9	286232
280	1904	154.4	293978
290	1972	153.0	301716
300	2040	151.7	309468
310	2108	150.5	317254
318.5	2165.8	149.5	323787
320	2176	149.6	325530
330	2244	150.4	337498
340	2312	151.2	349574
350	2380	151.8	361284
*** 353.9	2406.5	152.1	366032

Jet B, JP-4 (6.5 Lbs/Gal)

U.S. GALLON	WEIGHT (lbs.)	LONG CG (in.)	MOMENT (in-lb)
----------------	------------------	------------------	-------------------

10	65	139.5	9068
20	130	139.7	18161
30	195	139.8	27261
40	260	139.9	36374
50	325	139.9	45468
* 54.6	354.9	139.9	49651
60	390	143.1	55809
70	455	147.0	66885
80	520	149.8	77896
90	585	151.8	88803
100	650	153.3	99645
110	715	154.5	110468
120	780	155.6	121368
130	845	156.4	132158
140	910	157.2	143052
150	975	157.9	153953
160	1040	158.5	164840
168.9	1097.9	159.2	174778
170	1105	158.7	175364
180	1170	155.1	181467
190	1235	151.9	187597
199.9	1299.4	149.0	193603
200	1300	149.0	193700
210	1365	150.2	205023
220	1430	151.4	216502
230	1495	152.5	227988
240	1560	153.4	239304
250	1625	154.3	250738
260	1690	155.1	262119
** 269.9	1754.4	155.9	273503
270	1755	155.9	273605
280	1820	154.4	281008
290	1885	153.0	288405
300	1950	151.7	295815
310	2015	150.5	303258
318.5	2070.3	149.5	309502
320	2080	149.6	311168
330	2145	150.4	322608
340	2210	151.2	334152
350	2275	151.8	345345
*** 353.9	2300.4	152.1	349883

Most critical amount for most forward C.G. condition at a weight empty below 6580 pounds has no fuel.

* Most critical amount for most forward C.G. condition at a weight empty of 6580 pounds or greater.

** Most critical fuel amount for most aft C.G. condition at a weight empty below 6550 pounds.

*** Most critical fuel amount for most aft C.G. condition at a weight empty of 6550 pounds or greater.

Weights given are nominal weights at 15 C.

Table 5-13M. USABLE FUEL LOADING TABLE WITH SELF-SEALING TANKS AND 123 LITERS AUX FUEL
(Metric)

Jet A, A-1, JP-5, JP-8 (.815kg/l)

LITERS	WEIGHT (kg)	LONG CG (mm)	MOMENT (kg-mm)
40	32.6	3543	115502
80	65.2	3548	231330
120	97.8	3551	347288
160	130.4	3553	463311
200	163.0	3553	579139
* 206.7	168.5	3553	598540
240	195.6	3675	718830
280	228.2	3762	858488
320	260.8	3831	999125
360	293.4	3877	1137512
400	326.0	3913	1275638
440	358.6	3942	1413601
480	391.2	3967	1551890
520	423.8	3988	1690114
560	456.4	4006	1828338
600	489.0	4023	1967247
639.1	520.9	4043	2105863
640	521.6	4041	2107786
680	554.2	3943	2185211
720	586.8	3855	2262114
756.6	616.6	3784	2333324
760	619.4	3787	2345668
800	652.0	3830	2497160
840	684.6	3851	2636395
880	717.2	3879	2782019
920	749.8	3904	2927219
960	782.4	3927	3072485
1000	815.0	3949	3218435
** 1021.5	832.5	3959	3295957
1040	847.6	3940	3339544
1080	880.2	3902	3434540
1120	912.8	3866	3528885
1160	945.4	3833	3623718
1200	978.0	3802	3718356
1205.5	982.5	3797	3730486
1240	1010.6	3816	3856450
1280	1043.2	3836	4001715
1320	1075.8	3865	4157967
*** 1339.4	1091.6	3864	4217985

Jet B, JP-4 (.779 kg/l)

LITERS	WEIGHT (kg)	LONG CG (mm)	MOMENT (kg-mm)
40	31.2	3543	110400
80	62.3	3548	221111
120	93.5	3551	331947
160	124.6	3553	442846
200	155.8	3553	553557
* 206.7	161.0	3553	572102
240	187.0	3675	687078
280	218.1	3762	820567
320	249.3	3831	954992
360	280.4	3877	1087266
400	311.6	3913	1219291
440	342.8	3942	1351160
480	373.9	3967	1483341
520	405.1	3988	1615459
560	436.2	4006	1747577
600	467.4	4023	1880350
639.1	497.9	4043	2012844
640	498.6	4041	2014681
680	529.7	3943	2088686
720	560.9	3855	2162192
756.6	589.4	3784	2230257
760	592.0	3787	2242055
800	623.2	3830	2386856
840	654.4	3851	2519940
880	685.5	3879	2659132
920	716.7	3904	2797919
960	747.8	3927	2936768
1000	779.0	3949	3076271
** 1021.5	795.7	3959	3150368
1040	810.2	3940	3192030
1080	841.3	3902	3282831
1120	872.5	3866	3373008
1160	903.6	3833	3463652
1200	934.8	3802	3554110
1205.5	939.1	3797	3565704
1240	966.0	3816	3686103
1280	997.1	3836	3824952
1320	1028.3	3865	3974302
*** 1339.4	1043.4	3864	4031669

Most critical amount for most forward C.G. condition at a weight empty below 2984 kilograms has no fu

* Most critical amount for most forward C.G. condition at a weight empty of 2984 kilograms or greater.

** Most critical fuel amount for most aft C.G. condition at a weight empty below 2971 kilograms.

*** Most critical fuel amount for most aft C.G. condition at a weight empty of 2971 kilograms or greater.

Weights given are nominal weights at 15 C.

Table 5-14. USABLE FUEL LOADING TABLE WITH SELF-SEALING TANKS AND 32.6 GAL AUX FUEL
(English)

Jet A, A-1, JP-5, JP-8 (6.8 Lbs/Gal)

U.S. GALLON	WEIGHT (lbs.)	LAT CG (in.)	MOMENT (in-lb)
----------------	------------------	-----------------	-------------------

10	68	0	0
20	136	0	0
30	204	0	0
40	272	0	0
50	340	0	0
54.6	371.3	0	0
60	408	-0.06	-24
70	476	-0.04	-19
80	544	-0.05	-27
90	612	-0.04	-24
100	680	-0.04	-27
110	748	-0.03	-22
120	816	-0.03	-24
130	884	-0.03	-27
140	952	-0.03	-29
150	1020	-0.03	-31
160	1088	-0.03	-33
168.9	1148.5	-0.02	-23
170	1156	-0.18	-208
180	1224	-0.36	-441
190	1292	-0.48	-620
199.9	1359.3	-0.58	-788
200	1360	-0.58	-789
210	1428	-0.55	-785
220	1496	-0.53	-793
230	1564	-0.51	-798
240	1632	-0.48	-783
250	1700	-0.47	-799
260	1768	-0.44	-778
269.9	1835.3	-0.43	-789
270	1836	-0.43	-789
280	1904	-0.41	-781
290	1972	-0.40	-789
300	2040	-0.39	-796
310	2108	-0.37	-780
318.5	2165.8	-0.36	-780
320	2176	-0.36	-783
330	2244	-0.35	-785
340	2312	-0.34	-786
350	2380	-0.33	-785
353.9	2406.5	-0.33	-794

Jet B, JP-4 (6.5 Lbs/Gal)

U.S. GALLON	WEIGHT (lbs.)	LAT CG (in.)	MOMENT (in-lb)
----------------	------------------	-----------------	-------------------

10	65	0	0
20	130	0	0
30	195	0	0
40	260	0	0
50	325	0	0
54.6	354.9	0	0
60	390	-0.06	-23
70	455	-0.04	-18
80	520	-0.05	-26
90	585	-0.04	-23
100	650	-0.04	-26
110	715	-0.03	-21
120	780	-0.03	-23
130	845	-0.03	-25
140	910	-0.03	-27
150	975	-0.03	-29
160	1040	-0.03	-31
168.9	1097.9	-0.02	-22
170	1105	-0.18	-199
180	1170	-0.36	-421
190	1235	-0.48	-593
199.9	1299.4	-0.58	-754
200	1300	-0.58	-754
210	1365	-0.55	-751
220	1430	-0.53	-758
230	1495	-0.51	-762
240	1560	-0.48	-749
250	1625	-0.47	-764
260	1690	-0.44	-744
269.9	1754.4	-0.43	-754
270	1755	-0.43	-755
280	1820	-0.41	-746
290	1885	-0.40	-754
300	1950	-0.39	-761
310	2015	-0.37	-746
318.5	2070.3	-0.36	-745
320	2080	-0.36	-749
330	2145	-0.35	-751
340	2210	-0.34	-751
350	2275	-0.33	-751
353.9	2300.4	-0.33	-759

* Most critical fuel amount for left side most lateral C.G. condition.
Weights given are nominal weights at 15 C.

412-FMS-63-5-14-1

Table 5-14M. USABLE FUEL LOADING TABLE WITH SELF-SEALING TANKS AND 123 LITERS AUX FUEL
(Metric)

Jet A, A-1, JP-5, JP-8 (.815kg/l)

Jet B, JP-4 (.779 kg/l)

LITERS	WEIGHT (kg)	LAT CG (mm)	MOMENT (kg-mm)
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40	32.6	0	0
80	65.2	0	0
120	97.8	0	0
160	130.4	0	0
200	163.0	0	0
206.7	168.5	0	0
240	195.6	-1	-196
280	228.2	-1	-228
320	260.8	-1	-261
360	293.4	-1	-293
400	326.0	-1	-326
440	358.6	-1	-359
480	391.2	-1	-391
520	423.8	-1	-424
560	456.4	-1	-456
600	489.0	-1	-489
639.1	520.9	-1	-521
640	521.6	-1	-522
680	554.2	-9	-4988
720	586.8	-12	-7042
756.6	616.6	-15	-9249
760	619.4	-15	-9291
800	652.0	-14	-9128
840	684.6	-13	-8900
880	717.2	-13	-9324
920	749.8	-12	-8998
960	782.4	-12	-9389
1000	815.0	-11	-8965
1021.5	832.5	-11	-9158
1040	847.6	-11	-9324
1080	880.2	-10	-8802
1120	912.8	-10	-9128
1160	945.4	-10	-9454
1200	978.0	-9	-8802
1205.5	982.5	-9	-8842
1240	1010.6	-9	-9095
1280	1043.2	-9	-9389
1320	1075.8	-8	-8606
1339.4	1091.6	-8	-8733

LITERS	WEIGHT (kg)	LAT CG (mm)	MOMENT (kg-mm)
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40	31.2	0	0
80	62.3	0	0
120	93.5	0	0
160	124.6	0	0
200	155.8	0	0
206.7	161.0	0	0
240	187.0	-1	-187
280	218.1	-1	-218
320	249.3	-1	-249
360	280.4	-1	-280
400	311.6	-1	-312
440	342.8	-1	-343
480	373.9	-1	-374
520	405.1	-1	-405
560	436.2	-1	-436
600	467.4	-1	-467
639.1	497.9	-1	-498
640	498.6	-1	-499
680	529.7	-9	-4767
720	560.9	-12	-6731
756.6	589.4	-15	-8841
760	592.0	-15	-8881
800	623.2	-14	-8725
840	654.4	-13	-8507
880	685.5	-13	-8912
920	716.7	-12	-8600
960	747.8	-12	-8974
1000	779.0	-11	-8569
1021.5	795.7	-11	-8753
1040	810.2	-11	-8912
1080	841.3	-10	-8413
1120	872.5	-10	-8725
1160	903.6	-10	-9036
1200	934.8	-9	-8413
1205.5	939.1	-9	-8452
1240	966.0	-9	-8694
1280	997.1	-9	-8974
1320	1028.3	-8	-8226
1339.4	1043.4	-8	-8347

* Most critical fuel amount for left side most lateral C.G. condition.
Weights given are nominal weights at 15 C.



ROTORCRAFT FLIGHT MANUAL

SUPPLEMENT TEN CELL — SELF SEALING FUEL

412-899-377

S/N 33108 — 33213

AND

S/N 36001 — 36019

S/N 36020 — 36086

S/N 36087 AND SUB

CERTIFIED
22 JUNE 1998

This supplement shall be attached to Model 412 or 412EP Flight Manual (BHT-412-FM-2, BHT-412-FM-3 and BHT-412-FM-4), when SELF SEALING FUEL CELLS are installed.

Information contained herein supplements information of basic Flight Manual. For Limitations, Procedures, and Performance Data not contained in this supplement, or other applicable supplements, consult basic Flight Manual.

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NOTICE PAGE

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MANAGER

ROTORCRAFT CERTIFICATION OFFICE
FEDERAL AVIATION ADMINISTRATION
FT. WORTH, TX 76193-0170

GENERAL INFORMATION

Ten Cell-Self Sealing Fuel Cells (412-899-377) replaces standard fuel cells with cells which have self-sealing capability against ballistic damage by small objects. Exterior geometry of self-sealing fuel cells are identical to standard cells. Interior geometry is slightly different since self-sealing cells have greater wall thickness. Due to increased wall thickness of cell, total and usable fuel capacities are less than those of basic helicopter.

All other components including fuel quantity gauging system are identical to standard system.

Section 1

LIMITATIONS

1-6. WEIGHT AND CENTER OF GRAVITY

Actual weight change shall be determined after installation and ballast readjusted, if necessary, to retain gross weight CG within allowable limits.

1-20. INSTRUMENT MARKINGS AND PLACARDS

Refer to figure 1-1.

BASIC FUEL CAP 2121 LBS
WITH AUX FUEL KIT 412-706-007 3236 LBS
412-706-009 2343 LBS

Location: Instrument panel

Figure 1-1. Instrument Markings and Placards

Section 2

NORMAL PROCEDURES

No change from basic manual.

Section 3

EMERGENCY/MALFUNCTION PROCEDURES

No change from basic manual.

Section 4

PERFORMANCE

No change from basic manual.

Section 5

WEIGHT AND BALANCE

5-7. FUEL LOADING

Fuel loading tables lists usable fuel quantities in 10 gallon (40 liter) increments, with weights and moments in both english and metric units for balance computation. Critical fuel loading for computing most forward and aft CGs are denoted.

5-7-A. BASIC SYSTEM — SELF SEALING FUEL CELL.

Total capacity: 326.5 U.S. gallons (1235.9 liters).

Usable fuel: 317.3 U.S. gallons (1201.1 liters).

- Tables 5-1 - Provides longitudinal CG data for approved fuels.
- Tables 5-2 - Provides lateral CG data for approved fuels.

5-7-B. BASIC SYSTEM WITH BOTH SEAT TYPE AUXILIARY TANKS (16.3 GAL /61.7 LITRE EACH).

Refer to BHT-412-FMS-25.2/25.3/25.4.

Total capacity: 359.1 U.S. gallons (1359.3 liters).

Usable fuel: 349.9 U.S. gallons (1324.5 liters).

- Table 5-3 - Provides longitudinal CG data for approved fuels.
- Table 5-4 - Provides lateral CG data for approved fuels.

5-7-C. BASIC SYSTEM WITH ONE SEAT TYPE AUXILIARY FUEL TANK (LEFT OR RIGHT).

Refer to BHT-412-FMS-25.2/25.3/25.4.

Total capacity: 342.8 U.S. gallons (1297.6 liters).

Usable fuel: 333.6 U.S. gallons (1262.8 liters).

- Table 5-5 - Provides longitudinal CG data for left side installation.
- Table 5-6 - Provides lateral CG data for left side installation.
- Table 5-7 - Provides longitudinal CG data for right side installation.
- Table 5-8 - Provides lateral CG data for right side installation.

5-7-D. BASIC SYSTEM WITH BOTH LONG RANGE AUXILIARY FUEL TANKS (81.7 GAL /309.3 LITRE EACH).

Refer to BHT-412-FMS-17.2/17.3/17.4.

Total capacity: 489.9 U.S. gallons (1854.4 liters).

Usable fuel: 480.7 U.S. gallons (1819.6 liters).

- Table 5-9 - Provides longitudinal CG data for approved fuels.
- Table 5-10 - Provides lateral CG data for approved fuels.

**5-7-E. BASIC SYSTEM WITH ONE
LONG RANGE AUXILIARY FUEL
TANK (LEFT OR RIGHT).**

Refer to BHT-412-FMS-17.2/17.3/17.4.

**Total capacity: 408.2 U.S. gallons (1545.2
liters).**

**Usable fuel: 399.0 U.S. gallons (1510.4
liters).**

- **Table 5-11 - Provides longitudinal CG data for left side installation.**
- **Table 5-12 - Provides lateral CG data for left side installation.**
- **Table 5-13 - Provides longitudinal CG data for right side installation.**
- **Table 5-14 - Provides lateral CG data for right side installation.**

Table 5-1. Usable fuel loading – longitudinal – basic helicopter (English)

USABLE FUEL LOADING TABLE - LONGITUDINAL CG (ENGLISH)							
Basic System - 317.3 US Gal							
Jet A, A-1, JP-5, JP-8 (6.8 lb./Gal)				Jet B, JP-4 (6.5 lb./Gal)			
Quantity (US Gal)	Weight (lb.)	CG (Inches)	Moment (in-lb.)	Quantity (US Gal)	Weight (lb.)	CG (Inches)	Moment (in-lb.)
10	68	138.6	9424	10	65	138.6	9009
20	136	139.1	18921	20	130	139.1	18086
30	204	139.4	28442	30	195	139.4	27187
40	272	139.6	37962	40	260	139.6	36288
50	340	139.7	47483	50	325	139.7	45388
60	408	139.7	57005	60	390	139.7	54490
* 62.6	426	139.7	59486	* 62.6	407	139.7	56862
70	476	143.6	68334	70	455	143.6	65319
80	544	147.7	80322	80	520	147.7	76778
90	612	150.8	92289	90	585	150.8	88217
100	680	153.3	104245	100	650	153.3	99646
110	748	155.3	116199	110	715	155.3	111073
120	816	157.0	128152	120	780	157.0	122498
130	884	158.5	140108	130	845	158.5	133926
** 138.9	944	159.6	150706	** 138.9	903	159.6	144057
140	952	159.1	151425	140	910	159.1	144744
150	1020	154.7	157776	150	975	154.7	150815
160	1088	150.9	164126	160	1040	150.9	156885
170	1156	147.5	170481	170	1105	147.5	162960
* 170.4	1158	147.4	170708	* 170.4	1107	147.4	163177
180	1224	148.9	182237	180	1170	148.9	174197
190	1292	150.3	194191	190	1235	150.3	185624
200	1360	151.6	206144	200	1300	151.6	197050
210	1428	152.7	218097	210	1365	152.7	208475
220	1496	153.8	230048	220	1430	153.8	219899
230	1564	154.7	241999	230	1495	154.7	231322
** 237.8	1617	155.3	251064	** 237.8	1546	155.3	239988
240	1632	154.9	252760	240	1560	154.9	241609
250	1700	153.3	260527	250	1625	153.3	249033
260	1768	151.8	268295	260	1690	151.8	256458
270	1836	150.4	276062	270	1755	150.4	263883
280	1904	149.1	283824	280	1820	149.1	271302
* 286.4	1948	148.3	288808	* 286.4	1948	148.3	288808
290	1972	148.6	293083	290	1885	148.6	280153
300	2040	149.5	305032	300	1950	149.5	291575
310	2108	150.4	316981	310	2015	150.4	302996
** 317.3	2158	151.0	325703	** 317.3	2062	151.0	311334

* Critical fuel quantities for most forward CG condition (includes zero fuel)

** Critical fuel quantities for most aft CG condition

Weights given are nominal weights at 15° C. (59° F)

For calculation purposes, use Weight and Moment values, (CGs shown are derived from Weight and Moment values and rounded to one decimal place).

NOTE This table is invalid with auxiliary fuel tank(s) installed

412-FMS-65-5-1-1

Table 5-1. Usable fuel loading – longitudinal – basic helicopter (Metric)

USABLE FUEL LOADING TABLE - LONGITUDINAL CG (METRIC)							
Basic System - 1201.1 litres							
Jet A, A-1, JP-5, JP-8 (0.815 kg/l)				Jet B, JP-4 (0.799 kg/l)			
Quantity (litres)	Weight (kg)	CG (mm)	Moment (mm-kg)	Quantity (litres)	Weight (kg)	CG (mm)	Moment (mm-kg)
40	32.6	3520	114758	40	32.0	3520	112505
80	65.2	3534	230430	80	63.9	3534	225907
120	97.8	3541	346339	120	95.9	3541	339540
160	130.4	3545	462247	160	127.8	3545	453172
200	163.0	3547	578156	200	159.8	3547	566805
* 237.0	193.1	3548	685353	* 237.0	189.3	3548	671898
240	195.6	3560	696327	240	191.8	3560	682657
280	228.2	3690	842099	280	223.7	3690	825567
320	260.8	3788	987966	320	255.7	3788	968570
360	293.4	3864	1133564	360	287.6	3864	1111310
400	326.0	3924	1279108	400	319.6	3924	1253997
440	358.6	3973	1424637	440	351.6	3973	1396668
480	391.2	4014	1570171	480	383.5	4014	1539345
520	423.8	4048	1715733	520	415.5	4048	1682050
** 525.7	428.4	4053	1736323	** 525.7	420.0	4053	1702236
560	456.4	3950	1802684	560	447.4	3950	1767294
600	489.0	3845	1879979	600	479.4	3845	1843072
640	521.6	3753	1957351	640	511.4	3753	1918925
* 644.9	525.6	3742	1966775	* 644.9	515.3	3742	1928163
680	554.2	3779	2094599	680	543.3	3779	2053478
720	586.8	3818	2240135	720	575.3	3818	2196156
760	619.4	3852	2385658	760	607.2	3852	2338823
800	652.0	3882	2531170	800	639.2	3882	2481478
840	684.6	3910	2676669	840	671.2	3910	2624121
880	717.2	3935	2822156	880	703.1	3935	2766752
** 900.2	733.7	3943	2892574	** 900.2	719.3	3943	2835788
920	749.8	3920	2939298	920	735.1	3920	2881594
960	782.4	3878	3033864	960	767.0	3878	2974303
1000	815.0	3839	3128430	1000	799.0	3839	3067013
1040	847.6	3802	3222967	1040	831.0	3802	3159694
1080	880.2	3769	3317451	1080	862.9	3769	3252323
* 1084.2	883.6	3766	3327431	* 1084.2	866.3	3766	3262107
1120	912.8	3788	3457537	1120	894.9	3788	3389659
1160	945.4	3811	3603004	1160	926.8	3811	3532270
1200	978.0	3833	3748467	1200	958.8	3833	3674877
** 1201.1	978.9	3833	3752508	** 1201.1	959.7	3833	3678839

* Critical fuel quantities for most forward CG condition (includes zero fuel)

** Critical fuel quantities for most aft CG condition

Weights given are nominal weights at 15° C. (59° F)

For calculation purposes, use Weight and Moment values, (CGs shown are derived from Weight and Moment values and rounded to whole numbers).

NOTE This table is invalid with auxiliary fuel tank(s) installed

Table 5-2. Usable fuel loading – lateral – basic helicopter (English)

USABLE FUEL LOADING TABLE - LATERAL CG (ENGLISH)							
Basic System - 317.3 US Gal							
Jet A, A-1, JP-5, JP-8 (6.8 lb./Gal)				Jet B, JP-4 (6.5 lb./Gal)			
Quantity (US Gal)	Weight (lb.)	CG (inches)	Moment (in-lb.)	Quantity (US Gal)	Weight (lb.)	CG (inches)	Moment (in-lb.)
10	68	0.0	0	10	65	0.0	0
20	136	0.0	0	20	130	0.0	0
30	204	0.0	0	30	195	0.0	0
40	272	0.0	0	40	260	0.0	0
50	340	0.0	0	50	325	0.0	0
60	408	0.0	0	60	390	0.0	0
62.6	426	0.0	0	62.6	407	0.0	0
70	476	0.0	-13	70	455	0.0	-12
80	544	0.0	-13	80	520	0.0	-13
90	612	0.0	-13	90	585	0.0	-13
100	680	0.0	-13	100	650	0.0	-13
110	748	0.0	-13	110	715	0.0	-13
120	816	0.0	-13	120	780	0.0	-13
130	884	0.0	-13	130	845	0.0	-13
138.9	944	0.0	-13	138.9	903	0.0	-13
140	952	-0.2	-171	140	910	-0.2	-164
150	1020	-0.4	-407	150	975	-0.4	-389
160	1088	-0.5	-540	160	1040	-0.5	-516
170	1156	-0.6	-672	170	1105	-0.6	-642
* 170.4	1158	-0.6	-677	* 170.4	1107	-0.6	-647
180	1224	-0.6	-677	180	1170	-0.6	-647
190	1292	-0.5	-677	190	1235	-0.5	-647
200	1360	-0.5	-677	200	1300	-0.5	-647
210	1428	-0.5	-677	210	1365	-0.5	-647
220	1496	-0.5	-677	220	1430	-0.5	-647
230	1564	-0.4	-677	230	1495	-0.4	-647
237.8	1617	-0.4	-677	237.8	1546	-0.4	-647
240	1632	-0.4	-677	240	1560	-0.4	-647
250	1700	-0.4	-677	250	1625	-0.4	-647
260	1768	-0.4	-677	260	1690	-0.4	-647
270	1836	-0.4	-677	270	1755	-0.4	-647
280	1904	-0.4	-677	280	1820	-0.4	-647
286.4	1948	-0.3	-677	286.4	1862	-0.3	-647
290	1972	-0.3	-677	290	1885	-0.3	-647
300	2040	-0.3	-678	300	1950	-0.3	-648
310	2108	-0.3	-688	310	2015	-0.3	-658
317.3	2158	-0.3	-696	317.3	2062	-0.3	-665

* Critical fuel quantity for most lateral CG condition

Weights given are nominal weights at 15° C (59° F)

For calculation purposes, use Weight and Moment values, (CGs shown are derived from Weight and Moment values and rounded to one decimal place).

NOTE This table is invalid with auxiliary fuel tank(s) installed

412-FMS-65-5-2-1

Table 5-2. Usable fuel loading – lateral – basic helicopter (Metric)

USABLE FUEL LOADING TABLE - LATERAL CG (METRIC)							
Basic System - 1201.1 litres							
Jet A, A-1, JP-5, JP-8 (0.815 kg/l)				Jet B, JP-4 (0.799 kg/l)			
Quantity (litres)	Weight (kg)	CG (mm)	Moment (mm-kg)	Quantity (litres)	Weight (kg)	CG (mm)	Moment (mm-kg)
40	32.6	0	0	40	32.0	0	0
80	65.2	0	0	80	63.9	0	0
120	97.8	0	0	120	95.9	0	0
160	130.4	0	0	160	127.8	0	0
200	163.0	0	0	200	159.8	0	0
237.0	193.2	0	0	237.0	189.4	0	0
240	195.6	0	-24	240	191.8	0	-24
280	228.2	-1	-150	280	223.7	-1	-147
320	260.8	-1	-152	320	255.7	-1	-149
360	293.4	-1	-152	360	287.6	-1	-149
400	326.0	0	-152	400	319.6	0	-149
440	358.6	0	-152	440	351.6	0	-149
480	391.2	0	-152	480	383.5	0	-149
520	423.8	0	-152	520	415.5	0	-149
525.7	428.4	0	-152	526	420.0	0	-149
560	456.4	-10	-4388	560	447.4	-10	-4301
600	489.0	-12	-5989	600	479.4	-12	-5872
640	521.6	-15	-7593	640	511.4	-15	-7444
* 644.9	525.6	-15	-7796	* 644.9	515.3	-15	-7643
680	554.2	-14	-7796	680	543.3	-14	-7643
720	586.8	-13	-7796	720	575.3	-13	-7643
760	619.4	-13	-7796	760	607.2	-13	-7643
800	652.0	-12	-7796	800	639.2	-12	-7643
840	684.6	-11	-7796	840	671.2	-11	-7643
880	717.2	-11	-7796	880	703.1	-11	-7643
900.2	733.7	-11	-7796	900.2	719.3	-11	-7643
920	749.8	-10	-7796	920	735.1	-10	-7643
960	782.4	-10	-7796	960	767.0	-10	-7643
1000	815.0	-10	-7796	1000	799.0	-10	-7643
1040	847.6	-9	-7796	1040	831.0	-9	-7643
1080	880.2	-9	-7796	1080	862.9	-9	-7643
1084.2	883.6	-9	-7796	1084.2	866.3	-9	-7643
1120	912.8	-9	-7803	1120	894.9	-9	-7649
1160	945.4	-8	-7885	1160	926.8	-8	-7730
1200	978.0	-8	-8014	1200	958.8	-8	-7857
1201.1	978.9	-8	-8018	1201.1	959.7	-8	-7860

* Critical fuel quantity for most lateral CG condition

Weights given are nominal weights at 15° C. (59° F)

For calculation purposes, use Weight and Moment values, (CGs shown are derived from Weight and Moment values and rounded to whole numbers).

NOTE This table is invalid with auxiliary fuel tank(s) installed

412-FMS-65-5-2-2

Table 5-3. Usable fuel loading – longitudinal – w/32.6 gal aux fuel in LH and RH position (English)

USABLE FUEL LOADING TABLE - LONGITUDINAL CG (ENGLISH)							
Basic with LH and RH 16.3 US Gal Aux Tank - 349.9 US Gal							
Jet A, A-1, JP-5, JP-8 (6.8 lb./Gal)				Jet B, JP-4 (6.5 lb./Gal)			
Quantity (US Gal)	Weight (lb.)	CG (Inches)	Moment (in.-lb.)	Quantity (US Gal)	Weight (lb.)	CG (Inches)	Moment (in.-lb.)
10	68	138.6	9424	10	65	138.6	9009
20	136	139.1	18921	20	130	139.1	18086
30	204	139.4	28442	30	195	139.4	27187
40	272	139.6	37962	40	260	139.6	36288
50	340	139.7	47483	50	325	139.7	45388
60	408	139.7	57005	60	390	139.7	54490
* 62.6	426	139.7	59486	* 62.6	407	139.7	56862
70	476	143.2	68162	70	455	143.2	65155
80	544	146.3	79611	80	520	146.3	76099
90	612	148.9	91140	90	585	148.9	87119
100	680	150.7	102497	100	650	150.7	97975
110	748	152.2	113831	110	715	152.2	108809
120	816	153.4	125151	120	780	153.4	119630
130	884	154.4	136497	130	845	154.4	130475
140	952	155.3	147835	140	910	155.3	141313
150	1020	156.1	159172	150	975	156.1	152149
160	1088	156.7	170527	160	1040	156.7	163004
170	1156	157.5	182123	170	1105	157.5	174088
** 171.5	1166	157.7	183874	** 171.5	1115	157.7	175762
180	1224	154.7	189293	180.0	1170	154.7	180942
190	1292	151.4	195642	190	1235	151.4	187011
200	1360	148.5	201997	200	1300	148.5	193085
* 203.0	1380	147.7	203876	* 203.0	1319	147.7	194882
210	1428	148.7	212296	210	1365	148.7	202930
220	1496	149.9	224251	220	1430	149.9	214358
230	1564	151.0	236204	230	1495	151.0	225784
240	1632	152.1	248157	240	1560	152.1	237209
250	1700	153.0	260109	250	1625	153.0	248633
260	1768	153.9	272059	260	1690	153.9	260057
270	1836	154.6	283909	270	1755	154.6	271384
** 270.4	1839	154.6	284232	** 270.4	1758	154.6	271692
280	1904	153.2	291675	280	1820	153.2	278807
290	1972	151.8	299443	290	1885	151.8	286232
300	2040	150.6	307211	300	1950	150.6	293657
310	2108	149.4	314974	310	2015	149.4	301078
* 319.0	2169	148.4	321976	* 319.0	2074	148.4	307771
320	2176	148.5	323144	320	2080	148.5	308888
330	2244	149.3	335093	330	2145	149.3	320310
340	2312	150.1	347042	340	2210	150.1	331731
** 349.9	2379	150.8	358871	** 349.9	2274	150.8	343038

* Critical fuel quantities for most forward CG condition (includes zero fuel)

** Critical fuel quantities for most aft CG condition

Weights given are nominal weights at 15° C. (59° F)

For calculation purposes, use Weight and Moment values, (CGs shown are derived from Weight and Moment values and rounded to one decimal place).

Table 5-3. Usable fuel loading – longitudinal – w/123.4 litres aux fuel in LH and RH position (Metric)

USABLE FUEL LOADING TABLE - LONGITUDINAL CG (METRIC)							
Basic with LH and RH 61.7 litre Aux Tank - 1324.5 litres							
Jet A, A-1, JP-5, JP-8 (0.815 kg/l)				Jet B, JP-4 (0.799 kg/l)			
Quantity (litres)	Weight (kg)	CG (mm)	Moment (mm-kg)	Quantity (litres)	Weight (kg)	CG (mm)	Moment (mm-kg)
40	32.6	3520	114758	40	32.0	3520	112505
80	65.2	3534	230430	80	63.9	3534	225907
120	97.8	3541	346339	120	95.9	3541	339540
160	130.4	3545	462247	160	127.8	3545	453172
200	163.0	3547	578156	200	159.8	3547	566805
* 237.0	193.1	3548	685353	* 237.0	189.3	3548	671898
240	195.6	3560	696315	240	191.8	3560	682645
280	228.2	3672	837899	280	223.7	3672	821450
320	260.8	3749	977655	320	255.7	3749	958462
360	293.4	3807	1116831	360	287.6	3807	1094905
400	326.0	3850	1254959	400	319.6	3850	1230322
440	358.6	3884	1392755	440	351.6	3884	1365412
480	391.2	3913	1530831	480	383.5	3913	1500778
520	423.8	3938	1668888	520	415.5	3938	1636125
560	456.4	3959	1806874	560	447.4	3959	1771402
600	489.0	3978	1945065	600	479.4	3978	1906880
640	521.6	3998	2085479	640	511.4	3998	2044537
** 649.1	529.0	4005	2118458	** 649.1	518.6	4005	2076868
680	554.2	3930	2178241	680	543.3	3930	2135478
720	586.8	3844	2255532	720	575.3	3844	2211251
760	619.4	3766	2332900	760	607.2	3766	2287101
* 768.3	626.1	3751	2348909	* 768.3	613.9	3751	2302796
800	652.0	3780	2464345	800	639.2	3780	2415966
840	684.6	3812	2609883	840	671.2	3812	2558646
880	717.2	3842	2755407	880	703.1	3842	2701313
920	749.8	3869	2900920	920	735.1	3869	2843970
960	782.4	3894	3046421	960	767.0	3894	2986614
1000	815.0	3916	3191909	1000	799.0	3916	3129245
** 1023.6	834.3	3925	3274709	** 1023.6	817.9	3925	3210420
1040	847.6	3909	3313383	1040	831.0	3909	3248335
1080	880.2	3872	3407950	1080	862.9	3872	3341045
1120	912.8	3837	3502516	1120	894.9	3837	3433755
1160	945.4	3805	3597060	1160	926.8	3805	3526443
1200	978.0	3775	3691544	1200	958.8	3775	3619072
* 1207.6	984.2	3769	3709565	* 1207.6	964.9	3769	3636739
1240	1010.6	3787	3827290	1240	990.8	3787	3752153
1280	1043.2	3808	3972758	1280	1022.7	3808	3894765
1320	1075.8	3828	4118221	1320	1054.7	3828	4037373
** 1324.5	1079.5	3830	4134642	** 1324.5	1058.3	3830	4053471

* Critical fuel quantities for most forward CG condition (includes zero fuel)

** Critical fuel quantities for most aft CG condition

Weights given are nominal weights at 15° C. (59° F)

For calculation purposes, use Weight and Moment values, (CGs shown are derived from Weight and Moment values and rounded to whole numbers).

Table 5-4. Usable fuel loading – lateral – w/32.6 gal. aux fuel in LH and RH position (English)

USABLE FUEL LOADING TABLE - LATERAL CG (ENGLISH)							
Basic with LH and RH 16.3 US Gal Aux Tank - 349.9 US Gal							
Jet A, A-1, JP-5, JP-8 (6.8 lb./Gal)				Jet B, JP-4 (6.5 lb./Gal)			
Quantity (US Gal)	Weight (lb.)	CG (inches)	Moment (in.-lb.)	Quantity (US Gal)	Weight (lb.)	CG (inches)	Moment (in.-lb.)
10	68	0.0	0	10	65	0.0	0
20	136	0.0	0	20	130	0.0	0
30	204	0.0	0	30	195	0.0	0
40	272	0.0	0	40	260	0.0	0
50	340	0.0	0	50	325	0.0	0
60	408	0.0	0	60	390	0.0	0
62.6	426	0.0	0	62.6	407	0.0	0
70	476	0.0	-11	70	455	0.0	-10
80	544	0.0	-13	80	520	0.0	-13
90	612	0.0	-13	90	585	0.0	-13
100	680	0.0	-13	100	650	0.0	-13
110	748	0.0	-13	110	715	0.0	-13
120	816	0.0	-13	120	780	0.0	-13
130	884	0.0	-13	130	845	0.0	-13
140	952	0.0	-13	140.0	910	0.0	-13
150	1020	0.0	-13	150	975	0.0	-13
160	1088	0.0	-13	160	1040	0.0	-13
170	1156	0.0	-13	170	1105	0.0	-13
171.5	1166	0.0	-13	171	1115	0.0	-13
180	1224	-0.3	-374	180.0	1170	-0.3	-357
190	1292	-0.4	-504	190	1235	-0.4	-482
200	1360	-0.5	-637	200	1300	-0.5	-609
* 203.0	1380	-0.5	-677	* 203.0	1319	-0.5	-647
210	1428	-0.5	-677	210	1365	-0.5	-647
220	1496	-0.5	-677	220	1430	-0.5	-647
230	1564	-0.4	-677	230	1495	-0.4	-647
240	1632	-0.4	-677	240.0	1560	-0.4	-647
250	1700	-0.4	-677	250	1625	-0.4	-647
260	1768	-0.4	-677	260	1690	-0.4	-647
270	1836	-0.4	-677	270	1755	-0.4	-647
270.4	1839	-0.4	-677	270	1758	-0.4	-647
280	1904	-0.4	-677	280	1820	-0.4	-647
290	1972	-0.3	-677	290.0	1885	-0.3	-647
300	2040	-0.3	-677	300	1950	-0.3	-647
310	2108	-0.3	-677	310	2015	-0.3	-647
319.0	2169	-0.3	-677	319	2074	-0.3	-647
320	2176	-0.3	-677	320	2080	-0.3	-647
330	2244	-0.3	-678	330	2145	-0.3	-648
340	2312	-0.3	-685	340	2210	-0.3	-655
349.9	2379	-0.3	-696	350	2274	-0.3	-665

* Critical fuel quantity for most lateral CG condition

Weights given are nominal weights at 15° C. (59° F)

For calculation purposes, use Weight and Moment values, (CGs shown are derived from Weight and Moment values and rounded to one decimal place).

**Table 5-4. Usable fuel loading – lateral – w/123.4 litre aux fuel in LH and RH position
(Metric)**

USABLE FUEL LOADING TABLE - LATERAL CG (METRIC)							
Basic with LH and RH 61.7 litre Aux Tank - 1324.5 litres							
Jet A, A-1, JP-5, JP-8 (0.815 kg/l)				Jet B, JP-4 (0.799 kg/l)			
Quantity (litres)	Weight (kg)	CG (mm)	Moment (mm-kg)	Quantity (litres)	Weight (kg)	CG (mm)	Moment (mm-kg)
40	32.6	0	0	40	32.0	0	0
80	65.2	0	0	80	63.9	0	0
120	97.8	0	0	120	95.9	0	0
160	130.4	0	0	160	127.8	0	0
200	163.0	0	0	200	159.8	0	0
237.0	193.1	0	0	237.0	189.3	0	0
240	195.6	0	-24	240	191.8	0	-24
280	228.2	-1	-147	280	223.7	-1	-144
320	260.8	-1	-152	320	255.7	-1	-149
360	293.4	-1	-152	360	287.6	-1	-149
400	326.0	0	-152	400	319.6	0	-149
440	358.6	0	-152	440	351.6	0	-149
480	391.2	0	-152	480	383.5	0	-149
520	423.8	0	-152	520	415.5	0	-149
560	456.4	0	-152	560	447.4	0	-149
600	489.0	0	-152	600	479.4	0	-149
640	521.6	0	-152	640	511.4	0	-149
649.1	529.0	0	-152	649.1	518.6	0	-149
680	554.2	-8	-4256	680	543.3	-8	-4172
720	586.8	-10	-5842	720	575.3	-10	-5727
760	619.4	-12	-7458	760	607.2	-12	-7312
* 768.3	626.1	-12	-7796	* 768.3	613.9	-12	-7643
800	652.0	-12	-7796	800	639.2	-12	-7643
840	684.6	-11	-7796	840	671.2	-11	-7643
880	717.2	-11	-7796	880	703.1	-11	-7643
920	749.8	-10	-7796	920	735.1	-10	-7643
960	782.4	-10	-7796	960	767.0	-10	-7643
1000	815.0	-10	-7796	1000	799.0	-10	-7643
1023.6	834.3	-9	-7796	1023.6	817.9	-9	-7643
1040	847.6	-9	-7796	1040	831.0	-9	-7643
1080	880.2	-9	-7796	1080	862.9	-9	-7643
1120	912.8	-9	-7796	1120	894.9	-9	-7643
1160	945.4	-8	-7796	1160	926.8	-8	-7643
1200	978.0	-8	-7796	1200	958.8	-8	-7643
1207.6	984.2	-8	-7796	1207.6	964.9	-8	-7643
1240	1010.6	-8	-7800	1240	990.8	-8	-7647
1280	1043.2	-8	-7875	1280	1022.7	-8	-7720
1320	1075.8	-7	-8003	1320	1054.7	-7	-7846
1324.5	1079.5	-7	-8018	1324.5	1058.3	-7	-7860

* Critical fuel quantity for most lateral CG condition

Weights given are nominal weights at 15° C. (59° F)

For calculation purposes, use Weight and Moment values, (CGs shown are derived from Weight and Moment values and rounded to whole numbers).

Table 5-5. Usable fuel loading – longitudinal – w/16.3 gal aux fuel in LH position (English)

USABLE FUEL LOADING TABLE - LONGITUDINAL CG (ENGLISH)							
Basic with LH 16.3 US Gal Aux Tank - 333.6 US Gal							
Jet A, A-1, JP-5, JP-8 (6.8 lb./Gal)				Jet B, JP-4 (6.5 lb./Gal)			
Quantity (US Gal)	Weight (lb.)	CG (inches)	Moment (in.-lb.)	Quantity (US Gal)	Weight (lb.)	CG (inches)	Moment (in.-lb.)
10	68	138.6	9424	10	65	138.6	9009
20	136	139.1	18921	20	130	139.1	18086
30	204	139.4	28442	30	195	139.4	27187
40	272	139.6	37962	40	260	139.6	36288
50	340	139.7	47483	50	325	139.7	45388
60	408	139.7	57005	60	390	139.7	54490
* 62.6	426	139.7	59486	* 62.6	407	139.7	56862
70	476	143.4	68237	70	455	143.4	65227
80	544	146.9	79910	80	520	146.9	76385
90	612	149.7	91587	90	585	149.7	87547
100	680	151.7	103177	100	650	151.7	98625
110	748	153.4	114749	110	715	153.4	109686
120	816	154.8	126337	120	780	154.8	120763
130	884	156.0	137919	130	845	156.0	131834
140	952	157.0	149506	140	910	157.0	142910
150	1020	158.0	161115	150	975	158.0	154007
** 155.2	1055	158.6	167290	** 155.2	1009	158.6	159909
160	1088	156.6	170361	160	1040	156.6	162845
170	1156	152.9	176708	170	1105	152.9	168912
180	1224	149.6	183061	180	1170	149.6	174985
* 186.7	1269	147.6	187292	* 186.7	1213	147.6	179029
190	1292	148.1	191289	190	1235	148.1	182850
200	1360	149.4	203244	200	1300	149.4	194277
210	1428	150.7	215198	210	1365	150.7	205704
220	1496	151.8	227151	220	1430	151.8	217129
230	1564	152.9	239103	230	1495	152.9	228554
240	1632	153.8	251054	240	1560	153.8	239978
250	1700	154.7	263004	250	1625	154.7	251401
** 254.1	1728	154.9	267648	** 254.1	1652	154.9	255840
260	1768	154.0	272217	260	1690	154.0	260208
270	1836	152.5	279985	270	1755	152.5	267633
280	1904	151.1	287753	280	1820	151.1	275058
290	1972	149.9	295518	290	1885	149.9	282481
300	2040	148.7	303279	300	1950	148.7	289899
* 302.7	2059	148.4	305392	* 302.7	1968	148.4	291919
310	2108	149.0	314088	310	2015	149.0	300232
320	2176	149.8	326037	320	2080	149.8	311653
330	2244	150.6	337985	330	2145	150.6	323074
** 333.6	2268	150.9	342287	** 333.6	2168	150.9	327186

* Critical fuel quantities for most forward CG condition (includes zero fuel)

** Critical fuel quantities for most aft CG condition

Weights given are nominal weights at 15° C. (59° F)

For calculation purposes, use Weight and Moment values, (CGs shown are derived from Weight and Moment values and rounded to one decimal place).

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Table 5-5. Usable fuel loading – longitudinal – w/61.7 litre aux fuel in LH position (Metric)

USABLE FUEL LOADING TABLE - LONGITUDINAL CG (METRIC)							
Basic with LH 61.7 litre Aux Tank - 1262.8 litres							
Jet A, A-1, JP-5, JP-8 (0.815 kg/l)				Jet B, JP-4 (0.799 kg/l)			
Quantity (litres)	Weight (kg)	CG (mm)	Moment (mm-kg)	Quantity (litres)	Weight (kg)	CG (mm)	Moment (mm-kg)
40	32.6	3520	114758	40	32.0	3520	112505
80	65.2	3534	230430	80	63.9	3534	225907
120	97.8	3541	346339	120	95.9	3541	339540
160	130.4	3545	462247	160	127.8	3545	453172
200	163.0	3547	578156	200	159.8	3547	566805
* 237.0	193.1	3548	685353	* 237.0	189.3	3548	671898
240	195.6	3560	696321	240	191.8	3560	682651
280	228.2	3679	839499	280	223.7	3679	823018
320	260.8	3766	982117	320	255.7	3766	962837
360	293.4	3829	1123351	360	287.6	3829	1101298
400	326.0	3878	1264312	400	319.6	3878	1239491
440	358.6	3919	1405307	440	351.6	3919	1377718
480	391.2	3953	1546342	480	383.5	3953	1515985
520	423.8	3982	1687364	520	415.5	3982	1654238
560	456.4	4007	1828624	560	447.4	4007	1792725
** 587.4	478.7	4026	1927390	** 587.4	469.3	4026	1889552
600	489.0	3991	1951824	600	479.4	3991	1913506
640	521.6	3890	2029103	640	511.4	3890	1989268
680	554.2	3801	2106436	680	543.3	3801	2065083
* 706.6	575.9	3747	2157842	* 706.6	564.6	3747	2115479
720	586.8	3761	2206699	720	575.3	3761	2163377
760	619.4	3798	2352242	760	607.2	3798	2306063
800	652.0	3831	2497772	800	639.2	3831	2448736
840	684.6	3861	2643290	840	671.2	3861	2591397
880	717.2	3888	2788798	880	703.1	3888	2734048
920	749.8	3913	2934290	920	735.1	3913	2876684
960	782.4	3935	3079073	960	767.0	3935	3018625
** 961.9	784.0	3933	3083642	** 961.9	768.6	3933	3023104
1000	815.0	3894	3173624	1000	799.0	3894	3111319
1040	847.6	3856	3268190	1040	831.0	3856	3204029
1080	880.2	3820	3362756	1080	862.9	3820	3296739
1120	912.8	3788	3457256	1120	894.9	3788	3389383
* 1145.9	933.9	3767	3518498	* 1145.9	915.6	3767	3449423
1160	945.4	3776	3569678	1160	926.8	3776	3499599
1200	978.0	3799	3715147	1200	958.8	3799	3642212
1240	1010.6	3820	3860613	1240	990.8	3820	3784822
** 1262.8	1029.2	3832	3943575	** 1262.8	1009.0	3832	3866155

* Critical fuel quantities for most forward CG condition (includes zero fuel)

** Critical fuel quantities for most aft CG condition

Weights given are nominal weights at 15° C. (59° F)

For calculation purposes, use Weight and Moment values, (CGs shown are derived from Weight and Moment values and rounded to whole numbers).

Table 5-6. Usable fuel – lateral – w/16.3 gal aux fuel in LH position (English)

USABLE FUEL LOADING TABLE - LATERAL CG (ENGLISH)							
Basic with LH 16.3 US Gal Aux Tank - 333.6 US Gal							
Jet A, A-1, JP-5, JP-8 (6.8 lb./Gal)				Jet B, JP-4 (6.5 lb./Gal)			
Quantity (US Gal)	Weight (lb.)	CG (inches)	Moment (in-lb.)	Quantity (US Gal)	Weight (lb.)	CG (inches)	Moment (in-lb.)
10	68	0.0	0	10	65	0.0	0
20	136	0.0	0	20	130	0.0	0
30	204	0.0	0	30	195	0.0	0
40	272	0.0	0	40	260	0.0	0
50	340	0.0	0	50	325	0.0	0
60	408	0.0	0	60	390	0.0	0
62.6	426	0.0	0	62.6	407	0.0	0
70	476	-0.3	-127	70	455	-0.3	-122
80	544	-0.8	-427	80	520	-0.8	-409
90	612	-1.1	-691	90	585	-1.1	-660
100	680	-1.5	-1019	100	650	-1.5	-974
110	748	-1.8	-1345	110	715	-1.8	-1286
120	816	-2.0	-1670	120	780	-2.0	-1596
130	884	-2.3	-1995	130	845	-2.3	-1907
140	952	-2.4	-2316	140	910	-2.4	-2213
150	1020	-2.6	-2618	150	975	-2.6	-2502
155.2	1055	-2.5	-2618	155.2	1009	-2.5	-2502
* 157.8	1073	-2.7	-2906	* 157.8	1025	-2.7	-2778
160	1088	-2.7	-2932	160	1040	-2.7	-2803
170	1156	-2.6	-3059	170	1105	-2.6	-2924
180	1224	-2.6	-3194	180	1170	-2.6	-3053
186.7	1269	-2.6	-3281	186.7	1213	-2.6	-3137
190	1292	-2.5	-3281	190	1235	-2.5	-3137
200	1360	-2.4	-3281	200	1300	-2.4	-3137
210	1428	-2.3	-3281	210	1365	-2.3	-3137
220	1496	-2.2	-3281	220	1430	-2.2	-3137
230	1564	-2.1	-3281	230	1495	-2.1	-3137
240	1632	-2.0	-3281	240	1560	-2.0	-3137
250	1700	-1.9	-3281	250	1625	-1.9	-3137
254.1	1728	-1.9	-3281	254.1	1652	-1.9	-3137
260	1768	-1.9	-3281	260	1690	-1.9	-3137
270	1836	-1.8	-3281	270	1755	-1.8	-3137
280	1904	-1.7	-3281	280	1820	-1.7	-3137
290	1972	-1.7	-3281	290	1885	-1.7	-3137
300	2040	-1.6	-3281	300	1950	-1.6	-3137
302.7	2059	-1.6	-3281	302.7	1968	-1.6	-3137
310	2108	-1.6	-3281	310	2015	-1.6	-3137
320	2176	-1.5	-3286	320	2080	-1.5	-3141
330	2244	-1.5	-3297	330	2145	-1.5	-3151
333.6	2268	-1.5	-3301	333.6	2168	-1.5	-3155

* Critical fuel quantity for most lateral CG condition

Weights given are nominal weights at 15° C. (59° F)

For calculation purposes, use Weight and Moment values, (CGs shown are derived from Weight and Moment values and rounded to one decimal place).

Table 5-6. Usable fuel – lateral – w/61.7 litre aux fuel in LH position (Metric)

USABLE FUEL LOADING TABLE - LATERAL CG (METRIC)							
Basic with LH 61.7 litre Aux Tank - 1262.8 litres							
Jet A, A-1, JP-5, JP-8 (0.815 kg/l)				Jet B, JP-4 (0.799 kg/l)			
Quantity (litres)	Weight (kg)	CG (mm)	Moment (mm-kg)	Quantity (litres)	Weight (kg)	CG (mm)	Moment (mm-kg)
40	32.6	0	0	40	32.0	0	0
80	65.2	0	0	80	63.9	0	0
120	97.8	0	0	120	95.9	0	0
160	130.4	0	0	160	127.8	0	0
200	163.0	0	0	200	159.8	0	0
237.0	193.1	0	0	237.0	189.3	0	0
240	195.6	0	-33	240	191.8	0	-32
280	228.2	-13	-2927	280	223.7	-13	-2869
320	260.8	-23	-5895	320	255.7	-23	-5779
360	293.4	-34	-9903	360	287.6	-34	-9708
400	326.0	-43	-13871	400	319.6	-43	-13599
440	358.6	-50	-17833	440	351.6	-50	-17483
480	391.2	-56	-21788	480	383.5	-56	-21361
520	423.8	-61	-25718	520	415.5	-61	-25213
560	456.4	-65	-29468	560	447.4	-65	-28890
587.4	478.7	-63	-30162	587.4	469.3	-63	-29570
* 597.2	486.7	-69	-33480	* 597.2	477.2	-69	-32823
600	489.0	-69	-33580	600	479.4	-69	-32921
640	521.6	-67	-35106	640	511.4	-67	-34417
680	554.2	-66	-36744	680	543.3	-66	-36023
706.6	575.9	-66	-37806	706.6	564.6	-66	-37063
720	586.8	-64	-37806	720	575.3	-64	-37063
760	619.4	-61	-37806	760	607.2	-61	-37063
800	652.0	-58	-37806	800	639.2	-58	-37063
840	684.6	-55	-37806	840	671.2	-55	-37063
880	717.2	-53	-37806	880	703.1	-53	-37063
920	749.8	-50	-37806	920	735.1	-50	-37063
960	782.4	-48	-37806	960	767.0	-48	-37063
961.9	784.0	-48	-37806	961.9	768.6	-48	-37063
1000	815.0	-46	-37806	1000	799.0	-46	-37063
1040	847.6	-45	-37806	1040	831.0	-45	-37063
1080	880.2	-43	-37806	1080	862.9	-43	-37063
1120	912.8	-41	-37806	1120	894.9	-41	-37063
1145.9	933.9	-40	-37806	1145.9	915.6	-40	-37063
1160	945.4	-40	-37806	1160	926.8	-40	-37063
1200	978.0	-39	-37828	1200	958.8	-39	-37085
1240	1010.6	-38	-37953	1240	990.8	-38	-37208
1262.8	1029.2	-37	-38028	1262.8	1009.0	-37	-37281

* Critical fuel quantity for most lateral CG condition

Weights given are nominal weights at 15° C. (59° F)

For calculation purposes, use Weight and Moment values. (CGs shown are derived from Weight and Moment values and rounded to whole numbers)

412-FMS-65-5-6-2

Table 5-7. Usable fuel – longitudinal – w/16.3 gal aux fuel in RH position (English)

USABLE FUEL LOADING TABLE - LONGITUDINAL CG (ENGLISH)							
Basic with RH 16.3 US Gal Aux Tank - 333.6 US Gal							
Jet A, A-1, JP-5, JP-8 (6.8 lb./Gal)				Jet B, JP-4 (6.5 lb./Gal)			
Quantity (US Gal)	Weight (lb.)	CG (Inches)	Moment (in-lb.)	Quantity (US Gal)	Weight (lb.)	CG (Inches)	Moment (in-lb.)
10	68	138.6	9424	10	65	138.6	9009
20	136	139.1	18921	20	130	139.1	18086
30	204	139.4	28442	30	195	139.4	27187
40	272	139.6	37962	40	260	139.6	36288
50	340	139.7	47483	50	325	139.7	45388
60	408	139.7	57005	60	390	139.7	54490
* 62.6	426	139.7	59486	* 62.6	407	139.7	56862
70	476	143.4	68237	70	455	143.4	65227
80	544	146.9	79910	80	520	146.9	76385
90	612	149.7	91587	90	585	149.7	87547
100	680	151.7	103177	100	650	151.7	98625
110	748	153.4	114749	110	715	153.4	109686
120	816	154.8	126337	120	780	154.8	120763
130	884	156.0	137919	130	845	156.0	131834
140	952	157.0	149506	140	910	157.0	142910
150	1020	158.0	161115	150	975	158.0	154007
** 155.2	1055	158.6	167290	** 155.2	1009	158.6	159909
160	1088	156.6	170361	160	1040	156.6	162845
170	1156	152.9	176708	170	1105	152.9	168912
180	1224	149.6	183061	180	1170	149.6	174985
* 186.7	1269	147.6	187292	* 186.7	1213	147.6	179029
190	1292	148.1	191289	190	1235	148.1	182850
200	1360	149.4	203244	200	1300	149.4	194277
210	1428	150.7	215198	210	1365	150.7	205704
220	1496	151.8	227151	220	1430	151.8	217129
230	1564	152.9	239103	230	1495	152.9	228554
240	1632	153.8	251054	240	1560	153.8	239978
250	1700	154.7	263004	250	1625	154.7	251401
* 254.1	1728	154.9	267648	* 254.1	1652	154.9	255840
260	1768	154.0	272217	260	1690	154.0	260208
270	1836	152.5	279985	270	1755	152.5	267633
280	1904	151.1	287753	280	1820	151.1	275058
290	1972	149.9	295518	290	1885	149.9	282481
300	2040	148.7	303279	300	1950	148.7	289899
* 302.7	2059	148.4	305392	* 302.7	1968	148.4	291919
310	2108	149.0	314088	310	2015	149.0	300232
320	2176	149.8	326037	320	2080	149.8	311653
330	2244	150.6	337985	330	2145	150.6	323074
** 333.6	2268	150.9	342287	** 333.6	2168	150.9	327186

* Critical fuel quantities for most forward CG condition (includes zero fuel)

** Critical fuel quantities for most aft CG condition

Weights given are nominal weights at 15° C. (59° F)

For calculation purposes, use Weight and Moment values. (CGs shown are derived from Weight and Moment values and rounded to one decimal place).

412-FMS-65-5-7-1

Table 5-7. Usable fuel – longitudinal – w/61.7 litre aux fuel in RH position (Metric)

USABLE FUEL LOADING TABLE - LONGITUDINAL CG (METRIC)							
Basic with RH 61.7 litre Aux Tank - 1262.8 litres							
Jet A, A-1, JP-5, JP-8 (0.815 kg/l)				Jet B, JP-4 (0.799 kg/l)			
Quantity (litres)	Weight (kg)	CG (mm)	Moment (mm-kg)	Quantity (litres)	Weight (kg)	CG (mm)	Moment (mm-kg)
40	32.6	3520	114758	40	32.0	3520	112505
80	65.2	3534	230430	80	63.9	3534	225907
120	97.8	3541	346339	120	95.9	3541	339540
160	130.4	3545	462247	160	127.8	3545	453172
200	163.0	3547	578156	200	159.8	3547	566805
* 237.0	193.1	3548	685353	* 237.0	189.3	3548	671898
240	195.6	3560	696321	240	191.8	3560	682651
280	228.2	3679	839499	280	223.7	3679	823018
320	260.8	3766	982117	320	255.7	3766	962837
360	293.4	3829	1123351	360	287.6	3829	1101298
400	326.0	3878	1264312	400	319.6	3878	1239491
440	358.6	3919	1405307	440	351.6	3919	1377718
480	391.2	3953	1546342	480	383.5	3953	1515985
520	423.8	3982	1687364	520	415.5	3982	1654238
560	456.4	4007	1828624	560	447.4	4007	1792725
** 587.4	478.7	4026	1927390	** 587.4	469.3	4026	1889552
600	489.0	3991	1951824	600	479.4	3991	1913506
640	521.6	3890	2029103	640	511.4	3890	1989268
680	554.2	3801	2106436	680	543.3	3801	2065083
* 706.6	575.9	3747	2157842	* 706.6	564.6	3747	2115479
720	586.8	3761	2206699	720	575.3	3761	2163377
760	619.4	3798	2352242	760	607.2	3798	2306063
800	652.0	3831	2497772	800	639.2	3831	2448736
840	684.6	3861	2643290	840	671.2	3861	2591397
880	717.2	3888	2788798	880	703.1	3888	2734048
920	749.8	3913	2934290	920	735.1	3913	2876684
960	782.4	3935	3079073	960	767.0	3935	3018625
** 961.9	784.0	3933	3083642	** 961.9	768.6	3933	3023104
1000	815.0	3894	3173624	1000	799.0	3894	3111319
1040	847.6	3856	3268190	1040	831.0	3856	3204029
1080	880.2	3820	3362756	1080	862.9	3820	3296739
1120	912.8	3788	3457256	1120	894.9	3788	3389933
* 1145.9	933.9	3767	3518498	* 1145.9	915.6	3767	3449423
1160	945.4	3776	3569678	1160	926.8	3776	3499599
1200	978.0	3799	3715147	1200	958.8	3799	3642212
1240	1010.6	3820	3860613	1240	990.8	3820	3784822
** 1262.8	1029.2	3832	3943575	** 1262.8	1009.0	3832	3866155

* Critical fuel quantities for most forward CG condition (includes zero fuel)

** Critical fuel quantities for most aft CG condition

Weights given are nominal weights at 15° C. (59° F)

For calculation purposes, use Weight and Moment values, (CGs shown are derived from Weight and Moment values and rounded to whole numbers).

Table 5-8. Usable fuel – lateral – w/16.3 gal aux fuel in RH position (English)

USABLE FUEL LOADING TABLE - LATERAL CG (ENGLISH)							
Basic with RH 16.3 US Gal Aux Tank - 333.6 US Gal							
Jet A, A-1, JP-5, JP-8 (6.8 lb./Gal)				Jet B, JP-4 (6.5 lb./Gal)			
Quantity (US Gal)	Weight (lb.)	CG (Inches)	Moment (in-lb.)	Quantity (US Gal)	Weight (lb.)	CG (Inches)	Moment (in-lb.)
10	68	0.0	0	10	65	0.0	0
20	136	0.0	0	20	130	0.0	0
30	204	0.0	0	30	195	0.0	0
40	272	0.0	0	40	260	0.0	0
50	340	0.0	0	50	325	0.0	0
60	408	0.0	0	60	390	0.0	0
62.6	426	0.0	0	62.6	407	0.0	0
70	476	0.2	104	70	455	0.2	99
80	544	0.7	401	80	520	0.7	383
90	612	1.1	664	90	585	1.1	635
100	680	1.5	993	100	650	1.5	949
110	748	1.8	1319	110	715	1.8	1260
120	816	2.0	1644	120	780	2.0	1571
130	884	2.2	1969	130	845	2.2	1882
140	952	2.4	2289	140	910	2.4	2188
* 150.0	1020	2.5	2592	* 150.0	975	2.5	2477
155.2	1055	2.5	2592	155.2	1009	2.5	2477
160	1088	2.1	2277	160	1040	2.1	2177
170	1156	1.9	2151	170	1105	1.9	2056
180	1224	1.6	2016	180	1170	1.6	1927
186.7	1269	1.5	1928	186.7	1213	1.5	1843
190	1292	1.5	1928	190	1235	1.5	1843
200	1360	1.4	1928	200	1300	1.4	1843
210	1428	1.4	1928	210	1365	1.4	1843
220	1496	1.3	1928	220	1430	1.3	1843
230	1564	1.2	1928	230	1495	1.2	1843
240	1632	1.2	1928	240	1560	1.2	1843
250	1700	1.1	1928	250	1625	1.1	1843
254.1	1728	1.1	1928	254.1	1652	1.1	1843
260	1768	1.1	1928	260	1690	1.1	1843
270	1836	1.1	1928	270	1755	1.1	1843
280	1904	1.0	1928	280	1820	1.0	1843
290	1972	1.0	1928	290	1885	1.0	1843
300	2040	0.9	1928	300	1950	0.9	1843
302.7	2059	0.9	1928	302.7	1968	0.9	1843
310	2108	0.9	1928	310	2015	0.9	1843
320	2176	0.9	1923	320	2080	0.9	1838
330	2244	0.9	1913	330	2145	0.9	1828
333.6	2268	0.8	1909	333.6	2168	0.8	1825

* Critical fuel quantity for most lateral CG condition

Weights given are nominal weights at 15° C. (59° F)

For calculation purposes, use Weight and Moment values, (CGs shown are derived from Weight and Moment values and rounded to one decimal place)

Table 5-8. Usable fuel – lateral – w/61.7 litre aux fuel in RH position (Metric)

USABLE FUEL LOADING TABLE - LATERAL CG (METRIC)							
Basic with RH 61.7 litre Aux Tank - 1262.8 litres							
Jet A, A-1, JP-5, JP-8 (0.815 kg/l)				Jet B, JP-4 (0.799 kg/l)			
Quantity (litres)	Weight (kg)	CG (mm)	Moment (mm-kg)	Quantity (litres)	Weight (kg)	CG (mm)	Moment (mm-kg)
40	32.6	0	0	40	32.0	0	0
80	65.2	0	0	80	63.9	0	0
120	97.8	0	0	120	95.9	0	0
160	130.4	0	0	160	127.8	0	0
200	163.0	0	0	200	159.8	0	0
237.0	193.1	0	0	237.0	189.3	0	0
240	195.6	0	-16	240	191.8	0	-16
280	228.2	12	2631	280	223.7	12	2579
320	260.8	21	5591	320	255.7	21	5481
360	293.4	33	9599	360	287.6	33	9410
400	326.0	42	13567	400	319.6	42	13301
440	358.6	49	17529	440	351.6	49	17185
480	391.2	55	21484	480	383.5	55	21063
520	423.8	60	25414	520	415.5	60	24915
560	456.4	64	29164	560	447.4	64	28592
* 567.8	462.8	65	29864	* 567.8	453.7	65	29278
587.4	478.7	62	29858	587.4	469.3	62	29272
600	489.0	54	26439	600	479.4	54	25920
640	521.6	48	24914	640	511.4	48	24425
680	554.2	42	23276	680	543.3	42	22819
706.6	575.9	39	22214	706.6	564.6	39	21778
720	586.8	38	22214	720	575.3	38	21778
760	619.4	36	22214	760	607.2	36	21778
800	652.0	34	22214	800	639.2	34	21778
840	684.6	32	22214	840	671.2	32	21778
880	717.2	31	22214	880	703.1	31	21778
920	749.8	30	22214	920	735.1	30	21778
960	782.4	28	22214	960	767.0	28	21778
961.9	784.0	28	22214	961.9	768.6	28	21778
1000	815.0	27	22214	1000	799.0	27	21778
1040	847.6	26	22214	1040	831.0	26	21778
1080	880.2	25	22214	1080	862.9	25	21778
1120	912.8	24	22214	1120	894.9	24	21778
1145.9	933.9	24	22214	1145.9	915.6	24	21778
1160	945.4	23	22214	1160	926.8	23	21778
1200	978.0	23	22192	1200	958.8	23	21756
1240	1010.6	22	22066	1240	990.8	22	21633
1262.8	1029.2	21	21992	1262.8	1009.0	21	21560

* Critical fuel quantity for most lateral CG condition

Weights given are nominal weights at 15° C. (59° F)

For calculation purposes, use Weight and Moment values. (CGs shown are derived from Weight and Moment values and rounded to whole numbers).

**Table 5-9. Usable fuel – longitudinal – w/81.7 gal aux fuel in LH and RH position
(English)**

USABLE FUEL LOADING TABLE - LONGITUDINAL CG (ENGLISH)							
Basic with LH and RH 81.7 US Gal Aux Tank - 480.7 US Gal							
Jet A, A-1, JP-5, JP-8 (6.8 lb./Gal)				Jet B, JP-4 (6.5 lb./Gal)			
Quantity (US Gal)	Weight (lb.)	CG (inches)	Moment (in.-lb.)	Quantity (US Gal)	Weight (lb.)	CG (inches)	Moment (in.-lb.)
10	68	138.6	9424	10	65	138.6	9009
20	136	139.1	18921	20	130	139.1	18086
30	204	139.4	28442	30	195	139.4	27187
40	272	139.6	37962	40	260	139.6	36288
50	340	139.7	47483	50	325	139.7	45388
60	408	139.7	57005	60	390	139.7	54490
* 62.6	426	139.7	59486	* 62.6	407	139.7	56862
70	476	143.3	68219	70	455	143.3	65209
80	544	146.4	79665	80	520	146.4	76150
90	612	148.1	90662	90	585	148.1	86662
100	680	149.5	101671	100	650	149.5	97185
110	748	150.7	112690	110	715	150.7	107719
120	816	151.6	123714	120	780	151.6	118256
130	884	152.4	134738	130	845	152.4	128794
140	952	153.1	145762	140	910	153.1	139332
150	1020	153.7	156787	150	975	153.7	149870
160	1088	154.2	167811	160	1040	154.2	160408
170	1156	154.7	178836	170	1105	154.7	170946
180	1224	155.1	189870	180	1170	155.1	181493
190	1292	155.5	200912	190	1235	155.5	192048
200	1360	155.9	211960	200	1300	155.9	202609
** 207.9	1414	156.1	220685	** 207.9	1351	156.1	210949
210	1428	155.5	222020	210	1365	155.5	212225
220	1496	152.7	228369	220	1430	152.7	218294
230	1564	150.1	234720	230	1495	150.1	224364
* 239.4	1628	147.9	240688	* 239.4	1556	147.9	230069
240	1632	147.9	241362	240	1560	147.9	230713
250	1700	148.5	252410	250	1625	148.5	241274
260	1768	149.0	263458	260	1690	149.0	251835
270	1836	149.5	274506	270	1755	149.5	262395
280	1904	150.0	285553	280	1820	150.0	272955
290	1972	150.4	296601	290	1885	150.4	283516
300	2040	150.8	307649	300	1950	150.8	294076
310	2108	151.2	318696	310	2015	151.2	304636
320	2176	151.5	329743	320	2080	151.5	315196
330	2244	151.9	340790	330	2145	151.9	325755
340	2312	152.2	351836	340	2210	152.2	336314
350	2380	152.5	362883	350	2275	152.5	346874
360	2448	152.7	373930	360	2340	152.7	357433
370	2516	153.0	384976	370	2405	153.0	367992
** 371.3	2525	153.0	386245	** 371.3	2414	153.0	369205
380	2584	152.1	392968	380	2470	152.1	375631
390	2652	151.1	400735	390	2535	151.1	383056
400	2720	150.2	408503	400	2600	150.2	390481
410	2788	149.3	416267	410	2665	149.3	397902
* 420.0	2856	148.5	423989	* 420.0	2730	148.5	405284
430	2924	148.8	435090	430	2795	148.8	415895
440	2992	149.1	446135	440	2860	149.1	426453
450	3060	149.4	457180	450	2925	149.4	437010
460	3128	149.7	468224	460	2990	149.7	447567
470	3196	150.0	479267	470	3055	150.0	458123
480	3264	150.2	490310	480	3120	150.2	468679
** 480.7	3269	150.2	491083	** 480.7	3125	150.2	469418

* Critical fuel quantities for most forward CG condition (includes zero fuel)

** Critical fuel quantities for most aft CG condition

Weights given are nominal weights at 15° C. (59° F)

For calculation purposes, use Weight and Moment values, (CGs shown are derived from Weight and Moment values and rounded to one decimal place).

412-FMS-65-5-9-1

Table 5-9. Usable fuel – longitudinal – w/309.3 litre aux fuel in LH and RH position (Metric)

USABLE FUEL LOADING TABLE - LONGITUDINAL CG (METRIC)							
Basic with LH and RH 309.3 litre Aux Tank - 1819.6 litres							
Jet A, A-1, JP-5, JP-8 (0.815 kg/l)				Jet B, JP-4 (0.799 kg/l)			
Quantity (litres)	Weight (kg)	CG (mm)	Moment (mm-kg)	Quantity (litres)	Weight (kg)	CG (mm)	Moment (mm-kg)
40	32.6	3520	114758	40	32.0	3520	112505
80	65.2	3534	230430	80	63.9	3534	225907
120	97.8	3541	346339	120	95.9	3541	339540
160	130.4	3545	462247	160	127.8	3545	453172
200	163.0	3547	578156	200	159.8	3547	566805
* 237.0	193.1	3548	685353	* 237.0	189.3	3548	671898
240	195.6	3560	696327	240	191.8	3560	682657
280	228.2	3674	838476	280	223.7	3674	822015
320	260.8	3740	975359	320	255.7	3740	956210
360	293.4	3781	1109200	360	287.6	3781	1087424
400	326.0	3814	1243330	400	319.6	3814	1218921
440	358.6	3841	1377534	440	351.6	3841	1350490
480	391.2	3864	1511743	480	383.5	3864	1482065
520	423.8	3884	1645954	520	415.5	3884	1613641
560	456.4	3900	1780168	560	447.4	3900	1745220
600	489.0	3915	1914384	600	479.4	3915	1876801
640	521.6	3928	2048601	640	511.4	3928	2008383
680	554.2	3939	2182925	680	543.3	3939	2140070
720	586.8	3949	2317355	720	575.3	3949	2271861
760	619.4	3958	2451861	760	607.2	3958	2403727
** 787.0	641.4	3964	2542574	** 787.0	628.8	3964	2492658
800	652.0	3938	2567749	800	639.2	3938	2517339
840	684.6	3864	2645028	840	671.2	3864	2593101
880	717.2	3796	2722362	880	703.1	3796	2668917
* 906.2	738.5	3755	2773026	* 906.2	724.0	3755	2718586
920	749.8	3760	2819463	920	735.1	3760	2764112
960	782.4	3776	2953967	960	767.0	3776	2895975
1000	815.0	3790	3088469	1000	799.0	3790	3027837
1040	847.6	3802	3222970	1040	831.0	3802	3159697
1080	880.2	3814	3357469	1080	862.9	3814	3291556
1120	912.8	3826	3491967	1120	894.9	3826	3423413
1160	945.4	3836	3626463	1160	926.8	3836	3555268
1200	978.0	3846	3760955	1200	958.8	3846	3687121
1240	1010.6	3855	3895443	1240	990.8	3855	3818968
1280	1043.2	3863	4029929	1280	1022.7	3863	3950814
1320	1075.8	3871	4164415	1320	1054.7	3871	4082660
1360	1108.4	3878	4298900	1360	1086.6	3878	4214505
1400	1141.0	3886	4433380	1400	1118.6	3886	4346345
** 1405.7	1145.6	3884	4450035	** 1405.7	1123.1	3884	4362672
1440	1173.6	3861	4531132	1440	1150.6	3861	4442177
1480	1206.2	3835	4625698	1480	1182.5	3835	4534887
1520	1238.8	3810	4720265	1520	1214.5	3810	4627597
1560	1271.4	3787	4814772	1560	1246.4	3787	4720249
* 1589.7	1295.6	3770	4884891	* 1589.7	1270.2	3770	4788991
1600	1304.0	3773	4919568	1600	1278.4	3773	4822988
1640	1336.6	3781	5054044	1640	1310.4	3781	4954823
1680	1369.2	3789	5188511	1680	1342.3	3789	5086651
1720	1401.8	3797	5322967	1720	1374.3	3797	5218467
1760	1434.4	3805	5457415	1760	1406.2	3805	5350276
1800	1467.0	3812	5591861	1800	1438.2	3812	5482082
** 1819.6	1483.0	3815	5657896	** 1819.6	1453.9	3815	5546821

* Critical fuel quantities for most forward CG condition (includes zero fuel)

** Critical fuel quantities for most aft CG condition

Weights given are nominal weights at 15° C. (59° F)

For calculation purposes, use Weight and Moment values, (CGs shown are derived from Weight and Moment values and rounded to whole numbers).

412-FMS-65-5-9-2

Table 5-10. Usable fuel – lateral – w/163.4 gal aux fuel in LH and RH position (English)

USABLE FUEL LOADING TABLE - LATERAL CG (ENGLISH)							
Basic with LH and RH 81.7 US Gal Aux Tank - 480.7 US Gal							
Jet A, A-1, JP-5, JP-8 (6.8 lb /Gal)				Jet B, JP-4 (6.5 lb./Gal)			
Quantity (US Gal)	Weight (lb.)	CG (inches)	Moment (in.-lb.)	Quantity (US Gal)	Weight (lb.)	CG (inches)	Moment (in.-lb.)
10	68	0.0	0	10	65	0.0	0
20	136	0.0	0	20	130	0.0	0
30	204	0.0	0	30	195	0.0	0
40	272	0.0	0	40	260	0.0	0
50	340	0.0	0	50	325	0.0	0
60	408	0.0	0	60	390	0.0	0
62.6	426	0.0	0	62.6	407	0.0	0
70	476	0.0	-12	70	455	0.0	-11
80	544	0.0	-13	80	520	0.0	-13
90	612	0.0	-13	90	585	0.0	-13
100	680	0.0	-13	100	650	0.0	-13
110	748	0.0	-13	110	715	0.0	-13
120	816	0.0	-13	120	780	0.0	-13
130	884	0.0	-13	130	845	0.0	-13
140	952	0.0	-13	140	910	0.0	-13
150	1020	0.0	-13	150	975	0.0	-13
160	1088	0.0	-13	160	1040	0.0	-13
170	1156	0.0	-13	170	1105	0.0	-13
180	1224	0.0	-13	180	1170	0.0	-13
190	1292	0.0	-13	190	1235	0.0	-13
200	1360	0.0	-13	200	1300	0.0	-13
207.9	1414	0.0	-13	207.9	1351	0.0	-13
210	1428	-0.2	-257	210	1365	-0.2	-246
220	1496	-0.3	-419	220	1430	-0.3	-401
230	1564	-0.4	-554	230	1495	-0.4	-529
* 239.4	1628	-0.4	-677	* 239.4	1556	-0.4	-647
240	1632	-0.4	-677	240	1560	-0.4	-647
250	1700	-0.4	-677	250	1625	-0.4	-647
260	1768	-0.4	-677	260	1690	-0.4	-647
270	1836	-0.4	-677	270	1755	-0.4	-647
280	1904	-0.4	-677	280	1820	-0.4	-647
290	1972	-0.3	-677	290	1885	-0.3	-647
300	2040	-0.3	-677	300	1950	-0.3	-647
310	2108	-0.3	-677	310	2015	-0.3	-647
320	2176	-0.3	-677	320	2080	-0.3	-647
330	2244	-0.3	-677	330	2145	-0.3	-647
340	2312	-0.3	-677	340	2210	-0.3	-647
350	2380	-0.3	-677	350	2275	-0.3	-647
360	2448	-0.3	-677	360	2340	-0.3	-647
370	2516	-0.3	-677	370	2405	-0.3	-647
371.3	2525	-0.3	-677	371.3	2414	-0.3	-647
380	2584	-0.3	-677	380	2470	-0.3	-647
390	2652	-0.3	-677	390	2535	-0.3	-647
400	2720	-0.2	-677	400	2600	-0.2	-647
410	2788	-0.2	-677	410	2665	-0.2	-647
420.0	2856	-0.2	-677	420.0	2730	-0.2	-647
430	2924	-0.2	-677	430	2795	-0.2	-647
440	2992	-0.2	-677	440	2860	-0.2	-648
450	3060	-0.2	-680	450	2925	-0.2	-650
460	3128	-0.2	-685	460	2990	-0.2	-655
470	3196	-0.2	-690	470	3055	-0.2	-660
480	3264	-0.2	-696	480	3120	-0.2	-665
480.7	3269	-0.2	-696	480.7	3125	-0.2	-665

* Critical fuel quantity for most lateral CG condition

Weights given are nominal weights at 15° C. (59° F)

For calculation purposes, use Weight and Moment values. (CGs shown are derived from Weight and Moment values and rounded to one decimal place).

412-FMS-65-5-10-1

Table 5-10. Usable fuel – lateral – w/618.5 litre aux fuel in LH and RH position (Metric)

USABLE FUEL LOADING TABLE - LATERAL CG (METRIC)							
Basic with LH and RH 309.3 litre Aux Tank - 1819.6 litres							
Jet A, A-1, JP-5, JP-8 (0.815 kg/l)				Jet B, JP-4 (0.799 kg/l)			
Quantity (litres)	Weight (kg)	CG (mm)	Moment (mm-kg)	Quantity (litres)	Weight (kg)	CG (mm)	Moment (mm-kg)
40	32.6	0	0	40	32.0	0	0
80	65.2	0	0	80	63.9	0	0
120	97.8	0	0	120	95.9	0	0
160	130.4	0	0	160	127.8	0	0
200	163.0	0	0	200	159.8	0	0
237.0	193.1	0	0	237.0	189.3	0	0
240	195.6	0	-24	240	191.8	0	-24
280	228.2	-1	-147	280	223.7	-1	-144
320	260.8	-1	-152	320	255.7	-1	-149
360	293.4	-1	-152	360	287.6	-1	-149
400	326.0	0	-152	400	319.6	0	-149
440	358.6	0	-152	440	351.6	0	-149
480	391.2	0	-152	480	383.5	0	-149
520	423.8	0	-152	520	415.5	0	-149
560	456.4	0	-152	560	447.4	0	-149
600	489.0	0	-152	600	479.4	0	-149
640	521.6	0	-152	640	511.4	0	-149
680	554.2	0	-152	680	543.3	0	-149
720	586.8	0	-152	720	575.3	0	-149
760	619.4	0	-152	760	607.2	0	-149
787.0	641.4	0	-152	787.0	628.8	0	-149
800	652.0	-5	-3584	800	639.2	-5	-3514
840	684.6	-7	-5111	840	671.2	-7	-5010
880	717.2	-9	-6749	880	703.1	-9	-6617
* 906.2	738.5	-11	-7796	* 906.2	724.0	-11	-7643
920	749.8	-10	-7796	920	735.1	-10	-7643
960	782.4	-10	-7796	960	767.0	-10	-7643
1000	815.0	-10	-7796	1000	799.0	-10	-7643
1040	847.6	-9	-7796	1040	831.0	-9	-7643
1080	880.2	-9	-7796	1080	862.9	-9	-7643
1120	912.8	-9	-7796	1120	894.9	-9	-7643
1160	945.4	-8	-7796	1160	926.8	-8	-7643
1200	978.0	-8	-7796	1200	958.8	-8	-7643
1240	1010.6	-8	-7796	1240	990.8	-8	-7643
1280	1043.2	-7	-7796	1280	1022.7	-7	-7643
1320	1075.8	-7	-7796	1320	1054.7	-7	-7643
1360	1108.4	-7	-7796	1360	1086.6	-7	-7643
1400	1141.0	-7	-7796	1400	1118.6	-7	-7643
1405.7	1145.6	-7	-7796	1405.7	1123.1	-7	-7643
1440	1173.6	-7	-7796	1440	1150.6	-7	-7643
1480	1206.2	-6	-7796	1480	1182.5	-6	-7643
1520	1238.8	-6	-7796	1520	1214.5	-6	-7643
1560	1271.4	-6	-7796	1560	1246.4	-6	-7643
1589.7	1295.6	-6	-7796	1589.7	1270.2	-6	-7643
1600	1304.0	-6	-7796	1600	1278.4	-6	-7643
1640	1336.6	-6	-7796	1640	1310.4	-6	-7643
1680	1369.2	-6	-7810	1680	1342.3	-6	-7657
1720	1401.8	-6	-7856	1720	1374.3	-6	-7702
1760	1434.4	-6	-7919	1760	1406.2	-6	-7764
1800	1467.0	-5	-7985	1800	1438.2	-5	-7828
1819.6	1483.0	-5	-8018	1819.6	1453.9	-5	-7860

* Critical fuel quantity for most lateral CG condition

Weights given are nominal weights at 15° C. (59° F)

For calculation purposes, use Weight and Moment values. (CGs shown are derived from Weight and Moment values and rounded to whole numbers).

412-FMS-65-5-10-2

Table 5-11. Usable fuel – longitudinal – w/81.7 gal aux fuel in LH position (English)

USABLE FUEL LOADING TABLE - LONGITUDINAL CG (ENGLISH)							
Basic with LH 81.7 US Gal Aux Tank - 399.0 US Gal							
Jet A, A-1, JP-5, JP-8 (6.8 lb./Gal)				Jet B, JP-4 (6.5 lb./Gal)			
Quantity (US Gal)	Weight (lb.)	CG (inches)	Moment (in.-lb.)	Quantity (US Gal)	Weight (lb.)	CG (inches)	Moment (in.-lb.)
10	68	138.6	9424	10	65	138.6	9009
20	136	139.1	18921	20	130	139.1	18086
30	204	139.4	28442	30	195	139.4	27187
40	272	139.6	37962	40	260	139.6	36288
50	340	139.7	47483	50	325	139.7	45388
60	408	139.7	57005	60	390	139.7	54490
* 62.6	426	139.7	59486	* 62.6	407	139.7	56862
70	476	143.4	68272	70	455	143.4	65260
80	544	146.8	79846	80	520	146.8	76324
90	612	149.0	91159	90	585	149.0	87137
100	680	150.7	102488	100	650	150.7	97967
110	748	152.2	113821	110	715	152.2	108799
120	816	153.4	125153	120	780	153.4	119632
130	884	154.4	136486	130	845	154.4	130465
140	952	155.3	147819	140	910	155.3	141297
150	1020	156.0	159155	150	975	156.0	152133
160	1088	156.7	170503	160	1040	156.7	162981
170	1156	157.3	181857	170	1105	157.3	173834
** 173.4	1179	157.5	185696	** 173.4	1127	157.5	177503
180	1224	155.1	189899	180	1170	155.1	181521
190	1292	151.9	196246	190	1235	151.9	187588
200	1360	149.0	202600	200	1300	149.0	193662
* 204.9	1393	147.7	205698	* 204.9	1332	147.7	196623
210	1428	148.1	211519	210	1365	148.1	202187
220	1496	149.0	222872	220	1430	149.0	213040
230	1564	149.8	234226	230	1495	149.8	223892
240	1632	150.5	245579	240	1560	150.5	234745
250	1700	151.1	256932	250	1625	151.1	245597
260	1768	151.7	268285	260	1690	151.7	256449
270	1836	152.3	279637	270	1755	152.3	267300
280	1904	152.8	290988	280	1820	152.8	278151
290	1972	153.3	302340	290	1885	153.3	289001
300	2040	153.8	313691	300	1950	153.8	299852
** 304.6	2071	153.9	318655	** 304.6	1980	153.9	304596
310	2108	153.2	322864	310	2015	153.2	308620
320	2176	151.9	330631	320	2080	151.9	316045
330	2244	150.8	338399	330	2145	150.8	323469
340	2312	149.7	346165	340	2210	149.7	330893
350	2380	148.7	353926	350	2275	148.7	338311
* 353.2	2402	148.4	356399	* 353.2	2296	148.4	340675
360	2448	148.7	364133	360	2340	148.7	348068
370	2516	149.2	375483	370	2405	149.2	358917
380	2584	149.7	386832	380	2470	149.7	369765
390	2652	150.1	398180	390	2535	150.1	380613
** 399.0	2713	150.5	408393	** 399.0	2594	150.5	390376

* Critical fuel quantities for most forward CG condition (includes zero fuel)

** Critical fuel quantities for most aft CG condition

Weights given are nominal weights at 15° C. (59° F)

For calculation purposes, use Weight and Moment values. (CGs shown are derived from Weight and Moment values and rounded to one decimal place).

412-FMS-65-5-11-1

Table 5-11. Usable fuel – longitudinal – w/309.3 litre aux fuel in LH position (Metric)

USABLE FUEL LOADING TABLE - LONGITUDINAL CG (METRIC)							
Basic with LH 309.3 litre Aux Tank - 1510.4 litres							
Jet A, A-1, JP-5, JP-8 (0.815 kg/l)				Jet B, JP-4 (0.799 kg/l)			
Quantity (litres)	Weight (kg)	CG (mm)	Moment (mm-kg)	Quantity (litres)	Weight (kg)	CG (mm)	Moment (mm-kg)
40	32.6	3520	114758	40	32.0	3520	112505
80	65.2	3534	230430	80	63.9	3534	225907
120	97.8	3541	346339	120	95.9	3541	339540
160	130.4	3545	462247	160	127.8	3545	453172
200	163.0	3547	578156	200	159.8	3547	566805
* 237.0	193.1	3548	685353	* 237.0	189.3	3548	671898
240	195.6	3560	696327	240	191.8	3560	682657
280	228.2	3680	839843	280	223.7	3680	823355
320	260.8	3754	979007	320	255.7	3754	959787
360	293.4	3807	1116853	360	287.6	3807	1094927
400	326.0	3849	1254810	400	319.6	3849	1230176
440	358.6	3884	1392777	440	351.6	3884	1365434
480	391.2	3913	1530743	480	383.5	3913	1500692
520	423.8	3937	1668711	520	415.5	3937	1635951
560	456.4	3959	1806684	560	447.4	3959	1771215
600	489.0	3977	1944827	600	479.4	3977	1906646
640	521.6	3994	2083059	640	511.4	3994	2042165
** 656.3	534.9	4000	2139449	** 656.3	524.4	4000	2097447
680	554.2	3943	2185216	680	543.3	3943	2142316
720	586.8	3856	2262497	720	575.3	3856	2218080
760	619.4	3778	2339857	760	607.2	3778	2293921
* 775.5	632.1	3750	2369900	* 775.5	619.6	3750	2323374
800	652.0	3765	2454459	800	639.2	3765	2406273
840	684.6	3787	2592684	840	671.2	3787	2541785
880	717.2	3808	2730905	880	703.1	3808	2677292
920	749.8	3827	2869123	920	735.1	3827	2812796
960	782.4	3844	3007337	960	767.0	3844	2948297
1000	815.0	3860	3145545	1000	799.0	3860	3083792
1040	847.6	3874	3283747	1040	831.0	3874	3219281
1080	880.2	3888	3421946	1080	862.9	3888	3354766
1120	912.8	3900	3560142	1120	894.9	3900	3490249
** 1153.0	939.7	3907	3671304	** 1153.0	921.2	3907	3599230
1160	945.4	3901	3687938	1160	926.8	3901	3615537
1200	978.0	3868	3782498	1200	958.8	3868	3708240
1240	1010.6	3836	3877064	1240	990.8	3836	3800950
1280	1043.2	3807	3971622	1280	1022.7	3807	3893652
1320	1075.8	3780	4066111	1320	1054.7	3780	3986286
* 1337.0	1089.6	3768	4106161	* 1337.0	1068.2	3768	4025549
1360	1108.4	3776	4185772	1360	1086.6	3776	4103598
1400	1141.0	3790	4323952	1400	1118.6	3790	4239064
1440	1173.6	3802	4462119	1440	1150.6	3802	4374519
1480	1206.2	3814	4600277	1480	1182.5	3814	4509964
** 1510.4	1231.0	3822	4705202	** 1510.4	1206.8	3822	4612830

* Critical fuel quantities for most forward CG condition (includes zero fuel)

** Critical fuel quantities for most aft CG condition

Weights given are nominal weights at 15° C. (59° F)

For calculation purposes, use Weight and Moment values, (CGs shown are derived from Weight and Moment values and rounded to whole numbers).

Table 5-12. Usable fuel – lateral – w/81.7 gal aux fuel in LH position (English)

USABLE FUEL LOADING TABLE - LATERAL CG (ENGLISH)							
Basic with LH 81.7 US Gal Aux Tank - 399.0 US Gal							
Jet A, A-1, JP-5, JP-8 (6.8 lb./Gal)				Jet B, JP-4 (6.5 lb./Gal)			
Quantity (US Gal)	Weight (lb.)	CG (inches)	Moment (in.-lb.)	Quantity (US Gal)	Weight (lb.)	CG (inches)	Moment (in.-lb.)
10	68	0.0	0	10	65	0.0	0
20	136	0.0	0	20	130	0.0	0
30	204	0.0	0	30	195	0.0	0
40	272	0.0	0	40	260	0.0	0
50	340	0.0	0	50	325	0.0	0
60	408	0.0	0	60	390	0.0	0
62.6	426	0.0	0	62.6	407	0.0	0
70	476	-0.2	-88	70	455	-0.2	-85
80	544	-1.0	-565	80	520	-1.0	-540
90	612	-2.1	-1262	90	585	-2.1	-1206
100	680	-2.8	-1928	100	650	-2.8	-1843
110	748	-3.5	-2587	110	715	-3.5	-2473
120	816	-4.0	-3245	120	780	-4.0	-3102
130	884	-4.4	-3903	130	845	-4.4	-3730
140	952	-4.8	-4559	140	910	-4.8	-4358
150	1020	-5.1	-5213	150	975	-5.1	-4983
160	1088	-5.4	-5855	160	1040	-5.4	-5596
170	1156	-5.6	-6491	170	1105	-5.6	-6205
173.4	1179	-5.7	-6706	173.4	1127	-5.7	-6410
180	1224	-5.8	-7042	180	1170	-5.8	-6732
190	1292	-5.5	-7170	190	1235	-5.5	-6853
200	1360	-5.4	-7305	200	1300	-5.4	-6983
204.9	1393	-5.3	-7369	204.9	1332	-5.3	-7044
210	1428	-5.4	-7695	210	1365	-5.4	-7356
220	1496	-5.6	-8331	220	1430	-5.6	-7963
230	1564	-5.7	-8966	230	1495	-5.7	-8571
240	1632	-5.9	-9602	240	1560	-5.9	-9178
250	1700	-6.0	-10237	250	1625	-6.0	-9785
260	1768	-6.1	-10872	260	1690	-6.1	-10392
270	1836	-6.3	-11507	270	1755	-6.3	-10999
280	1904	-6.4	-12142	280	1820	-6.4	-11606
290	1972	-6.5	-12777	290	1885	-6.5	-12213
300	2040	-6.6	-13411	300	1950	-6.6	-12820
* 304.6	2071	-6.6	-13661	* 304.6	1980	-6.6	-13058
310	2108	-6.5	-13661	310	2015	-6.5	-13058
320	2176	-6.3	-13661	320	2080	-6.3	-13058
330	2244	-6.1	-13661	330	2145	-6.1	-13058
340	2312	-5.9	-13661	340	2210	-5.9	-13058
350	2380	-5.7	-13661	350	2275	-5.7	-13058
353.2	2402	-5.7	-13661	353.2	2296	-5.7	-13058
360	2448	-5.8	-14093	360	2340	-5.8	-13471
370	2516	-5.9	-14729	370	2405	-5.9	-14080
380	2584	-5.9	-15370	380	2470	-5.9	-14692
390	2652	-6.0	-16014	390	2535	-6.0	-15308
399.0	2713	-6.1	-16594	399.0	2594	-6.1	-15862

* Critical fuel quantity for most lateral CG condition

Weights given are nominal weights at 15° C. (59° F)

For calculation purposes, use Weight and Moment values, (CGs shown are derived from Weight and Moment values and rounded to one decimal place).

412-FMS-65-5-12-1

Table 5-12. Usable fuel – lateral – w/309.3 litre aux fuel in LH position (Metric)

USABLE FUEL LOADING TABLE - LATERAL CG (METRIC)							
Basic with LH 309.3 litre Aux Tank - 1510.4 litres							
Jet A, A-1, JP-5, JP-8 (0.815 kg/l)				Jet B, JP-4 (0.799 kg/l)			
Quantity (litres)	Weight (kg)	CG (mm)	Moment (mm-kg)	Quantity (litres)	Weight (kg)	CG (mm)	Moment (mm-kg)
40	32.6	0	0	40	32.0	0	0
80	65.2	0	0	80	63.9	0	0
120	97.8	0	0	120	95.9	0	0
160	130.4	0	0	160	127.8	0	0
200	163.0	0	0	200	159.8	0	0
237.0	193.1	0	0	237.0	189.3	0	0
240	195.6	0	-24	240	191.8	0	-24
280	228.2	-13	-2885	280	223.7	-13	-2829
320	260.8	-39	-10202	320	255.7	-39	-10002
360	293.4	-63	-18472	360	287.6	-63	-18109
400	326.0	-81	-26520	400	319.6	-81	-25999
440	358.6	-96	-34536	440	351.6	-96	-33858
480	391.2	-109	-42542	480	383.5	-109	-41707
520	423.8	-119	-50541	520	415.5	-119	-49549
560	456.4	-128	-58527	560	447.4	-128	-57378
600	489.0	-136	-66357	600	479.4	-136	-65054
640	521.6	-142	-74102	640	511.4	-142	-72647
656.3	534.9	-144	-77260	656.3	524.4	-144	-75743
680	554.2	-146	-81083	680	543.3	-146	-79491
720	586.8	-141	-82633	720	575.3	-141	-81011
760	619.4	-136	-84280	760	607.2	-136	-82625
775.5	632.1	-134	-84904	775.5	619.6	-134	-83237
800	652.0	-137	-89639	800	639.2	-137	-87879
840	684.6	-142	-97377	840	671.2	-142	-95466
880	717.2	-147	-105113	880	703.1	-147	-103049
920	749.8	-151	-112846	920	735.1	-151	-110631
960	782.4	-154	-120578	960	767.0	-154	-118210
1000	815.0	-157	-128309	1000	799.0	-157	-125790
1040	847.6	-161	-136042	1040	831.0	-161	-133371
1080	880.2	-163	-143771	1080	862.9	-163	-140949
1120	912.8	-166	-151498	1120	894.9	-166	-148524
* 1153.0	939.7	-167	-157387	* 1153.0	921.2	-167	-154297
1160	945.4	-166	-157387	1160	926.8	-166	-154297
1200	978.0	-161	-157387	1200	958.8	-161	-154297
1240	1010.6	-156	-157387	1240	990.8	-156	-154297
1280	1043.2	-151	-157387	1280	1022.7	-151	-154297
1320	1075.8	-146	-157387	1320	1054.7	-146	-154297
1337.0	1089.6	-144	-157387	1337.0	1068.2	-144	-154297
1360	1108.4	-146	-161840	1360	1086.6	-146	-158663
1400	1141.0	-149	-169585	1400	1118.6	-149	-166256
1440	1173.6	-151	-177387	1440	1150.6	-151	-173905
1480	1206.2	-154	-185225	1480	1182.5	-154	-181589
1510.4	1231.0	-155	-191181	1510.4	1206.8	-155	-187427

* Critical fuel quantity for most lateral CG condition

Weights given are nominal weights at 15° C. (59° F)

For calculation purposes, use Weight and Moment values, (CGs shown are derived from Weight and Moment values and rounded to whole numbers).

412-FMS-65-5-12-2

Table 5-13. Usable fuel – longitudinal – w/81.7 gal aux fuel in RH position (English)

USABLE FUEL LOADING TABLE - LONGITUDINAL CG (ENGLISH)							
Basic with RH 81.7 US Gal Aux Tank - 399.0 US Gal							
Jet A, A-1, JP-5, JP-8 (6.8 lb./Gal)				Jet B, JP-4 (6.5 lb./Gal)			
Quantity (US Gal)	Weight (lb.)	CG (inches)	Moment (in-lb.)	Quantity (US Gal)	Weight (lb.)	CG (inches)	Moment (in-lb.)
10	68	138.6	9424	10	65	138.6	9009
20	136	139.1	18921	20	130	139.1	18086
30	204	139.4	28442	30	195	139.4	27187
40	272	139.6	37962	40	260	139.6	36288
50	340	139.7	47483	50	325	139.7	45388
60	408	139.7	57005	60	390	139.7	54490
* 62.6	426	139.7	59486	* 62.6	407	139.7	56862
70	476	143.4	68272	70	455	143.4	65260
80	544	146.8	79846	80	520	146.8	76324
90	612	149.0	91159	90	585	149.0	87137
100	680	150.7	102488	100	650	150.7	97967
110	748	152.2	113821	110	715	152.2	108799
120	816	153.4	125153	120	780	153.4	119632
130	884	154.4	136486	130	845	154.4	130465
140	952	155.3	147819	140	910	155.3	141297
150	1020	156.0	159155	150	975	156.0	152133
160	1088	156.7	170503	160	1040	156.7	162981
170	1156	157.3	181857	170	1105	157.3	173834
** 173.4	1179	157.5	185696	** 173.4	1127	157.5	177503
180	1224	155.1	189899	180	1170	155.1	181521
190	1292	151.9	196246	190	1235	151.9	187588
200	1360	149.0	202600	200	1300	149.0	193662
* 204.9	1393	147.7	205698	* 204.9	1332	147.7	196623
210	1428	148.1	211519	210	1365	148.1	202187
220	1496	149.0	222872	220	1430	149.0	213040
230	1564	149.8	234226	230	1495	149.8	223892
240	1632	150.5	245579	240	1560	150.5	234745
250	1700	151.1	256932	250	1625	151.1	245597
260	1768	151.7	268285	260	1690	151.7	256449
270	1836	152.3	279637	270	1755	152.3	267300
280	1904	152.8	290988	280	1820	152.8	278151
290	1972	153.3	302340	290	1885	153.3	289001
300	2040	153.8	313691	300	1950	153.8	299852
** 304.6	2071	153.9	318655	** 304.6	1980	153.9	304596
310	2108	153.2	322864	310	2015	153.2	308620
320	2176	151.9	330631	320	2080	151.9	316045
330	2244	150.8	338399	330	2145	150.8	323469
340	2312	149.7	346165	340	2210	149.7	330893
350	2380	148.7	353926	350	2275	148.7	338311
* 353.2	2402	148.4	356399	* 353.2	2296	148.4	340675
360	2448	148.7	364133	360	2340	148.7	348068
370	2516	149.2	375483	370	2405	149.2	358917
380	2584	149.7	386832	380	2470	149.7	369765
390	2652	150.1	398180	390	2535	150.1	380613
** 399.0	2713	150.5	408393	** 399.0	2594	150.5	390376

* Critical fuel quantities for most forward CG condition (includes zero fuel)

** Critical fuel quantities for most aft CG condition

Weights given are nominal weights at 15° C. (59° F)

For calculation purposes, use Weight and Moment values, (CGs shown are derived from Weight and Moment values and rounded to one decimal place).

412-FMS-65-5-13-1

Table 5-13. Usable fuel – longitudinal – w/309.3 litre aux fuel in RH position (Metric)

USABLE FUEL LOADING TABLE - LONGITUDINAL CG (METRIC)							
Basic with RH 309.3 litre Aux Tank - 1510.4 litres							
Jet A, A-1, JP-5, JP-8 (0.815 kg/l)				Jet B, JP-4 (0.799 kg/l)			
Quantity (litres)	Weight (kg)	CG (mm)	Moment (mm-kg)	Quantity (litres)	Weight (kg)	CG (mm)	Moment (mm-kg)
40	32.6	3520	114758	40	32.0	3520	112505
80	65.2	3534	230430	80	63.9	3534	225907
120	97.8	3541	346339	120	95.9	3541	339540
160	130.4	3545	462247	160	127.8	3545	453172
200	163.0	3547	578156	200	159.8	3547	566805
* 237.0	193.1	3548	685353	* 237.0	189.3	3548	671898
240	195.6	3560	696327	240	191.8	3560	682657
280	228.2	3680	839843	280	223.7	3680	823355
320	260.8	3754	979007	320	255.7	3754	959787
360	293.4	3807	1116853	360	287.6	3807	1094927
400	326.0	3849	1254810	400	319.6	3849	1230176
440	358.6	3884	1392777	440	351.6	3884	1365434
480	391.2	3913	1530743	480	383.5	3913	1500692
520	423.8	3937	1668711	520	415.5	3937	1635951
560	456.4	3959	1806684	560	447.4	3959	1771215
600	489.0	3977	1944827	600	479.4	3977	1906646
640	521.6	3994	2083059	640	511.4	3994	2042165
** 656.3	534.9	4000	2139449	** 656.3	524.4	4000	2097447
680	554.2	3943	2185216	680	543.3	3943	2142316
720	586.8	3856	2262497	720	575.3	3856	2218080
760	619.4	3778	2339857	760	607.2	3778	2293921
* 775.5	632.1	3750	2369900	* 775.5	619.6	3750	2323374
800	652.0	3765	2454459	800	639.2	3765	2406273
840	684.6	3787	2592684	840	671.2	3787	2541785
880	717.2	3808	2730905	880	703.1	3808	2677292
920	749.8	3827	2869123	920	735.1	3827	2812796
960	782.4	3844	3007337	960	767.0	3844	2948297
1000	815.0	3860	3145545	1000	799.0	3860	3083792
1040	847.6	3874	3283747	1040	831.0	3874	3219281
1080	880.2	3888	3421946	1080	862.9	3888	3354766
1120	912.8	3900	3560142	1120	894.9	3900	3490249
** 1153.0	939.7	3907	3671304	** 1153.0	921.2	3907	3599230
1160	945.4	3901	3687938	1160	926.8	3901	3615537
1200	978.0	3868	3782498	1200	958.8	3868	3708240
1240	1010.6	3836	3877064	1240	990.8	3836	3800950
1280	1043.2	3807	3971622	1280	1022.7	3807	3893652
1320	1075.8	3780	4066111	1320	1054.7	3780	3986286
* 1337.0	1089.6	3768	4106161	* 1337.0	1068.2	3768	4025549
1360	1108.4	3776	4185772	1360	1086.6	3776	4103598
1400	1141.0	3790	4323952	1400	1118.6	3790	4239064
1440	1173.6	3802	4462119	1440	1150.6	3802	4374519
1480	1206.2	3814	4600277	1480	1182.5	3814	4509964
** 1510.4	1231.0	3822	4705202	** 1510.4	1206.8	3822	4612830

* Critical fuel quantities for most forward CG condition (includes zero fuel)

** Critical fuel quantities for most aft CG condition

Weights given are nominal weights at 15° C. (59° F)

For calculation purposes, use Weight and Moment values, (CGs shown are derived from Weight and Moment values and rounded to one decimal place).

412-FMS-65-5-13-2

Table 5-14. Usable fuel – lateral – w/81.7 gal aux fuel in RH position (English)

USABLE FUEL LOADING TABLE - LATERAL CG (ENGLISH)							
Basic with RH 81.7 US Gal Aux Tank - 399.0 US Gal							
Jet A, A-1, JP-5, JP-8 (6.8 lb./Gal)				Jet B, JP-4 (6.5 lb./Gal)			
Quantity (US Gal)	Weight (lb.)	CG (inches)	Moment (in-lb.)	Quantity (US Gal)	Weight (lb.)	CG (inches)	Moment (in-lb.)
10	68	0.0	0	10	65	0.0	0
20	136	0.0	0	20	130	0.0	0
30	204	0.0	0	30	195	0.0	0
40	272	0.0	0	40	260	0.0	0
50	340	0.0	0	50	325	0.0	0
60	408	0.0	0	60	390	0.0	0
62.6	426	0.0	0	62.6	407	0.0	0
70	476	0.1	64	70	455	0.1	61
80	544	1.0	539	80	520	1.0	515
90	612	2.0	1235	90	585	2.0	1181
100	680	2.8	1902	100	650	2.8	1818
110	748	3.4	2561	110	715	3.4	2448
120	816	3.9	3219	120	780	3.9	3077
130	884	4.4	3876	130	845	4.4	3705
140	952	4.8	4533	140	910	4.8	4333
150	1020	5.1	5186	150	975	5.1	4958
160	1088	5.4	5828	160	1040	5.4	5571
170	1156	5.6	6465	170	1105	5.6	6179
173.4	1179	5.7	6680	173.4	1127	5.7	6385
180	1224	5.2	6343	180	1170	5.2	6063
190	1292	4.8	6216	190	1235	4.8	5942
200	1360	4.5	6080	200	1300	4.5	5812
204.9	1393	4.3	6016	204.9	1332	4.3	5751
210	1428	4.4	6342	210	1365	4.4	6062
220	1496	4.7	6978	220	1430	4.7	6670
230	1564	4.9	7613	230	1495	4.9	7277
240	1632	5.1	8248	240	1560	5.1	7884
250	1700	5.2	8883	250	1625	5.2	8492
260	1768	5.4	9518	260	1690	5.4	9099
270	1836	5.5	10154	270	1755	5.5	9706
280	1904	5.7	10789	280	1820	5.7	10313
290	1972	5.8	11423	290	1885	5.8	10919
300	2040	5.9	12058	300	1950	5.9	11526
* 304.6	2071	5.9	12307	* 304.6	1980	5.9	11764
310	2108	5.8	12307	310	2015	5.8	11764
320	2176	5.7	12307	320	2080	5.7	11764
330	2244	5.5	12307	330	2145	5.5	11764
340	2312	5.3	12307	340	2210	5.3	11764
350	2380	5.2	12307	350	2275	5.2	11764
353.2	2402	5.1	12307	353.2	2296	5.1	11764
360	2448	5.2	12740	360	2340	5.2	12178
370	2516	5.3	13374	370	2405	5.3	12784
380	2584	5.4	14005	380	2470	5.4	13388
390	2652	5.5	14635	390	2535	5.5	13989
399.0	2713	5.6	15202	399.0	2594	5.6	14531

* Critical fuel quantity for most lateral CG condition

Weights given are nominal weights at 15° C. (59° F)

For calculation purposes, use Weight and Moment values, (CGs shown are derived from Weight and Moment values and rounded to one decimal place).

412-FMS-65-5-14-1

Table 5-14. Usable fuel – lateral – w/309.3 litre aux fuel in RH position (Metric)

USABLE FUEL LOADING TABLE - LATERAL CG (METRIC)							
Basic with RH 309.3 litre Aux Tank - 1510.4 litres							
Jet A, A-1, JP-5, JP-8 (0.815 kg/l)				Jet B, JP-4 (0.799 kg/l)			
Quantity (litres)	Weight (kg)	CG (mm)	Moment (mm-kg)	Quantity (litres)	Weight (kg)	CG (mm)	Moment (mm-kg)
40	32.6	0	0	40	32.0	0	0
80	65.2	0	0	80	63.9	0	0
120	97.8	0	0	120	95.9	0	0
160	130.4	0	0	160	127.8	0	0
200	163.0	0	0	200	159.8	0	0
237.0	193.1	0	0	237.0	189.3	0	0
240	195.6	0	-24	240	191.8	0	-24
280	228.2	11	2589	280	223.7	11	2538
320	260.8	38	9898	320	255.7	38	9704
360	293.4	62	18168	360	287.6	62	17811
400	326.0	80	26216	400	319.6	80	25701
440	358.6	95	34232	440	351.6	95	33560
480	391.2	108	42239	480	383.5	108	41409
520	423.8	119	50237	520	415.5	119	49251
560	456.4	128	58223	560	447.4	128	57080
600	489.0	135	66053	600	479.4	135	64756
640	521.6	141	73798	640	511.4	141	72349
656.3	534.9	144	76956	656.3	524.4	144	75446
680	554.2	132	73133	680	543.3	132	71698
720	586.8	122	71583	720	575.3	122	70178
760	619.4	113	69937	760	607.2	113	68564
775.5	632.1	110	69313	775.5	619.6	110	67952
800	652.0	114	74048	800	639.2	114	72594
840	684.6	119	81786	840	671.2	119	80181
880	717.2	125	89522	880	703.1	125	87764
920	749.8	130	97255	920	735.1	130	95345
960	782.4	134	104986	960	767.0	134	102925
1000	815.0	138	112718	1000	799.0	138	110505
1040	847.6	142	120451	1040	831.0	142	118086
1080	880.2	146	128180	1080	862.9	146	125663
1120	912.8	149	135907	1120	894.9	149	133239
* 1153.0	939.7	151	141796	* 1153.0	921.2	151	139012
1160	945.4	150	141796	1160	926.8	150	139012
1200	978.0	145	141796	1200	958.8	145	139012
1240	1010.6	140	141796	1240	990.8	140	139012
1280	1043.2	136	141796	1280	1022.7	136	139012
1320	1075.8	132	141796	1320	1054.7	132	139012
1337.0	1089.6	130	141796	1337.0	1068.2	130	139012
1360	1108.4	132	146249	1360	1086.6	132	143378
1400	1141.0	135	153970	1400	1118.6	135	150947
1440	1173.6	138	161656	1440	1150.6	138	158482
1480	1206.2	140	169323	1480	1182.5	140	165999
1510.4	1231.0	142	175145	1510.4	1206.8	142	171707

* Critical fuel quantity for most lateral CG condition

Weights given are nominal weights at 15° C. (59° F)

For calculation purposes, use Weight and Moment values, (CGs shown are derived from Weight and Moment values and rounded to one decimal place).

412-FMS-65-5-14-2



ROTORCRAFT FLIGHT MANUAL

SUPPLEMENT PT6T-3BF ENGINE (30 MINUTE OEI RATING)

412-706-054
S/N 33001 — 33107
33108 — 33213
AND
36001 — 36019
CERTIFIED
18 DECEMBER 1998

This supplement shall be attached to Model 412 Flight Manual (BHT-412-FM-1, BHT-412-FM-2, or BHT-412-FMS-19.1) when PT6T-3BF Engine, 30 minute OEI rating kit is installed.

Information contained herein supplements information of basic Flight Manual. For Limitations, Procedures, and Performance Data not contained in this supplement, or other applicable supplements; consult basic Flight Manual.

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18 DECEMBER 1998

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MANAGER

ROTORCRAFT CERTIFICATION OFFICE
FEDERAL AVIATION ADMINISTRATION
FT. WORTH, TX 76193-0170

GENERAL INFORMATION

The PT6T-3B engine, in accordance with 412-706-054, may be redesignated as PT6T-3BF to offer an increased 30 minute OEI rating. This 30 minute OEI rating represents improved OEI

capabilities while maximum continuous OEI performance is reduced from that presented in the basic Flight Manual or applicable supplement.

Section 1

LIMITATIONS

INTRODUCTION

PT6T-3BF engine offers an increased 30 minute OEI rating.

WEIGHT

Actual weight changes shall be determined after kit is installed and ballast readjusted, if necessary, to return empty weight CG to within allowable limits.

POWER PLANT

Pratt and Whitney Aircraft of Canada, Ltd.
PT6T-3BF.

NOTE

Operation in an OEI range is intended for emergency use only when one engine becomes inoperative due to an actual malfunction.

Anytime an engine is operated in an OEI range, an entry shall be made in the helicopter logbook detailing the extent of operation in excess of twin engine takeoff power limits. This does not apply to approved ITT limits for starting or for power assurance checks.

GAS PRODUCER RPM (N₁)

NOTE

Gas producer indicator 212-075-037-113 (or equivalent) must be installed prior to or concurrent with kit.

TWIN ENGINE OPERATION

No change from basic manual.

ONE ENGINE INOPERATIVE (OEI)

Continuous OEI	101.8%
30 minute OEI range	101.8 to 103.4%
Maximum OEI (30 minute)	103.4%

INTERTURBINE TEMPERATURE

TWIN ENGINE OPERATION

Maximum continuous	765°C
5 minute range	765 to 810°C
Maximum	810°C
Maximum start (2 seconds maximum above 960°C)	1090°C

ONE ENGINE INOPERATIVE (OEI)

Maximum continuous OEI	810°C
30 minute OEI range	810 to 850°C
Maximum OEI	850°C

ENGINE OIL PRESSURE

No change from basic manual.

ENGINE OIL TEMPERATURE

Minimum	0°C
Continuous operation	0 to 115°C
Maximum for MIL-L-7808 oil	115°C
Maximum for MIL-L-23699 oil	120°C
Maximum for DOD-L-85734 oil	120°C

COMBINING GEARBOX OIL PRESSURE

No change from basic manual.

COMBINING GEARBOX OIL TEMPERATURE

Minimum	0°C
Continuous operation	0 to 115°C
Maximum for MIL-L-7808 oil	115°C
Maximum for MIL-L-23699 oil	120°C
Maximum for DOD-L-85734 oil	120°C

INSTRUMENT MARKINGS AND PLACARDS

Refer to figure 1-1 for instrument range markings and figure 1-2 for placards and decals.



INTERTURBINE TEMPERATURE (ITT)

	300 TO 765°C	Continuous operation
	765 to 810°C	5 minute range
	810°C	Maximum
	810 TO 850°C	30 minute OEI range
	850°C	Maximum 30 minute OEI
	1090°C	Maximum for starting (2 seconds maximum above 960°C)

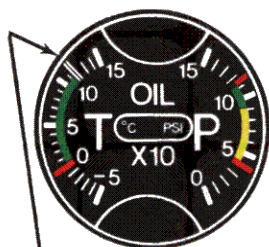


GAS PRODUCER RPM (N1)

	12%	Minimum for opening throttle during start
	61%	Idle RPM
	61 to 101.8%	Continuous operation
	101.8%	Maximum continuous/twin engine and OEI operations
	101.8% to 103.4%	30 minute OEI range
	103.4%	Maximum OEI (30 minutes)

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Figure 1-1. Instrument markings (Sheet 1 of 2)



ENGINE OIL TEMPERATURE

	0°C	Minimum
	0 TO 115°C	Continuous operation
	115°C	Maximum for MIL-L-7808 oil
	120°C	Maximum for MIL-L-23699 or DOD-L-85734 oil

ENGINE OIL PRESSURE

No change from basic manual

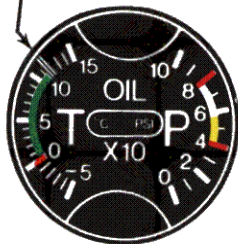


COMBINING GEARBOX OIL TEMPERATURE

	0°C	Minimum
	0 TO 115°C	Continuous operation
	115°C	Maximum for MIL-L-7808 oil
	120°C	Maximum for MIL-L-23699 or DOD-L-85734 oil

COMBINING GEARBOX OIL PRESSURE

No change from basic manual



412FS67.1-1-1-2

Figure 1-1 Instrument markings (Sheet 2 of 2)

TWIN & OEI 101.8%
30 MIN OEI 103.4%

LOCATION: INSTRUMENT PANEL

412FS67.1-1-2

Figure 1-2 Placards and decals

Section 2

NORMAL PROCEDURES

No change from basic manual.

Section 3

EMERGENCY/MALFUNCTION PROCEDURES

No change from basic manual.

Section 4

PERFORMANCE

INTRODUCTION

Performance data presented herein are derived from engine manufacturer's specification power for PT6T-3BF engine less installation losses.

CLIMB AND DESCENT

Refer to figure 4-1 for increased single engine rate of climb - 30 minute power for helicopters with maximum gross weight of 11,600 pounds (refer to BHT-412-FM-1).

Refer to figure 4-2 for single engine rate of climb - 30 minute power for helicopters with maximum gross weight of 11,900 pounds (refer to BHT-412-FM-2 and BHT-412-FMS-19.1).

PROBLEM:

What is maximum rate of climb for following conditions?

Helicopter gross weight — 10,500 pounds.

Pressure altitude — 5,500 feet.

OAT — 0°C.

EXAMPLE:

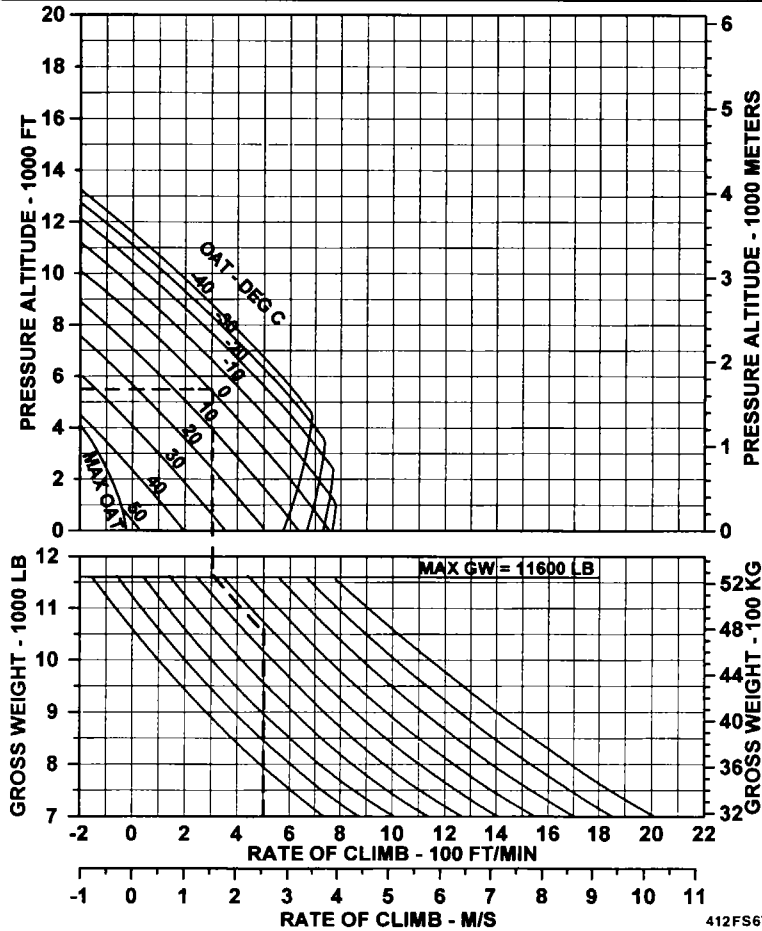
1. Enter rate of climb chart at 5,500 feet pressure altitude.
2. Move right, horizontally, to intersect 0° OAT line.
3. Descend vertically to intersect MAX GW line in lower portion of chart.
4. Follow curvature of trend lines to intersect actual helicopter gross weight line of 10,500 pounds.
5. Descend vertically to bottom of chart and read 500 feet per minute rate of climb.

SINGLE ENGINE RATE OF CLIMB

30 MINUTE POWER
ENGINE RPM 97%
GENERATOR 150 AMPS

70 KIAS
HEATER OFF

- WITH ALL DOORS OPEN OR REMOVED:
- 1. CLIMB SPEED IS 60 KIAS
 - 2. RATE OF CLIMB WILL DECREASE 275 FT/MIN



412FS67.1-4-1-1

Figure 4-1. Single engine rate of climb - 30 minute power (11,600 pounds) (Sheet 1 of 2)

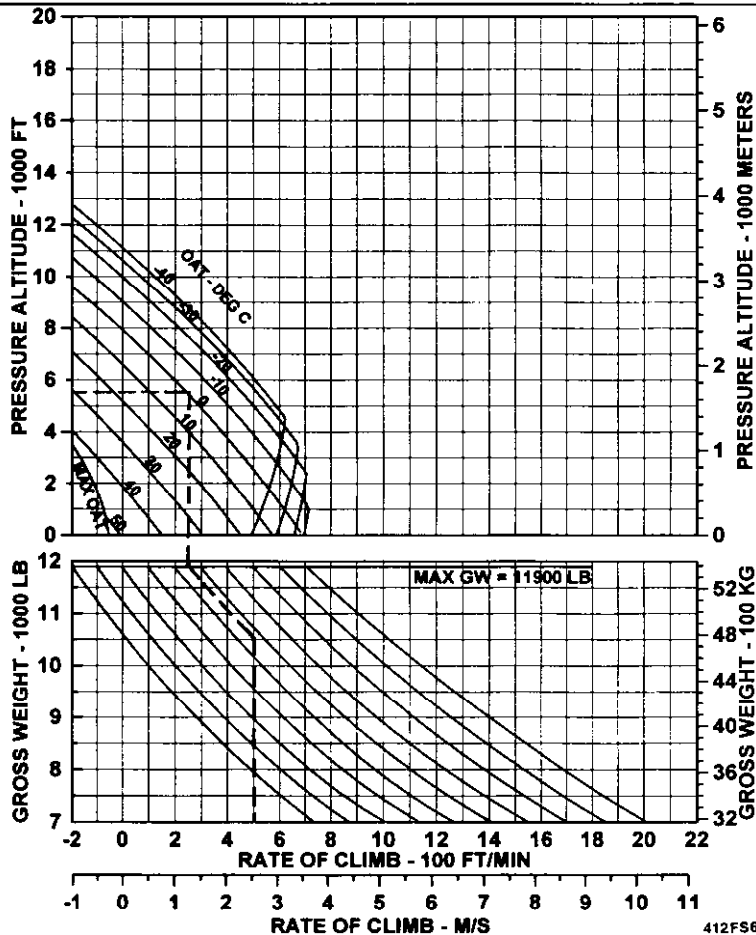
SINGLE ENGINE RATE OF CLIMB

30 MINUTE POWER
ENGINE RPM 97%
GENERATOR 150 AMPS

70 KIAS
HEATER OFF

WITH ALL DOORS OPEN OR REMOVED:

1. CLIMB SPEED IS 60 KIAS
2. RATE OF CLIMB WILL DECREASE 275 FT/MIN



412FS67.1-4-1-2

Figure 4-1. Single engine rate of climb - 30 minute power (11,900 pounds) (Sheet 2 of 2)

Section 6

CATEGORY A OPERATIONS

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Section 6

CATEGORY A OPERATIONS

GENERAL INFORMATION

ORGANIZATION

The information contained in this section is for category "A" operations. For limitations, normal procedures, emergency

and malfunction procedures, and performance data not contained in this section, consult the appropriate sections of this flight manual.

DEFINITIONS:

CATEGORY "A" TAKEOFF

- Operation of the helicopter in such a manner that if one engine fails at any time after the start of the takeoff, the helicopter can:

1. At or prior to CDP, return to and safely stop on the takeoff area; or
2. At or after CDP, climb out from point of failure and attain single engine forward flight.

CATEGORY "A" LANDING

- Operation of the helicopter in such a manner that if one engine fails at any time after the start of a landing approach the helicopter can:

1. At or after LDP, continue the approach and safely land and stop on the clear heliport; or
2. At or prior to LDP, climb out from point of failure and attain single engine forward flight.

CRITICAL DECISION POINT

- The last point in the takeoff path at which a rejected takeoff can be assured, and the first point at which a completed takeoff can be assured.

LANDING DECISION POINT

- That point on the landing profile after which the helicopter is committed to landing.

**COMPLETED TAKEOFF
DISTANCE REQUIRED**

- The horizontal distance from the start of the prescribed takeoff procedure to a point at least 35 feet above the takeoff surface where V_{TOSS} and a positive rate of climb are attained following an engine failure occurring at or after CDP.

**REJECTED TAKEOFF
DISTANCE REQUIRED**

- The horizontal distance from the start of the prescribed takeoff procedure to the point where the helicopter is brought to a safe stop on the designated surface following an engine failure occurring at or prior to CDP.

TAKEOFF FLIGHT PATH

- The distance traveled from where the aircraft reaches V_{TOSS} at or above 35 feet AGL to 1000 feet AGL.

TAKEOFF SAFETY SPEED

- The airspeed that will assure the required climb performance with one engine inoperative.

**LANDING DISTANCE
REQUIRED**

- The horizontal distance necessary to achieve a takeoff flight path at V_{TOSS} and an altitude of 35 feet or higher, with one engine inoperative at or prior to LDP; or the horizontal distance necessary to land the helicopter without further incident, with one engine inoperative at or after LDP.

BALKED LANDING

- The discontinuation of a landing approach and the initiation of a climbout. Category "A" balked landing capability following an engine failure is assured at or prior to LDP.

ABBREVIATIONS:

AGL — Above Ground Level
 CDP — Critical Decision Point
 CDT — Critical Decision Time
 GROC — Gross Rate of Climb
 LDP — Landing Decision Point

V_{MIN} IFR — Minimum Airspeed for IFR
 V_{TOSS}
 (V_2) — Takeoff Safety Speed
 V_Y — Best Rate of Climb Speed
 WAT — Weight-Altitude-Temperature

LIMITATIONS

TAKEOFF AND LANDING WEIGHT VS ALTITUDE LIMITATIONS

Refer to Gross Weight-Altitude-Ambient Air Temperature Limits Charts (figure 6-1). Charts designated part A may be used for gross weights to 10,000 pounds (4636 kg). Part B charts may be used for gross weights to 10,800 pounds (4899 kg). Part C charts may be used for gross 1 weights to 11,900 pounds (5398 kg).

Interpolation of data between charts for different parts is not permitted. Testing has not been conducted in areas between Parts A, B, and C.

ALTITUDE LIMIT FOR TAKEOFF AND LANDING

4000 feet pressure altitude.

CROSSWIND LIMITATIONS

The crosswind limit for takeoff and landing is 20 knots. Refer to the unfactored Headwind Component Chart in PERFORMANCE subsection.

Takeoff or landing downwind or with quartering tailwinds is prohibited.

CONFIGURATION

Standard landing gear or high skid gear with or without emergency floats (floats stowed).

WEIGHT — ALTITUDE — TEMPERATURE FOR TAKEOFF AND LANDING PART A

$V_{TOSS} = 40 \text{ KIAS}$

GW TO 10,000 LBS (4536 kg)

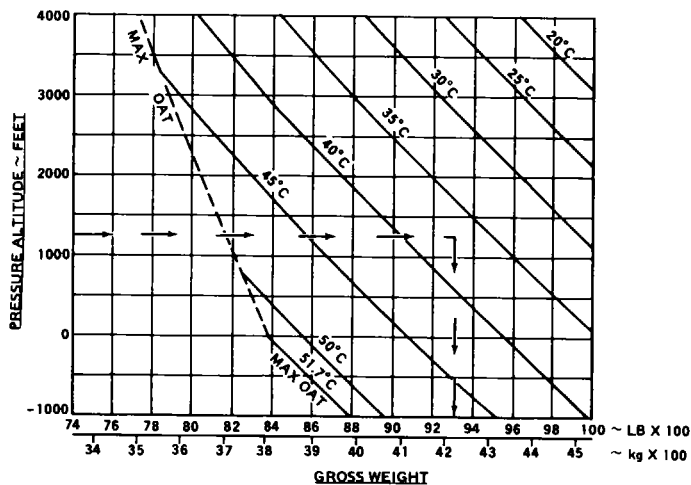


Figure 6-1. Gross weight-altitude-ambient air temperature limits charts — takeoff and landing (Sheet 1 of 3)

**WEIGHT — ALTITUDE — TEMPERATURE FOR TAKEOFF AND LANDING
PART B** $V_{TOSS} = 45 \text{ KIAS}$

GW TO 10,800 LBS (4899 kg)

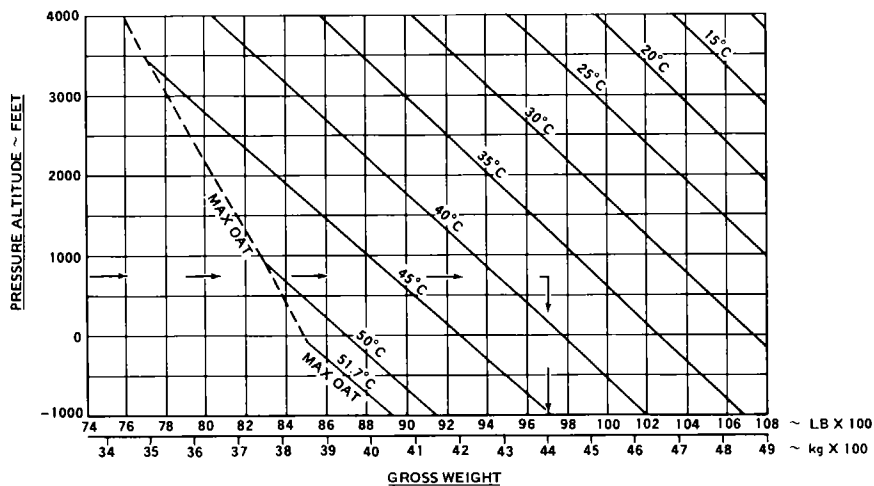


Figure 6-1. Gross weight-altitude-ambient air temperature limits charts — takeoff and landing (Sheet 2 of 3)

WEIGHT — ALTITUDE — TEMPERATURE LIMITATIONS FOR TAKEOFF AND LANDING PART C

$V_{TOSS} = 55 \text{ KIAS}$

GW TO 11,900 LBS (5398 kg)

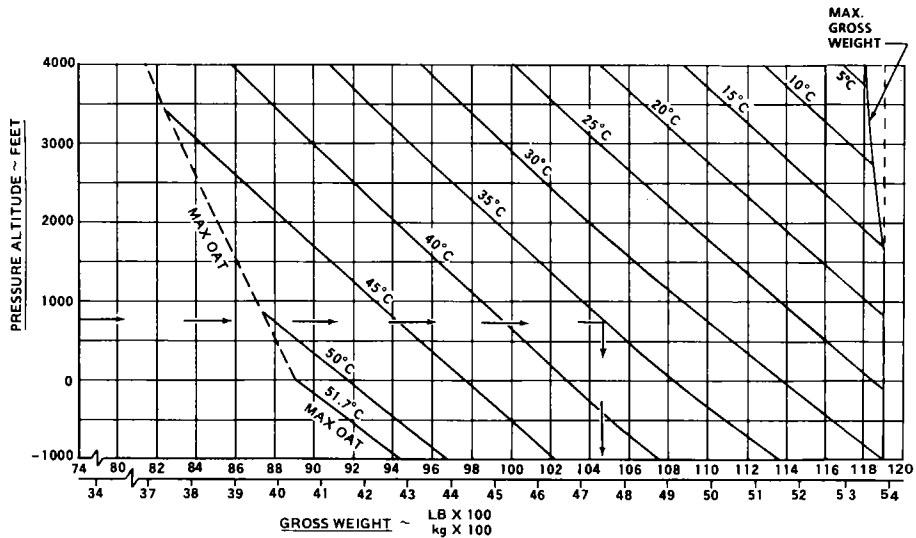


Figure 6-1. Gross weight-altitude-ambient air temperature limits charts — takeoff and landing (Sheet 3 of 3)

NORMAL PROCEDURES

PRIOR TO TAKEOFF

POWER ASSURANCE CHECK (refer to category "A" PERFORMANCE data).

STANDARD TYPE TAKEOFF

Obtain CDP information — Refer to figure 6-1 and 6-2.

Collective — Flat pitch.

ENG — 100% RPM (N2).

Altimeter — Set, note indication with collective fully down.

Instruments — Normal operating range.

SEAT BELT and NO SMOKE switches — As desired.

Area — Clear.

Hover at approximately 4 feet (1.2 meters) skid height and note torque.

Adjust ADI pitch bar to indicate level.

Initiate a takeoff from hover using a TRANSMISSION TORQUE of 10% above that required to hover and ten degrees nose down attitude.

NOTE

Do not exceed TRANSMISSION TORQUE, ITT, or GAS PROD RPM (N1) limits.

Maintain pitch attitude as the helicopter moves forward to achieve the correct Critical Decision Point (CDP) shown on the takeoff flight path profile diagram (figure 6-2).

NOTE

CDP height is determined by reference to the pilots barometric altitude. Indicated altitude with collective full down on the takeoff surface is used as a ground level reference.

After attaining CDP, accelerate the helicopter to 65 KIAS and continue the climb.

STANDARD TYPE LANDING

NOTE

A standard type landing is initiated from a Landing Decision Point (LDP) of 40 KIAS and an altitude of 100 feet (30.5 meters) above the runway, either in level flight or with a rate of descent of not more than 500 feet per minute (figure 6-3).

Flight controls — Adjust friction to desired level.

GOV switches — AUTO.

Throttles — Fully open.

ENG — 100% RPM (N2).

FORCE TRIM switch — As desired.

STEP switch — As desired.

Altimeter — Set to nearest reporting station.

SEAT BELT and NO SMOKE sign — As desired.

TAKE-OFF FLIGHT PATH PROFILE

2½ MINUTE POWER RATING
AFTER POWER FAILURE, THEN REDUCE
TO 30 MINUTE POWER AT 500 FT (152 m) AGL.

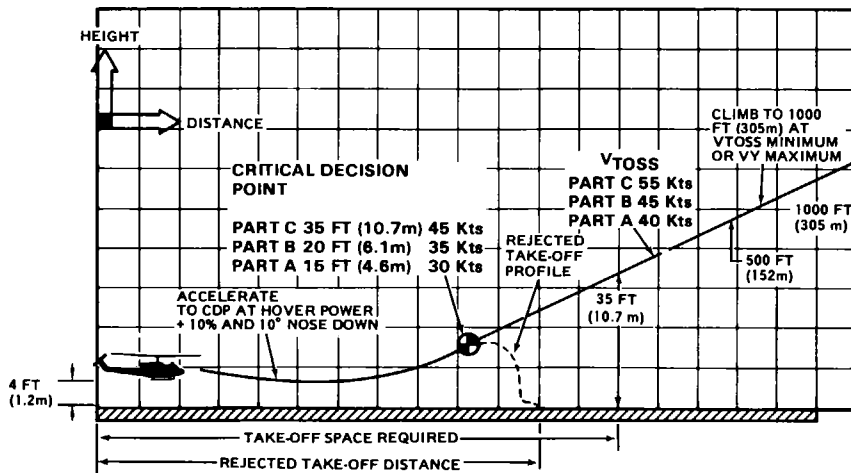


Figure 6-2. Takeoff flight path profile

LANDING FLIGHT PATH PROFILE

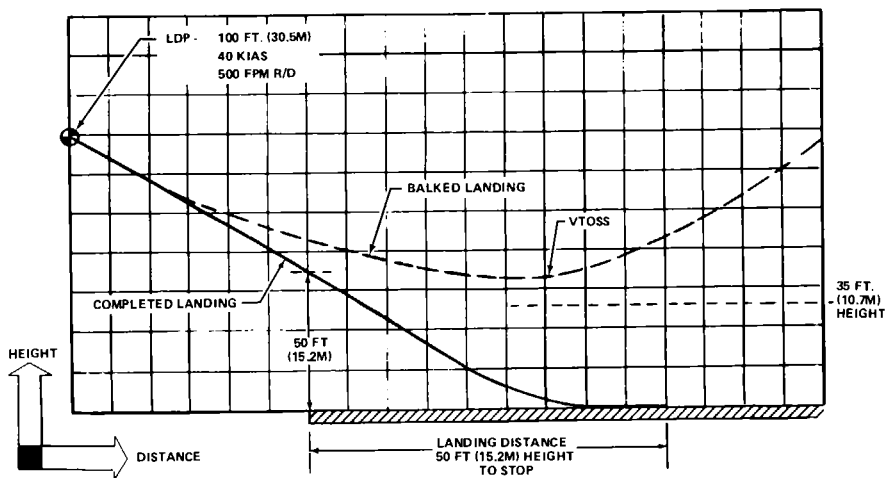


Figure 6-3. Landing flight path profile

EMERGENCY AND MALFUNCTION PROCEDURES

INTRODUCTION

Tables 6-1 through 6-4 list panel wording, fault conditions, and corrective actions for emergencies and malfunctions that might

occur during takeoff prior to CDP, during takeoff after CDP, during landing prior to LDP, and during landing after LDP.

Table 6-1. Warning lights — Takeoff prior to CDP

PANEL WORDING	FAULT CONDITION	CORRECTIVE ACTION
MASTER CAUTION	Warning or caution light(s) illuminated.	Land immediately.
FIRE PULL (1 or 2)	Fire indication in No. 1 or No. 2 engine compartment.	Land immediately. Pull affected FIRE PULL handle. Select MAIN fire extinguisher; if necessary, select RESERVE fire extinguisher.
BAGGAGE FIRE	Smoke in baggage compartment.	Land immediately. Inspect tailboom area for damage.
ENG OUT (1 or 2)	GAS PROD abnormally low, below $53 \pm 2\%$ RPM (N1), on No. 1 or No. 2 engine.	Land immediately. Refer to ENGINE OUT procedures.
X M S N PRESSURE	O I L Transmission oil pressure below limit.	Land immediately.
XMSN OIL TEMP	T r a n s m i s s i o n oil temperature above limit.	Land immediately.
C B O X PRESSURE	O I L Combining gearbox oil pressure below normal.	Land immediately.
C BOX TEMP	C o m b i n i n g gearbox temperature above limit.	Land immediately.

Table 6-1. Warning lights — Takeoff prior to CDP (Cont)

PANEL WORDING	FAULT CONDITION	CORRECTIVE ACTION
BATTERY TEMP	Battery case temperature above limits.	Land immediately. BATTERY BUS 1 and BUS 2 switches — OFF.
WARNING BATTERY SHALL NOT BE USED FOR ENGINE START AFTER ILLUMINATION OF <i>BATTERY TEMP</i> LIGHT. BATTERY SHALL BE REMOVED AND SERVICED IN ACCORDANCE WITH MANUFACTURER'S INSTRUCTIONS PRIOR TO RETURN TO SERVICE.		
ROTOR BRAKE	Rotor brake linings not retracted.	Land immediately.

Table 6-2. Warning lights — Takeoff after CDP

PANEL WORDING	FAULT CONDITION	CORRECTIVE ACTION
MASTER CAUTION	Warning or Caution light(s) illuminated.	Accelerate to V_{TOSS} . Reset MASTER CAUTION light; take appropriate corrective action as required by illuminated segment(s).
FIRE PULL (1 or 2)	Fire indication in No. 1 or No. 2 engine compartment.	Accelerate to V_{TOSS} . Pull affected FIRE PULL handle. Select MAIN fire extinguisher. Close throttle of affected engine. Select RESERVE fire extinguisher if necessary. Land as soon as possible.
BAGGAGE FIRE	Smoke in baggage compartment.	Land immediately. Inspect tailboom area for damage.
ENG OUT (1 or 2)	GAS PROD abnormally low, below $53 \pm 2\%$ RPM (N1), on No. 1 or No. 2 engine.	Accelerate to V_{TOSS} . Secure appropriate engine. Refer to ENGINE OUT procedure. Land as soon as possible.

Table 6-2. Warning lights — Takeoff after CDP (Cont)

PANEL WORDING	FAULT CONDITION	CORRECTIVE ACTION
X M S N O I L PRESSURE	Transmission oil pressure below limit.	Accelerate to V_{TOSS} . Reduce power; verify fault on XMSN OIL pressure gage. Land immediately.
XMSN OIL TEMP	Transmission oil temperature above limit.	Accelerate to V_{TOSS} . Reduce power; verify fault on XMSN OIL temperature gage. Land as soon as possible.
C B O X O I L PRESSURE	Combining gearbox oil pressure below normal.	Accelerate to V_{TOSS} . Reduce power; verify fault on GEAR BOX pressure gage. Land immediately.
C BOX TEMP	Combining gearbox oil temperature above limit.	Accelerate to V_{TOSS} . Reduce power; verify fault on GEAR BOX temperature gage. Land as soon as possible.
BATTERY TEMP	Battery case temperature above limits.	Accelerate to V_{TOSS} . BATTERY BUS 1 and BUS 2 switches — OFF. Land as soon as practical.
WARNING BATTERY SHALL NOT BE USED FOR ENGINE START AFTER ILLUMINATION OF BATTERY TEMP LIGHT. BATTERY SHALL BE REMOVED AND SERVICED IN ACCORDANCE WITH MANUFACTURER'S INSTRUCTIONS PRIOR TO RETURN TO SERVICE.		
ROTOR BRAKE	Rotor brake linings not retracted.	Accelerate to V_{TOSS} . Check rotor brake handle fully up in detent. If light remains on, land as soon as possible.

Table 6-3. Warning lights — Landing prior to LDP

PANEL WORDING	FAULT CONDITION	CORRECTIVE ACTION
MASTER CAUTION	Warning or caution light(s) illuminated.	Reset MASTER CAUTION light; take appropriate corrective action as required by illuminated segment.
FIRE PULL (1 or 2)	Fire indication in No. 1 or No. 2 engine compartment.	Pull affected FIRE PULL handle. Select MAIN fire extinguisher. Close throttle of affected engine. Select RESERVE fire extinguisher, if necessary. Land as soon as possible.
BAGGAGE FIRE	Smoke in baggage compartment.	Land immediately. Inspect tailboom area for damage.
ENG OUT (1 or 2)	GAS PROD abnormally low, below $53 \pm 2\%$ RPM (N1), on No. 1 or No. 2 engine.	Maintain V_{TOSS} . Secure appropriate engine. Land as soon as possible. Refer to ENGINE OUT procedure.
X M S N PRESSURE	O I L Transmission oil pressure below limit.	Reduce power. Verify fault on XMSN OIL pressure gage. Land immediately.
XMSN OIL TEMP	Transmission oil temperature above limit.	Reduce power. Verify fault on XMSN OIL temperature gage. Land as soon as possible.
C B O X PRESSURE	O I L Combining gearbox oil pressure below normal.	Reduce power. Verify fault on GEAR BOX pressure gage. Land as soon as possible.
C BOX OIL TEMP	Combining gearbox oil temperature above limit.	Reduce power. Verify fault on GEAR BOX temperature gage. Land as soon as possible.

Table 6-3. Warning lights — Landing prior to LDP (Cont)

PANEL WORDING	FAULT CONDITION	CORRECTIVE ACTION
BATTERY TEMP	Battery case temperature above limits.	BATTERY BUS 1 and BUS 2 switches — OFF. Land as soon as practical.
WARNING BATTERY SHALL NOT BE USED FOR ENGINE START AFTER ILLUMINATION OF <i>BATTERY TEMP</i> LIGHT. BATTERY SHALL BE REMOVED AND SERVICED IN ACCORDANCE WITH MANUFACTURER'S INSTRUCTIONS PRIOR TO RETURN TO SERVICE.		
ROTOR BRAKE	Rotor brake linings not retracted.	Check rotor brake handle fully up in detent. If light remains on, land as soon as possible.

Table 6-4. Warning lights — Landing after LDP

PANEL WORDING	FAULT CONDITION	CORRECTIVE ACTION
MASTER CAUTION	Warning or caution light(s) illuminated.	Land immediately.
FIRE PULL (1 or 2)	Fire indication in No. 1 or No. 2 engine compartment.	Land immediately. Pull affected FIRE PULL handle. Close throttle of affected engine. Select MAIN fire extinguisher; if necessary, select RESERVE fire extinguisher.
BAGGAGE FIRE	Smoke in baggage compartment.	Land immediately. Inspect tailboom area for damage.
ENG OUT (1 or 2)	GAS PROD abnormally low, below $53 \pm 2\%$ RPM (N1), on No. 1 or No. 2 engine.	Land immediately. Refer to ENGINE OUT procedures.
X M S N PRESSURE	O I L Transmission oil pressure below limit.	Land immediately.

Table 6-4. Warning lights — Landing after LDP (Cont)

PANEL WORDING	FAULT CONDITION	CORRECTIVE ACTION
XMSN OIL TEMP	Transmission oil temperature above limit.	Land immediately.
C BOX OIL PRESSURE	Combining gearbox oil pressure below normal.	Land immediately.
C BOX TEMP	Combining gearbox temperature above limit.	Land immediately.
BATTERY TEMP	Battery case temperature above limits.	Land immediately. BATTERY BUS 1 and BUS 2 switches — OFF.

WARNING

BATTERY SHALL NOT BE USED FOR ENGINE START AFTER ILLUMINATION OF BATTERY TEMP LIGHT. BATTERY SHALL BE REMOVED AND SERVICED IN ACCORDANCE WITH MANUFACTURER'S INSTRUCTIONS PRIOR TO RETURN TO SERVICE.

ROTOR BRAKE	Rotor brake linings not retracted.	Land immediately.
-------------	------------------------------------	-------------------

ENGINE OUT**DURING TAKEOFF PRIOR TO CRITICAL DECISION POINT (CDP)**

An engine failure prior to reaching CDP will necessitate a landing back to the takeoff surface. If height permits, a positive deceleration to reduce forward airspeed is required. As the helicopter descends, it should be leveled and the collective should be used as required to cushion the landing. Some forward ground speed is normally required at touchdown.

Maintain control of the helicopter.

Collective — Adjust to maintain ROTOR RPM and OEI power limits.

Flare to reduce ground speed.

Assume landing attitude before touchdown.

Throttle (affected engine) — Closed.

Complete shutdown of affected engine.

DURING TAKEOFF AFTER CRITICAL DECISION POINT (CDP)

In the event of an engine failure following CDP, airspeed should be increased to the

takeoff safety speed (V_{TOSS}) or maintained, whichever is higher. Climb out to 500 feet (152 meters) above the takeoff surface and accelerate to 65 KIAS. Reduce power to 30 minute limit.

CAUTION

DURING COLD WEATHER OPERATIONS, CAREFULLY MONITOR *TORQUE* OF THE NORMAL ENGINE WHEN ONE ENGINE FAILS OR IS SHUT DOWN IN FLIGHT.

NOTE

During takeoff, after CDP, it is permissible to droop ROTOR RPM to 91% during the transition from twin engine to single engine flight following an engine failure. ROTOR RPM should be regained to normal operating range at or before attaining appropriate best rate of climb speed.

Maintain control of the helicopter.

Collective — Adjust to maintain ROTOR RPM and OEI power limits.

Airspeed — If below V_{TOSS} , smoothly increase to V_{TOSS} and initiate a climb.

Throttle (affected engine) — Close.

Complete shutdown of affected engine.

ENG (unaffected engine) — Set to 100% RPM (N2).

DURING LANDING PRIOR TO LANDING DECISION POINT (LDP)

Execute the same procedures as for single engine failure on takeoff after CDP or proceed to LDP and use the procedure below.

DURING LANDING AFTER THE LANDING DECISION POINT (LDP)

The helicopter, with an emergency, is committed to land after LDP. The landing is accomplished using up to the maximum power of the remaining engine while maintaining rotor speed within limits.

Maintain control of the helicopter.

Collective — Adjust to maintain ROTOR RPM and OEI power limits.

Flare to reduce speed.

Assume landing attitude before touchdown.

Throttle (affected engine) — Closed.

Complete shutdown of affected engine.

PERFORMANCE

CATEGORY PERFORMANCE

“ A ”

headwind component chart, is applied to parts A, B, and C of the Takeoff Space Required charts.

The power performance data presented in this section is based on engine manufacturers minimum specification power for the PT6T-3B engine with installation losses.

The takeoff and landing data presented in this section is based on tests performed on a level asphalt running 75 feet wide. The minimum runway length for standard takeoff and landing procedures varies with wind, gross weight, pressure altitude, and temperature.

POWER ASSURANCE CHECKS

Refer to Section 4 for power assurance charts to determine if the engine (power sections) can produce installed specification power.

The hover check is performed prior to takeoff. The in-flight check is provided for in-flight monitoring of engine performance. If either engine (power section) does not meet the requirements of the hover or in-flight power assurance check, category “A” performance will not be achievable. The cause of engine power loss, or excessive interturbine temperature (ITT) or GAS PROD RPM (N1) shall be determined as soon as practical. Refer to appropriate engine maintenance manual.

HEADWIND COMPONENT

The Unfactored Headwind Component chart (figure 6-4) is provided with an example to determine critical crosswind and corrected headwind for category “A” takeoff and landings. The headwind component, as calculated from the

REJECTED TAKEOFF DISTANCE REQUIRED

The rejected takeoff distance required is the space necessary to takeoff, climb to CDP, encounter an engine failure at CDP, return to takeoff surface, and stop safely. The rejected takeoff distance required is obtained from either part A, B, or C of the Rejected Takeoff Distance Required charts (figure 6-5).

TAKEOFF SPACE REQUIRED

The takeoff space required is the horizontal distance required to takeoff, climb to CDP, encounter an engine failure, accelerate to V_{TOSS} , and climb to 35 feet (10.7 meters) AGL above the takeoff space. Takeoff space required is obtained from either part A, B, or C of the Takeoff Space Required charts (figure 6-6) using the headwind component from the Unfactored Headwind Component chart (figure 6-4).

TAKEOFF FLIGHT PATH

The takeoff flight path begins at the end of Takeoff Space Required, at 35 feet (10.7 meters) AGL or higher, above the takeoff space and at V_{TOSS} . Parts A, B, and C of the Takeoff Flight Path charts (figure 6-7) provide data for 35 to 500 feet (10.7 to 152 meters) and 500 to 1,000 feet (152 to 305 meters) AGL. These charts provide altitude gain for each 100 feet (30.5 meters) horizontal distance traveled.

LANDING SPACE REQUIRED

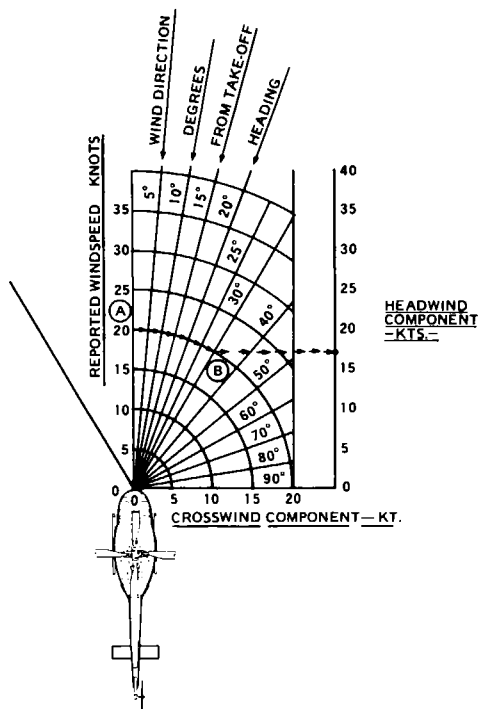
Landing space required is the distance necessary to come to a stop from LDP with one engine inoperative.

Landing space required is obtained from the Landing Space Required chart (figure 6-8).

The headwind component is obtained from calculation of the Unfactored Headwind Component chart (figure 6-4), and applied to the Landing Space Required chart (figure 6-8) to obtain corrected landing distance.

CORRECTED LANDING DISTANCE

Corrected landing distance from LDP is landing distance corrected for wind factor.



EXAMPLE

1. TAKE OFF HEADING 170°
2. REPORTED WIND DIRECTION 200°
3. WIND DIRECTION, DEGREES FROM TAKE-OFF HEADING 30°
4. REPORTED WIND SPEED 20 KNOTS
5. ENTER CHART AT REPORTED WIND SPEED, POINT A
6. FOLLOW THE SHAPE OF THE CURVED LINES, TO WIND DIRECTION, DEGREES FROM TAKE-OFF HEADING, POINT B.
7. PROCEED HORIZONTALLY TO THE HEADWIND COMPONENT SCALE AND READ HEADWIND COMPONENT 17 KNOTS
8. TAILWINDS HAVE NOT BEEN DEMONSTRATED

Figure 6-4. Unfactored headwind component chart

REJECTED TAKE-OFF DISTANCE REQUIRED

PART A

GW 7,500 TO 10,000 LB (3402 TO 4536 Kg)

 $V_{TOSS} = 40$ KIAS

CDP = 30 KIAS AT 15 FT (4.6m)

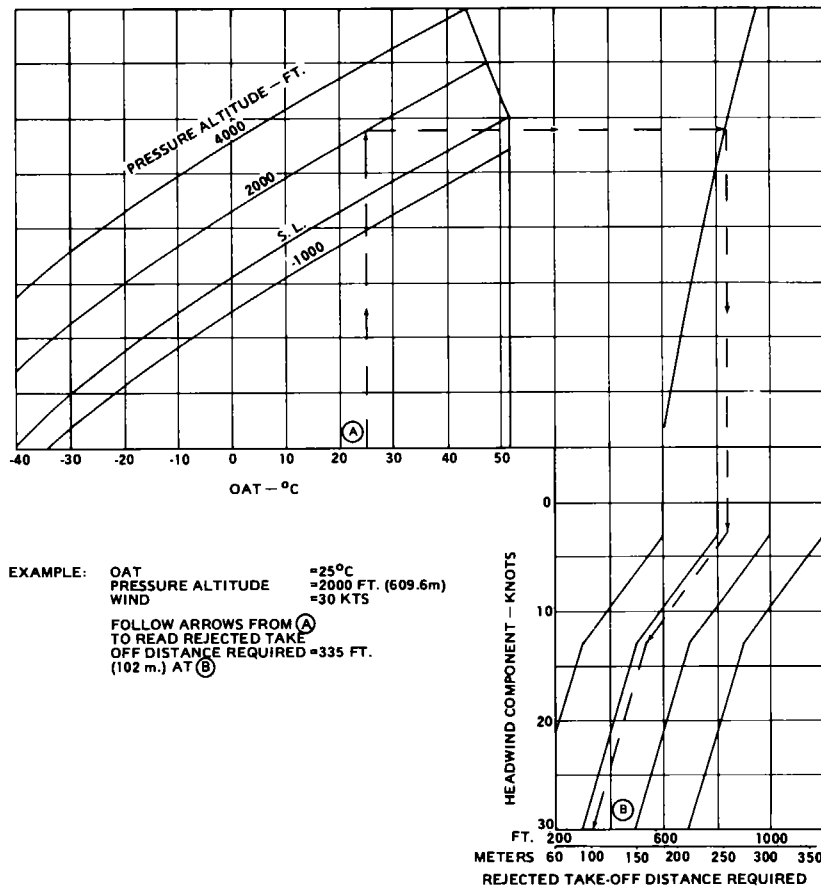


Figure 6-5. Rejected takeoff distance required (Sheet 1 of 3)

REJECTED TAKE-OFF DISTANCE REQUIRED

PART B

GW 7,500 TO 10,800 LB (3402 TO 4899 Kg)

V_{TOSS} = 45 KIAS

CDP = 35 KIAS AT 20 FT (6.1m)

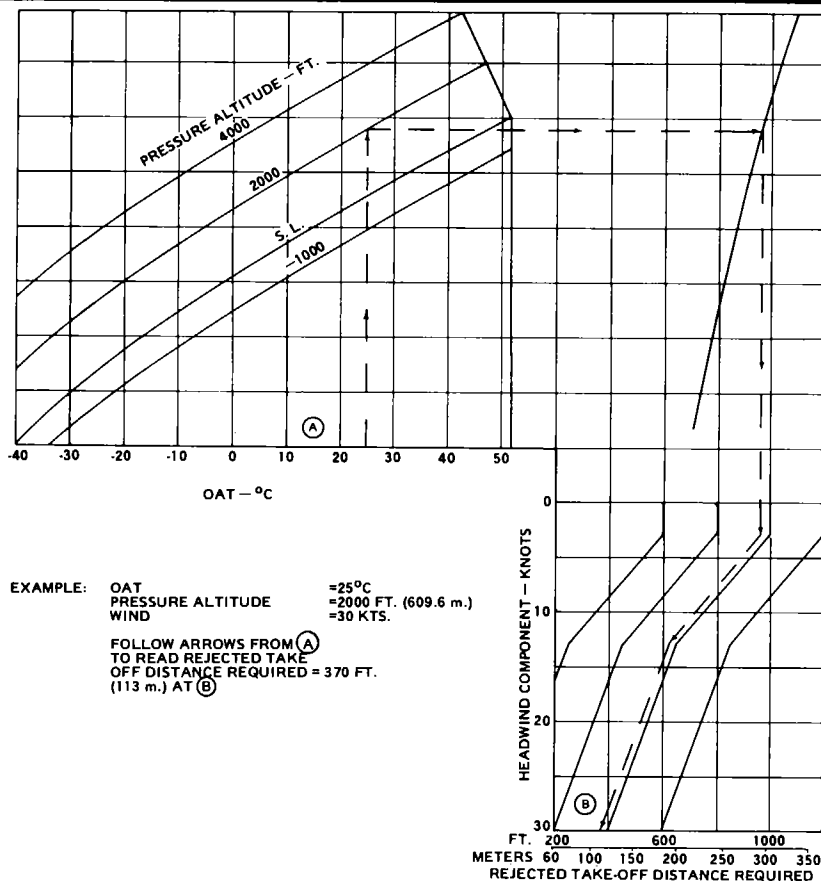


Figure 6-5. Rejected takeoff distance required (Sheet 2 of 3)

REJECTED TAKE-OFF DISTANCE REQUIRED

PART C

GW 7,500 TO 11,900 LB (3402 TO 5398 Kg)

V_{TOSS} = 55 KIAS

CDP = 45 KIAS AT 35 FT (10.7m)

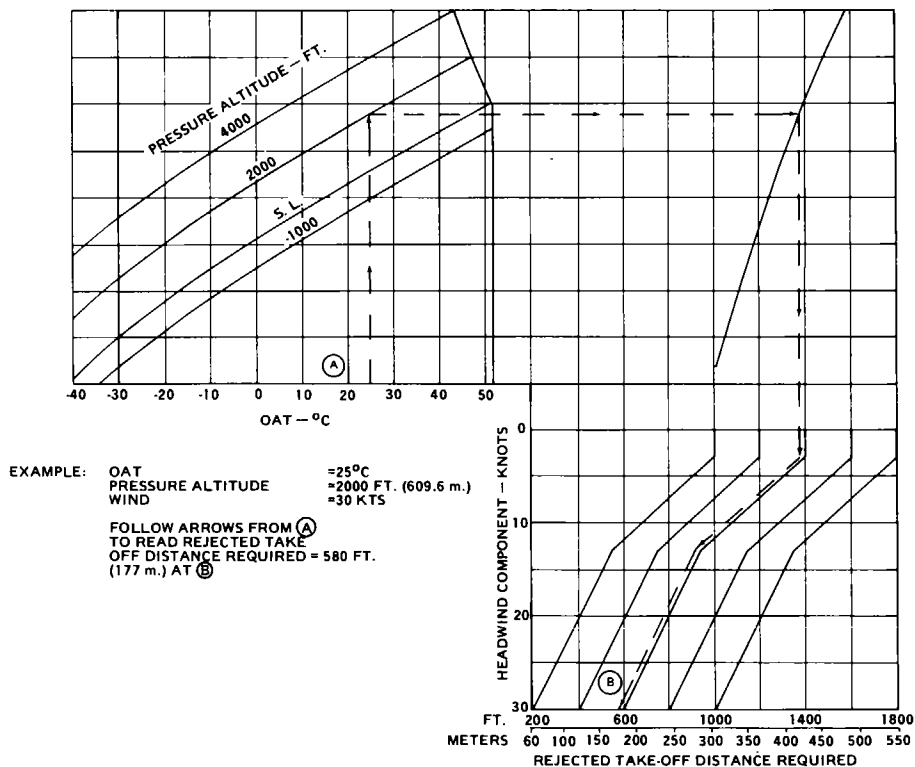


Figure 6-5. Rejected takeoff distance required (Sheet 3 of 3)

TAKE-OFF SPACE REQUIRED

PART A

GW 7,500 TO 10,000 LB (3402 TO 4536 Kg)

 $V_{TOSS} = 40$ KIAS

CDP = 30 KIAS AT 15 FT (4.6m)

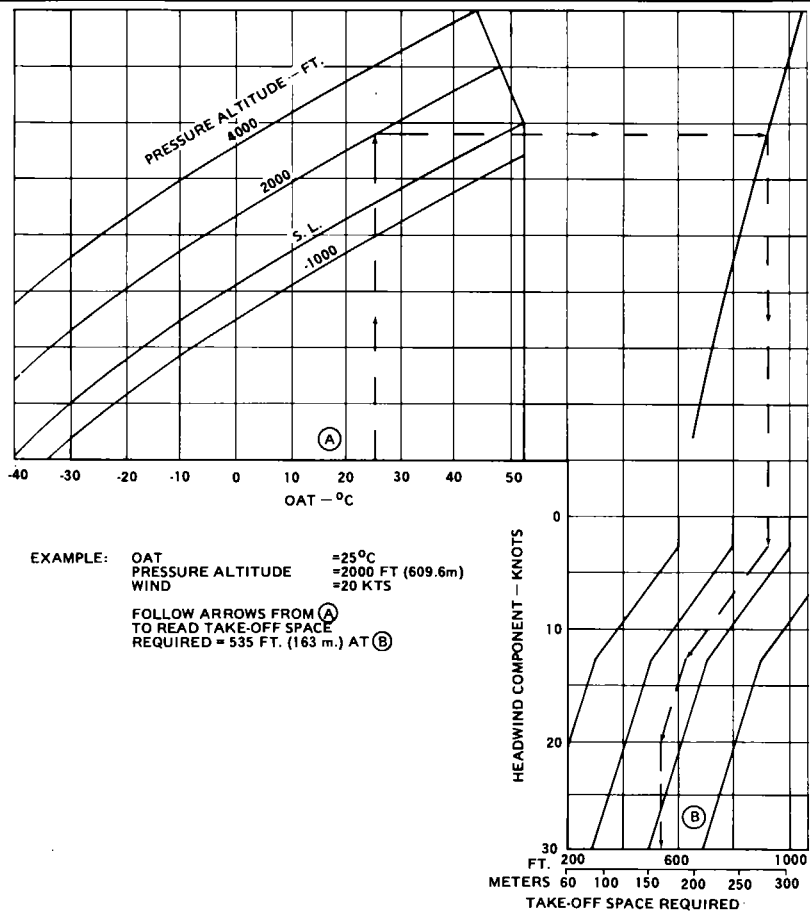


Figure 6-6. Takeoff space required chart (Sheet 1 of 3)

TAKE-OFF SPACE REQUIRED

PART B

GW 7,500 TO 10,800 LB (3402 TO 4899 Kg)

 $V_{TOSS} = 45$ KIAS

CDP = 35 KIAS AT 20 FT (6.1m)

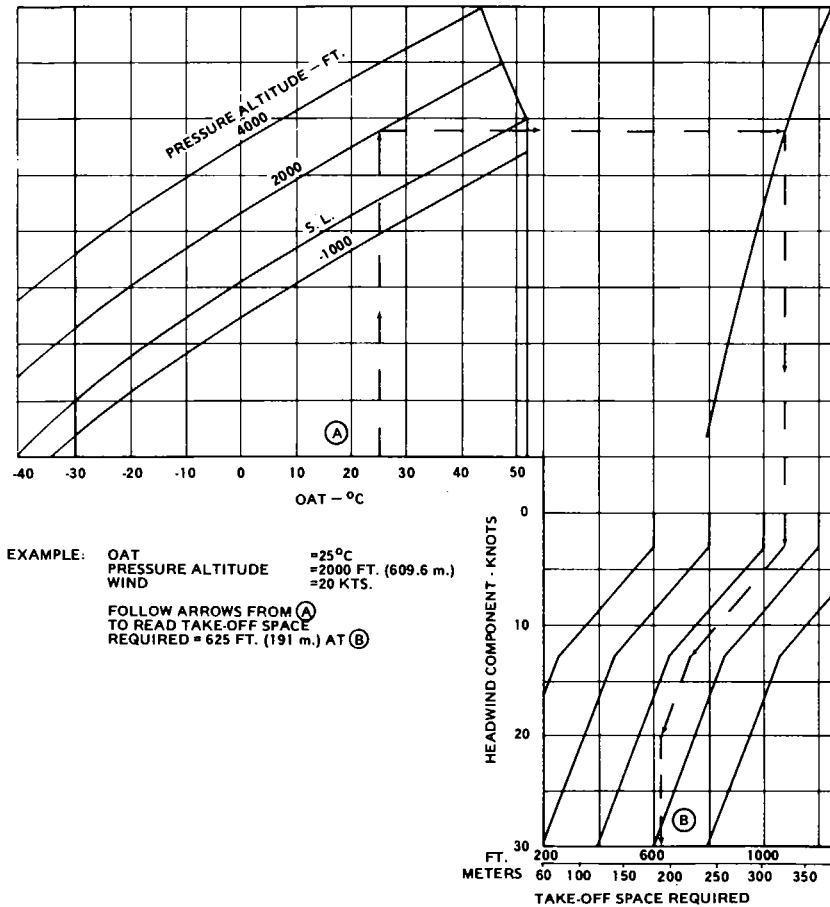


Figure 6-6. Takeoff space required chart (Sheet 2 of 3)

TAKE-OFF SPACE REQUIRED

PART C

GW 7,500 TO 11,900 LB (3402 TO 5398 Kg)

 $V_{TOSS} = 55$ KIAS

CDP = 45 KIAS AT 35 FT (10.7m)

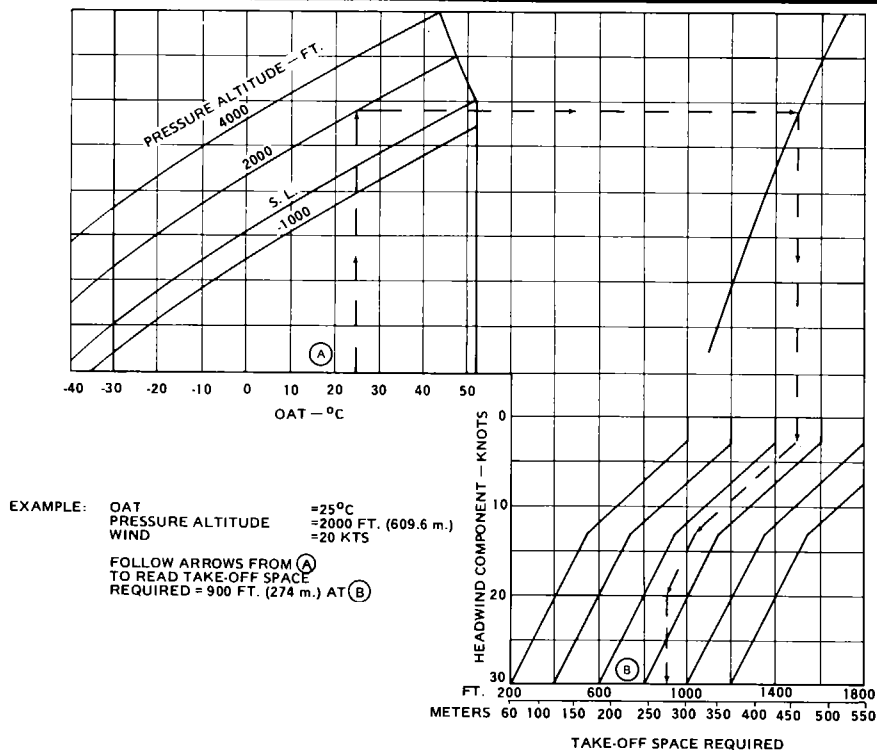


Figure 6-6. Takeoff space required chart (Sheet 3 of 3)

TAKEOFF FLIGHT PATH 35 TO 500 FEET AGL **PART A**

2.5 MINUTE OEI POWER
ENG - 97% RPM (N2)
GENERATOR 150 AMPS

$V_{\text{toss}} = 40 \text{ KIAS}$
HEATER AND ANTI-ICE OFF
INOPERATIVE ENGINE SECURED

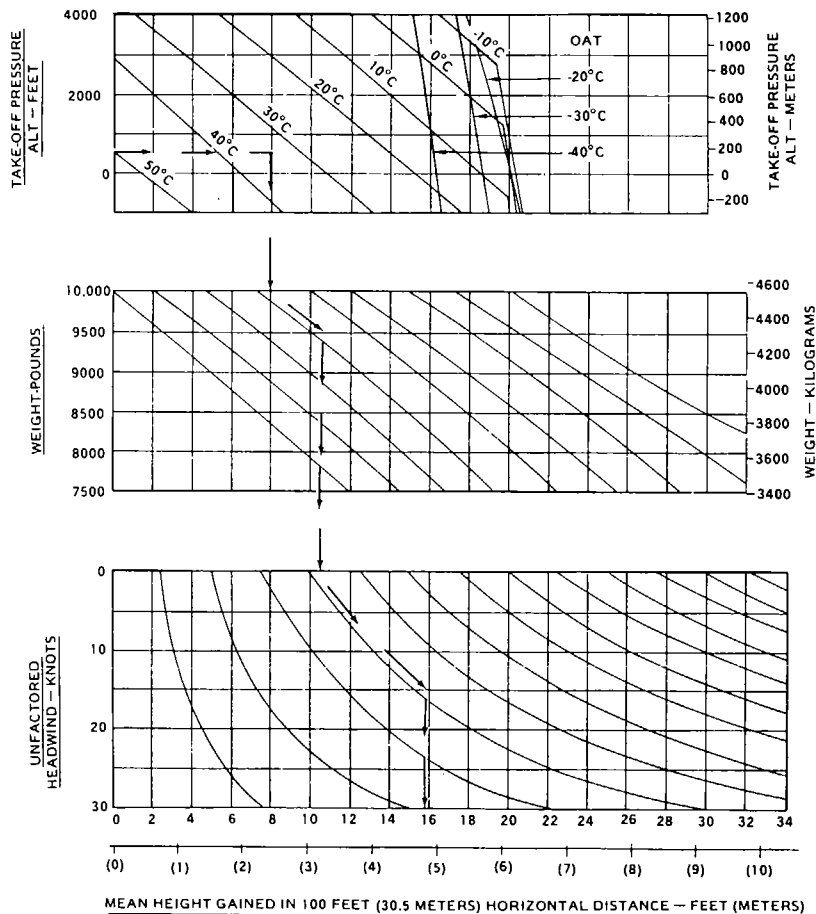


Figure 6-7. Takeoff flight path (Sheet 1 of 6)

TAKEOFF FLIGHT PATH 500 TO 1000 FEET AGL PART A

30 MINUTE OEI POWER
ENG - 97% RPM (N2)
GENERATOR 150 AMPS

$V_{\text{loss}} = 40 \text{ KIAS}$
HEATER AND ANTI-ICE OFF
INOPERATIVE ENGINE SECURED

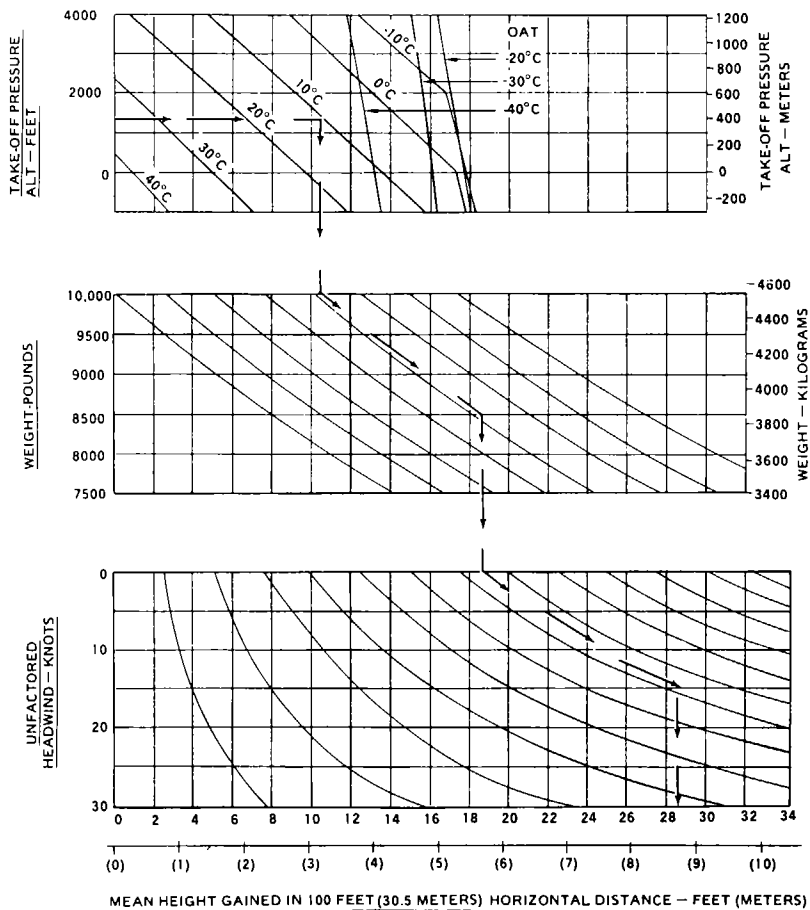


Figure 6-7. Takeoff flight path (Sheet 2 of 6)

TAKEOFF FLIGHT PATH 35 TO 500 FEET AGL **PART B**

2.5 MINUTE OEI POWER
ENG — 97% RPM (N2)
GENERATOR 150 AMPS

$V_{\text{loss}} = 45 \text{ KIAS}$
HEATER AND ANTI-ICE OFF
INOPERATIVE ENGINE SECURED

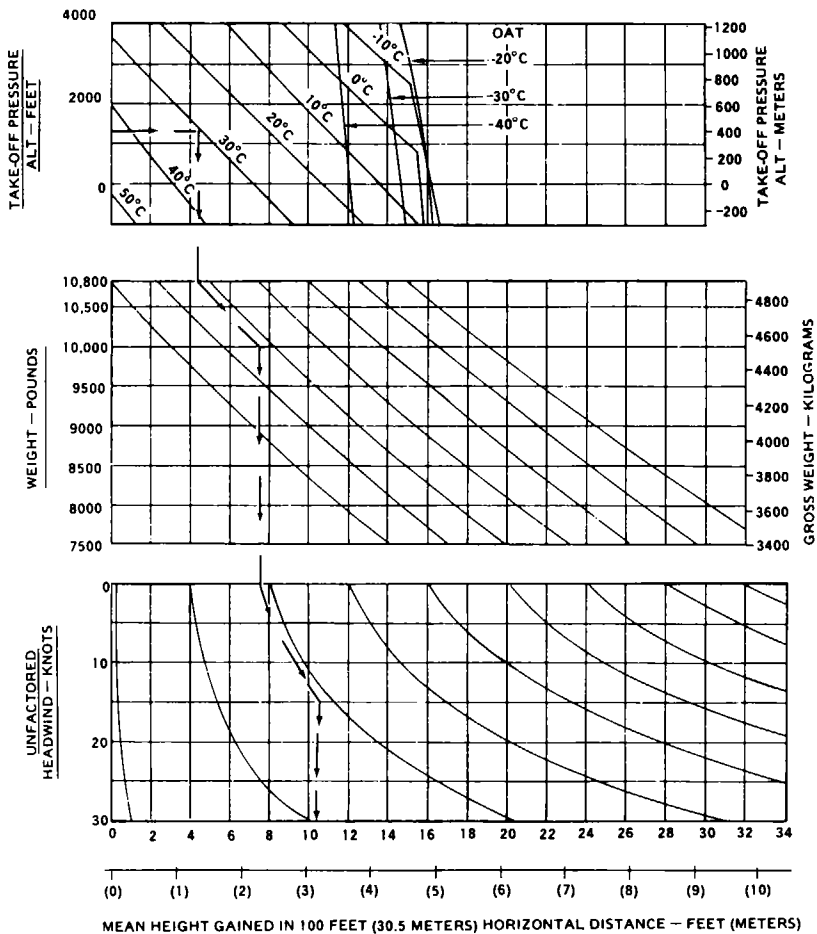


Figure 6-7. Takeoff flight path (Sheet 3 of 6)

**TAKEOFF FLIGHT PATH 500 TO 1000 FEET AGL
PART B**

30 MINUTE OEI POWER
ENG — 97% RPM (N2)
GENERATOR 150 AMPS

$V_{\text{loss}} = 45 \text{ KIAS}$
HEATER AND ANTI-ICE OFF
INOPERATIVE ENGINE SECURED

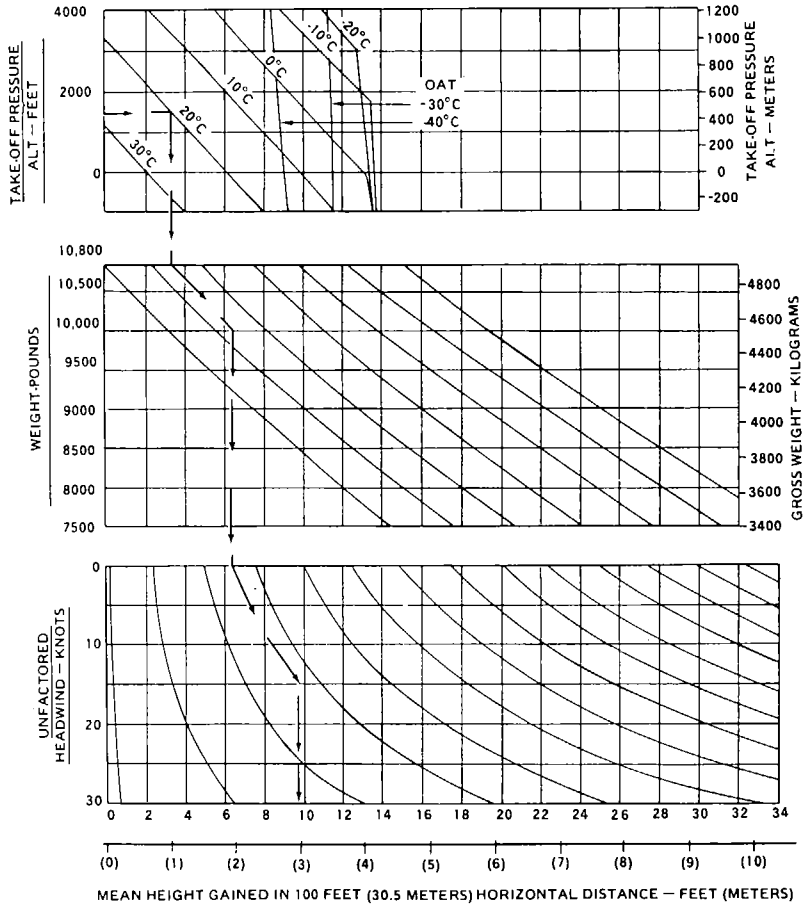


Figure 6-7. Takeoff flight path (Sheet 4 of 6)

TAKEOFF FLIGHT PATH 35 TO 500 FEET AGL **PART C**

2.5 MINUTE OEI POWER
ENG — 97% RPM (N2)
GENERATOR 150 AMPS

$V_{\text{toss}} = 55 \text{ KIAS}$
HEATER AND ANTI-ICE OFF
INOPERATIVE ENGINE SECURED

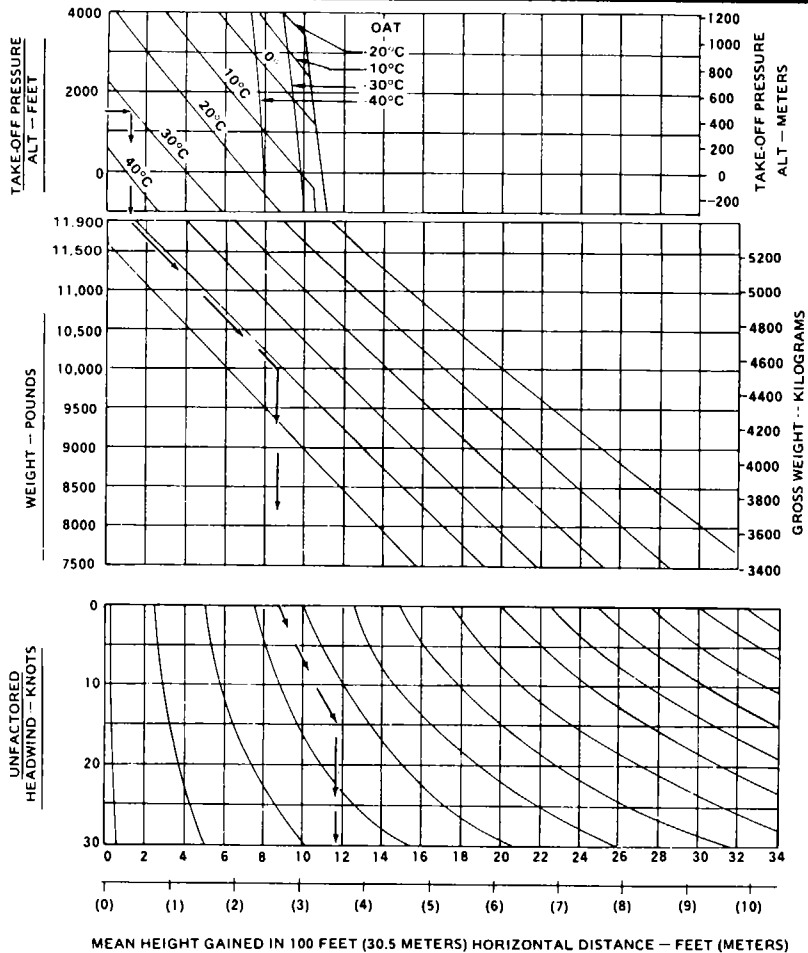


Figure 6-7. Takeoff flight path (Sheet 5 of 6)

TAKEOFF FLIGHT PATH 500 TO 1000 FEET AGL PART C

30 MINUTE OEI POWER
ENG - 97% RPM (N2)
GENERATOR 150 AMPS

$V_{\text{toss}} = 55 \text{ KIAS}$
HEATER AND ANTI-ICE OFF
INOPERATIVE ENGINE SECURED

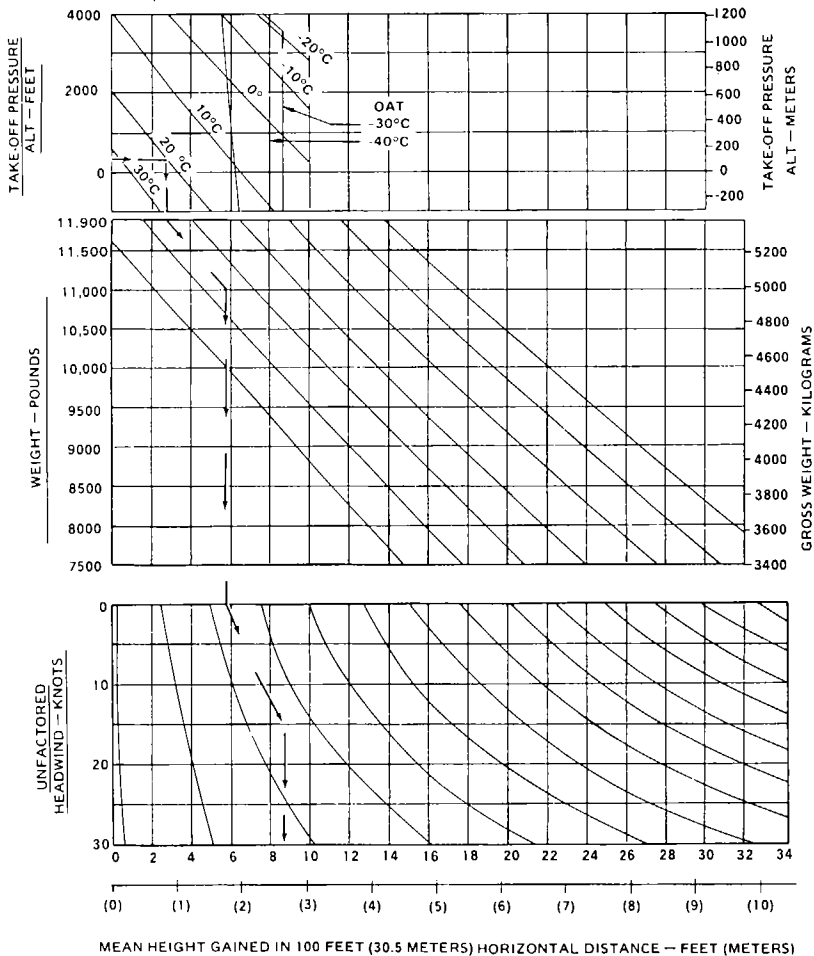


Figure 6-7. Takeoff flight path (Sheet 6 of 6)

LANDING SPACE REQUIRED — 50 FT. HEIGHT TO STOP

GW 7,500 TO 11,900 LB (3402 TO 5398 Kg)

LANDING DECISION POINT 100 FEET (30.5 METERS) @ 40 KIAS

RATE OF DESCENT 500 FT/MIN.

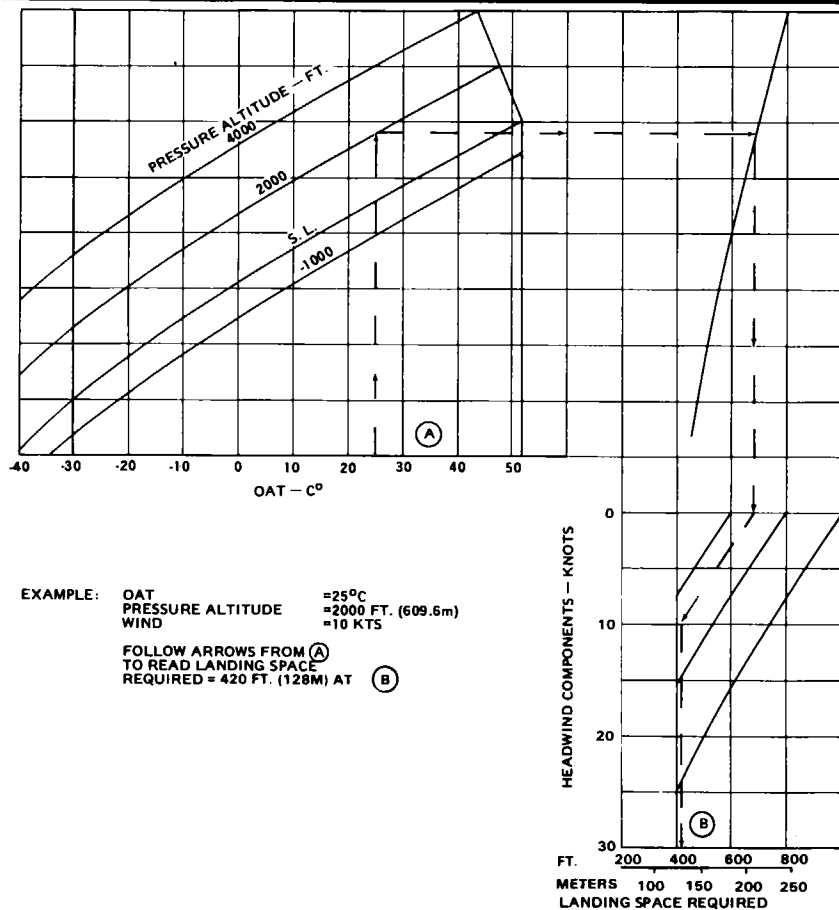


Figure 6-8. Landing space required chart